IEEE P802.1AE–2018-Rev/D1.2
December 10, 2025
(Revision of IEEE Std 802.1AE™-2018 as amended by IEEE Std 802.1AE-2018/Cor1-2020,
IEEE Std 802.1AEdk™-2023 and IEEE P802.1AEef/D0.1)

Draft Standard for
Local and Metropolitan Area Networks—

Media Access Control (MAC) Security

Prepared by the
IEEE 802.1 Working Group

Developed by

LAN/MAN Standards Committee
of the
IEEE Computer Society

This and the following cover pages are not part of the draft. They provide revision and other information for IEEE 802.1 Working Group members and will be updated as convenient. New participants: Please read these cover pages, they contain information that should help you contribute effectively to this standards development project. The Introduction to the current draft should be useful to all readers.

The text proper of this draft begins with the Title page.

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Introduction to the current draft[1]

This document is a revision in progress of IEEE Std 802.1AE-2018, incorporating published amendments, draft amendments (if, and as, noted on the Title page), the agreed or proposed resolution of Maintenance items (see below), and additional Technical corrections and Editorial corrections to the description of existing functionality.

The Scope of the Revision project includes, in addition to technical and editorial corrections to the description of the functionality provided by the base standard., IEEE Std 802.1AE-2018 and the approved IEEE Std 802.1AEdk-2023 amendment, any amendments and corrigenda approved at the time of project completion. At present the intention is to roll in the text developed for the P802.1AEef 'Ascon Cipher Suite', anticipating that project will complete prior to the Revision. The production of ongoing roll up drafts is intended to avoid the problems previously experienced with the editorial merging of approved amendments with the base standard, by facilitating review of the real effect of the amendment project as it proceeds rather than after the fact. That being said, comments on the text added or changed by the P802.1AEef amendment project should be directed to that project, its ballots, and its drafts. This cover note (in the editor's Introduction to the P802.1-2018-Rev drafts) will focus on changes additional to those introduced in P802.1AEef.

Beginning with P802.1AE-2018-Rev/D1.2, drafts include P802.1AEef changes. The intent is to show, in each draft, the final text resulting from the corrections and the roll up of an identified P802.1AEef draft. If useful, convenient, and feasible, a change version showing all changes (whether corrections made by this project or resulting from the roll up) may be produced, but whether or not that will be feasible is to be determined.

These cover pages also include an introduction to Participation in 802.1 standards development, a summary of the PAR (Project Authorization Request),and a general discussion of Draft development.

P802.1AE-2018-Rev/D1.2

This draft rolls in P802.1AEef/D0.1.

Maintenance items

This draft includes the following proposed or agreed resolutions of maintenance items that were not addressed by rolled-up amendments.

— None to date.

Technical corrections

This draft does not include any technical corrections beyond those addressed by the rolled-up amendments.

Editorial corrections

This draft includes the following editorial changes that do not appear in the amendments:

— During the development of P802.1AEdk, updates to Figure 13-2 were prepared, but not included in the draft. IEEE Std 802.1dk-2023 did update Figure 13-1, but that was unnecessary (the replacement figure did not make any change). This draft includes the missing Figure 13-2 update, which is editorial in that it simply summarizes what is in the MIB and makes no normative change.

— Text was added in pre-publication editing to add a reference to Figure 20-2 (IEEE editorial policy requires that any Figure in the document be referenced in text). The addition appeared a little

[1]The whole or parts of the introduction, possibly updated, to past drafts may be retained at the Editor's discretion, with the most recent introduction first. The introduction to each draft may solicit input on specific subjects.

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awkward as it was immediately under a heading introducing the figure an just above the titled Figure itself, but met the requirement of clearly making no normative change. This draft relocates that reference to the start of the following text.

— In the third paragraph of Clause 23 of the published IEEE Std 802.1AEdk, there were references to clauses 23.8.2 through 23.8.5. The referenced material had been consolidated into 23.8.1 after draft 1.5. Since 802.1AEdk was developed prior to using consolidated sources for amendments and roll-ups this error was not spotted, as unsatisfied references are common in stand-alone amendments. The unwanted cross-references have been removed.

Keywords

The following Keywords have been added or removed to the Keywords in the frontmatter of the standard (following the Title Page proper which lists the base standard for this revision plus incorporated amendments), following the abstract. The starting point for the Keyword list was IEEE Std 802.1AE-2018.

— No changes to date.

MIB modules

The MIB modules specified by this standard are attached to the draft pdf as plain text (UTF-8) .mib files.

YANG modules

The MIB modules specified by this standard are attached to the draft pdf as plain text (UTF-8) .mib files.

As noted above, the following YANG modules in this draft have been updated as a result of maintenance activity not addressed by the included amendments:

— No changes to date.

A Revision statement has been added to each (if any) of these modules indicating that they are “TO BE Published ..” as part of the Revision, together with an indication of the change. The “TO BE ” is to be removed and the revision date updated at pre-publication time [the revision statements in published standard reflects only the historical derivation of each published (i.e. standardized) module version from its published predecessors. The description of the change from the prior published version should be retained. Note that these updated modules are not ‘the standard’ at this time.

Sources

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The IEEE SA PAR (Project Authorization Request) is summarized in these cover pages and a link provided to the full text. As part of the IEEE 802® process, the PAR of each project is reviewed regularly to ensure its continued validity. A vote of “Approve” on this draft is also an affirmation that the PAR and CSD is still valid.

Comments on this draft are encouraged. NOTE: All issues related to IEEE standards presentation style, formatting, spelling, etc. are routinely handled between the 802.1 Editor and the IEEE Staff Editors prior to publication, after balloting and the process of achieving agreement on the technical content of the standard is complete. Readers are urged to devote their valuable time and energy only to comments that materially affect either the technical content of the document or the clarity of that technical content. Comments should not simply state what is wrong, but also what might be done to fix the problem.

Full participation in the work of IEEE 802.1 requires attendance at IEEE 802 meetings. Information on 802.1 activities, working papers, and email distribution lists etc. can be found on the 802.1 Website:

http://ieee802.org/1/

Use of the email distribution list is not presently restricted to 802.1 members, and the working group has a policy of considering comments from all who are interested and willing to contribute to the development of the draft. Individuals not attending meetings have helped to identify sources of misunderstanding and ambiguity in past projects. The email lists exist primarily to allow the members of the working group to develop standards, and are not a general forum. All contributors to the work of 802.1 should familiarize themselves with the IEEE patent policy and anyone using the email distribution list will be assumed to have done so. Information can be found at http://standards.ieee.org/db/patents/

Comments on this draft may be sent to the 802.1 email exploder, to the Editor, or to the Chair of the 802.1 Working Group..

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As part of our IEEE 802 process, the text of the PAR and CSD (Criteria for Standards Development, formerly referred to as the 5 Criteria or 5C's) is reviewed on a regular basis in order to ensure their continued validity. A vote of “Approve” on this draft is also an affirmation by the balloter that the PAR is still valid.

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PAR (Project Authorization Request)

The PAR for the Revision of IEEE Std 802.1AE-2018 was approved by IEEE NesCom, 10th September, 2025:

https://www.ieee802.org/1/files/public/docs2025/ae-2018-rev-approved-PAR-0925.pdf

This is a maintenance PAR so did not require an IEEE 802 CSD (Criteria for Standards Development).

The Scope and Purpose remains unchanged from IEEE Std 802.1AE-2018.

The Need for the Project is as follows:

“Revision of this successful standard, published in 2018, is timely. This project will incorporate the existing approved amendment, IEEE Std 802.1AEdk-2023. It will also address editorial and technical corrections to existing functionality and incorporate any approved amendments and corrigenda at the time of project completion.”

Draft development

During the early stages of draft development, 802.1 editors have a responsibility to attempt to craft technically coherent drafts from the resolutions of ballot comments and from the other discussions that take place in the working group meetings. Preparation of drafts often exposes inconsistencies in editor’s instructions or exposes the need to make choices between approaches that were not fully apparent in the meeting. Choices and requests by the editors’ for contributions on specific issues will be found in the editors’ Introduction to the current draft and at appropriate points in the draft.

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The ballot comments received on each draft, and the editors’ proposed and final disposition of comments on working group drafts, are part of the audit trail of the development of the standard and are available, along with all the revisions of the draft on the 802.1 website (for address see above).

During the early stages of draft development the proposed text can be moved around a great deal, and even minor rearrangement can lead to a lot of ‘change’, not all of which is noteworthy from the point of the reviewer, so the use of automatic change bars is not very effective. In early drafts change bars may be omitted or applied manually, with a view to drawing the readers attention to the most significant areas of change. Readers interested in viewing every change are encouraged to use Adobe Acrobat to compare the document with their selected prior draft. Note that the FrameMaker change bar feature is useless when it comes to indicating changes to Figures.

This draft has been prepared from a set of Framemaker files with conditional text that supports the production of an amendment draft and a preliminary roll up of that amendment draft into the text of the base standard, i.e. IEEE Std 802.1AE as of the last Revision as amended by prior amendments (usually as of the close of their successful SA ballots) as noted on the Title Page and the first Cover Page. The editor may make preliminary roll ups available to check consistency with the base standard and cross-references to text that does not appear in this amendment. Roll ups may also be recorded as part of the approved P802.1AE Revision project.

For a description of the use of conditional text and other FrameMaker and Style considerations applicable to this draft see the EDITOR-PLEASE-READ-ME file in the FrameMaker books used to generate these drafts.

Records of participants in the development of the standard are added after SA Ballot, as part of pre-publication editing by IEEE Staff.

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IEEE Standard for Local and Metropolitan Area Networks—

Media Access Control (MAC) Security

Prepared by the
Security Task Group of IEEE 802.1

Developed by the
LAN/MAN Standards Committee
of the
IEEE Computer Society

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Abstract: This standard specifies how all or part of a network can be secured transparently to peer protocol entities that use the MAC Service provided by IEEE 802® LANs to communicate. MAC security (MACsec) provides connectionless user data confidentiality, frame data integrity, and data origin authenticity.

Keywords: Ascon, authorized port, bridged networks, confidentiality, data origin authenticity, IEEE 802.1AE™, IEEE 802.1AEbn™, IEEE 802.1AEbw™, IEEE 802.1AEcg™, IEEE 802.1AEdk™, GCM, integrity, LANs, local area networks, MAC Bridges, MAC security, MAC Service, MANs, metropolitan area networks, port-based network access control, secure association, security, transparent bridging

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The following members of the individual balloting committee voted on this standard. Balloters may have voted for approval, disapproval, or abstention.

A.N. Other

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Introduction

This introduction is not part of IEEE Std 802.1AE™-202X, IEEE Standard for Local and metropolitan area networks—Media Access Control (MAC) Security

The first edition of IEEE Std 802.1AE was published in 2006.

IEEE Std 802.1AEbn™-2011 added the option of using the GCM-AES-256 Cipher Suite.

IEEE Std 802.1AEbw™-2013 added the GCM-AES-XPN-128 and GCM-AES-XPN-256 Cipher Suites. These extended packet numbering (XPN) Cipher Suites allow more than 2^{32} frames to be protected with a single Secure Association Key (SAK) and so ease the timeliness requirements on key agreement protocols for very high speed (100 Gb/s plus) operation.

IEEE Std 802.1AEcg™-2017 specified Ethernet Data Encryption devices (EDEs) that provide transparent secure connectivity while supporting provider network service selection and provider backbone network selection as specified in IEEE Std 802.1Q™.

The IEEE Std 802.1AE-2018™ revision incorporated the text of IEEE Std 802.1AE-2006, amendments IEEE Std 802.1AEbn-2011, IEEE Std 802.1AEbw-2013, and IEEE Std 802.1AEcg-2017.

The IEEE Std 802.1AE-2018/Cor 1-2020 corrigendum corrected an unintentional change to Figure 9-4, replacing it with Figure 9-4 of IEEE_Std_802.1AE-2006.

IEEE Std 802.1AEdk™-2023 specified MAC privacy protection, and added YANG modules for MAC Security and MAC Privacy protection.

IEEE Std 802.1AEef-202X added the Ascon-XPN-128 Cipher Suite.

This edition, IEEE Std 802.1AE-202X incorporates the text of IEEE Std 802.1AE-2018, Corrigendum 1 to that standard, IEEE Std 802.1AEdk™-2023, IEEE Std 802.1AEef-202X, and technical and editorial corrections to those standards.

Relationship between IEEE Std 802.1AE and other IEEE Std 802® standards

IEEE Std 802.1X™-2010 specifies Port-based Network Access Control, provides a means of authenticating and authorizing devices attached to a LAN, and includes the MACsec Key Agreement protocol (MKA) necessary to make use of the MAC Security protocol (MACsec) specified in IEEE Std 802.1AE.

MACsec is not intended as a substitute for the security mechanisms specified by IEEE Std 802.11™ Wireless LAN Medium Access Control. That standard also uses IEEE Std 802.1X, thus facilitating the use of a common authentication and authorization framework for LAN media to which this standard applies and for Wireless LANs.

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IEEE Standard for
Local and Metropolitan Area Networks—

Media Access Control (MAC) Security

1. Overview

1.1 Introduction

IEEE 802® Local Area Networks (LANs) are often deployed in networks that support mission-critical applications. These include corporate networks of considerable extent, and public networks that support many customers with different economic interests. The protocols that configure, manage, and regulate access to these networks typically run over the networks themselves. Preventing disruption and data loss arising from transmission and reception by unauthorized parties is highly desirable, since it is not practical to secure the entire network against physical access by determined attackers.

The MAC Security protocol (MACsec), as defined by this standard, allows authorized systems that attach to and interconnect LANs in a network to maintain confidentiality of transmitted data and to take measures against frames transmitted or modified by unauthorized devices.

MACsec facilitates

a) Maintenance of correct network connectivity and services
b) Isolation of denial of service attacks
c) Localization of any source of network communication to the LAN of origin
d) The construction of public networks, offering service to unrelated or possibly mutually suspicious customers, using shared LAN infrastructures
e) Secure communication between organizations, using a LAN for transmission
f) Incremental and non-disruptive deployment, protecting the most vulnerable network components.

To deliver these benefits, MACsec has to be used in conjunction with appropriate policies for higher-level protocol operation in networked systems, an authentication and authorization framework, and network management. IEEE Std 802.1X™ provides authentication and cryptographic key distribution.[1]

MACsec protects communication between trusted components of the network infrastructure, thus protecting the network operation. MACsec cannot protect against attacks facilitated by the trusted components themselves, and is complementary to, rather than a replacement for, end-to-end application-to-application security protocols. The latter can secure application data independent of network operation, but cannot necessarily defend the operation of network components, or prevent attacks using unauthorized communication from reaching the systems that operate the applications.

MAC Privacy protection protocol, as defined by this standard, can be used in conjunction with MACsec to reduce the ability of adversaries to correlate the MAC addresses, sizes, and transmission timing of user data frames with individual persons, network applications, details of those applications, and levels of application activity.

[1] Information on other references can be found in Clause 2.

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1.2 Scope

The scope of this standard is to specify provision of connectionless user data confidentiality, frame data integrity, and data origin authenticity by media access independent protocols and entities that operate transparently to MAC Clients.

NOTE—The MAC Clients are as specified in IEEE Std 802®, IEEE Std 802.1Q™, and IEEE Std 802.1X™.[2]

To this end, it

a) Specifies the requirements to be satisfied by equipment claiming conformance to this standard.
b) Specifies the requirements for MAC Security in terms of provision of the MAC Service and the preservation of the semantics and parameters of service requests and indications.
c) Describes the threats, both intentional and accidental, to correct provision of the service.
d) Specifies security services that prevent, or restrict, the effect of attacks that exploit these threats.
e) Examines the potential impact of both the threats and the use of MACsec on the Quality of Service (QoS), specifying constraints on the design and operation of MAC Security entities and protocols.
f) Models support of the secure MAC Service in terms of the operation of media access control method independent MAC Security Entities (SecYs) within the MAC Sublayer.
g) Specifies the format of the MACsec Protocol Data Unit (MPDUs) used to provide secure service.
h) Identifies the functions to be performed by each SecY, and provides an architectural model of its internal operation in terms of Processes and Entities that provide those functions.
i) Specifies each SecY's use of an associated and collocated Port Access Entity (PAE, IEEE Std 802.1X) to discover and authenticate MACsec protocol peers and its use of that PAE's Key Agreement Entity (KaY) to agree and update cryptographic keys.
j) Specifies performance requirements and recommends default values and applicable ranges for the operational parameters of a SecY.
k) Specifies how SecYs are incorporated within the architecture of end stations, bridges, and two-port Ethernet Data Encryption devices (EDEs).
l) Establishes the requirements for management of MAC Security, identifying the managed objects and defining the management operations for SecYs.
m) Specifies a Management Information Base (MIB) module for SecY management.
n) Specifies a YANG configuration and operational state model for SecY management.
o) Specifies requirements, criteria, and choices of Cipher Suites for use with this standard.
p) Describes threats to individual privacy that can result from an adversary's observation of individual frames, even if those frames are integrity protected and their data confidentiality protected.
q) Models support of a privacy protected secure MAC Service in terms of the operation of MAC Privacy protection Entities (PrYs) that encapsulate user data frames in MAC Privacy protection Protocol Data Units (MPPDUs) to hide the user source and destination MAC addresses and to reduce any correlation of the sizes and transmission timing of frames with user identities and communication purposes, applications, or content.
r) Specifies the addressing, encoding, and decoding of MPPDUs.
s) Identifies the functions to be performed by each PrY, and provides an architectural model of its internal operation in terms of Processes and Entities that provide those functions.
t) Specifies performance requirements and recommends default values and applicable ranges for the operational parameters of a PrY.
u) Specifies how PrYs can be incorporated within the architecture of end stations, bridges, two-port Ethernet Data Encryption devices (EDEs), and bridged networks.

[2] Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

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v) Describes the requirements for management of MAC Privacy protection, identifying the managed objects and defining the manged objects for PrYs.

w) Specifies a Management Information Base (MIB) module for PrY management.

x) Specifies a YANG configuration and operational state model for PrY management.

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2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used; therefore, each referenced document is cited in text, and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 802®, IEEE Standard for Local and Metropolitan Area Networks: Overview and Architecture.[3,4]

IEEE Std 802.1Q™, IEEE Standard for Local and Metropolitan Area Networks: Bridges and Bridged Networks.

IEEE Std 802.1X™, IEEE Standard for Local and Metropolitan Area Networks: Port-Based Network Access Control.

IEEE Std 802.1AB™, IEEE Standard for Local and Metropolitan Area Networks: Station and Media Access Control Connectivity and Discovery.

IEEE Std 802.1AC™, IEEE Standard for Local and metropolitan area networks—Media Access Control (MAC) Service Definition.

IEEE Std 802.3™, IEEE Standard for Ethernet.

IETF RFC 1213: Management Information Base for Network Management of TCP/IP-based internets: MIB-II, McCloghrie, K., and Rose, M. T., March 1991.[5]

IETF RFC 2578, STD 58, Structure of Management Information for Version 2 of the Simple Network Management Protocol (SNMPv2), McCloghrie, K., Perkins, D., Schoenwaelder, J., Case, J., Rose, M., and Waldbusser, S., April 1999.

IETF RFC 2579, STD 58, Textual Conventions for Version 2 of the Simple Network Management Protocol (SNMPv2), McCloghrie, K., Perkins, D., Schoenwaelder, J., Case, J., Rose, M., and Waldbusser, S., April 1999.

IETF RFC 2580, STD 58, Conformance Statements for SMIv2, McCloghrie, K., Perkins, D., Schoenwaelder, J., Case, J., Rose, M., and Waldbusser, S., April 1999.

IETF RFC 2863, The Interfaces Group MIB using SMIv2, McCloghrie, K., and Kastenholz, F., June 2000.

IETF RFC 3418, Management Information Base (MIB) for the Simple Network Management Protocol (SNMP), Preshun, R., editor, December 2002.

IETF RFC 7317, A YANG Data Model for System Management, Bierman, A., Bjorklund, M., August 2014.

IETF RFC 7950, The YANG 1.1 Data Modeling Language, Bjorklund, M., August 2016.

IETF RFC 8343, A YANG Data Model for Interface Management, Bjorklund, M., March 2018.

ISO/IEC 14882, Information Technology—Programming languages—C++.[6]

[3] IEEE publications are available from The Institute of Electrical and Electronics Engineers (https://www.standards.ieee.org).
[4] The IEEE standards or products referred to in this clause are trademarks of The Institute of Electrical and Electronics Engineers, Inc.
[5] IETF RFCs are available from the Internet Engineering Task Force (https://www.ietf.org/rfc.html).

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NIST Special Publication 800-38D, Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC, November 2007.[7]

NIST Special Publication 800-232 ipd, Ascon-Based Lightweight Cryptography Standards for Constrained Devices, Authenticated Encryption, Hash, and Extendable Output Functions, November 2024

[6] ISO/IEC documents are available from the International Organization of Standardization (https://www.iso.org/) and from the International Electrotechnical Commission (http://www.iec.ch). These documents are also available from the American National Standards Institute (https://www.ansi.org/).

[7] NIST Special Publications are available from the National Institute of Standards and Technology (https://csrc.nist.gov/).

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3. Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.[8]

access priority: The priority that a client of the MAC Service or MAC Internal Sublayer Service (ISS) associates with a given transmit request.

NOTE—In this standard a MAC Security Entity (SecY) is a **shim** that can be a client of a service access point supported (at its lower interface) by a specific media access method, e.g., IEEE Std 802.3, and can provide (at its upper interface) a service access point that is used by its own client. From the point of view of the SecY, the priority associated with each transmit request received from its client is the **user priority** for that request and the priority associated with the corresponding transmit request that it makes of the underlying service is the **access priority**.

Association Number (AN): A number that is concatenated with the Secure Channel Identifier (SCI) to identify a Secure Association (SA).

black-side: Identifies the Ethernet Data Encryption device (EDE) port that uses MACsec to protect transmitted frames and verify received frames.

bounded receive delay: A guarantee that a frame will not be delivered after a known bounded time.

NOTE—In the case of the MAC Service, this is typically assumed to be less than two seconds.

Bridged Local Area Network: A concatenation of individual IEEE 802 Local Area Networks (LANs) interconnected by MAC Bridges.

NOTE—Unless explicitly specified the use of the word *network* in this standard refers to a Bridged Local Area Network. The term *Bridged Local Area Network* is not otherwise abbreviated. The term *Local Area Network* and the abbreviation *LAN* are used exclusively to refer to an individual LAN specified by a MAC technology without the inclusion of Bridges. This precise use of terminology within this specification allows a Bridged Local Area Network to be distinguished from an individual LAN that has been bridged to other LANs in the network. In more general usage such precise terminology is not required, as it is an explicit goal of this standard that MAC Security is transparent to the users of the MAC Service.

Cipher Suite: A set of one or more algorithms, designed to provide any number of the following: data confidentiality, data authenticity, data integrity.

Common Port: An instance of the MAC Internal Sublayer Service (ISS) used by the SecY to provide transmission and reception of frames for both the controlled and uncontrolled ports.

Controlled Port: The access point used to provide the secure MAC Service to a client of a MAC Security Entity (SecY).

cryptographic key: A parameter that determines the operation of a cryptographic function such as

a) The transformation from plain text to cipher text and vice versa
b) Synchronized generation of keying material
c) Digital signature computation or validation.[9]

cryptographic mode of operation: An algorithm for the cryptographic transformation of data that features a symmetric key block cipher algorithm. *Syn:* **mode.**[10]

[8] *IEEE Standards Dictionary Online* is available at https://dictionary.ieee.org.

[9] This and some other definitions in this clause have been drawn from ASC TR1/X9, Technical Report for ABA ASC/X9 Standards Definitions, Acronyms, and Symbols, 2002.

[10] This and some other definitions in this clause have been drawn from NIST Special Publication 800-38A, Recommendation for Block Cipher Modes of Operation: Methods and Techniques, 2001.

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Customer Edge Port: The red-side port of an Ethernet Data Encryption device (EDE-CS, EDE-CC, or EDE-SS).

NOTE—The terms *customer* and *provider* applied to the external and internal ports of an EDE-CS are those used by IEEE Std 802.1Q in its description of Provider Edge Bridges (PEBs) and reflect the role of those ports in the layered network architecture. They do not indicate control or ownership of the equipment. In this standard it is convenient to extend the use of those terms to ports that have the same relative relationship to the edge and network components of an EDE-CC or EDE-SS. This is not a suggestion that further variants of PEBs and Backbone Edge Bridges (BEBs) be specified, as the existence of additional variants would complicate interoperability, service provision, and the task of this standard.

Customer Network Port (CNP): A port on the network component of an Ethernet Data Encryption device (EDE-CS, EDE-CC, or EDE-SS) that provides internal connectivity to the edge component of that EDE.

data integrity: A property whereby data has not been altered in an unauthorized manner since it was created, transmitted or stored.[11]

edge component: The bridge component in an Ethernet Data Encryption device (EDE-CS, EDE-CC, or EDE-SS) that is attached to the red-side port.

Ethernet Data Encryption device (EDE): A two-port bridge that transmits and receives frames that are assumed to be unprotected to and from one red-side port, and conditionally relays those frames to and from its other black-side port, protecting and verifying frames transmitted and received on the black-side port using MACsec.

express frame: A frame that a protocol entity identifies as a candidate for early transmission using **preemption** capabilities.

NOTE—Not all protocol entities that forward a given frame need identify that frame as an express frame or a preemptable frame. In this standard that identification uses the priority of the frame.

IEEE 802 Local Area Network (LAN): LAN technologies that provide a MAC Service equivalent to the MAC Service defined in IEEE Std 802.1AC. IEEE 802 LANs include IEEE Std 802.3 (Ethernet) and IEEE Std 802.11 [B2] (Wireless).

NOTE—IEEE 802 LANs are also referred to in the text of this standard simply as LANs.

initialization vector (IV): A vector used in defining the starting point of an encryption process within a cryptographic algorithm.[12]

integrity: *See:* **data integrity.**

integrity check value (ICV): A value that is derived by performing an algorithmic transformation on the data unit for which data integrity services are provided. The ICV is sent with the protected data unit and is recalculated and compared by the receiver to detect data modification.

key: *See:* **cryptographic key.**

key management: The generation, storage, distribution, deletion, archiving, and application of keys in accordance with a security policy.

[11] This and some other definitions in this clause have been drawn from NIST Special Publication 800-57, Recommendation for Key Management, 2005.

[12] This and some other definitions in this clause have been drawn from Federal Information Processing Standard (FIPS) 140-2, Security Requirements for Cryptographic Modules, 2001.

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Layer Management Interface (LMI): The interface between a protocol entity in a system and the system management, providing for the exchange of parameters with other system entities that are not attached to the service access points used and provided by the protocol entity.

Local Area Network (LAN): *See:* **IEEE 802 Local Area Network (LAN).**

MAC Security Entity (SecY): The entity that operates the MAC Security protocol within a system.

MAC Security TAG (SecTAG): A protocol header, comprising a number of octets and beginning with an EtherType, that is prepended to the service data unit supplied by the client of the protocol and is used to provide security guarantees.

MAC service data unit (MSDU): A sequence of zero or more octets that compose the data to be communicated with a single MAC Service request or indication.

master key: A secret key that is used to derive one or more cryptographic keys that are used directly to protect data transfer.

message authentication: If the message arrives authenticated, the cryptographic guarantee is that the message was not modified in transit and that the message originated from an entity with the proper cryptographic credentials.

mode: *See:* **cryptographic mode of operation.**

multipoint: Involving or potentially involving more than one participant in the role of receiver, or in the role of transmitter, in a single data transfer or set of related data transfers.

network component: The bridge component in an Ethernet Data Encryption device (EDE-CS, EDE-CC, or EDE-SS) that is attached to the black-side port.

nonce: A non-repeating value, such as a counter, used in key management protocols to thwart replay and other types of attack.[13]

Packet Number (PN): A monotonically increasing value that is guaranteed unique for each MACsec frame transmitted using a given Secure Association Key (SAK).

Port Identifier: A 16-bit identifier that uniquely identifies each of a system's transmit Secure Channels (SCs) that uses the same MAC address as a component of its Secure Channel Identifier (SCI).

NOTE—The Port Identifier is not constrained to correspond to any other port number, identifier, or index. There can be more than one SC for a physical port, identifying frames transmitted by separate virtual ports, and more than one SC for a physical or virtual port if that port uses different SCs to transmit frames of different priorities.

preemptable frame: A frame that a protocol entity identifies as a candidate for suspension by **preemption** capabilities, so as to allow the earlier transmission of an **express frame**.

preemption: The temporary suspension of the transmission (or encoding for transmission) of a **preemptable frame** to allow the earlier transmission of an **express frame**.

NOTE—The preemption capabilities specified by IEEE Std 802.3 can be used in conjunction with the MAC Security protocol (MACsec) to expedite the transmission of an **express frame** that becomes available for transmission after transmission of a **preemptable frame** has begun. The MAC Privacy protection protocol also supports preemption, allowing the encoding of an express frame prior to the encoding of the remaining fragment(s) of a preemptable frame.

[13] This and some other definitions in this clause have been drawn from ASC TR1/X9, Technical Report for ABA ASC/X9 Standards Definitions, Acronyms, and Symbols, 2002.

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Privacy Channel: A sequence of frames with the same MAC source and destination addresses each conveying a single MAC Privacy Protocol Data Unit (MPPDU), with the sequence conveying a sequence of entire or fragmented user data frames and padding.

NOTE—In this standard all unqualified references to “fragments” and “fragmentation” are to MPPDU encoding.

Privacy Frame: A frame that conveys a single MAC Privacy Protocol Data Unit (MPPDU) that includes a single, unfragmented, user data frame followed by zero or more octets of padding.

Private Port: The access point used to provide the privacy protected secure MAC Service to a client of a MAC Privacy protection Entity (PrY).

Protocol Data Unit (PDU): A unit of data specified in a protocol and consisting of protocol information and, possibly, user data.

Provider Edge Port (PEP): A port on the edge component of an Ethernet Data Encryption device (EDE-CS, EDE-CC, or EDE-SS) that provides internal connectivity to the network component of that EDE.

Provider Network Port (PNP): The black-side port of an Ethernet Data Encryption device (EDE-CS, EDE-CC, or EDE-SS).

red-side: Identifies the Ethernet Data Encryption device (EDE) port that does not use MACsec to protect transmitted frames or verify received frames.

Reserved Address: A group address filtered by a bridge component to restrict the scope of the control protocols using that Destination Address (DA).

secret key: A cryptographic key used with a secret key cryptographic algorithm that is uniquely associated with one or more entities and should not be made public.[14]

Secure Association (SA): A security relationship that provides security guarantees for frames transmitted from one member of a Connectivity Association (CA) to the others. Each SA is supported by a single secret key, or a single set of keys where the cryptographic operations used to protect one frame require more than one key.

Secure Association Identifier (SAI): An identifier for an Secure Association (SA), comprising the Secure Channel Identifier (SCI) concatenated with the Association Number (AN).

Secure Association Key (SAK): The secret key used by a Secure Association (SA).

Secure Channel (SC): A security relationship used to provide security guarantees for frames transmitted from one member of a Connectivity Association (CA) to the others. An SC is supported by a sequence of Secure Associations (SAs) thus allowing the periodic use of fresh keys without terminating the relationship.

Secure Channel Identifier (SCI): A unique identifier for a Secure Channel (SC), composed of a MAC Address and a Port Identifier.

NOTE—Key agreement protocols such as the MACsec Key Agreement protocol (MKA) (see IEEE Std 802.1X) are responsible for ensuring that each SCI used with a given Secure Association Key (SAK) is unique where a Cipher Suite requires that for nonce construction, as does the Default Cipher Suite (14.5). SCI uniqueness does not rely on MAC Address allocation procedures.

[14] From FIPS 140-2.

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secure Connectivity Association (CA): A security relationship, established and maintained by key agreement protocols, that comprises a fully connected subset of the service access points in stations attached to a single Local Area Network (LAN) that are to be supported by MACsec.

shim: A protocol entity that uses the same service as it provides.

NOTE—Shims specified or referenced in this standard secure the ISS, enhance privacy, or provided multiplexing over separate instances of the ISS.

Short Secure Channel Identifier (SSCI): A 32-bit value, managed by the key agreement protocol, that is unique for each SCI within the context of all MAC Security Entities (SecYs) using a given Secure Association Key (SAK).

spoofing: Claiming a fraudulent identity for purposes of mounting an attack.

traffic: A sequence of frames forwarded in a network.

Uncontrolled Port: The access point used to provide the insecure MAC Service to a client of a MAC Security Entity (SecY).

user priority: The priority associated with a transmit request accepted by an entity that provides the MAC Service or MAC Internal Sublayer Service (ISS).

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4. Abbreviations and acronyms

AEAD	Authenticated Encryption with Associated Data
AES	Advanced Encryption Standard
AN	Association Number
ATS	Asynchronous Traffic Shaping
BEB	Backbone Edge Bridge
BPDU	Bridge Protocol Data Unit
CA	secure Connectivity Association
CFM	Connectivity Fault Management
CNP	Customer Network Port
CRC	Cyclic Redundancy Check
C-TAG	C-VLAN tag
C-VID	Customer VLAN Identifier
C-VLAN	Customer Virtual Local Area Network
DA	Destination Address
EDE	Ethernet Data Encryption device
EDE-CC	Ethernet Data Encryption device with red-side recognition of C-TAGs and black-side addition and removal of C-TAGs
EDE-CS	Ethernet Data Encryption device with red-side recognition of C-TAGs and black-side addition and removal of S-TAGs
EDE-M	VLAN-unaware Ethernet Data Encryption device operating as a Customer Bridge
EDE-SS	Ethernet Data Encryption device with red-side recognition of S-TAGs and black-side addition and removal of S-TAGs
EISS	Enhanced Internal Sublayer Service
EPON	Ethernet Passive Optical Network
ES	end station
FCS	frame check sequence
GCM	Galois Counter Mode
ICV	integrity check value
ISS	Internal Sublayer Service
IV	initialization vector
KaY	MAC Security Key Agreement Entity
LACP	Link Aggregation Control Protocol
LAN	IEEE 802 Local Area Network
LLC	Logical Link Control (ISO/IEC/IEEE 8802.2™ [B16])
LLDP	Link Layer Discovery Protocol

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LMI	Layer Management Interface
MAC	Media Access Control
MACsec	Media Access Control Security
MIB	Management Information Base
MKA	MACsec Key Agreement protocol (IEEE Std 802.1X)
MKPDU	MACsec Key Agreement Protocol Data Unit
MPDU	MACsec Protocol Data Unit
MPP	MAC Privacy protection[15]
MPPDU	MAC Privacy protection Protocol Data Unit
MPPCI	MAC Privacy protection Protocol Component Identifier
MSDU	MAC Service Data Unit
MSTP	Multiple Spanning Tree Protocol
OLT	Optical Line Terminal
ONU	Optical Network Unit
PAE	Port Access Entity
PBN	Provider Bridged Network
PCI	Personal Correlatable Information
PCP	Priority Code Point (IEEE Std 802.1Q)
PDU	Protocol Data Unit
PEB	Provider Edge Bridge
PEP	Provider Edge Port
PII	Personally Identifiable Information
PN	Packet Number
PNP	Provider Network Port
PrY	MAC Privacy protection Entity
PSFP	Per-Stream Filtering and Policing
PVID	port VLAN Identifier
QoS	Quality of Service
RSTP	Rapid Spanning Tree Algorithm and Protocol
SA	Secure Association or Source Address, as applicable
SAI	Secure Association Identifier
SAK	Secure Association Key
SC	Secure Channel
SCB	Single Copy Broadcast
SCI	Secure Channel Identifier

[15] The acronym MPP is used in figures.

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SecTAG	MAC Security TAG
SecY	MAC Security Entity
SL	Short Length
SMI	Structure of Management Information
SNMP	Simple Network Management Protocol
SSCI	Short Secure Channel Identifier
STP	Spanning Tree Protocol
S-VLAN	Service Virtual Local Area Network
TPMR	Two-Port MAC Relay
UNI	User Network Interface (IEEE Std 802.1Q)
VID	VLAN Identifier
VLAN	Virtual Local Area Network
XPN	Extended Packet Numbering
YANG	Yet Another Next Generation[16]

[16] YANG is best viewed as a name, not an acronym.

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5. Conformance

A claim of conformance to this standard for the implementation of MAC Security is a claim that the behavior of an implementation of a MAC Security Entity (SecY) meets the requirements of this standard (5.3, 5.4) as they apply to the operation of the MACsec protocol, management of its operation, and provision of service to the protocol clients of the SecY, as revealed through externally observable behavior of the system of which the SecY forms a part.

A claim of conformance for the implementation of MAC Security may be a claim of full conformance, or a claim of conformance with Cipher Suite variance, as specified in 5.4.

Conformance to this standard does not ensure that the system of which a MAC Security implementation forms a part is secure, or that the operation of other protocols used to support MAC Security, such as key management and network management do not provide a way for an attacker to breach that security.

Conformance to this standard does not require any restriction as to the nature of the system of which a SecY forms part other than as constrained by the SecY's required and optional capabilities (5.3, 5.4). Clause 11 describes the use of SecYs within a number of different types of systems. These include, but are not limited to, systems specified in IEEE Std 802.1Q and those that make use of IEEE Std 802.1X. Successful interoperable use of MACsec in those systems also requires conformance to those standards. In addition Clause 15 of this standard makes use of components specified in IEEE Std 802.1Q to define further systems, Ethernet Data Encryption devices (EDEs), whose purpose is to secure the MAC Service within networks comprising bridging systems specified by IEEE Std 802.1Q in a way that is transparent to the operation of those bridging systems. Additional claims of conformance can be made to this standard in respect of EDEs (5.5–5.7).

A claim of conformance to this standard for the implementation of MAC Privacy protection is a claim that the behavior of an implementation of a MAC Privacy protection entity (PrY) meets the requirements of this standard (5.10, 5.11) as they apply to the operation of the MAC Privacy protection protocol, management of its operation, and provision of service to the protocol clients of the PrY, as revealed through externally observable behavior of the system of which the PrY forms a part.

Conformance to this standard does not require any restriction as to the nature of the system of which a PrY forms part other than as constrained by the PrY's required and optional capabilities. Clause 21 describes the deployment of PrYs in a number of different types of system and network scenarios.

5.1 Requirements terminology

For consistency with existing IEEE and IEEE 802.1 standards, requirements placed upon conformant implementations of this standard are expressed using the following terminology:

a) ***shall*** is used for mandatory requirements.
b) ***may*** is used to describe implementation or administrative choices ("may" means "is permitted to", and hence, "may" and "may not" mean precisely the same thing).
c) ***should*** is used for recommended choices (the behaviors described by "should" and "should not" are both permissible but not equally desirable choices).

The PICS proformas (5.2) reflects the occurrences of the words *shall, may,* and *should* within the standard.

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The standard avoids needless repetition and apparent duplication of its formal requirements by using ***is***, ***is not***, ***are***, and ***are not*** for definitions and the logical consequences of conformant behavior. Behavior that is permitted but is neither always required nor directly controlled by an implementor or administrator, or whose conformance requirement is detailed elsewhere, is described by ***can***. Behavior that never occurs in a conformant implementation or system of conformant implementations is described by ***cannot***.

5.2 Protocol Implementation Conformance Statements (PICS)

The supplier of a MAC Security Entity (SecY) implementation that is claimed to conform to this standard shall complete a copy of the PICS proforma provided in Annex A and shall provide the information necessary to identify both the supplier and the implementation.

The supplier of an EDE that is claimed to conform to this standard shall complete a copy of the PICS proforma provided in Annex D and shall provide the information necessary to identify both the supplier and the implementation. The supplier of an EDE implementation shall also complete or provide copies of the following PICS proforma(s) adhering to any restrictions required by conformance to this standard and marking any exceptions required by conformance to this standard:

a) For all types of EDE, the PICS proforma for each SecY implementation provided in Annex A of this standard.
b) For all types of EDE, the PICS proforma specified by IEEE Std 802.1X.
c) For an EDE-M: the IEEE 802.1Q PICS proforma as required for a VLAN-unaware MAC Bridge.
d) For an EDE-CS: the IEEE 802.1Q PICS proforma as required for a Provider Edge Bridge.
e) For an EDE-CC: the IEEE 802.1Q PICS proforma as required for each of the two C-VLAN components.
f) For an EDE-SS: the IEEE 802.1Q PICS proforma as required for each of the two S-VLAN components.

The supplier of a MAC Privacy protection Entity (PrY) implementation that is claimed to conform to this standard shall complete a copy of the PICS proforma provided in Annex H (normative) and shall provide the information necessary to identify both the supplier and the implementation.

5.3 MAC Security Entity requirements

An implementation of a MAC Security Entity (SecY) for which conformance to this standard is claimed shall

a) Support the Controlled and Uncontrolled Ports, and use a Common Port as specified in Clause 10.
b) Support the MAC status and point-to-point parameters for the Controlled and Uncontrolled Ports as specified in 6.4, 6.5, and 10.7.
c) Process transmit requests from the Controlled Port as required by the specification of Secure Frame Generation (10.5).
d) Process receive indications from the Common Port as required by the specification of Secure Frame Verification (10.6), prior to causing receive indications at the Controlled Port.
e) Encode and decode MACsec Protocol Data Units (MPDUs) as specified in Clause 9.
f) Use a 48-bit MAC Address and a 16-bit Port Identifier unique within the scope of that address assignment to identify each transmit SCI, as specified in 8.2.1.
g) Satisfy the performance requirements specified in Table 10-4 and 8.2.2.
h) Support the Layer Management Interface (LMI) operations required by the Key Agreement Entity as specified in Clause 10.
i) Provide the management functionality specified in 10.7.

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j) Protect and validate MPDUs by using Cipher Suites as specified in 14.1.
k) Support Integrity Protection using either or both of the mandatory Cipher Suites (GCM-AES-128 and Ascon-XPN-128) specified in Clause 14.
l) For each Cipher Suite implemented, support a minimum of
    1) One receive SC
    2) Two receive SAKs
    3) One transmit SC
    4) One of the two receive SAKs at a time for transmission, with the ability to change from one to the other within the time specified in Table 10-4.
m) Specify the following parameters for each Cipher Suite implemented
    1) The maximum number of receive SCs supported
    2) The maximum number of receive SAKs
    3) The maximum number of transmit SCs supported.

An implementation of a SecY for which conformance to this standard is claimed shall not

n) Introduce an undetected frame error rate greater than that achievable by preserving the original frame check sequence (FCS), as required by 10.4.
o) Implement any Cipher Suite that is additional to those specified in Clause 14 and does not meet all the criteria specified in 14.2, 14.3, and 14.4.1.
p) Support access to MACsec parameters by a management agent using any version of SNMP prior to v3.

An implementation of a SecY for which full conformance to this standard is claimed shall not

q) Implement Cipher Suites other than those specified in Clause 14.

NOTE—Conformance with Cipher Suite variance is allowed, as specified in 5.4 and in 14.4.1.

5.4 MAC Security Entity options

An implementation of a MAC Security Entity (SecY) for which conformance to this standard is claimed may

a) Support access to MACsec parameters by a management agent using SNMP version v3 and the MIB module specified in Clause 13.
b) Support more than one receive SC.
c) Support more than two receive SAKs.
d) Support more than one transmit SC.
e) Support Confidentiality Protection using the Default Cipher Suite without a confidentiality offset, as specified in Clause 14.
f) Support Confidentiality Protection using the Default Cipher Suite with a confidentiality offset, as specified in Clause 14.
g) Support Confidentiality Protection using the Ascon-XPN-128 Cipher Suite, as specified in Clause 14.
h) Include Cipher Suites that are specified in Clause 14 in addition to the Default Cipher Suite.

An implementation of a SecY that supports more than one transmit SC shall

i) Support a Traffic Class Table and an Access Priority Table as specified in 10.7.17.

An implementation of a SecY for which conformance with Cipher Suite variance is claimed may

j) Use Cipher Suites not specified in Clause 14, but meeting the criteria specified in 14.2, 14.3, and 14.4.1.

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5.5 EDE conformance

Ethernet Data Encryption devices (EDEs) of various types comprise bridging systems and bridge components used as specified in this standard. Clause 15 provides a taxonomy, specification, and a description of the intended use of each type (EDE-M, EDE-CC, EDE-CS, or EDE-SS).

The bridging system and/or bridge components that comprise an implementation of an EDE that is claimed to conform to this standard shall conform to the requirements of (and may use any of the options and recommendations permitted by) IEEE Std 802.1Q, IEEE Std 802.1X, and the provisions of this standard for each MAC Security Entity (SecY) that is part of the EDE implementation, with the restrictions, additions, exceptions, and clarifications specified in this standard for that type of EDE.

An implementation of any type of EDE that is claimed to conform to this standard shall

a) Have two and only two externally accessible Bridge Ports, a red-side port and a black-side port.

   NOTE—A red-side port can also be referred to as the *customer* or *edge* port and the black-side port as a *provider* or *network* port. The use of either or both of the pair of terms, *customer/provider* and *edge/network* to refer to an EDE-M's ports is consistent with the relative roles played by ports in multicomponent bridges and EDEs.

b) Associate a Port Access Entity (PAE) that includes a MACsec Key Agreement Entity (KaY) capable of operating MKA with each SecY required by this standard for the particular type of EDE (15.2, 15.4, 15.5, 15.6, 15.7).

An implementation of any type of EDE that is claimed to conform to this standard may

c) Provide MAC Privacy protection for black-side port interface stack clients with a MAC Privacy protection entity (PrY) that uses the secure MAC Service provided by that port's SecY (21.5).

5.6 EDE-M conformance

An implementation of an EDE-M (15.2, 15.4) that is claimed to conform to this standard shall

a) Comprise a VLAN-unaware MAC Bridge as specified by IEEE Std 802.1Q (5.14 of IEEE Std 802.1Q-2022) with the constraints and exceptions specified in this standard.

b) Incorporate a SecY in the black-side port interface stack (15.2, 15.4).

c) Be capable of being configured to secure connectivity within a customer or provider network (as specified in 15.2), using the Nearest non-TPMR group address for group-addressed EAPOL PDUs, and filtering frames whose destination MAC Address is a TPMR component Reserved Address or the Nearest non-TPMR Bridge group address.

d) Be capable of being configured to secure connectivity across a PBN to a peer EDE-M, MACsec-capable Customer Bridge, or an EDE-CS (as specified in 15.4 and 15.5), using the Nearest Customer Bridge group address for group addressed EAPOL PDUs and filtering frames whose destination MAC Address is a C-VLAN component Reserved Address with the exception of the Nearest Customer Bridge group address.

and may

e) Be capable of being configured to secure connectivity across a PBN to a peer EDE-CC (as specified in 15.6), using the EDE-CC PAE group address for group addressed EAPOL PDUs, and filtering frames whose destination MAC Address is a C-VLAN component Reserved Address or the EDE-CC PAE group address.

f) Be capable of being configured to recover signaled priority from a C-VLAN tag and to priority tag (or not) frames transmitted by the black-side port as specified in 15.4.

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5.7 EDE-CS conformance

An implementation of an EDE-CS (15.5) that is claimed to conform to this standard shall

a) Comprise a Provider Edge Bridge as specified by IEEE Std 802.1Q (5.10.2 of IEEE Std 802.1Q-2022) including one and only one C-VLAN component (identified by this standard as the edge component of the EDE) and an S-VLAN component (the network component of the EDE).

b) Incorporate a SecY in each of the internal Provider Edge Port interface stacks (15.5).

c) Be capable of being configured to use the Nearest Customer Bridge group address for group addressed EAPOL PDUs (15.5).

5.8 EDE-CC conformance

An implementation of an EDE-CC (15.6) that is claimed to conform to this standard shall

a) Comprise two C-VLAN bridge components, each as specified by IEEE Std 802.1Q (5.5 of IEEE Std 802.1Q-2022)—an edge component and a network component—internally connected as specified in 15.6.

b) Incorporate a SecY in each of the internal Provider Edge Port interface stacks (15.6).

c) Be capable of being configured to use the EDE-CC PAE group address as the destination MAC address of group addressed EAPOL PDUs for group addressed EAPOL PDUs (as specified in 15.6).

d) Filter and not forward all frames whose destination MAC address is either one of the addresses identified by IEEE Std 802.1Q as a C-VLAN component Reserved Address or the EDE-CC PAE group address.

An EDE-CC that is not capable of providing MAC Privacy protection shall

e) Transmit frames received from the red-side customer port and relayed to the black-side network port untagged if they were received untagged and C-tagged with the same C-VID if they were C-tagged on receipt (15.6).

An EDE-CC that is capable of providing MAC Privacy protection [5.5c)] shall

f) Be capable of being configured to transmit frames received from the red-side customer port and relayed to the black-side network port untagged if they were received untagged and C-tagged with the same C-VID if they were C-tagged on receipt (15.6).

An EDE-CC that is capable of providing MAC Privacy protection [5.5c)] may

g) Be capable, when relaying a frame received on the red-side customer port, of using the received C-VID to select the C-VID to be used when tagging the MACsec protected frame for transmission by the black-side network port (15.4).

NOTE—This optional capability was added by the IEEE Std 802.1AEdk MAC Privacy protection amendment to this standard to hide the use of two or more customer VLANs for traffic destined for the same peer PrY. Its use requires additional management and care in deployment. The frame transmitted through a peer red-side port, after protected transmission over a service provider network, is that originally received by the EDE.

5.9 EDE-SS conformance

An implementation of an EDE-SS (15.7) that is claimed to conform to this standard shall

a) Comprise two S-VLAN components, each as specified by IEEE Std 802.1Q (5.6 of IEEE Std 802.1Q-2022)—an edge component and a network component—internally connected as specified in 15.7.

b) Incorporate a SecY in each of the internal Provider Edge Port interface stacks (15.7).

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c) Be capable of being configured to use the EDE-SS PAE group address as the destination MAC address of group addressed EAPOL PDUs for group addressed EAPOL PDUs (as specified in 15.7).

d) Filter and not forward all frames whose destination MAC address is either one of the addresses identified by IEEE Std 802.1Q as an S-VLAN component Reserved Address (Table 8-2 of IEEE Std 802.1Q-2022) or the EDE-SS PAE group address.

e) Transmit frames received from the red-side customer port and relayed to the black-side network port untagged if they were received untagged and S-tagged with the same S-VID if they were S-tagged on receipt (15.7).

5.10 MAC Privacy protection Entity requirements

An implementation of a MAC Privacy protection Entity (PrY) for which conformance to this standard is claimed shall

a) Support the Private and Controlled Ports as specified in Clause 20.

b) Support the MAC status and point-to-point parameters for the Private Port as specified in 20.4.

c) Transmit only MPPDUs whose encoding satisfies the constraints specified in 19.6.

d) Support requests to transmit user data frames through the Private Port as specified in 20.5 through 20.10.

e) Support per priority privacy selection (unprotected, as an individual Privacy Frame, or in a Privacy Channel) of user data frames transmitted by the client of the PrY's Private Port, as specified in 20.5.

f) Support unprotected user data frame transmission, with the ability to manage the access priority used to transmit that frame, as specified in 20.6.

g) Support individual Privacy Frame transmission, with the ability to manage the access priority used to transmit that frame, as specified in 20.7.

h) Support the configuration of two, one, or no transmit Privacy Channels as specified in 20.8.

i) Support the generation of Privacy Channel MPPDUs as specified in 20.9.

j) Be capable of being configured to use the default Privacy Channel MPPDU generation algorithm specified in 20.9.4.

k) Accept user data frames for Privacy Channel encapsulation as specified in 20.10.

l) Be capable of being configured to use the default Privacy Channel MPPDU encapsulation algorithm specified in 20.10.1.

m) Transmit all MPPDUs using the same destination MAC Address and the same source MAC Address as specified in 18.1.

n) Use the PAE Group Address that is being used by MKA as the destination MAC Address of transmitted MPPDUs and to recognize received MPPDUs when the PrY's Controlled Port is directly supported by a SecY (17.4, 18.1, 21.1.1).

o) Use the MAC Address used to generate MACsec SCI(s) as the source MAC Address of transmitted MPPDUs when the PrY's Controlled Port is directly supported by a SecY, as specified in 18.1.

p) Support administrative configuration of the destination MAC Address used to transmit MPPDUs if and when the PrY's Controlled Port is not directly supported by a SecY as an individual or group MAC address (18.1, 20.14).

q) Use a MAC Address associated with the PrY's interface stack as the source MAC Address of transmitted MPPDUs if and when the PrY is not directly supported by a SecY, as specified in 18.1.

r) Recognize received MPPDUs as specified in 20.11.

s) Validate and extract received MPPDU components as specified in 20.12 and 19.7.

t) Support simultaneous reassembly of Express and Preemptable frames from received Frame Fragments as specified in 20.13.

u) Deliver user data frames received in Encapsulated Frame components and reassembled from received Frame Fragments in the order specified in 20.13.

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v) Be capable of being configured to use the default reassembly algorithm as specified in 20.13.1 to simultaneously reassemble Express Frame Fragments and Preemptable Frame Fragments transmitted by a single peer.

5.11 MAC Privacy protection Entity options

An implementation of a MAC Privacy protection Entity (PrY) for which conformance to this standard is claimed may

a) Be capable of being configured to use the default reassembly algorithm as specified in 20.13.1. for reception from a specified number (greater than one) of peer PrYs, each transmitting Express Frame Fragments and Preemptable Frame Fragments.

k) Use Private Port transmit request stream_handles to select Privacy Selection Table entries (20.5.2).

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6. Secure provision of the MAC Service

MACsec provides secure communications between stations that are attached to the same LAN. An authenticated and authorized peer MAC Security Entity (SecY) within each station uses the insecure MAC Service provided by the LAN to provide the secure MAC Service to its client (see Figure 6-1). The requirements for MACsec discussed in this clause are informed by the goal of preserving the MAC Service.

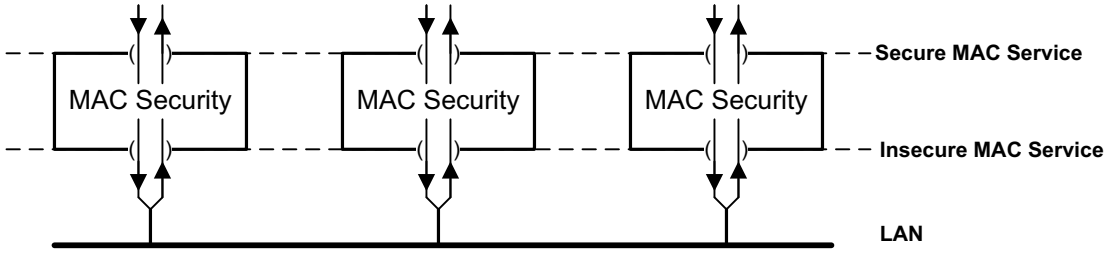


Figure 6-1—MACsec secured LAN with three stations

This clause discusses the

a) Primitives, parameters, connectivity, and status parameters provided by the MAC Service
b) Security threats posed by abuses of the MAC Service
c) Connectivity used and provided by the MACsec protocol
d) Service guarantees provided by MACsec and the security services they support
e) Quality of Service (QoS) issues addressed in the design, implementation, and use of MACsec.

NOTE 1—In order to introduce the concepts used in this standard, this clause can repeat or summarize the specification in other clauses; however, it contains no normative provisions that apply either to the subject matter of those other clauses or to the other standards referenced. For conformance to this standard see Clause 5.

NOTE 2—MACsec does not itself guarantee the security of a Bridged Local Area Network, as that security depends on the security of the individual LANs that comprise the network, on the policies adopted by clients of the secure MAC Service (7.3), and on the security of the MAC Bridges that interconnect those LANs.

NOTE 3—Authentication and authorization are outside the scope of this standard, which ensures secure communication between mutually authenticated and authorized service access points.

NOTE 4—The MAC Service and the secure MAC Service are provided at a service access point to a single client. The client is either a logical link control (LLC) Entity or an entity that in turn provides the MAC Service or a MAC Internal Sublayer Service (IEEE Std 802.1AC, IEEE Std 802.1Q).

6.1 MAC Service primitives and parameters

The MAC Service (IEEE Std 802.1AC) provides unconfirmed connectionless-mode data transfer between source and destination stations. The invocation of a request primitive at a service access point within a source station results, with a high probability, in a corresponding indication primitive at selected service access points in destination stations. A single service request at one service access point results in no more than one service indication at each of the other service access points.

Each request and indication primitive has the following four parameters:

— Destination Address
— Source Address
— Priority
— MAC Service Data Unit (MSDU)

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The MAC Service can be provided by a single LAN or by a Bridged Local Area Network. The service provided to an LLC Client in an end station is specified in IEEE Std 802.1AC. The service provided by a LAN to a MAC Bridge is the MAC Internal Sublayer Service (ISS), which includes parameters necessary to the bridge relay function including the frame check sequence (FCS). Except as otherwise explicitly noted, the term *MAC Service* as used in the remainder of this clause refers both to the provision of the MAC Service to an LLC client and to provision of the ISS. Multiple instances of the MAC Service can be provided using a single instance of the ISS and supported in VLAN-aware Bridges using the Enhanced Internal Sublayer Service (EISS), (see 6.6 of IEEE Std 802.1Q-2022). When a VLAN TAG (IEEE Std 802.1Q) is used to distinguish the service instances supported, the additional parameters of the EISS are all encoded within the ISS MSDU.

NOTE 1—The MAC Service defined in IEEE Std 802.1AC is an abstraction of the features common to a number of specific media access control methods and is a guide to the development of client protocols.

NOTE 2—The priority parameter described in this clause is also referred to as the user_priority in some specifications. The functions that support the ISS can calculate an access_priority for use on a LAN in local support of the user_priority.

The MAC Service provided by a single LAN preserves the relative order of service requests and corresponding service indications with the same requested priority. Each instance of the MAC Service provided using an instance of the EISS preserves the relative order of requests and indications with the same destination address, source address, and priority if the destination address is an individual address, and the relative order of requests and indications with the same destination address and priority if the destination address is a group address.

NOTE 3—A Provider Bridged Network can use the EISS to provide instances of the MAC Service that appear, with the exception of ordering constraints, to be a single LAN and are used as such by MACsec.

The address and MSDU parameters delivered with a service indication have identical values to those supplied with the corresponding service request. The MAC Service does not validate the parameters supplied with the request; for example, it does not provide any assurance that the source address used by an LLC client is the individual address previously allocated to the station.

NOTE 4—For example, in the absence of policies that require authorization to use an address, or check that a MACsec participant does not change its address, the use of MACsec will not protect against ARP spoofing. MACsec can form part of a solution that prevents ARP spoofing provided that suitable client policies are used in conjunction with MACsec, as mentioned in 1.1, in Clause 6, and in Clause 7 (particularly 7.3). In the case of ARP spoofing, appropriate policies include a selection of not accepting frames from an end station with a different source address than that used in the authentication dialogue, not accepting frames from bridges that are not known to obey that rule, requiring use of device identification to ensure use of a legitimate MAC address, and filtering certain higher-layer protocol frames unless received (using MACsec) from a system allowed to send them.

The priority parameter delivered with a service indication has an identical value to that supplied with the corresponding service request, where the media access control method used supports communication of priority. The access to the LAN granted by the media access control method can take the requested priority value into account, but can also be based on other factors. The IEEE 802.3™ media access control method does not convey priority, and the priority value delivered with the service indication is determined by management of the receiving station. However, where the EISS is used to support an instance of the MAC Service, the priority parameter of the EISS is encoded within a VLAN TAG (IEEE Std 802.1Q) that forms the initial octets of the MSDU accompanying a service request to the instance of the ISS used to support the EISS. Figure 6-2 shows the priority field encapsulated in the VLAN TAG within the Secure Data portion of the MACsec frame. In this case, the value of the priority parameter delivered with the service indication will be identical to that provided with the corresponding request.

6.2 MAC Service connectivity

The MAC Service provided by a single point-to-point or shared media LAN provides symmetric and transitive connectivity between all the stations connected to that LAN. Following a service request at one service access point, a corresponding service indication occurs, with high probability, at all the MAC ISS

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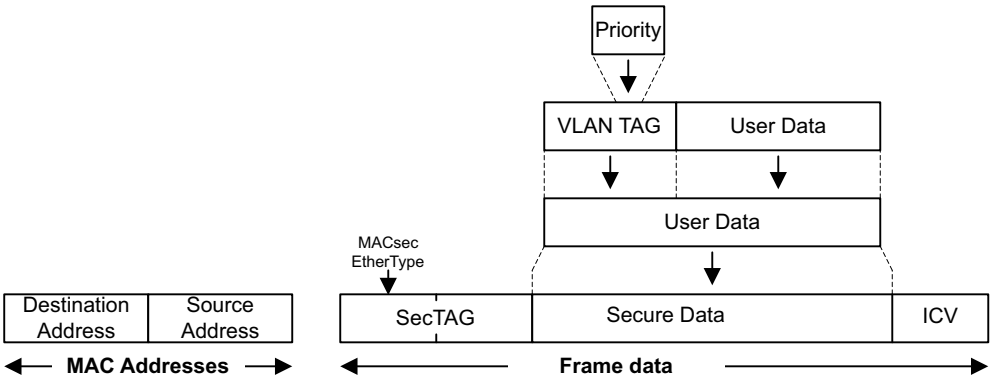


Figure 6-2—MACsec Frame, VLAN TAG, and QoS

access points in other stations attached to the same LAN. Service indications at service access points that provide the MAC Service to an LLC Entity are filtered by media access functions, generally within each receiving station, to exclude frames that are not destined to an individual or group address used by the client.

NOTE 1—Symmetric connectivity means that if station A can communicate with station B, then B can also communicate with A. Transitive connectivity means that if station A can communicate with B, and B with C, then A can also communicate with C.

NOTE 2—Some media access control devices or methods are capable of not delivering some or all frames with unwanted destination addresses to some or all stations.

MAC Service clients and client protocols can operate incorrectly if the connectivity provided is not as expected. While some protocols can detect the lack of symmetric connectivity, they can simply deny service as a result. Protocols can operate inefficiently if transitive connectivity is not provided. While MAC Bridges can filter frames to restrict provision of service, the use of Virtual LANs (VLANs), with each VLAN providing full connectivity, is preferred to lessen the administrative burden of ensuring correct connectivity.

NOTE 3—The original Spanning Tree Protocol (STP) could create loops in the network if symmetric connectivity was not provided. The Rapid Spanning Tree Algorithm and Protocol (RSTP), (see IEEE Std 802.1Q), detects non-symmetric connectivity between Bridges, but will deny service until the problem is resolved, and intermittent non-symmetric connectivity can result in data loops. For example, the operation of the OSPF routing protocol on a LAN is inefficient unless all participants can receive frames sent by each other. If a LAN that provides the ISS to attached MAC Bridges merely delivers frames to their intended destination instead of providing full connectivity, learning of source addresses can be inhibited and frames flooded throughout the bridged network for an indefinite period.

6.3 Point-to-multipoint LANs

A point-to-multipoint LAN provides connectivity from a single distinguished station to one or more other stations, i.e., from one point to multiple other points, and from each of the other stations to the distinguished station. The point-to-multipoint LAN does not provide direct connectivity between pairs of stations unless the distinguished station is one of the pair. Efficient multicast and broadcast from the distinguished station to all the others is provided: a single service request with a given destination address can result in service indications at each multipoint station wishing to receive frames with that destination address. Communication between the other stations occurs via the distinguished station, as specified by the relevant standard for the particular technology.

NOTE—Examples of point-to-multipoint LANs include IEEE 802.3 Ethernet Passive Optical Network (EPON) (see Clause 12), IEEE Std 802.11™ [B2], and certain provider network VLAN configurations. Depending on the particular LAN or network technology the distinguished stations are variously referred to as optical line terminals (OLTs), wireless access points (WAPs), head-ends, concentrators, star concentrators, or root nodes, and the other stations as optical network units (ONUs), stations, modems, or leaf nodes.

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6.4 MAC status parameters

Each service access point can make available status parameters that reflect the operational state and administrative controls for the service instance provided at that access point.

The **MAC_Enabled** parameter is TRUE if use of the service is permitted and is otherwise FALSE. The value of this parameter is determined by administrative controls specific to the entity providing the service.

The **MAC_Operational** parameter is TRUE if, and only if, service requests can be made and service indications can occur.

The value of the MAC_Enabled and MAC_Operational parameters are determined by the specific entity providing the MAC Service. IEEE Std 802.1AC and IEEE Std 802.1Q specify how that determination is made for provision of the insecure ISS by specific media access control methods, and for provision of the insecure EISS. This standard specifies how these parameters are determined for the secure MAC Service.

NOTE—Correct provision and use of the MAC_Operational parameter is essential for high performance implementation of RSTP (IEEE Std 802.1Q), Multiple Spanning Tree Protocol (MSTP, IEEE Std 802.1Q), and Link Aggregation Control Protocol (LACP, IEEE Std 802.1AX™ [B4]). In the absence of this parameter, loss of connectivity is determined by repetitive loss of protocol frames that are normally transmitted at intervals of a few seconds, and it is assumed that frames transmitted immediately after a medium availability transition have a high probability of not being received by protocol peers.

6.5 MAC point-to-point parameters

Each service access point can make available status parameters that reflect the point-to-point status for the service instance provided, and that allow administrative control over the use of that information.

If the **operPointToPointMAC** parameter is TRUE, the service is used as if it provides connectivity to at most one other system; if FALSE, the service is used as if it can provide connectivity to a number of systems.

The **adminPointToPointMAC** parameter can take one of three values. If it is

a) **ForceTrue**, operPointToPointMAC shall be TRUE, regardless of any indications to the contrary generated by the service providing entity.
b) **ForceFalse**, operPointToPointMAC shall be FALSE.
c) **Auto**, operPointToPointMAC is as currently determined by the service providing entity.

IEEE Std 802.1AC and IEEE Std 802.1Q specify how the point-to-point status determination is made for provision of the insecure ISS by specific media access control methods, and for provision of the insecure EISS. This standard specifies how it is determined for the secure MAC Service.

NOTE—RSTP and MSTP (IEEE Std 802.1Q) require the use of operPointToPointMAC to facilitate rapid reconfiguration in some network failure scenarios. LACP (IEEE Std 802.1AX [B4]) does not aggregate links that are not point-to-point and can use operPointToPointMAC to make this determination.

6.6 Security threats

The expected features of the MAC Service described in 6.1 through 6.5—the relationships between service requests and indications, preservation of the parameters of these primitives, the connectivity provided, and the relationship of the MAC status parameters to the connectivity—can be accidentally and unintentionally distorted through misconfiguration or deliberately abused. Misconfiguration or abuse can result in

a) Inability to issue service requests
b) Indiscriminate loss of service indications

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c) Specifically targeted loss of service indications
d) Repeated service indications at the intended destinations
e) Service indications with modified address or data parameters
f) Additional service indications with unmodified or selectively modified parameters
g) Service indications at unintended recipients
h) Delayed service indications that can disrupt network operation
i) Disclosure of the MSDU to unauthorized parties.

Deliberate abuse can serve as a basis for an attack upon the resources accessible from a LAN through attacks on the protocols that use the service and provide access to or control over those resources. The effort required by an attacker to abuse the service in any particular way depends in general on the media access control method used by the LAN, and the particular devices and components that support it.

The MAC Service does not guarantee the origin or authenticity of service requests and the accompanying parameters. Since a station's sole use of its source address and the restriction of service indications to intended recipients can depend on cooperative behavior from other stations, it is usually easy for an attacker that can attach a station to a LAN to receive any service indication and to issue additional service requests with parameters based on those indications. Other service abuses can require physical access to inconveniently located components.

It is beyond the scope of this standard to enumerate all the ways in which abuses of the service can be exploited. They include techniques commonly referred to as passive wiretapping, masquerading, and man-in-the-middle attacks. The latter is facilitated by source address spoofing, usually after another station with that source address has been observed to have been granted access to some resource. Attacks can include

j) Denial of service, to all or to selected stations
k) Theft of service
l) Access to confidential information
m) Modification of confidential information
n) Access to or control over restricted resources.

MACsec does not protect against brute force denial of service attacks that can be mounted by abusing the operation of particular media access control methods through degrading the communication channel or transmitting erroneous media access method specific control frames.

6.7 MACsec connectivity

The connectivity provided (6.2) between the MAC Internal Sublayer Service (ISS) access points of stations connected to a single LAN composes an insecure association between communicating stations. Key agreement protocols as defined in IEEE Std 802.1X establish and maintain a secure Connectivity Association (CA), which is a fully (i.e., symmetric and transitive) connected subset of the ISS service access points. Each instance of MACsec operates within a single CA.

NOTE 1—ISO/IEC 15802, the MAC Service definition, introduces the term *Connectivity Association* to discuss the relationship between service access points without referring to the details of particular media access control methods or to terms such as *physical connection* or *logical connection* that have other associated attributes and meanings.

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MACsec itself does not provide comprehensive monitoring of the connectivity provided by a CA, although it can detect and will signal certain failures to the local MAC Security Key Agreement Entity (KaY). Together, operation of key agreement protocols and MACsec ensures that the status parameters provided by an instance of the secure MAC Service correctly reflect both the current connectivity and changes in the connectivity of the CA. Specifically

a) MAC_Operational is only True if the CA is complete (i.e., is symmetric and transitive), and the local MACsec Entity (SecY) can both receive and transmit.

NOTE 2—EPON is discussed in Clause 12.

b) If MAC_Operational is True in stations wanting to join a new CA and in stations already in the target CA, and if stations are added to the CA, MAC_Operational transitions to False in either all the stations originally participating in the CA or in all those added, for sufficient duration such that clients of the service are aware of the transition.

Determining which group transitions MAC_Operational to False is outside the scope of this specification and is determined by the KaY and signaled through the LMI.

c) If MAC_Operational is False in stations wanting to join a CA, and if these stations are added to a CA, there is no change in the MAC_Operational status of the stations already in the target CA, and MAC_Operational will transition to True in the joining stations after some period of time. This is the typical case for a single station joining a CA, in which its MAC_Operational is False until the join is accomplished when its state transitions to MAC_Operational True.

d) If adminPointToPointMAC is set to Auto and MAC_Operational is True, then operPointToPointMAC is True only if at most one other station is participating in the CA. If adminPointToPointMAC is set to forceFalse, then operPointToPointMAC must be False, regardless of the number of stations in the CA.

NOTE 3—Communication between KaYs in stations that compose a CA does not depend on the operation of MACsec.

6.8 MACsec guarantees

At each service access point that is a member of a CA, MACsec ensures that any service indication

a) Is the result of a service request at a service access point that is also a member of the same CA
b) Has the parameter values that are identical to those supplied with the service request

and can also ensure that

c) No more than one indication results from one service request
d) A service indication does not occur after a known bounded time has elapsed since the service request was made
e) The values of the octets that comprise the MAC Service Data Unit (MSDU) parameter cannot be ascertained except by members of the CA.

MACsec does not

f) Conceal the following from stations that are not members of the CA:
    1) Service requests
    2) Values of service request address parameters
    3) The number of octets that comprises the MSDU
g) Validate the parameters provided with a service request.

MACsec provides guarantees to within known bounds that are derived from the cryptographic methods and other mechanisms used.

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The known bounded time in item d) is typically longer than required to enforce the maximum transit delay requirements of the MAC Service.

NOTE—The addition of explicit time indications to the MAC Security TAG (SecTAG) to provide tight bounds for transit delay was considered in the development of this standard, but the value delivered is small for the added complexity and the burden imposed on key agreement protocols. Higher-layer protocols that have tight timing requirements typically add their own timing markers. As these markers are carried within the MSDU, their integrity is protected by MACsec.

6.9 Security services

The guarantees provided by MACsec support the following security services for stations participating in MACsec:

a) Connectionless data integrity [6.8a), 6.8b)]
b) Data origin authenticity [6.8a)]. If the connectivity model is point-to-point, the originator is authenticated, but if the connectivity model is multipoint, then the authenticated originator is any member of the CA, rather than a particular individual station.
c) Confidentiality [6.8d)]
d) Replay protection [6.8c), 6.8d)]
e) Bounded receive delay

and can be used to limit the nature and extent of

f) Denial of service attacks.

MACsec does not support

g) Non-repudiation (ISO/IEC 7498-2)

or protect against

h) Traffic analysis (ISO/IEC 7498-2).

A MAC Bridge that forwards a frame with an erroneous source MAC address can unintentionally facilitate a denial of service or other attack on other LANs within a Bridged Local Area Network. The MAC Bridge should use MACsec in conjunction with an appropriate policy to verify the binding of the source MAC address to access to resources.

6.10 Quality of Service maintenance

The quality of the MAC Service can be lowered by direct attacks on the operation of particular media access control methods and indirect attacks on their resource allocation procedures facilitated by masquerading and unauthorized data modification. MACsec does not provide guarantees for frames, known as MAC control frames, that are internal to the operation of a particular media access method and cannot defend against abuses that use or affect such frames. MAC control frames are not forwarded by MAC Bridges, so attacks that exploit them can be localized to particular LANs.

NOTE—Where, within the operation of a particular media access control method, it is possible to establish secure Connectivity Associations (CAs) prior to performing certain control functions, those functions should be supported by frames transmitted using an instance of the ISS. The parameters of those frames can then be protected by MACsec, and the scope for abuse restricted. It is not a requirement of the Open Systems Interconnection (OSI) layer model (ISO 7498) that management and control of a particular layer be carried out purely within that layer by protocols whose identifiers and formats are specific to that layer. For example, SNMP can be used to manage MAC Bridges.

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Operation of a security protocol has the potential to lower some aspects of QoS. The operation and design of MACsec is discussed below as it relates to

a) Service availability
b) Frame loss
c) Frame misordering
d) Frame duplication
e) Frame transit delay
f) Frame lifetime
g) Undetected frame error rate
h) Maximum service data unit size supported
i) Frame priority
j) Throughput.

Use of MACsec can lower service availability if delays occur in the creation of Connectivity Associations or in the distribution and maintenance of cryptographic keying material. Failures or attacks upon the system that supports authentication and authorization can result in denial of service.

The operation of MACsec introduces no additional frame loss on individual LAN segments other than that expected for a specific media access control method as a consequence of a small increase in frame size. The operation of MACsec between two customer systems across a provider bridged network can introduce additional frame loss caused by possible frame reordering from expedited forward or link aggregations within the provider bridged network. The reception of misordered frames can cause MACsec implementations to discard additional frames depending upon the configuration of replay protection parameters. MACsec can use separate secure channels to transit frames with different access priorities and thus reduce or eliminate undesirable frame discard resulting from the mutual reordering of those frames.

Conforming implementations of MACsec are capable of applying a new keying material starting with any frame in a sequence that is received with the minimum intervening spacing specified by the specific media access control method in use. Each frame protected by MACsec remains independent of its predecessors and successors, so loss of a single frame does not imply loss of additional frames.

MACsec does not introduce any additional potential for duplicating or misordering frames. No retransmission mechanisms are added to relax requirements for distribution and use of MACsec related information. MACsec implementations are required to preserve the sequence of requests and indications between the secure service access point supported and the insecure service access point used.

Since the MAC Service is provided at an abstract interface within an end station, it is not possible to specify the total frame transit delay precisely. It is, however, possible to measure the additional transit delay introduced by an additional component or intermediate system.

The minimum additional transit delay introduced by MACsec is due to the increase in the MSDU size required to convey security information and essential buffering requirements required to meet the processing requirements of particular Cipher Suites. Specific limits are placed on the additional delays allowed to MACsec implementations (Table 10-4). The permitted delay is short compared with the upper bound mandated by the MAC Service, so it does not threaten the correct operation of higher-layer protocols.

Frame lifetime can be increased by MACsec if additional delay is introduced by providing security. The typical bound on frame lifetime is approximately two seconds.

MACsec does not increase the undetected frame error rate for frames received and transmitted on a single LAN. The frame check sequence (FCS) method used by each specific media access control method protects the entire frame including information added by MACsec. The integrity check added by MACsec can

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increase the probability of detecting unintentional frame modifications, particularly where those do not correspond to the expected noise characteristics for which the FCS was originally designed, but equally is not a substitute for the FCS since it is designed to ensure that an attacker has an exceedingly low chance of predicting how to make an undetected modification to the frame's parameters rather than to efficiently detect burst noise characteristics.

Use of MACsec on each of a MAC Bridge's Ports will force a change in the data covered by an FCS, even if the frame is being relayed between LANs that use the same media access control method. Application of the techniques described in Annex O of IEEE Std 802.1Q-2022 allow an implementation to achieve an arbitrarily small increase in undetected frame error rate, even in cases where the data that is within the coverage of the FCS is changed.

The MSDU size that can be supported by an IEEE 802 LAN varies with the media access control method and its associated parameters (speed, electrical characteristics, etc.), and can be constrained by the owner of the LAN. MACsec adds security information to a transmitted MSDU, and thus the secure MAC service offers a smaller MSDU size than the insecure MAC service that it employs.

Where MACsec is used to support an instance of the ISS that in turn supports the EISS, encoding of the priority parameter of the EISS within the ISS MSDU ensures that priority can be communicated unchanged between service access points that are attached to a single LAN. Since MACsec is terminated at each of those service access points, a bridge that makes use of that ISS instance to support one of its ports can access or change the priority even if the two instances of MACsec encrypt the MSDU in order to provide confidentiality. An Access Priority Table (10.7.17) can be used to derive the access priority requested from the medium supporting the ISS from the priority requested by the user of the EISS.

Cryptography can be computationally intensive, and the operation of MACsec has the potential to lower throughput. The Cipher Suite(s) mandated and recommended by this standard have been chosen, in part, for their ability to support economic implementation across the range of LAN MAC data rates.

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7. Principles of secure network operation

This clause establishes the principles and a model of secure network operation. It describes the security relationships used to support the secure MAC Service (Clause 6), and how that service is used to provide overall network security. It provides the context necessary to understand the operation of the MAC Security protocol (MACsec) (see Clause 8) and individual MAC Security Entities (SecYs) (see Clause 10).

Secure network operation comprises use of the secure MAC Service on each of the individual LANs that compose the network together with the application of appropriate security policies by the MAC Service clients in end stations and in intermediate systems that forward frames. This clause defines

a) The security relationships that support secure MAC Service

and describes how the secure MAC Service is

b) Supported on each of the individual LANs that compose the network (7.1)
c) Used by the protocol entities that are its Clients (7.3)

and delineates the responsibilities of the

d) Key Agreement Entities (KaYs, IEEE Std 802.1X)
e) MAC Security Entities
f) Clients of the secure MAC Service.

Security relationships and the terms that identify them have been defined, in various ways, by a number of publicly available documents. This standard has deliberately chosen new terms to minimize confusion whenever differences could exist between previously used terms and the requirements. For example, the attributes associated with an Secure Association Identifier (SAI) (see Figure 7-7) are similar to but not exactly the same as those associated with the Security Parameter Index (SPI) defined by IPsec (IETF RFC 4303 [B10]). The normative properties of all terms used in this standard are as defined by this standard.

NOTE 1—The use of the term *secure network* in this clause refers to a network of end stations, LANs, bridges, routers and similar systems, and the servers and services that support these. The description and specification in this clause are limited to use of the secure MAC Service to contribute to overall system security (see Clause 1).

NOTE 2—In order to introduce the concepts used in this standard, this clause can repeat or summarize the specification in other clauses; however, it contains no normative provisions that apply either to the subject matter of those other clauses or to the other standards referenced. For conformance to this standard (see Clause 5).

NOTE 3—The term *individual LAN* is used in this standard to refer explicitly to an instance of media access method-specific technologies providing the MAC Service directly. The term excludes larger networks or subsets of a network that are created by aggregation or concatenation of individual LANs by Link Aggregation or bridges.

NOTE 4—The examples presented in this clause are intended to serve as a guide to best practice; however, the use of MAC Security is not limited to the examples given. Limits to the use of MAC Security that are required for the successful operation of network configuration and other protocols are made explicit.

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7.1 Support of the secure MAC Service by an individual LAN

Each port that is capable of participating in an instance of the secure MAC Service comprises both a KaY and a SecY. Each KaY discovers or is made aware of the KaYs present in other stations attached to the same LAN, ensures that those stations are mutually authenticated and authorized, and creates and maintains the secure relationships between the stations that are used by the SecYs to transmit and receive frames. Specifically

a) A secure Connectivity Association (CA) is created to meet the requirements of the MAC Service (6.2) and MACsec (6.7) for connectivity between the stations attached to an individual LAN.
b) Each CA is supported by unidirectional Secure Channels (SCs), each SC supporting secure transmission of frames through the use of symmetric key cryptography, from one of the systems to all the others in the CA.
c) Each SC is supported by an overlapped sequence of Security Associations (SAs).
d) Each SA uses a fresh Secure Association Key (SAK) to provide the MACsec service guarantees (6.8) and security services (6.9) for a sequence of transmitted frames.

NOTE—An SC can be required to last for many years without interruption, since interrupting the MAC Service can cause client protocols to re-initialize and recalculate aggregations, spanning trees, and routes (for example). An SC lasts through a succession of SAs, each using a new SAK, to defend against a successful attack on a key while it is still in use. In contrast it is desirable to use a new SAK at periodic intervals to defend against a successful attack on a key while it is still in use. In addition, the MACsec protocol (Clause 8 and Clause 9) only allows 2^{32}–1 frames to be protected with a single key unless a Cipher Suite that supports extended packet numbering is used. Since 2^{32} minimum-sized IEEE 802.3 frames can be sent in approximately 5 min at 10 Gb/s, this can force the use of a new SA.

These security relationships (CAs, SCs, and SAs) and the information associated with each of them are further discussed in 7.1.1, 7.1.2, and 7.1.3. Their mutual relationship, and the insecure connectivity provided by the LAN that supports them, are illustrated in Figure 7-1 through Figure 7-3 for a point-to-point LAN and in Figure 7-4 through Figure 7-6 for stations attached to a shared media LAN.

Figure 7-1 shows two stations, A and B, connected to a point-to-point LAN that provides insecure bi-directional connectivity.

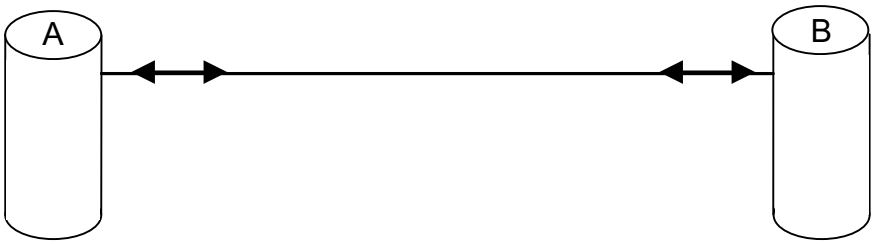


Figure 7-1—Two stations connected by a point-to-point LAN

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Figure 7-2 depicts the CA created by MACsec Key Agreement following mutual authentication and authorization of A and B.

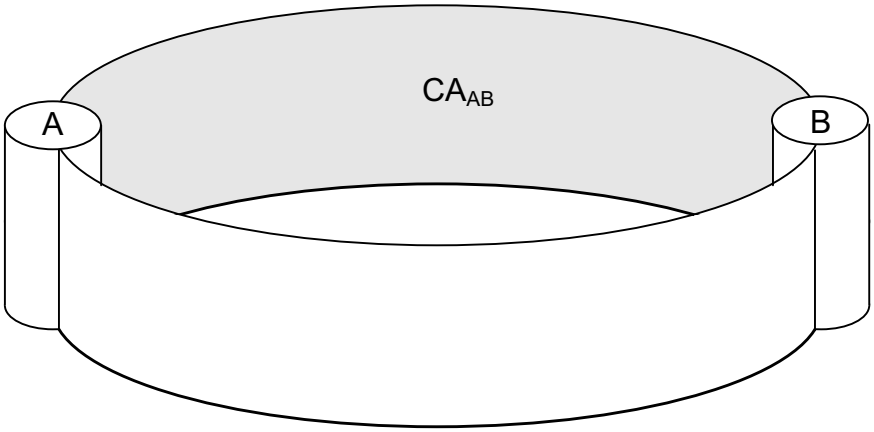


Figure 7-2—Two stations in a CA created by MACsec Key Agreement

Figure 7-3 shows the two SCs that support the CA.

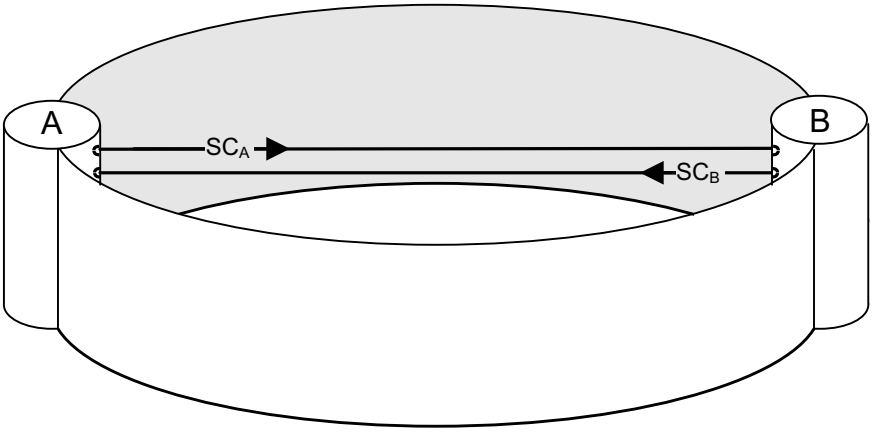


Figure 7-3—Secure communication between two stations

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Figure 7-4 shows four stations, A, B, C, and D, attached to a shared media LAN that provides full but insecure connectivity between the stations.

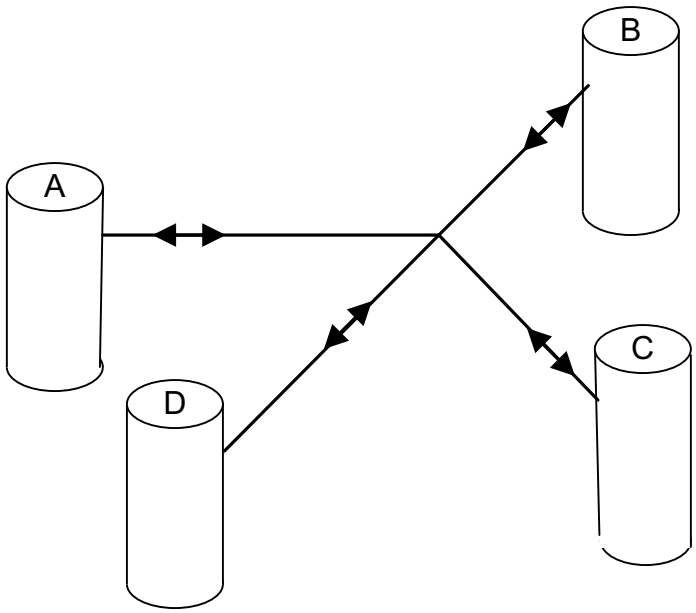


Figure 7-4—Four stations attached to a shared media LAN

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Figure 7-5 depicts a CA created by MACsec Key Agreement following mutual authentication and authorization of A, B, and C. The CA excludes D.

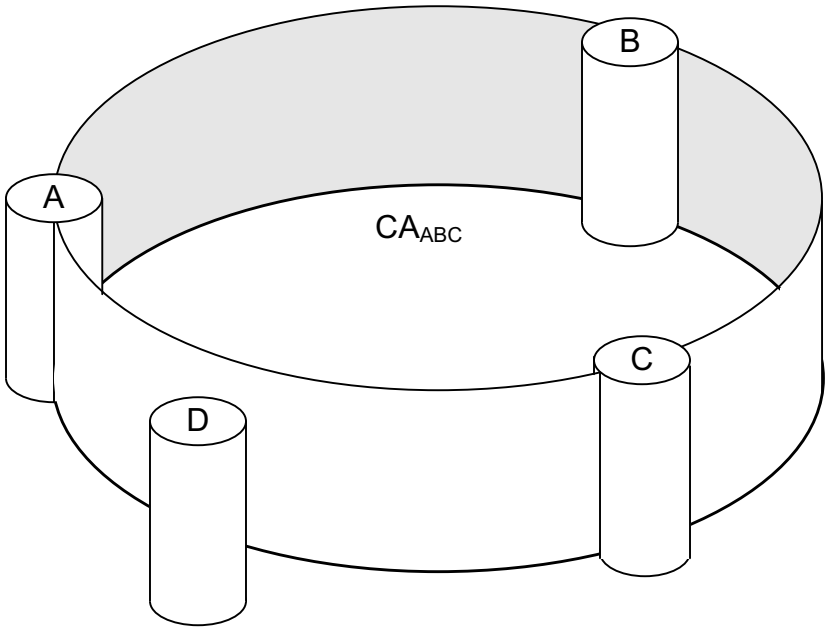


Figure 7-5—A CA including ports A, B, and C

Figure 7-6 shows the three SCs that support the CA, one for transmission by each of A, B, and C.

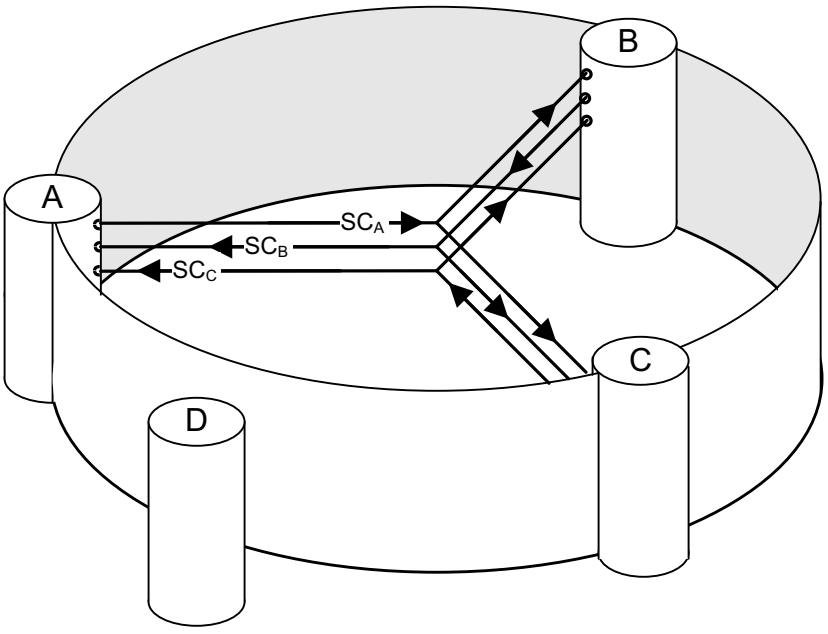


Figure 7-6—Secure communication between three stations

While D can send and receive frames using the insecure connectivity provided by the shared LAN, it does not have SAKs that would allow it to participate in any of the SAs that currently support SC_A, SC_B, or SC_C; therefore, D cannot compromise the integrity, confidentiality, or origin of any of the frames being exchanged by A, B, and C.

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In the above examples (Figure 7-3, Figure 7-6), each station transmits frames using a single SC. A station can also use multiple transmit SCs, using each transmitted frame's priority to allocate to it one of the SCs. Each of these transmit SCs supports secure transmission for frames of one or more priorities from the transmitting station to all the others in the CA. This use of multiple transmit SCs allows MACsec to enforce in-order delivery (or the use of a smaller replay window than might otherwise be the case) for frames of any given priority.

7.1.1 Connectivity Association (CA)

MACsec Key Agreement is responsible for discovering, authenticating, and authorizing the potential participants in a CA. A SecY, as specified in this standard, does not need to be aware of the CA, except as a list of SCs in which it needs to participate. Since all the SCs in a CA use the same Cipher Suite at any one time, the Cipher Suite can be considered a property of the CA. A change in the Cipher Suite necessitates an interruption to the service provided by the CA.

Each SecY participates in only a single CA at any one time. There is a limit, readable by management and by the KaY, on the number of peer SecYs that can participate in a CA (10.7.7, 13.5).

NOTE—If this specification had allowed different SCs to use different Cipher Suites, a SecY implementing more than one Cipher Suite would have to be capable of simultaneous transmitting using one Cipher Suite and receiving using one or more other Cipher Suites.

7.1.2 Secure Channel (SC)

Each SecY transmits frames conveying secure MAC Service requests of any given priority on a single SC. Each SC provides unidirectional point-to-multipoint communication, and it can be long lived, persisting through SAK changes. Each SC is identified by a Secure Channel Identifier (SCI) comprising a 48-bit MAC address concatenated with a 16-bit Port Identifier.

NOTE 1—Including the Port Identifier component would appear to be unnecessary in the case of a simple system with a MAC address and a single SecY for each port. However, some systems require support for more SecYs than they have uniquely allocated addresses because they make use of technologies that support virtual MACs, or because their interface stacks include the possibility of including multiple SecYs at different sublayers (as do Provider Bridges [IEEE Std 802.1Q], for example), or because they transmit frames of different priorities using different SCs.

NOTE 2—An EPON Optical Line Terminal (OLT) can use a distinct SC to support the Single Copy Broadcast (SCB) capability (Clause 12). The formal identifier for this SC comprises a System Identifier for the OLT and a reserved Port Identifier. Both can be represented in the secured frame by a single SCB bit (Clause 9).

MACsec Key Agreement is responsible for informing each SecY of the identifier to be used for each transmitting SC and of the existence and identifier of each of the SCs for which the SecY is to receive frames. While the structure of the communication facilitated by each SC is point-to-multipoint (which encompasses point-to-point as a special case), the SecY does not have to be aware that its transmissions can reach multiple receivers, that the frames that it receives could be received by other SecYs, or of any relationship or lack of relationship between the inbound SCs (except in determining the value of the operPointToPointMAC status parameter, 6.5, 10.7.4). The operation of the MACsec Key Agreement protocol (MKA, specified in IEEE Std 802.1X) is defined in terms of the behavior of participants, each representing a single KaY and a single transmit SCI. When a SecY uses multiple transmit SCIs, each SCI is represented by a separate participant that sends and receives MACsec Key Agreement PDUs (MKPDUs) to and from all the other participants just as if it were representing a separate SecY in the same group CA, see Annex E.

NOTE 3—The point-to-multipoint nature of the SC does have technical consequences, in particular the decision to change from one SA to another is made by the transmitter using the SC, not by one or some number of the receivers.

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7.1.3 Secure Association (SA)

Each SC comprises a succession of SAs, each with a different SAK. Each SA is identified by the SC identifier concatenated with a two-bit Association Number (AN, Figure 7-7). The Secure Association Identifier (SAI) thus created allows the receiving SecY to identify the SA and thus the SAK used to decrypt and authenticate the received frame. The SAI is only unique for the SAs used by SecYs participating in a given CA at any instant.

MACsec Key Agreement is responsible for creating and distributing SAKs to each of the SecYs in a CA. This key creation and distribution is independent of the cryptographic operation of each of the SecYs. The same SAK can be used for SAs that compose different SCs, provided that every initialization vector (IV) used with the SAK is unique. When the Default Cipher Suite (14.5) is used, the SCI is included in the IV to ensure uniqueness across SCs.

The decision to replace one SA with its successor is made by the SecY that transmits using the SC, after MACsec Key Agreement has informed it that all the other SecYs are prepared to receive using that SA. No notification, other than receipt of a secured frame with a different SAI is sent to the receiver. At any one instant a SecY has to be capable of storing SAKs for two SAs for each inbound SC, and of swapping from one SA to another without notice. Certain LAN technologies can reorder frames of different priority, so reception of frames on a single SC can use interleaved SAs. The time bound within which a receiver can accept interleaved SAs is 0.5 s.

The transmitting SecY does not interleave frames for different SAs on a given SC.

NOTE—When MKA (see IEEE Std 802.1X) is used to distribute the SAKs used by each of the SCs supporting a given CA, the same SAK is used for all SAs with a given AN (at any given time). The transmit SA for each SC is replaced with its successor at approximately the same time so that there is no need for any SecY participating in the CA to support more than two SAKs at a time.

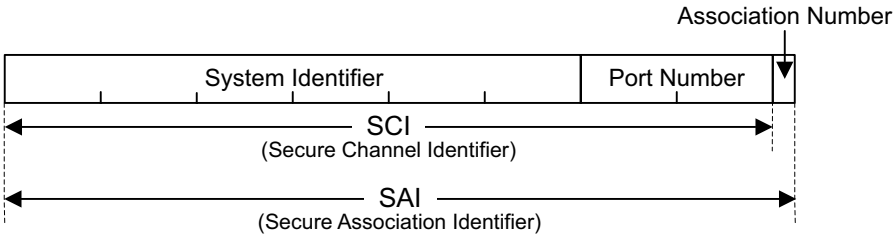


Figure 7-7—Secure Channel and Secure Association Identifiers

If a SecY does not have a usable SA for its outbound SC, i.e., an SA that can be used at no notice for frame transmission with a PN value that is not exhausted, or any of the current SAs for inbound SCs are not usable, then the MAC_Operational status parameter (6.4) will transition to FALSE.

7.2 Multiple instances of the secure MAC Service on a single LAN

Each service access point for an instance of the secure MAC Service is supported by a service access point for an instance of an insecure MAC Internal Sublayer Service (ISS). Multiple instances of the secure MAC Service can be provided by a single LAN, provided that each instance is uniquely identified by unencrypted fields contained in each received frame. These fields identify separate instances of the insecure MAC ISS, each capable of supporting a distinct service access point for MACsec.

Identification of each insecure service instance, and multiplexing and demultiplexing to and from the transmission capabilities provided by the LAN, can be performed wholly below the ISS by a media-specific or media-dependent function. Some media are defined to support such a multiplexing function, e.g., the

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LLID used by IEEE 802.3 EPON (see Clause 12). Where such media-specific multiplexing functions are not available, the source address or SCI may be used (11.8). Provider Bridges are also capable of supporting multiple instances of the ISS over a network of individual LANs (see 11.6).

MACsec should not be used to support multiple instances of the secure MAC Service on a single physical LAN without the use of unencrypted frame fields to identify separate instances of insecure service, each supporting a single instance of secure service. While the use of security to provide multiplexing is impossible to prevent (since different cryptographic keys can be used to separate connectivity), relying solely on security to define the connectivity makes deployment and fault management difficult—the topology of an entire network could change as security was enabled or disabled on a single LAN. Key agreement protocols that use the insecure MAC service can require a matching instance of that service for each secure service instance.

NOTE 1—The service access point for the secure MAC Service is referred to as Controlled Port of the SecY (Clause 10) and the service access point for the insecure MAC Service as the SecY's Common Port. Access to the insecure service for protocol entities above MAC Security is provided at the Uncontrolled Port.

NOTE 2—Although the field or fields used to provide service instance multiplexing are not parameters of the ISS, and thus are not protected, the integrity of the secure MAC Service is not compromised. If the unprotected fields are modified, the frame can be delivered to the wrong SecY, but will subsequently fail integrity checks. Different SecYs use different security associations, keys, and cryptographic nonces. Additional management parameters are (cryptographically) bound to individual SecYs, not to the values of frame fields.

The secure MAC Service requirements for symmetric and transitive connectivity ensure that two or more service instances on the same LAN will appear as separate LANs to the clients of the SecYs. There is therefore no conflict between the use of bridges and the provision of multiple secure service instances.

When clients that are connected to a first service instance change and connect to a second service instance, the secure connectivity alters. MAC_Operational temporarily transitions to False for a minimum amount of time to allow the CA to re-establish its membership (as determined by the KaY). In particular, each time membership of a CA changes, MAC_Operational transitions False for at least one of each pair of SecYs whose connectivity has changed. For example, if members of CA_x leave CA_x and join CA_y and if CA_y has MAC_Operational True, then MAC_Operational transitions to False for either the members of CA_x who are joining CA_y or for the original members of CA_y. MAC_Operational transitions to True once all the new members have joined CA_y.

NOTE 3—Two SecYs that connect to the same LAN and participate in the same CA appear connected to the same LAN (as one would expect) and appear connected to different LANs as they participate in distinct CAs. The effect is similar to partitioning a LAN by switching a repeater on or off.

Distinct instances of the secure point-to-point MAC Service can be provided by a bridge to different end stations connected to the same shared media by using the source address of the frames transmitted by each end station to identify one of a number of SecYs in the receiving bridge (11.8). Where the source address is not sufficient to select the receiving SecY, the SCI can be used to provide service instance multiplexing for both secured and unsecured frames (11.8, 9.5, 10.6).

NOTE 4—IEEE Std 802.11 [B2] specifies its own mechanisms for identifying separate secure associations.

7.3 Use of the secure MAC Service

The secure MAC Service guarantees (6.8) the integrity of the parameters of each service indication, and that each indication is a result of a request made by a SecY that is a member of the same CA as the receiver, though not by any particular member. Management controls associated with each KaY can require certain authentication and key management methods to ensure these guarantees. However, the degree of trust placed in the security of the communication does not imply the degree of trust associated with the communicating peers. Accordingly, the MACsec Key Agreement framework facilitates authorization of each potential member and allows management of the acceptable authorization for inclusion in the CA.

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The secure MAC service does not itself provide any means to label or distinguish different levels of authorization, and does not associate different levels of authorization with individual invocations of the service. A station either participates in a service instance or it does not.

To ensure correct operation of client protocols, secure service indications are not filtered or modified by a SecY except as specified in Clause 8 and Clause 9. Each protocol entity that is a client of the secure MAC Service should implement suitable policies (7.3.1) to support overall network security.

NOTE 1—For example, correct operation of Spanning Tree Protocol depends on the delivery of BPDUs to the Spanning Tree Protocol Entity of a given bridge from all the other bridges attached to the LAN that transmit frames that can be relayed by the given bridge. If a SecY were to require a higher level of authorization to pass received BPDUs through the Controlled Port, data loops in the network could result. However, the STP Entity can adopt a policy of discarding frames rather than permit another system that is not authorized as a bridge to be the Designated Bridge for the CA.

The client policies in use at any time should reflect the intersection of the capabilities permitted to the members of the CA. Policies can be

a) Selected by the client on the basis of the level of authorization, as provided by the PAE through a Layer Management Interface (LMI) (see 10.7) or
b) Selected by a central server that forms part of the management framework for the network, and
    1) Securely downloaded or
    2) Communicated to the client using a secure connection

NOTE 2—If one of the members of the CA is a bridge (strictly speaking the Bridge Port is the CA member), the other members should adopt policies that reflect their confidence in the policies applied by the bridge to forward frames. In this case the trust is partly transitive—the question to be answered by each member of the CA is the degree of trust to place in the bridge's trust of systems that originate frames that the bridge will forward.

7.3.1 Client policies

Client policies, which are not specified in this standard, can include but are not limited to

a) Limiting the set of protocol procedures that can be invoked by the peer
b) Segregating communications between different sets of peer users of the MAC Service
c) Filtering, i.e., discarding, received frames

Clients of the secure MAC Service can also record any exceptional policy actions taken, so as to initiate further administrative action, outside the scope of this standard, with the entities responsible for the operation of the authenticated peer systems.

NOTE—A VLAN-aware Bridge that assigns frames that have been received from a specific Bridge Port (the bridge's point of attachment to a service instance) to a VLAN on the basis of the authorization associated with the Port provides an example of policy of segregating communications, as described in item b) above.

7.3.2 Use of the secure MAC Service by bridges

Each Bridge Port uses the service provided by an individual LAN (see Clause 11), which is not dependent for its connectivity on the operation of other bridges. This ensures that the configuration protocols used by bridges, including the spanning tree protocol, operate over a physical topology (comprising a bipartite graph of bridges and individual LANs connected by Bridge Ports) that is not itself dependent on the active topology, or subsets of the active topology, calculated by those same configuration protocols.

NOTE 1—The apparent exception to this configuration restriction, which does not permit the creation of security associations to create "secure tunnels" through selected bridges in a Bridged Local Area Network, is the use of a Provider Bridged Network as specified in IEEE Std 802.1Q. However, a Provider Bridged Network appears to Customer Bridges as a single LAN providing full connectivity independent of the operation of Customer Bridge protocols.

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PAEs and KaYs use group addressed frames to identify and communicate with peers whose SecYs are potential participants in the same CA. Frames with the group addresses used for this purpose are filtered by certain bridges and EDEs (Clause 15) to restrict each instance of the secure MAC Service to the appropriate LAN scope. By default PAEs and KaYs for MAC Bridges, VLAN Bridges, Provider Bridges, and Provider Backbone Bridges use the PAE group address specified by IEEE Std 802.1X (also identified as the Nearest Non-TMPR Bridge group address by IEEE Std 802.1Q). Use of this address restricts each instance of the secure MAC Service to an individual customer LAN.

NOTE 2—Use of Reserved Group MAC Addresses helps to ensure that the physical topology as perceived by spanning tree and other configuration protocols aligns with that provided by MAC Security.

The policies applied by the Bridge Forwarding Process that is a client of each MAC service instance can include but are not limited to

a) Use of static Filtering Database Entries
b) Use of the RSTP and MSTP restrictedRole parameters
c) The PVID for the port
d) Configuration of the VLAN Translation Table
e) Inclusion in the Member Set for any given VLAN and the setting of Enable Ingress Filtering
f) Identification of the Port as a Provider Edge Port
g) Port priority
h) Priority remapping tables.

NOTE 3—A Bridge Port is one of the bridge's points of attachment to an instance of the ISS and is used by the MAC Relay Entity and associated Higher-Layer Entities as specified in IEEE Std 802.1Q.

NOTE 4—The RSTP and MSTP restrictedRole parameters in IEEE Std 802.1Q ensure that the spanning tree active topology for other Bridge Ports is unaffected by BPDUs received on the Port, while continuing to protect against data loops and allowing the peer system to use the BPDUs it receives to select between redundant service instances. The restrictedRole parameter should be set if the authorization (see also 7.3) of the peer system(s) is not sufficient to allow full participation in determining the active topology of the network.

In response to a limited authorization on the Bridge Port, a bridge can be configured to discard frames other than from a specified number of MAC addresses and to use additional services provided by the network administrator to ensure that these permitted addresses are not used by other end stations in the network.

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8. MAC Security protocol (MACsec)

MACsec provides the secure MAC Service (Clause 6) on a frame-by-frame basis, using cryptographic methods within the context of security relationships maintained by MACsec Key Agreement.

This clause

a) Sets out requirements for the design (8.1) and support (8.2) of MACsec
b) Provides an overview of its operation (8.3).

NOTE 1—The operation of MACsec Key Agreement Entity (KaY) and the protocols it uses are outside the scope of this standard. However, the security relationships (Clause 7) it establishes are essential to the operation of MACsec and form part of the support requirements.

Conformance to this standard is in terms of the observable protocol arising from the operation of a MAC Security Entity (SecY, Clause 10), including management of MACsec and the service provided to client protocols that use the secure MAC service.

Each of the possible sets of cryptographic algorithms used by MACsec to provide connectionless frame integrity and data confidentiality compose a Cipher Suite. This clause describes the result of Cipher Suite use by the SecY, illustrated in Figure 8-1. The normative specification of each Cipher Suite is provided in Clause 14. The Cipher Suite is selected as part of the establishment of the CA (7.1.1).

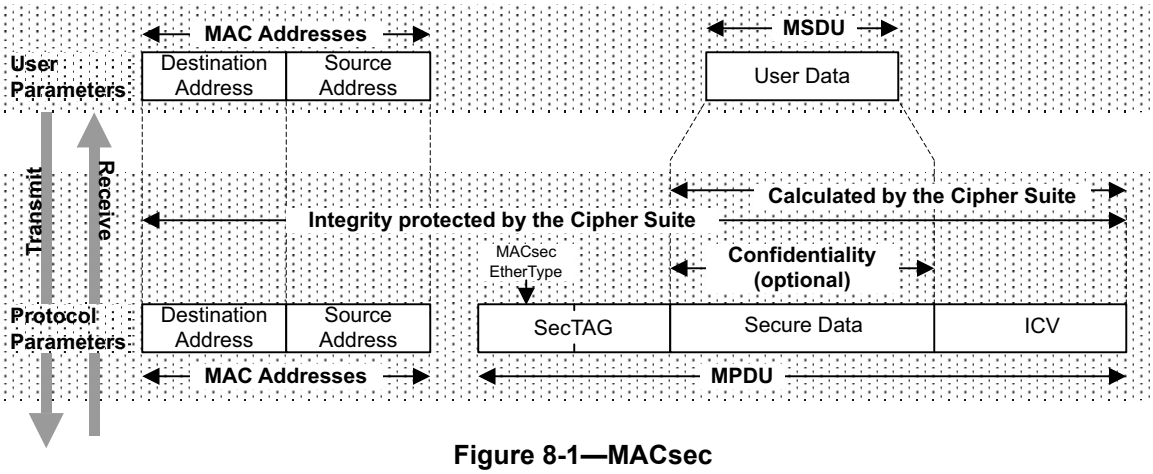


Figure 8-1—MACsec

NOTE 2—The Destination Address and Source Address parameters are shown as separate from the MPDU in Figure 8-1, as they are separate parameters of each service request. The encoding of these parameters into a transmitted frame on a medium is accomplished by the supporting service, which can interpose additional octets between those of the addresses and the MSDU. In the strict sense of externally visible transmission, this standard deals with parameters of service primitives, not with frames. However, it is often convenient to talk of these parameters as a frame.

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8.1 Protocol design requirements

MACsec operates in networks comprising end stations and individual point-to-point or shared media LANs, arbitrarily interconnected by intermediate systems, such as MAC Bridges, VLAN-aware Bridges, and routers. MACsec supports, preserves, and maintains the quality of the secure MAC Service in all its aspects as specified by Clause 6, meeting requirements for

a) Connectivity (6.7)
b) Security (6.8, 6.9, 8.1.1)
c) Manageability (8.1.2)
d) Interoperability (8.1.3)
e) Deployment (8.1.4)
f) Coexistence (8.1.5)
g) Scalability (8.1.6)
h) Intrusion detection (8.1.7)
i) Localization and isolation of attacks (8.1.8)
j) Implementation (8.1.9).

These requirements are met by the operation of MACsec (8.3) together with requirements placed on

k) The architecture that specifies how MAC Security Entities (SecYs) are placed within LAN stations and communicate with selected peers (Clause 11)
l) The choice of cryptographic methods that compose each MACsec Cipher Suite (Clause 14)
m) Support of the protocol by each SecY, and on the system that contains it (8.2)
n) The operation of the protocols that support MACsec Key Agreement, including aspects of authentication, authorization, and distribution of keys.

8.1.1 Security requirements

In addition to providing the security guarantees (6.8) and services (6.9) required for support of the secure MAC service, the design of MACsec

a) Enables a succession of SAs, each with its own Secure Association Key (SAK) to be used to support the connectivity provided by each SC. The succession of SAKs (7.1), together with the use of Key Agreement protocols that provide Perfect Forward Secrecy, protects against the compromise of any single SAK, without disrupting service.
b) Ensures that a fresh SA, supporting an existing CA, can be used within a known bounded time (1 s, see 8.1.9) at intervals that are also bounded (keys can be changed as frequently as once every 10 s) after Key Agreement provides the associated SAK.
c) Allows operation of the Key Agreement protocol to be independent of MACsec. In particular allows fresh SAKs to be supplied at any time, without unnecessarily disrupting communication.

The security provided by each SAK rests on the security provided by the Cipher Suite (see Clause 14 for requirements), which in turn depends on the guarantees provide by the cryptographic mode of operation and its underlying block cipher, and on the protocols and procedures used to ensure that keys remain secret.

8.1.2 Manageability requirements

The design of MACsec ensures that the protocols that configure, and that run over media, individual LANs, and Bridged or Virtual Bridged Local Area Networks as a whole, can continue to operate with no diminution in the capabilities available to and customarily used by network administrators. Existing firewall and forwarding filters can still be applied to specific protocols.

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When the Default Cipher Suite is used for integrity protection without confidentiality protection, protocol analyzers and other tools that support MACsec parsing can understand the User Data transmitted, but cannot modify that data without the receiving SecY being aware of the intrusion. This capability is also available whenever the Secure Data remains the same as the User Data and the integrity check value (ICV) length is the same as that of the Default Cipher Suite.

Where MACsec supports a shared media CA, or a point-to-point CA that uses shared transmission facilities, MACsec can convey the SCI (7.1.2, 8.2.1, 9.9), thus identifying the secure system that transmitted the MPDU both to the intended recipient and to network management systems.

8.1.3 Interoperability requirements

Change the first paragraph of 8.1.3 as follows:

Interoperability between independent implementations of MACsec is facilitated by mandatory implementation of the Default Cipher Suite (GCM-AES-128, 14.5) or the Ascon-XPN-128 Cipher Suite (14.9).

The use of Cipher Suites as a specification tool reduces the number of permutations of cryptographic algorithms and their parameters. Clause 14 mandates elements of Cipher Suite specification.

Where the underlying MAC Service used by MACsec is supported by a Provider Bridged Network (IEEE Std 802.1Q), communicating SecYs can be attached to different media operating (locally) at different transmission rates. Interoperability between, for example, 10 Gb/s and 1 Gb/s, and between 1 Gb/s and 100 Mb/s requires interoperability across the speed range. The design of MACsec facilitates interoperability from 1 Mb/s to 100 Gb/s without modification or negotiation of protocol formats and parameters. Operation at higher transmission rates depends on the capabilities of the Cipher Suite. The mandatory default Cipher Suite has been selected (Clause 14) in part because of its ability to perform across this range.

NOTE—Clearly additional ways of interconnecting different media access control methods could be standardized in the future. The above requirement mandates that interoperability be preserved between SecYs attached to a wide range of media operating over a wide speed range.

Communication between SecYs using different media access methods requires that MACsec not make use of any media-specific additions to the MAC Service, or rely on any deficiencies in support of the service being common to all communicating participants. MACsec includes an explicit indication of the length of the Secure Data to avoid imposing the minimum frame size and padding requirements of IEEE Std 802.3 on all other media access methods that make use of MACsec.

8.1.4 Deployment requirements

Change the first paragraph of 8.1.3 as follows:

The design of MACsec allows security to be introduced into a network one LAN at a time. Additionally the controls provided by a SecY (Clause 10) allow the deployment of MACsec capable systems one by one on a LAN, prior to enabling security. Integrity checking of MPDUs that are integrity but not confidentiality protected can be disabled to facilitate testing of Key Agreement protocols prior to enabling security. Management counters allow a network system administrator to confirm that the connectivity provided by a SecY is complete and that enabling security will not disrupt existing required connectivity.

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8.1.5 Coexistence requirements

The design of MACsec allows coexistence with other protocols on the same insecure LAN. This

a) Supports incremental deployment (8.1.4).
b) Allows fresh keys to be derived, using Key Agreement protocols that can be independently specified and use different frame formats, while MACsec is operating.
c) Supports use of shared media providing independent services.

8.1.6 Scalability requirements

The resources required to support MACsec in any single LAN station (an end station or a Bridge Port) are a function of the number of the SecY peers on the same LAN, but are independent of other systems attached to the same network but not the same LAN.

8.1.7 Unauthorized access attempts

Detecting attempts at unauthorized access is facilitated by integrity and replay protection, and the management counters (10.7) that record the receipt of invalid (presumably modified) and repeated and misordered (likely to be replayed) frames. Management for client policies (7.3) that use the guaranteed connectivity provided by MACsec should also record attempted violations.

8.1.8 Localization and isolation of attacks

MACsec discards frames sent by systems that are not authenticated and authorized members of the CA, thus localizing the traffic sent by those stations to a single LAN. The authorization accorded by the policies enforced by clients of MACsec (7.3.1) can restrict unauthorized attempts to affect protocols that control the network infrastructure. Where communication that does result in unauthorized behavior elsewhere in the network has been permitted, the use of MACsec by the intervening systems allows tracing of the source of that communication.

8.1.9 Implementation

The design of MACsec allows the SecY to function asynchronously with respect to other processes in the system. Key Agreement protocols and changes of SAKs are not tightly synchronized to the service requests and indications processed by the SecY. Delays in communication and variations in scheduling between the SecY and KaY can be as much as one second, allowing autonomous processing of frames in real time by the SecY while the KaY can operate as a normally scheduled software process. Time is also allowed for the KaY to compute keys and for the SecY to compute key schedules, and perform other preprocessing.

8.2 Protocol support requirements

The support of MACsec places requirements on

a) The secure system of which the SecY forms a part for
    1) SC identification (8.2.1)
    2) Support of transmit and receive SAKs (8.2.2).
b) The functionality provided by Key Agreement protocols, and the operation of the KaY for
    1) Independence of KaY operation from MACsec operation and state (8.2.3)
    2) Discovering connectivity (8.2.4)
    3) Authentication (8.2.5)
    4) Authorization (8.2.6)
    5) Key exchange and maintenance (8.2.7).

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8.2.1 SC identification requirements

Each SecY shall be capable of identifying each of its transmit SCs with an SCI that comprises a unique 48-bit MAC Address and a 16-bit Port Identifier that is unique within the scope of that address (7.1.2, 9.9).

NOTE—MKA (IEEE Std 802.1X) verifies that each participant in any given CA has a unique SCI, as part of satisfying Cipher Suite requirements prior to establishing secure communication.

8.2.2 SA Key requirements

On transmit the Cipher Suite implementation shall be able to

a) Prepare a new SAK for use within one second (8.1.9) of being given it by the KaY.
b) Change from the use of one installed SAK to the next within the time normally taken to transmit one minimum sized frame, and shall not discard any frames as a result of the change.

NOTE—Elsewhere in this standard, the requirement for switching between SAKs is modeled as a requirement to support two SAKs for transmission, allowing management counters to reflect the continued use of a key after its successor has been provided by the KaY. The behavior of an implementation capable of accepting the new key and using it within one frame time is fully conforming, and will not cause any apparent management anomalies.

On receive the Cipher Suite implementation shall be able to

c) Receive any frame and its immediate successor using any one of two SAKs, allowing the selection of different keys to switch without missing a frame.
d) Prepare a new SAK for use within one second (8.1.9) of being given it by the KaY.

The system does not need to be able to seamlessly switch between Cipher Suites.

8.2.3 KaY independence of MACsec

The KaY is aware of the required connectivity, identifying the SCs that compose the CA, independently of the design and state of MACsec.

The KaY operates resiliently in face of specifically identified denial of service attacks (as identified by the key agreement protocol specification).

These requirements are met in part by distinguishing key agreement frames from MACsec frames by using a different EtherType.

8.2.4 Discovering connectivity

The KaY discovers connections between peer stations and recognizes potential connections.

NOTE 1— The MAC status parameters (6.4) indicate when connectivity changes. The status parameters provided by the KaY can also temporarily transition False to indicate a change in the authentication or authorization of its peers, preventing attacks that secretly degrade the trust.

The KaY accepts indications of which Cipher Suites are supported by the SecY via the LMI.

NOTE 2—The negotiation of which Cipher Suite is to be used on a connection is based on what Cipher Suites are available locally and at the peer SecY.

The KaY accepts indications of which connectivity capabilities are supported by the SecY via the LMI. The KaY delivers the connectivity selection to the SecY via the LMI.

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8.2.5 Authentication requirements

The PAE supports mutual authentication of peer stations, and the SecY assumes that such authentication has taken place.

8.2.6 Authorization requirements

The PAE provides authorization of services to be delivered to a peer station.

The PAE provides information to local services about the currently selected Cipher Suite.

8.2.7 Key exchange and maintenance

The KaY delivers SAKs via the LMI (10.7.28).

The KaY creates, manages, and maintains one CA that connects two or more KaYs and their corresponding SecYs. The KaY creates and maintains all of the point-to-multipoint SCs and SAs between itself and all the stations within the CA (10.2, 10.7.11–10.7.15, 10.7.21–10.7.24). An SAK delivered by a given KaY is not shared with any other KaY, is not used by the given KaY to support more than one CA, and once used to support an SA for a given SC is not re-used to support any other SA for that SC. A KaY can (and in the MACsec Key Agreement protocol (MKA) specified in IEEE Std 802.1X does) use a single SAK to support multiple SCs within a CA. It is recognized that two SAKs can have the same value with a probability of no less than 1 in $2^{keysize}$ when generated by an approved pseudorandom function.

The KaY monitors the use of PNs by the SecY via the LMI in order to identify impending exhaustion of the transmitting SA (10.7.23). IEEE Std 802.1X-2010 specifies the distribution of a fresh SAK when the value of the PN exceeds that of the constant PendingPNExhaustion (0xC000 0000 for 32-bit PNs). If extended packet numbering (a 64-bit PN) is used in conjunction with IEEE Std 802.1X, PendingPNExhaustion takes the value 0xC000 0000 0000 0000.

The KaY accepts indications that one SA is retired and a new one is started, in other words, when an overlapping pair of SAs is provisioned and the SecY switches from one to the next.

8.3 MACsec operation

MACsec comprises modification and additions to the MAC Service Data Unit (MSDU) conveyed by each frame transmitted by a user of the protocol, and illustrated in Figure 8-1. The MAC Security TAG (SecTAG) conveys parameters that identify the protocol, identify the key to be used to validate the received frame, and provide replay protection. The Secure Data field conveys the User Data, encrypted if confidentiality is provided. The ICV ensures the integrity of the MAC Destination Address, MAC Source Address, SecTAG, and User Data.

NOTE 1—The addition of the SecTAG and ICV to the MPDU, together with possible expansion of the User Data when conveyed in the Secure Data field can increase the size of a frame to the point that it no longer conforms to the maximum frame size specified by the media access method standard. If the implementation of the service used by MACsec cannot transmit the resulting MPDU, it is discarded.

MACsec does not transmit additional frames, such as keep alives or key exchanges. Each frame is delivered unmodified to peer users, subject to validation of the origin, destination and source address, and user data.

Figure 8-2 illustrates the transmission and reception of a frame by MACsec.

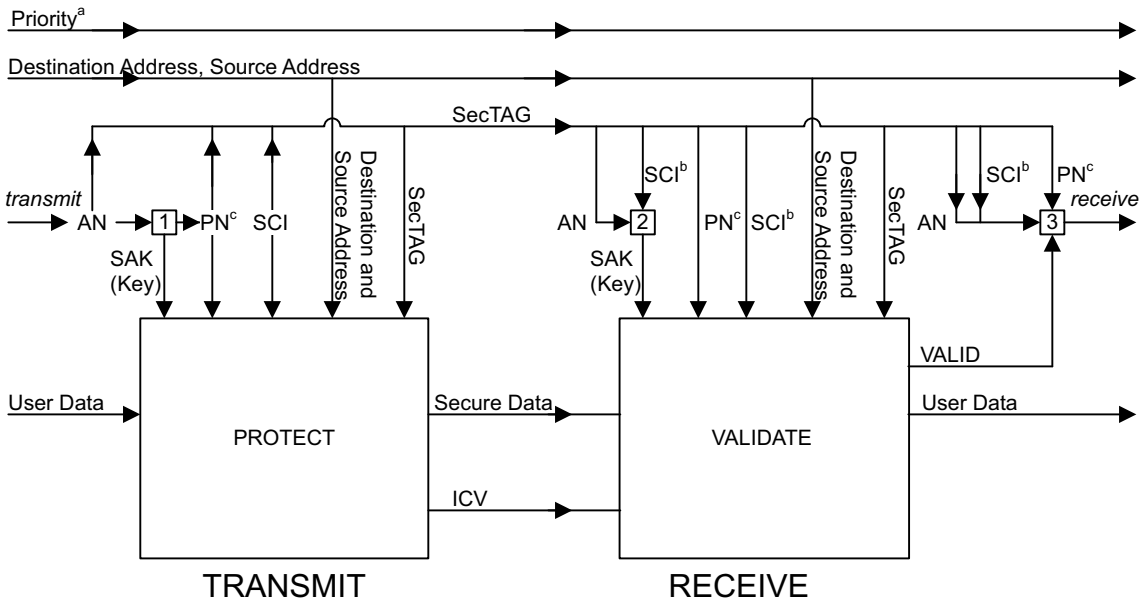
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On transmission, the frame is first assigned to a transmit SC and to the SA (7.1.3) identified by its Association Number (AN) (7.1.3, 9.6) that will be used by that SC to protect the transmitted frame. The AN is used to identify the SAK (7.1.3) and the next PN (9.8) for that SA. The AN, the SCI (7.1.2), and the 32 least significant bits of the PN are encoded in the SecTAG (the SCI can be omitted for point-to-point CAs if only one transmit SC is in use) along with the MACsec EtherType (9.8) and the number of octets in the frame following the SecTAG if less than 48 [8.1.3; Short Length (SL) field in 9.7].

The protection function (14.1) of the Current Cipher Suite is presented with the SAK, the PN and SCI, the destination and source addresses of the frame together with the octets of the SecTAG, and the User Data. It returns the Secure Data and the ICV.

On receipt of a MACsec frame, the AN, SCI, PN, and SL field (if present) are extracted from the SecTAG. If the CA is point-to-point and the SCI is not present, the value previously communicated by the KaY will be used. The AN and SCI are used to assign the frame to an SA, and hence to identify the SAK. If the Current Cipher Suite uses extended packet numbering (a 64-bit PN), the full PN is recovered (as specified in 10.6) using the 32 least significant bits conveyed in the SecTAG and the 32 most significant bits used in a prior successful frame validation.

The validation function of the Current Cipher Suite is presented with the SAK, the PN and SCI, the destination and source addresses of the frame together with the octets of the SecTAG, and the Secure Data and ICV. If the integrity of the frame has been preserved and the User Data can be successfully decoded from the Secure Data, a VALID indication and the octets of the User Data are returned.



[a] Priority can be changed by media access method or receiving system and is not protected

[b] The SCI is extracted from the SCI field of the SecTAG if present. A value conveyed by key agreement (point-to-point only) is used otherwise.

[c] The SecTAG carries only the least significant 32 bits of the PN. When a 64 bit PN (extended packet numbering) is used, the most significant 32 bits are recovered on receipt, and the complete 64 bit PN is presented to PROTECT, VALIDATE, and the replay check.

Functions
1 Lookup Key and next PN for transmit SA identified by AN
2 Lookup Key PN for receive SA identified by SCI, AN
3 Discard if received frame not VALID. Discard if replay check of PN for receive SA identified by SCI, AN fails. Updated replay check.

Figure 8-2—MACsec operation

NOTE 2—If the Current Cipher Suite supports extended packet numbering, the PN comprises 64 bits. The validation functions of the GCM-AES-XPN Cipher Suites (14.7, 14.8) use the SCI to identify a 32-bit Short Secure Channel Identifier (SSCI) supplied by the KaY and construct a 96-bit IV using that SSCI and the PN.

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If the receive frame is valid, replay protection (if enabled) is applied, by checking that the received PN is not less than the lowest acceptable PN for the SA. If the check succeeds the parameters of the frame, unchanged from those transmitted, are presented to the MACsec client, and the lowest acceptable PN updated. The lowest acceptable PN can lag behind the received PN values, providing a window in which replay is tolerated, to allow receipt of frames that have been misordered by the network.

The format and encoding of each of the fields that comprise the SecTAG, including the support of different MACsec protocol versions is specified in Clause 9. The operation of the SecY that operates the MACsec protocol, the service that it provides, and the management control variables, error handling, and diagnostic information recorded is described in Clause 10.

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9. Encoding of MACsec Protocol Data Units

This clause specifies the structure and encoding of the MACsec Protocol Data Units (MPDUs) exchanged between MAC Security Entities (SecYs). It

a) Specifies rules for the representation and encoding of protocol fields
b) Specifies the major components of each MPDU and the fields they comprise
c) Reviews the purpose of each field and the functionality provided
d) Specifies validation of the MPDU on reception
e) Documents the allocation of an EtherType value, the MACsec EtherType, to identify MPDUs.

NOTE—The MPDU validation checks specified in this clause are deliberately limited to ensuring successful decoding, and do not overlap with the specification of SecY operation (Clause 10).

9.1 Structure, representation, and encoding

All MPDUs shall contain an integral number of octets.

The octets in a MPDU are numbered starting from 1 and increasing in the order they are put into the MAC Service Data Unit (MSDU) that accompanies a request to or indication from the instance of the MAC Internal Sublayer Service (ISS) used by a SecY.

The bits in an octet are numbered from 1 to 8 in order of increasing bit significance, where 1 is the least significant bit in the octet.

Where octets and bits within a MPDU are represented using a diagram, octets shown higher on the page than subsequent octets and octets shown to the left of subsequent octets at the same height on the page are lower numbered, bits shown to the left of other bits within the same octet are higher numbered.

Where two or more consecutive octets are represented as hexadecimal values, lower numbered octet(s) are shown to the left and each octet following the first is preceded by a hyphen, e.g., 01-80-C2-00-00-00.

When consecutive octets are used to encode a binary number, the lower octet number has the more significant value.When consecutive bits within an octet are used to encode a binary number, the higher bit number has the most significant value. When bits within consecutive octets are used to encode a binary number, the lower octet number composes the more significant bits of the number. A flag is encoded as a single bit, and is set (True) if the bit takes the value 1, and clear (False) otherwise. The remaining bits within the octet can be used to encode other protocol fields.

9.2 Major components

Each MPDU comprises

a) A MAC Security TAG (SecTAG) (9.3)
b) Secure Data (9.10)
c) An integrity check value (ICV) (9.11).

Each of these components comprises an integral number of octets and is encoded in successive octets of the MPDU as illustrated in Figure 9-1.

NOTE—The MPDU does not include the source and destination MAC addresses, as these are separate parameters of the service requests and indications to and from the insecure service that supports MACsec.

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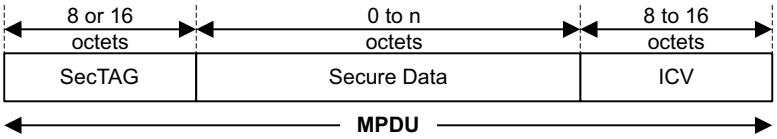


Figure 9-1—MPDU components

9.3 MAC Security TAG

The SecTAG is identified by the MACsec EtherType (9.4), and conveys the

a) TAG Control Information (TCI, 9.5)
b) Association Number (AN, 9.6)
c) Short Length (SL, 9.7)
d) Packet Number (PN, 9.8)
e) Optionally encoded Secure Channel Identifier (SCI, 9.9).

The format of the SecTAG is illustrated in Figure 9-2.

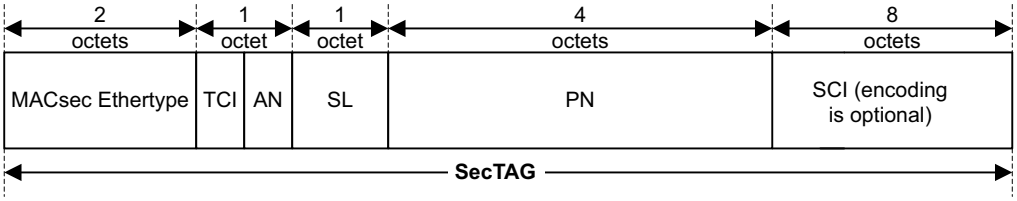


Figure 9-2—SecTAG format

9.4 MACsec EtherType

The MACsec EtherType (Table 9-1) comprises octet 1 and octet 2 of the SecTAG. It is included to allow

a) Coexistence of MACsec capable systems in the same environment as other systems
b) Incremental deployment of MACsec capable systems
c) Peer SecYs to communicate using the same media as other communicating entities
d) Concurrent operation of Key Agreement protocols that are independent of the MACsec protocol and the Current Cipher Suite
e) Operation of other protocols and entities that make use of the service provided by the SecY's Uncontrolled Port to communicate independently of the Key Agreement state.

Table 9-1—MACsec EtherType allocation

Tag Type	Name	Value
IEEE 802.1AE Security TAG	MACsec EtherType	88-E5

The encoding of the MACsec EtherType in the MPDU is illustrated in Figure 9-3.

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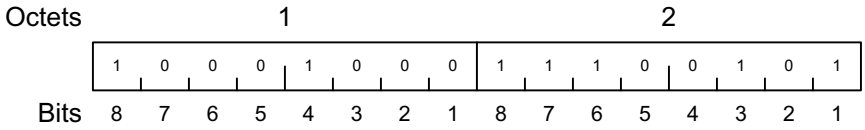


Figure 9-3—MACsec EtherType encoding

9.5 TAG Control Information (TCI)

The TCI field comprises bits 8 through 3 of octet 3 (Figure 9-4) of the SecTAG. These bits facilitate

a) Version numbering of the MACsec protocol without changing the MACsec EtherType
b) Optional use of the MAC Source Address parameter to convey the SCI
c) Optional inclusion of an explicitly encoded SCI (7.1.2, Figure 7-7)
d) Use of the EPON (Clause 12) Single Copy Broadcast (SCB) capability, without requiring an explicit SCI to distinguish the SCB Secure Channel
e) Extraction of the User Data from MPDUs by systems that do not possess the SAK (8.1.2, 8.1.4) when confidentiality is not being provided
f) Determination of whether confidentiality or integrity alone are in use.

The encoding of the MACsec TCI in the MPDU is illustrated in Figure 9-4.

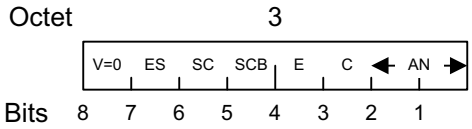


Figure 9-4—MACsec TCI and AN Encoding

The version number shall be 0 and is encoded in bit 8.

NOTE—Future versions of the MACsec protocol may use additional bits of the TCI to encode the version number. The fields and format of the remainder of the MPDU may change if the version number changes.

If the MPDU is transmitted by an end station and the first 6 octets of the SCI are equal to the value of the octets of MAC Source Address parameter of the ISS request in canonical format order, bit 7 [the End Station (ES) bit] of the TCI may be set. If the ES bit is set, bit 6 (the SC bit) shall not be set and an SCI shall not be explicitly encoded in the SecTAG. The ES bit is clear if the Source Address is not used to determine the SCI.

If an SCI (9.9, 7.1.2) is explicitly encoded in the SecTAG, bit 6 (the SC bit) of the TCI shall be set. The SC bit shall be clear if an SCI is not present in the SecTAG.

If and only if the MPDU is associated with the Secure Channel that supports the EPON Single Copy Broadcast capability, bit 5 (the SCB bit) of the TCI may be set. If the SCB bit is set, bit 6 (the SC bit) shall not be set and an SCI shall not be explicitly included in the SecTAG.

If the ES bit is set and the SCB is not set, the SCI comprises a Port Identifier (7.1.2) component of 00-01. If the SCB bit is set, the Port Identifier (7.1.2) component has the reserved SCB value of 00-00.

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If the Encryption (E) bit is set and the Changed Text (C) bit is clear, the frame is not processed by the SecY (10.6) but is reserved for use by the KaY. Otherwise, the E bit is set if and only if confidentiality is being provided and is clear if integrity only is being provided and the C bit is clear if and only if the Secure Data is exactly the same as the User Data and the ICV is 16 octets long.

When the Default Cipher Suite (14.5) is used for integrity protection only, the Secure Data is the unmodified User Data, and a 16 octet ICV is used. Both the E bit and the C bit are therefore clear, and the data conveyed by MACsec is available to applications, such as network management, that need to see the data but are not trusted with the SAK that would permit its modification. Other Cipher Suites may also integrity protect data without modifying it, and use a 16 octet ICV, enabling read access to the data by other applications. The E and C bits are also clear for such Cipher Suites when integrity only is provided.

Some cryptographic algorithms modify or add to the data even when integrity only is being provided, or use an ICV that is not 16 octets long. The C bit is never clear for such an algorithm, even if the E bit is clear to indicate that confidentiality is not provided. Recovery of the data from a MACsec frame with the E bit clear and the C bit set requires knowledge of the Cipher Suite at a minimum. That information is not provided in the MACsec frame.

If both the C bit and E bit are set, confidentiality of the original User Data is being provided.

9.6 Association Number (AN)

The AN is encoded as an integer in bits 1 and 2 of octet 3 of the SecTAG (Figure 9-4) and identifies up to four different SAs within the context of an SC.

NOTE—Although each receiving SecY only needs to maintain two SAs per SC, the use of a 2-bit AN simplifies the design of protocols that update values associated with each of the SAs.

9.7 Short Length (SL)

SL is an integer encoded in bits 1 through 6 of octet 4 of the SecTAG and is set to the number of octets in the Secure Data (9.10) field, i.e., the number of octets between the last octet of the SecTAG and the first octet of the ICV, if that number is less than 48. Otherwise, SL is set to zero. If the number is zero then the frame is deemed not to have been short. The Secure Data field always comprises at least one octet.

Bits 7 and 8 of octet 4 shall be zero.

9.8 Packet Number (PN)

The 32 least significant bits of the PN are encoded in octets 5 through 8 of the SecTAG to

a) Provide a unique IV PDU for all MPDUs transmitted using the same SA
b) Support replay protection.

NOTE 1—The IV used by the Default Cipher Suite GCM-AES-128 (14.5) and the GCM-AES-256 Cipher Suite (14.6) comprises the SCI (even if the SCI is not transmitted in the SecTAG) and a 32-bit PN. Subject to proper unique MAC Address allocation procedures, the SCI is a globally unique identifier for a SecY. To satisfy the IV uniqueness requirements of Counter mode of operation, a fresh key is used before PN values are reused.

NOTE 2—If the Current Cipher Suite provides extended packet numbering, i.e., uses a 64-bit PN, the 32 least significant bits of the PN are conveyed in this SecTAG field and the 32 most significant bits are recovered on receipt as specified in 10.6. The IV used by the GCM-AES-XPN Cipher Suites (14.7, 14.8) is constructed from a 32-bit SSCI distributed by key agreement protocol and unique for each SCI within the scope of the CA (and hence within potential users of the same SAK) and the 64-bit non-repeating PN.

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9.9 Secure Channel Identifier (SCI)

If the SC bit in the TCI is set, the SCI (7.1.2, 8.2.1) is encoded in octets 9 through 16 of the SecTAG and facilitates

a) Identification of the SA where the CA comprises three or more SCs
b) Network management identification of the SecY that has transmitted the frame.

Octets 9 through 14 of the SecTAG encode the System Identifier component of the SCI. This comprises the six octets of a MAC address uniquely associated with the transmitting SecY. The octet values and their sequence conform to the Canonical Format specified by IEEE Std 802.

Octets 15 and 16 of the SecTAG encode the Port Identifier component of the SCI, as an integer.

The 64-bit value FF-FF-FF-FF-FF-FF-FF-FF is never used as an SCI and is reserved for use by implementations to indicate the absence of an SC or an SCI in contexts where an SC can be present.

An explicitly encoded SCI field in the SecTAG is not required on point-to-point links, which are identified by the operPointToPointMAC status parameter of the service provider, if the transmitting SecY uses only one transmit SC. In that case, the secure association created by the SecY for the peer SecYs, together with the direction of transmission of the secured MPDU, can be used to identify the transmitting SecY. Therefore, an explicitly encoded SCI is unnecessary. Although the SCI does not have to be repeated in each frame when only two SecYs participate in a CA (see Clause 8, Clause 9, and Clause 10), the SCI (for Cipher Suites using a 32-bit PN) or the SSCI (for Cipher Suites using a 64-bit PN) still forms part of the cryptographic computation.

9.10 Secure Data

The Secure Data comprises all the octets that follow the SecTAG and precede the ICV. The Secure Data field is never of zero length, since the primitives of the MAC Service require a non-null MSDU (User Data) parameter.

NOTE 1—In practice, if the MSDU composed by the operation of the current Cipher Suite following MPDU reception contains less than two octets, it will be discarded by the user of the SecY's controlled port, since it is too short to contain an EtherType or an LLC length field. Such discard is, however, determined by the user of the Controlled Port and not by the SecY itself.

NOTE 2—Ethernet transports frames of a minimum size, and provides no explicit indication of PDU length if the PDU is composed of fewer octets. The SL field allows the originator of the frame, which is not necessarily aware of the need of an intervening Ethernet component to pad the frame, to specify the number of octets in the MPDU, thus allowing the receiver to unambiguously locate the ICV.

9.11 Integrity check value (ICV)

The length of the ICV is Cipher Suite dependent, but is not less than 8 octets and not more than 16 octets, depending on the Cipher Suite.

NOTE—The ICV protects the destination and source MAC address parameters, as well as all the fields of the MPDU.

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9.12 PDU validation

A received MPDU is valid if and only if it comprises a valid SecTAG, one or more octets of Secure Data, and an ICV, i.e.,

a) It comprises at least 17 octets.
b) Octets 1 and 2 compose the MACsec EtherType.
c) The V bit in the TCI is clear.
d) If the ES or the SCB bit in the TCI is set, then the SC bit is clear.
e) Bits 7 and 8 of octet 4 of the SecTAG are clear.
f) If the C and SC bits in the TCI are clear, the MPDU comprises 24 octets plus the number of octets indicated by the SL field if that is non-zero and at least 72 octets otherwise.
g) If the C bit is clear and the SC bit set, then the MPDU comprises 32 octets plus the number of octets indicated by the SL field if that is non-zero and at least 80 octets otherwise.
h) If the C bit is set and the SC bit clear, then the MPDU comprises 8 octets plus the minimum length of the ICV as determined by the Cipher Suite in use at the receiving SecY, plus the number of octets indicated by the SL field if that is non-zero and at least 48 additional octets otherwise.
i) If the C and SC bits are both set, the frame comprises at least 16 octets plus the minimum length of the ICV as determined by the Cipher Suite in use at the receiving SecY, plus the number of octets indicated by the SL field if that is non-zero and at least 48 additional octets otherwise.

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10. Principles of MAC Security Entity (SecY) operation

This clause

a) Provides an overview of the SecY (10.1), the service that it provides, and its relationship to other entities in a secure system including its associated MACsec Key Agreement Entity (KaY).
b) Describes the functionality of the SecY (10.2).
c) Provides a model of operation (10.3) comprising an architecture (10.4) and its constituent processes (10.5 through 10.7) that supports the detailed functionality including management controls.
d) Details the addressing requirements and specifies the addressing of SecYs (10.8).

NOTE—Clause 6 defines the properties of the secure MAC Service, Clause 7 describes the security relationships used to support the service and how the service is used, providing the context within which each SecY operates, Clause 8 sets out requirements for the MACsec protocol and introduces the operation of the protocol, and Clause 9 specifies the encoding of parameters in MPDUs. This clause does not repeat all the information provided in those prior clauses, but includes sufficient reference to facilitate an understanding of SecY operation. Clause 7 of IEEE Std 802.1AC-2016 describes the basic architectural concepts and terms used in this clause, including service, service access point, service primitive, and ports.

10.1 SecY overview

Each SecY uses the MAC Service provided by a Common Port (10.4) to provide one instance of the secure MAC Service (Clause 6) to the user of its Controlled Port and one instance of insecure service to the user of its Uncontrolled Port (Figure 10-1).

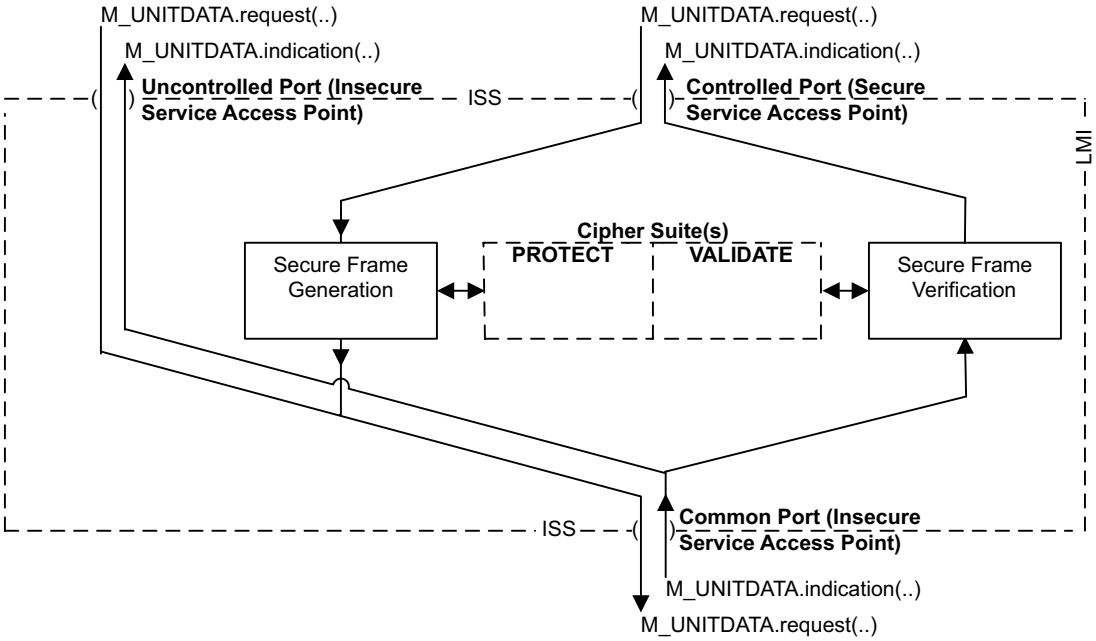


Figure 10-1—SecY

The integrity and origin of the parameters of each service request and indication accepted from and delivered to the Controlled Port are protected and validated by the SecY. The SecY may also encrypt to provide user data confidentiality. If the parameters that accompany a service indication at the Common Port are not successfully validated as required by management controls, no service indication will occur at the Controlled Port and the received parameters will be discarded.

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Each service request made by the user of a SecY's Uncontrolled Port results in an identical request at the Common Port, and each service indication received from the Common Port results in an identical indication to the user of its Uncontrolled Port in addition to any indication at the Controlled Port.

NOTE 1—Some frames received at the Uncontrolled Port will be discarded because they can only be useful to a SecY supporting the associated Controlled Port.

The relative order of Common Port indications and the corresponding indications at the Uncontrolled Port and the Controlled Port is not defined, save that the order of indications from one Port to another Port is preserved. Similarly the relative order of user requests at the Uncontrolled and Controlled Ports does not define the order of requests to the Common Port. The interval between any request or indication and the SecY making a corresponding request or indication shall not exceed the bounds specified in Table 10-4.

The specification of the cryptographic algorithms used at any time to provide integrity and confidentiality, together with the values of parameters (for example, key size) used by those algorithms, compose a Cipher Suite (Clause 14). This standard mandates a default Cipher Suite that can provide integrity protection only or both integrity and confidentiality. A SecY may implement additional Cipher Suites. This standard only permits the use of Cipher Suites that meet well defined criteria (14.2, 14.3).

The KaY is part of the Port Access Entity (PAE, IEEE Std 802.1X) associated with the SecY and uses the service provided by the Uncontrolled Port to transmit and receive frames that support key agreement protocols. These frames are distinguished by EtherType, so other selected protocol entities can also communicate using insecure frames by making use of the Uncontrolled Port.

The KaY determines the value of the MAC_Operational parameter (IEEE Std 802.1AC) associated with Controlled Port (10.7.4, 10.7.5) consistent with the provisions of this standard (6.4, 6.5, 6.7, 7.1.3, 7.2, 10.5.1, 10.5.2, 10.7.14, 10.7.2, 10.7.25).

The KaY communicates transmit and receive keys and other information (10.2) to the SecY through its Layer Management Interface (LMI). The LMI is also used to exchange information with local protocol entities responsible for network management, such as an SNMP Agent.

NOTE 2—The term *local* refers to any other entity residing within the same system. Information exchange with a local entity can be modeled as occurring through its LMI (10.1, 10.3, 10.4, Figure 10-1), thus facilitating information exchange between entities not necessarily adjacent in a protocol layer reference model. No constraints are placed on the information exchanged, but there is no synchronization with any particular invocation of service at a service access point, so LMI exchanges do not effectively add to the parameters of a service such as the MAC service.

10.2 SecY functions

Each SecY supports

a) Secure transmission of the parameters of service requests made by the user of its Controlled Port
b) Insecure transparent transmission from the Uncontrolled Port
c) Reception, verification, and delivery of secure service indications to the Controlled Port
d) Reception and transparent delivery of service indications to the Uncontrolled Port
e) MAC Status (6.4) and point-to-point parameters (6.5) for the Uncontrolled and Controlled Ports.

Management controls that support deployment (8.1.4) of MACsec include

f) Transmission and reception by the user of the Controlled Port without frame modifications
g) Reception without integrity checking
h) Use of multiple transmit SCs and a configurable replayWindow to support media access control methods and provider networks that can misorder frames with different priorities and/or addresses.

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Selection of a Cipher Suite, CA establishment, and SA support, is supported by allowing the KaY to

i) Discover which Cipher Suites are implemented and how many receive SCs each can support
j) Select the Current Cipher Suite
k) Identify the SCs to be used to support reception for the CA
l) Provide transmit and receive SAKs for identified SAs
m) Confirm that SAKs have been installed, i.e., are ready for use
n) Monitor the PN used for transmission, in order to provide new SAKs prior to PN exhaustion.

Operational and diagnostic controls and statistics provide

o) Administrative control over the optional security tagging capabilities of the SecY
p) A count of frames intended for transmission but discarded as too long for the Common Port
q) Counts of received frames without the MACsec EtherType, discarded by validation checks, without SCIs when the LAN connectivity is not restricted to point-to-point communication, identified as belonging to unknown SCs, identified as belonging to an SA that is not in use, failing the replay check, failing the integrity check, and delivered to the user.

NOTE—Except where explicitly specified otherwise, throughout this standard the term *user* refers to the user of the MAC service instance provided by the Controlled Port, and the term *provider* refers to the instance of protocol and procedures that provides the MAC service instance to the SecY at the Common Port.

10.3 Model of operation

The model of operation in this clause is simply a basis for describing the functionality of a SecY. It is in no way intended to constrain real implementations; these may adopt any internal model of operation compatible with the externally visible behavior that this standard specifies. Conformance of equipment to this standard is purely in respect of observable protocol.

10.4 SecY architecture

A SecY uses an instance of the MAC Internal Sublayer Service (ISS, 6.1), referred to as the Common Port, to provide a secured instance of the ISS, the Controlled Port, and an insecure instance of the ISS, the Uncontrolled Port, that provides transparent transmission and reception through the Common Port.

The architecture of a SecY is illustrated in Figure 10-2 and comprises

a) The Controlled, Uncontrolled, and Common Ports together with their MAC Status parameters
b) The Secure Frame Generation process (10.5)
c) The Secure Frame Verification process (10.6)
d) Cipher Suite protection of transmitted frames and validation of received frames (8.2, Clause 14)
e) A Transmit Multiplexer and a Receive Demultiplexer
f) Optional transmit and receive frame check sequence (FCS) regenerators
g) A SecY Management process (10.7).

The Transmit Multiplexer accepts transmit requests from the Uncontrolled Port and the Secure Frame Generation process for the Controlled Port and submits corresponding requests to the Common Port. The Receive Demultiplexer submits each indication from the Common Port to the Uncontrolled Port and to the Secure Frame Verification process for the Controlled Port.

NOTE 1—This specification most clearly sets out the resulting behavior of a conforming implementation. Real implementations can implement the behavior in any way that yields the same externally visible behavior (including the values of management counters). For example, examination of the specification in this clause shows that there need be no implementation burden corresponding to duplication of the received frame if validateFrames is Strict and none of the users of the Uncontrolled Port make use of the MACsec EtherType. .

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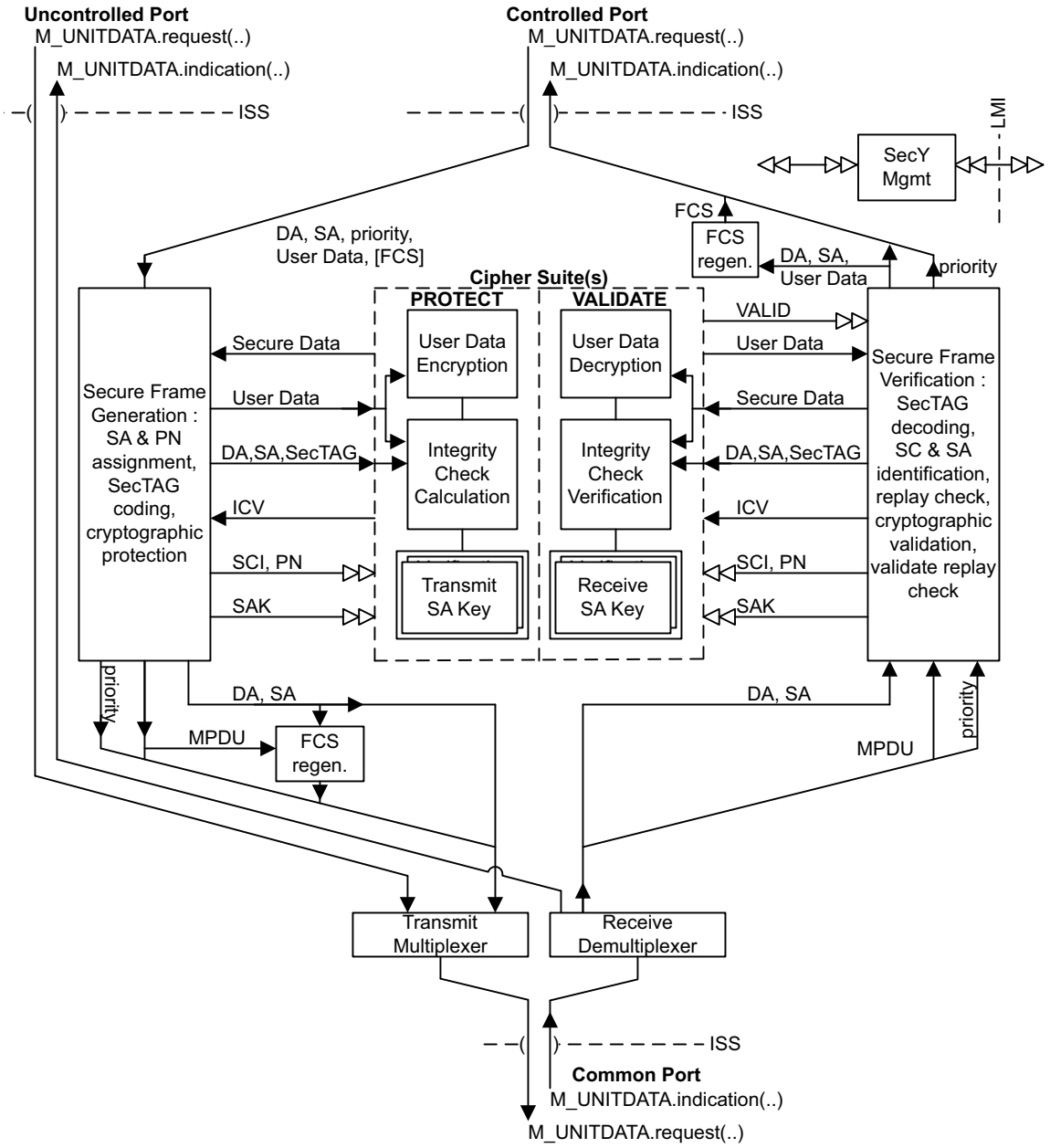


Figure 10-2—SecY architecture and operation

A Layer Management Interface (LMI) is used by the SecY Management process to communicate the capabilities of the SecY, its controls, status, protocol, management events, and counters to and from other entities that compose the secure system containing the SecY.

Management controls are provided to allow a SecY to be incorporated in a network system before MACsec is deployed, and to facilitate staged deployment. If protectFrames is not set, frames submitted to the Controlled Port are transmitted without modification. The validateFrames control allows untagged frames to be received, and Cipher Suite validation of tagged frames to be disabled or its result simply counted without frame discard. The replayProtect and replayWindow controls allows replay protection to be disabled, to

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operate on a packet number window, or to enforce strict frame order. If replayProtect is set but the replayWindow is not zero, frames within the window can be received out of order; however, they are not replay protected. Management counters allow configuration and operational errors to be identified and rectified before enabling secure operation. The effect of the controls, and the counters maintained, are summarized in Figure 10-3 and Figure 10-4.

The FCS can be included as a parameter of an M_UNITDATA.request or M_UNITDATA.indication primitive. When the data that is within the FCS coverage is modified by the addition of an ICV or encryption of the user data, the FCS changes. The SecY shall not introduce an undetected frame error rate greater than that which would have been achieved by preserving the original FCS (6.10).

NOTE 2—There are number of possibilities for changing FCS without diminishing the coverage provided. One is to generate a new FCS by algorithmically modifying the received FCS, based on knowledge of the FCS algorithm and the transformations that the frame has undergone between reception and transmission.

10.5 Secure frame generation

For each transmit request at the Controlled Port, the Secure Frame Generation process

a) Assigns the frame to an SA (10.5.1).
b) Assigns the nextPN variable for that SA to be used as the value of the PN for that protected frame (10.5.2).
c) Encodes the octets of the SecTAG including the least significant 32 bits of the PN in the PN field (10.5.3).
d) Provides the protection function (14.1, 10.5.4) of the Current Cipher Suite with
    1) The SAK
    2) The SCI for the SC used by the SecY to transmit
    3) The PN
    4) The SecTAG
    5) The sequence of octets that compose the User Data.
e) Receives the following parameters from the Cipher Suite protection operation:
    6) The sequence of octets that compose the Secure Data
    7) The ICV
f) Issues a request to the Transmit Multiplexer with the destination and source MAC addresses and an MPDU comprising the octets of the SecTAG, Secure Data, and the ICV concatenated in that order (10.5.5). If the SecY does not implement an Access Priority Table (10.7.17), the priority of the request is the same as that received from the Controlled Port; otherwise, it is the access priority given by the table for the received priority.

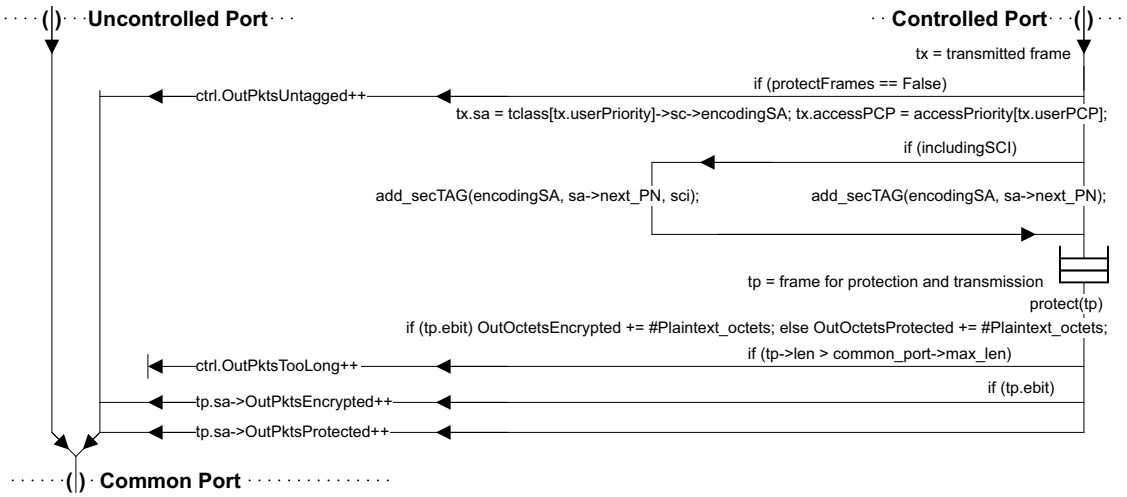
If the management control protectFrames is False, the preceding steps are omitted, an identical transmit request is made to the Transmit Multiplexer, and the OutPktsUntagged counter incremented.

NOTE—This model of operation supports the externally observable behavior that can result when the Cipher Suite implementation calculates the Secure Data and ICV parameters for a number of frames in parallel, and the responses to protection and validation requests are delayed. Transmitted frames are not misordered.

10.5.1 Transmit SA assignment

Each frame is assigned to the SA identified by the current value of the encodingSA variable for the selected transmit SC. If the SecY does not implement a Traffic Class Table it uses a single transmit SC. If implemented, the Traffic Class Table specifies the value of the most significant four bits of the SCI's Port Identifier component for each possible transmit request user priority, allowing selection of one of up to eight distinct SCs (see 10.7.17).

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Tests and their consequences are annotated in this diagram using the computer language 'C++' (ISO/IEC 14882), with variable names corresponding to abbreviations of the text of this clause (10), which takes precedence.

NOTE—Secure generation frame counters are identified as reported by management. Confidentiality or integrity only protection is selected for an SA when it is created, so either but not both of the OutOctetsEncrypted or OutOctetsProtected counts and either OutPktsEncrypted or the OutPktsProtected will be incremented while that SA is in use, and the current value of the packet counter can be derived from nextPN for the SA less any change in the value of OutPktsTooLong since that SA has been used for protection, allowing an implementation to optimize counter resources.

Figure 10-3—Management controls and counters for secure frame generation

The encodingSA is updated following an LMI request from the KaY to start transmitting using the SA and can be read but not written by network management. Frames will be protected using the encodingSA immediately after the last frame assigned to the previous SA has been protected. If the SA is not available for use, and the management control protectFrames is set, MAC_Operational transitions to False for the Controlled Port, and frames are neither accepted or delivered using the port.

10.5.2 Transmit PN assignment

The frame's PN is set to the value of nextPN for the SA, and nextPN is incremented. If the nextPN variable for the encodingSA is zero (or 2^{32} if the Current Cipher Suite does not support extended packet numbering, 2^{64} if it does) and the protectFrames control is set, MAC_Operational transitions to False for the Controlled Port and frames are neither accepted or delivered. The initial value of nextPN is set by the KaY via the LMI prior to use of the SA, and its current value can be read both while and after the SA is used to transmit frames. The value of nextPN can be read, but not written, by network management.

10.5.3 SecTAG encoding

The SecTAG is encoded as specified in Clause 9.

The SC bit in the SecTAG shall be set and the SCI explicitly encoded in the SecTAG, and the management status parameter includingSCI set to True, if and only if

a) The management control alwaysIncludeSCI is True,
   or
b) The number of transmit SCs is greater than one,
   or
c) The number of receive SCs enabled for reception is greater than one, and
   1) The management control useES is False,
      and
   2) The management control useSCB is False.

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If the management control useES is True and includingSCI is False, the ES bit in the SecTAG shall be set. Otherwise, if useES is False or includingSCI is True, the ES bit shall be clear.

If the management control useSCB is True and includingSCI is False, the SCB bit in the SecTAG shall be set. Otherwise, if useSCB is False or includingSCI is True, the SCB bit shall be clear.

NOTE—These rules cover the case where useSCB is True and the number of active receive channels is greater than one. However, SCB bit use is currently restricted to supporting a transmit only EPON interface (see Clause 12).

Table 10-1 summarizes the rules [a) through c) above], with each of the columns to the right representing a valid combination of controls, number of SCs, and SecTAG encoding.

Table 10-1—Management controls and SecTAG encoding

Mgmt controls	alwaysIncludeSCI	T[a]	F	F	F	F	F
	useES	—	—	F	T	T	F
	useSCB	—	—	F	T	F	T
#SCs	#transmitSCs > 1	—	T	—	F	F	F
	#receiveSCs enabled for reception > 1	—	—	T	—	—	—
Mgmt status	includingSCI	T	T	T	F	F	F
SecTAG encoding	SC bit set? (SCI explicitly encoded)	Y	Y	Y	N	N	N
	ES bit set?	N	N	N	Y	Y	N
	SCB bit set?	N	N	N	Y	N	Y

[a]T = True, F = False, — = don't care, Y= Yes, N = No

The values of useES, useSCB, and alwaysIncludeSCI can be written and read by management. The read-only management status parameter includingSCI is True if an SCI is explicitly encoded in each SecTAG, and False otherwise. The number of active receive SCs is controlled by the KaY but can be read by management.

If a frame is to be integrity protected, but not encrypted, with the number and value of the octets of the Secure Data exactly the same as those of the User Data, and an ICV of 16 octets, then the E bit shall be clear and the C bit clear. The E bit shall be clear and the C bit set if the frame is not encrypted but the octets of the Secure Data differ from those of the User Data or the ICV is not 16 octets.

If both confidentiality (through encryption) and integrity protection are applied to a frame then both the E bit and the C bit shall be set. The SecY shall not encode a SecTAG that has both the E bit set and the C bit clear for any frame received from the Controlled Port for transmission.

10.5.4 Cryptographic protection

If the Cipher Suite is currently protecting frames using the previous SA and its SA Key, as reflected by the value of the encipheringSA, the frame can be queued awaiting protection. The value of the encipheringSA is updated, and protection of the frame parameters is started within a minimum frame size transmission delay, after the last frame has been protected using the previous key.

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The use of each of the Cipher Suites specified by this standard is specified in Clause 14, which takes precedence over any explanation in this or other clauses.

The appropriate octet counter is incremented by the number of octets in the User Data (OutOctetsEncrypted if confidentiality protection was provided, and OutOctetsProtected otherwise).

10.5.5 Transmit request

If the MPDU composed of the concatenated octets of the SecTAG, Secure Data, and ICV exceeds the size of the MSDU supported by the Common Port, the frame is discarded and a counter incremented. Details of the discarded frame may be recorded to assist network management resolution of the problem. Otherwise, the parameters of the service request are submitted to the Transmit Multiplexer.

10.6 Secure frame verification

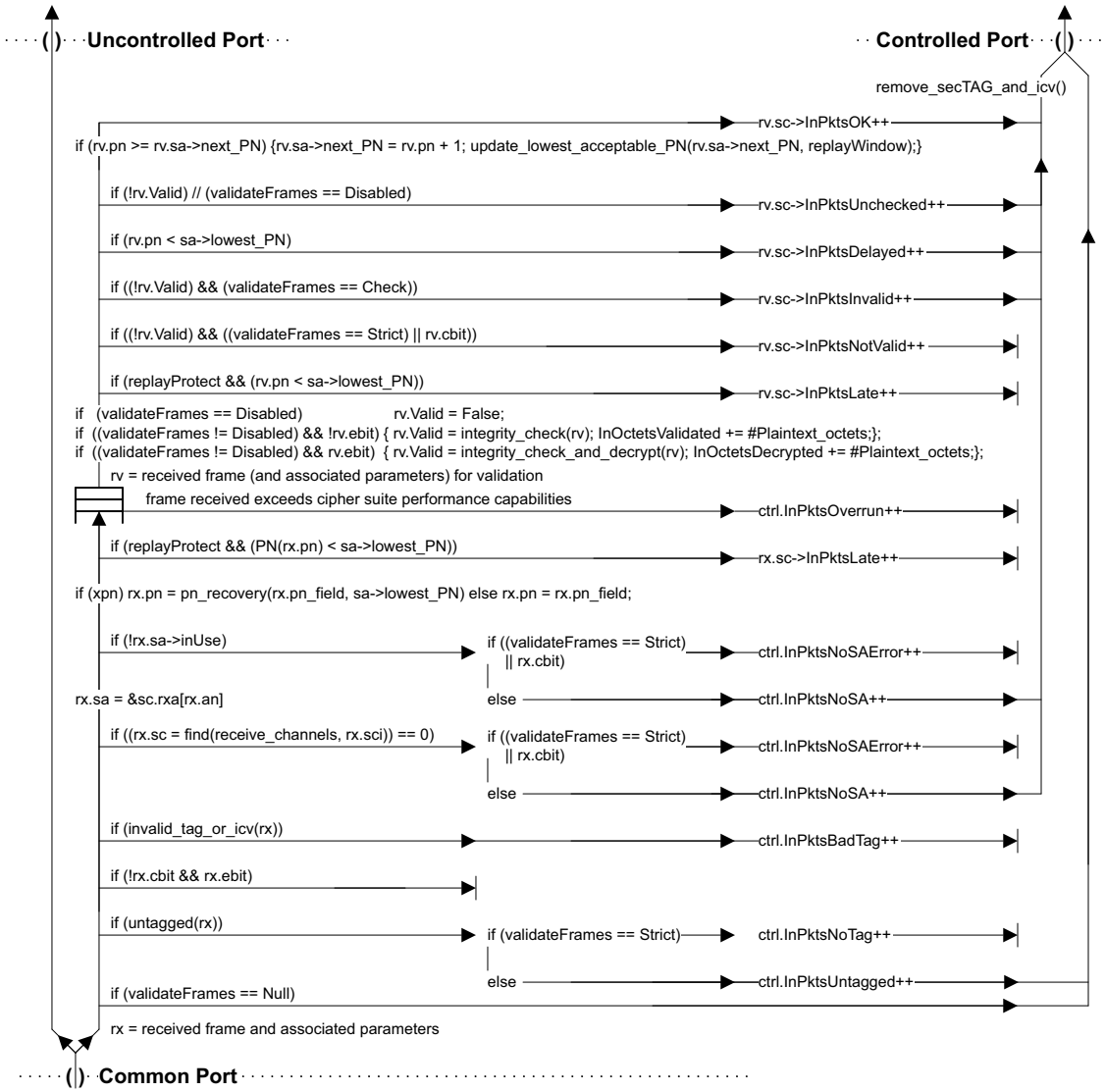
For each receive indication from the Receive Demultiplexer, the Secure Frame Verification process

a) Examines the user data for a SecTAG.
b) Validates frames with a SecTAG as specified in 9.12.
c) Extracts and decodes the SecTAG as specified in 9.3 through 9.9.
d) Extracts the User Data and ICV as specified in 9.10 and 9.11.
e) Assigns the frame to an SA (10.6.1).
f) Recovers the PN and performs a preliminary replay check against the last validated PN for the SA (10.6.2).
g) Provides the validation function (14.1, 10.6.3) of the Current Cipher Suite with
    1) The SAK
    2) The SCI for the SC used by the SecY to transmit
    3) The PN
    4) The SecTAG
    5) The sequence of octets that compose the Secure Data
    6) The ICV.
h) Receives the following parameters from the Cipher Suite validation operation
    1) A Valid indication, if the integrity check was valid and the User Data could be recovered
    2) The sequence of octets that compose the User Data.
i) Updates the replay check (10.6.4).
j) Issues an indication to the Controlled Port with the DA, SA, and priority of the frame as received from the Receive Demultiplexer, and the User Data provided by the validation operation (10.6.5).

If the management control validateFrames is not Strict, frames without a SecTAG are received, counted, and delivered to the Controlled Port; otherwise, they are counted and discarded. If validateFrames is Disabled, cryptographic validation is not applied to tagged frames, but frames whose original service user data can be recovered are delivered. Frames with a SecTAG that has the TCI E bit set but the C bit clear are discarded, as this reserved encoding is used to identify frames with a SecTAG that are not to be delivered to the Controlled Port. If validateFrames is Null, all received frames are delivered to the Controlled Port without modification, irrespective of the absence, presence, or validity of a SecTAG, and the processing described in a) through j) above and in 10.6.1 through 10.6.5 is not performed. Figure 10-4 summarizes the operation of secure frame verification management controls and counters.

Setting validateFrames to Null shall also cause the secure frame generation control protectFrames (10.5) to become False, thus allowing a port that includes a SecY to behave as if the SecY were not present. In particular, it allows a MACsec-capable bridge or EDE to forward frames that have a SecTAG but no other outer tag (such as a VLAN tag).

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Tests and their consequences are annotated in this diagram using the computer language ‘C++’ (ISO/IEC 14882), with variable names corresponding to abbreviations of the text of this clause (10), which takes precedence.

NOTE—Secure verification frame counters are identified as reported by management. Whether a given counter can be incremented depends on the management control validateFrames and on whether received frames are confidentiality protected, allowing an implementation to optimize resources. As shown in the figure, only one counter for each of the sets {InPktsUntagged, InPktsNoTag} and {InPktsNoSA, InPktsNoSAError} for the Controlled Port as a whole and only one counter for each of the sets {InPktsLate, InPktsDelayed}, {InPktsInvalid, InPktsNotValid}, and {InPktsUnchecked, InPktsOK} for each received SC can be incremented while validateFrames and confidentiality policy remain unchanged.

Figure 10-4—Management controls and counters for secure frame verification

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10.6.1 Receive SA assignment

An SCI is associated with the received frame and used to locate the receive SC. If an SCI is not explicitly encoded in the SecTAG, the value established by the KaY for a single peer is used.

If the SC is not found, the received SCI may be recorded to assist network management resolution of the problem, and

a) If validateFrames is Strict or the C bit in the SecTAG is set, the InPktsNoSAError counter is incremented and the frame is discarded; otherwise
b) The InPktsNoSA counter is incremented and the frame (with the SecTAG and ICV removed) is delivered to the Controlled Port.

If the receive SC has been identified, the frame's AN is used to locate the receive SA received frame and processing continues with the preliminary replay check. If the SA is not in use:

c) If validateFrames is Strict or the C bit is set, the frame is discarded and the InPktsNoSAError counter incremented; otherwise
d) The InPktsNoSA counter is incremented and the frame delivered to the Controlled Port.

NOTE—The short phrase "the frame is discarded" is commonly used to express the more formal notion of not processing a service primitive (an indication or request) further and recovering the resources that embody the parameters of that service primitive. No further processing is applied. However, if a duplicate of the primitive has been submitted to another process (by the Receive Demultiplexer in this case) processing of that duplicate is unaffected.

10.6.2 PN recovery and preliminary replay check

If the Current Cipher Suite does not use extended packet numbering, i.e., the PN comprises 32 bits, the value of the PN is that decoded from the 4 octet PN field in the SecTAG of the received frame (9.1, 9.8).

If the Current Cipher Suite supports extended packet numbering, the PN comprises 64 bits. The least significant 32 bits of the PN are those decoded from the PN field in the SecTAG of the received frame. The 32 most significant bits of the PN are recovered for each received frame by applying the assumption that they have remained unchanged since their use in the frame with the lowest acceptable PN—unless the most significant of the 32 least significant bits of the lowest acceptable PN is set and the corresponding bit of the received PN is not set, in which case the value of the 32 most significant bits of the PN is one more than the value of the 32 most significant bits of the lowest acceptable PN. Table 10-2 provides examples.

The recovered PN value is not guaranteed to be the same as that used by the transmitter to protect the frame, but all PN values in the range lowest acceptable PN to lowest acceptable PN plus 2^{31} will be recovered correctly. If the recovered PN value is incorrect, the Cipher Suite validation operation will not return VALID and the frame will be discarded if validateFrames is Strict (10.6.5, 10.7.8). A recovered PN value is used to update the lowest acceptable PN only if the validation operation with that PN value returns VALID.

NOTE 1—For a discussion of the PN recovery algorithm, its incidental properties and alternatives, that goes beyond the normative requirements of this standard, see Seaman [B21].

NOTE 2—If a large number of successive frames were to be lost (2^{30}–1, corresponding to approximately 9 seconds of full utilization of a 400 Gb/s link by minimum sized Ethernet frames) subsequent receipt of MACsec frames might fail to establish a correct PN value. The MACsec Key Agreement protocol (MKA) specified in IEEE Std 802.1X and its amendments communicates the value of the high order bits periodically to recover from this eventuality.

If replayProtect control is enabled and the PN recovered from the received frame is less than the lowest acceptable packet number (see 10.6.5) for the SA, the frame is discarded and the InPktsLate counter incremented.

NOTE 3—If the SC is supported by a network that includes buffering with priority queuing, such as a provider bridged network, delivered frames can be reordered.

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Table 10-2—Extended packet number recovery (examples)

SecTAG PN field value	0x 2A2B 5051
Lowest acceptable PN	0x 0000 0007 1234 DEF0
PN	0x 0000 0007 2A2B 5051
SecTAG PN field value	0x 2A2B 5051
Lowest acceptable PN	0x 0000 0007 8234 DEF0
PN	0x 0000 0008 2A2B 5051
SecTAG PN field value	0x 9A2B 5051
Lowest acceptable PN	0x 0000 0007 8234 DEF0
PN	0x 0000 0007 9A2B 5051
SecTAG PN field value	0x 9A2B 5051
Lowest acceptable PN	0x 0000 0007 2234 DEF0
PN	0x 0000 0007 9A2B 5051

10.6.3 Cryptographic validation

The frame can be queued awaiting validation. If the frame reception rate exceeds the Cipher Suite's validation capabilities, the frame may be discarded and the InPktsOverrun counter incremented.

If the validateFrames control is Disabled, the Cipher Suite validation is not used to validate the frame.

If validateFrames is not Disabled, and the E bit in the SecTAG is set, the Cipher Suite is used to validate and decrypt the frame. If the Cipher Suite does not provide confidentiality protection, it shall not return VALID. The InOctetsDecrypted counter is incremented by the number of octets in the resulting User Data (or an estimate of that number, if VALID is not returned).

If validateFrames is not Disabled, and the E bit in the SecTAG is clear, the Cipher Suite is used to validate the frame. If the Cipher Suite does not provide integrity protection without confidentiality it shall not return VALID. The InOctetsValidated counter is incremented by the number of octets in the resulting User Data (or an estimate of that number, if VALID is not returned).

The frame is marked valid if the Cipher Suite is used and returns VALID and is marked invalid otherwise. The use of each of the Cipher Suites specified by this standard is specified in Clause 14, which takes precedence over any explanation in this or other clauses.

10.6.4 Replay check update

If the PN of the received frame is less than the lowest acceptable packet number for the SA, and replayProtect is enabled, the frame is discarded and the InPktsLate counter incremented.

NOTE—This model of operation assumes that any queuing within the verification process occurs prior to frame validation, and the check described uses the lowest acceptable PN updated by prior frames as described in 10.6.5. Implementations can process frames as convenient, provided the externally observable result is the same.

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10.6.5 Receive indication

If the received frame is marked as invalid, and the validateFrames control is Strict or the C bit in the SecTAG was set, the frame is discarded and the InPktsNotValid counter incremented. Otherwise, the frame is delivered to the Controlled Port, and the appropriate counter incremented as follows:

a) If the frame is not valid and validateFrames is set to Check, InPktsInvalid; otherwise,
b) If the received PN is less than the lowest acceptable PN (treating a 32-bit PN value of zero as 2^{32} and a 64-bit PN value of zero as 2^{64}), InPktsDelayed; otherwise,
c) If the frame is not valid, InPktsUnchecked; otherwise,
d) InPktsOK.

If the PN for the frame was equal to or greater than the nextPN variable for the SA and the frame is valid, nextPN is set to the value for the received frame, incremented by one. The lowest acceptable PN variable is set to the greater of its existing value and the value of nextPN minus the replayWindow variable.

NOTE—The lowest acceptable packet number can also be set or incremented by the KaY to ensure timely delivery.

10.7 SecY management

The SecY management process controls, monitors, and reports on the operation of the SecY, providing access to operational controls and statistics for network management and the KaY through the LMI. It

a) Reports the value of the SCI for the SecY's default traffic class SC (10.7.1).
b) Maintains the MAC Status (6.4) parameters and point-to-point MAC parameters (6.5) for the Uncontrolled (10.7.2) and Controlled (10.7.4) Ports.
c) Provides interface statistics for the Uncontrolled (10.7.3) and Controlled Ports (10.7.6), deriving the latter from the detailed statistics maintained by the SecY.
d) Provides information on the frame verification (10.7.7) and generation (10.7.16) capabilities.
e) Supports control of frame verification (10.7.8) and generation (10.7.17), including management of a Traffic Class Table that allows the user priority associated with the Controlled Port transmit request to select one of a number of transmit SCs, and an Access Priority Table.
f) Supports creation of transmit SCs (10.7.20), each corresponding to one of the values appearing in Traffic Class Table entries.
g) Supports creation of transmit SAs (10.7.22), each associated with an SAK, for the transmit SC.
h) Supports creation of receive SCs (10.7.11), each corresponding to potential member of the CA.
i) Supports creation of receive SAs (10.7.13) for each receive SC, each associated with an SAK.
j) Supports control over reception (10.7.15) and transmission (10.7.24) using individual SAs, and allows the lowest acceptable PN to be set and updated for reception.
k) Maintains statistics for receive and transmit SCs and SAs, accumulating statistics from past SAs.
l) Provides a list of the Cipher Suites with their basic capabilities and properties, and a list of those Cipher Suites implemented by the SecY with management control over their use (10.7.25).
m) Allows selection of the current Cipher Suite, from those implemented.
n) Supports installation of SAKs for the current Cipher Suite, for transmission, reception, or both.

Figure 10-5 illustrates the management information that represents a SecY's capabilities and provides control over and reporting on its operation. For convenience the figure uses UML 2.0 conventions together with C++ language constructs. For an explanation of these conventions, see Fowler [B1]. The containment relationships in Figure 10-5 have been chosen primarily to reflect the necessary relationships between

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lifetimes of potentially transient objects. For example, a receive SC can contain a succession of SAs, but never more than one per AN at a time, and all receive SAs for an SC are deleted when the receive SC ceases to exist. A paradigm of object creation and deletion is used, instead of one of data structure reuse, to express the required bounding of the lifetime of key information.

Figure 10-5 includes data made available, through the LMI, to a SecY by other components (notably the KaY) within the same system, and data and operations that a SecY makes available to those components, in addition to operations and parameters accessible by one or more remote management protocols. Security considerations, and the utility of transient data, restrict management access to some parameters and operations. For particular managed object specifications based on this clause and figure, see Clause 13 (MAC Security Entity MIB) and Clause 23 (YANG Data Models).

NOTE 1—Figure 10-5 omits parameters specific to extended packet numbering [used by some but not all Cipher Suites (14.7, 14.8)] and not accessible by network management. Specifically: 1) the createReceiveSA(), ReceiveSA(), createTransmitSA(), and TransmitSA() procedures all take an additional SSCI parameter, whose value becomes a parameter of the created SA; 2) the install_key() procedure takes an additional Salt parameter, whose value becomes an inaccessible parameter of the Data_key object. These parameters are specified in 10.7.13, 10.7.22, and 10.7.28.

NOTE 2—Figure 10-5 was revised by IEEE Std 802.1AEdk-2022 to provide cross-references and to indicate the accessibility, read (r) or write (w) status of parameters that can be managed by remote management protocol, as specified for the SecY MIB module (Clause 13) or the YANG module (Clause 23). Whether access to particular management parameters is permitted is also subject to security considerations, as described for each of those modules.

In Figure 10-5 the management information for each SecY is shown as indexed by controlledPortNumber within a SecY System. This containment relationship complements that specified in IEEE Std 802.1X, where the management information for each PAE is indexed by portNumber (12.9.2 of IEEE Std 802.1X-2020) within a PAE System and includes the controlledPortNumber that identifies the Controlled Port of the associated SecY. The containment relationship also matches that specified in Clause 13 (MAC Security Entity MIB) and Clause 23 (YANG Data Models). In the MIB module a SecY System corresponds to a SecY MIB module instance, and each controlledPortNumber to the ifIndex (IETF RFC 2863) value used to identify a SecY within that module (13.3.2, 13.2). In the YANG module the management information for each PrY is indexed by an interfaceName.

If a Bridge Port is supported by a SecY (11.3) the ifIndex value used to identify the SecY's Controlled Port will be that identifying the ISS interface (service access point) used by the Bridge Port. IEEE Std 802.1Q specifies Bridge Port Numbers that identify Bridge Ports from the point of view of a bridge's MAC Relay Entity, and port numbers in general to identify ISS interfaces. In simple, common, cases (11.3) each Bridge Port Number can and most likely will be the same as the port number (and ifIndex value) identifying the Controlled Port, though an optional mapping table is specified (12.5.1 of IEEE Std 802.1Q-2022).

IEEE Std 802.1Q can constrain the relationship between Bridge Port Numbers and other bridging parameters (see, for example, 12.13 of IEEE Std 802.1Q-2022) and if RSTP or MSTP are implemented the maximum number of Bridge Ports is 4095 (17.3.2.2 of IEEE Std 802.1Q-2022). In a system comprising multiple bridge components, each port is uniquely identified by a ComponentID and Port Number pair. The SCI values used by a SecYs supporting Bridge Ports do not have to be derived from the Bridge Port Numbers or (possibly different) controlledPortNumbers so do not further constrain those port numbers. However, the least significant 12 bits (if a SecY supports multiple traffic class SCs) and all 16 bits (otherwise) of the Port Identifier can be assigned—subject only to the requirement for SCI uniqueness (8.2.1), so that in the simple case of a bridge component with 4095 or fewer ports, each SCI's Port Identifier can convey the Bridge Port Number and use the Bridge Address for the MAC Address-based component of each SCI, if so desired.

NOTE 2—IEEE Std 802.1AEcg-2017 added the SecY System to Figure 10-5 and clarified the management use of port numbers and ifIndex values, but did not change any related normative provisions.

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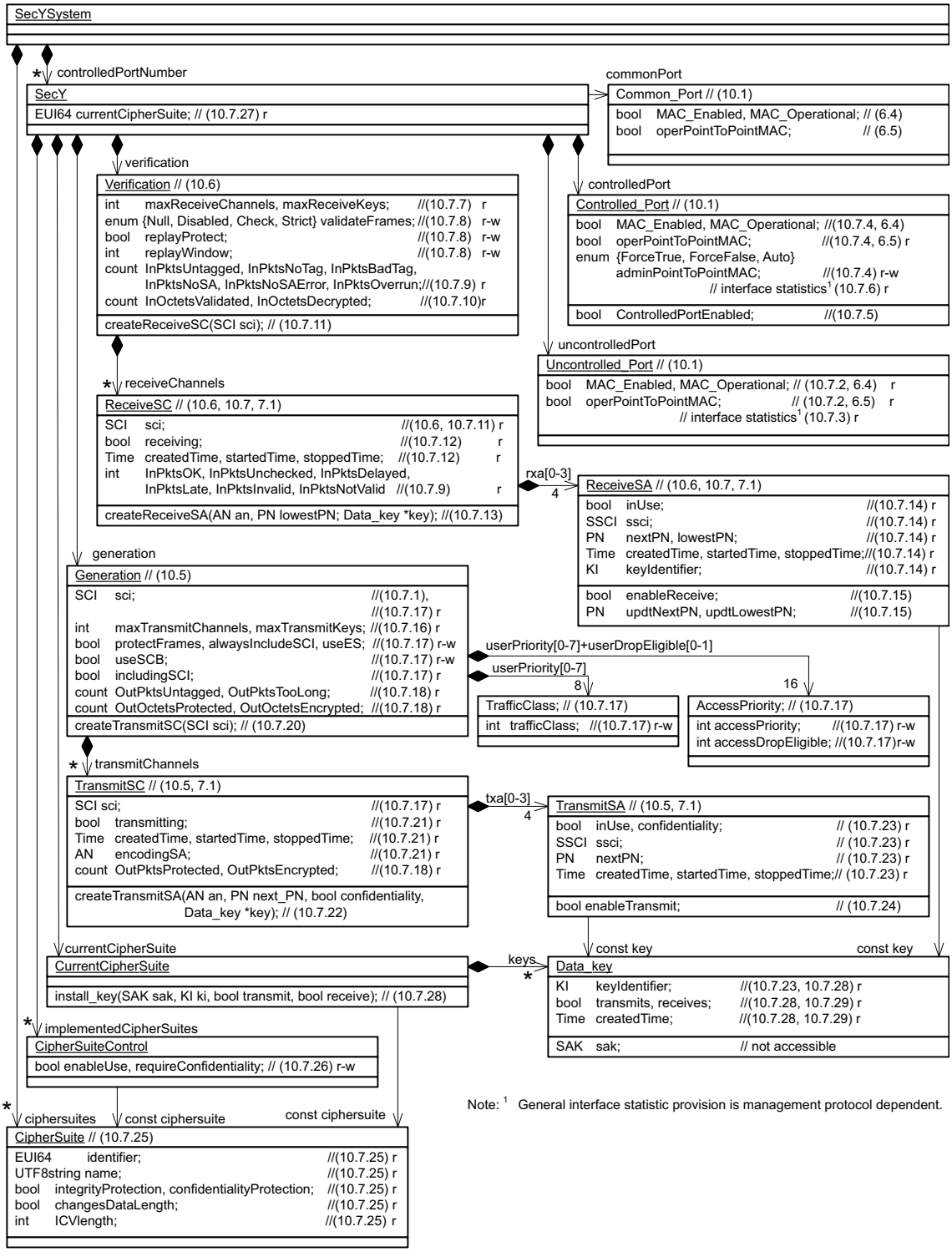


Figure 10-5—SecY managed objects

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The representation of the time of the occurrence of an event, such as the creation time for a receive SC (see 10.7.12, 10.7.14, 10.7.21, 10.7.23, 10.7.28), can depend on the management protocol used to convey that information. In SNMP sysUpTime, the time for which the SNMP agent has been running, is used. In YANG the wall-clock date-and-time is used.

Conformance to this standard is strictly in terms of the external behavior required by this standard, as revealed through the relationship of the operation of the SecY to the operations supported by the SMIv2 MIB module (Clause 13) and to the specifications of protocols operated by the KaY. Interactions with the KaY through the LMI are wholly contained within the secure system, and there is no conformance with respect to syntactic elements that are used to describe that interface in this clause. Table 10-4 specifies performance requirements for SecY operation, including maximum delays for the execution of management operations.

10.7.1 SCI

The SCI for the SecY's default traffic class (7.1.2, 8.2.1) can be read but not written by management.

If the SecY supports more than one transmit SC [5.4(e), 10.7.1, 10.7.17], the four most significant bits of the Port Identifier component of this SCI are zero.

10.7.2 Uncontrolled Port status

The following status parameters are provided to the user(s) of the Uncontrolled Port, including the KaY:

a) MAC_Enabled
b) MAC_Operational
c) operPointToPointMAC

Their values are identical to those for the Common Port. They can be read but not written by management.

10.7.3 Uncontrolled Port statistics

The following statistics are provided to support IETF RFC 2863 interface MIB Counters:

a) ifInOctets
b) ifInUcastPkts, ifInMulticastPkts, and ifInBroadcastPkts
c) ifInDiscards
d) ifInErrors
e) ifOutOctets
f) ifOutUcastPkts, ifOutMulticastPkts, and ifOutBroadcastPkts
g) ifOutErrors

The ifInOctets, ifInUcastPkts, ifInMulticastPkts, and ifInBroadcastPkts counts are identical to those of Common Port and are not separately recorded. The ifInDiscards and ifInErrors counts are zero, as the operation of the Uncontrolled Port provides no error checking or occasion to discard packets, beyond that provided by its users or by the entity supporting the Common Port.

The ifOutErrorscount is zero, as no checking is applied to frames transmitted by the Uncontrolled Port.The ifOutOctets, ifOutUcastPkts, ifOutMulticastPkts, and ifOutBroadcastPkts counts are the same as those for the user of the Uncontrolled Port.

NOTE—The names of 32-bit counters specified by IETF RFC 2863 have been used to identify statistics, avoiding the need for a mapping from this standard's terminology to that specified for the Interfaces Group MIB. The use of 64-bit counters can be required to support that MIB, as specified in 3.1.6 of IETF RFC 2863, with the lower 32-bits reported when object is accessed using the names specified in a), b), e), and f) above. The corresponding 64-bit counter names are

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ifHCInOctets, ifHCInUcastPkts, ifHCInMulticastPkts, ifHCInBroadcastPkts, ifHCIOutOctets, ifHCOutUcastPkts, ifHCOutMulticastPkts, and ifHCOutBroadcastPkts. Corresponding 64-bit YANG data nodes are specified by IETF RFC 8343.

10.7.4 Controlled Port status

The following status parameters are provided to the user of the Controlled Port, and can be read but not directly written by management:

a) MAC_Enabled, True if and only if
   1) ControlledPortEnabled (10.7.5) is True, and
   2) MAC_Enabled is True for the Common Port, and
   3) transmitting (10.7.21) is True for the transmit SC, and
   4) receiving (10.7.12) is True for at least one receive SC.

b) MAC_Operational, True if and only if
   1) MAC_Enabled is True, and
   2) MAC_Operational is True for the Common Port.

c) operPointToPointMAC. If adminPointToPointMAC is Auto (6.5), operPointToPointMAC is True if and only if
   1) validateFrames (10.7.8) is Strict, and receiving is enabled for receive SCs from at most one peer SecY, or
   2) validateFrames is not Strict, and operPointToPointMAC is True for the Common Port.

   Receive SCs are assumed to originate from the same peer SecY if their SCIs are the same with the exception of the four most significant bits of the Port Identifier component.

The following status parameter may be read and written by management:

d) adminPointToPointMAC (6.5)

NOTE—Prior to IEEE Std 802.1AEcg-2017, each SecY used a single transmit SC. The adminPointToPointMAC variable can be used to configure operPointToPointMAC in the event that an earlier implementation of this standard does not recognize two receive SCs as being from the same SecY or configures two distinct SecYs (in the same CA) with SCIs that differ only in the most significant bits of the Port Identifier.

10.7.5 Controlled Port controls

The KaY uses the following parameter(s):

a) ControlledPortEnabled

By setting ControlledPortEnabled False, the KaY can prohibit use of the Controlled Port until the secure connectivity required has been configured.

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10.7.6 Controlled Port statistics

The following statistics are provided to support IETF RFC 2863 interface MIB Counters:

a) ifInOctets
b) ifInUcastPkts, ifInMulticastPkts, and ifInBroadcastPkts
c) ifInDiscards
d) ifInErrors
e) ifOutOctets
f) ifOutUcastPkts, ifOutMulticastPkts, and ifOutBroadcastPkts
g) ifOutErrors

The ifInOctets count is the sum of all the octets of the MSDUs delivered to the user of the Controlled Port by the Secure Frame Verification process (10.6), plus the octets of the destination and source MAC addresses.

The ifInDiscards count is the sum of all the InPktsNoTag, InPktsLate, and InPktsOverrun counts. The ifInErrors count is the sum of all the InPktsBadTag, InPktsNoSA, and InPktsNotValid counts (10.6, Figure 10-4).

The ifOutOctets count is the sum of the all octets of the MSDUs delivered by the user of the Controlled Port to the Secure Frame Generation process (10.5), plus the octets of the destination and source MAC addresses.

The ifOutErrors count is equal to the OutPktsTooLong count (Figure 10-3). If ifOutDiscards is reported as part of IETF RFC 2863 counts, it is zero.

NOTE 1—The names of 32-bit counters specified by IETF RFC 2863 have been used to identify statistics, avoiding the need for a mapping from this standard's terminology to that specified for the Interfaces Group MIB. The use of 64-bit counters can be required to support that MIB, as specified in 3.1.6 of IETF RFC 2863, with the lower 32-bits reported when object is accessed using the names specified in a), b), e), and f) above. The corresponding 64-bit counter names are ifHCInOctets, ifHCInUcastPkts, ifHCInMulticastPkts, ifHCInBroadcastPkts, ifHCIOutOctets, ifHCOutUcastPkts, ifHCOutMulticastPkts, and ifHCOutBroadcastPkts. Corresponding 64-bit YANG data nodes are specified by IETF RFC 8343.

NOTE 2—Separate per SecY 64-bit counters for InPktsNoTAG, InPktsOverrun, InPktsBadTag, and InPktsNoSA, and separate per SC 64-bit counters for InPktsLate, and inPktsNotValid, are required to support the MIB and YANG modules specified in this standard (10.7.9, 13.6, Clause 23).

10.7.7 Frame verification capabilities

The SecY's frame verification capabilities are represented by the following parameters:

a) Maximum number of receive channels
b) Maximum number of keys in simultaneous use for reception

These parameters can be read but not written by management.

10.7.8 Frame verification controls

Frame verification is subject to the following controls, as specified in 10.6:

a) validateFrames, taking values of Null, Disabled, Check, or Strict, with a default of Strict
b) replayProtect, True or False, with a default of True
c) replayWindow, taking values between 0 and 2^{32}–1, with a default of 0

The validateFrames and replayProtect controls are provided to facilitate deployment. They can be read by management. Each may be written by management, but a conformant implementation shall provide a mechanism to allow write access by network management to be disabled for each parameter individually. If management access is prohibited to any of these parameters, its default value should be used.

If the Current Cipher Suite uses extended packet numbering, i.e., a 64-bit PN, the maximum value of replayWindow used in the Secure Frame Verification process (10.6) is 2^{30}–1, thus ensuring that the replayWindow does not encompass more than half of the range of PNs that can be correctly recovered

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(10.6.2). Any higher value set by network management is retained for possible subsequent use with a different Cipher Suite and will be reported if read by network management. This provision provides compatibility with prior revisions of this standard, though it is unlikely that such a high value of replayWindow would have been used.

10.7.9 Frame verification statistics

Any given received frame increments (10.6) exactly one of the following counts [item a) through item l)]. The following counts are maintained for the frame verification process as a whole:

a) InPktsUntagged
b) InPktsNoTag
c) InPktsBadTag
d) InPktsNoSA
e) InPktsNoSAError
f) InPktsOverrun

The following counts are maintained only for each receive SC and are discarded if the record of the SC is deleted by the KaY:

g) InPktsOK
h) InPktsUnchecked
i) InPktsInvalid
j) InPktsNotValid
k) InPktsDelayed
l) InPktsLate

The counts reported for each SC include those for current and prior SAs, with ANs that have since been reused. This allows useful counts to be maintained on high-speed LANs where an SA may be used for little more than 5 min, and an AN reused after 20 min.The times at which each SC and SA were, or are, in use are recorded (10.7.12, 10.7.14) and assist correlation of the statistics collected with network events.

10.7.10 Frame validation statistics

Investigation or validation of the performance of the cryptographic functions is supported by maintaining counts of packets (InPktsOverrun, 10.6.3, 10.7.9) that have been discarded due to inability to validate frames at the received rate, and by accumulation of the following counts:

a) InOctetsValidated, the number of octets of User Data recovered from received frames that were integrity protected but not encrypted.
b) InOctetsDecrypted, the number of octets of User Data recovered from received frames that were both integrity protected and encrypted.

These counts are incremented even if the User Data recovered failed the integrity check or could not be recovered. In the latter case, an estimate of the number of User Data octets is used, as judged by the load imposed on the validation function.

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10.7.11 Receive SC creation

A receive SC, with a given SCI that remains unchanged for the life of the SC, is created following a request from the KaY. Each SC has a unique SCI.

Receive SCs and SAs (10.7.13) may also be created and controlled by management, but a conformant implementation shall provide a mechanism to allow creation and setting of control parameters by network management to be disabled.

10.7.12 Receive SC status

The following status parameters can be read, but not written, by management:

a) receiving, True if inUse (10.7.14) is True for any of the SAs for the SC, and False otherwise
b) createdTime, the system time when the SC was created
c) startedTime, the system time when receiving last became True for the SC
d) stoppedTime, the system time when receiving last became False for the SC

When the SC is created, receiving is False, and startedTime and stoppedTime are equal to createdTime.

The record of the SC should be retained after it is no longer used, subject to the availability of system resources, to provide information about immediate past operation.

10.7.13 Receive SA creation

A receive SA is created for an existing SC on request from the KaY, with the following parameters:

a) The association number, AN, for the SA
b) nextPN (10.6, 10.6.5)
c) lowestPN, the lowest acceptable PN value for a received frame (10.6, 10.6.2, 10.6.4, 10.6.5)
d) A reference to an SAK that is unchanged for the life of the SA

and, if the Current Cipher Suite uses extended packet numbering (14.7, 14.8), the KaY also supplies the following parameter:

e) SSCI for the SA

   Each SA that uses the same SAK has a different SSCI when these Cipher Suites are used. When the SA is created, its SCI and SSCI are provided (for use in subsequent validation operations) to the instance of the Current Cipher Suite identified by the referenced SAK. A receive SA will not be created if the SSCI supplied duplicates that for a different SCI (for the same SAK, for transmission or reception).

Frame verification statistics (10.7.9) for the SA are set to zero when the SA is created. Any prior SA with the same AN for the SC is deleted. Creation of the SA fails unless the referenced SAK exists and is installed (i.e., is available for use). A management protocol dependent reference is associated with each SA. This reference allows each SA to be distinguished from any previously created for the same SCI and AN.

MKA, specified in IEEE Std 802.1X, does not distribute SSCIs explicitly. A KaY assigns SSCI values as follows. The KaY with numerically greatest SCI uses the SSCI value 0x00000001, the KaY with the next to the greatest SCI uses the SSCI value 0x00000002, and so on. This assignment procedure is not necessarily applicable to any other key agreement protocol.

NOTE—At any given time (when configured by a KaY using MKA as specified in IEEE Std 802.1X), this and other Cipher Suites (including those specified in 14.5, 14.6, and 14.7) use the same SAK for all SAs (each with a different SCI) within the same CA and with the same AN. MKA guarantees that each KaY that uses a given SAK has a unique SCI, and these SCIs are present in every MKPDU that conveys a (key-wrapped) SAK.The number of SCIs (and hence

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the number of SSCIs) is ultimately limited by the maximum number of current members in a group CA that MKA can support (less than 100) but is likely to be further limited by the port-based network control application (see Clause 7 of IEEE Std 802.1X-2010).

10.7.14 Receive SA status

The following parameters can be read, but not directly written, by management:

a) inUse
b) nextPN (10.6, 10.6.5)
c) lowestPN, the lowest acceptable PN value for a received frame (10.6, 10.6.2, 10.6.4, 10.6.5)
d) createdTime, the system time when the SA was created
e) startedTime, the system time when inUse last became True for the SA
f) stoppedTime, the system time when inUse last became False for the SA
g) keyIdentifier (10.7.28), identifying the SAK used by the SA

and, if the Current Cipher Suite uses extended packet numbering (14.7, 14.8), the following parameter:

h) ssci, the SSCI for this receive SA

If inUse is True, and MAC_Operational is True for the Common Port, the SA can receive frames.

The keyIdentifier is an octet string, whose format and interpretation depends on the key agreement protocol in use. It does not contain any information about the SAK other than that explicitly chosen by the key agreement protocol to publicly identify the key. If MKA is being used, it is the 128-bit Key Identifier (KI) specified by IEEE Std 802.1X encoded in an octet string as specified by that standard.

10.7.15 Receive SA control

The KaY uses the following parameters to control the use of each receive SA:

a) enableReceive
b) updtNextPN
c) updtLowestPN

When the SA is created, enableReceive and inUse are False and the SA cannot be used to receive frames. The SA shall be able to receive, and inUse shall be True, when enableReceive is set. The SA shall stop receiving, and inUse shall be False, when enableReceive is reset.

The value of nextPN (or lowestPN as appropriate) shall be set to the greater of its existing value and the supplied of updtNextPN (or updtLowestPN). Initially, following creation, the values of nextPN and lowestPN will have been set to the values supplied by KaY.

10.7.16 Frame generation capabilities

The SecY's frame generation capabilities are represented by the following parameter(s):

a) Maximum number of transmit channels
b) Maximum number of keys in simultaneous use for transmission

These parameters can be read but not written by management.

NOTE—An individual SecY can support multiple traffic class SCs (10.7.17). When MKA is used (see Annex E), an SAK distributed by the Key Server is used by all newly created SAs (each supporting one of the SCs in the CA) so a SecY need only support two keys for transmission and reception at a time (allowing for rollover without frame loss, from one SAK to its successor), irrespective of the number of its traffic class SCs and peers in the CA.

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10.7.17 Frame generation controls

Frame generation is subject to the following controls:

a) protectFrames (10.5), True or False, with a default of True
b) alwaysIncludeSCI (10.5.3), True or False, with a default of False
c) useES (10.5.3), True or False, with a default of False
d) useSCB (10.5.3), True or False, with a default of False

The protectFrames control is provided to facilitate deployment. The protectFrames, alwaysIncludeSCI, useES, and useSCB controls can be read by management and may be written, but a conformant implementation shall provide a mechanism to allow write access by network management to be disabled. If management access is prohibited, the default or a value determined by the KaY should be used.

The following status parameter can be read, but not written, by management:

e) includingSCI (10.5.3), True if and only if the SC bit is set and the SCI explicitly encoded in each SecTAG transmitted

The SecY may map each frame to a transmit SC using a Traffic Class Table and the frame's user priority. Up to eight transmit SCs may be implemented, allowing separate transmit SCs for each possible user priority. However, the reason for the possible use of multiple transmit SCs is to take advantage of the fact that their separate SAs use different PN values and thus to minimize the size of the replayWindow, and in particular to facilitate strict reception ordering and replay protection when the Common Port is supported by a service (such as a Provider Bridged Network, see 11.7) that can reorder frames of different priority. In such cases, the useful number of traffic classes might be two or three, corresponding to the differentiated classes of service provided. While the Traffic Class Table mirrors that specified by IEEE Std 802.1Q for the management of bridge queues, a SecY has a minimal implementation dependent buffering requirement and there is no reason to suppose that any given implementation might provide more timely service if the Common Port does not provide priority differentiated services.

NOTE 1—IEEE Std 802.1AEcg-2017, introducing the use of multiple transmit SCs, was developed contemporaneously with IEEE Std 802.3br™-2016 [B6], which added a capability that allows a high priority Ethernet frame to preempt one of lower priority and thus be received in its entirety prior to the latter. This provides another example of a service that can reorder frames on the basis of priority and for which the use of a separate transmit SC with separate PN number spaces can be used to allow strict ordering and strict replay protection for preemptible and preempting frames separately.

Each entry in the Traffic Class Table is a traffic class, represented by an integer from 0 (default) through 7 that also comprises the numeric value of the four most significant bits of the Port Identifier component of the SCI for the selected SC.

The SecY may map the user priority of each frame's transmit request at the Controlled Port to the access priority to be used for the corresponding transmit request at the Common Port using the Access Priority Table. The table index and its output both comprise 4 bits, representing both the priority (most significant three bits) and drop_eligible (least significant bit) of the user priority and access priority. The default value of each table entry is that of its index, thus leaving the priority and drop_eligible bits unchanged. This default is appropriate if the service provided by the Common Port already implements its own mapping from requested priority to its own priority or other parameters used to make decisions that affect frame reordering, and that mapping matches the Traffic Class Table's mapping of user priority to transmit SC. The default is also appropriate if the administrator is willing to tolerate the degree of misordering, and the replayWindow size that implies, resulting from allocating frames of different access priority to the same SC in the interest of providing a differentiated service to the higher priority frames without using additional transmit SCs. Otherwise, it is recommended that the Access Priority Table be configured so that frames allocated to the same transmit SC use the same access priority.

NOTE 2—Where MACsec is used to support an instance of the ISS that in turn supports the EISS, the priority originally requested by the EISS user is encoded in the VLAN tag within the ISS MSDU and is thus protected by MACsec and is communicated unchanged to the peer EISS user, unaffected by local access priority mapping decisions.

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10.7.18 Frame generation statistics

Any given transmitted frame (10.5) increments exactly one of the following counts [item a) through item d)]. The following counts are maintained for the frame generation process as a whole:

a) OutPktsUntagged
b) OutPktsTooLong

The following counts are maintained for each transmit SC:

c) OutPktsProtected
d) OutPktsEncrypted

The counts reported for each SC include those for current and prior SAs, with ANs that have since been reused. This allows useful counts to be maintained on high-speed LANs where an SA may be used for little more than 5 min, and an AN reused after 20 min.The times at which each SC and SA were, or are, in use are recorded (10.7.21, 10.7.23) and assist correlation of the statistics collected with network events.

NOTE—The OutPktsProtected and OutPktsEncrypted counts can be correctly reported, without the need for each frame to increment separate real-time counters. The packets for a given SA are either all encrypted (confidentiality protected) or all only integrity protected, so the counts for active SAs can be derived from the nextPN values (less any contribution to OutPktsTooLong made after PN assignment to discarded frames) and summed with that those previously accumulated for the SC. When an SA is replaced by a successor with the same AN, its counts are added to those accumulated for the SC.

10.7.19 Frame protection statistics

Investigation or validation of the performance of the cryptographic functions is supported by accumulation of the following counts:

a) OutOctetsProtected, the number of octets of User Data in transmitted frames that were integrity protected but not encrypted.
b) OutOctetsEncrypted, the number of octets of User Data in transmitted frames that were both integrity protected and encrypted.

10.7.20 Transmit SC creation

A transmit SC, with a given SCI that remains unchanged for the life of the SC, is created, as requested by the KaY, for the default traffic class SC and for each of the other SCs identified by the Traffic Class Table (if implemented). The KaY is responsible for ensuring the uniqueness of the SCI of any SC in a CA that might use the same SAK.

Transmit SCs and SAs (10.7.22) may also be created and controlled by management, but a conformant implementation shall provide a mechanism to allow creation and setting of control parameters by network management to be disabled.

10.7.21 Transmit SC status

The following status parameters can be read, but not directly written, by management:

a) transmitting, True if inUse (10.7.23) is True for any of the SAs for the SC, and False otherwise
b) encodingSA (10.5.1)
c) createdTime, the system time when the SC was created
d) startedTime, the system time when transmitting last became True for the SC
e) stoppedTime, the system time when transmitting last became False for the SC

When the SC is created, transmitting is False and startedTime and stoppedTime are equal to createdTime.

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10.7.22 Transmit SA creation

An SA is created for a transmit SC on request from the KaY, with the following parameters:

a) AN, the association number for the SA
b) nextPN, the initial value of Transmit PN (10.5.2) for the SA
c) confidentiality, True if the SA is to provide confidentiality as well as integrity for transmitted frames
d) A reference to an SAK that is unchanged for the life of the SA

and, if the Current Cipher Suite uses extended packet numbering (14.7, 14.8), the KaY also supplies the following parameter:

e) SSCI for the SA

Each SA that uses the same SAK has a different SSCI when these Cipher Suites are used. When the SA is created, its SCI and SSCI are provided (for use in subsequent protection operations) to the instance of the Current Cipher Suite identified by the referenced SAK. A transmit SA will not be created if the SSCI supplied duplicates that for a different SCI (for the same SAK, for transmission or reception).

Frame generation statistics (10.7.18) for the SA are set to zero when the SA is created. Any prior SA with the same AN is deleted. Creation of the SA fails unless the referenced SAK exists and is installed (i.e., is available for use). A management protocol dependent reference is associated with each SA. This reference allows the transmit SA to be distinguished from any previously created with the same AN.

MKA, specified in IEEE Std 802.1X, does not distribute SSCIs explicitly. A KaY assigns SSCI values as specified in 10.7.13.

10.7.23 Transmit SA status

The following parameters can be read, but not directly written, by management:

a) inUse
b) createdTime, the system time when the SA was created
c) startedTime, the system time when inUse last became True for the SA
d) stoppedTime, the system time when inUse last became False for the SA
e) nextPN (10.5, 10.5.2)
f) confidentiality, True if the SA is providing confidentiality as well as integrity for transmitted frames
g) keyIdentifier (10.7.28), identifying the SAK used by the SA

and, if the Current Cipher Suite uses extended packet numbering (14.7, 14.8), the following parameter:

h) ssci, the SSCI for this transmit SA

If inUse is True, and MAC_Operational is True for the Common Port, the SA can transmit frames.

The keyIdentifier is an octet string, whose format and interpretation depends on the key agreement protocol in use. It does not contain any information about the SAK other than that explicitly chosen by the key agreement protocol to publicly identify the key. If MKA is being used it is the 128-bit Key Identifier (KI) specified by IEEE Std 802.1X encoded in an octet string as specified by that standard.

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10.7.24 Transmit SA controls

The KaY uses the following parameters to control the use of each transmit SA:

a) enableTransmit

When the SA is created, enableTransmit and inUse are False, and the SA is not used to transmit frames. The SC parameter encodingSA shall be set to the value of the AN for the SA and inUse set True, when enableTransmit is set. The SA shall stop transmitting, and inUse reset, when enableTransmit is reset.

10.7.25 Implemented Cipher Suites

The following per Cipher Suite read-only capability information is provided by the system of which the SecY is a part:

a) Cipher Suite Identifier, a globally unique 64-bit (EUI-64) identifier
b) Cipher Suite Name, a human readable and displayable UTF-8 (IETF RFC 2279 [B7]) string
c) integrityProtection, True if integrity protection without confidentiality can be provided
d) confidentialityProtection, True if confidentiality with integrity protection can be provided
e) offsetConfidentiality, True if a selectable offset for confidentiality can be provided
f) changesDataLength, True if the data length is changed
g) ICVlength, number of octets in the ICV

The Cipher Suite Identifier and Cipher Suite Name are both assigned by the document that specifies use of the Cipher Suite with this standard. If the Cipher Suite provides integrityProtection and confidentialityProtection, the SecY shall be capable of receiving frames with either, as signaled by the E and C bits in the SecTAG.

The confidentialityProtection parameter shall be True if and only if the Cipher Suite implementation is capable of being configured so that, when confidentiality is selected, all the octets of the MSDU are integrity and confidentiality protected.

The offsetConfidentiality parameter shall be True if and only if the Cipher Suite implementation is capable of both integrityProtection and confidentialityProtection, and of being configured so that, when confidentiality is selected, a selectable number (0, 30, or 50) of the initial octets of the MSDU are only integrity protected, and appear in the MPDU immediately after the SecTAG in the order and with the values in the MSDU (Figure 8-1), while the remaining octets are confidentiality and integrity protected.

NOTE—IEEE Std 802.1AE-2006 specified the confidentiality offset option to facilitate early MACsec deployment on systems that needed to examine the initial octets of IP version 4 or version 6 frames to decide where to store received frames, before decrypting the frame. The XPN Cipher Suites do not support confidentiality offsets.

10.7.26 SecY Cipher Suite use

The Cipher Suite capabilities implemented for each SecY can be read by management. The following controls may be written by management, but a conformant implementation shall provide a mechanism to allow write access by network management to be disabled for each parameter individually:

a) enableUse, True if use of the Cipher Suite is permitted
b) requireConfidentiality, True if the Cipher Suite can only be used to provide both confidentiality and integrity (and not integrity only, or confidentiality with an offset)

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The MKA Key Server selects the Cipher Suite to be used to protect communication within a CA. If enableUse is False for the selected Cipher Suite, the SecY does not participate in the CA and MAC_Operational for the Controlled Port remains false. If the MKA Key Server has selected integrity protection and enableUse and requireConfidentiality are both True for the selected Cipher Suite, confidentiality protection is used.

NOTE—A system might contain distinct SecY implementations with differing detailed Cipher Suite capabilities. Each of the latter can be represented by a distinct set of Cipher Suite implementation capability information (10.7.25), with each SecY's capabilities represented by a list of references (each with separate use controls) to some of those sets.

10.7.27 Cipher Suite selection

The KaY uses the following parameter to select the Current Cipher Suite:

a) currentCipherSuite, the Cipher Suite Identifier (10.7.25) for the cipher suite

If offsetConfidentiality (10.7.25) is not False for the Cipher Suite, the following parameter is specified:

b) confidentialityOffset, the number of initial octets of each MSDU without confidentiality protection

The CurrentCipherSuite is selected by the KaY. The Current Cipher Suite may also be selected and keys created by management, but a conformant implementation shall provide a mechanism to allow such selection and creation by network management to be disabled. The confidentialityOffset applies to all frames transmitted and received with confidentiality protection. If both confidentialityProtection and offsetConfidentiality are supported, then it takes the values 0, 30, and 50.

If the Current Cipher Suite is changed, all keys created for that Cipher Suite are deleted, and (as a consequence) inUse will become False for all SAs, with the further consequence that MAC_Operational will become False for the Controlled Port.

10.7.28 SAK creation

An SAK is installed, i.e., an instance of the Current Cipher Suite for a given SAK is created, on request from the KaY with the following parameters:

a) The SAK value
b) keyIdentifier, used by network management to reference the key
c) transmit, True if the key is to be installed for transmission
d) receive, True if the key is to be installed for reception

and, if the Current Cipher Suite uses extended packet numbering, the following parameter:

e) a Salt (McGrew [B17])

If a MACsec Key Agreement protocol (MKA) Key Server does not distribute an explicit Salt, each KaY computes this parameter.

GCM-AES-XPN-128 (14.7) and GCM-AES-XPN-256 (14.8) use a 96-bit Salt. The 64 least significant bits are the 64 least significant bits of the MKA Key Server's Member Identifier (MI), the 16 next most significant bits comprise the exclusive-or of the 16 next most significant bits of that MI with the 16 most significant bits of the 32-bit MKA Key Number (KN), and the 16 most significant bits comprise the exclusive-or of the 16 most significant bits of that MI with the 16 least significant bits of the KN.

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Ascon-XPN-128 (14.9) uses a 128-bit Salt, computed as follows:

— The 48 least significant bits are the 48 least significant bits of the MKA Key Server's Member Identifier (MI).

— The 16 next most significant bits are the exclusive-or of the 16 next most significant bits of that MI with the 16 least significant bits of the 32-bit MKA Key Number (KN).

— The 32 next most significant bits are the 32 most significant bits of that MI.

— The 16 next most significant bits are the 16 least significant bits of that MI.

— The 8 next most significant bits are the 8 most significant bits of the 24 least significant bits of the Key Server's MI exclusive or'd with 8 most significant bits of the KN.

— The 8 most significant bits of the Salt are the 8 most significant bits of the 32 least significant bits of that MI exclusive or'd with the 8 least significant bits of the 16 most significant bits of the KN.

These ways of obtaining a Salt are not necessarily applicable to any other key agreement protocol. Table 10-3 provides examples.

Table 10-3—MKA Salt construction (examples)

Key Number (KN)	$1234\ 5678_{16}$[a]
Key Server MI	$1122\ 3344\ 5566\ 7788\ 99AA\ BBCC_{16}$
96-bit Salt	$475A\ 2170\ 5566\ 7788\ 99AA\ BBCC_{16}$
128-bit Salt	$ADB8\ BBCC\ 1122\ 3344\ 031E\ 7788\ 99AA\ BBCC_{16}$
Key Number (KN)	$0001\ 2853_{16}$
Key Server MI	$E630\ E81A\ 48DE\ 85B4\ 6A21\ C66F_{16}$
96-bit Salt	$CE63\ E81B\ 48DE\ 85B4\ 6A21\ C66F_{16}$
128-bit Salt	$6B21\ C66F\ E630\ E81A\ 608D\ 85B4\ 6A21\ C66F_{16}$

[a] All the entries in this table are hexadecimal integers, with the most significant digits to the left. Spaces have been inserted, breaking each integer into 16-bit blocks, for ease of comprehension.

10.7.29 SAK status

The following parameters can be read, but not directly written, by management:

a) transmits, True if the key has been installed for transmission, i.e., can be used by a transmit SA
b) receives, True if the key has been installed for reception, i.e., can be used by a receive SA
c) createdTime, the system time when the SAK record was created

10.8 Addressing

Frames transmitted between end stations using the MAC Service carry the MAC Address of the source and destination peer end stations in the source and destination address fields of the frames, respectively. Communicating peer SecYs can secure communication for all or part of the path used by such frames, and are not directly addressed by the communicating peers, nor are the frames modified to include additional addresses. Each SecY does not have a MAC Address of its own, but is associated with a local entity that forms part of the secure system.

The addressing used by Key Agreement Entities and the means they use to identify SecYs within the same secure system are outside the scope of this specification.

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While destination and source MAC addresses are not required to identify SecYs, they are parameters of the MAC Internal Sublayer Service (ISS) used and provided by a SecY, and are covered by the ICV, generated by a Cipher Suite implementation while remaining unencrypted. To facilitate ICV calculation and verification, all frames processed by SecYs use 48-bit MAC addresses.

10.9 Priority

While priority is a parameter of both an ISS M_UNITDATA.request and corresponding M_UNITDATA.indications, end-to-end communication of the requested priority is not a service attribute (6.1). Protocols supporting the ISS can use the requested priority to perform local actions in the originating station, and do not necessarily attempt to communicate the parameter. Accordingly, the requested and indicated priorities do not contribute to the ICV, and are not explicitly included in the encoded MSDU by a transmitting SecY.

NOTE—If communication of priority is desired, either guaranteed unchanged or available to a service provider for possible modification to meet the admission control and service characteristics of a particular network, use of the EISS in conjunction with the ISS is indicated. See Clause 7.

10.10 SecY performance requirements

Table 10-4 places requirements on SecY performance to ensure that MACsec operates correctly.

Table 10-4—SecY performance requirements

Parameter	Permitted values
SecY transmit delay	< Wire transmit time for maximum sized MPDU + (4 times wire transmit time for 64 octet MPDUs)
SecY transmit delay variance	< SecY transmit delay
SecY receive delay	< Wire transmit time for maximum sized MPDU + (4 times wire transmit time for 64 octet MPDUs)
SecY receive delay variance	< SecY receive delay
SC and SA creation and control delay	< 0.1 second
Transmit SAK install delay	< 1 second (8.2.2)
Transmit SAK switch delay	< Wire transmit time for 64 octet MPDU (8.2.2)
Receive SAK install delay	< 1 second
Receive SAK switch delay	No frame loss

All times are in seconds.

Time-sensitive networking (TSN) applications can benefit from or further constrain delays and delay variances experienced by relayed and transmitted frames (see IEEE Std 802.1AS™ [B3], IEEE Std 802.1Q).

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11. MAC Security in systems

This clause specifies how MAC Security is incorporated within the architecture of

a) End stations (11.2)
b) MAC Bridges (11.3)
c) VLAN-aware Bridges (11.4)
d) Systems that incorporate Link Aggregation (11.5)
e) Systems that incorporate Link Layer Discovery Protocol (LLDP, 11.6)
f) Provider Bridges and VLAN-aware Bridges attached to Provider Bridged Networks (11.7)
g) LANs that provide independently secured access for multiple end stations (11.8).

The figures in this clause illustrate the relative position of components within the MAC Service interface stacks (11.1) of each of these systems. Both the secure MAC Service provided by the Controlled Port and the insecure service provided to the Uncontrolled Port are shown.

NOTE—For more information on the Controlled and Uncontrolled Ports and the operation of the SecY, see Clause 10.

11.1 MAC Service interface stacks

Each LAN MAC, e.g., that is specified in IEEE Std 802.3, is capable of providing the MAC Service directly to LLC and its clients, as illustrated in Figure 11-1.

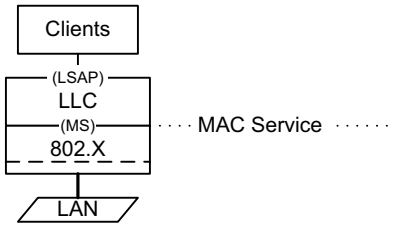


Figure 11-1—Direct support of the MAC Service by a media access method

NOTE 1—The term *802.X* refers to any one of the IEEE 802 LAN media access control method technologies.

Alternatively, media access method independent functions, such as VLAN tagging of frames (IEEE Std 802.1Q) and MAC Security (as specified by this standard), can be used to support the MAC Service (IEEE Std 802.1AC), the MAC Internal Sublayer Service (ISS, IEEE Std 802.1AC), or the Enhanced Internal Sublayer Service (EISS, IEEE Std 802.1Q). These functions use an ISS access point provided by media access method independent or media access method dependent convergence functions. See Figure 11-2.

Each SecY uses an ISS access point and provides the ISS at its Controlled and Uncontrolled Ports. This allows use of MAC Security with other media-independent functions. However, interoperability between systems using MAC Security requires not only interoperability between SecY implementations and use of the same LAN MAC technology, but also that the same, or compatible, media interface functions are used with the same relative position within the interface stack, as specified in this clause.

NOTE 2—MAC Bridges and VLAN-aware Bridges provide interoperability between access points for the MAC Service, the ISS, and the EISS, using the following common elements of those service specifications. The MAC Service, the ISS, and the EISS all use the same request and indication primitives. The parameters used by the ISS for each primitive are a superset of those of the MAC Service. An EISS access point effectively provides access to multiple ISS instances.

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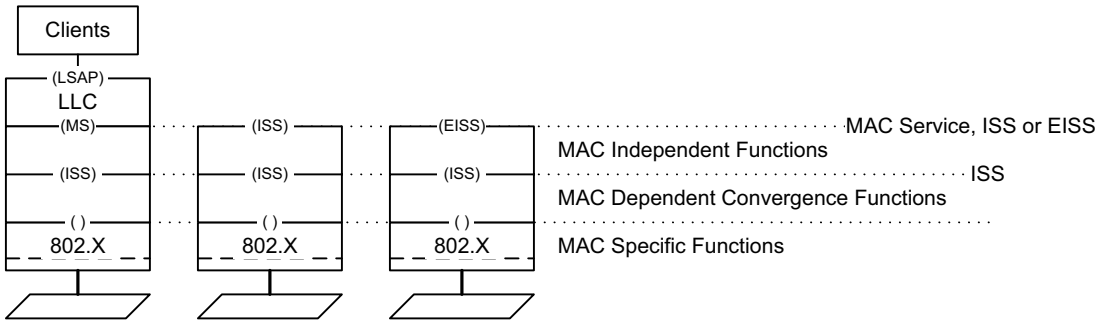


Figure 11-2—Provision of MAC Service with media-independent functions

11.2 MACsec in end stations

The ISS provided by the SecY is trivially mapped to and from the MAC Service provided within an end station. Service indications for unwanted destination MAC addresses are discarded, and the source MAC address of service requests is that of the station. Figure 11-3 shows MAC Security as the sole media-independent function within a station.

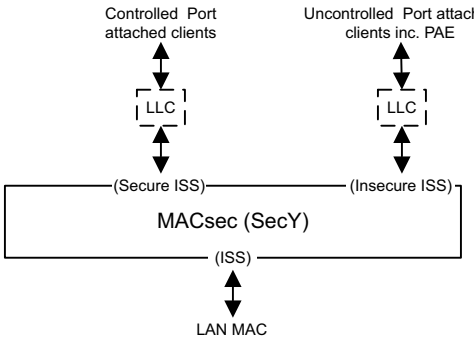


Figure 11-3—MACsec in an end station

11.3 MACsec in MAC Bridges

MAC Bridges are specified in IEEE Std 802.1Q. The MAC Relay Entity forwards frames between the ISS access points supported by each of the Bridge Ports. To provide MAC Security for such a system, each of the insecure interfaces presented by a LAN supports MACsec, which in turn supports the functions described in 8.5 of IEEE Std 802.1Q-2022. Figure 11-4 shows a bridge with and without MACsec.

NOTE—If the MAC Bridge aggregates multiple LANs to support a single Bridge Port, each individual LAN supports its own SecY, which provides the secure MAC Service to the Link Aggregation sublayer, as specified in 11.5. Each aggregated port then provides secure service to the Bridge Port transmit and receive functions.

Figure 11-5 shows the interface stack for each of the Bridge Ports.

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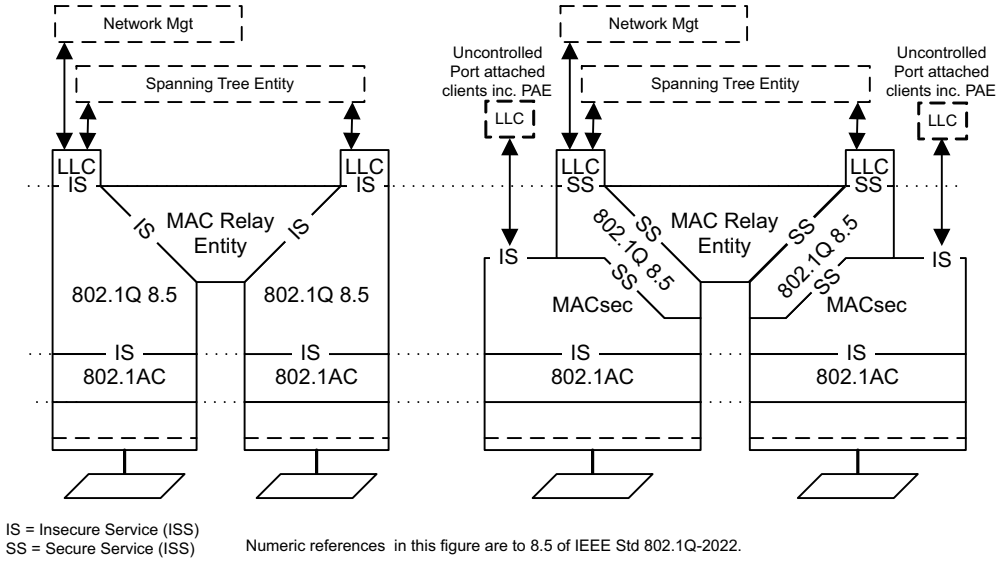


Figure 11-4—MACsec in a VLAN-unaware MAC Bridge

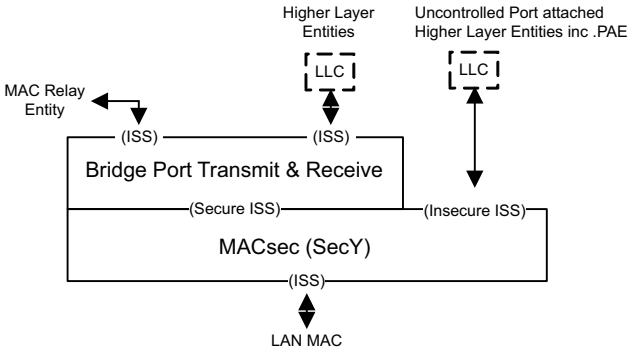


Figure 11-5— VLAN-unaware MAC Bridge Port with MACsec

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11.4 MACsec in VLAN-aware Bridges

VLAN-aware Bridges are specified in IEEE Std 802.1Q. Figure 11-6 illustrates the addition of MAC Security.

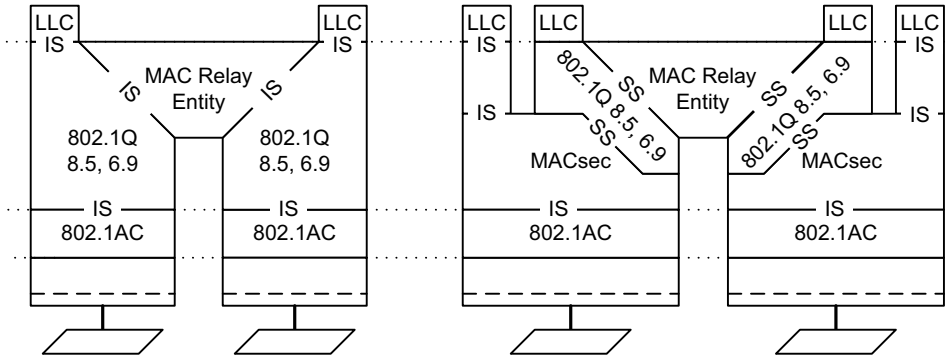


Figure 11-6—Addition of MAC Security to a VLAN-aware MAC Bridge

Figure 11-7 shows the interface stack for each of the VLAN-aware Bridge Ports.

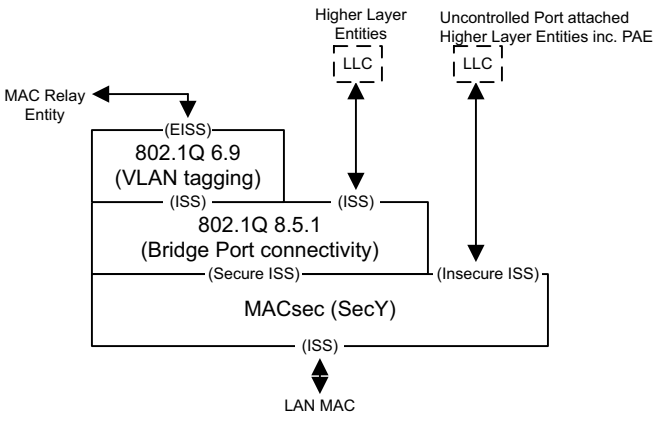


Figure 11-7— IEEE 802.1Q VLAN-aware Bridge Port with MACsec

Figure 6-2 shows the frame format and placement of the VLAN tag within the frame relative to MACsec. Thus if there is encryption, the VLAN tag is not in the clear.

NOTE—If the use of a protocol analyzer and other monitoring tools based on capture and analysis of packets on the wire is desired, integrity protection only, without confidentiality, should be used.

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The position of MACsec, below both the Bridge Port connectivity and VLAN tagging functions, has the following consequences:

a) Each Bridge Port uses a single SecY, with a single transmit SC and a single receive SC for each of the other bridges and stations attached to the LAN, to support all VLANs.
b) Interoperability with MAC Bridges, that are not VLAN-aware, is supported in the same way as VLAN-aware and unaware bridges without MAC Security.
c) Higher-layer entities attached to the Bridge Port, such as the Spanning Tree Protocol Entity and protocol stacks for network management, do not need to be supported by separate SecYs. In particular a MACsec protected point-to-point link between two bridges continues to function as a point-to-point link despite the end station functions associated with each Bridge Port.
d) Changes in the operation of MAC Security do not cause differences in the network connectivity used by the MAC Relay Entity and in the network connectivity perceived by the Controlled Port attached higher-layer entities that execute control protocols for the relay function.

11.5 MACsec and Link Aggregation

Link Aggregation is specified in IEEE Std 802.1AX [B4]. The service provided by two separate point-to-point LANs is combined to provide a single service interface. To provide MAC Security for such a system, two independent SecYs operate below the link aggregation sublayer. If the two links are being aggregated dynamically, as provided for by the Link Aggregation Control Protocol (LACP), the operation of LACP will be protected. In addition, if the authentication provided by the KaYs determines that the two links do not connect to the same partner system, local system management can change the aggregation keys. Changes in link aggregation do not cause changes to the MACsec CAs, SCs, SAs, or SAKs.

NOTE 1—LACP aggregation keys have nothing to do with cryptography. See IEEE Std 802.1AX [B4] for details.

Figure 11-8 shows part of an interface stack with MAC Security and Link Aggregation. The insecure service access points for each of the SecYs are independently provided to the KaY associated with each SecY, and may or may not be aggregated separately.

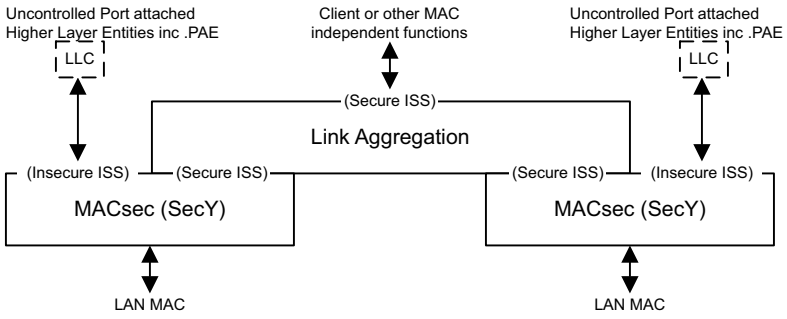


Figure 11-8—MACsec and Link Aggregation in an interface stack

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Figure 11-9 shows the addition of link aggregation to the interface stack for a VLAN-aware Bridge Port that also uses MACsec.

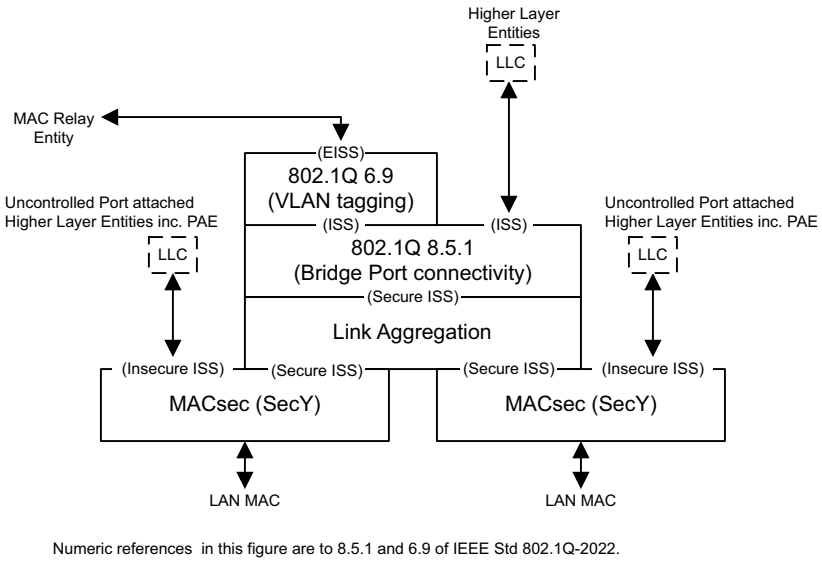


Figure 11-9—IEEE 802.1Q VLAN-aware Bridge Port with MACsec and Link Aggregation

11.6 Link Layer Discovery Protocol (LLDP)

LLDP is specified in IEEE Std 802.1AB. When used in conjunction with MACsec each LLDP Agent should make use of the Secure ISS provided by MACsec for the attached LAN as shown in Figure 11-10.

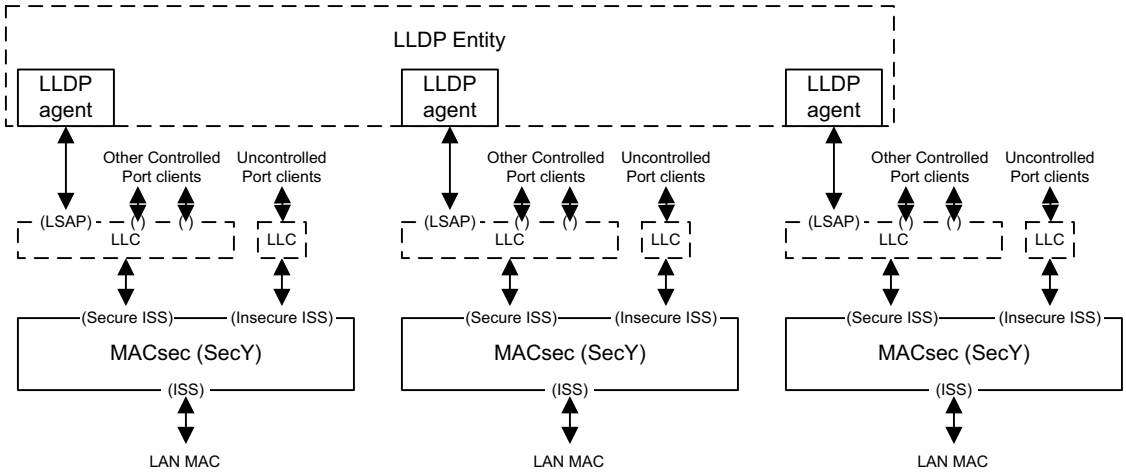


Figure 11-10—MACsec with LLDP

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11.7 MACsec in Provider Bridged Networks

Provider Bridges (IEEE Std 802.1Q) enable service providers to use VLANs to offer the equivalent of separate LANs to different users. Data for each of the virtual LANs is segregated within the provider's network by using a Service VLAN TAG (S-TAG) that is distinguished, by EtherType, from the Customer VLAN-TAGs (C-TAGs) used within each customer's network. See Figure 11-11.

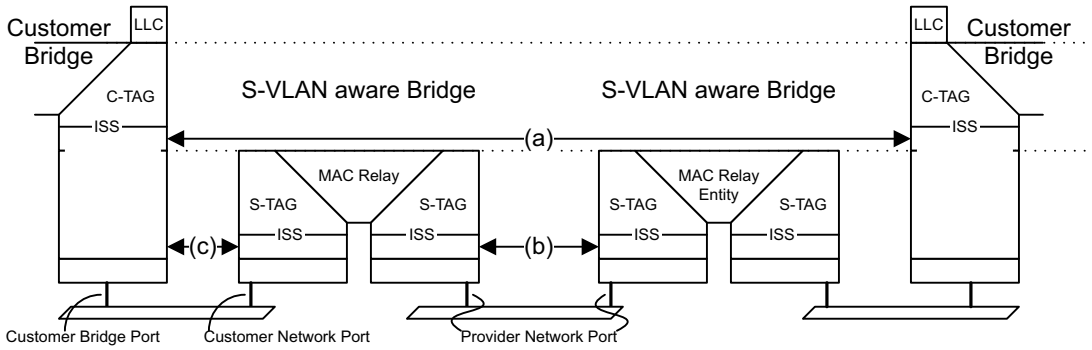


Figure 11-11—Internal organization of the MAC sublayer in a Provider Bridged Network

NOTE—Figure 11-11 is based on Figure 15-1 of IEEE Std 802.1Q-2022.

MACsec can be used to secure communication between

a) A customer's bridges or other equipment, across the provider's network
b) Adjacent S-VLAN aware Bridges, within the provider's network
c) A customer's bridge and the provider's network.

If it is the customer's intention to secure only one of item a) or item c), then the use of one of the interface stacks illustrated in Figure 11-3 (for an end station), Figure 11-5 (for a MAC Bridge), or Figure 11-7 (for a VLAN-aware Bridge) within the customer equipment is sufficient.

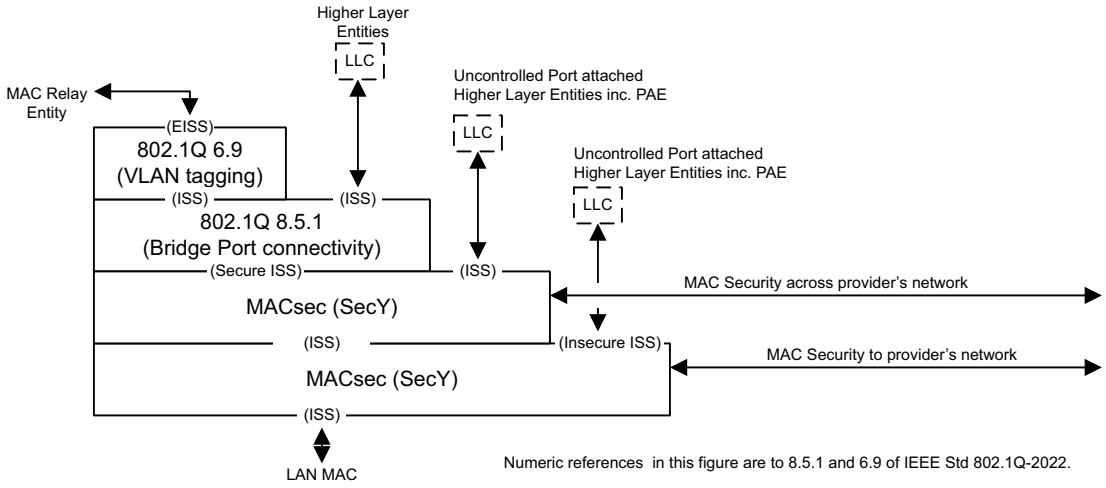


Figure 11-12—Interface stack for MAC Security to and across provider's network

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Use of the interface stack illustrated in Figure 11-7 within the provider's S-VLAN aware Bridge Ports is sufficient to secure either item b) or item c) as required. If item c) is not to be secured, MACsec is either omitted from the interface stack for the Customer Network Port (see Figure 11-11), or the Bridge Port connectivity function (8.5.1 of IEEE Std 802.1Q-2022) uses the service provided by the Uncontrolled Port.

If it is the intention to secure both item a) and item c) from the Customer Bridge Port, then the use of two independent SecY's within the port's interface stack is required as shown in Figure 11-12.

Figure 11-13 shows the addition of the service access priority selection function described in 6.13 of IEEE Std 802.1Q-2022 to the interface stack of Figure 11-12, together with the use of Link Aggregation to support attachment to the provider's network with two LANs.

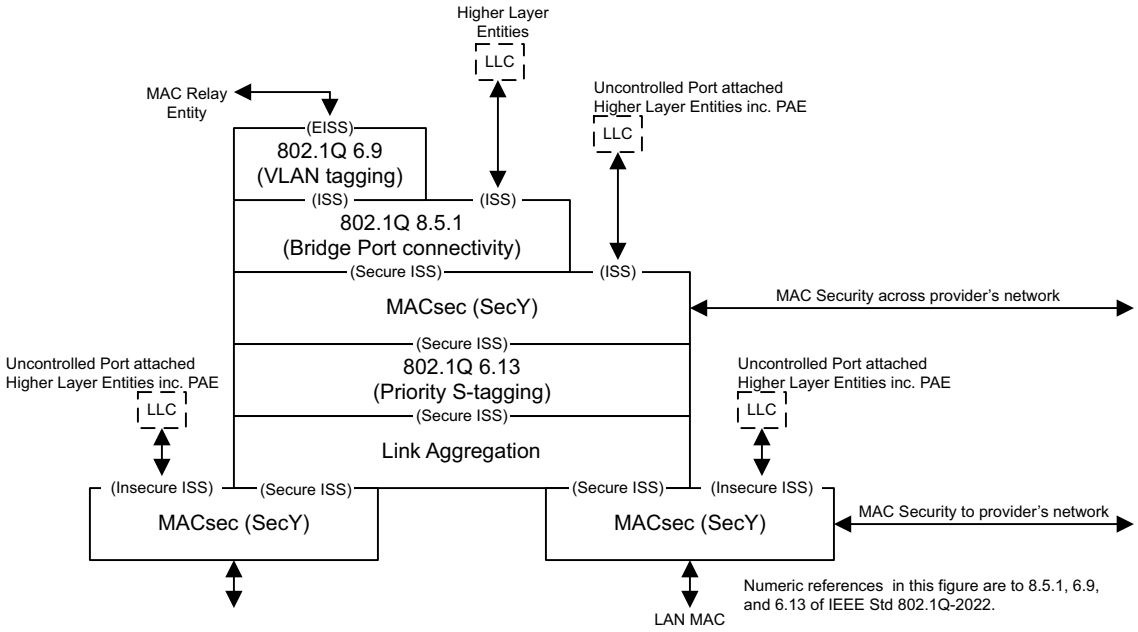


Figure 11-13—Provider network with priority selection and aggregation

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11.8 MACsec and multi-access LANs

MACsec can be used to support the equivalent of multiple LANs from one station to each of a number of others using the service provided by a single LAN. Each station that connects to more than one of the multiple LANs does so by using a distinct SecY for each of those connections. MACsec frames for each of the multiple LANs are distinguished from frames for the others by the SCI of the originating SecY. If a station has more than one SecY, the SCIs for each SecY's transmit SC or SCs are based on the MAC Address allocated to that station but use a different Port Identifier component (9.9). Figure 11-14 shows one station (A in the figure) with two connections, one to each of two others (B, C).

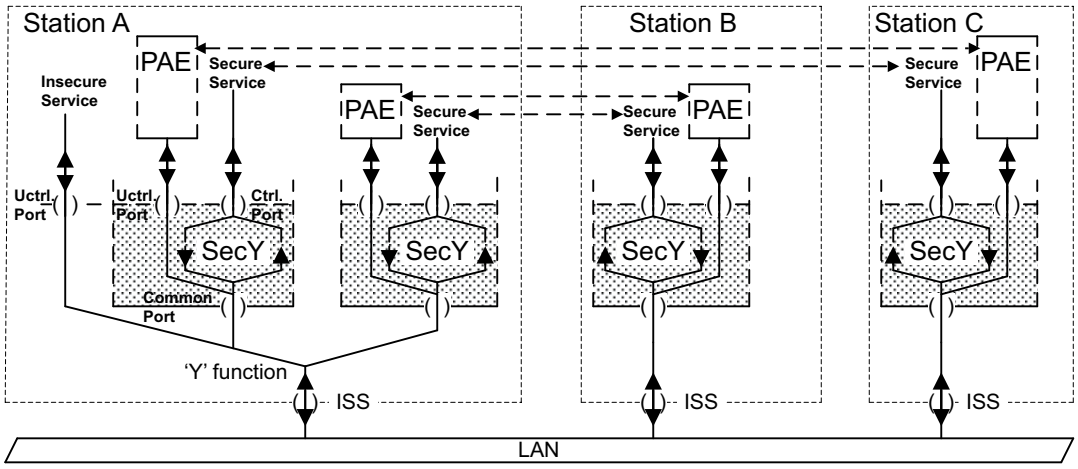


Figure 11-14—An example multi-access LAN

Frames transmitted by each SecY's Uncontrolled Port can include a SecTAG, with an SCI value used by the SecY's Controlled Port. These frames are distinguished by setting the E bit in the SecTAG TCI True and the C bit False, and are discarded by the frame verification process for the Controlled Port (10.6). The connectivity between Uncontrolled Ports using the SecTAG thus matches the secure connectivity provided between the corresponding Controlled Ports. The protocol entities attached to the SecY's Uncontrolled Port add and remove this SecTAG as required.

NOTE—Frames including a SecTAG and E bit True and C bit False were not used by any standard protocol at the time of the development of IEEE Std 802.1AEcg-2017, but this normative provision remains for possible future use by protocols that need to associate Uncontrolled Port frames with individual SCIs.

Frames transmitted through a SecY's Uncontrolled Port to a multi-access LAN can omit the SecTAG, provided that only one bi-directional unicast communication is supported between any pair of stations. The recipient uses the source address of the frame to identify the peer SecY.

Each multi-access capable station also supports an Uncontrolled Port (shown to the left in station A in Figure 11-14) that allows arbitrary frames to be transmitted on the LAN and received, if they are not MACsec frames, by any of the systems. These Uncontrolled Ports support the protocols required to discover peer multi-access capable systems, and to associate SCIs (and hence SecYs and KaYs) with each connection. The entities that operate such discovery and association protocols in stations, such as station A, that are capable of supporting multiple SecYs on a single LAN, are typically capable of instantiating some number of SecYs and associated entities on demand. The Controlled Ports thus provided to higher-layer entities can be transient, and are referred to as "virtual Ports".

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Where a protocol entity for each SecY's Uncontrolled Port transmits frames without a SecTAG, it is possible for there to be no externally observable difference between the operation of entities attached to those ports and of an equivalent entity or entities attached to the Uncontrolled Port for the station as a whole. Whether to emphasize common functions or peer relationships is a choice for each protocol's specification.

Figure 11-15 shows part of an interface stack for a multi-access capable system. The 'Y' function can simply copy all indications from its lower service access point to all upper access points, and any request from an upper service access point to the lower access point. Each KaY and SecY will discard indications for SCIs that do not match one of their receive SCs. Alternatively, the 'Y' function can selectively deliver indications for known SCIs to the appropriate SecY, as instructed by the higher-layer entity responsible for virtual port creation and its association. Its detailed specification is determined by the specification of that entity.

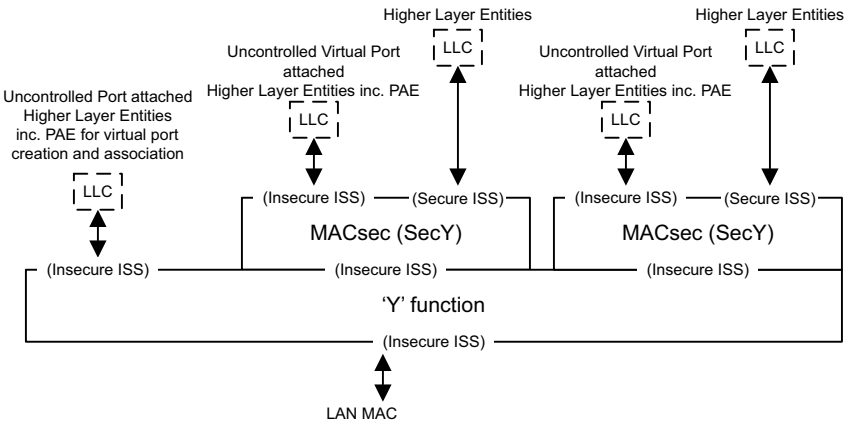


Figure 11-15—Multi-access LAN interface stack

The connectivity provided by a multi-access LAN depends on the security provided and can change as security is deployed, enabled, or disabled. Because this can lead to difficulties in the management of bridged networks, multi-access LANs should not be used to support LANs with two or more attached bridges. They are appropriate for the attachment of end stations or hosts at the periphery of the network.

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12. MACsec and EPON

Clause 64 and Clause 65 of IEEE Std 802.3-2018 specify an Ethernet passive optical network (EPON) that uses a physical fiber tree topology to provide efficient point-to-multipoint connectivity from a single OLT to one or more ONUs. Clause 64 specifies the instantiation of multiple MAC entities within the OLT, each with an associated service access point that provides point-to-point connectivity to a specific ONU separate from the connectivity provided to other ONUs. An additional MAC instance provides a Single Copy Broadcast (SCB) service access point that allows a single copy of a frame to be received by all ONUs.

MACsec provides a separate instance of the secure MAC Service to provide bi-directional connectivity between each ONU and the OLT, as illustrated in Figure 12-1, and thus ensures the confidentiality, integrity, and origin authenticity of each data frame sent and received by the OLT and each ONU. These guarantees are provided irrespective of the ability of an attacker to transmit or receive frames to or from the OLT or any ONU, even if that attacker can exactly mimic the EPON media access method specific behavior of any of the securely communicating participants.

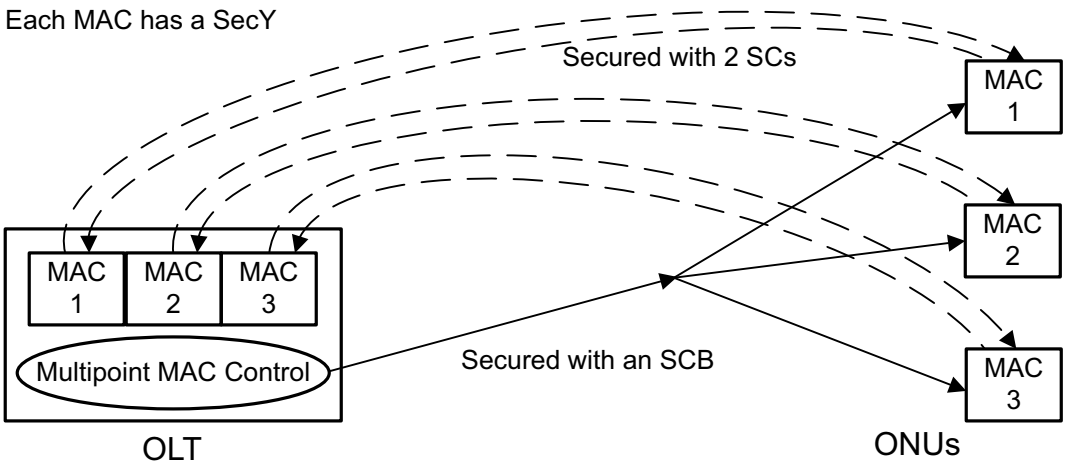


Figure 12-1—MACsec with EPON, showing SCs and SCB

In the OLT, each instance of the secure MAC Service is provided by a distinct SecY that uses the insecure instance of the MAC Service provided by one of the point-to-point MAC entities in the OLT.

MACsec can support the SCB service access point with a dedicated SC. Appropriate distribution to the ONUs of the encryption and authentication keys for the sequence of SAs that compose the SC ensures the confidentiality, integrity, and origin of each frame sent using the SCB.

NOTE 1—Since the SCB MAC interfaces in the OLT lacks a peer interface in each ONU, the keys for the sequence of SAs that support them are distributed to the Key Agreement Entities of all authorized ONUs using the insecure bi-directional MAC Service associated with each of the point-to-point MAC instances.

NOTE 2—An ONU can elect to discard frames from the SCB as these are readily identifiable by the EPON MAC. However, if such frames are received, their integrity and origin should be secured, particularly if the system comprising the ONU bridges or routes such frames. Otherwise, an attacker could use frames that appear to be sent using the SCB to penetrate the attached network, even if the point-to-point EPON connectivity has been correctly secured.

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13. MAC Security Entity MIB

13.1 Introduction

This clause contains an SMIv2 Management Information Base (MIB) for managing the operation of a MAC Security Entity (SecY), based on the specifications contained in Clause 10 and Clause 11. This clause includes a MIB module that is compliant to SMIv2.

NOTE—Annex G (informative) provides a detailed history and rationale for changes to SecY management and this MIB module. Normative changes to compliance and conformance statements are also described in 13.5.

13.2 The Internet-Standard Management Framework

For a detailed overview of the documents that describe the current Internet-Standard Management Framework, please refer to section 7 of IETF RFC 3410 [B8].

Managed objects are accessed via a virtual information store, termed the Management Information Base or MIB. MIB objects are generally accessed through the Simple Network Management Protocol (SNMP). Objects in the MIB are defined using the mechanisms defined in the Structure of Management Information (SMI). This memo specifies a MIB module that is compliant to the SMIv2, which is described in IETF RFC 2578, IETF RFC 2579, and IETF RFC 2580.

13.3 Relationship to other MIBs

13.3.1 System MIB Group

It is assumed that a system implementing this MIB will also implement the “system” group defined in IETF RFC 3418 (or at least that subset of the system group defined in IETF RFC 1213).

13.3.2 Relationship to the Interfaces MIB

It is assumed that a system implementing this MIB module will implement the “interfaces” group defined in IETF RFC 2863, the Interfaces Group MIB. This MIB includes the clarifications mandated by IETF RFC 2863 for any MIB that is medium-specific or an adjunct of the Interfaces Group MIB.

The MACsec defines a secure shim layer, SecY, in the interface stack. The IEEE SecY MIB specifies the detail attributes of the secure shim layer in the interface stack. As such, it needs to integrate with IF-MIB. For interface stack diagram, refer to Figure 13-1.

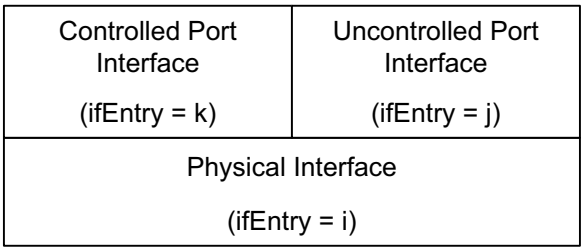


Figure 13-1—MACsec Interface Stack

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The SecY's Controlled Port is a service access point that provides one instance of the secure MAC Service, and the SecY's Uncontrolled Port is a service access point providing one instance of the insecure MAC Service. According to IETF RFC 2863 these two service access points should be defined as sublayers in the interface stack. Each should have its own conceptual row in the ifTable, though these should be created together. The two interfaces co-exist without interference—one is not "on top of " the other. The interface type for Controlled Port is defined as macSecControlledIF(231) and the interface type for Uncontrolled Port is defined as macSecUncontrolledIF(232).

The MAC_Enabled and MAC_Operational parameters (6.4) could be mapped to the ifAdminStatus and ifOperStatus objects. The ifAdminStatus object for Controlled Port interface and Uncontrolled port interface should be read only.

MIB tables in this SecY MIB represent information of the Controlled Port interface and are indexed by the interface number pointing to the Controlled Port sublayer interface in the interface stack.

The attributes in Table 13-1 are part of the required ifGeneralInformationGroup object group specified in IETF RFC 2863, and are not duplicated in the SecY MIB.

Table 13-1—Use of ifGeneralInformationGroup Objects

ifGeneralInformationGroup Objects	Use for MACsec
ifDescr	See interfaces MIB (IETF RFC 2863).
ifType	controlled port: macSecControlledIF(231). uncontrolled port: macSecUncontrolledIF(232).
ifSpeed	controlled port: same as the physical interface ifSpeed. uncontrolled port: same as the physical interface ifSpeed.
ifPhysAddress	This object should have an octet string with zero length for a SecY's controlled port and uncontrolled port.
ifAdminStatus	See interfaces MIB, read only for controlled port and uncontrolled port.
ifOperStatus	See interfaces MIB.
ifLastChange	See interfaces MIB.
ifName	See interfaces MIB.
ifLinkUpDownTrapEnable	See interfaces MIB, Default set as follows: controlled port: disabled(2). uncontrolled port: disabled(2).
ifHightSpeed	See interfaces MIB controlled port: same as the physical interface's ifHighSpeed. uncontrolled port: same as the physical interface's ifHighSpeed.
ifConnectorPresent	See interfaces MIB. Default set as follows: controlled port: false(2). uncontrolled port: false(2).
ifAlias	See interfaces MIB.
ifTableLastChange	See interfaces MIB.

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The attributes in Table 13-2 are part of the required ifCounterDiscontinuityGroup object group specified in IETF RFC 2863 and are not duplicated in the SecY MIB.

Table 13-2—Use of ifCounterDiscontinuityGroup Object

ifCounterDiscontinuityGroup Object	Use for MACsec
ifCounterDiscontinuityTime	See interfaces MIB. controlled port: always 0, no discontinuity. uncontrolled port: always 0, no discontinuity.

ifStackTable will be used to identify the layer relationships between Controlled Port interface and physical interface and between Uncontrolled Port interface and physical interface. Use of ifStackTable is necessary to represent the interface stack with MACsec service capability. Refer to Figure 13-1.

The ifStackTable is then used to show the relationships between the various MACsec interfaces, as illustrated Table 13-3.

Table 13-3—Use of ifStackTable

HigherLayer	LowerLayer
j	i
k	i

The attributes in Table 13-4 are part of the required ifStackGroup2 object group specified in IETF RFC 2863, and are not duplicated in the SecY MIB.

Table 13-4—Use of ifStackGroup2 Objects

ifStackGroup2 Objects	Use for MACsec
ifStackStatus	See interfaces MIB.
ifStackLastChange	See interfaces MIB.

The use of the ifPacketGroup object group specified in IETF RFC 2863 is described in 10.7.6 for the Controlled Port interface and in 10.7.3 for the Uncontrolled Port interface.

The ifRcvAddressTable is not applicable for Controlled Port and Uncontrolled Port interfaces.

13.4 Security considerations

There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of read-write and/or read-create. All such objects are sensitive or vulnerable in some network environments. The support for SET operations in a non-secure environment without proper protection can have a negative effect on network operations. These are the tables and objects and their sensitivity/vulnerability:

a) secyIfTable, secyIfCipherTable, secyIfTCTable, and secyIfAPTable contain system level information for each interface supported by the SecY. SET access to these tables by unauthorized

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persons can disable the MAC security protection functions, block network connectivity, and impact network performance. A comparison of the secyIfTable and the IF-MIB can identify which ports are not protected by a SecY.

b) secyRxSANextPN (deprecated) in secyRxSATable provides the capability to change the replay protection window. SET access to this object by unauthorized persons can affect the MACsec replay protection function, block network connectivity, and impact network performance.

Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than not-accessible) are sensitive or vulnerable in some network environments. It is thus important to control even GET and/or NOTIFY access to these objects and to encrypt the values of these objects when sending them over the network via SNMP.

The MIB module provides statistics from the interface level (SecY) to each secure association (SA). These statistics provide information for the diagnosis or debugging of the migration from a non-secure environment to a secure environment and can be used to observe the activities of MACsec operation. This information is useful for security monitoring by authorized personnel, but is also potentially useful to attackers; therefore, it needs to be protected against unauthorized access.

These are the tables and objects and their sensitivity/vulnerability:

c) secyTSATable (and secyTxSATable deprecated) provides information on each transmitting SA. secyTSAConfidentiality exposes whether confidentiality is supported or not for the SA. This information could help an attacker focus their attacks on traffic without confidentiality protection.

d) secyRxSATable contains information about receiving SAs. secyRxSANextPN is used in replay protection to determine which frames should be discarded. Read access to these related parameters could allow an attacker to know the PN range that an attempted replay must fall within.

e) secyCipherSuiteTable provides information about the capabilities of the cipher suites supported by the implementation. Access to this information could allow an attacker to focus their attacks on implementations with specific cipher suites and specific weaknesses.

f) secyRxSAStatsTable (deprecated) and secyRxSCStatsTable contain statistics for each receiving SA and each receiving SC. Read access could allow an attacker to compare these statistics with Figure 10-5 to determine which aspect of their attack failed, and to modify their attack until a different counter is incremented, indicating that they have succeeded in meeting a particular requirement.

g) secyStatsTable contains statistics about the SecY. This information is SecY interface level statistics information, and also read access to this information can help an attacker determine if a system might be vulnerable.

h) The global parameters secyIfMaxPeerSCs, secyIfRxMaxKeys, and secyIfTxMaxKeys might be used by an attacker when attempting to overload the system capabilities to cause a denial of service attack.

SNMP versions prior to SNMPv3 did not include adequate security. Even if the network itself is secure (for example by using IPSec), there is no control as to who on the secure network is allowed to access and GET/SET (read/change/create/delete) the objects in this MIB module.

If SNMP is to be used, activation of the security features provided by the SNMPv3 framework (see IETF RFC 3410 [B8], section 8), including the SNMPv3 cryptographic mechanisms (for authentication and privacy) are required for the security goals of this standard to be met.

Further, implementers should not deploy SNMP versions prior to SNMPv3. Instead, implementers should deploy SNMPv3 to enable cryptographic security. It is then a customer/operator responsibility to ensure that the SNMP entity giving access to an instance of this MIB module is properly configured to give access to the objects only to those principals (users) that have legitimate rights to indeed GET or SET (change/create/delete) them.

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13.5 Structure of the MIB module

A single MIB module is defined in this clause. Within the MIB module, each SecY is identified by the InterfaceIndex used by the Interfaces MIB (13.3.2, Figure 13-1) for the Controlled Port sublayer interface. This facilitates identification of the SecY when investigating an interface stack, and discovery of the other entities (via the Interfaces MIB) that are related to a particular SecY, including the associated PAE.

At the top level, the MIB module identifies MIB notifications, MIB objects, and MIB conformance information, though no notifications are defined. Figure 13-1 illustrates the structure of the MIB module, as described in this clause.

NOTE 1—MIB conformance is represented by objects, with an initial OID (object identifier) of secyMIBConformance, but in this MIB description the term MIB objects refers specifically to objects with an initial OID of secyMIBObjects.

MIB objects are arranged in a number of tables (in MIB terms a SEQUENCE OF entries) with each entry in the table comprising an number of basic objects (in MIB terms a SEQUENCE OF objects such as truth values, integers, text strings). MIB objects are further classified into management and statistics objects.

The entries in each of the management object tables are indexed in one of the following ways:

a) By the interface index, if the objects in each entry are for the SecY identified by that index).
b) By the interface index and SCI, if the objects are for an SCI used by the SecY.
c) By the interface index, SCI, and AN, if the objects are for an SA used by the SecY.
d) By a cipher suite index, defined in the module, for information specific to a given cipher suite but applicable to all SecYs in the system.
e) By a cipher suite index and SCI, for cipher suite information for a given SecY.

Each of the statistics object table entries augments a particular management object table entry, effectively using the same index. The per SecY, per SC, and per SA management tables for Controlled Port transmission and reception are summarized in Table 13-5 and Table 13-6, and the corresponding statistics tables in Table 13-7. Management tables for system-wide and per SecY Cipher Suite information are summarized in Table 13-8.

The MIB conformance objects are organized into compliance statements and conformance groups.

The compliance statement (secyMIBCompliance) for the original revision of this MIB module, published in IEEE Std 802.1AE-2006, mandated implementation of each of the groups specified in that revision, and each of the management object tables and statistics object tables specified has a corresponding group that includes all the objects for each entry in the table, though read-only access is permitted for a number of objects. This compliance statement and some of the tables have now been deprecated, and implementation of all the tables and objects it specifies is no longer a requirement of this standard for a conformance claim of network management MIB support. Implementations may continue to support this original compliance statement to support interoperability, either as an additional or as the only supported compliance statement, but it is expected to become obsolete in some future revision.

The current revision of this MIB module adds support for multiple traffic classes (with a complementary reduction in per SC statistics collection) and extended packet number reporting, with an additional compliance statement (secyMIBTcCompliance) and new conformance groups for additional tables. An implementation of a SecY for which support of network management using the MIB module is claimed should support secyMIBTcCompliance. An implementation that does not require multiple traffic class support can still benefit from reduced statistics collection and from the resolution of minor inconsistencies with the normative text elsewhere in the standard (which takes precedence).

NOTE 2—Annex G provides additional information on the management and MIB revisions.

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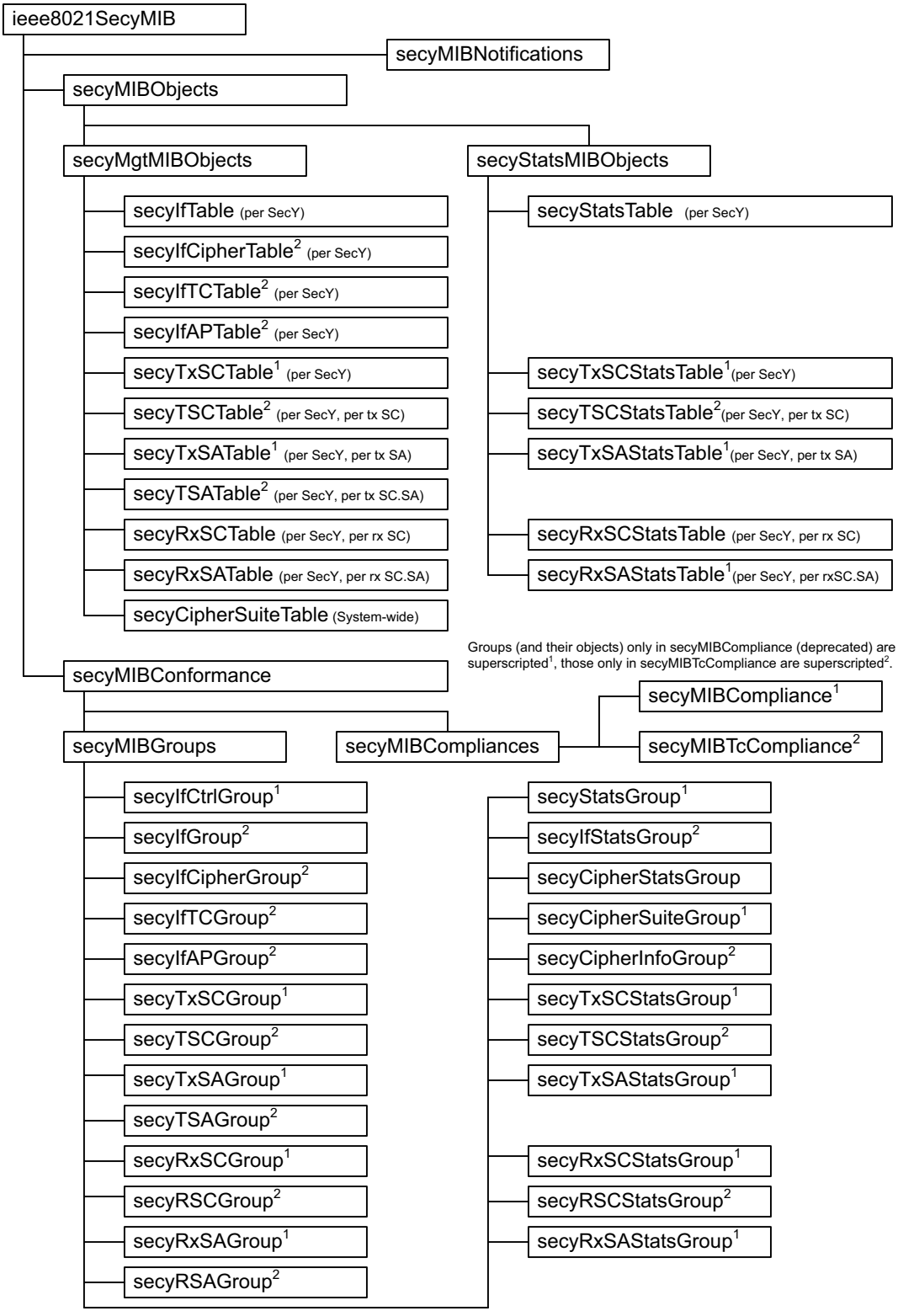


Figure 13-2—SecY MIB structure

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Table 13-5—Controlled Port service management

Table	Table Entry objects	Figure 10-5 reference and definition
secyIfTable secyIfEntry [secyIfInterfaceIndex] secyIfCtrlGroup[1a] secyIfGroup[2]	secyIfInterfaceIndex secyIfSCI[2] secyIfMaxPeerSCs secyIfRxMaxKeys secyIfTxMaxKeys secyIfProtectFramesEnable secyIfValidateFrames secyIfReplayProtectEnable secyIfReplayProtectWindow secyIfCurrentCipherSuite secyIfAdminPt2PtMAC secyIfOperPt2PtMAC secyIfIncludeSCIEnable secyIfUseESEnable secyIfUseSCBEnable secyIfIncludingSCI[2] secyIfMaxTSCs[2]	Aligned with IF-MIB InterfaceIndex (10.1) Generation.sci (7.1.2, 8.2.1, 10.7.1) Verification.maxReceiveChannels (10.7.7) Verification.maxReceiveKeys (10.7.7) Generation.maxTransmitKeys (10.7.16) Generation.protectFrames (10.7.17, Fig 10-3) Verification.validateFrames (10.7.8, Fig 10-4) Verification.replayProtect (10.7.8, Fig 10-4) Verification.replayWindow (10.7.8, Fig 10-4) CurrentCipherSuite.ciphersuite (10.7.25) controlledPort.adminPointToPointMAC (6.5, 10.7.4) controlledPort.operPointToPointMAC (6.5, 10.7.4) Generation.alwaysIncludeSCI (10.5.3,10.7.17) Generation.useES (10.5.3,10.7.17) Generation.useSCB (10.5.3, 10.7.17) Generation.includingSCI (10.5.3, 10.7.17, Fig 10-3) Generation.maxTransmitChannels (10.7.16)
secyIfCipherTable[2] secyIfCipherEntry[2] [secyIfInterfaceIndex, secyCipherSuiteIndex] secyIfCipherGroup[2]	secyIfCipherImplemented[2b] secyIfCipherEnableUse[2] secyIfCipherRqConfidentiality[2]	implementedCipherSuites (10.7.26) CipherSuiteControl.enableUse (10.7.26) CipherSuiteControl.requireConfidentiality (10.7.26)
secyIfTCTable[2] secyIfTCEntry[2] [secyIfInterfaceIndex, secyTCUserPriority[2]] secyTCGroup[2]	secyIfTCUserPriority[2] secyIfTCTrafficClass[2]	Generation.userpri (10.5.1, 10.7.17) TrafficClass.trafficClass (10.5.1, 10.7.17)
secyIfAPTable[2] secyAPEntry[2] [secyIfInterfaceIndex, secyAPUserPCP[2]] secyIfAPGroup[2]	secyIfAPUserPCP[2] secyIfAPAccessPCP[2]	Generation.userpcp (10.5, 10.7.17) AccessPriority.accessPCP (10.5, 10.7.17)

[a] Tables, table entries, groups and objects that appear only in secyMIBCompliance (deprecated) are superscripted[1], and those that appear only in secyTcMIBCompliance (recommended) are superscripted[2]. Those that are used in both are not superscripted.

[b] If the CipherSuite referenced by secyCipherSuiteIndex is not implemented for the SecY identified by the secyIfInterfaceIndex, the corresponding instance of this object is not required, if it is present secyCipherSuiteAvailable (read-only) and secyCipherSuiteEnable (normally read-write) will be False and not writable.

1

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Table 13-6—Transmit and receive SC management

Table	Table Entry objects	Figure 10-5 reference and definition
secyTxSCTable[1a] secyTxSCEntry[1] [secyIfInterfaceIndex] secyTxSCGroup[1]	secyTxSCI[1] secyTxSCState[1] secyTxSCEncodingSA[1] secyTxSCEncipheringSA[1] secyTxSCCreatedTime[1] secyTxSCStartedTime[1] secyTxSCStoppedTime[1]	TransmitSC.sci (7.1.2, 10.7.1) TransmitSC.transmitting (10.7.21, 10.7.23) TransmitSC.encodingSA (10.5.1, 10.7.21) deprecated (10.5.4) TransmitSC.createdTime (10.7.21) TransmitSC.startedTime (10.7.21) TransmitSC.stoppedTime (10.7.21)
secyTSCTable secyTSCEntry[2] [secyIfInterfaceIndex, secyTSCI] secyTSCGroup[2]	secyTSCI[2] secyTSCState[2] secyTSCEncodingSA[2] secyTSCCreatedTime[2] secyTSCStartedTime[2] secyTSCStoppedTime[2]	TransmitSC.sci (7.1.2, 10.7.17, 10.7.20) TransmitSC.transmitting (10.7.21, 10.7.23) TransmitSC.encodingSA (10.5.1, 10.7.21) TransmitSC.createdTime (10.7.21) TransmitSC.startedTime (10.7.21) TransmitSC.stoppedTime (10.7.21)
secyTxSATable[1] secyTxSAEntry[1] [secyIfInterfaceIndex, secyTxSA] secyTxSAGroup[1]	secyTxSA[1] secyTxSAState[1] secyTxSANextPN[1] secyTxSAConfidentiality[1] secyTxSASAKUnchanged[1] secyTxSACreatedTime secyTxSAStartedTime secyTxSAStoppedTime	TransmitSC.txa (10.7.22) TransmitSA.inUse (10.7.23) TransmitSA.nextPN (10.5, 10.7.23) TransmitSA.confidentiality (10.7.23) deprecated TransmitSA.createdTime (10.7.23) TransmitSA.startedTime (10.7.23) TransmitSA.stoppedTime (10.7.23)
secyTSATable[2] secyTSAEntry[2] [secyIfInterfaceIndex, secyTSCI, secyTSA] secyTSAGroup[2]	secyTSA[2] secyTSAState[2] secyTSANextXPN[2] secyTSAConfidentiality[2] secyTSACreatedTime[2] secyTSAStartedTime[2] secyTSAStoppedTime[2] secyTSAKeyIdentifier[2] secyTSASSCI[2]	TransmitSC.txa (10.7.22) TransmitSA.inUse (10.7.23) TransmitSA.nextPN (10.5, 10.7.23) TransmitSA.confidentiality (10.7.23) TransmitSA.createdTime (10.7.23) TransmitSA.startedTime (10.7.23) TransmitSA.stoppedTime (10.7.23) TransmitSA.keyIdentifier (10.7.23) TransmitSA.ssci (10.7.23)
secyRxSCTable secyRxSCEntry [secyIfInterfaceIndex, secyRxSCI] secyRxSCGroup[1] secyRcSCGroup[2]	secyRxSCI secyRxSCState secyRxSCCurrentSA[1] secyRxSCCreatedTime secyRxSCStartedTime secyRxSCStoppedTime	ReceiveSC.sci (10.7.11) ReceiveSC.receiving (10.7.12, 10.7.14, 10.7.15) deprecated ReceiveSC.createdTime (10.7.12) ReceiveSC.startedTime (10.7.12) ReceiveSC.stoppedTime (10.7.12)
secyRxSATable secyRxSAEntry [secyIfInterfaceIndex, secyRxSCI, secyRxSA] secyRxSAGroup[1] secyRSAGroup[2]	secyRxSA secyRxSAState secyRxSANextPN[1] secyRxSANextXPN[2] secyRxSALowestXPN[2] secyRxSASAKUnchanged[1] secyRxSACreatedTime secyRxSAStartedTime secyRxSAStoppedTime secyRxSAKeyIdentifier[2] secyRxSASSCI[2]	ReceiveSC.rxa (10.7.13) ReceiveSA.inUse (10.7.14) deprecated ReceiveSA.nextPN (10.7.14) ReceiveSA.lowestPN (10.6.2, 10.6.4, 10.6.5, 10.7.14, Fig 10-4) deprecated ReceiveSA.createdTime (10.7.14) ReceiveSA.startedTime (10.7.14) ReceiveSA.stoppedTime (10.7.14) ReceiveSA.keyIdentifier (10.7.14) ReceiveSA.ssci (10.7.14)

[a] Tables, table entries, groups and objects that appear only in secyMIBCompliance (deprecated) are superscripted[1], and those that appear only in secyTcMIBCompliance (recommended) are superscripted[2]. Those that are used in both are not superscripted.

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Table 13-7—Transmit and receive statistics

Table	Table Entry objects	Figure 10-5 reference and definition
secyStatsTable secyStatsEntry augments secyIfEntry secyStatsGroup[1a] secyIfStatsGroup[2] secyCipherStatsGroup[2]	secyStatsTxUntaggedPkts secyStatsTxTooLongPkts secyStatsRxUntaggedPkts secyStatsRxNoTagPkts secyStatsRxBadTagPkts secyStatsRxUnknownSCIPkts[1] secyStatsRxNoSCIPkts[1] secyStatsRxOverrunPkts secyStatsRxNoSAPkts[2] secyStatsRxNoSAErrorPkts[2] secyStatsTxOctetsProtected[2] secyStatsTxOctetsEncrypted[2] secyStatsRxOctetsValidated[2] secyStatsRxOctetsDecrypted[2]	Generation.OutPktsUntagged (10.7.18, Fig 10-3) Generation.OutPktsTooLong (10.7.18, Fig 10-3) Verification.InPktsUntagged (10.7.18, Fig 10-4) Verification.InPktsNoTag (10.7.9, Fig 10-4) Verification.InPktsBadTag (10.7.9, Fig 10-4) deprecated deprecated Verification.InPktsOverrun (10.7.9, Fig 10-4) Verification.InPktsNoSA (10.7.9, Fig 10-4) Verification.InPktsNoSAError(10.7.9, Fig 10-4) Generation.OutOctetsProtected (10.7.9, Fig 10-3) Generation.OutsOctetsEncrypted (10.7.9, Fig 10-3) Verification.InOctetsValidated (10.6.3, Fig 10-4) Verification.InOctetsValidated (10.6.3, Fig 10-4)
secyTxSCStatsTable[1] secyTxSCStatsEntry[1] augments secyTxSCEntry secyTxSCStatsGroup[1]	secyTxSCStatsProtectedPkts [1] secyTxSCStatsEncryptedPkts[1] secyTxSCStatsOctetsProtected[1] secyTxSCStatsOctetsEncrypted[1]	TransmitSC.OutPktsProtected (10.7.18, Fig 10-3) TransmitSC.OutPktsEncrypted (10.7.18, Fig 10-3) deprecated deprecated
secyTSCStatsTable[2] secyTSCStatsEntry[2] augments secyTSCEntry secyTSCStatsGroup[2]	secyTSCStatsProtectedPkts[2] secyTSCStatsEncryptedPkts[2]	TransmitSC.OutPktsProtected (10.7.18, Fig 10-3) TransmitSC.OutPktsEncrypted (10.7.18, Fig 10-3)
secyTxSAStatsTable[1] secyTxSAStatsEntry[1] augments secyTxSAEntry[1] secyTxSAStatsGroup[1]	secyTxSAStatsProtectedPkts[1] secyTxSAStatsEncryptedPkts[1]	deprecated deprecated
secyRxSCStatsTable secyRxSCStatsEntry augments secyRxSCEntry secyRxSCStatsGroup[1] secyRSCStatsGroup[2]	secyRxSCStatsUnusedSAPkts[1] secyRxSCStatsNoUsingSAPkts[1] secyRxSCStatsLatePkts secyRxSCStatsNotValidPkts secyRxSCStatsInvalidPkts secyRxSCStatsDelayedPkts secyRxSCStatsUncheckedPkts secyRxSCStatsOKPkts secyRxSCStatsOctetsValidated[1] secyRxSCStatsOctetsDecrypted[1]	deprecated deprecated ReceiveSC.InPktsLate (10.7.9, Fig 10-4) ReceiveSC.InPktsNotValid (10.7.9, Fig 10-4) ReceiveSC.InPktsInvalid (10.7.9, Fig 10-4) ReceiveSC.InPktsDelayed (10.7.9, Fig 10-4) ReceiveSC.InPktsUnchecked (10.7.9, Fig 10-4) ReceiveSC.InPktsOK(10.7.9, Fig 10-4) deprecated deprecated
secyRxSAStatsTable[1] secyRxSAStatsEntry augments secyRxSAEntry secyRxSAStatsGroup[1]	secyRxSAStatsUnusedSAPkts[1] secyRxSAStatsNotUsingSAPkts[1] secyRxSAStatsNotValidPkts[1] secyRxSAStatsInvalidPkts[1] secyRxSAStatsOKPkts[1]	deprecated deprecated deprecated deprecated deprecated

[a] Tables, table entries, groups and objects that appear only in secyMIBCompliance (deprecated) are superscripted[1], and those only in secyTcMIBCompliance (recommended) are superscripted[2]. Those that are used in both are not superscripted.

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Table 13-8—Cipher Suite information

Table Table Entry [Index]	Table Entry Objects	Figure 10-5 reference and definition
secyCipherSuiteTable	secyCipherSuiteIndex	
secyCipherSuiteEntry	secyCipherSuiteId	CipherSuite.identifier (10.7.25, Table 14-1)
[secyCipherSuiteIndex]	secyCipherSuiteName	CipherSuite.name (10.7.25, Table 14-1)
	secyCipherSuiteCapability	CipherSuite.integrityProtection, confidentialityProtection (10.7.25)
secyCipherSuiteGroup[1]		
secyCipherInfoGroup[2]	secyCipherSuiteProtection[1]	deprecated
	secyCipherSuiteProtectionOffset[1]	deprecated
	secyCipherSuiteDataLengthChange	CipherSuite.changesDataLength (10.7.25)
	secyCipherSuiteICVLength	CipherSuite.ICVlength (10.7.25)
	secyCipherSuiteRowStatus[1]	deprecated

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13.6 MAC Security Entity (SecY) MIB definition[17] [18]

```
IEEE8021-SECY-MIB DEFINITIONS ::= BEGIN

-- ============================================================================
-- IEEEE802.1AE MAC Security Entity (SecY) MIB
-- ============================================================================

IMPORTS
    MODULE-IDENTITY, OBJECT-TYPE,
    Unsigned32, Integer32, Counter32,Counter64
        FROM SNMPv2-SMI
    TEXTUAL-CONVENTION, RowPointer, TimeStamp, TruthValue, RowStatus
        FROM SNMPv2-TC
    SnmpAdminString
        FROM SNMP-FRAMEWORK-MIB
    MODULE-COMPLIANCE, OBJECT-GROUP
        FROM SNMPv2-CONF
    InterfaceIndex, ifCounterDiscontinuityGroup
        FROM IF-MIB
    ;

ieee8021SecyMIB MODULE-IDENTITY
LAST-UPDATED "202206060000Z" -- June 6, 2022
    ORGANIZATION "IEEE 802.1 Working Group"
    CONTACT-INFO
        "  WG-URL: http://www.ieee802.org/1/
         WG-EMail: stds-802-1-l@ieee.org
          Contact: IEEE 802.1 Working Group Chair
           Postal: C/O IEEE 802.1 Working Group
                   IEEE Standards Association
                   445 Hoes Lane
                   Piscataway, NJ 08854
                   USA
           E-mail: stds-802-1-chairs@ieee.org"
    DESCRIPTION
    "The MAC security entity (SecY) MIB module. A SecY is a shim in an interface
     stack that uses the MAC Security (MACsec) protocol.

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     This version of this MIB module is part of IEEE Std 802.1AEdk-2022;
     see that standard for full legal notices.

     Unless otherwise indicated, the references in this MIB module are to
     IEEE Std 802.1AE-2018 as amended by IEEE Std 802.1AE-2018/Cor 1-2020 and
     IEEE Std 802.1AEdk-2022.

     Each SecY transmits MACsec protected frames on one or more
     Secure Channels (SCs) to each of the other SecYs attached to the
     same LAN and participating in the same Secure Connectivity
     Association (CA). The CA is a security relationship, that is
     established and maintained by key agreement protocols and supported
     by MACsec to provide full connectivity between its participants.
     Each SC provides unidirectional point to multipoint connectivity
     from one participant to all the others and is supported by a
     succession of similarly point to multipoint Secure Associations
     (SAs). The Secure Association  Key (SAK) used to protect frames is
     changed as an SA is replaced by its (overlapping) successor so
     fresh keys can be used without disrupting a long lived SC and CA.

     Two different upper interfaces, a Controlled Port (for frames
     protected by MACsec, providing an instance of the secure MAC
     service) and an Uncontrolled Port (for frames not requiring
     protection, like the key agreement frames used to establish the
```

[17] *Copyright release for MIBs*: Users of this standard may freely reproduce the MIB modules in this standard so that they can be used for their intended purpose.

[18] The MIB text in this clause includes clickable cross-references to MIB objects (highlighted). A plain text (UTF-8) version of this MIB is attached to the PDF version of this standard, and can be obtained by Web browser from the IEEE 802.1 Website at https://1.ieee802.org/mib-modules/.

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```
     CA and distribute keys) are associated with a SecY  shim. For each
     instance of a SecY two ifTable rows (one for each interface) run on
     top of an ifTable row representing the 'Common Port' interface,
     such as a row with ifType ='ethernetCsmacd(6)'.

     ___________________________________|_____________________________________
     |                                  |                                    |
     |  Controlled Port Interface       |   Uncontrolled Port Interface      |
     |  (ifEntry = j,ifType =           |   (ifEntry = k, ifType =           |
     |   macSecControlledIF(231))       |    macSecUncontrolledIF(232))      |
     |__________________________________|____________________________________|
     |                                                                       |
     |                     Physical Interface                                |
     |                       (ifEntry = i)                                   |
     |                (ifType = ethernetCsmacd(6))                           |
     |_______________________________________________________________________|
        Example MACsec Interface Stack. i, j, k are ifIndexes each
        indicating a row in the ifTable.
    "
    REVISION "202206060000Z" -- June 6, 2022
    DESCRIPTION
    "Published as part of IEEE Std 802.1AEdk-2022.
     Cross references, contact information, and descriptions updated."

    REVISION    "201712071816Z"
    DESCRIPTION
    "Published as part of IEEE Std 802.1AE-2018.
     Updated CONTACT-INFO."

    REVISION    "201605102049Z"
    DESCRIPTION
    "Updated by the IEEE Std 802.1AEcg amendment. Object DESCRIPTIONs
     and references aligned with text of the standard (including prior
     amendments). IEEE 802.1AEcg Annex G details changes.
     The initial version of this ieee8021SecyMIB used the object
     name prefix 'secy' rather than 'ieee8021secy' (recommended by
     RFC 4181). The 'secy' prefix has been retained in this revision for
     for backwards compatibility and internal consistency."

    REVISION    "200601100000Z"
    DESCRIPTION "Initial version of this MIB in IEEE 802.1AE-2006"
    ::= { iso(1) std(0) iso8802(8802) ieee802dot1(1)
          ieee802dot1mibs(1) 3 }
-- ============================================================================
-- Textual Conventions

SecySCI ::= TEXTUAL-CONVENTION
    STATUS current
    DESCRIPTION
    "Textual convention for a Secure Channel Identifier (SCI).
    Each SC is identified by an SCI comprising a 48-bit MAC Address,
    allocated to the transmitting system and a 16-bit Port Identifier."
    REFERENCE   "7.1.2, Figure 7-7"
    SYNTAX OCTET STRING (SIZE (8))

SecyAN ::= TEXTUAL-CONVENTION
    DISPLAY-HINT "d"
    STATUS current
    DESCRIPTION
    "Textual convention for an Association Number (AN).
    Each SC is comprised of a succession of SAs, each with a different SAK,
    identified by a Secure Association Identifier (SAI) comprising an SCI
    concatenated with a two-bit AN.  The SAI is unique for SAs used by SecYs
    participating in a given CA at any instant."
    REFERENCE   "7.1.3, Figure 7-7"
    SYNTAX Unsigned32 (0..3)

-- ============================================================================
-- subtrees in the SecY MIB
```


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```
secyMIBNotifications    OBJECT IDENTIFIER ::= { ieee8021SecyMIB 0 }
secyMIBObjects          OBJECT IDENTIFIER ::= { ieee8021SecyMIB 1 }
    secyMgmtMIBObjects  OBJECT IDENTIFIER ::= {  secyMIBObjects 1 }
    secyStatsMIBObjects OBJECT IDENTIFIER ::= {  secyMIBObjects 2 }
secyMIBConformance      OBJECT IDENTIFIER ::= { ieee8021SecyMIB 2 }
    secyMIBCompliances  OBJECT IDENTIFIER ::= { secyMIBConformance 1 }
    secyMIBGroups       OBJECT IDENTIFIER ::= { secyMIBConformance 2 }
-- ============================================================================
--secyMgmtMIBObjects
--  secyIfTable
--  secyTSCTable   --
--  secyTSATable   --
--  secyIfTCTable  --
--  secyIfAPTable  --
--  secyRxSCTable
--  secyRxSATable
--  secyCipherSuiteTable
--  secyIfCipherTable
--The following are historic following approval of IEEE Std 802.1AEcg-2017,
--even if their STATUS remains 'current'. They do not include any objects
--that are part of a current conformance OBJECT-GROUP, and lack traffic
--class transmit SC and XPN support:
--  secyTxSCTable, secyTxSATable
-- ============================================================================
--  secyIfTable
secyIfTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyIfEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
    "A table with an entry for each MAC Security protocol (MACsec) capable
    interface in the system, i.e. for each SecY. Configured value of writable
    objects in each table entry MUST be persistent and remain unchanged across
    re-initialization of the system's management entity."
    REFERENCE "10.7, Table 13-1"
    ::= { secyMgmtMIBObjects 1 }

--secyIfEntry
secyIfEntry    OBJECT-TYPE
    SYNTAX      SecyIfEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
    "A table entry with service information for a particular SecY."
    INDEX   { secyIfInterfaceIndex }
    ::= { secyIfTable 1 }

--SecyIfEntry
SecyIfEntry ::= SEQUENCE {
    secyIfInterfaceIndex           InterfaceIndex,
    secyIfMaxPeerSCs               Unsigned32,
    secyIfRxMaxKeys                Unsigned32,
    secyIfTxMaxKeys                Unsigned32,
    secyIfProtectFramesEnable      TruthValue,
    secyIfValidateFrames           INTEGER,
    secyIfReplayProtectEnable      TruthValue,
    secyIfReplayProtectWindow      Unsigned32,
    secyIfCurrentCipherSuite       Unsigned32,
    secyIfAdminPt2PtMAC            INTEGER,
    secyIfOperPt2PtMAC             TruthValue,
    secyIfIncludeSCIEnable         TruthValue,
    secyIfUseESEnable              TruthValue,
    secyIfUseSCBEnable             TruthValue,
    secyIfSCI                      SecySCI,
    secyIfIncludingSCI             TruthValue,
    secyIfMaxTSCs                  Unsigned32
}
```


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```
--secyIfInterfaceIndex
secyIfInterfaceIndex    OBJECT-TYPE
    SYNTAX      InterfaceIndex
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "The Interface Index for this SecY's Controlled Port."
    REFERENCE   "10.1"
    ::= { secyIfEntry 1 }

--secyIfMaxPeerSCs
secyIfMaxPeerSCs    OBJECT-TYPE
    SYNTAX      Unsigned32
    UNITS       "security connections"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The maximum number of peer SCs for this SecY."
    REFERENCE   "10.7.7"
    ::= { secyIfEntry 2 }

--secyIfRxMaxKeys
secyIfRxMaxKeys    OBJECT-TYPE
    SYNTAX      Unsigned32
    UNITS       "keys"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The maximum number of keys in simultaneous use for
                 reception for this SecY."
    REFERENCE   "10.7.7"
    ::= { secyIfEntry 3 }

--secyIfTxMaxKeys
secyIfTxMaxKeys    OBJECT-TYPE
    SYNTAX      Unsigned32
    UNITS       "keys"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The maximum number of keys in simultaneous use for
                 transmission for this SecY."
    REFERENCE   "10.7.16"
    ::= { secyIfEntry 4 }

--secyIfProtectFramesEnable
secyIfProtectFramesEnable    OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "Enables or disables protection of transmitted frames."
    REFERENCE   "10.7.17, Figure 10-3"
    DEFVAL { true }
    ::= { secyIfEntry 5 }

--secyIfValidateFrames
secyIfValidateFrames    OBJECT-TYPE
    SYNTAX      INTEGER {
                    disabled(1),
                    check(2),
                    strict(3),
                    null(4)    -- 802.1AEcg
                }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
    "Controls validation of received frames.
     disabled(1) : disable validation, remove SecTAGs and ICVs (if present.
     check(2)    : enable validation, do not discard invalid frames.
     strict(3)   : enable validation and  discard invalid frames.
     null(4)     : no processing, do not remove SecTAGs or ICVs."
    REFERENCE   "10.7.8, Figure 10-4"
    DEFVAL { strict }
    ::= { secyIfEntry 6 }
```


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```
--secyIfReplayProtectEnable
secyIfReplayProtectEnable    OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "Enables or disables replay protection."
    REFERENCE   "10.7.8, Figure 10-4"
    DEFVAL { true }
    ::= { secyIfEntry 7 }

--secyIfReplayProtectWindow
secyIfReplayProtectWindow    OBJECT-TYPE
    SYNTAX      Unsigned32
    UNITS       "Packets"
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "The replay protection window size."
    REFERENCE   "10.7.8, Figure 10-4"
    DEFVAL { 0 }
    ::= { secyIfEntry 8 }

--secyIfCurrentCipherSuite
secyIfCurrentCipherSuite    OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "The secyCipherSuiteTableindex for this SecY's in use Cipher
                 Suite. Should be read-only if secyIfCipherTable implemented."
    REFERENCE   "10.7.25"
    ::= { secyIfEntry 9 }

--secyIfAdminPt2PtMAC
secyIfAdminPt2PtMAC    OBJECT-TYPE
    SYNTAX      INTEGER {
                    forceTrue(1),
                    forceFalse(2),
                    auto(3)
                }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
    "Controls the value of operPointToPointMAC (secyOperPt2PtMAC)
     reported to the user(s) of this SecY's Controlled Port:

     forceTrue(1)  : operPointToPointMAC is True, regardless of the
                     configuration and status of the SecY.
     forceFalse(2) : operPointToPointMAC is False, regardless of the
                     configuration and status of the SecY.
     auto(3)       : OperPointMAC is True if secyIfvalidateFrames is
                     strict and reception is from at most one peer SecY,
                     or if secyIfvalidateFrames is not strict and
                     operPointToPointMAC is True for the Common Port,
                     and is False otherwise."
    REFERENCE   "6.5, 10.7.4"
    DEFVAL { auto }
    ::= { secyIfEntry 10 }

--secyIfOperPt2PtMAC
secyIfOperPt2PtMAC    OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
    "Reflects the current service connectivity to be assumed by the
     user(s) of the SecY's Controlled Port:

     true(1)  : connectivity is to at most one other system.
     false(2) : connectivity is to one or more other systems."
    REFERENCE   "6.5, 10.7.4"
    ::= { secyIfEntry 11 }
```


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```
--secyIfIncludeSCIEnable
secyIfIncludeSCIEnable    OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "Mandates inclusion of an explicit SCI in the SecTAG
                 when transmitting protected frames."
    REFERENCE   "10.5.3 alwaysIncludeSCI, 10.7.17"
    DEFVAL { false }
    ::= { secyIfEntry 12 }

--secyIfUseESEnable
secyIfUseESEnable    OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "Enables use of the ES bit in the SecTAG when
                 transmitting protected frames."
    REFERENCE   "10.5.3 useES, 10.7.17"
    DEFVAL { false }
    ::= { secyIfEntry 13 }

--secyIfUseSCBEnable
secyIfUseSCBEnable    OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "Enables use of the SCB bit in the SecTAG when
                 transmitting protected frames."
    REFERENCE   "10.5.3 useSCB, 10.7.17"
    DEFVAL { false }
    ::= { secyIfEntry 14 }

--secyIfSCI
secyIfSCI       OBJECT-TYPE
    SYNTAX      SecySCI
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The SCI for the SecY's default traffic class."
    REFERENCE   "7.1.2, 10.7.1"
    ::= { secyIfEntry 15 }

--secyIfIncludingSCI
secyIfIncludingSCI    OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "True if an explicit SCI is included in the SecTAG when
                 transmitting protected frames."
    REFERENCE   "10.5.3 includingSCI, 10.7.17"
    DEFVAL { false }
    ::= { secyIfEntry 16 }

--secyIfMaxTSCs
secyIfMaxTSCs    OBJECT-TYPE
    SYNTAX      Unsigned32
    UNITS       "security connections"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The maximum number of transmit SCs for this SecY."
    REFERENCE   "10.7.16"
    ::= { secyIfEntry 17 }

```


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```
-- ============================================================================
--secyTSCTable
secyTSCTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyTSCEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A transmit SC management table for systems with SecY's capable
                of supporting  traffic class SCs."
    REFERENCE   "7.1.2, 10.7.17, 10.7.20"
    ::= { secyMgmtMIBObjects 10 }

--secyTSCEntry
secyTSCEntry    OBJECT-TYPE
    SYNTAX      SecyTSCEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "An entry with transmit SC information for one of the system's
                 SecYs and one of its traffic classes."
    INDEX { secyIfInterfaceIndex, secyTSCI }
    ::= { secyTSCTable 1 }

--SecyTSCEntry
SecyTSCEntry ::= SEQUENCE {
    secyTSCI              SecySCI,
    secyTSCState          INTEGER,
    secyTSCEncodingSA     RowPointer,
    secyTSCCreatedTime    TimeStamp,
    secyTSCStartedTime    TimeStamp,
    secyTSCStoppedTime    TimeStamp
}
--secyTSCI
secyTSCI        OBJECT-TYPE
    SYNTAX      SecySCI
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "The SCI for the transmit SC for this SecY and traffic class."
    REFERENCE   "7.1.2, 10.7.17, 10.7.20"
    ::= { secyTSCEntry 1 }

--secyTSCState
secyTSCState    OBJECT-TYPE
    SYNTAX      INTEGER { inUse(1), notInUse(2)}
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The state of the transmit SC for this SecY and traffic class:
                 inUse(1)    : one or more SAs are in use.
                 notInUse(2) : no SAs are in use for this SC."
    REFERENCE   "10.7.20"
    ::= { secyTSCEntry 2 }

--secyTSCEncodingSA
secyTSCEncodingSA    OBJECT-TYPE
    SYNTAX      RowPointer
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
    "The SA currently used to encode the SecTAG. The row pointer points to an
     entry in the secyTSATable. If no such information is available, the value
     shall be the OBJECT IDENTIFIER { 0 0 }."
    REFERENCE   "10.5.1, 10.7.21"
    ::= { secyTSCEntry 3 }

--secyTSCCreatedTime
secyTSCCreatedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this transmitting SC was created."
    REFERENCE   "10.7.21"
    ::= { secyTSCEntry 4 }

```


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```
--secyTSCStartedTime
secyTSCStartedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this SC last started transmitting."
    REFERENCE   "10.7.21"
    ::= { secyTSCEntry 5 }

--secyTSCStoppedTime
secyTSCStoppedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this SC last stopped transmitting."
    REFERENCE   "10.7.21"
    ::= { secyTSCEntry 6 }

-- ============================================================================
--secyTSATable
secyTSATable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyTSAEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A transmit SA management table for systems with SecY's capable
                 of supporting  traffic class SC's."
    REFERENCE   "10.7.22, Table 13-2"
    ::= { secyMgmtMIBObjects 11 }

--secyTSAEntry
secyTSAEntry    OBJECT-TYPE
    SYNTAX      SecyTSAEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "An entry for a transmit SA."
    INDEX   { secyIfInterfaceIndex, secyTSCI, secyTSA }
    ::= { secyTSATable 1 }

--SecyTSAEntry
SecyTSAEntry ::= SEQUENCE {
    secyTSA                   SecyAN,
    secyTSAState              INTEGER,
    secyTSANextXPN            Counter64,
    secyTSAConfidentiality    TruthValue,
    secyTSAKeyIdentifier      SnmpAdminString,
    secyTSASSCI               Integer32,
    secyTSACreatedTime        TimeStamp,
    secyTSAStartedTime        TimeStamp,
    secyTSAStoppedTime        TimeStamp
}
--secyTSA
secyTSA     OBJECT-TYPE
    SYNTAX      SecyAN
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "The association number (AN) for this transmit SA."
    REFERENCE   "10.7.22"
    ::= { secyTSAEntry 1 }

--secyTSAState
secyTSAState   OBJECT-TYPE
    SYNTAX      INTEGER { inUse(1), notInUse(2)}
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The transmit SA current status: inUse(1) , notInUse(2)."
    REFERENCE   "10.7.23"
    ::= { secyTSAEntry 2 }

```


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```
--secyTSANextXPN
secyTSANextXPN  OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The next packet number (PN) for this SA."
    REFERENCE   "10.5, 10.7.23"
    ::= { secyTSAEntry 3 }

--secyTSAConfidentiality
secyTSAConfidentiality    OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "True if the SA provides confidentiality."
    REFERENCE   "10.7.23"
    ::= { secyTSAEntry 4 }

--secyTSAKeyIdentifier
secyTSAKeyIdentifier    OBJECT-TYPE
    SYNTAX      SnmpAdminString (SIZE (1..32))
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The Key Identifier (KI) for the SAK for this SA."
    REFERENCE   "IEEE 802.1X, 10.7.23"
    ::= { secyTSAEntry 5 }

--secyTSASSCI
secyTSASSCI     OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The SSCI for this SA, 0 if not using an XPN Cipher Suite."
    REFERENCE   "IEEE 802.1X, 10.7.23"
    ::= { secyTSAEntry 6 }

--secyTSACreatedTime
secyTSACreatedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this transmit SA was created."
    REFERENCE   "10.7.23"
    ::= { secyTSAEntry 7 }

--secyTSAStartedTime
secyTSAStartedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this SA last started transmitting."
    REFERENCE   "10.7.23"
    ::= { secyTSAEntry 8 }

--secyTSAStoppedTime
secyTSAStoppedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this SA last stopped transmitting."
    REFERENCE   "10.7.23"
    ::= { secyTSAEntry 9 }

```


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```
-- ============================================================================
--secyRxSCTable
secyRxSCTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyRxSCEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A table of receive SCs for the system."
    REFERENCE   "10.7.11, Table 13-2"
    ::= { secyMgmtMIBObjects 4 }

--secyRxSCEntry
secyRxSCEntry   OBJECT-TYPE
    SYNTAX      SecyRxSCEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A table entry for a receive SC."
    INDEX    { secyIfInterfaceIndex, secyRxSCI }
    ::= { secyRxSCTable 1 }

--SecyRxSCEntry
SecyRxSCEntry ::= SEQUENCE {
    secyRxSCI              SecySCI,
    secyRxSCState          INTEGER,
    secyRxSCCurrentSA      RowPointer, -- deprecated
    secyRxSCCreatedTime    TimeStamp,
    secyRxSCStartedTime    TimeStamp,
    secyRxSCStoppedTime    TimeStamp
}

--secyRxSCI
secyRxSCI       OBJECT-TYPE
    SYNTAX      SecySCI
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "The SCI for the receive SC."
    REFERENCE   "10.7.11"
    ::= { secyRxSCEntry 1 }

--secyRxSCState
secyRxSCState   OBJECT-TYPE
    SYNTAX      INTEGER {inUse(1), notInUse(2)}
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The receive SCs current state:
                 inUse(1) : one or more SAs for this SC are in use.
                 notInUse(2) : no SAs for this SC is in use."
    REFERENCE   "10.7.12 receiving, 10.7.14 inUse, 10.7.15"
    ::= { secyRxSCEntry 2 }

--secyRxSCCurrentSA
secyRxSCCurrentSA    OBJECT-TYPE
    SYNTAX      RowPointer
    MAX-ACCESS  read-only
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION
    "The current receiving association number for the SC in use.
     The row pointer points to an entry in the secyRxSATable.  If no
     such information can be identified, the value of this object shall
     be the OBJECT IDENTIFIER { 0 0 }."
    REFERENCE   "10.7.15, 10.7.13"
    ::= { secyRxSCEntry 3 }

--secyRxSCCreatedTime
secyRxSCCreatedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this receiving SC was created."
    REFERENCE   "10.7.12"
    ::= { secyRxSCEntry 4 }

```


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```
--secyRxSCStartedTime
secyRxSCStartedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this SC last started receiving."
    REFERENCE   "10.7.12"
    ::= { secyRxSCEntry 5 }

--secyRxSCStoppedTime
secyRxSCStoppedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this SC last stopped receiving."
    REFERENCE   "10.7.12"
    ::= { secyRxSCEntry 6 }

-- ============================================================================
--secyRxSATable
secyRxSATable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyRxSAEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A table with entries for the system's receive SAs."
    REFERENCE   "10.7.13"
    ::= { secyMgmtMIBObjects 5 }

--secyRxSAEntry
secyRxSAEntry   OBJECT-TYPE
    SYNTAX      SecyRxSAEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "An entry for one of the SAs used by one of the system's
                 SecY's to receive protected frames."
    INDEX    { secyIfInterfaceIndex, secyRxSCI, secyRxSA }
    ::= { secyRxSATable 1 }

--SecyRxSAEntry
SecyRxSAEntry ::= SEQUENCE {
    secyRxSA               SecyAN,
    secyRxSAState          INTEGER,
    secyRxSANextPN         Unsigned32,      -- deprecated
    secyRxSASAKUnchanged   TruthValue,      -- deprecated
    secyRxSACreatedTime    TimeStamp,
    secyRxSAStartedTime    TimeStamp,
    secyRxSAStoppedTime    TimeStamp,
    secyRxSANextXPN        Counter64,
    secyRxSALowestXPN      Counter64,
    secyRxSAKeyIdentifier  SnmpAdminString,
    secyRxSASSCI           Integer32
}

--secyRxSA
secyRxSA        OBJECT-TYPE
    SYNTAX      SecyAN
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "This receive SA's association number (AN)."
    REFERENCE   "10.7.13"
    ::= { secyRxSAEntry 1 }

--secyRxSAState
secyRxSAState   OBJECT-TYPE
    SYNTAX      INTEGER { inUse(1), notInUse(2)}
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "This receive SA's current state."
    REFERENCE   "10.7.14"
    ::= { secyRxSAEntry 2 }

```


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```
--secyRxSANextPN
secyRxSANextPN  OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  read-write
    STATUS      deprecated
    DESCRIPTION "Use secyRxSANextXPN for both 32-bit PN and 64-bit XPN values.
                 If implemented, this object contains the lower 32 bits."
    REFERENCE   "10.6.5, 10.7.14, Figure 10-4"
    ::= { secyRxSAEntry 3 }

--secyRxSASAKUnchanged
secyRxSASAKUnchanged   OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION "An SAK reference, unchanged for the receiving SA's life."
    REFERENCE   "10.7.13"
    ::= { secyRxSAEntry 4 }

--secyRxSACreatedTime
secyRxSACreatedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this receiving SA was created."
    REFERENCE   "10.7.14"
    ::= { secyRxSAEntry 5 }

--secyRxSAStartedTime
secyRxSAStartedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this SA last started receiving."
    REFERENCE   "10.7.14"
    ::= { secyRxSAEntry 6 }

--secyRxSAStoppedTime
secyRxSAStoppedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this SA last stopped receiving frames."
    REFERENCE   "10.7.14"
    ::= { secyRxSAEntry 7 }

--secyRxSANextXPN
secyRxSANextXPN        OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "One more than the highest PN conveyed in the SecTAG of
                 successfully validates frames received on this SA."
    REFERENCE   "10.6.5, 10.7.14, Figure 10-4"
    ::= { secyRxSAEntry 8 }

--secyRxSALowestXPN
secyRxSALowestXPN    OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The lowest acceptable packet number. A received frame
                 with a lower PN is  discarded if
                 secyIfReplayProtectEnable is enabled."
    REFERENCE   "10.6.2, 10.6.4, 10.6.5, 10.7.14,
        Figure 10-4"
    ::= { secyRxSAEntry 9 }
```


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```
--secyRxSAKeyIdentifier
secyRxSAKeyIdentifier    OBJECT-TYPE
    SYNTAX      SnmpAdminString (SIZE (1..32))
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The Key Identifier (KI) for the SAK for this SA."
    REFERENCE   "IEEE 802.1X, 10.7.14"
    ::= { secyRxSAEntry 10 }

--secyRxSASSCI
secyRxSASSCI    OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The SSCI for this SA, 0 if an XPN Cipher Suite is not in use."
    REFERENCE   "IEEE 802.1X, 10.7.14"
    ::= { secyRxSAEntry 11 }

-- ============================================================================
--secyCipherSuiteTable
secyCipherSuiteTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyCipherSuiteEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
    "A table of the system's Cipher Suite capabilities, which can differ
     by Cipher Suite implementation, so there can be more than one entry
     with the same secyCipherSuiteId. The secyIfCipherTable lists
     available entries by SecY, avoiding the need for remote network
     management to  write objects or create rows in this table. Any
     configured values shall be stored in persistent memory and remain
     unchanged across a re-initialization of the management system."
    REFERENCE   "10.7.25"
    ::= { secyMgmtMIBObjects 6 }

--secyCipherSuiteEntry
secyCipherSuiteEntry    OBJECT-TYPE
    SYNTAX      SecyCipherSuiteEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "An entry for a Cipher Suite implementation."
    INDEX { secyCipherSuiteIndex }
    ::= { secyCipherSuiteTable 1 }

--SecyCipherSuiteEntry
SecyCipherSuiteEntry ::= SEQUENCE {
    secyCipherSuiteIndex               Unsigned32,
    secyCipherSuiteId                  OCTET STRING,
    secyCipherSuiteName                SnmpAdminString,
    secyCipherSuiteCapability          BITS,
    secyCipherSuiteProtection          BITS,
    secyCipherSuiteProtectionOffset    INTEGER,
    secyCipherSuiteDataLengthChange    TruthValue,
    secyCipherSuiteICVLength           Unsigned32,
    secyCipherSuiteRowStatus           RowStatus
}

--secyCipherSuiteIndex
secyCipherSuiteIndex    OBJECT-TYPE
    SYNTAX      Unsigned32 (1..4294967295)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "The CipherSuiteTable entry index."
    ::= { secyCipherSuiteEntry 1 }

```


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```
--secyCipherSuiteId
secyCipherSuiteId    OBJECT-TYPE
    SYNTAX      OCTET STRING (SIZE (8))
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION "A unique 64-bit (EUI-64) identifier for the Cipher
                 Suite."
    REFERENCE   "10.7.25, Table 14-1"
    ::= { secyCipherSuiteEntry 2 }

--secyCipherSuiteName
secyCipherSuiteName    OBJECT-TYPE
    SYNTAX      SnmpAdminString (SIZE (1..128))
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION "The Cipher Suite Name, 128 octets or fewer."
    REFERENCE   "10.7.25, Table 14-1"
    ::= { secyCipherSuiteEntry 3 }

--secyCipherSuiteCapability
secyCipherSuiteCapability    OBJECT-TYPE
    SYNTAX      BITS {
                    integrity(0),
                    confidentiality(1),
                    offsetConfidentiality(2)
                }
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION "Cipher Suite implementation capability information.

                 integrity(0)             : integrity protection.
                 confidentiality(1)       : confidentiality protection.
                 offsetConfidentiality(2) : offset confidentiality
                                            protection."
    REFERENCE   "10.7.25, 10.7.25"
    ::= { secyCipherSuiteEntry 4 }

--secyCipherSuiteProtection
secyCipherSuiteProtection    OBJECT-TYPE
    SYNTAX      BITS {
                    integrity(0),
                    confidentiality(1),
                    offsetConfidentiality(2)
                }
    MAX-ACCESS  read-create
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION
    "The secyIfCipherSuite table supports per SecY configuration and should be
     used instead of this object.If the secyCipherSuiteCapability integrity
     bit is on, it can be turned on for this object. If the integrity and
     confidentiality bits of the secyCipherSuiteCapability are both on, the
     confidentiality bit of this object can be turned on provided that the
     integrity bit is also turned on, and the offsetConfidentiality bit can
     also be turned on if the secyCipherSuiteCapability has that bit on.

     integrity(0)       : enable (on) or disable integrity protection.
     confidentiality(1) : enable (on) or disable confidentiality protection.
     offsetConfidentiality(2) : enable (on) or disable offset confidentiality."
    REFERENCE   "10.7.25"
    DEFVAL { { integrity } }
    ::= { secyCipherSuiteEntry 5 }

```


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```
--secyCipherSuiteProtectionOffset
secyCipherSuiteProtectionOffset    OBJECT-TYPE
    SYNTAX      Integer32 (0 | 30 | 50)
    UNITS       "bytes"
    MAX-ACCESS  read-create
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION
    "The confidentiality protection offset options provided by the cipher
     suite. Can only be non-zero if the secyCipherSuiteProtection offset
     confidentiality bit is on, and then can only be 0 if the confidentiality
     bit is on."
    REFERENCE   "10.7.25, 10.7.26"
    DEFVAL { 0 }
    ::= { secyCipherSuiteEntry 6 }

--secyCipherSuiteDataLengthChange
secyCipherSuiteDataLengthChange    OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION "True if cipher suite changes the length of the data."
    REFERENCE   "10.7.25, Figure 9-1"
    ::= { secyCipherSuiteEntry 7 }

--secyCipherSuiteICVLength
secyCipherSuiteICVLength    OBJECT-TYPE
    SYNTAX      Unsigned32 (8..16)
    UNITS       "octets"
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION "The length of the integrity check value (ICV) field."
    REFERENCE   "10.7.25, Figure 9-1"
    ::= { secyCipherSuiteEntry 8 }

--secyCipherSuiteRowStatus
secyCipherSuiteRowStatus    OBJECT-TYPE
    SYNTAX      RowStatus
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION
    "The secyIfCipherTable (if implemented) avoids the need for
     network manager creation of entries in the secyCipherSuiteTable,
     and RowStatus should always be valid(1), with any per SecY
     unavailability indicated by an absence of a corresponding
     secyIfCipherTable entry or one with secyCipherSuiteAvailable
     false (the latter can indicate temporary unavailability)."
    REFERENCE   "10.7.25"
    ::= { secyCipherSuiteEntry 9 }

-- ============================================================================
--secyIfCipherTable
secyIfCipherTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyIfCipherEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
       "A table with an entry for the Cipher Suite capabilities
        implemented for each SecY in this system, providing per SecY
        control of Cipher Suite use.

        The configured value of writable objects in each table entry
        shall be stored in persistent memory and remain unchanged across
        a re-initialization of the system's management entity."
    REFERENCE   "10.7.26, Table 13-1"
    ::= { secyMgmtMIBObjects 7 }

```


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```
--secyIfCipherEntry
secyIfCipherEntry    OBJECT-TYPE
    SYNTAX      SecyIfCipherEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A table entry with Cipher Suite control for a SecY."
    INDEX   { secyIfInterfaceIndex, secyCipherSuiteIndex }
    ::= { secyIfCipherTable 1 }

--SecyIfCipherEntry
SecyIfCipherEntry ::= SEQUENCE {
    secyIfCipherImplemented         TruthValue,
    secyIfCipherEnableUse           TruthValue,
    secyIfCipherRqConfidentiality   TruthValue
}

--secyIfCipherImplemented
secyIfCipherImplemented    OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "True if the Cipher Suite implementation can be used by
                 this SecY (if secIfCipherEnableUse is true)."
    REFERENCE   "10.7.26"
    DEFVAL { true }
    ::= { secyIfCipherEntry 1 }

--secyIfCipherEnableUse
secyIfCipherEnableUse    OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "Enables use of the Cipher Suite by this SecY."
    REFERENCE   "10.7.26"
    DEFVAL { true }
    ::= { secyIfCipherEntry 2 }

--secyIfCipherRqConfidentiality
secyIfCipherRqConfidentiality    OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "True if confidentiality protection (without an offset)
                 is required if this Cipher Suite is used."
    REFERENCE   "10.7.26"
    DEFVAL { true }
    ::= { secyIfCipherEntry 3 }

-- ============================================================================
--secyIfTCTable
secyIfTCTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyIfTCEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
       "The Traffic Class Table for each SecY in this system.
        The configured value of writable objects in each table entry
        MUST be stored in persistent memory and remain unchanged across
        a re-initialization of the system's management entity."
    REFERENCE   "10.5.1, 10.7.17, Table 13-1"
    ::= { secyMgmtMIBObjects 8 }

--secyIfTCEntry
secyIfTCEntry   OBJECT-TYPE
    SYNTAX      SecyIfTCEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A table entry providing Traffic Class selection for a given
                 SecY and user priority."
    INDEX   { secyIfInterfaceIndex, secyIfTCUserPriority }
    ::= { secyIfTCTable 1 }

```


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```
--SecyIfTCEntry
SecyIfTCEntry ::= SEQUENCE {
    secyIfTCUserPriority           Integer32,
    secyIfTCTrafficClass           Integer32
}

--secyIfTCUserPriority
secyIfTCUserPriority    OBJECT-TYPE
    SYNTAX      Integer32 (0..7)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "One of the possible user priority values for a frame."
    REFERENCE   "10.7.17"
    ::= { secyIfTCEntry 1 }

--secyIfTCTrafficClass
secyIfTCTrafficClass    OBJECT-TYPE
    SYNTAX      Integer32 (0..7)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
       "The Traffic Class for this SecY and user priority, as
        transmitted in the four most significant bits of the Port
        Identifier component of the SCI of protected frames."
    REFERENCE   "10.7.17"
    DEFVAL { 0 }
    ::= { secyIfTCEntry 2 }

-- ============================================================================
--secyIfAPTable
secyIfAPTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyIfAPEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
       "The Access Priority Table for each SecY in this system.
        The configured value of writable objects in each table entry
        MUST be stored in persistent memory and remain unchanged across
        a re-initialization of the system's management entity."
    REFERENCE   "10.5.1, 10.7.17, Table 13-1"
    ::= { secyMgmtMIBObjects 9 }

--secyIfAPEntry
secyIfAPEntry   OBJECT-TYPE
    SYNTAX      SecyIfAPEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A table entry for a given SecY, selecting the access priority
                 and the drop_eligible parameter value used for a given user
                 priority and drop_eligible parameter value."
    INDEX   { secyIfInterfaceIndex, secyIfAPUserPCP }
    ::= { secyIfAPTable 1 }

--SecyIfAPEntry
SecyIfAPEntry ::= SEQUENCE {
    secyIfAPUserPCP             Integer32,
    secyIfAPAccessPCP           Integer32
}

--secyIfAPUserPCP
secyIfAPUserPCP OBJECT-TYPE
    SYNTAX      Integer32 (0..15)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "The user priority (3 more significant bits) and drop_eligible
                 parameter (least significant bit) values."
    REFERENCE   "10.5, 10.7.17"
    ::= { secyIfAPEntry 1 }

```


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```
--secyIfAPAccessPCP
secyIfAPAccessPCP    OBJECT-TYPE
    SYNTAX      Integer32 (0..15)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "The access priority (3 more significant bits) and drop_eligible
                 parameter (least significant bit) values."
    REFERENCE   "10.5, 10.7.17"
    ::= { secyIfAPEntry 2 }
-- ============================================================================
--secyTxSCTable
secyTxSCTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyTxSCEntry
    MAX-ACCESS  not-accessible
    STATUS      current  -- ??
    DESCRIPTION "A transmit SC management table for systems not supporting
                 traffic class SC's, with an entry for each SecY."
    REFERENCE   "10.7.17, 10.7.20, Table 13-2"
    ::= { secyMgmtMIBObjects 2 }

--secyTxSCEntry
secyTxSCEntry    OBJECT-TYPE
    SYNTAX      SecyTxSCEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "An entry with transmit SC information for a SecY."
    INDEX { secyIfInterfaceIndex }
    ::= { secyTxSCTable 1 }

--SecyTxSCEntry
SecyTxSCEntry ::= SEQUENCE {
    secyTxSCI              SecySCI,
    secyTxSCState          INTEGER,
    secyTxSCEncodingSA     RowPointer,
    secyTxSCEncipheringSA  RowPointer, -- deprecated
    secyTxSCCreatedTime    TimeStamp,
    secyTxSCStartedTime    TimeStamp,
    secyTxSCStoppedTime    TimeStamp
}
--secyTxSCI
secyTxSCI       OBJECT-TYPE
    SYNTAX      SecySCI
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The SCI for the SecY's transmit SC."
    REFERENCE   "7.1.2, 10.7.1"
    ::= { secyTxSCEntry 1 }

--secyTxSCState
secyTxSCState   OBJECT-TYPE
    SYNTAX      INTEGER { inUse(1), notInUse(2)}
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The transmitting state of the SecY's transmit SC."
    REFERENCE   "10.7.21 transmitting, 10.7.23"
    ::= { secyTxSCEntry 2 }

--secyTxSCEncodingSA
secyTxSCEncodingSA    OBJECT-TYPE
    SYNTAX      RowPointer
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
    "The SA currently used to encode the SecTAG for frames awaiting transmission.
     The row pointer points to an entry in the secyTxSATable. If no such
     information is available, the value shall be the OBJECT IDENTIFIER { 0 0 }."
    REFERENCE   "10.5.1, 10.7.21"
    ::= { secyTxSCEntry 3 }

```


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```
--secyTxSCEncipheringSA
secyTxSCEncipheringSA    OBJECT-TYPE
    SYNTAX      RowPointer
    MAX-ACCESS  read-only
    STATUS      deprecated
    DESCRIPTION
    "The SA currently used to encipher frames for transmission.
     The row pointer points to an entry in the secyTxSATable. If no such
     information is available, the value shall be the OBJECT IDENTIFIER { 0 0 }."
    REFERENCE   "10.5.4"
    ::= { secyTxSCEntry 4 }

--secyTxSCCreatedTime
secyTxSCCreatedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this transmitting SC was created."
    REFERENCE   "10.7.21"
    ::= { secyTxSCEntry 5 }

--secyTxSCStartedTime
secyTxSCStartedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this SC last started transmitting."
    REFERENCE   "10.7.21"
    ::= { secyTxSCEntry 6 }

--secyTxSCStoppedTime
secyTxSCStoppedTime    OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this SC last stopped transmitting."
    REFERENCE   "10.7.21"
    ::= { secyTxSCEntry 7 }

-- ============================================================================
--secyTxSATable
secyTxSATable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyTxSAEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A transmit SA management table for a system with no SecYs
                 capable of supporting traffic class transmit SCs."
    REFERENCE   "10.7.22, Table 13-2"
    ::= { secyMgmtMIBObjects 3 }

--secyTxSAEntry
secyTxSAEntry    OBJECT-TYPE
    SYNTAX      SecyTxSAEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "An entry for a transmit SA."
    INDEX   { secyIfInterfaceIndex, secyTxSA }
    ::= { secyTxSATable 1 }

--SecyTxSAEntry
SecyTxSAEntry ::= SEQUENCE {
    secyTxSA                    SecyAN,
    secyTxSAState               INTEGER,
    secyTxSANextPN              Unsigned32,
    secyTxSAConfidentiality     TruthValue,
    secyTxSASAKUnchanged        TruthValue, -- deprecated
    secyTxSACreatedTime         TimeStamp,
    secyTxSAStartedTime         TimeStamp,
    secyTxSAStoppedTime         TimeStamp
}

```


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```
--secyTxSA
secyTxSA        OBJECT-TYPE
    SYNTAX      SecyAN
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "The association number (AN) for this transmit SA."
    REFERENCE   "10.7.22"
    ::= { secyTxSAEntry 1 }

--secyTxSAState
secyTxSAState   OBJECT-TYPE
    SYNTAX      INTEGER {inUse(1), notInUse(2)}
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The transmit SAs current status: inUse(1), notInUse(2)."
    REFERENCE   "10.7.22"
    ::= { secyTxSAEntry 2 }

--secyTxSANextPN
secyTxSANextPN  OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The next packet number (PN) for this SA."
    REFERENCE   "10.5, 10.7.23"
    ::= { secyTxSAEntry 3 }

--secyTxSAConfidentiality
secyTxSAConfidentiality   OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "True if the SA provides confidentiality as well as
                 integrity for transmitted frames."
    REFERENCE   "10.7.23"
    ::= { secyTxSAEntry 4 }

--secyTxSASAKUnchanged
secyTxSASAKUnchanged      OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      deprecated
    DESCRIPTION "A reference to an SAK that is unchanged for the life
                 of the transmitting SA."
    REFERENCE   "10.7.22"
    ::= { secyTxSAEntry 5 }

--secyTxSACreatedTime
secyTxSACreatedTime       OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this transmit SA was created."
    REFERENCE   "10.7.23"
    ::= { secyTxSAEntry 6 }

--secyTxSAStartedTime
secyTxSAStartedTime       OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this SA last started transmitting."
    REFERENCE   "10.7.23"
    ::= { secyTxSAEntry 7 }
```


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```
--secyTxSAStoppedTime
secyTxSAStoppedTime       OBJECT-TYPE
    SYNTAX      TimeStamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The system time when this SA last stopped transmitting."
    REFERENCE   "10.7.23"
    ::= { secyTxSAEntry 8 }

-- ============================================================================
--secyStatsMIBObjects
-- secyTSCStatsTable
-- secyRxSCStatsTable
-- secyRxSAStatsTable
-- secyStatsTable
--The following are historic following approval of IEEE Std 802.1AEcg-2017,
--even if their STATUS remains 'current'. They do not include any objects
--that are part of a current conformance OBJECT-GROUP, and lack traffic
--class transmit SC and XPN support:
-- secyTxSCStatsTable, secyTxSAStatsTable
-- ============================================================================
--secyTSCStatsTable
secyTSCStatsTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyTSCStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A table of statistics for each SecY's transmit SCs."
    REFERENCE   "10.7.18, 10.7.19, Figure 10-3"
    ::= { secyStatsMIBObjects 12 }

--secyTSCStatsEntry
secyTSCStatsEntry    OBJECT-TYPE
    SYNTAX      SecyTSCStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A entry containing counts for a transmit SC, since SA counters
         are reset when the SA's AN  is reused these are a summation for
         all current and prior SAs belonging to the SC."
    AUGMENTS { secyTSCEntry }
    ::= { secyTSCStatsTable 1 }

--SecyTSCStatsEntry
SecyTSCStatsEntry ::= SEQUENCE {
    secyTSCStatsProtectedPkts      Counter64,
    secyTSCStatsEncryptedPkts      Counter64
}

--secyTSCStatsProtectedPkts
secyTSCStatsProtectedPkts    OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The number of integrity protected but not encrypted packets
         for this transmit SC."
    REFERENCE   "10.7.18, Figure 10-3"
    ::= { secyTSCStatsEntry 1 }

```


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```
--secyTSCStatsEncryptedPkts
secyTSCStatsEncryptedPkts    OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The number of integrity protected and encrypted packets for
         this transmit SC."
    REFERENCE   "10.7.18, Figure 10-3"
    ::= { secyTSCStatsEntry 2 }

-- ============================================================================
--secyRxSAStatsTable
secyRxSAStatsTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyRxSAStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      deprecated
    DESCRIPTION
        "A table that contains the statistics objects for each
         receiving SA in the MAC security entity."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyStatsMIBObjects 3 }

--secyRxSAStatsEntry
secyRxSAStatsEntry    OBJECT-TYPE
    SYNTAX      SecyRxSAStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION
       "An entry with statistics for a receive SA.  The AN that
        identifies an SA (for a given SC) and this corresponding entry
        can be reused. When creating the SA and before (re)using the
        entry, the SA counters are (re)set to 0. When the SA is stopped
        (secyRxSA notInuse) the counters stop incrementing.

        The secyRxSATable timestamps SA creation, start, and stop."
    AUGMENTS { secyRxSAEntry }
    ::= { secyRxSAStatsTable 1 }

--SecyRxSAStatsEntry
SecyRxSAStatsEntry ::= SEQUENCE {
    secyRxSAStatsUnusedSAPkts    Counter32, -- deprecated
    secyRxSAStatsNoUsingSAPkts   Counter32, -- deprecated
    secyRxSAStatsNotValidPkts    Counter32, -- deprecated
    secyRxSAStatsInvalidPkts     Counter32, -- deprecated
    secyRxSAStatsOKPkts          Counter32  -- deprecated
}

--secyRxSAStatsUnusedSAPkts
secyRxSAStatsUnusedSAPkts    OBJECT-TYPE
    SYNTAX      Counter32
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      deprecated
    DESCRIPTION
        "For this SA which is not currently in use, the number of
         received, unencrypted, packets with secyValidateFrames
         not in the strict mode."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyRxSAStatsEntry 1 }

```


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```
--secyRxSAStatsNoUsingSAPkts
secyRxSAStatsNoUsingSAPkts   OBJECT-TYPE
    SYNTAX      Counter32
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      deprecated
    DESCRIPTION
        "For this SA which is not currently in use, the number of
         received packets that have been discarded, and have
         either the packets encrypted or secyValidateFrames set to
         strict mode."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyRxSAStatsEntry 4 }

--secyRxSAStatsNotValidPkts
secyRxSAStatsNotValidPkts     OBJECT-TYPE
    SYNTAX      Counter32
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      deprecated
    DESCRIPTION
        "For this SA, the number discarded packets with the
         condition that the packets are not valid and one of the
         following conditions are true: either secyValidateFrames in
         strict mode or the packets encrypted."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyRxSAStatsEntry 13 }

--secyRxSAStatsInvalidPkts
secyRxSAStatsInvalidPkts     OBJECT-TYPE
    SYNTAX      Counter32
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      deprecated
    DESCRIPTION
        "For this SA, the number of packets with the condition
         that the packets are not valid and secyValidateFrames is in
         check mode."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyRxSAStatsEntry 16 }

--secyRxSAStatsOKPkts
secyRxSAStatsOKPkts          OBJECT-TYPE
    SYNTAX      Counter32
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      deprecated
    DESCRIPTION
        "For this SA, the number of validated packets."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyRxSAStatsEntry 25 }

-- ============================================================================
--secyRxSCStatsTable
secyRxSCStatsTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyRxSCStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A table of statistics for each receive SC for each of
                 the system's SecYs."
    REFERENCE   "10.7.9, 10.7.9, Figure 10-4"
    ::= { secyStatsMIBObjects 4 }

```


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```
--secyRxSCStatsEntry
secyRxSCStatsEntry    OBJECT-TYPE
    SYNTAX      SecyRxSCStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
       "An entry containing counts for a receive SC. SA counters are
        reset when the SA's AN  is reused, so these SC counts are a
        summation for all current and prior SAs belonging to the SC."
    AUGMENTS { secyRxSCEntry }
    ::= { secyRxSCStatsTable 1 }

--SecyRxSCStatsEntry
SecyRxSCStatsEntry ::= SEQUENCE {
    secyRxSCStatsUnusedSAPkts         Counter64, -- deprecated
    secyRxSCStatsNoUsingSAPkts        Counter64, -- deprecated
    secyRxSCStatsLatePkts             Counter64,
    secyRxSCStatsNotValidPkts         Counter64,
    secyRxSCStatsInvalidPkts          Counter64,
    secyRxSCStatsDelayedPkts          Counter64,
    secyRxSCStatsUncheckedPkts        Counter64,
    secyRxSCStatsOKPkts               Counter64,
    secyRxSCStatsOctetsValidated      Counter64, -- deprecated
    secyRxSCStatsOctetsDecrypted      Counter64  -- deprecated
}

--secyRxSCStatsUnusedSAPkts
secyRxSCStatsUnusedSAPkts    OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION "The sum of secyRxSAStatsUnusedSAPkts counts for all
                 current and prior SAs belonging to this SC."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyRxSCStatsEntry 1 }

--secyRxSCStatsNoUsingSAPkts
secyRxSCStatsNoUsingSAPkts    OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION "The sum of secyRxSAStatsNoUsingSAPkts counts for all
                 current and prior SAs belonging to this SC."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyRxSCStatsEntry 2 }

--secyRxSCStatsLatePkts
secyRxSCStatsLatePkts        OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Count of packets discarded for this SC, with a received PN
                 lower than the lowest acceptable PN (secyRxSALowestXPN) while
                 secyIfReplayProtectEnable was true."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyRxSCStatsEntry 3 }

--secyRxSCStatsNotValidPkts
secyRxSCStatsNotValidPkts    OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Count of packets discarded for this SC, because validation
                 failed and they were encrypted (unrecoverable) or
                 secyIfvalidateFrames was 'strict'."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyRxSCStatsEntry 4 }
```


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```
--secyRxSCStatsInvalidPkts
secyRxSCStatsInvalidPkts     OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Count of packets received for this SC, that failed validation
                 but were received unencrypted while secyIfvalidateFrames
                 was 'check'."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyRxSCStatsEntry 5 }

--secyRxSCStatsDelayedPkts
secyRxSCStatsDelayedPkts     OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Count of packets received for this SC, with PN lower than
                 the lowest acceptable PN (secyRxSALowestXPN) while
                 secyIfReplayProtectEnable was false."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyRxSCStatsEntry 6 }

--secyRxSCStatsUncheckedPkts
secyRxSCStatsUncheckedPkts   OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Count of packets received for this SC while
                 secyValidateFrames was 'disabled'."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyRxSCStatsEntry 7 }

--secyRxSCStatsOKPkts
secyRxSCStatsOKPkts          OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Count of packets received for this SC that were
                 successfully validated and within the replay window."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyRxSCStatsEntry 8 }

--secyRxSCStatsOctetsValidated
secyRxSCStatsOctetsValidated     OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Octets"
    MAX-ACCESS  read-only
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION "Count of plaintext octets recovered from packets that were
                 integrity protected but not encrypted."
    REFERENCE   "Deprecated, the secyIsStatsTable has per SecY counts
                 for cryptographic performance management."
    ::= { secyRxSCStatsEntry 9 }

--secyRxSCStatsOctetsDecrypted
secyRxSCStatsOctetsDecrypted    OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Octets"
    MAX-ACCESS  read-only
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION "Count of plaintext octets recovered from packets
                 that were integrity protected and encrypted."
    REFERENCE   "Deprecated, the secyIsStatsTable has per SecY counts
                 for cryptographic performance management."
    ::= { secyRxSCStatsEntry 10 }

```


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```
-- ============================================================================
--secyStatsTable
secyStatsTable  OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A table of statistics for each of the system's SecYs."
    REFERENCE   "10.7.9, 10.7.18, Figure 10-3, 10.5"
    ::= { secyStatsMIBObjects 5 }

--secyStatsEntry
secyStatsEntry  OBJECT-TYPE
    SYNTAX      SecyStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry containing counts for a SecY."
    AUGMENTS { secyIfEntry }
    ::= { secyStatsTable 1 }

--SecyStatsEntry
SecyStatsEntry ::= SEQUENCE {
    secyStatsTxUntaggedPkts     Counter64,
    secyStatsTxTooLongPkts      Counter64,
    secyStatsRxUntaggedPkts     Counter64,
    secyStatsRxNoTagPkts        Counter64,
    secyStatsRxBadTagPkts       Counter64,
    secyStatsRxUnknownSCIPkts   Counter64, -- deprecated
    secyStatsRxNoSCIPkts        Counter64, -- deprecated
    secyStatsRxOverrunPkts      Counter64,
    secyStatsRxNoSAPkts         Counter64, -- 802.1AEcg
    secyStatsRxNoSAErrorPkts    Counter64, -- 802.1AEcg
    secyStatsTxOctetsProtected  Counter64, -- 802.1AEcg
    secyStatsTxOctetsEncrypted  Counter64, -- 802.1AEcg
    secyStatsRxOctetsValidated  Counter64, -- 802.1AEcg
    secyStatsRxOctetsDecrypted  Counter64  -- 802.1AEcg
}

--secyStatsTxUntaggedPkts
secyStatsTxUntaggedPkts    OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of packets transmitted without a SecTAG
                 because secyProtectFramesEnable is configured false."
    REFERENCE   "10.7.18, Figure 10-3"
    ::= { secyStatsEntry 1 }

--secyStatsTxTooLongPkts
secyStatsTxTooLongPkts    OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of transmit packets discarded because their
                 length is greater than the ifMtu of the Common Port."
    REFERENCE   "10.7.18, Figure 10-3"
    ::= { secyStatsEntry 2 }

--secyStatsRxUntaggedPkts
secyStatsRxUntaggedPkts    OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of packets without the MACsec tag (SecTAG)
                 received while secyValidateFrames was not 'strict'."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyStatsEntry 3 }

```


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```
--secyStatsRxNoTagPkts
secyStatsRxNoTagPkts       OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of received packets  without a SecTAG
                 discarded because secyValidateFrames was 'strict'."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyStatsEntry 4 }

--secyStatsRxBadTagPkts
secyStatsRxBadTagPkts      OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of received packets discarded with an
                 invalid SecTAG, zero value PN, or invalid ICV."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyStatsEntry 5 }

--secyStatsRxUnknownSCIPkts
secyStatsRxUnknownSCIPkts  OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION "The number of received packets with an unknown SCI."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyStatsEntry 6 }

--secyStatsRxNoSCIPkts
secyStatsRxNoSCIPkts       OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION "The number of discarded packets with an unknown SCI."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyStatsEntry 7 }

--secyStatsRxOverrunPkts
secyStatsRxOverrunPkts       OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of packets discarded because they exceeded
                 cryptographic performance capabilities."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyStatsEntry 8 }

--secyStatsRxNoSAPkts
secyStatsRxNoSAPkts       OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of received packets with an unknown SCI
                 or for an unused SA."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyStatsEntry 9 }

```


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```
--secyStatsRxNoSAErrorPkts
secyStatsRxNoSAErrorPkts       OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of packets discarded because the received
                 SCI is unknown or the SA is not in use."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyStatsEntry 10 }

--secyStatsTxOctetsProtected
secyStatsTxOctetsProtected       OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Octets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of plain text octets integrity protected
                 but not encrypted in transmitted frames."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyStatsEntry 11 }

--secyStatsTxOctetsEncrypted
secyStatsTxOctetsEncrypted       OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Octets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of plain text octets integrity protected
                 and encrypted in transmitted frames."
    REFERENCE   "10.7.9, Figure 10-4"
    ::= { secyStatsEntry 12 }

--secyStatsRxOctetsValidated
secyStatsRxOctetsValidated       OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Octets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of plaintext octets recovered from packets
                 that were integrity protected but not encrypted."
    REFERENCE   "10.6.3, Figure 10-4"
    ::= { secyStatsEntry 13 }

--secyStatsRxOctetsDecrypted
secyStatsRxOctetsDecrypted       OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Octets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of plaintext octets recovered from packets
                 that were integrity protected and encrypted."
    REFERENCE   "10.6.3, Figure 10-4"
    ::= { secyStatsEntry 14 }
-- ============================================================================
--secyTxSCStatsTable
secyTxSCStatsTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyTxSCStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A transmit SC statistics table for systems without traffic
                 class SC support, with an entry for each SecY."
    REFERENCE   "10.7.18, 10.7.19, Figure 10-3"
    ::= { secyStatsMIBObjects 2 }

```


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```
--secyTxSCStatsEntry
secyTxSCStatsEntry    OBJECT-TYPE
    SYNTAX      SecyTxSCStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A transmit SC statistics table entry (for systems without
                 traffic class SC support) with cumulative counts for a given
                 SecY's current and prior SAs."
    AUGMENTS { secyTxSCEntry }
    ::= { secyTxSCStatsTable 1 }

--SecyTxSCStatsEntry
SecyTxSCStatsEntry ::= SEQUENCE {
    secyTxSCStatsProtectedPkts      Counter64,
    secyTxSCStatsEncryptedPkts      Counter64,
    secyTxSCStatsOctetsProtected    Counter64, -- deprecated
    secyTxSCStatsOctetsEncrypted    Counter64  -- deprecated
}
--secyTxSCStatsProtectedPkts
secyTxSCStatsProtectedPkts    OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Count of integrity protected but unencrypted packets for this
                 transmit SC."
    REFERENCE   "10.7.18, Figure 10-3"
    ::= { secyTxSCStatsEntry 1 }

--secyTxSCStatsEncryptedPkts
secyTxSCStatsEncryptedPkts    OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Count of integrity protected and encrypted packets for this
                 transmit SC."
    REFERENCE   "10.7.18, Figure 10-3"
    ::= { secyTxSCStatsEntry 4 }

--secyTxSCStatsOctetsProtected
secyTxSCStatsOctetsProtected    OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Octets"
    MAX-ACCESS  read-only
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION "Count of integrity protected plain text octets that are
integrity
                 protected but not encrypted for this transmit SC."
    REFERENCE   "10.7.19, Figure 10-3"
    ::= { secyTxSCStatsEntry 10 }

--secyTxSCStatsOctetsEncrypted
secyTxSCStatsOctetsEncrypted    OBJECT-TYPE
    SYNTAX      Counter64
    UNITS       "Octets"
    MAX-ACCESS  read-only
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION
        "The number of plain text octets that are integrity protected
         and encrypted on the transmit SC."
    REFERENCE   "10.7.19, Figure 10-3"
    ::= { secyTxSCStatsEntry 11 }

```


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```
-- ============================================================================
--secyTxSAStatsTable
secyTxSAStatsTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF SecyTxSAStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      deprecated
    DESCRIPTION "A table of statistics for each transmit SA for each of
                 the system's SecYs."
    REFERENCE   "10.7.18, Figure 10-4"
    ::= { secyStatsMIBObjects 1 }

--secyTxSAStatsEntry
secyTxSAStatsEntry    OBJECT-TYPE
    SYNTAX      SecyTxSAStatsEntry
    MAX-ACCESS  not-accessible
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION
       "An entry with statistics for a transmit SA.  The AN that
        identifies an SA (for a given SC) and this corresponding entry
        can be reused. When creating the SA and before (re)using the
        entry, the SA counters are (re)set to 0. When the SA is stopped
        (secyTxSA notInuse) the counters stop incrementing.

        The secyTxSATable timestamps SA creation, start, and stop."
    AUGMENTS { secyTxSAEntry }
    ::= { secyTxSAStatsTable 1 }

--SecyTxSAStatsEntry
SecyTxSAStatsEntry ::= SEQUENCE {
    secyTxSAStatsProtectedPkts    Counter32,
    secyTxSAStatsEncryptedPkts    Counter32
}

--secyTxSAStatsProtectedPkts
secyTxSAStatsProtectedPkts    OBJECT-TYPE
    SYNTAX      Counter32
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION "Count of integrity protected but unencrypted packets for this
                 transmit SA. Zero if secyTxSAConfidentiality is True, and one
                 less than secyTxSANextPN otherwise."
    REFERENCE   "10.7.18, Figure 10-4"
    ::= { secyTxSAStatsEntry 1 }

--secyTxSAStatsEncryptedPkts
secyTxSAStatsEncryptedPkts    OBJECT-TYPE
    SYNTAX      Counter32
    UNITS       "Packets"
    MAX-ACCESS  read-only
    STATUS      deprecated -- 802.1AEcg
    DESCRIPTION "Count of integrity protected and encrypted packets for this
                 transmit SA. Zero if secyTxSAConfidentiality is False, and
                 one less than secyTxSANextPN otherwise."
    REFERENCE   "10.7.18, Figure 10-4"
    ::= { secyTxSAStatsEntry 2 }

```


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```
-- ============================================================================
--secyMIBCompliances
--secyMIBTcCompliance
secyMIBTcCompliance MODULE-COMPLIANCE
    STATUS current -- 802.1AEcg
    DESCRIPTION
    "The compliance statement for an IEEE8021-SECY-MIB supporting
     traffic class transmit SCs, added by IEEE 802.1AEcg."
    MODULE IF-MIB
        MANDATORY-GROUPS {
            ifCounterDiscontinuityGroup
            }
    MODULE  -- this module
        MANDATORY-GROUPS {
            secyIfGroup,
            secyIfCipherGroup,
            secyIfTCGroup,
            secyIfAPGroup,
            secyTSCGroup,
            secyTSAGroup,
            secyRSCGroup,
            secyRSAGroup,
            secyCipherInfoGroup,
            secyCipherStatsGroup,
            secyTSCStatsGroup,
            secyRSCStatsGroup,
            secyIfStatsGroup
        }
        OBJECT secyIfCurrentCipherSuite
            MIN-ACCESS    read-only
            DESCRIPTION "should be read-only, use the secyIfCipherTable
                         to control ciper suite use."
        OBJECT    secyCipherSuiteId
            MIN-ACCESS    read-only
            DESCRIPTION "read-create not required, may be read-only."
        OBJECT    secyCipherSuiteName
            MIN-ACCESS    read-only
            DESCRIPTION "read-create not required, should be read-only."
        OBJECT    secyCipherSuiteCapability
            MIN-ACCESS    read-only
            DESCRIPTION "read-create not required, should be read-only."
        OBJECT    secyCipherSuiteDataLengthChange
            MIN-ACCESS    read-only
            DESCRIPTION "read-create not required, should be read-only."
        OBJECT    secyCipherSuiteICVLength
            MIN-ACCESS    read-only
            DESCRIPTION "read-create not required, should be read-only."
    ::= { secyMIBCompliances 2 }
```


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```
-- ============================================================================
--secyMIBCompliance
secyMIBCompliance MODULE-COMPLIANCE
    STATUS deprecated -- 802.1AEcg
    DESCRIPTION
    "The compliance statement for the IEEE8021-SECY-MIB as specified in
     IEEE Std 802.1AE-2006."
    MODULE  -- this module
        MANDATORY-GROUPS {
            secyIfCtrlGroup,
            secyTxSCGroup,
            secyTxSAGroup,
            secyRxSCGroup,
            secyRxSAGroup,
            secyCipherSuiteGroup,
            secyTxSAStatsGroup,
            secyTxSCStatsGroup,
            secyRxSAStatsGroup,
            secyRxSCStatsGroup,
            secyStatsGroup
        }
        OBJECT secyIfCurrentCipherSuite
            MIN-ACCESS    read-only
            DESCRIPTION "write access not required, may be read-only."
        OBJECT    secyCipherSuiteId
            MIN-ACCESS    read-only
            DESCRIPTION "read-create not required, may be read-only."
        OBJECT    secyCipherSuiteName
            MIN-ACCESS    read-only
            DESCRIPTION "read-create not required, may be read-only."
        OBJECT    secyCipherSuiteCapability
            MIN-ACCESS    read-only
            DESCRIPTION "read-create not required, may be read-only."
        OBJECT    secyCipherSuiteProtection
            MIN-ACCESS    read-only
            DESCRIPTION "read-create not required, may be read-only."
        OBJECT    secyCipherSuiteProtectionOffset
            MIN-ACCESS    read-only
            DESCRIPTION "read-create not required, may be read-only."
        OBJECT    secyCipherSuiteDataLengthChange
            MIN-ACCESS    read-only
            DESCRIPTION "read-create not required, may be read-only."
        OBJECT    secyCipherSuiteICVLength
            MIN-ACCESS    read-only
            DESCRIPTION "read-create not required, may be read-only."
        OBJECT    secyCipherSuiteRowStatus
            MIN-ACCESS    read-only
            DESCRIPTION "read-create not required, may be read-only."
    ::= { secyMIBCompliances 1 }

```


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```
-- ============================================================================
--secyMIBGroups
--secyIfGroup
secyIfGroup    OBJECT-GROUP
    OBJECTS {
        secyIfMaxPeerSCs,
        secyIfRxMaxKeys,
        secyIfTxMaxKeys,
        secyIfProtectFramesEnable,
        secyIfValidateFrames,
        secyIfReplayProtectEnable,
        secyIfReplayProtectWindow,
        secyIfCurrentCipherSuite,
        secyIfAdminPt2PtMAC,
        secyIfOperPt2PtMAC,
        secyIfIncludeSCIEnable,
        secyIfUseESEnable,
        secyIfUseSCBEnable,
        secyIfSCI,                  -- 802.1AEcg
        secyIfIncludingSCI,         -- 802.1AEcg
        secyIfMaxTSCs               -- 802.1AEcg
    }
    STATUS      current --- Updated secyIfCtrlGroup
    DESCRIPTION "SecY service management (secyIfTable objects) for
                 systems supporting traffic class SCs."
    ::= { secyMIBGroups 12 }

--secyIfCtrlGroup
secyIfCtrlGroup    OBJECT-GROUP
    OBJECTS {
        secyIfMaxPeerSCs,
        secyIfRxMaxKeys,
        secyIfTxMaxKeys,
        secyIfProtectFramesEnable,
        secyIfValidateFrames,
        secyIfReplayProtectEnable,
        secyIfReplayProtectWindow,
        secyIfCurrentCipherSuite,
        secyIfAdminPt2PtMAC,
        secyIfOperPt2PtMAC,
        secyIfIncludeSCIEnable,
        secyIfUseESEnable,
        secyIfUseSCBEnable
    }
    STATUS      deprecated
    DESCRIPTION "SecY service management (secyIfTable) objects."
    ::= { secyMIBGroups 1 }

--secyIfTCGroup
secyIfTCGroup    OBJECT-GROUP
    OBJECTS {
        secyIfTCTrafficClass
    }
    STATUS      current --- 802.1AEcg
    DESCRIPTION "Traffic class control (secyIfTCTable)."
    ::= { secyMIBGroups 14 }

--secyIfAPGroup
secyIfAPGroup    OBJECT-GROUP
    OBJECTS {
        secyIfAPAccessPCP
    }
    STATUS      current
    DESCRIPTION "Access Priority Code Point control (secyIfAPTable)."
    ::= { secyMIBGroups 15 }


```


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```
--secyTSCGroup
secyTSCGroup    OBJECT-GROUP
    OBJECTS {
        secyTSCState,
        secyTSCEncodingSA,
        secyTSCCreatedTime,
        secyTSCStartedTime,
        secyTSCStoppedTime
    }
    STATUS      current --- Updated secyTxSCGroup
    DESCRIPTION "Transmit SC management (secyTSCTable objects) for
                 systems supporting traffic class SCs."
    ::= { secyMIBGroups 16 }

--secyTxSCGroup
secyTxSCGroup    OBJECT-GROUP
    OBJECTS {
        secyTxSCI,
        secyTxSCState,
        secyTxSCEncodingSA,
        secyTxSCEncipheringSA,
        secyTxSCCreatedTime,
        secyTxSCStartedTime,
        secyTxSCStoppedTime
    }
    STATUS      deprecated
    DESCRIPTION "Transmit SC management objects (for systems without
                 traffic class SC capabilities)."
    ::= { secyMIBGroups 2 }

--secyTSAGroup
secyTSAGroup    OBJECT-GROUP
    OBJECTS {
        secyTSAState,
        secyTSANextXPN,
        secyTSAConfidentiality,
        secyTSAKeyIdentifier,
        secyTSASSCI,
        secyTSACreatedTime,
        secyTSAStartedTime,
        secyTSAStoppedTime
    }
    STATUS      current --- 802.1AEcg, updates secyTxSAGroup
    DESCRIPTION "Transmit SA management (secyTSATable objects) for
                 systems supporting traffic class SCs."
    ::= { secyMIBGroups 17 }

--secyTxSAGroup
secyTxSAGroup    OBJECT-GROUP
    OBJECTS {
        secyTxSAState,
        secyTxSANextPN,
        secyTxSAConfidentiality,
        secyTxSASAKUnchanged,
        secyTxSACreatedTime,
        secyTxSAStartedTime,
        secyTxSAStoppedTime
    }
    STATUS      deprecated
    DESCRIPTION "Transmit SA management objects (for systems without
                 traffic class SC capabilities)."
    ::= { secyMIBGroups 3 }

```


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```
--secyRSCGroup
secyRSCGroup    OBJECT-GROUP
    OBJECTS {
        secyRxSCState,
        secyRxSCCreatedTime,
        secyRxSCStartedTime,
        secyRxSCStoppedTime
    }
    STATUS      current --- 802.1AEcg, updates secyRxSCGroup
    DESCRIPTION "Receive SC management (secyRxSCTable objects)."
    ::= { secyMIBGroups 18 }

--secyRxSCGroup
secyRxSCGroup    OBJECT-GROUP
    OBJECTS {
        secyRxSCState,
        secyRxSCCurrentSA,
        secyRxSCCreatedTime,
        secyRxSCStartedTime,
        secyRxSCStoppedTime
    }
    STATUS      deprecated
    DESCRIPTION "Receive SC management objects."
    ::= { secyMIBGroups 4 }

--secyRSAGroup
secyRSAGroup    OBJECT-GROUP
    OBJECTS {
        secyRxSAState,
        secyRxSANextXPN,
        secyRxSALowestXPN,
        secyRxSAKeyIdentifier,
        secyRxSASSCI,
        secyRxSACreatedTime,
        secyRxSAStartedTime,
        secyRxSAStoppedTime
    }
    STATUS      current --- Updated secyRxSAGroup
    DESCRIPTION "Receive SA (secyRxSATable objects)."
    ::= { secyMIBGroups 19 }

--secyRxSAGroup
secyRxSAGroup    OBJECT-GROUP
    OBJECTS {
        secyRxSAState,
        secyRxSANextPN,
        secyRxSASAKUnchanged,
        secyRxSACreatedTime,
        secyRxSAStartedTime,
        secyRxSAStoppedTime
    }
    STATUS      deprecated
    DESCRIPTION "Receive SA management objects."
    ::= { secyMIBGroups 5 }

-- Cipher information, use, and statistics MIB Groups

--secyCipherInfoGroup
secyCipherInfoGroup    OBJECT-GROUP
    OBJECTS {
        secyCipherSuiteId,
        secyCipherSuiteName,
        secyCipherSuiteCapability,
        secyCipherSuiteDataLengthChange,
        secyCipherSuiteICVLength
    }
    STATUS      current --- Updated secyCipherSuiteGroup
    DESCRIPTION "Cipher Suite implementation information
                 (secyCipherSuiteTable objects)."
    ::= { secyMIBGroups 21 }

```


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```
--secyCipherSuiteGroup
secyCipherSuiteGroup    OBJECT-GROUP
    OBJECTS {
        secyCipherSuiteId,
        secyCipherSuiteName,
        secyCipherSuiteCapability,
        secyCipherSuiteProtection,
        secyCipherSuiteProtectionOffset,
        secyCipherSuiteDataLengthChange,
        secyCipherSuiteICVLength,
        secyCipherSuiteRowStatus
    }
    STATUS      deprecated
    DESCRIPTION "Cipher Suite information objects."
    ::= { secyMIBGroups 6 }

--secyIfCipherGroup
secyIfCipherGroup    OBJECT-GROUP
    OBJECTS {
        secyIfCipherImplemented,
        secyIfCipherEnableUse,
        secyIfCipherRqConfidentiality
    }
    STATUS      current --- 802.1AEcg
    DESCRIPTION "Cipher Suite use control (secyIfCipherTable  objects)."
    ::= { secyMIBGroups 13 }

--secyCipherStatsGroup
secyCipherStatsGroup    OBJECT-GROUP
    OBJECTS {
        secyStatsTxOctetsProtected,
        secyStatsTxOctetsEncrypted,
        secyStatsRxOctetsValidated,
        secyStatsRxOctetsDecrypted
    }
    STATUS      current
    DESCRIPTION
        "Cipher Suite performance statistics (from secyStatsTable)."
    ::= { secyMIBGroups 24 }

--secyTxSAStatsGroup
secyTxSAStatsGroup    OBJECT-GROUP
    OBJECTS {
        secyTxSAStatsProtectedPkts,
        secyTxSAStatsEncryptedPkts
    }
    STATUS      deprecated
    DESCRIPTION "Transmit SA statistics objects."
    ::= { secyMIBGroups 7 }

--secyRxSAStatsGroup
secyRxSAStatsGroup    OBJECT-GROUP
    OBJECTS {
        secyRxSAStatsUnusedSAPkts,
        secyRxSAStatsNoUsingSAPkts,
        secyRxSAStatsNotValidPkts,
        secyRxSAStatsInvalidPkts,
        secyRxSAStatsOKPkts
    }
    STATUS      deprecated
    DESCRIPTION "Receive SA statistics objects."
    ::= { secyMIBGroups 8 }
```


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```
--secyTSCStatsGroup
secyTSCStatsGroup    OBJECT-GROUP
    OBJECTS {
        secyTSCStatsProtectedPkts,
        secyTSCStatsEncryptedPkts
    }
    STATUS      current --- Updated secyTxSCStatsGroup
    DESCRIPTION "Transmit SC statistics (secyTSCStatsTable objects)."
    ::= { secyMIBGroups 22 }

--secyTxSCStatsGroup
secyTxSCStatsGroup    OBJECT-GROUP
    OBJECTS {
        secyTxSCStatsProtectedPkts,
        secyTxSCStatsEncryptedPkts,
        secyTxSCStatsOctetsProtected,
        secyTxSCStatsOctetsEncrypted
    }
    STATUS      deprecated
    DESCRIPTION "Transmit SC statistics objects."
    ::= { secyMIBGroups 9 }

--secyRSCStatsGroup
secyRSCStatsGroup    OBJECT-GROUP
    OBJECTS {
        secyRxSCStatsLatePkts,
        secyRxSCStatsNotValidPkts,
        secyRxSCStatsInvalidPkts,
        secyRxSCStatsDelayedPkts,
        secyRxSCStatsUncheckedPkts,
        secyRxSCStatsOKPkts
    }
    STATUS      current --- Updated secyRxSCStatsGroup
    DESCRIPTION "Receive SC statistics (secyRxSCStatsTable objects)."
    ::= { secyMIBGroups 23 }

--secyRxSCStatsGroup
secyRxSCStatsGroup    OBJECT-GROUP
    OBJECTS {
        secyRxSCStatsUnusedSAPkts,
        secyRxSCStatsNoUsingSAPkts,
        secyRxSCStatsLatePkts,
        secyRxSCStatsNotValidPkts,
        secyRxSCStatsInvalidPkts,
        secyRxSCStatsDelayedPkts,
        secyRxSCStatsUncheckedPkts,
        secyRxSCStatsOKPkts,
        secyRxSCStatsOctetsValidated,
        secyRxSCStatsOctetsDecrypted
    }
    STATUS      deprecated
    DESCRIPTION
        "Receive SC statistics objects."
    ::= { secyMIBGroups 10 }

--secyIfStatsGroup
secyIfStatsGroup    OBJECT-GROUP
    OBJECTS {
        secyStatsTxUntaggedPkts,
        secyStatsTxTooLongPkts,
        secyStatsRxUntaggedPkts,
        secyStatsRxNoTagPkts,
        secyStatsRxBadTagPkts,
        secyStatsRxNoSAPkts,
        secyStatsRxNoSAErrorPkts,
        secyStatsRxOverrunPkts
    }
    STATUS      current --- 802.1AEcg, updates secyRxSCStatsGroup
    DESCRIPTION
        "SecY statistics (secyStatsTable objects)."
    ::= { secyMIBGroups 20 }

```


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```
--secyStatsGroup
secyStatsGroup    OBJECT-GROUP
    OBJECTS {
        secyStatsTxUntaggedPkts,
        secyStatsTxTooLongPkts,
        secyStatsRxUntaggedPkts,
        secyStatsRxNoTagPkts,
        secyStatsRxBadTagPkts,
        secyStatsRxUnknownSCIPkts,
        secyStatsRxNoSCIPkts,
        secyStatsRxOverrunPkts
    }
    STATUS      deprecated
    DESCRIPTION
        "SecY statistics objects."
    ::= { secyMIBGroups 11 }

END

```


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14. Cipher Suites

A Cipher Suite is an interoperable specification of cryptographic algorithms together with the values of parameters (for example, key size) to be used by those algorithms. Specification of the cryptographic functions required by MACsec in terms of Cipher Suites increases interoperability by providing a clear default and a limited number of alternatives.

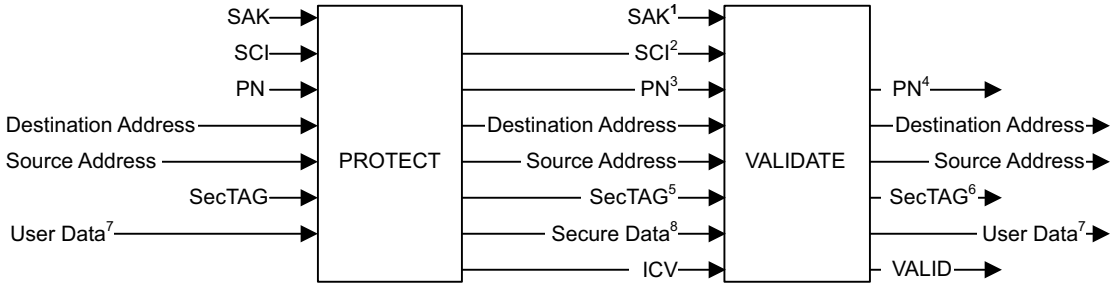
This clause specifies

a) Terms that describe the use of each Cipher Suite by the MAC Security Entity (SecY)
b) Capabilities required of each Cipher Suite
c) Requirements this standard places on Cipher Suite specification
d) Mandatory and optional Cipher Suites for use in conjunction with this standard
e) Criteria for the use of additional Cipher Suites in conjunction with MACsec for implementations for which a claim of conformance to this standard is made.

NOTE—The choice and combination of cryptographic methods is notorious for the introduction of unexpected security exposures. Each Cipher Suite uses an algorithm or combination of algorithms whose interactions have been studied by the professional security community. Each Cipher Suite specification (14.5–14.8) in this clause comprises the necessary combination (e.g., concatenation of named strings) and mapping of parameters and parameter names used in the other clauses of this standard to the parameters and parameter names used by a public established standard that specifies the cryptographic operations.

14.1 Cipher Suite use

A Cipher Suite is initialized with one or more Cipher Suite dependent keys, and then used to protect protocol parameters. Any implementation of the same Cipher Suite, initialized with the same key values, can be used to validate and recover the protected parameters. The protect and validate operations are illustrated in Figure 14-1, and their inputs and outputs specified after the figure.



[1] The SAK to be used on receipt of the frame is identified by the SCI and the AN.
[2] The SCI is extracted from the SCI field of the SecTAG if present. A value conveyed by key agreement (point-to-point only) is used otherwise.
[3] The PN is conveyed in the SecTAG
[4] The validated PN can be used for replay protection.
[5] All the transmitted octets of the SecTAG are protected, including the optional SCI field if present
[6] The validated received SecTAG contains bits of the TCI, and optionally the SCI, these can be used for service multiplexing (11.7).
[7] The length, in octets, of the User Data is conveyed by the User Data parameter, and is protected by Cipher Suite operation.
[8] The length, in octets, of the Secure Data is conveyed by the MACsec frame, unless it is short, when it is conveyed by the SL parameter in the SecTAG TCI

Figure 14-1—Cipher Suite Protect and Validate operations

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```
Protect ( SAK,                                  Validate ( SAK,
          SCI, PN,                                         SCI, PN,
          Destination Address, Source Address,             Destination Address, Source Address,
          SecTAG,                                          SecTAG,
          User Data                                        Secure Data, ICV
        ) Secure Data, ICV                               ) User Data, VALID
```

The SAK (Secure Association Key, 7.1) is the value of the Cipher Suite dependent key(s).

The SCI (Secure Channel Identifier, 7.1.2) is a 64-bit identifier that is globally unique amongst all correctly configured Cipher Suite implementation instances protecting MACsec protocol parameters.

The PN (Packet Number, 8.3) is a number that is never zero, is incremented each time a protect request is made for a given SCI, and is never repeated for an SCI unless the SAK is changed. The size of the PN depends on the Cipher Suite, and is 32 bits unless otherwise specified. Cipher Suites that provide extended packet numbering use a 64-bit PN. Irrespective of the size of the PN, only the least significant 32 bits are conveyed in the SecTAG. If extended packet numbering is used, the most significant 32 bits are recovered for each received frame as specified in 10.6.2.

The Destination Address and Source Address are the MAC addresses of the frame. MAC Addresses are specified as octet strings, using the canonical format specified in IEEE Std 802.

The SecTAG is as specified in Clause 9.

The ICV (integrity check value, 8.3) is a string of octets. VALID is a Boolean parameter. If TRUE the validation was successful.

Given the SAK, SCI, PN, Source Address, Destination Address, SecTAG, and the User Data, the Protect operation returns the Secure Data and ICV.

Given the same SAK, SCI, PN, Source Address, Destination Address, and SecTAG, and the Secure Data and ICV, the Verify operation returns the original User Data and VALID. If any of the parameters were modified, VALID is returned False.

14.2 Cipher Suite capabilities

Any Cipher Suite used with MACsec shall

a) Provide integrity protection for the SCI, PN, Source Address, Destination Address, SecTAG, and from 0 through 2^{16}– 1 octets of User Data on each invocation.

b) Provide integrity and confidentiality (if specified) for at least 2^{32}– 1 invocations, each with a different PN, without requiring a fresh SAK.

c) Given any specific number of octets of User Data, generate a predictable number of octets of Secure Data and ICV.

and may

d) Provide confidentiality protection for all the octets of the User Data.

e) Provide confidentiality protection for all the octets of the User Data following an initial number of octets, as specified in 10.7.25.

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and shall not

f) Generate Secure Data that when added to the number of octets in the ICV contains over 896 octets more than the User Data.

NOTE—A Cipher Suite may introduce additional fields into the Secure Data even if confidentiality is not provided.

g) Modify or constrain the values of the SCI, PN, Source Address, Destination Address, or SecTAG fields, other than as specified in this clause (Clause 14).

h) Require an SAK exceeding 1024 bits long (in total for all keys that compose the SAK).

i) Require different keys for the protect and validate operations.

An implementation of MACsec for which conformance to this standard is claimed includes at least one Cipher Suite that provides integrity without confidentiality, with the Secure Data the same as the User Data, and the ICV comprising 16 octets. This requirement is met by the each of the Cipher Suites specified in this clause (Clause 14).

14.3 Cipher Suite specification

Each Cipher Suite specification shall comprise an interoperable specification of the protection and verification procedures in terms of the parameters specified in 14.1 and shall state

a) Whether confidentiality of the User Data is provided

b) The maximum difference in the lengths of the User Data and Secure Data

c) The length of the ICV

d) The length and properties of the keys required, including assumptions of the scope of uniqueness.

NOTE—While this standard provides definitive specifications of the Cipher Suites that support full conformance, those specifications make the greatest possible use of other public and established standards, and are principally concerned with ensuring unambiguous application of those standards in the context of MACsec.

14.4 Cipher Suite conformance

An implementation of MACsec that claims full conformance to this standard shall implement either or both of the mandatory Cipher Suites in Table 14-1, may implement one or more of the associated Optional Cipher Suites in the table, and shall not implement any other Cipher Suite. Every conformant implementation shall include at least one Cipher Suite that does not encrypt User Data.

Table 14-1 assigns a Cipher Suite reference number for use in protocol identification within a MACsec context, provides a short name for use in this standard, indicates the type of cryptographic algorithm used and the security services provided, and references the clause of this standard that provides the definitive description of the Cipher Suite.

NOTE—In IEEE Std 802.1AE-2006 (the first edition of this standard), the Cipher Suite Identifier for GCM-AES-128 was incorrectly shown as 00-80-02-00-01-00-00-01 in Table 14-1. Prior to the inclusion of GCM-AES-256, GCM-AES-128 was the only conformant Cipher Suite. IEEE Std 802.1X uses a reserved encoding for the Default Cipher Suite rather than the Cipher Suite Identifier to identify GCM-AES-128.

14.4.1 Conformance with Cipher Suite variance

An implementation of MACsec that claims conformance to this standard with Cipher Suite variance, shall implement either or both of the mandatory Cipher Suites in Table 14-1, may implement one or more of the associated optional Cipher Suites in Table 14-1, and may implement alternate Cipher Suites that meet the

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Change Table 14-1 adding a new row as follows (all rows shown):

Table 14-1—MACsec Cipher Suites

Cipher Suite Identifier	Cipher Suite Name	Services provided		Mandatory/Optional	Defining Clause
		Integrity without confidentiality	Integrity and confidentiality		
00-80-C2-00-01-00-00-01	GCM-AES-128	Yes	Yes	Mandatory [a]	14.5
00-80-C2-00-01-00-00-02	GCM-AES-256	Yes	Yes	Optional [b]	14.6
00-80-C2-00-01-00-00-03	GCM-AES-XPN-128	Yes	Yes	Optional [b]	14.7
00-80-C2-00-01-00-00-04	GCM-AES-XPN-256	Yes	Yes	Optional [b]	14.8
00-80-C2-00-01-00-00-10	Ascon-XPN-128	Yes	Yes	Mandatory [a]	14.9

[a] At least one of the Mandatory Cipher Suites is to be implemented by any conformant implementation.
[b] May be implemented if, and only if, GCM-AES-128 (the Default Cipher Suite) is implemented.

requirements of 14.2 and 14.3, and the following guidelines, and shall not implement any other Cipher Suite, or other combination of cryptographic algorithms and parameters.

The use of additional Cipher Suites shall meet the following guidelines:

a) Algorithms chosen have an effective key length of at least 128 bits. In schemes built on block ciphers, the underlying block cipher has a block width of at least 128 bits.

b) If serviced by separate algorithms, the properties of the authentication and confidentiality mechanisms are combinable in accordance with well-established security results. Either the encryption happens before authentication, or the encryption is performed through keystream generation.

c) Either of the following holds true:

   1) The underlying cryptographic cipher is approved by either a national or international standards body or a government agency; or

   2) The following conditions i) through iv) apply:

      i) The Cipher Suite provides message authentication using a message authentication algorithm with a publicly available proof of security against forgery attacks, even in a model where the attacker has the ability to choose messages for the sender.

      ii) If confidentiality is provided, the confidentiality mechanism has a publicly available proof of security in a model where the attacker has the ability to adaptively choose both plaintext and cipher text.

      iii) Mechanisms for confidentiality and message authentication are used in a way that is consistent with their proof of security. For example, if using the Cipher Block Chaining (AES-CBC) mode of operation the IV is performed through keystream generation.

      iv) Mechanisms for confidentiality and message authentication are used in a way that is consistent with their proof of security. For instance, if using the Cipher Block Chaining

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(AES-CBC) mode of operation, the IV is randomly selected with each message, and not sequentially.

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14.5 Default Cipher Suite (GCM-AES-128)

The Default Cipher Suite uses the Galois/Counter Mode of Operation with the AES-128 symmetric block cipher, as specified in this clause by reference to the terms *K, IV, A, P, C, T* used in NIST SP 800-38D.

K is the 128 bit SAK. The 64 most significant bits of the 96-bit *IV* are the octets of the SCI, encoded as a binary number (9.1). The 32 least significant bits of the 96-bit *IV* are the octets of the PN, encoded as a binary number (9.1). *T* is the ICV and is 128 bits long. When the bit-strings *A*, *P*, and *C* are specified in terms of octet strings, earlier octets compose earlier bits, and more significant bits in each octet are earlier.

NOTE 1—IETF RFC 5116 [B11], McGrew [B17], and McGrew and Viega [B18] provide additional information about GCM and its security properties and use.

NOTE 2—The bit strings obtained by transforming MAC Address and data octets using these rules do not correspond to IEEE 802.3 “wire order” for frame transmission.

When the Default Cipher Suite is used for Integrity Protection

— *A* is the Destination MAC Address, Source MAC Address, and the octets of the SecTAG and User Data concatenated in that order.
— *P* is null.
— The Secure Data is the octets of the User Data, without modification.

When the Default Cipher Suite is used for Confidentiality Protection without a confidentiality offset

— *A* is the Destination MAC Address, Source MAC Address, and the octets of the SecTAG concatenated in that order.
— *P* is the octets of the User Data.
— The Secure Data is *C*.

When the Default Cipher Suite is used for Confidentiality Protection with a confidentiality offset

— *A* is the Destination MAC Address, Source MAC Address, and the octets of the SecTAG and the first confidentialityOffset (10.7.27) octets of the User Data (or all the octets of the User Data if that comprises fewer than confidentialityOffset octets) concatenated in that order.
— *P* is the remaining octets (if any) of the User Data.
— The Secure Data is the first confidentialityOffset octets of the User Data concatenated with *C*, in that order (or all the octets of the User Data if that comprises fewer than confidentialityOffset octets).

NOTE 3—IEEE Std 802.1AE-2006 specified the confidentiality offset option to facilitate early MACsec deployment on systems that needed to examine the initial octets of IP version 4 or version 6 frames to decide where to store received frames, before decrypting the frame. The XPN Cipher Suites standardized in IEEE Std 802.1AEbw-2013 do not support confidentiality offsets.

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14.6 GCM-AES-256

GCM-AES-256 uses the Galois/Counter Mode of operation with the AES-256 symmetric block cipher, as specified in this clause by reference to the terms *K, IV, A, P, C, T* used in NIST SP 800-38D.

K is the 256 bit SAK. The 64 most significant bits of the 96-bit *IV* are the octets of the SCI, encoded as a binary number (9.1). The 32 least significant bits of the 96-bit *IV* are the octets of the PN, encoded as a binary number (9.1). *T* is the ICV, and is 128 bits long. When the bit-strings *A*, *P*, and *C* are specified in terms of octet strings, earlier octets compose earlier bits, and more significant bits in each octet are earlier.

NOTE 1—The bit strings obtained by transforming MAC Address and data octets using these rules do not correspond to IEEE 802.3 “wire order” for frame transmission.

When this Cipher Suite is used for Integrity Protection

— *A* is the Destination MAC Address, Source MAC Address, and the octets of the SecTAG and User Data concatenated in that order.

— *P* is null.

— The Secure Data is the octets of the User Data, without modification.

When this Cipher Suite is used for Confidentiality Protection without a confidentiality offset

— *A* is the Destination MAC Address, Source MAC Address, and the octets of the SecTAG concatenated in that order.

— *P* is the octets of the User Data.

— The Secure Data is *C*.

When this Cipher Suite is used for Confidentiality Protection with a confidentiality offset

— *A* is the Destination MAC Address, Source MAC Address, and the octets of the SecTAG and the first confidentialityOffset (10.7.27) octets of the User Data (or all the octets of the User Data if that comprises fewer than confidentialityOffset octets) concatenated in that order.

— *P* is the remaining octets (if any) of the User Data.

— The Secure Data is the first confidentialityOffset octets of the User Data concatenated with *C*, in that order (or all the octets of the User Data if that comprises fewer than confidentialityOffset octets).

NOTE 2—IEEE Std 802.1AE-2006 specified the confidentiality offset option to facilitate early MACsec deployment on systems that needed to examine the initial octets of IP version 4 or version 6 frames to decide where to store received frames, before decrypting the frame. The XPN Cipher Suites standardized in IEEE Std 802.1AEbw-2013 do not support confidentiality offsets.

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14.7 GCM-AES-XPN-128

Each instance of the GCM-AES-XPN-128 Cipher Suite, i.e., the protection and validation capabilities created for a given SAK at the request of the KaY (10.7.26, Figure 10-5) maintains an instance of the following parameter as specified in 10.7.28:

a) Salt, a 96-bit value distributed by key agreement protocol to all members of the CA.

and an instance of the following parameter for each SCI, as supplied by the KaY when an SA that uses the SCI and the given SAK is created (10.7.13, 10.7.21):

b) SSCI, a 32-bit value that is unique for each SCI using a given SAK.

NOTE 1—The maximum number of SSCIs for a given SAK is thus limited by the maximum number of SCIs (equivalently, by the maximum number of simultaneous members in a CA as requirements placed on the KaY (8.2.7) prohibit the use of the same SAK in multiple CAs). A claim of conformance to this standard requires a statement of the maximum number of receive SCs supported (5.3m, A.5, A.12, A.13). The total number of SCIs will be one greater (to include the transmit SC) or two greater [for an EPON OLT supporting an SCB (Clause 12)]. Whether and to what extent the same SAK is used by different SAs (each with a different SCI, and hence a different SSCI for that SAK) depends on the key agreement protocol, and the number of members in a CA will also be ultimately limited by the capabilities of the key agreement protocol. The practical requirements of the port-based network control application (see Clause 7 of IEEE Std 802.1X-2010) are likely to be more limited.

GCM-AES-XPN-128 uses the Galois/Counter Mode of operation with the AES-128 symmetric block cipher, as specified in this clause by reference to the terms *K, IV, A, P, C, T* used in NIST SP 800-38D.

K is the 128-bit SAK. The 32 most significant bits of the 96-bit *IV* are the octets of the SSCI for the SCI, encoded as a binary number (9.1) and exclusive-or'd with the 32 most significant bits of the Salt. The 64 least significant bits of the 96-bit *IV* are the octets of the PN, encoded as a binary number (9.1) and exclusive-or'd with the 64 least significant bits of the Salt. *T* is the ICV, and is 128 bits long. When the bit-strings *A*, *P*, and *C* are specified in terms of octet strings, earlier octets compose earlier bits, and more significant bits in each octet are earlier.

NOTE 2—The bit strings obtained by transforming MAC Address and data octets using these rules do not correspond to IEEE 802.3 "wire order" for frame transmission.

When this Cipher Suite is used for Integrity Protection

— *A* is the Destination MAC Address, Source MAC Address, and the octets of the SecTAG and User Data concatenated in that order.

— *P* is null.

— The Secure Data is the octets of the User Data, without modification.

When this Cipher Suite is used for Confidentiality Protection without a confidentiality offset

— *A* is the Destination MAC Address, Source MAC Address, and the octets of the SecTAG concatenated in that order.

— *P* is the octets of the User Data.

— The Secure Data is *C*.

This Cipher Suite does not provide Confidentiality Protection with a confidentiality offset.

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14.8 GCM-AES-XPN-256

Each instance of the GCM-AES-XPN-256 Cipher Suite, i.e., the protection and validation capabilities created for a given SAK at the request of the KaY (10.7.26, Figure 10-5) maintains an instance of the following parameter as specified in 10.7.28:

a) Salt, a 96-bit value distributed by key agreement protocol to all members of the CA.

and an instance of the following parameter for each SCI, as supplied by the KaY when an SA that uses the SCI and the given SAK is created (10.7.13, 10.7.21):

b) SSCI, a 32-bit value that is unique for each SCI using a given SAK.

NOTE 1—The maximum number of SSCIs for a given SAK is thus limited by the maximum number of SCIs (equivalently, by the maximum number of simultaneous members in a CA as requirements placed on the KaY (8.2.7) prohibit the use of the same SAK in multiple CAs). A claim of conformance to this standard requires a statement of the maximum number of receive SCs supported (5.3m, A.5, A.12, A.13). The total number of SCIs will be one greater (to include the transmit SC) or two greater [for an EPON OLT supporting an SCB (Clause 12)]. Whether and to what extent the same SAK is used by different SAs (each with a different SCI, and hence a different SSCI for that SAK) depends on the key agreement protocol, and the number of members in a CA will also be ultimately limited by the capabilities of the key agreement protocol. The practical requirements of the port-based network control application (see Clause 7 of IEEE Std 802.1X-2010) are likely to be more limited.

GCM-AES-XPN-256 uses the Galois/Counter Mode of operation with the AES-256 symmetric block cipher, as specified in this clause by reference to the terms *K, IV, A, P, C, T* used in NIST SP 800-38D.

K is the 256-bit SAK. The 32 most significant bits of the 96-bit *IV* are the octets of the SSCI for the SCI, encoded as a binary number (9.1) and exclusive-or'd with the 32 most significant bits of the Salt. The 64 least significant bits of the 96-bit *IV* are the octets of the PN, encoded as a binary number (9.1), and exclusive-or'd with the 64 least significant bits of the Salt. *T* is the ICV, and is 128 bits long. When the bit-strings *A*, *P*, and *C* are specified in terms of octet strings, earlier octets compose earlier bits, and more significant bits in each octet are earlier.

NOTE 2—The bit strings obtained by transforming MAC Address and data octets using these rules do not correspond to IEEE 802.3 "wire order" for frame transmission.

When this Cipher Suite is used for Integrity Protection

— *A* is the Destination MAC Address, Source MAC Address, and the octets of the SecTAG and User Data concatenated in that order.

— *P* is null.

— The Secure Data is the octets of the User Data, without modification.

When this Cipher Suite is used for Confidentiality Protection without a confidentiality offset

— *A* is the Destination MAC Address, Source MAC Address, and the octets of the SecTAG concatenated in that order.

— *P* is the octets of the User Data.

— The Secure Data is *C*.

This Cipher Suite does not provide Confidentiality Protection with a confidentiality offset.

14.9 Ascon-XPN-128

The Ascon-XPN-128 Cipher Suite protects each frame using nonce-based authentication encryption with associated data as specified in this clause by reference to the Ascon-AEAD128 algorithm and the associated terms *K*, *N*, *A*, *P*, *C*, *T* specified in NIST SP 800-232.

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K is the 128-bit SAK.

N is the 128-bit nonce, and is constructed as specified in 14.9.3.

T is the 128-bit ICV.

When this Cipher Suite is used for Integrity Protection

— *A* is the Destination MAC Address, Source MAC Address, and the first four octets of the SecTAG and the octets of the User Data concatenated in that order.

— *P* and *C* are null.

— The Secure Data is the octets of the User Data, without modification.

When this Cipher Suite is used for Confidentiality Protection

— *A* is the Destination MAC Address, Source MAC Address, and the first four octets of the SecTAG concatenated in that order.

— *P* is the octets of the User Data.

— The Secure Data is *C*.

This Cipher Suite does not provide Confidentiality Protection with a confidentiality offset.

NOTE—The specification of associated data, A, for this Cipher Suite differs from that for the GCM Cipher Suites.

14.9.1 Packet Number limitation

To provide 128-bit security strength, no more than 2^{54} octets (2^{50} blocks) can be protected by a single key (4.4.1 of NIST SP 800-232). This standard limits the Packet Numbers used with this Cipher Suite to the range 1 through $2^{48} – 1$. The 16 most significant bits of the extended Packet Number's possible 64 bits are not used. MAC Operational (6.4) shall be set False if the value of nextPN for secure frame generation is 2^{48} or greater (10.5.2), and no further frames transmitted using that SA. When this Cipher Suite is in use PendingPNExhaustion (8.2.7) takes the value 0xC000 0000 0000. It is the responsibility of key agreement protocol to further limit PN use to satisfy security strength requirements through timely distribution of fresh SAKs.

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14.9.2 Endian issues

The NIST SP 800-232 specification is little-endian (Appendix A of NIST SP 800-232), whereas the structure and encoding of MACsec Protocol Data Units (MPDUs,) uses big-endian (network byte order) conventions and is Cipher Suite independent. The following provisions of this standard are made explicit to avoid any confusion that might lead to a lack of interoperability between implementations or difficulty in network management. In MPDUs:

— User Data comprises a sequence of octets. When Integrity Protection is provided their transmission order is the same as the order in which they are provided in MSDUs to the transmitting SecY's Controlled Port, and their reception order is the same as the order in which they are delivered to the SecY's transmission port. When presented to or received from the Ascon-AEAD128 protection or validation functions in multi-octet registers or fields, earlier octets are the less bit significant components of those fields.

— Secure Data comprises a sequence of octets. When presented to or received from the Ascon-AEAD128 protection or validation functions in multi-octet registers or fields, earlier octets are the less bit significant components of those fields.

— The MPDU's Destination and Source MAC Addresses and the MAC Address component of an SCI are sequences of octets, with the octet containing the I/G bit and the U/L bit first in transmission order. When presented to or received from the Ascon-AEAD128 protection or validation functions in multi-octet registers or fields, earlier octets are the less bit significant components of those fields. However when MAC Addresses are to be numerically compared they are treated as binary numbers, with earlier octets being more significant as specified in 9.1.

— The two octet Port Identifiers in SCIs are similarly treated by this Cipher Suite as a sequence of two octets. However when they are to be numerically compared they are treated as binary numbers, with earlier octets being more significant as specified in 9.1.

— The bit significance of the fields of the first 128-bit block of 'associated data', *A*, for both Integrity and Confidentiality protection as presented to the Ascon-AEAD128 protection or validation functions is illustrated in Figure 14-2.

NOTE—The fields in this block are treated as a sequence of octets. While the two octets that compose an EtherType are encoded in network byte order, the difference in imputed octet significance arising from their little-endian treatment for Ascon-AEAD128 processing is not material since both the transmitter, protecting the MPDU, and the receiver, validating it, invoke little-endian processing.

— The lowest 32-bits of the Packet Number (PN) are encoded in the MPDU most significant octet first, so the interpretation, by a network administrator, of the MPDU PN field is consistent with 9.1 and does not depend on knowledge of the current Cipher Suite. If the PN value is to be used in a little-endian register for comparison or arithmetical operations, the octet order of the least significant 32 bits needs to be reversed prior to transmission and on reception. The PN contribution to the Cipher Suite nonce, *N*, does not use transmission order. (14.9.3, Table 14-2).

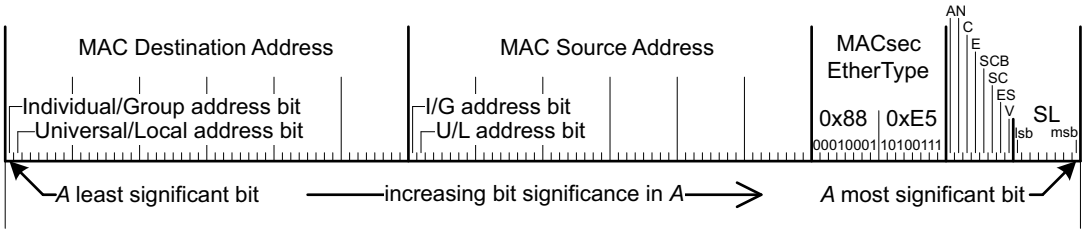


Figure 14-2—Initial MPDU associated data Ascon-AEAD128 block

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14.9.3 Nonce construction

The Cipher Suite nonce, *N*, is constructed from the protected frame's extended Packet Number and the transmitter's SCI (as encoded in the SecTAG, if present), bit-wise exclusive or'd with a 128-bit Salt (10.7.28) distributed by key agreement protocol to all members of the CA.

NOTE 1—The Salt is not a secret nonce mask as described in 4.2.2 of NIST SP 800-232, however its use does ensure that the nonce is not repeated for up to 2^{16} successive Key Numbers for SAKs distributed by a given Key Server instance (more if no CA participant uses the more significant bits of the Port Identifier as part of its SCI). The Salt also allows fuller use of the nonce space than might otherwise occur, prior to fresh SAK distribution, for lower data rates.

Figure 14-3 shows the PN and SCI contributions to the nonce, and the 128-bit Salt that to be used with MKA unless otherwise specified by IEEE Std 802.1X (10.7.28).

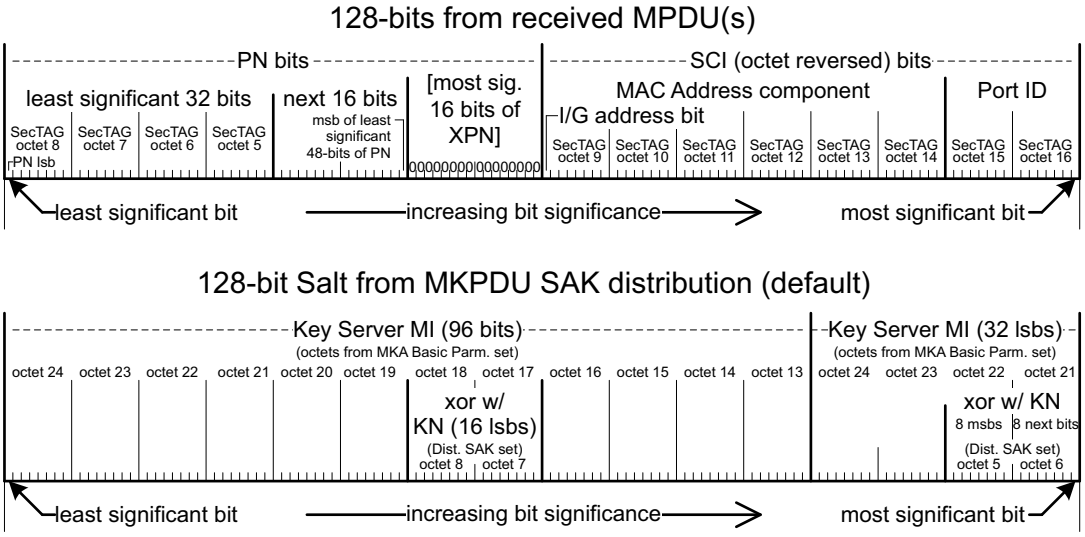


Figure 14-3—Ascon-XPN-128 Nonce construction

The 128 bits contributed to the nonce by the extended Packet Number and the transmitter's SCI are shown in the upper part of Figure 14-3, and are (reading from right to left, most to least significant) as follows:

— The 8 most significant bits are the 8 least significant bits of the SCI's Port Identifier.

— The next 8 significant bits are the 8 most significant bits of the SCI's Port Identifier.

— The next 6 octets comprise the SCI's MAC Address, with the most significant bits being those of the least significant octet of the SCI when that is treated as a binary number as specified in 9.1.

— The next 16 bits are zero, and correspond to the 16 most significant bits of the extended Packet Number, which are restricted to zero for this Cipher Suite.

— The next 48 bits are the 48 least significant bits of the extended Packet Number.

NOTE 2—The SCI contribution to the nonce corresponds to a little endian interpretation of its reception order (if present in the SecTAG), chosen to avoid the need to reverse the octets on reception for Cipher Suite processing. The PN contribution to the nonce corresponds to the numeric evaluation of the PN field as specified in 9.1 since arithmetic operations (range comparison and increment) are required for PN processing.

Table 14-2 provides an example nonce construction. Sequences of octets are shown as hexadecimal values separated by spaces, in transmission order left (first) to right. Numerical values are shown in hexadecimal, most significant digits to the left, with leading zero padding as appropriate to the notional size of the integer, with spaces breaking integers into 16-bit blocks for ease of comprehension.

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Table 14-2—Nonce construction (example)

Frame fields	Value in protected frame [a]
MAC DA	E2 01 06 D7 CD 0D
MAC SA	F0 76 1E 8D CD 3D
MACsec EtherType	88 E5
TCI and AN	2C [b]
SL	00 [c]
PN field octets	76D4 57ED [e]
SCI	68 F2 E7 76 96 CE 00 01 [d]
SecY parameters	**Value (applicable to protected frame) [e]**
lowest acceptable PN	0000 0025 76D4 57DD
PN	0000 0025 76D4 57ED
Key Number (KN)	0001 2853
Key Server MI	E630 E81A 48DE 85B4 6A21 C66F
Salt	6B21 C66F E630 E81A 608D 85B4 6A21 C66F
Nonce prior to Salt	0100 CE96 76E7 F268 0000 0025 76D4 57ED
Nonce	6A21 08F9 90D7 1A72 608D 8591 1CF5 9182

[a] Octets (apart from PN) in transmission order, left to right, hexadecimal.

[b] SCI included in SecTAG, Confidentiality protection, AN = 0.

[c] Not short, frame contains 48 or more octets of User Data.

[d] Comprising a MAC Address with a Port Identifier of 0001

[e] Integers, hexadecimal, most significant digits to the left, leading zero padding to notional size, spaces added to break into 16-bit blocks do not denote transmission order.

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15. Ethernet Data Encryption devices

An Ethernet Data Encryption device (EDE) is a frame forwarding device with two physical ports that uses IEEE Std 802.1Q, IEEE Std 802.1AE, and IEEE Std 802.1X to provide integrity and confidentiality for frames forwarded on network hops open to attack. One port (red side) receives and transmits unprotected frames, while frames transmitted and received on the other (black side) are protected by MACsec.

This clause describes and specifies the following:

a) The common characteristics of EDEs—the rationale for identifying some of the many possible MACsec capable bridging systems as EDEs—and provides an EDE taxonomy (15.1).
b) How connectivity between adjacent bridges in a customer bridged network or a Provider Bridged Network (PBN) can be secured by an EDE-M (15.2) comprising a two-port VLAN-unaware MAC Bridge with a MAC Security Entity (SecY) supporting one port.
c) Requirements for securing connectivity across a PBN (15.3).
d) How connectivity across a PBN can be secured by an EDE-M (15.4), or by an EDE-CS (15.5), EDE-CC (15.6), or EDE-SS (15.7)—each comprising two VLAN bridge components, and each using MACsec to preserve frame data integrity, data origin authenticity, and confidentiality, while allowing the provider to use the frame's VLAN tag to perform service selection and to convey Priority Code Point (PCP) and drop-eligible (DEI) information.
e) Interoperability between EDEs and other MACsec-capable bridging systems (15.8).
f) Considerations applicable to User Network Interface (UNI) access and Connectivity Fault Management (CFM) use when EDEs are used (15.9).

An understanding of architectural concepts common to this and other IEEE 802.1 standards is essential to understanding this standard's specification of EDEs. The reader is encouraged to review Clause 7 of IEEE Std 802.1AC-2016 and IEEE Std 802.1Q's use of bridge components in its specification of Provider Edge Bridges (PEBs), Backbone Edge Bridges (BEBs), PBNs, and provider network service interfaces (see, in particular, Clause 15, Clause 16, and Clause 25 of IEEE Std 802.1Q-2022).

15.1 EDE characteristics

The specification of EDEs arises from the desire to separate, so far as is possible, the implementation and use of MAC security within bridged networks from other implementation and management concerns. Reduction of the scope (in terms both of quantity and variety) of the functionality co-resident with the implementation and management of MACsec and its associated authentication, authorization, and key agreement functions can have two benefits. First, there is less software to validate, and fewer people and organizations might have to be involved in the development of each EDE. Second, it might be possible to assign management responsibility for EDEs and other bridging systems to separate smaller administrative organizations, each with its own particular expertise. Against these benefits are to be set the costs arising from additional items of equipment and the operational coordination necessary. This standard does not attempt to judge the balance of these benefits and costs, which are implementation and deployment specific. In particular there is no suggestion that the specification of EDEs means that these are always preferred to the use of other bridging systems specified in IEEE Std 802.1Q with MACsec supporting particular ports.

Restricting an EDE to having two and only two physical ports reduces the requirement for traffic class processing, particularly if the ports operate at close to the same data rate. A two-port EDE might also have no need to appear as a node in certain protocols at all, with frames for those protocols (or specific instances of those protocols as identified by destination group MAC address) being relayed simply from one port to another. Since other types of bridging systems will be usually attached to an EDE, there is no need to learn from the source address of frames.

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NOTE 1—IEEE Std 802.1Q provides for some functional simplification in two-port systems or components. Provider Edge Bridges can simply forward frames addressed to the Nearest Customer Bridge Address if the C-VLAN component of a PEB has a single Provider Edge Port (PEP), i.e., it provides connectivity to a single provider network service instance (stated in terms of the equivalent condition of connecting to the S-VLAN component through a single Customer Network Port in 13.41 of IEEE Std 802.1Q-2022). If the enhanced filtering utility criteria (8.7.2 of IEEE Std 802.1Q-2022) can never be met (a common condition for a two-port component supporting point-to-point connectivity), no source address learning need ever occur, and the size of the Filtering Database can be restricted to that necessary to accommodate the Permanent Database.

This standard specifies various types of EDE, distinguishing them by their bridging components. An EDE-M comprises a single VLAN-unaware MAC Bridge component, an EDE-CS (both a C-VLAN and an S-VLAN component), an EDE-CC two C-VLAN components, and an EDE-SS two S-VLAN components. The architecture and use of each type is explained in 15.2 through 15.7.

NOTE 2—The first-time reader of this specification is encouraged to read 15.2 through 15.7 before attempting to understand this taxonomy and its inherent possibilities in detail. In brief, describing each type of EDE in terms of existing components (just as IEEE Std 802.1Q specifies a Provider Edge Bridge in terms of the C-VLAN and S-VLAN components that comprise Customer Bridges and Provider Bridges respectively) makes for economy of specification and simplifies analysis. Since the existing components can already be connected in a network, and any part of a valid network can be considered a valid (if large) system, no new interoperability challenges arise.

15.2 Securing LANs with EDE-Ms

In the simplest EDE network configuration, the protected (black-side) of each of a pair of EDE-Ms is attached to a single LAN providing point-to-point connectivity between the EDEs as shown in Figure 15-1.

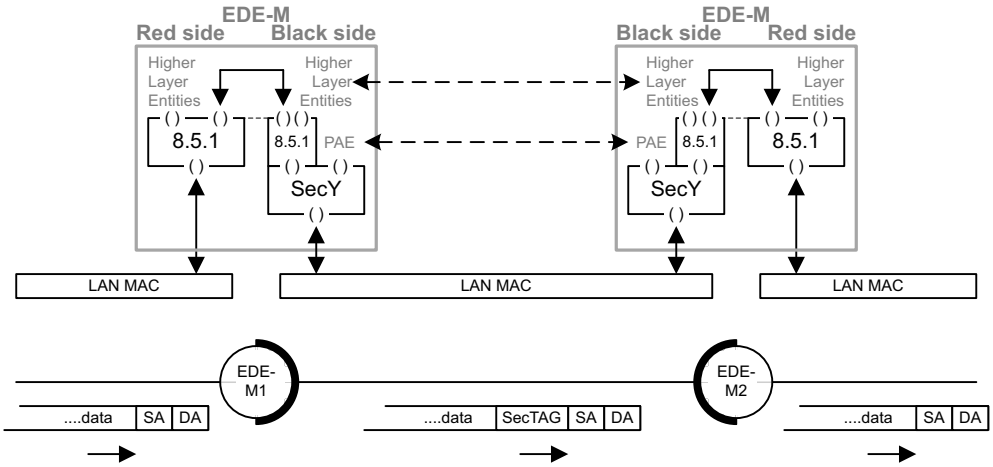


Figure 15-1—EDE-Ms connected by a point-to-point LAN

The upper part of Figure 15-1 shows the interface stacks, each attached to a LAN, in two EDE-Ms. The lower part shows the path, from the red side of one of the EDEs through to the red side of the other, and the change to a frame transmitted along that path.

NOTE 1—The architecture of a bridge is often drawn as in Figure 11-4, showing the MAC Relay entity below the MAC Service boundary to show that the relay is transparent to higher layer protocols. In this clause (Clause 15) it is convenient to use figures that focus on the interface stacks supporting relay and higher layer entities as in Figure 11-5. Numbers in the figures in this clause (e.g., 8.5.1 in Figure 15-1, 6.9 in Figure 15-2) refer to relevant clauses of IEEE Std 802.1Q-2022. The representation of EDE-M1 and EDE-M2 in Figure 15-1 provides a simple indication that frames transmitted and received by their black-side ports have had SecTAGs (and MACsec processing) applied. Similarly, other figures in this clause use light and dark patterns to indicate the possible presence of a C-VLAN or an S-VLAG tag respectively. For example, in Figure 15-3 an untagged or C-tagged frame from B1 is Sec-tagged by EDE-1, has an S-TAG added by PB1, and passes through PB2 before the outer tags are processed and removed en-route to B2. This notation can be used in larger network diagrams where showing the frame on each connecting link is inconvenient.

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A pair of EDE-Ms can secure a point-to-point LAN connecting the ports of two Customer Bridges or Provider Bridges, as in Figure 15-2.

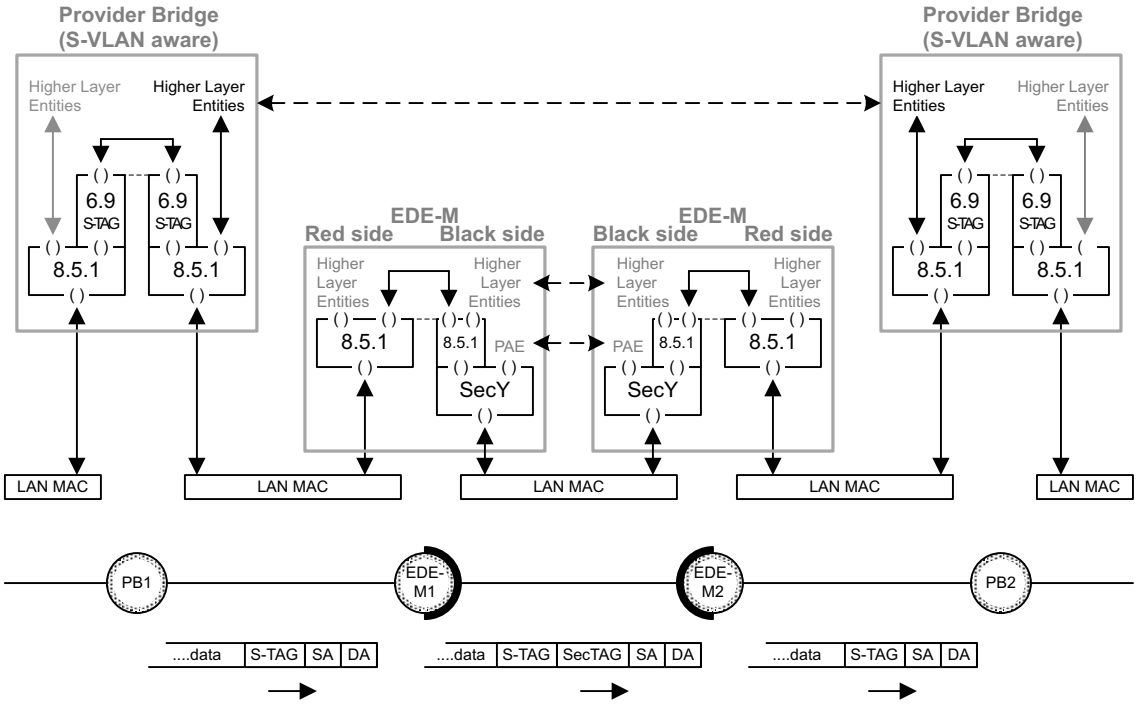


Figure 15-2—EDE-Ms securing a point-to-point LAN between Provider Bridges

Each EDE-M's PAE uses the Nearest non-TPMR group address[19] (01-80-C2-00-00-03) as the destination address for EAPOL PDUs. Frames with this destination address are filtered by all bridges specified in IEEE 802.1 standards (including EDEs) other than TPMRs. A PAE receiving a frame with this destination address can be certain that no IEEE 802.1 standards-conformant bridge lies between itself and the originator of the frame. If such a bridge is interposed between the EDE-Ms, they will not exchange EAPOL PDUs and will not, as a consequence, use MACsec to protect the frames (if any) that they forward. This avoids unintended use of the EDE-Ms in a misconfigured network. Similarly neither EDE-M shown can receive EAPOL PDUs from other EDEs connected to PB1 or PB2. This helps to prevent accidental creation of MACsec protected connectivity through (but without the participation of) an intervening bridge, and the undesirable consequence of making it impossible for that bridge to understand frames that it was intended to receive.

NOTE 2—EAPOL PDUs are used to initiate and reinitiate EAP authentication exchanges, convey EAP PDUs in support of those exchanges, and convey MACsec Key Agreement PDUs (MKPDUs). At the time of the development of IEEE Std 802.1AEcg-2017, EAPOL destination addresses were specified in 11.1 and Table 11-1 of IEEE Std 802.1X-2010. Filtering of Reserved Addresses by bridges was specified in 8.6.3, 8.13.4, and Table 8-1, Table 8-2, and Table 8-3 of IEEE Std 802.1Q-2022.

Each EDE-M also filters the addresses specified by IEEE Std 802.1Q as TMPR component Reserved Addresses. When an EDE-M's PAE is configured to use the Nearest non-TPMR group address, the other Reserved Addresses specified for MAC Bridge, C-VLAN, and S-VLAN components are forwarded, making the EDE-Ms and the connection they protect transparent to protocols using those addresses.

[19]This address was identified as the "IEEE Std 802.1X PAE address" in IEEE Std 802.1Q-2005 and as the "PAE group address" in IEEE Std 802.1X-2010.

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15.3 Securing connectivity across PBNs

IEEE Std 802.1Q specifies support of the MAC Service by Provider Bridged Networks and their principles of operation. Individual instances of the MAC Service are segregated within the PBN by S-VLAN tag, and access to those instances are provided by port-based, C-tagged, or S-tagged service interfaces. A given service instance can support service interfaces of more than one type. For example, in a hub-and-spoke configuration, it might be convenient to use a C-tagged service interface at the central site but a port-based interface at the remote sites, since the latter communicate directly only with the central site. Such an arrangement avoids having to configure each remote site differently. Alternately, each remote site might be configured to use the same identical S-TAG value. In the first of these alternatives, the PBN adds or removes the tag at the service interface; in the latter, it is translated. S-VLAN tag translation within the PBN also allows the service provider to re-allocate service instances without changing the customers' service instance selection. The VLAN tags used for service selection also carry priority and drop_eligible fields that can be policed and changed by the service provider.

NOTE—This summary of PBN service instance selection, segregation, and priority handling serves only to provide context for the provisions for this standard. For an authoritative specification, refer to IEEE Std 802.1Q.

Since VLAN tags used for communicating priority information and for service instance selection can be modified by the PBN service provider, they cannot be protected by MACsec, as any change would then result in discard on verification failure. It is also preferable for the service provider not to have to make any change to accommodate MACsec-protected frames. From the service provider's point of view, a frame whose initial EtherType is the MACsec EtherType is untagged (unless the provider's interface is itself a member of the CA that is protecting frames between the interface and customer's equipment). Figure 15-3 illustrates the passage of two MACsec-protected frames from one customer bridge network to another through port-based interfaces provided by a PBN. The first frame comprises a MAC DA, SA, and arbitrary data, with each of these frame fields reaching its destination unmodified. The second comprises a MAC DA and SA, a C-VLAN tag, and arbitrary data. Each of these frame fields also reaches its destination unmodified—the C-VLAN tag is just one possibility for the initial octets of the arbitrary data of the first example frame—and its protection ensures that an attacker cannot simply change the VLAN assignment of the frame.

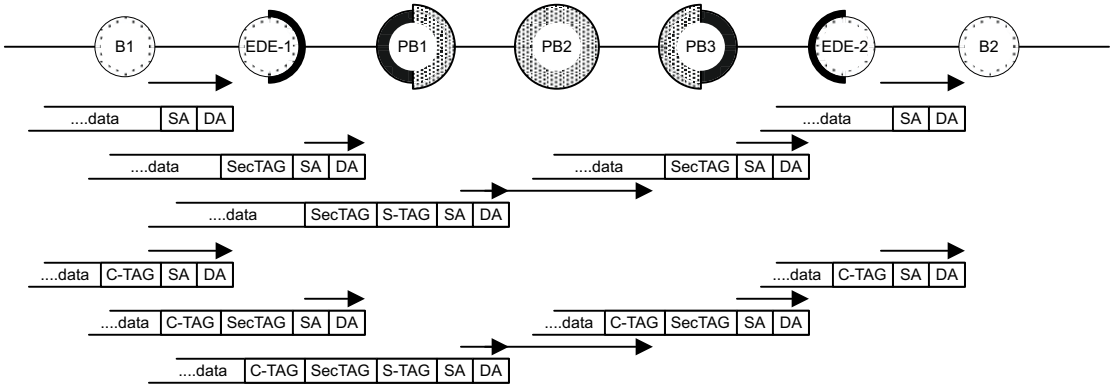


Figure 15-3—MACsec protected frame traversing a PBN

Since the PBN service provider (operating PB1) sees frames received from the customer as untagged, Figure 15-3 does not provide a way for EDE-1 to communicate each frame's priority to PB1. This can be done by priority tagging the frame after protecting it with MACsec. This requirement to communicate priority is so common as to be a required EDE-M capability (15.4, Figure 15-4). Other types of EDE, including the EDE-CS (15.5, Figure 15-6), add a full VLAN tag that depends on the protected VLAN tag so that the service provider can support service selection without requiring access to the protected data.

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15.4 Securing PBN connectivity with an EDE-M

The point-to-point connectivity between the EDEs shown in Figure 15-1 could equally be provided by a Provider Bridged Network (PBN). Figure 15-4 illustrates the use of a pair of EDE-Ms to secure connectivity between Customer Bridges attached to port-based service interfaces provided by a PBN. (Numeric references in the figure are to 8.5.1, 6.9, and 6.13 in IEEE Std 802.1Q-2022.)

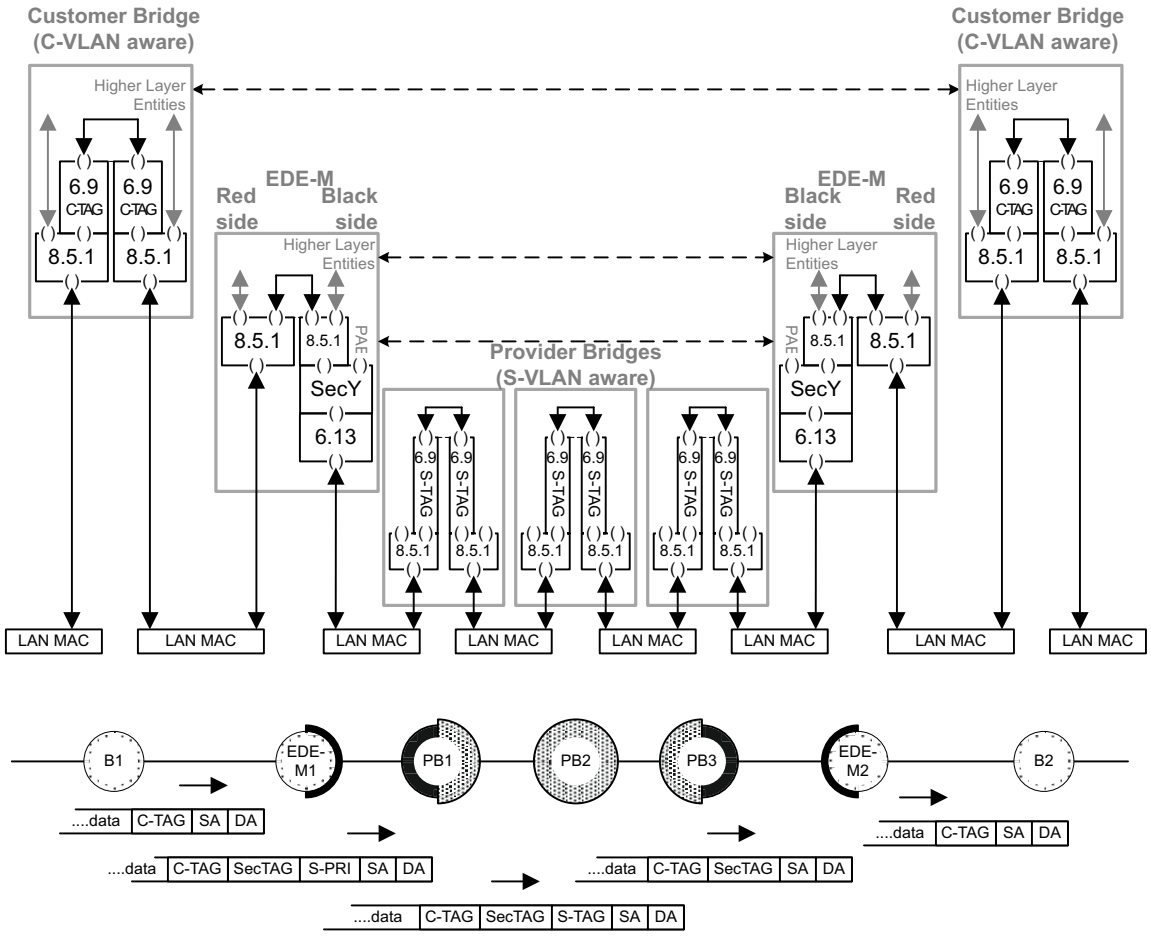


Figure 15-4—EDE-Ms securing point-to-point LAN connectivity across a PBN

As shown in the figure, EDE-M1 (on the left) may recover the signaled priority from the C-VLAN tag (if present) of the received frame as specified by IEEE Std 802.1Q [see 6.20 (Support of the ISS with signaled priority) of IEEE Std 802.1Q-2022] before protecting the frame using MACsec. If the EDE-M1 is capable of recovering signaled priority, it shall also be capable of being configured to priority tag [see 6.13 (Support of the ISS for attachment to a PBN) of IEEE Std 802.1Q-2022] or not priority tag frames transmitted by the black-side port. EDE-M2 (on the left) receives the frame and (if it is capable of recognizing signaled priority and priority tagging frames) removes any S-TAG immediately following the MAC SA and DA (not shown in the figure, which assumes the service provider interface has been configured to deliver frames that are not S-tagged). It validates the frame using MACsec before recovering the priority originally signaled by B1 in the C-VLAN tag (if present, and if capable of recovering signaled priority) and forwarding the frame.

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In this scenario, each EDE-M's PAE uses the Bridge Group Address (01-80-C2-00-00-00) as the destination address of EAPOL PDUs transmitted to its peer PAE. The use of this address is specified in IEEE Std 802.1X, and it is also identified as the Nearest Customer Bridge group address by IEEE Std 802.1Q. Each EDE-M filters any frame with a destination address that is one of the Reserved Addresses specified by IEEE Std 802.1Q as filtered by MAC Bridge and C-VLAN components (Table 8-1 of IEEE Std 802.1Q-2022) with the exception of the Nearest Customer Bridge group address.

NOTE—A frame that is received on the red-side port and not forwarded by the MAC Relay Entity will not be received by the PAE for the black-side port or transmitted on the black-side port. A frame that is received on the black-side port and not forwarded will not be transmitted on the red-side port but can be received by the PAE for the black-side port.

The PBN service is not necessarily limited to point-to-point connectivity, Figure 15-5 illustrates the secure use of a multi-point service to connect three customer bridges.

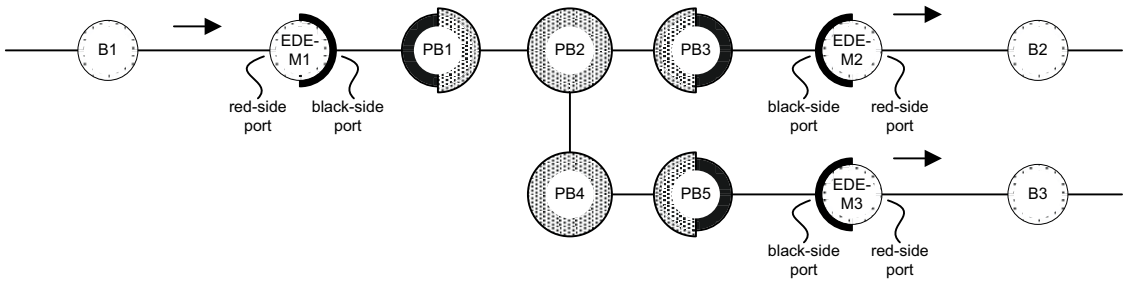


Figure 15-5—EDE-Ms securing multi-point PBN connectivity

In the figure the provider bridge ports attached to the black-side ports of each of the EDEs are assumed to be configured to provide connectivity between each of the latter.

15.5 Securing PBN connectivity with an EDE-CS

An EDE-CS provides a red-side C-tagged service interface and uses a black-side S-tagged service interface. It allows an attached customer network to use C-VIDs to select between provider service instances, protects each of those service instances with a separate CA, and identifies frames for each with a single S-VID. An EDE-CS comprises both a C-VLAN and an S-VLAN component, just as in a Provider Edge Bridge (PEB), with the following additions and restrictions. A single C-VLAN component provides a single red-side Customer Edge Port and one or more Provider Edge Ports, each supported by a SecY. Each of the Provider Edge Ports is attached to one of the Customer Network Ports of the S-VLAN component, which supports a single black-side Provider Network Port.

NOTE—The terms *customer* and *provider* applied to the external and internal ports of an EDE-CS are those used by IEEE Std 802.1Q in its description of PBs, PEBs, and BEBs and reflect the role of those ports in the layered network architecture. They do not indicate control or ownership of the equipment.

Figure 15-6 depicts an example network, with a single provider operated bridge (PB) that provides S-tagged service interfaces to two customer operated EDE-CSs and a customer operated PEB, and a port-based interface to a customer owned EDE-M. Consider EDE-CS1, on the left of the figure. This has a C-VLAN component, with a single Customer Edge Port connected to bridge B1 and three Provider Edge Ports, and an S-VLAN component, with a Provider Network Port attached to the provider's S-tagged service interface and three Customer Network Ports. Each of the three Provider Edge Ports has a SecY that protects transmitted and received frames and connects to one of the Customer Network Ports. EDE-CS1's C-VLAN component

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is constrained, as required by IEEE Std 802.1Q's specification of PEB operation, to forward frames received on its Customer Edge Port for a given VLAN to at most one of its Provider Edge Ports, and hence to at most one Customer Network Port. Each of the Customer Network Ports does not have a SecY, and thus treats each received frame as untagged and assigns it to the S-VLAN identified by the port's PVID as follows:

— The upper port's PVID is configured with the S-VID used by the point-to-point service instance that provides connectivity to EDE-M2 (thus protecting communication between B1 and B2).

— The middle port's PVID provides connectivity to the upper Provider Edge Port in EDE-CS3 (protecting communication between B1 and B3).

— The lower port's PVID provides connectivity to the upper C-VLAN component in PEB4 (with a SecY on its Provider Edge Port, protecting communication between B1 and B4.1).

Not all the protected traffic has to pass through EDE-CS1 or be associated with one of its protected service instances; the figure also shows EDE-CS3 and PEB4 (with a SecY on the Provider Edge Port of its lower C-VLAN component) protecting traffic between B3 and B4.2.

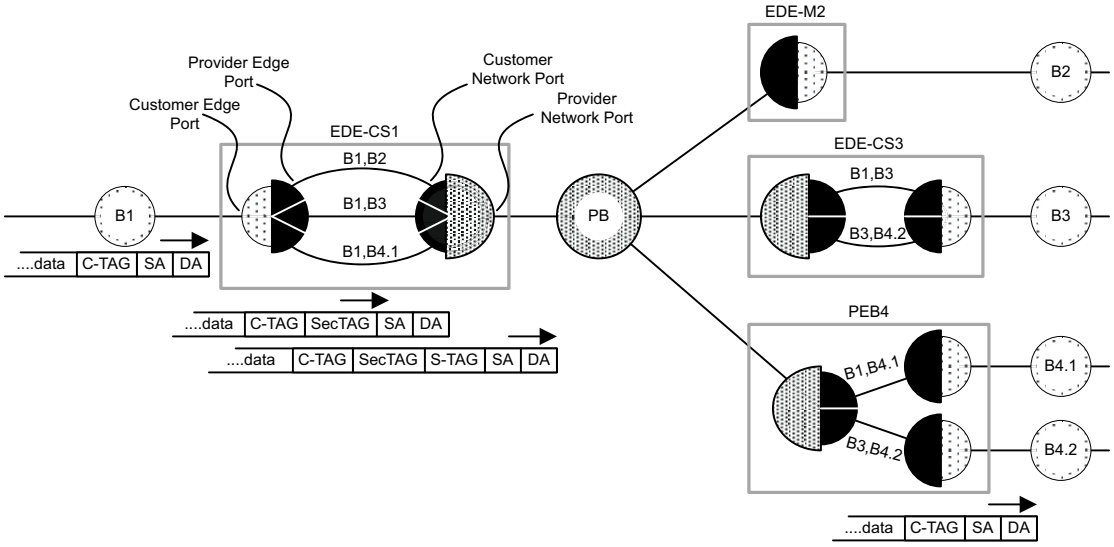


Figure 15-6—Example network with an EDE-CS

Figure 15-7 shows the internal architecture of an EDE-CS together with that of a Customer Bridge and a Provider Bridge (attached to the EDE-CS's Customer Edge Port and Provider Network Port respectively). This view of the interface stacks involved in the connection of the EDE-CS to an S-tagged interface depicts just one path through the network.

The PAE of each EDE-CS's Provider Edge Port shall be capable of being configured to use the Bridge Group Address (01-80-C2-00-00-00, also known as the Nearest Customer Bridge group address) as the destination address of EAPOL PDUs that it transmits and receives. The C-VLAN component of an EDE-CS filters any frame with a destination address that is one of the Reserved Addresses specified by IEEE Std 802.1Q as filtered by MAC Bridge and C-VLAN components (Table 8-1 of IEEE Std 802.1Q-2022), including the Nearest Customer Bridge group address. The S-VLAN component filters any frame with a destination address that is one of the Reserved Addresses specified by IEEE Std 802.1Q as filtered by S-VLAN components.

NOTE—Because the EDE-CS's C-VLAN component provides connectivity to multiple service instances, it does not offer the same level of transparency to protocols using the Nearest Customer Bridge group address as does an EDE-M.

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In Figure 15-6 the PAE associated with the uppermost Provider Edge Port on EDE-CS1 is connected (over an S-VLAN supported by the Provider Bridge PB) to the PAE for the black-side port of EDE-M2, and exchanges EAPOL PDUs with that PAE. The path between the PAEs is supported by the S-VLAN components (the network component in EDE-CS1 and the Provider Bridge PB) that do not filter frames with the destination MAC address used by the EAPOL PDUs. Similarly EAPOL PDUs are exchanged between the PAE for the middle Provider Edge Port of EDE-CS1 exchanges EAPOL PDUs with the PAE for the upper Provider Edge Port of EDE-CS3 (over another S-VLAN), between the PAE for lower Provider Edge Port of EDE-CS1 and that for the upper Provider Edge Port of PEB4, and between the PAEs for the lower Provider Edge Ports for EDE-CS3 and PEB4.

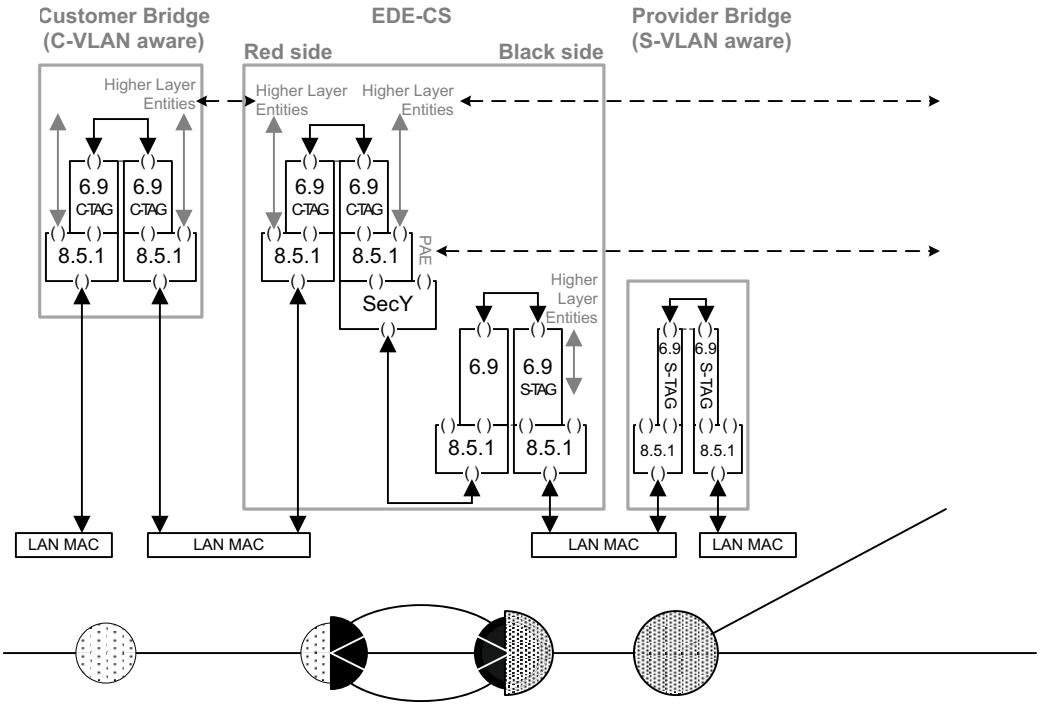


Figure 15-7—EDE-CS connected to a PBN S-tagged interface

15.6 Securing PBN connectivity with an EDE-CC

A service provider might offer C-tagged service interfaces but not S-tagged interfaces. Equally it might be desirable to secure existing network connectivity by adding an EDE between a Customer Bridge and the provider network while retaining the C-VLAN service selection capability. Figure 15-8 illustrates this before and after scenario.

The upper half of the figure depicts a Customer Bridge attached to a PEB that provides access to three point-to-point service instances, each selected by one or more C-VIDs or by the PVID used by the PEB's Customer Edge Port to classify frames received untagged. Frames with any given VID are forwarded through at most one of the PEB's Provider Edge Ports and hence through at most one Customer Network Port. The PEB's S-VLAN component sees each of these frames as untagged (as it does not recognize a C-TAG) and assigns each to the S-VLAN (and thus to the point-to-point service instance) identified by the Customer Network Port's PVID. The frame is S-tagged with the selected S-VID as it passes through the PEB's Provider Network Port into the PBN. In the other direction, each frame received from the PBN is directed by the PEB's S-VLAN component to Customer Network Port whose PVID matches the received S-VID. The S-TAG is removed on transmission through the Customer Network Port, revealing the C-tagged

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frame, which is then forwarded by the PEB's C-VLAN component. Within the PEB, one C-VLAN can be carried without a tag on each of the *internal LANs* that connects a Provider Edge Port with a Customer Network Port, which leads to the possibility of frames for each of these C-VLANs being carried without a C-TAG within the PBN.

In the lower half of Figure 15-8 an EDE-CC has been added between the Customer Bridge and the PEB. In the figure the PEB is shown as supporting three service provider instances, protecting communication between B1 and B2, B1 and B3, and between B1 and B4 (B2, B3, and B4 lying elsewhere in the network), and the EDE-CC's edge component has three Provider Edge Ports, each participating in a CA that protects one of the service instances. In this scenario, it is unnecessary to carry the additional C-TAG, added by the EDE-CC's network component, over the PBN—it can be removed by the PEB's Provider Edge Port as the attached internal LAN suffices to identify the provider service instance within the PEB. The PBN frame format for each of the service instances is then as shown in the bottom right of the figure. If the PEB had been configured to map these CAs to one or two service instances, a C-TAG would be required to distinguish those carried over a common service instance.

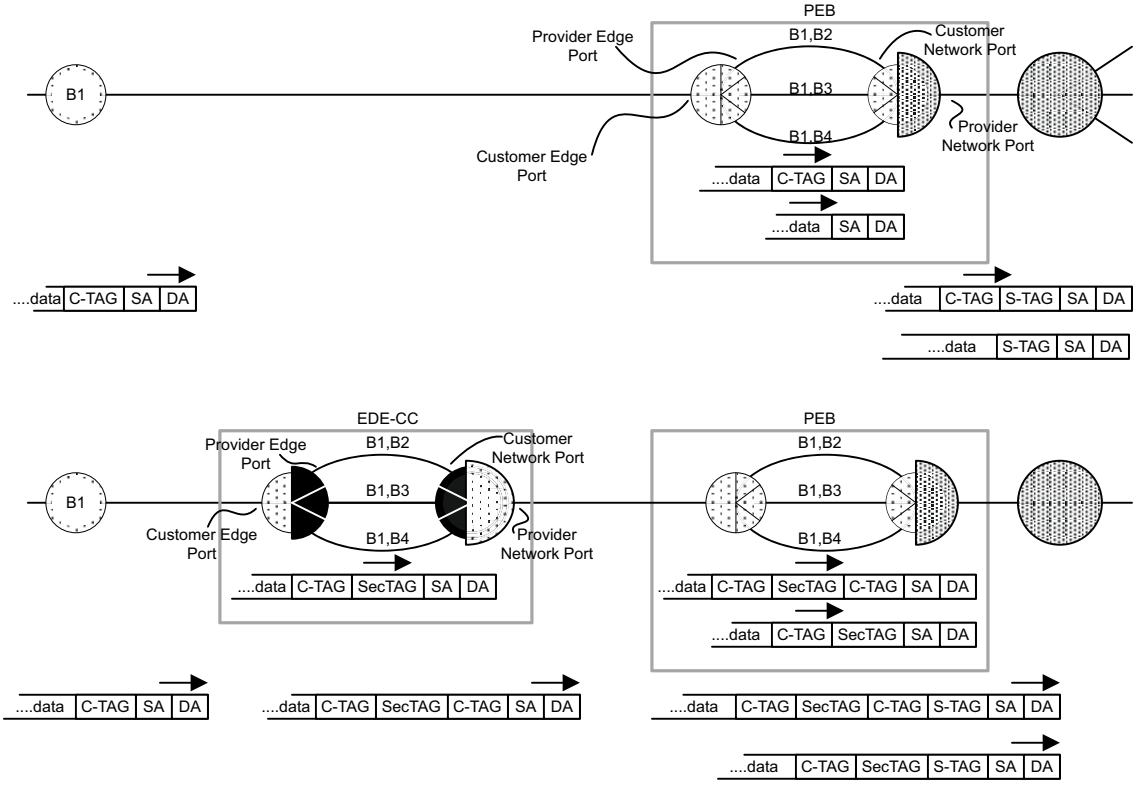


Figure 15-8—Using an EDE-CC with a C-tagged provider service interface

Figure 15-9 shows the architecture of the EDE-CC in more detail. This standard extends the use of the terms Customer Edge Port, Provider Edge Port, Customer Network Port, and Provider Network Port (initially defined in IEEE Std 802.1Q for Provider Edge Bridges) to identify ports that play similar roles in EDE-CSs, EDE-CCs, and EDE-SSs. However, if a SecY associated with a EDE-CC's Provider Edge Port is configured not to protect frames (as might be done to facilitate initial deployment), a SecTAG will not be added to transmit frames, and the EDE-CC's Customer Network Port component will see the received frames as already C-VLAN-tagged and will not add a further tag. The externally observable behavior of the EDE-CC would then resemble that of a single Customer Bridge, not a Provider Edge Bridge.

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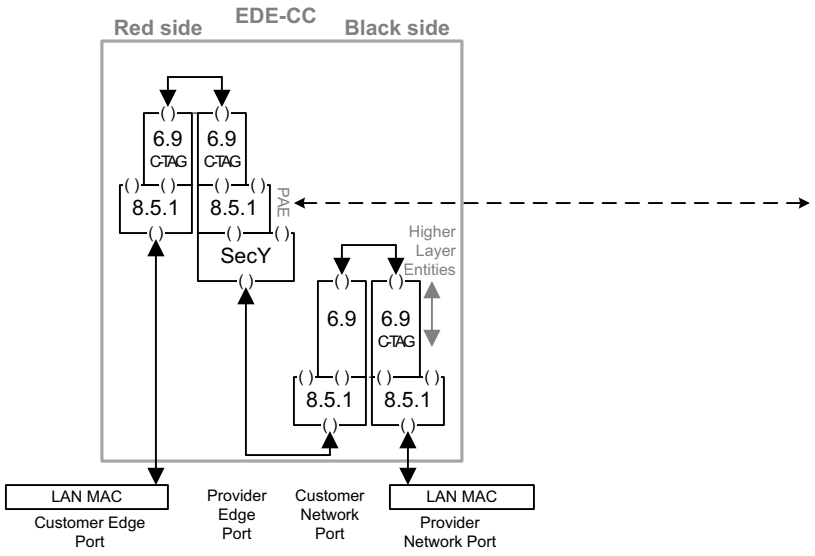


Figure 15-9—EDE-CC architecture

The PAE of each Provider Edge Port for an EDE-CC's edge component shall be capable of being configured to use the EDE-CC PEP Address (see Table 15-1) as the destination address of EAPOL PDUs that it transmits and receives. In Figure 15-9, the EDE-CC's C-VLAN network component (shown on the right in the figure, with a Customer Network Port and a Provider Network Port) is shown at a lower level than its accompanying edge component to emphasize the fact that it is transparent to the operation of the Provider Edge Port PAEs and other edge component protocol entities. It filters any frame with a destination address that is one of the Reserved Addresses specified by IEEE Std 802.1Q as filtered by S-VLAN components. The EDE-CC's C-VLAN edge component (shown on the left in the figure, with a Customer Edge Port and a Provider Edge Port) filters any frame with a destination address that is either one of the Reserved Addresses specified by IEEE Std 802.1Q as filtered by MAC Bridge and C-VLAN components (Table 8-1 of IEEE Std 802.1Q-2022) or the EDE-CC PEP Address.

The configuration of an EDE-CC that does not provide MAC Privacy protection is constrained to restrict egress for each Provider Edge Port to a single C-VID and to restrict the PVID for the internally connected Customer Network Port to the same value, with the consequence that the outer C-VID always matches the inner C-VID. The PVID for the Customer Edge Port is constrained to be the same as that for the Provider Network Port and the Static VLAN Registration Entry (8.8.2 of IEEE Std 802.1Q-2022) for that and other VIDs are constrained so that frames for that VID are transmitted untagged on both ports, with the consequence that frames received untagged on either port are forwarded (if at all) untagged on the other. These restrictions simplify EDE management, supporting the desired separation of concerns (15.1) and maintaining the scope of address learning within each C-VLAN. If the desired secured connectivity between the EDE-CC and its potential (provider network attached) peers depends only on their characteristics and does not vary by C-VLAN, an EDE can create that secure connectivity on demand—initiating EAP or starting MKA instances to authenticate and authorize the VLAN connectivity as frames for each VLAN are received—reducing the need to communicate VLAN specific details between administrative organizations. Further restrictions on the use of EAPOL and MKA to support such dynamically created connectivity—including use of pre-shared or cached secure Connectivity Association Keys (CAKs) and announcements—are beyond the scope of this specification (see IEEE Std 802.1X for detailed capabilities).

NOTE—The descriptive advantage of the two component architecture of PEBs and EDEs is not limited to VLAN multiplexing over service instances. It allows existing and developing port-based queue servicing specifications to be applied in the context of the resources available to each provider service instance, for example.

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An EDE-CC that provides MAC Privacy protection may be capable of using the C-VID of a user data frame received on the red-side Customer Edge Port (or the absence of a C-TAG on that frame) to select the C-VID (or the absence of a tag) when tagging the MACsec protected frame for transmission on the black-side Provider Network Port. This multiplexing functionality hides the otherwise visible distinction between user data frames assigned to different C-VLANs but destined for the same PrY. The required tagging is configured using the management controls specified by IEEE Std 802.1Q for each of the EDE's bridge components (edge and network) as follows:

a) On the network component :
   1) A distinct Customer Network Port (CNP, see Figure 15-8 and Figure 15-9) is configured for each desired outer, black-side visible, C-VID value with a PVID of that value.
   2) A Static VLAN Registration Entry (8.8.2 of IEEE Std 802.1Q) for that C-VID, with fixed registration that includes only the CNP and the Provider Network Port (PNP) in the member set and specifies untagged transmission on the CNP and tagged transmission on the PNP (with the possible exception of allowing untagged transmission for a single C-VID, allocated for that purpose, on the PNP).

b) On the edge component:
   1) A Provider Edge Port (PEP) for each of the above CNPs, connected to that CNP within the EDE.
   2) A Static VLAN Registration Entry for each red-side visible C-VID, with a fixed registration with a member set that includes only the Customer Edge Port (CEP) PVID and the PEP connected to the CNP whose PVID is the desired outer, black-side visible, C-VID, with tagged transmission on the CEP and PEP.
   3) A Static VLAN Registration Entry for a single C-VID value allocated for untagged transmission, with a member set that includes only the CEP and one PEP.

NOTE 2—The multiplexing functionality provided by this configuration [a) and b), above] can be configured in an EDE-CS by using the Customer Edge Port Configuration managed object (12.13.2 of IEEE Std 802.1Q-2022).

15.7 Securing PBN connectivity with an EDE-SS

An EDE-SS addresses a similar requirement to that for an EDE-CC—securing existing network connectivity with minimal change to existing systems, in this case retaining S-VLAN service selection capability, using the same architecture with two S-VLAN components instead of two C-VLAN components. The same configurations restrictions apply: the value of the outer S-VID added and removed by the EDE's network component matches that of the inner S-VID, and frames received untagged by the Customer Edge Port are transmitted untagged by the Provider Network Port and vice versa.

The PAE of each EDE-SS's Provider Edge Port shall be capable of being configured to use the EDE-SS PEP Address (see Table 15-1) as the destination address of EAPOL PDUs that it transmits and receives. The S-VLAN network component filters any frame with a destination address that is one of the Reserved Addresses specified by IEEE Std 802.1Q as filtered by S-VLAN components. The S-VLAN edge component filters any frame with a destination address that is one of these addresses or the EDE-SS PEP Address.

NOTE—An EDE device with red-side recognition of S-TAGs and black-side addition and removal of I-TAGs and B-TAGs, used to secure connectivity across a Provider Bridged Backbone Network (PBBN), would not differ from a BEB with an EDE-SS on the customer side and is therefore not described in this standard.

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15.8 EDE Interoperability

The PAEs specified for each EDE type (in 15.2, 15.4, 15.6, and 15.7) can be configured to use the group MAC address typically used by a potential peer MACsec capable system. Table 15-1 summarizes the group addresses specified by this standard and IEEE Std 802.1X, and their filtering by bridge components.

Table 15-1—PAE Group Addresses

Address assignment	Address value	Address filtering				
EDE-CC PEP Address	01-80-C2-00-00-1F	Y[a]				
Bridge Group Address, Nearest Customer Bridge group address	01-80-C2-00-00-00	Y	Y			
EDE-SS PEP Address	01-80-C2-00-00-0B	Y	Y	Y		
Nearest non-TPMR Bridge group address, IEEE Std 802.1X PAE address[b]	01-80-C2-00-00-03	Y	Y	Y	Y	
Individual LAN Scope group address, Nearest Bridge group address[c]	01-80-C2-00-00-0E	Y	Y	Y	Y	Y
		EDE-CC Edge components				
		MAC Bridge & C-VLAN components (Customer Bridges, PEB w/multiple PEPs[d])				
		PEB C-VLAN components w/ single PEP				
		S-VLAN components (Provider Bridges, Provider Backbone Bridges, PEBs)				
		TPMR components				

[a]Y indicates, Yes, this address is filtered by the component.

[b]Identified as the "IEEE Std 802.1X PAE address" in IEEE Std 802.1Q-2003 and IEEE Std 802.1Q-2005 and as the "PAE group address" in IEEE Std 802.1X-2010.

[c]It is intended that no IEEE 802.1 relay device will be defined that will forward frames that carry this destination address.

[d]A PEB's C-VLAN component with multiple PEPs supports more than one provider network service instance.

Table 15-2 summarizes the use of these addresses in various scenarios. In each case, the choice of address is constrained by the need for it to be forwarded (and not filtered) by intervening components. For example, a PAE for an EDE-M connected via a port-based interface providing access to a single provider service instance can use the Nearest Customer Bridge group address to communicate to a peer PAE in a similarly connected EDE-M or Customer Bridge, but cannot use that address to communicate to an EDE-CC connected to a C-tagged interface. Where connectivity is impossible or undesirable, an address is not given. For example, securing connectivity between a EDE-CC connected to a provider network and a TPMR connected to a distant LAN might be possible using the EDE-CC PAE address, but could render traffic relayed by the TPMR unintelligible to neighboring Customer Bridges, which operate at a higher sublayer in the network's connectivity. The network administrator should take care not to introduce a similar (sub)layering violation by configuring the PAE of an EDE-M that is not directly connected to a PBN with the EDE-CC PAE address, as that could interfere with the operation of configuration protocols between the EDE-M and its immediate neighbors.

In addition to agreeing on the group addresses to be used by their PAEs, MACsec-capable systems connected to a PBN can only interoperate if the use, addition, removal, or modification of VLAN tags by the provider network is appropriate. EDE-CCs (for example) rely on the presence of VLAN tags to distinguish provider network service instances. EDE-Ms (on the other hand) need to receive frames from the attached single provider network service instance untagged. When EDE-Ms (or MACsec-capable Customer Bridges)

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are connected across a PBN in a hub-and-spoke configuration to an EDE-CC acting a hub, the PBN has to be configured to add a C-VLAN tag prior to EDE-CC reception so that the latter can separate frames from each spoke and has to remove the outer C-VLAN tag from frames transmitted by the EDE-CC before they are delivered to each spoke.

Table 15-2—PAE Group Address use

System	Connectivity	Address[a,b]								
EDE-CC	C-tagged PBN i/f	-1F	-1F	-1F						
EDE-M	Port-based PBN i/f	-1F	-00	-00	-00					
Customer Bridge	Port-based PBN i/f	-1F	-00	-00	-00					
EDE-CS	S-tagged PBN i/f		-00	-00	-00					
EDE-SS	S-tagged PBN i/f					-0B				
Provider Bridge	Individual LAN						-03	-03	-03	-0E
EDE-M	Individual LAN						-03	-03	-03	-0E
Customer Bridge	Individual LAN						-03	-03	-03	-0E
TPMR	Individual LAN						-0E	-0E	-0E	-0E

[a]The system connected as shown in the *i*th row of the table interoperates with that in the *j*th row (also connected as shown in that row) using the group address in *i*th row and *j*th column.

[b]For convenience the EDE-CC PEP Address is shown as -1F, the Nearest Customer Bridge group address as -00, the EDE-SS PEP Address as -0B, the Nearest non-TPMR Bridge group address as -03, and the Nearest Bridge group address as 0E.

NOTE—Table 15-2 is not intended to cover all MACsec interoperability scenarios.

Any outer C-VLAN (or S-VLAN in the case of an EDE-SS) tag only facilitates service selection and connectivity across the PBN. This tag is neither used or trusted by systems attached to red-side ports. The level of trust associated with connectivity between EDEs is based on contents of each MACsec-protected frame, including (where appropriate) the VLAN tag. For example, red-side traffic can be separated by VLAN according to the use made of each VLAN. If the PAE of an EDE-CC's Provider Edge Port establishes connectivity across a PBN and the level of authorization (based on authentication attributes) associated with the remote EDE is sufficient only to permit limited access (perhaps to a *guest VLAN*), then the EDE-CC's Edge Components VLAN ingress controls for the PEP can be configured to deny access to other VLANs.

15.9 EDEs, CFM, and UNI Access

Provider network operators typically define a User Network Interface (UNI) that enables autoconfiguration of devices attached to provider network services and provides service status information. One set of protocols that supports these UNI functions is the Ethernet Local Management Interface (E-LMI) specified by the MEF Forum (MEF 16 [B20]). Connectivity Fault Management (CFM) as specified by Clause 18 through Clause 22 of IEEE Std 802.1Q-2022 can be used to monitor provider service instances, detecting and isolating connectivity failures. CFM supports the hierarchical nesting of maintenance domains, naturally complementing the service (sub)layering enforced by the use of MACsec. CFM PDUs that are interpreted, modified, or generated by the provider network are not forwarded through EDE red-side ports, but are sourced and sunk by maintenance end points within an EDE. An EDE-CC PEP can be configured to relay unprotected traffic for a given VLAN so that a red-side attached system can access the provider network UNI directly, but the red-side network then depends on those red-side systems not to forward frames for those VLANs further, and those systems would themselves depend on filtering of unprotected frames within the EDE for their own protection. The specification of such filtering or the substitution of equivalent proxy functions is specific to UNI operations and beyond the scope of this specification.

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16. Using MIB modules to manage EDEs

Ethernet Data Encryption devices (EDEs), specified in Clause 15, can be managed using the MIB modules specified in IEEE Std 802.1Q, IEEE Std 802.1X, and Clause 13 (MAC Security Entity MIB) of this standard. This clause specifies how EDEs are managed using these MIB modules, drawing particular attention to how constraints on EDE behavior are reflected in the MIBs and their use.

16.1 Security considerations

The user of this standard is encouraged to review the security considerations described by the defining standard for each of the MIB modules referenced by this clause. An EDE implementation can restrict remote management use of the management controls identified as sensitive. When used to manage EDE functionality, a greater weight might be placed on their sensitivity than in other uses.

16.2 EDE-M Management

An EDE-M is managed as a VLAN-unaware MAC Bridge (5.14 of IEEE Std 802.1Q-2022) using the MIB modules specified in IEEE Std 802.1Q, with the black-side port interface stack including a SecY managed by the MIB module specified in this standard and supported by a PAE managed by the MIB module specified in IEEE Std 802.1X. The Bridge Name (the sysDescr object in the SNMPv2-MIB) should identify the bridge as an EDE-M, but this standard does not constrain the syntax of that identification. Similarly the Port Name (the ieee8021BridgeBasePortName in the IEEE8021-BRIDGE-MIB) should identify the red-side and black-side ports, but this standard does not constrain the syntax of that identification.

16.3 EDE-CS Management

An EDE-CS is managed as a Provider Edge Bridge (5.10.2 of IEEE Std 802.1Q-2022) using the MIB modules specified in IEEE Std 802.1Q, with each Provider Edge Port (PEP) stack including a SecY managed by the MIB module specified in this standard and supported by a PAE managed by the MIB module specified in IEEE Std 802.1X. The Bridge Name (the sysDescr object in the SNMPv2-MIB) should identify the bridge as an EDE-CS, but this standard does not constrain the syntax of that identification. Similarly the Port Name (the ieee8021BridgeBasePortName in the IEEE8021-BRIDGE-MIB) should identify the red-side Customer Edge Port and the black-side Provider Network Port, but this standard does not constrain the syntax of that identification.

16.4 EDE-CC and EDE-SS Management

The sysDescr object in the SNMPv2-MIB should identify the system as an EDE-CC or EDE-SS as appropriate, but this standard does not constrain the syntax of that identification.

The edge component of an EDE-CC or EDE-SS, its Customer Edge Port, and its Provider Edge Ports may be managed using the MIB modules specified in IEEE Std 802.1Q. The edge component has a ComponentID of 1. The red-side port is the only Bridge Port for that component that is identified as a Customer Edge Port.

NOTE—The applicable MIB modules specified in IEEE Std 802.1Q include, at a minimum, the IEEE8021-TC-MIB, the IEEE8021-BRIDGE-MIB, and the IEEE8021-Q-BRIDGE-MIB. The ComponentId is used as the ieee8021BridgeBaseComponentId in the ieee8021BridgeBaseTable of the IEEE8021-BRIDGE-MIB, and the ieee802QBridgeComponentId in the ieee8021QBridgeTable of the IEEE8021-QBRIDGE MIB.

An EDE-CC that restricts egress for each Provider Edge Port (PEP) to a single red-side C-VLAN as identified by a single C-VID value and the PVID for the Customer Network Port to the same value or an EDE-SS can be managed without explicitly managing its network component. The static or dynamic instantiation of each Provider Edge Port results in the instantiation of a matching Customer Network Port

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(CNP) and an internal connection between the PEP and the CNP. The PVID values, egress, ingress, and tagging parameters associated with each of the network component's ports are determined by values for the edge component and the EDE configuration restrictions (15.6, 15.7).

An EDE-CC that supports MAC Privacy protection (21.5, Figure 21-4) also restricts egress for each red-side C-VLAN/C-VID to a single PEP but may [5.8 g)] allow egress for multiple red-side C-VIDs over the same PEP, multiplexing those C-VLANs over the outer, black-side, C-VLAN that supports transmission to the PEP's peer PrY(s). Configuration of this capability requires configuration of both edge and network components of the EDE-CC (15.6, 21.5) using the MIB capabilities provided by IEEE Std 802.1Q.

The network component of an EDE-CC or EDE-SS is identified by a ComponentID of 2 to support the use of additional capabilities, such as CFM or queue service disciplines applied to the Provider Network Port as a whole, which can require or make use of network component management. The red-side port is the only Bridge Port for that component that is identified as a Provider Network Port.

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17. MAC Privacy protection

This clause provides an overview of MAC Privacy protection. It provides the context necessary to understand the MAC Privacy protection protocol (Clause 18), the encoding of MAC Privacy protection Protocol Data Units (MPPDUs, Clause 19), and the detailed operation of individual MAC Privacy protection Entities (PrYs, Clause 20), and describes the following:

a) The need for MAC Privacy protection (17.1).
b) How individual user data frames are protected (17.2).
c) Quality of Service impacts and their mitigation (17.3).
d) How MAC Privacy protection is configured (17.4).

Interoperability, deployment, and network configuration requirements are detailed in 18.2 (Data origin authenticity, frame data integrity and confidentiality), 18.5 (Coexistence and use), and Clause 21 (MAC Privacy protection in Systems). Should any conflict be apparent between the text of this clause (Clause 17) and that of Clauses 18 through 20, the latter take precedence.

17.1 The need for MAC Privacy protection

Privacy protection is associated with the rights of individual persons to control the disclosure of information associated with themselves and their activities (personally identifiable information, PII). PII can be carried in the user data fields of IEEE 802 MAC data frames and can be hidden from adversaries (persons or organizations attempting to gain unauthorized access to information) by the confidentiality protection provided by MACsec or higher layer protocols (e.g. IPsec). However adversaries can also correlate multiple items of information (personal correlatable information, PCI) in order to construct a ‘fingerprint’ of that person or to correlate their activities with previously computed fingerprints of known activities.

NOTE—IEEE Std 802E [B5] further describes privacy considerations and the use of the terms PII, PCI, adversary, correlation, fingerprint, personal device, and shared service device in the context of IEEE 802 networks.

17.1.1 Privacy and confidentiality

Any potential adversary can still observe the MAC source and destination addresses of a data frame that has been confidentiality protected by MACsec and therefore might, for example, be able to associate the source MAC address of a particular device (a personal device) with an individual and subsequently track that individual and his or her interactions with other individuals or network based services. The sizes of data frames and their transmission timing can also be correlated with particular network applications or the details of those applications (e.g. the size of financial transactions). These adversaries might gain important information simply from the amount of information being sent.

17.1.2 Privacy and organizations

The need to assure privacy is not limited to individuals. Organizations are under an obligation to protect and therefore not to disclose certain information and need to be aware of the extent to which adversaries can draw conclusions by combining multiple observations, each of which might convey little information when considered separately. Organizations that transmit data on behalf of individuals or other organizations are also under an obligation to keep all such data private.

17.1.3 Privacy and network operation

While privacy protection is desired, the operation of networks depends on access by the systems that make up that network (bridges, routers, and switches) to the destination and source MAC addresses of frames to be forwarded. Forwarding systems can require access to information in the frames’ user data to associate some frames with data flows and bandwidth reservations. Annex I describes privacy considerations related to the

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use, design, and deployment of bridged networks in more detail. The requirement for intermediate system access to addresses and user data fields means that all user PCI cannot be simply cryptographically confidentiality protected from its original source to its original destination. Just as for MACsec without additional privacy protection, addresses and data are protected hop-by-hop, although a single hop can extend over a service provider connection with Ethernet Data Encryption devices (EDEs, Clause 15). Authenticated and authorized systems that protect user data frames with MACsec over potentially exposed network connections can also provide MAC Privacy protection as described in this clause. The MAC addresses of the privacy protection systems are exposed to external observation, but these devices can be shared service devices or at least (if they are personal devices) be at fixed locations so (when privacy protection is applied) there is little or no observable correlation between the flow of protected frames and PII.

17.2 Protecting user data frames

In the absence of privacy protection MACsec secures communication while minimizing its impact on the MAC Service's Quality of Service (QoS) parameters (6.10). Individual frames are cryptographically protected and transmitted with minimal delay including only the addition of only those octets required to support cryptographic integrity and confidentiality protection. Each frame's source and destination MAC Addresses remain unmodified.

This deliberately limited impact on the transmission and reception of frames can allow a potentially adversarial observer to correlate some or all of the following:

— source and destination MAC addresses

— the patterns of frame sizes

— transmission timing and transmission frequency

with:

— the identities of communicating users

— the reason they are communicating

— the content (in some cases) of confidentiality protected communication.

When protecting privacy is paramount, QoS and simplicity of network configuration can be less important. MAC Privacy protection Entities (PrYs) can enhance privacy by the following:

a) Encapsulating user data frames, and their source and destination MAC addresses, within MAC Privacy protection Data Units (MPPDUs).

b) Padding MPPDUs to fixed sizes before their contents are confidentiality protected by MACsec.

c) Controlling the timing of MPPDU transmissions.

The MAC Source Address of each MPPDU identifies its encapsulating PrY, and the MAC Destination Address identifies its decapsulating PrY(s) with a unicast or multicast address. When MPPDUs are confidentiality protected by MACsec, the encapsulated user data frames' source and destination MAC addresses and frame sizes are hidden from an observer who lacks the protecting secret key. Figure 17-1 shows the addition of PrYs to the interface stacks of two bridges protecting frames exchanged with MACsec.

NOTE—The formal term MPPDU is used to avoid confusion with terminology defined elsewhere, and for consistency with similar terms.

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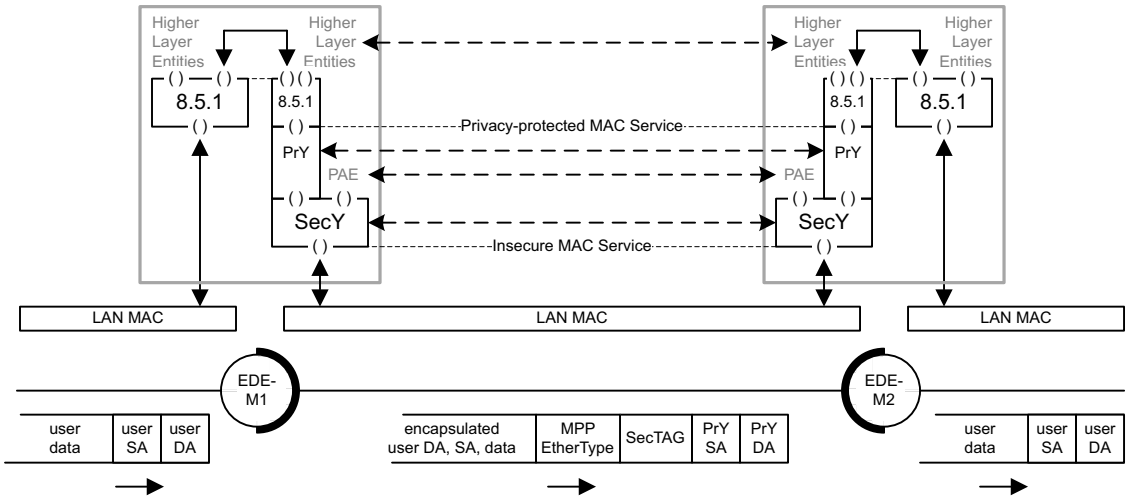
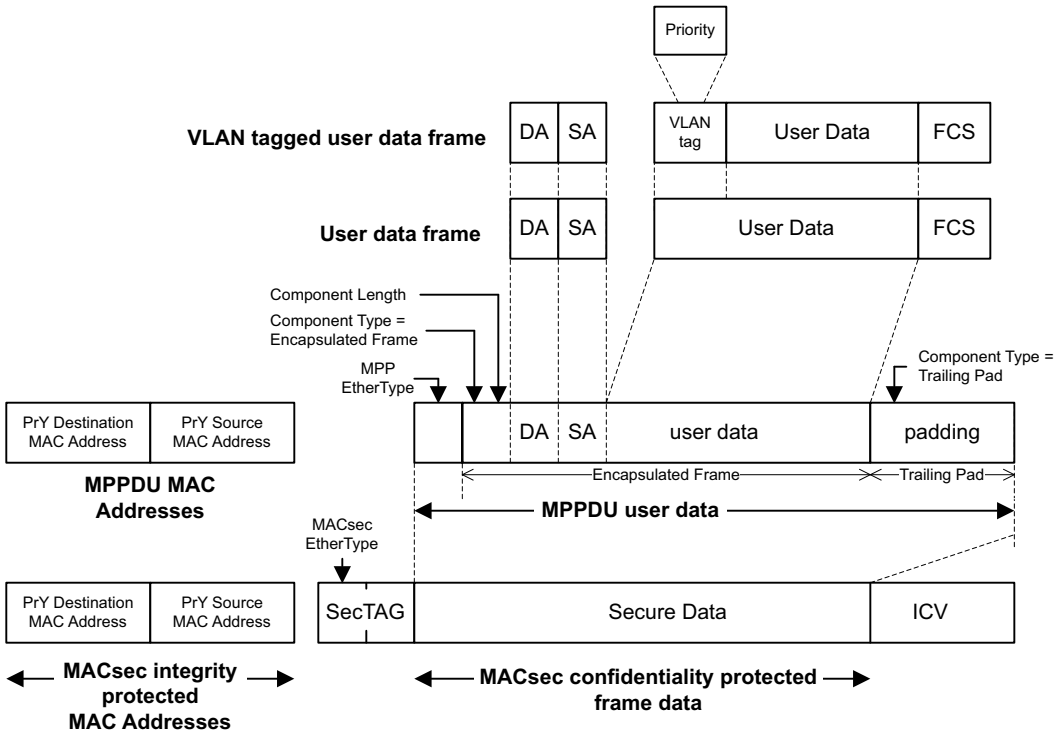


Figure 17-1—Privacy protected communication between bridges

Figure 17-2 shows a user data frame (DA, SA, and user data) followed by padding, in an MPPDU that is then protected by MACsec.



Source and destination MAC addresses are shown as separate from user data. The supporting service encodes these separate ISS service request parameters into a frame and can add octets between them. Strictly this standard specifies service primitive parameters, not frames. However, it is often convenient to talk of these parameters as a frame.

The MPPDU shown encapsulates a single encapsulated user data frame. More than one user data frame can be encapsulated in a single MPPDU and user data frames can be fragmented and carried in successive MPPDUs (19.5).

Figure 17-2—A privacy protected user data frame

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If the original user data frame included a VLAN tag, the user priority information in that tag is carried in the frame's MAC Service user data and will be unchanged when the receiving PrY decapsulates the user data frame. The received user data frame’s user priority does not depend on the access priority used to transmit the confidentiality protected MPPDU.

The length of the encapsulated data frame component is carried explicitly, so padding can be added to prevent observation of the original data frame size once the MPPDU has been protected by MACsec. This removes or reduces the contribution that observation of MPPDU frame sizes can make to inferences about the communication. An MPPDU can also be sent without any user data, including only padding, to guard against observation of user activity.

The media-dependent FCS of the user data frame is not carried within the MPPDU. The MACsec Integrity Check Value (ICV) provides superior protection against deliberate or inadvertent modification of data.

Hiding an original user data frame's MAC destination and source addresses is not necessarily a privacy requirement since the addresses can identify shared service devices, such as routers, and can be the same for all the original user data frames transiting a given link. However MPPDUs always encapsulate these addresses. Encapsulated data frames are independent of one another and, once received, of other user frames or information carried in the same MPPDU. The self-describing format of MPPDUs permits a range of PrY transmission strategies compatible with mandatory PrY reception behavior. However, fine tuning of MPPDU sizes and transmission timing can itself reveal information about the intended mix of user protocols, as can any temporary adjustments to those parameters to better accommodate those protocols.

MAC privacy protection is most effective when all the user data frames passing between two PrYs are protected. A network administrator can decide that MACsec integrity and confidentiality protection is sufficient for some user protocols, and that privacy protection is not to be applied to those protocols. Such protocols might already use frames of fixed sizes and MAC addresses that provide an external observer with little information (e.g. use of the Nearest Customer Bridge group address as the MAC Destination Address). A receiving PrY accepts such frames without modification.

17.3 Quality of Service impact and mitigation

MAC Privacy protection can impact quality of service by:

a) Increasing the number of octets required to encode individual user data frames within MPPDUs.

b) Adding padding to MPPDUs.

c) Requesting MACsec protected MPPDU transmission using a different priority (the access priority) from the original user priority.

NOTE—The user priority accompanying each transmit request (see IEEE Std 802.1AC) is locally determined or derived from the value associated with a received and subsequently forwarded frame as encoded, e.g., in the received frame’s VLAN tag (see IEEE Std 802.1Q).

d) Hiding the value of the drop eligibility parameter associated with individual user data frames

e) Transmitting MPPDUs at intervals that are fixed, or at least uncorrelated with the user data frame sizes or transmission queue occupancy.

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17.3.1 MPPDU encoding and padding

The encoding and padding octets added by privacy protection can be viewed either as delaying the completion of transmission or as delaying the completion of reception. On reception, a privacy protected user data frame cannot be passed to the receiving service user before the MACsec ICV protecting the entire MPPDU has been validated. The additional transmission delay is not necessarily of significant concern on network connections that are most exposed to unauthorized observers and have a high bandwidth delay product, e.g. across a Provider Bridge Network (PBN), where the effect of the delay can be equivalent to a modest increase in the distance between MAC Privacy protection systems.

Additional octets also reduce the total bandwidth available, potentially delaying subsequent frames. Multiple user data frames can be encapsulated in a single MPPDU, using space that might otherwise be occupied by padding octets. Encapsulating multiple data frames in this way can improve bandwidth efficiency as the media-dependent overhead of sending separate frames, as well as the number of octets required for MACsec protection, is amortized over several user data frames.

17.3.2 MPPDU transmission priority

The access priority (i.e. the priority used to request transmission) of the MPPDU encapsulating a user data frame can be the same as that of the frame's user priority. However, configuring a PrY to transmit fixed sized MPPDUs of several different access priorities on a fixed schedule for several access priorities could degrade the effectiveness of user priority: transmission of a scheduled low priority MPPDU might postpone the transmission of a high priority user data frame. If the next user data frame to be transmitted is to be encapsulated in an MPPDU with a different access priority, the opportunity to encapsulate an additional user data frame in the MPPDU (instead of adding padding) would be lost; and the number of MPPDUs transmitted without conveying a user data frame (i.e. containing only padding) can increase.

NOTE—A PrY does not dictate the transmission order of user data frames or buffer those frames to improve MPPDU packing. In a bridge, the next frame to transmit through an interface is determined by the transmission selection process (see IEEE Std 802.1Q) when an opportunity to transmit arises.

17.3.3 Encoding multiple user data frames in a single MPPDU

If successive user data frames are to be encoded in MPPDUs with the same access priority they can also be encoded in the same MPPDU. All PrY implementations are capable of encoding multiple successive user data frames in the same MPPDU (subject to MPPDU size constraints), if those user data frames are available for transmission before transmission of the MPPDU begins. A PrY can also encapsulate user data frames that become available for transmission after MPPDU transmission has started, and can add padding at the beginning of the MPPDU's data field (if scheduled transmission has begun before a user data frame is available) or between encapsulated data frames.

The late addition of a user data frame to an in-progress MPPDU transmission reduces the transmit delay that would otherwise be experienced by that user data frame. A series of pads can be added between encapsulated frames to allow for the earliest addition of unscheduled frames. To limit the implementation-dependent workload imposed on a receiving PrY that has to scan through multiple pads, any two back to back pads should have a combined length of greater than 64 octets (see Clause 19).

User data frames can be fragmented, using space in fixed sized MPPDUs that would otherwise be padded, reducing the fluctuations in available bandwidth that might otherwise result from the interaction of separate user flows. Frames that can be fragmented are identified as preemptable frames or (more urgent) express frames. One or more express frames that become available for transmission after a preemptable Frame

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Fragment (19.5.4) in one MPPDU has been transmitted and before a following scheduled MPPDU transmission can preempt subsequent preemptable Frame Fragments. Express Frame Fragments can also be conveyed in MPPDUs transmitted with a higher access priority so their reception can be interleaved with the reception of previously transmitted preemptable Frame Fragments.

NOTE—In this standard all unqualified references to “fragments” and “fragmentation” are to MPPDU encoding.

The individual fragments that compose a given user data frame are transmitted with an incrementing sequence number, with separate sequence number spaces for express and preemptable Frame Fragments (19.5.4), and Initial and Final fragment marking of the first and last fragments. A conformant PrY is required to be able to reassemble a frame from a sequence of fragments received with successively incrementing sequence numbers, with Initial and Final marking. Sequence numbering is not required for unfragmented frames.

NOTE 1— A fragment marked as both Initial and Final is a complete valid frame.

NOTE 2—When a MAC Privacy protection system uses link aggregation (11.5, 21.4), PrYs (like SecYs, see 11.5) are positioned below the Link Aggregation Collection and Distribution functions so all the fragments of any given frame are transmitted through a single interface, and are received in order if the underlying service is order preserving.

NOTE 3—The MPPDU encoding described does permit reassembly of frame fragments received out of order.

17.3.4 Drop eligibility

A single MPPDU can convey multiple user data frames (17.4.2) with different values of the drop_eligible parameter (see IEEE Std 802.1AC). In that case the values of the drop_eligible parameter of those user data frames does not influence the choice of MPPDU access priority, and the drop_eligible parameter associated with the MPPDU transmission is always False.

If an encapsulated user data frame contains a VLAN tag, the priority and drop_eligible information in that tag is available to the receiving PrY’s service user.

NOTE—Frames can be subject to traffic shaping, e.g. by the Forwarding Process of a bridge component (see IEEE Std 802.1Q). Traffic shaping can also be performed after user data frames are encapsulated in MPPDUs, e.g. by the network side component of an EDE-CS or EDE-CC.

17.3.5 MPPDU transmission scheduling

The transmission of MPPDUs can be delayed to allow their transmission at regular intervals (17.4), but is not otherwise delayed (e.g. in the expectation of further data frames becoming available for transmission in the same MPPDU). Explicit Pads (i.e. pads whose length is specified) can be encoded in an MPPDU so user data frames that become available for transmission after part of an MPPDU has been transmitted can be encoded in that MPPDU, reducing the delay experienced by user data frames associated with time-sensitive streams. If the PrY is part of an interface stack supporting the transmission of user data frames subject to the operation of a stream gate control list, MPPDU transmissions can be scheduled to allow the encoding of those user data frames towards the end of the MPPDU. Similarly, eligibility times for user data frame transmission can reflect the fact that those frames will be received after the MPPDU has been completely received, validated, and decoded.

NOTE—IEEE Std 802.1Q specifies stream gate control lists in support of Per-Stream Filtering and Policing (PSFP) and the calculation of eligibility times for Asynchronous Traffic Shaping (ATS).

17.4 Configuring MAC Privacy protection

MAC Privacy protection can be used wherever MAC Security can be deployed, with point-to-point, multipoint, or point-to-multipoint connectivity between peer PrYs. The use of privacy protection is an administrative decision, resulting in management configuration of each transmitting PrY. A PrY is always capable of transparently receiving user data frames that have not been privacy protected, so systems that incorporate PrYs can be deployed before privacy protected transmission is enabled.

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A PrY is also always capable of receiving MPPDUs addressed to that PrY. A transmitting PrY can choose privacy protection and encapsulation strategies (e.g. protection or not, padding, MPPDU sizes, control over transmission timing) that depend on user data frame attributes without having to communicate those choices to the receiving PrY(s). This standard facilitates network management by specifying management controls that each conformant PrY is required to support (i.e. can be configured to use). A configurable Privacy Selection Table (17.4.3) allows per user priority selection of one of the following ways of transmitting a user data frame:

a) Unmodified, without MAC Privacy protection.

b) In a Privacy Frame (17.4.1).

   A Privacy Frame encapsulates a single user data frame, hiding that frame's MAC addresses, and can be padded to obscure its size.

c) In a Privacy Channel (17.4.2).

   A Privacy Channel provides control over transmission timing, can carry multiple user data frames in a single MPPDU, and can fragment user data frames over successive MPPDUs to reduce the bandwidth inefficiencies and variations that can result from using fixed size MPPDUs.

NOTE 1—User priority commonly distinguishes frames associated with broad application classes (see IEEE Std 802.1Q) such as network critical traffic or flows requiring bandwidth allocation.

A receiving PrY processes all MPPDUs alike: the use of any given MPPDU as a Privacy Frame or to support a Privacy Channel is not encoded in the MPPDU. Each MPPDU comprises a MAC Privacy protection EtherType followed by one or more MPPDU components (19.5). Each of these components (conveying entire user data frames or fragments of those frames) is self describing, and can be separately extracted from the received stream of MPPDUs before each user data frame (reassembled from fragments if necessary) is passed to the PrY's user. A transmitting PrY may support additional MPPDU encapsulation and scheduling algorithms in addition to those that can be configured using the Privacy Selection Table, Privacy Frames, and Privacy Channels as specified in this standard.

NOTE 2—The specified PrY reception behavior permits a range of transmitting PrY encapsulation, MPPDU sizing, and scheduling algorithms beyond that specified for standardized management. If additional privacy protection enhancements require a transmitting PrY to know a receiving PrYs capabilities, the MACsec Key Agreement protocol (MKA) provides a suitable secure transport for standardized parameters.

Each user data frame can include priority information encoded in a VLAN tag, allowing its recovery by, for example, the Bridge Port Transmit and Receive process of a VLAN Bridge (IEEE Std 802.1Q). The priority used to transmit an MPPDU is not used on receipt and could have been modified as the MPPDU was forwarded to the receiving PrY.

The destination MAC Address of MPPDUs transmitted by a PrY located in the same interface stack as its associated SecY is the PAE Group Address configured for use by MKA in support of that SecY (Table 15-1, Table 15-2), and does not need to be separately configured. If MKA does not operate and discover that SecY's peers, MPPDU encapsulation of user data frames is disabled. MPPDU encapsulation of data frames is enabled (if administratively configured) if the SecY's peers are present, even if confidentiality protection is not being provided. The destination MAC Address used by a transmitting PrY that is in a separate system from its associated SecY (21.8) can be configured to be an individual address or a group address.

NOTE 3—The use of the PAE Group Address with the check that peer SecYs are using that address, defends (in the case of SecY and PrY collocation) against the inadvertent broadcast of MPPDUs throughout the connected network.

A PrY relies on its associated SecY's operation of MACsec for the data integrity, data confidentiality, and data origin authenticity of the MPPDUs transmitted by peer PrYs. No additional parameters are required for MAC Privacy protection protocol operation.

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17.4.1 Privacy Frames

A Privacy Frame encapsulates a single user data frame.

A Privacy Frame's MPPDU components comprise an Encapsulated Frame (19.5.1) and, if required, a Trailing Pad (19.5.2). The PrY management controls specified by this standard provide for a user priority dependent constraint (privacyPadding, 20.7) on the size of the MPPDU to four octets [to allow for the MAC Privacy protection EtherType and the MAC Privacy protection Protocol Component Identifier (MPPCI)] plus a multiple of 16, 32, or 64 octets. The specified size does not include the original user data frame's FCS (not carried in the MPDU), the MPPDU's own source and destination MAC addresses, and SecTAG and ICV added by MACsec when confidentiality protecting the frame, nor does it include additional tags that can be subsequently added by other components of a system prior to physical transmission such as a VLAN tag added by the network component of an EDE (Clause 15). The maximum size of the MPPDU (for a given user priority) is the maximum size of the user data frame specified for the PrY's user's Private Port interface (20.1) plus the number of octets required to encode and pad that maximum sized frame.

NOTE—The number of octets that the PrY uses to encapsulate a given user data frame in a Privacy Frame can be calculated from minimum protected frame size and the size increment specified in the Privacy Selection Table. Those parameters are available to other system components through the Pry's LMI (there is little reason to limit their visibility within the system of which the PrY is a part, since they could probably be deduced by an adversary observing the transmitted traffic) and can be used, e.g., by a bridge's MAC Relay Entity (8.6.8), as part of scheduling transmissions.

The use of Privacy Frames as opposed to Privacy Channels to convey individual user data frames reflects a decision on the part of the network administrator that any leakage of PCI resulting from changes in their sizes or transmission timing is unimportant (e.g. they might encapsulate network routing traffic, rather than packets sent by network applications). Where the underlying service provided to communicating PrYs does not usefully distinguish between frames transmitted with different priorities, Privacy Frames that encapsulate user data frames of different user priority can be sent with the same access priority, concealing the original priority from any observer. Where the original user priority is encoded in a VLAN tag in the user data frame, that priority can be recovered by the receiving PrY's user.

Privacy Frames, and their component Encapsulated Frames, are not queued for transmission by a PrY. Their transmission, via a transmit request at the PrY's supporting Controlled Port, is a direct result of the transmit request made by the PrY's user at its Private Port interface.

17.4.2 Privacy Channels

A Privacy Channel facilitates the regular transmission of fixed sized MPPDUs.

A Privacy Channel's MPPDU's components comprise zero or more Encapsulated Frames, zero or more Encapsulated Frame Fragments, zero or more Explicit Pads, and zero or one Trailing Pad (19.5). A conformant PrY can be configured to support a Preemptable Privacy Channel, and/or an Express Privacy Channel. If both Privacy Channels are enabled, user data frames identified by the Privacy Selection Table (17.4.3) as allocated to a given Privacy Channel are conveyed by that Privacy Channel. If a single Privacy Channel is enabled frames allocated to either Privacy Channel are conveyed by the enabled Privacy Channel, and if neither Privacy Channel is enabled they are transmitted as Privacy Frames using the other parameters (accessPriority, revealDE, and privacyPadding) configured for each user priority in the Privacy Selection Table.

Each Privacy Channel is characterized by the following parameters:

a) The user data frame size (userDataFrameSize) in octets.

   This parameter specifies the maximum sized user data frame that can be encapsulated in a Privacy Channel MPPDU as an Encapsulated Frame (19.5.1). It includes the MAC addresses, user data, and FCS of the user data frame. The fixed size MPPDU used (prior to the application of MACsec confidentiality protection, and the FCS added by the specific media access method used to transmit

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the protected frame) includes 12 additional octets (including the PrY MAC Addresses, MAC Privacy protection EtherType, Encapsulated Frame component identifier, but excluding the original FCS which is not encoded in the MPPDU).

b) The requested bit rate (requestedKbitRate) in kilobits per second.

In the absence of individual user data frames that are not privacy protected or that are conveyed as Privacy Frames, MPPDUs for the Privacy Channel are transmitted at regular intervals determined by the userDataFrameSize and requestedKbitRate when the default Channel MPPDU generation algorithm is used. See 20.9.4 for detailed calculations.

NOTE 1—Individual frames that can compete for transmission bandwidth can be sent by the PrY's user, but can also be sent by other protocol entities that insert frames for transmission lower in the interface stack, e.g. periodic and as required MACsec Key Agreement (MKA) transmissions through the SecY's Uncontrolled Port. Any resulting delay in the transmission of Privacy Channel MPPDUs does not reveal information about the content of those MPPDUs.

NOTE 2—The requestedKbitRate and userBurstOctets parameters are used to provide compatibility with the specifications for handling time-sensitive traffic in IEEE Std 802.1Q, including Asynchronous Traffic Shaping (ATS). Privacy Channel MPPDU transmission can also be scheduled to support the use of transmission gates and a gate control list (20.9.5). A bridge that supports per-stream filtering and policing (PSFP, IEEE Std 802.1Q) can use a number of criteria including priority to gate the transmission of frames of individual streams by bridge ports, and thus remove or reduce the impact of potential PrY scheduling conflicts on time-sensitive streams.

c) The permitted burst size (userBurstOctets).

This parameter allows temporarily rapid transmission of Privacy Channel MPPDUs to recover bandwidth lost to competing frames (20.9.4).

d) The access priority used to transmit the Privacy Channel's MPPDUs.

NOTE 3—If the PrY and SecY form part of the Provider Edge Port of an EDE-CS or EDE-CC (see Figures 15-6, 15-7, and 15-8) the SecY's transmitted access priority is encoded by the black-side bridge component in the outer VLAN tag added to the protected frames transmitted by the Provider Network Port.

The underlying service used by the PrY is expected to preserve the transmission order of frames of any given priority. If the interface stack (or the interface stacks of frame forwarding devices on the path between peer PrYs) supports IEEE Std 802.3 frame preemption or other class of service differentiated forwarding capabilities, the access priority of the Express Privacy Channel can be configured to take advantage of those capabilities.

17.4.3 Privacy Selection Table

A configurable Privacy Selection Table has an entry for each of the eight possible user priority values (0 through 7). Each entry indicates whether user data frames of that priority are transmitted unmodified, in a Privacy Frame, or in a Privacy Channel (17.4). If the frames are to be transmitted in a Privacy Channel they are classified as preemptable frames or express frames, in case they need to be fragmented (17.4).

Figure 17-3 illustrates the use of a PrY's Privacy Selection Table in conjunction with the Traffic Class and Access Priority Tables specified for a SecY (10.7.17). Each user data frame transmission request to the PrY (from, for example, the Bridge Port Transmit and Receive function as illustrated in Figure 17-3 and specified by IEEE Std 802.1Q) is shown at the top of the figure. Frames with a user priority of 0 or 1 are assigned to the Preemptable Privacy Channel, with access priority 0. The PrY is the user of the service provided by the SecY, so that communicated priority value is the SecY's user priority, and is used (by the SecY's Traffic Class Table) to assign the MPPDUs conveying to Secure Channel (SC) 0 requesting priority 0 (as specified by the SecY's Access Priority Table) from the underlying medium when the MACsec protected frames are transmitted.

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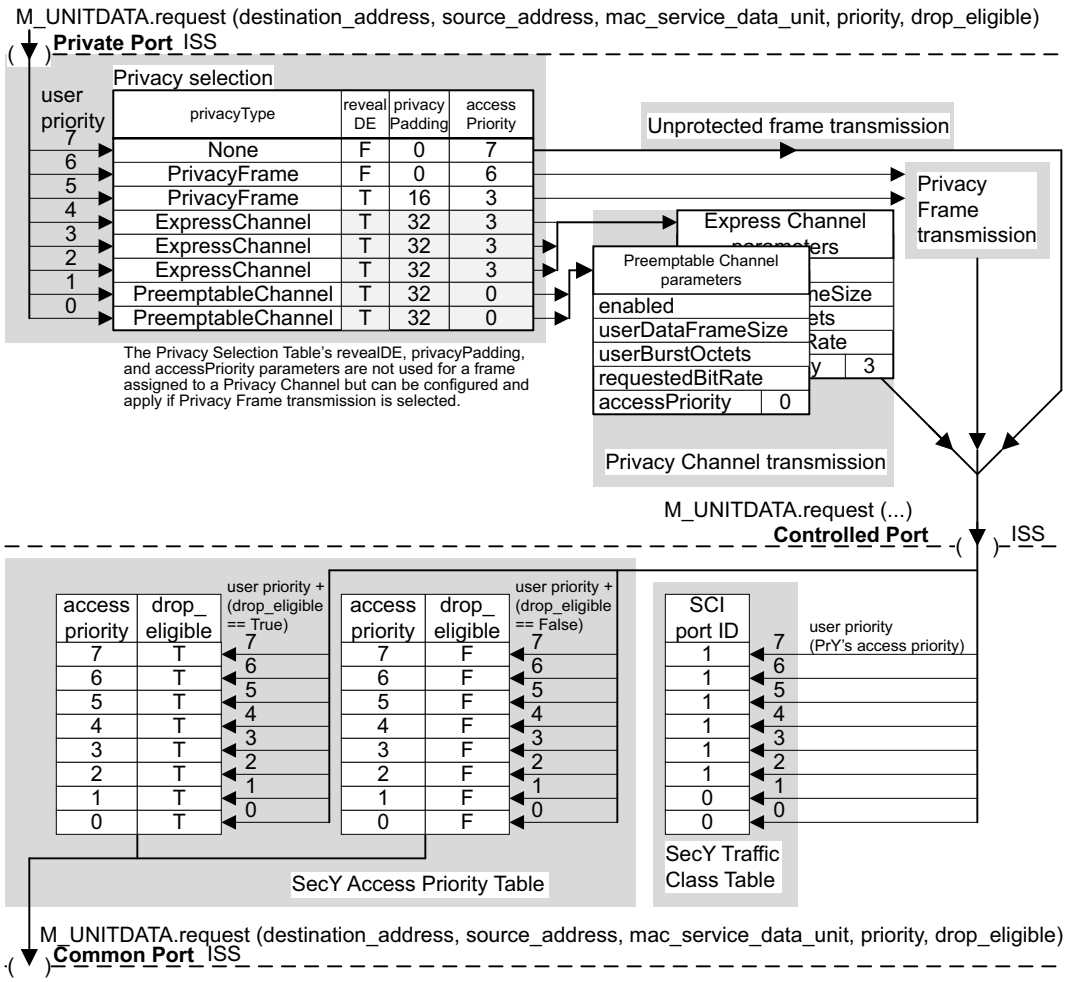


Figure 17-3—Privacy selection, priority and traffic class mapping

Similarly, the PrY assigns frames with user priorities of 2 through 4 to the Express Privacy Channel. They are conveyed by that Privacy Channel (since it has been configured) and transmitted in MPPDUs with access priority 3. The SecY assigns these to SC 1.

NOTE—Different values of PrY access priority are only relevant if they select differentiated services from a supporting SecY (between two transmit SCs, for example) or from the media access method or other protocol entity that supports the SecY's Controlled Port (providing frame preemption, for example).

In Figure 17-3 user data frames with a user priority of 5, 6, or 7 are not assigned to a Privacy Channel. Their transmission is not constrained by a channel schedule and can occur at any time, even though (with the configuration shown) they are assigned to the same SC. Their transmission can delay a Privacy Channel's scheduled transmissions, but this timing conflict and delay does not expose Privacy Channel PCI, unless the original source(s) of those frames correlated their transmission with frames carried by the Privacy Channel.

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18. MAC Privacy protection protocol

The MAC Privacy protection protocol comprises rules for the following:

a) Encapsulation of user data frames in MAC Privacy protection Protocol Data Units (MPPDUs)
b) Addressing of MPPDUs
c) MPPDU decoding, and the recovery of user data frames from a stream of MPPDUs.

A given user data frame can be encoded in one or more MPPDUs, each with the same source and the same individual or group destination MPPDU MAC Address. Once protected by MACsec, these MPPDUs enhance the privacy of user communication between that source and destination(s). The structure of MPPDUs is self-describing and allows a recipient of MPPDUs to recover encapsulated user data frames without the need to share additional parameters with their source.

Clause 17 provides an overview of MAC Privacy protection, and explains why user data frames need to be encapsulated in MPPDUs as well as being integrity and confidentiality protected by MACsec. MPPDU encoding is specified in Clause 19 and their transmission and reception by Privacy protection entities (PrYs) in Clause 20. Clause 21 describes how MAC Privacy protection is supported in a number of interoperable interface stack and system configurations.

This clause adds details of design and support requirements not already described in Clause 17 and provides additional context for other clauses in this standard, as follows:

— Addressing (18.1)
— Data origin authenticity, frame data integrity and confidentiality (18.2)
— Applicability (18.3)
— Bandwidth utilization, fragmentation, and transit delay (18.4)
— Coexistence and use (18.5)

18.1 Addressing

The intended recipient of an MPPDU is identified by the destination MAC Address (DA) of the transmitted MPPDU, and the transmitter by its source MAC Address (SA). The recipient identifies those MPPDUs that can convey a given fragmented user data frame by their DA, SA pair. The DA can be a group address and can identify one or more recipients. All MPPDUs transmitted by a given PrY use the same DA and SA.

A receiving PrY only decodes MPPDUs addressed to that PrY (i.e. MPPDUs with an individual or group MAC Destination Address associated with that PrY). MPPDUs destined for other addresses are passed to the PrY's user without modification. This forwarding permits system by system deployment of privacy protection capability in a network, and layered deployment as illustrated for MACsec in Figure 11-12 and Figure 11-13.

NOTE 1—Bridge components within the transmitting system can add SecTAGs or VLAN tags to the transmitted MPPDUs, e.g. as described in Figure 17-2 and Clause 21. Network components between the transmitter and the intended recipient(s) can also add tags, and use those tags to direct frames to a particular network region, potentially avoiding MAC address allocation conflicts, e.g. between different customer networks attached to a Provider Bridged Network (PBN). Peer network components are responsible for removing those added tags before the MPPDU is delivered to the intended recipient(s).

NOTE 2—A SecY can use SecTAGs with different SCIs to support MACsec with multiple access priorities while retaining strict delivery and replay protection within an access priority. The normative text above (18.1) deliberately excludes both the use of the SCI to distinguish streams of MPPDUs between the same DA, SA pair and any suggestion that an implementation needs to be capable of reassembling fragmented user data frames on a finer granularity than provided for by the combination of DA, SA, and the Express/Preemptable classification (19.5.4). All Frame Fragments with the same value for Express/Preemptable transmitted between a given DA, SA pair are conveyed in MPPDUs transmitted with the same access priority.

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When a PrY is directly supported by a SecY, the destination MAC Address (DA) used to transmit MPPDUs and to recognize received MPPDUs sent to that PrY shall be the PAE Group Address configured for use by MKA (see IEEE Std 802.1X) in support of that SecY.

When a PrY is directly supported by a SecY, the source address of each MPPDU that it transmits shall be the MAC Address used to transmit MKPDUs and to generate the SCI(s) used by that SecY. If the secure Connectivity Association (CA) established for MACsec use is a group CA, MKA checks that each SCI is unique within the scope of the CA, and so establishes the required uniqueness for use of the MPPDU SA to identify the reassembly context for fragments received from peer PrYs. Thus neither the MPPDU DA nor the MPPDU SA need be configured. If the receiving interface stack is a Bridge Port that does not include an active PrY, received MPPDUs are discarded and not relayed by the Bridge.

When a PrY is separated from the supporting SecY (see 21.8), the destination MAC Address of the MPPDUs that the PrY transmits needs to be configured. Care needs to be taken to avoid the use of an unrecognized individual or group address that could be flooded throughout a network. Successful configuration of privacy protected connectivity can be verified, if necessary, by sending a suitable CFM message (see IEEE Std 802.1Q) in Privacy Frames to elicit a response from each peer.

When a PrY is separated from the supporting SecY, the destination MAC Address of the MPPDUs that it recognizes for reception shall be either the individual MAC address that it uses as the source MAC Address (SA) of the MPPDUs that the PrY transmits or a configured group address (20.14).

When a PrY is separated from the supporting SecY, the source address of each MPPDU that it transmits shall be an individual address that is associated with the interface stack of which the PrY is a part and is persistent across reinitialization of system components.

See 21.1.1 for additional addressing considerations.

18.2 Data origin authenticity, frame data integrity and confidentiality

MAC Privacy protection relies on the use of MACsec and its supporting authentication, authorization, and key agreement protocols to ensure that:

a) MPPDUs are only sent to and received from authenticated and authorized peer PrYs.

b) The MPPDU destination and source MAC addresses and data are received as sent, without modification.

c) The MPPDU data is kept confidential, and only available to the intended recipient PrYs.

18.3 Applicability

Privacy protection can be provided by encapsulating user data frames in MPPDUs wherever MACsec is applicable and does not compromise MACsec's ability to meet the requirements described in 8.1.

Additional considerations do apply when privacy protection is supported by shared media, as follows:

a) True shared media—where each frame transmission is available to a set of potential recipients without the possibility of intermediate network systems selectively forwarding any given frame to a subset of those recipients.

b) Virtual shared media—where intermediate systems can deliver group addressed frames to one or more sets of potential recipients while selectively delivering frames to individual destination MAC addresses.

NOTE 1—In the case of the Ethernet Data Encryption (EDEs) described in Clause 15, virtual shared media considerations do not apply as the VLAN tagged service selection restricts each EDE PEP to a single peer.

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In both cases a system transmitting on the shared media could choose to encapsulate some user data frames in group addressed MPPDUs (for reception by all of its peers) and some (each destined for a specific peer) in individually addressed frames. A requirement to keep the nature and level of current activity private would then require a constant level of Privacy Channel transmission for each DA, SA pair.

In the case of true shared media as, e.g., originally envisaged for Ethernet using CSMA/CD, using Privacy Channels with user data frame fragmentation requires each recipient to be capable of supporting the simultaneous reassembly of at least one Express user data frame and one Preemptable user data frame from each of its peers. The use of individually addressed MPPDUs does not reduce that requirement, and is not recommended.

In the case of virtual shared media the use of individual MPPDU destination addresses can significantly reduce intermediate network resource usage, even when there is a privacy requirement to maintain a constant level of activity. The applicable considerations are similar to those for Provider Backbone Bridged Networks (PBBNs, see IEEE Std 802.1Q) with the additional restriction that successive fragments of any given user data frame need to be conveyed in MPPDUs with the same DA. User data frames with any given destination address should be consistently encapsulated in MPPDUs with the same DA. User data frames can be replicated by a bridge component and transmitted through multiple ports on that component, in separate MPPDUs, if they need to be flooded to multiple destinations.

NOTE 2—Bridged networks can flood frames to individual end stations whose addresses have not been learnt, but a newly active station usually transmits one or more group addressed frames to seek peers or acquire configuration information and waits for a response, allowing other bridges to learn its location, before it consumes much bandwidth.

NOTE 3—A Privacy protection entity (PrY, Clause 20) uses the same DA to transmit all MPPDUs.

18.4 Bandwidth utilization, fragmentation, and transit delay

The MAC Privacy protocol allows frames to be padded to arbitrary lengths, and thus allows transmission of fixed sized MPPDUs that conceal the size of user data frames. Use of fixed sized MPPDUs carries with it the possibility of inefficient bandwidth utilization, as the next user data frame to be transmitted might be at or close to the maximum size permitted and not quite fit. MPPDU encoding allows fragmentation, to allow most if not all of the MPPDU to carry user data. The resulting improvement in bandwidth utilization can also reduce the transit delay experienced by individual user data frames.

The MAC Privacy protocol does not constrain the timing of MPPDU transmission, MPPDUs can be transmitted at fixed intervals to conceal the timing characteristics of network applications and the level of network activity. In the interval between MPPDU transmissions one or more high priority user data frames (identified as Express frames) can become available for transmission. Those Express frames, possibly fragmented, can be transmitted before any remaining fragment of lower priority frames (identified as Preemptable frames).

The MPPDU encoding rules (Clause 19) constrain the use of fragmentation and padding to facilitate interoperability, allowing externally observable implementation characteristics to be specified and tested, as follows:

a) The minimum size of each fragment is restricted, as is the minimum size of pads other than the last.

b) The fragments of Express and Preemptable user data frames are independently sequence numbered, and the initial and final fragments of each user data frame identified.

c) A conformant receiver can rely on in-order delivery of MPPDUs, need only be capable of both reassembling one Express and reassembling one Preemptable user data frame from each of its peers, and delivering each user data frame in the order that its reception is complete.

NOTE 1—A conformant implementation using the mandatory to implement default MPPDU encapsulation algorithm (20.10.1, Figure 20-3) encodes at most two fragments in any given MPPDU as follows: a final fragment (Express or Preemptable), followed by zero or more complete user data frames, followed by an initial fragment (Express or

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Preemptable) followed by zero or more octets of pad. A conformant implementation that fragments Preemptable frames to allow for the latest possible addition of Express frames can transmit MPPDUs that encode minimum sized Preemptable frame fragments interspersed with minimum sized Express frames, followed by zero or more octets of pad.

NOTE 2—An MPPDU that conveys multiple small user data frames can use less bandwidth than the independent transmission of those frames.

NOTE 3—The MAC Privacy protocol does not enforce in-order delivery of user data frames, but a supporting SecY can be configured to provide strict replay protection and in-order delivery of MPPDUs and other frames, resulting in transmission order reception of all user data frames of a given traffic class.

Express and Preemptable fragments are independently numbered, each using a 24-bit sequence number, that is incremented by the transmitter for each fragment sent. A conformant transmitter does not restart or change the order of the sequence numbers used unless at least 2 seconds has elapsed since the MPPDU with the prior sequence number has been transmitted. A conformant receiver retains fragments pending reassembly of a user data frame for at most 0.1 second. If the PrY's Controlled Port's MAC_Operational parameter becomes false, any retained fragments are discarded (20.13, 20.13.1).

NOTE 4—The 24-bit sequence number space does not wrap within the specified fragment reassembly time for a 1 Tb/s transmitter using the default MPPDU encapsulation algorithm and an MPPDU capable of conveying a maximum sized user data frame, even if the traffic load constantly forces the transmission of two fragments in each MPPDU. User data frames that are encapsulated without fragmentation are not sequence numbered.

18.5 Coexistence and use

Privacy protected user data frames can be transmitted between the same pair of stations as, and interspersed with, user data frames that are not privacy protected. The choice of whether any particular frame is privacy protected or not is made by the transmitter and can be changed without reference to the receiver, and without reordering user data frames.

A system that transmits MPPDUs can encode user data frames in those MPPDUs in any way permitted by the encoding rules (Clause 19), encoding one or multiple user data frames in each MPPDU and transmitting each MPPDU with any value of access priority, subject only to the rules for fragmenting user data frames which limit the complexity of the receiving reassembly process. The transmission of a sequence of MPPDUs that convey fragments of any given user data frame (classified as fragments of a preemptable frame or an express frame) can be interspersed with the transmission of user data frames that are not privacy protected or of MPPDUs that do not convey fragments of the same classification, up to the fragment reassembly time limit.

NOTE 1—This standard supports a common approach to transmission behavior, and management of that behavior, with a PrY specification that support transmission of Privacy Frames (MPPDUs that convey a single user data frame) and up to two Privacy Channels (sequences of MPPDUs that convey one or more user data frames with the possibility of fragmentation). However the encoding of MPPDUs does not distinguish Privacy Frames and Privacy Channel MPPDUs, and PrY specification supports the reception and recovery of user data frames from any MPPDU or sequence of MPPDUs encoded as specified in Clause 19.

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19. Encoding of MAC Privacy protection Protocol Data Units

This clause specifies the structure and encoding of the MAC Privacy protection Protocol Data Units (MPPDUs) exchanged between MAC Privacy protection Entities (PrYs). It specifies and describes the following:

a) Rules for the structure, representation, and encoding of protocol fields (19.1)

b) The general format of MPPDUs (19.2)

c) The allocation of the MAC Privacy protection EtherType to identify MPPDUs (19.3)

d) The strategy for interoperability with possible future protocol versions (19.4).

e) The encoding of MPPDU components (19.5)

f) Rules for the generation of MPPDUs using the specified components (19.6)

g) Validation and extraction of MPPDU components on reception (19.7)

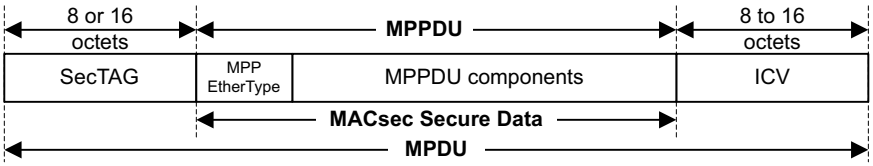
NOTE—The MPPDU validation checks specified do not overlap with the specification of PrY operation (Clause 20).

19.1 Structure, representation, and encoding

All MPPDUs contain an integer number of octets. Octets and bits in the text and figures in this specification are represented and numbered, and values are encoded, using the conventions specified in 9.1.

19.2 MPPDU Format

Figure 19-1 shows a MPPDU that has been confidentiality protected by MACsec. The MACsec Secure Data includes the MAC Privacy protection EtherType (19.3) and one or more MPPDU components (19.5). Each component encodes a user data frame (or a fragment of a user data frame) or padding.



The source and destination MAC addresses of the communicating PrYs are not shown, as these are separate parameters of the service requests and indications that support transmission and reception of the MACsec protected MPPDU (see Figure 8-1).

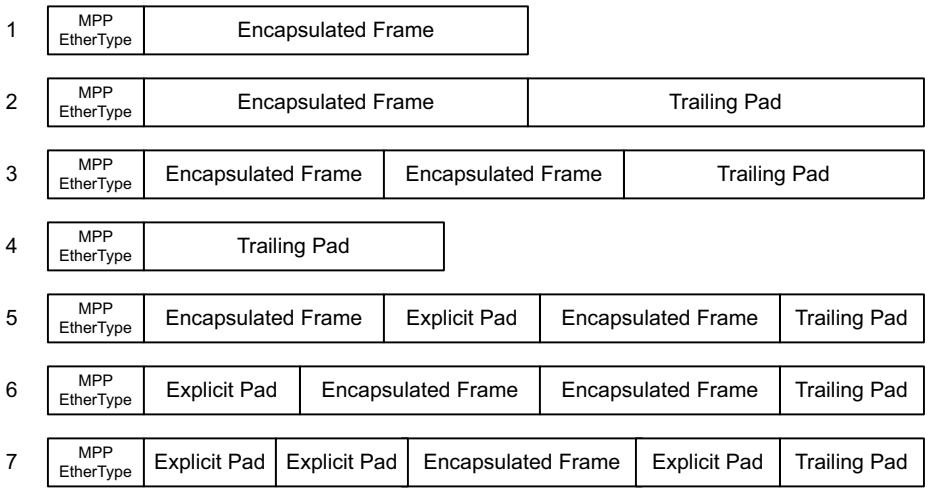
Figure 19-1—MACsec protected MPPDU

Figure 19-2 illustrates some possible MPPDU component combinations.

The first two examples show a single Encapsulated Frame, concealing (once the MPPDU has been MACsec confidentiality protected) the encapsulated user data frame's source and destination MAC addresses. The addition of a Trailing Pad in the second example obscures the length of that frame.

The third example shows multiple Encapsulated Frames in a single MPPDU, again with a Trailing Pad. An observer of the confidentiality protected frame is unaware of how many user data frames have been encoded or how much padding has been used. The only MPPDU component in the protected frame might be a Trailing Pad, as shown in the fourth example, sent to obscure changes in network activity when no user data frame is available for transmission. The remaining examples show the use of Explicit Pads, each with a specified length, used to allow the possible later addition of one or more Encapsulated Frames (19.5.1).

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In each of these examples one or more of the Encapsulated Frame components could have been a Frame Fragment (19.5.4)

Figure 19-2—MPPDU Examples

NOTE 1—Not all PrYs and PrY configurations need to be capable of transmitting MPPDUs with all the component possibilities shown. A conformant PrY (5.10, 5.11, Clause 20) need only be capable of transmitting Privacy Frames and sequences of MPPDUs that support Privacy Channels. A Privacy Frame (17.4.1) is an MPPDU that includes a single Encapsulated Frame, conveying a single unfragmented user data frame, with a possible Trailing Pad. A Privacy Channel (17.4.2, 20.8, 20.10) can be supported without the use of Explicit Pads. However all conformant PrYs are required to be capable of receiving all correctly encoded MPPDUs (17.4).

NOTE 2—The use of any given MPPDU as a Privacy Frame, or to support a Privacy Channel, or as part of any other encapsulation strategy is not encoded in the MPPDU (17.4).

19.3 MAC Privacy protection EtherType

The MAC Privacy protection EtherType (Table 19-1) comprises octet 1 and octet 2 of each MPPDU. This EtherType is included within the MPPDU to allow:

a) Coexistence of MAC Privacy protection capable systems in the same environment as other systems.
b) Incremental deployment of MAC Privacy protection capable systems.
c) Transmission of privacy protected and unprotected user data frames on the same media and between the same systems, using the same system addresses.

Table 19-1—MAC Privacy protection EtherType allocation

Name	Value
MAC Privacy protection EtherType	E2-3B

The encoding of the MAC Privacy protection EtherType in the MPDU is illustrated in Figure 19-3.

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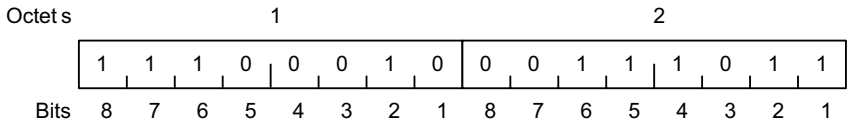


Figure 19-3—MAC Privacy protection EtherType encoding

19.4 Protocol Version strategy

This standard specifies basic MAC Privacy protection protocol capabilities and the MPPDU components that support those capabilities. MPPDUs do not include a protocol version identifier. Future revisions of this standard can specify additional MPPDU component types while supporting interoperability with implementations conformant to this standard by correctly recognizing and processing the MPPDU components specified in this clause (Clause 19).

The general format of MPPDU components is extensible (19.5), allowing future revisions of this standard to specify additional MPPDU components if necessary. This general extensible format includes a component length, intelligible to implementations conformant to the initial specification of the protocol and all future versions. All conformant implementations are thus capable of discarding unrecognized component types, while continuing to process other components that can be encoded later in an MPPDU.

This protocol version strategy is not limited to supporting the incremental addition of capabilities and component types in future revisions of this standard. It permits the addition of independent optional sets of capabilities, if that proves desirable. However the set of basic capabilities specified in this standard is conveniently identified as version 0 for management purposes.

NOTE—Any future need to communicate protocol capabilities between PrYs can be met by adding new MPPDU component types, by using an encoding that is discarded by this initial protocol version, or by adding attributes to MKA (see IEEE Std 802.1X).

19.5 MPPDU component encoding

An MPPDU component is a Trailing Pad (19.5.2) if the first octet of the component is zero and there are no more octets in the MPPDU, or the first two octets are both zero. Otherwise the component has the general format shown in Figure 19-4, with a component identifier (MPPCI) followed by zero or more octets.

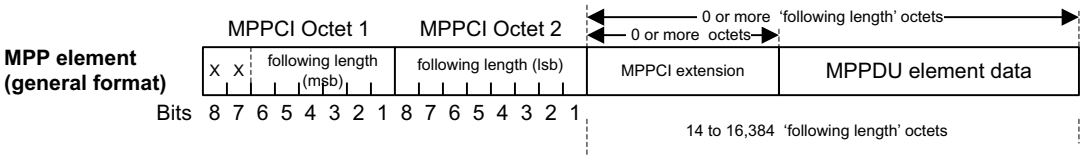
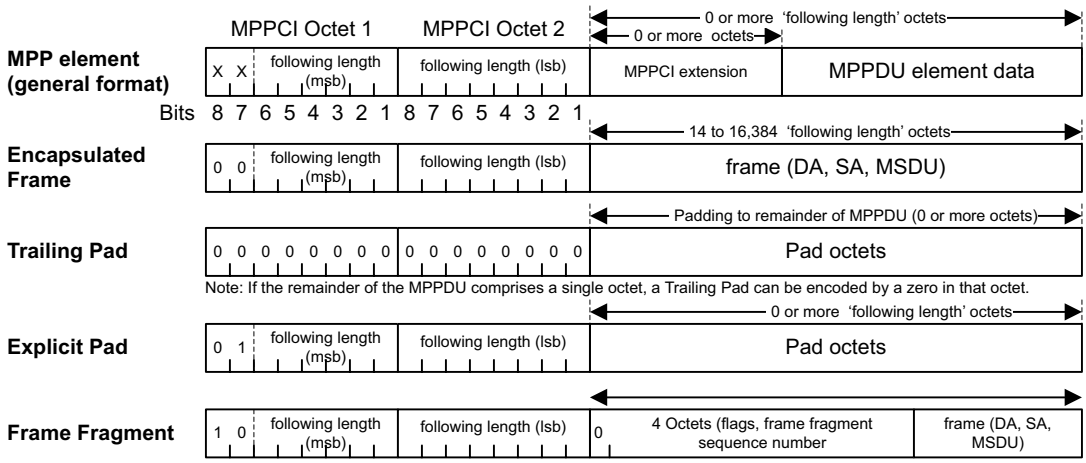


Figure 19-4—MPPDU component format

In the general format, bit 6 (most significant) through bit 1 of the first octet of the MPPDU and bits 8 through 1 (least significant) of the second octet always encode the length of the component following the first two octets. This length field allows a receiver to skip any unrecognized MPPDU component and process any following components present in the MPPDU. The components and component identifiers specified by this standard are shown in Figure 19-5 and specified in detail in 19.5.1—19.5.4.

Encapsulated Frames, Explicit Pads, and Frame Fragments can each be present zero or more times in an MPPDU and their relative position(s) in the MPPDU are not constrained. If a Trailing Pad is present, it is the last component in the MPPDU. Constraints on the use of these components in the generation of MPPDUs are specified in 19.6. Validation of received MPPDUs and of the components they contain, and the extraction of those components is specified in 19.7.

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An Encapsulated Frame has a following length of 14 octets or greater (19.5.1). While this encoding is capable of encapsulating user data frames with lengths from 14 to 16,383 octets in length, that does not imply that any specific media access control method can support frames with that entire range of lengths.

Each Frame Fragment conveys a fragment of a frame that is at least 64 octets in length (19.5.4).

The use of the bit pattern 11 in bits 8 and 7 of Octet 1 is reserved for future specification, as is the use of 10 in those bits with 1 in bit 8 of the third octet of the component.

Figure 19-5—MPPDU component encoding

19.5.1 Encapsulated Frame

An MPPDU component is identified as an Encapsulated Frame if:

a) Bit 8 and bit 7 of the first octet of the component are both zero, and

b) Bits 6 (most significant) through 1 of the first octet and bits 8 through 1 (least significant) of the second octet (least significant) encode a *following length* value of 14 or greater.

The encapsulated frame follows, beginning in the third octet of the MPPDU component. The first 6 octets of the frame encode a 48-bit MAC Destination Address in the standard encoding, i.e. in the canonical format (also Figure 10, 8.2, and Annex C of IEEE Std 802-2014), and the next 6 octets encode a 48-bit MAC source address in the standard encoding. The two following octets are the initial octets of the MSDU (the Length/Type field of IEEE Std 802.3 frames), and any subsequent octets are the remaining octets of the MSDU. The original user data frame's FCS is not encoded in the component.

The use of bit 8 and bit 7 of the first octet of an MPPDU component, together with a following *length value* of 1 through 13 is reserved for future specification. The reception of such a component is specified in 19.7.

19.5.2 Trailing Pad

An MPPDU component is a Trailing Pad if:

a) The first and second octets of the component are both zero, or

b) The first octet is zero and there are no more octets in the MPPDU.

Any octets in the MPPDU following the second octet are part of that Trailing Pad. A Trailing Pad indicates that no further components are present in the MPPDU (19.6, 19.7).

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19.5.3 Explicit Pad

An MPPDU component is an Explicit Pad if:

a) Bit 8 of the first octet is zero and bit 7 is one.

Bits 6 through 1 of octet 1 (more significant) and bits 8 through 1 of octet 2 (less significant) encode the *following length*, i.e. the number of octets following in the Explicit Pad after octet 2.

The octets in the MPPDU following octet 1 of the Explicit Pad, including octet 2 up to the *following length* count octets, are part of that Explicit Pad.

NOTE—The specification of the Explicit Pad component accommodates the possibility that the MPPDU includes fewer remaining octets than encoded in the length field in octets 1 and 2 when it is the last component in a MPPDU

19.5.4 Frame Fragment

An MPPDU component is a Frame Fragment if:

a) Bit 8 of the first octet is one and bit 7 is zero, and

b) Bit 8 of the third octet is zero.

Figure 19-6 illustrates the format of Frame Fragments.

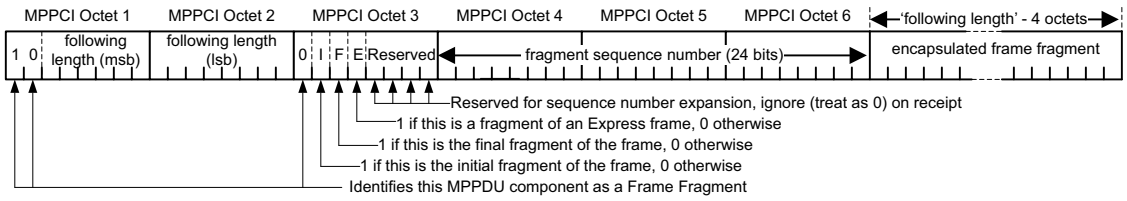


Figure 19-6—Frame Fragment

A user data frame can be conveyed in a sequence of Frame Fragments, with each successive Frame Fragment incrementing (modulo 2^{24}) a fragment sequence number encoded in the fourth (most significant) through sixth octets of the component. Two independent fragment sequence number spaces are supported, each identified by one of the two possible values (zero or one) of bit 5 of the third octet [the Express (E) bit]. The independent sequence number spaces allows the sequence of Frame Fragments encoding a given user data frame to be interrupted by, or interleaved with, Frame Fragments encoding another frame. By convention fragmented Express frames are encoded with the E bit set (i.e. when it is one) and fragmented Preemptable frames are encoded with the E bit clear.

Each encapsulated frame fragment comprises 64 or more octets (see 19.6, 19.7) i.e. the Frame Fragment's *following length* is greater than or equal to 68. A user data frame of less than 128 octets (MAC DA, SA, and MSDU) in length cannot be fragmented.

When bit 7 of the third octet [the Initial (I) bit] is set (i.e. when it is one), the encapsulated fragment is the initial fragment of a user data frame and the initial octets of the fragment (beginning with the seventh octet of the component) comprise that frame's MAC DA, SA, and Length/Type encoded protocol identifier as specified for an Encapsulated Frame (19.5.1).

When bit 6 of the third octet [the Final (F) bit] is set, the encapsulated fragment is the final fragment of a user data frame. A Frame Fragment can be encoded with both the I bit and the F bit set, in which case the entire user data frame (MAC DA, SA, and MSDU, but not any FCS) is present in the seventh octet and following octets of the Frame Fragment component.

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The MPPDU format does not identify any given MPPDU as a Privacy Frame (17.4.1) or as supporting a Privacy Channel (17.4.2). This standard distinguishes Privacy Frames and Privacy Channels simply as part of specifying mandatory to implement PrY transmission capabilities. Any given received MPPDU can include Frame Fragments with the Express (E) bit set and Frame Fragments with the E bit clear.

Bits 4 through 1 of the third octet are reserved for future specification of an extended sequence number space, and are ignored on receipt by protocol implementations conformant to this standard.

NOTE—If encoded MACsec protected MPPDUs are received from an Ethernet link running at speeds up to 400 Gb/s, with each MPPDU capable of conveying a non-fragmented 1518 octet user data frame but in fact including two Frame Fragments, the 2^{24} fragment sequence number space specified by this standard takes at least 0.25 seconds to wrap. This remains true even if bits 4 through 1 of the component's third octet are used to encode the more significant bits of an extended sequence number space.

19.5.5 Unrecognized components

The MPPDU validation process (19.7) specified for this edition of this standard treats components with any of the following MPPCI bit patterns as unrecognized:

a) In the first octet bits 8 and 7 are both one, and a second octet is present in the MPPDU.
b) In the first octet bit 8 is one and bit 7 is zero, and in the third octet bit 8 is one.
c) The first octet has the value zero, and the value of the *following length* is between 1 and 13 (inclusive).

These MPPCI values are reserved for specification by future revisions of this standard.

19.5.6 Incorrectly encoded components

The MPPDU validation process (19.7) identifies components with any of the following bit patterns as incorrectly encoded:

a) In the first octet bit 8 is one and bit 7 is zero, a second octet is present in the MPPDU, but there is no third octet.
b) In the first octet bits 8 and 7 are both zero (matching the encoding of an Encapsulated Frame) and the *following length* is greater than the number of octets remaining in the MPPDU.

19.6 MPPDU generation

Mandatory to implement MPPDU generation capability is specified for PrY operation (Clause 20). While not specifying desired behavior, the following rules constrain and permit aspects of the valid generation of MPPDUs:

a) All MPPDUs shall include the MAC Privacy protection EtherType as the first two octets.
b) An MPPDU shall only contain MPPDU components specified in this standard.
c) Any Encapsulated Frame or Frame Fragment component included in the MPPDU shall be present in its entirety as indicated by an accurate *following length*.
d) An Explicit Pad component can be encoded with a *following length* that exceeds the number of octets remaining in the MPPDU.
e) All pad octets in a Trailing Pad or an Explicit Pad shall have the value zero.

NOTE 1—This requirement guards against the accidental inclusion of data in an MPPDU, even if that data was previously transmitted to the same PrY(s). It is not validated on receipt (19.7), but is required for conformance.

f) A user data frame that is fragmented for transmission shall be encoded in a sequence of Frame Fragments having the same value of the E bit and consecutively numbered fragment sequence numbers, with the I bit set in the first Frame Fragments and the F bit set in the last.

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g) Any type of MPPDU component can be present multiple times in a MPPDU (subject to the size of the MPPDU), and the order of individual components is not constrained with the following exceptions:

1) Any Trailing Pad is the last component in the MPPDU. A receiver does not extract any further components or data from the MPPDU once the initial octets of a Trailing Pad are encountered.

2) If more than one Frame Fragment with a given value of the E bit is present in an MPPDU, those components are encoded in fragment sequence number order.

NOTE 2—A solitary Explicit Pad can provide padding that is as short as two octets (including the MPPCI). This permits, for example, the transmission of Encapsulated Frames starting on specific octet boundaries within an MPPDU for a best fit with stream gates (IEEE Std 802.1Q) with some variation in user data frame size. An immediately following Explicit Pad can occupy just those MPPDU octets that could have been taken by an Encapsulated Frame if a user data frame had been available for transmission. User data frames originating from stations attached to some media can be shorter than those sent by stations using IEEE Std 802.3, and do not have be padded before being encapsulated in an MPPDU.

The management counters specified by this standard for MPPDU transmission (20.14.1) reflect the mandatory to implement support of unprotected, Privacy Frame, or Privacy Channel transmission (17.4, 20.6–20.10). Separate counters are specified for Privacy Frames and for each Privacy Channel. If additional MPPDU generation and user data frame encapsulation algorithms are implemented, their configuration should be supported by appropriate performance monitoring.

19.7 MPPDU validation

A PrY recognizes a frame that is addressed to that PrY as an MPPDU if the first two octets of the frame's MSDU compose the MAC Privacy protection EtherType.

NOTE 1—A frame that is addressed to a PrY has a MAC Destination Address that is the individual address associated with the interface stack of which the PrY is part or is a Group Address that the PrY uses to receive frames (Clause 20). A PrY also verifies the MAC Source Address (SA) of each MPPDU. MPPDUs received with SAs that a PrY is not configured to receive are discarded and not validated or otherwise processed (Clause 20).

Each of the following possible MPPDU components is validated in the order they are encoded in the MPPDU, and the management counters specified in 20.14.2 updated, as follows:

a) If no octets remain to be processed, the validation of the MPPDU is complete.

b) If only one octet remains to be processed, InPadOctets is incremented, the octet is discarded and validation of the MPPDU is complete.

NOTE 4—The value of the octet is not checked. If the MPPDU was correctly encoded it is zero, identifying the octet as a Trailing Pad.

c) If the component is incorrectly encoded (19.5.6) the octets of that component and any octets following in the MPPDU are discarded, inErroredMppdus is incremented, and validation of the MPPDU is terminated.

d) If the component is unrecognized (19.5.5) inUnknownMppcis is incremented, and the component (including any octets within the *following length* and in the MPPDU) is discarded.

e) If the component is an Encapsulated User Data Frame (19.5.1), inEncapsulatedFrames is incremented, the number of octets indicated by the following length is added to InUserOctets, and the component is extracted.

NOTE 5—Case c) or d) above applies if the component has bits 8 and 7 of the first MPPCI octet both zero but has a *following length* that is too short for an encapsulated frame or too long for the MPPDU.

f) If component is a Trailing Pad (19.5.2), the number of pad octets (the Trailing Pad MPPCI and the remaining octets in the MPPDU) is added to InPadOctets and validation of the MPPDU is complete.

g) If the component is an Explicit Pad (19.5.3), the length of the Explicit Pad plus the number of octets following in that component (the *following length* or the number remaining in the MPPDU if less) is added to InPadOctets and the component is discarded.

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h) If the component is a Frame Fragment (19.5.4), inExpressFragments (if the E bit is set) or inPreemptableFragments (if the E bit is clear) is incremented, the number of octets indicated by the following length is added to inUserOctets, and the component is extracted.

NOTE 4—The InUserOctets counter is incremented (even though the fragment can be discarded due to the loss of other fragments of the original user data frame) to allow link usage to be assessed by comparing InUserOctets and InPadOctets.

Those components that are unrecognized or recognized as padding are discarded, i.e. identified as requiring no further processing. Encapsulated Frame and Frame Fragment components are specified as being extracted, i.e. identified as requiring further processing as specified in Clause 18 and Clause 20. Components from each peer PrY that are extracted are retained in the order received, until processed. The model of PrY operation (Clause 20) is consistent with that processing taking place immediately after each component is validated and before components that follow in the same MPPDU are extracted, or with processing after components from several received MPPDUs have been extracted. If the processing of an MPPDU is terminated during validation due to an error in the encoding of a component it is unspecified as to whether earlier components encoded in that MPPDU are retained for further processing or not.

NOTE 2—The validation process makes no assumptions about the validity of unrecognized components, though any component with a following length that exceeds the number of octets remaining in the MPPDU is identified as the last component in the MPPDU.

The MPPDU reception and validation process, and the associated management counters specified in 20.14.2, are independent of the MPPDU transmission algorithms used by peer PrYs.

NOTE 3—A system can derive octet counts and counts of the number of frames received from other counters maintained for the interface stack, where those are suitable for this purpose. As specified in 20.2 conformant implementations exhibit the externally observable behavior specified by this standard.

This standard specifies constraints on the sizes of MPPDUs and of individual MPPDU components or combination of those components that can be generated (19.6). If any of those constraints are violated for a given MPPDU, a receiving PrY can terminate, but is not required to terminate, validation of that MPPDU. Terminating the validation of any given MPPDU shall not affect the processing of other MPPDUs.

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20. MAC Privacy protection Entity (PrY) operation

This clause

a) Provides an overview of the MAC Privacy protection Entity (PrY) and the service it provides (20.1).

b) Provides a model of operation (20.2) comprising an architecture (20.3) and its constituent processes that supports the PrY's functionality including management (20.14).

Clause 17 provides an overview of MAC Privacy protection and an introduction to PrY functionality. Clause 18 describes MAC Privacy protection protocol design, support, and addressing requirements, and Clause 19 describes the encoding of the MAC Privacy protection Protocol Units (MPPDUs) transmitted and received by PrYs. Clause 21 describes the position of PrYs in the interface stacks of particular systems.

20.1 PrY overview

A PrY is a shim (see IEEE Std 802.1AC) that transmits and receives MPPDUs conveying user data frames to and from peer PrYs in other systems. Each PrY is supported by a SecY (Clause 10) that confidentiality protects the MPPDUs transmitted by the PrY. That SecY is typically positioned immediately below the PrY in an interface stack (Figure 20-1, 21.1), but can be located in a separate system (21.8).

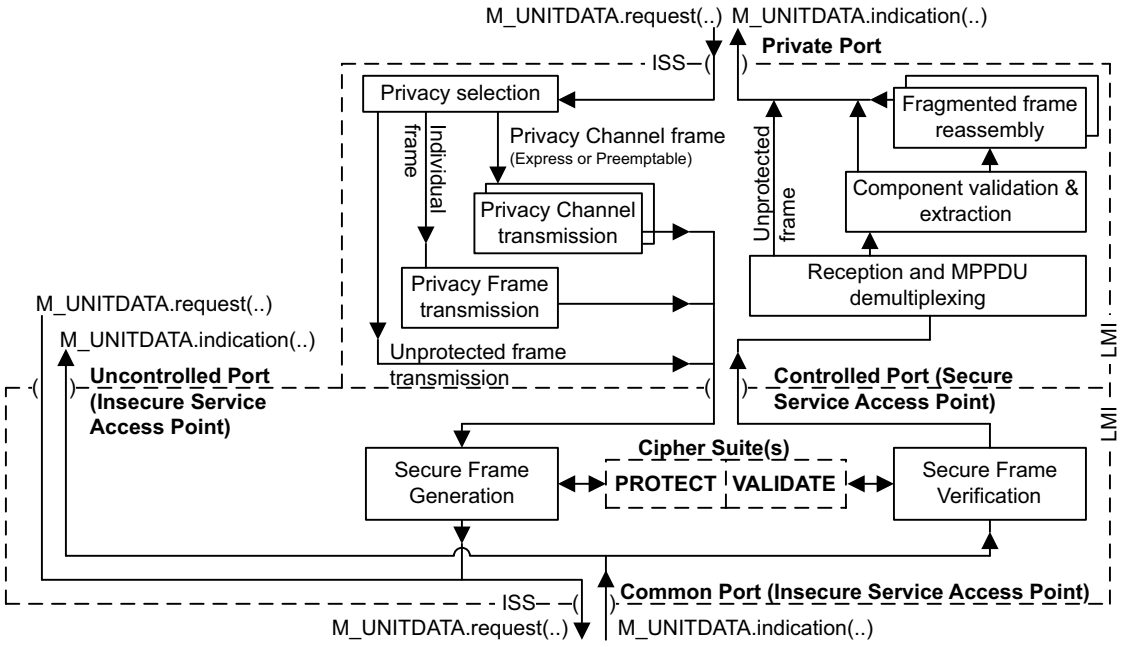


Figure 20-1—PrY and SecY

Each PrY provides the MAC Internal Sublayer Service (ISS, IEEE Std 802.1AC) at its Private Port which accepts user data frames for transmission and delivers received user data frames. The PrY uses an instance of the ISS at its Controlled Port to transmit and receive both MPPDUs that encapsulate those user data frames and user data frames that are not privacy protected. When the PrY is directly supported by a SecY, that Controlled Port is the SecY's Controlled Port, as shown in Figure 20-1. If the PrY is in a separate system or interface stack from its associated SecY, a physically secure data path between their respective Controlled Ports is required (21.8).

NOTE 1—Throughout this clause (Clause 20) the term *user* refers to the user of the PrY's Private Port.

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20.2 Model of operation

The model of operation in this clause is simply a basis for describing the functionality of a PrY. It is not intended to over constrain real implementations; these may adopt any internal model of operation compatible with the externally visible behavior that this standard specifies. Conformance of equipment to this standard is purely in respect of observable protocol.

NOTE 1—PrYs use and complement the QoS mechanisms specified for bridges in IEEE Std 802.1Q.

20.3 PrY architecture

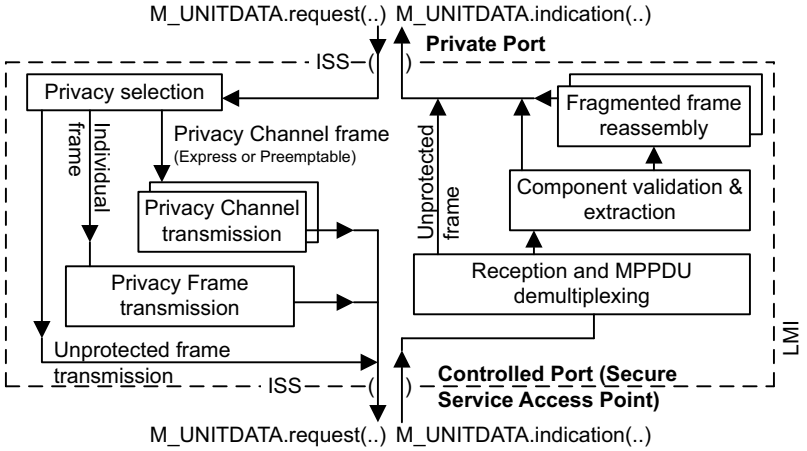


Figure 20-2—PrY architecture

The PrY architecture is shown in Figure 20-2. The transmit and receive data paths are independent of one another, but both are supported by:

a) The PrY's Private Port and Controlled Port and their MAC status (6.4) and point-to-point (6.5) parameters (20.4).

b) The PrY's Management Entity (20.14) and Layer Management Interface (LMI).

The PrY's transmit data path is supported by the following:

c) A Privacy Selection process that uses a manageable Privacy Selection Table to determine if, and how, user data frames are protected (20.5).

d) Transmission of selected user data frames without privacy protection (20.6).

e) An individual frame transmission function that can transmit user data frames as individual Privacy Frames i.e. encapsulated in an MPPDU that conveys only that user data frame in an MPPDU, padded as specified by the Privacy Selection Table (20.7).

f) A Privacy Channel transmission process that can encapsulate multiple user data frames within a single MPPDU, can fragment individual user data frames so they are conveyed in more than one MPPDU, and can pad those MPPDUs to a fixed size (20.8).

The PrY's receive data path is supported by the following::

g) A reception process that recognizes and demultiplexes MPPDUs addressed to this PrY (20.11) for validation and component extraction, submitting other received frames directly to the PrY's user.

h) MPPDU validation (20.12), removing padding and unrecognized components and extracting Encapsulated Frames (19.5.1) and Frame Fragments (19.5.4) for reassembly.

i) A user data frame Reassembly process (20.13)

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20.4 MAC status and point-to-point parameters

The MAC status parameters [MAC_Enabled and MAC_Operational, (6.4)] and point-to-point parameters [operPointToPointMAC and adminPointToPointMAC, (6.5)] for the Private Port are not independently controlled. Their values are the same as those for the PrY's Controlled Port, and are determined by the protocol entity supporting that port.

20.5 Privacy Selection

The PrY's user requests transmission of a single user data frame by issuing an ISS transmit request at the PrY's Private Port. The PrY uses the priority that the user has associated with that request, in conjunction with a manageable Privacy Selection Table (17.4, 17.4.3), to select one of the following ways of transmitting that user data frame through the Controlled Port:

a) Without privacy protection (20.6).
b) As an individual Privacy Frame (20.7, 17.4.1), i.e. in an MPPDU that encapsulates only that data frame (with the possible addition of padding).
c) In a Privacy Channel (20.8, 17.3.6) as an Express frame or as a Preemptable frame, with the possibility of fragmentation into successive MPPDUs.

The encapsulation and transmission of user data frames in MPPDUs can also be enabled and disabled for the PrY as a whole. If MPPDU encapsulation and transmission is enabled, the Privacy Selection Table is used as specified in a) through c) above. If disabled, all frames are transmitted without MAC Privacy protection (20.6). The Privacy Selection Table can be configured before MPPDU encapsulation is enabled. MPPDU encapsulation can also be disabled, at the risk of privacy information exposure, to support transparent operation and troubleshooting. MPPDU reception and demultiplexing (20.11) can be independently enabled and disabled, so a pair of communicating PrYs can be configured to receive MPPDU before either PrY transmits MPPDUs.

20.5.1 Accepting user data frames for transmission

A PrY provides minimal buffering for transmitted user data frames, sufficient only to allow user data frames to be encoded in a Privacy Channel MPPDU awaiting transmission, and any remaining frame fragment to be retained for encapsulation in a following MPPDU for its selected Privacy Channel. A transmit request is only accepted if one of the following applies:

a) The user data frame is to be transmitted as an individual frame (without privacy protection or as a Privacy Frame), the immediately prior user data frame has been transmitted through the Controlled Port, and transmission of a previously generated Privacy Channel MPPDU (20.9) is not pending.
b) The user data frame is to be conveyed by a Privacy Channel, an MPPDU has been generated and not transmitted for that Privacy Channel, and the user data frame (or a fragment of that frame) can be encoded in that MPPDU.

A PrY's Private Port can accept a transmit request for a user data frame for transmission without privacy protection or as an individual Privacy Frame even if the effect of accepting that request is to delay the generation of a Privacy Channel MPPDU (see 20.9.4).

NOTE 1—The use of minimal PrY buffering allows the Transmission Selection processes specified for end stations and bridge (see IEEE Std 802.1Q) to retain control over the use of transmission bandwidth, so far as efficient use of Privacy Channel bandwidth permits. Conditional per-user priority acceptance of user data frames for transmission is similar to the use of Priority-based Flow Control (see IEEE Std 802.1Q), but without external dependencies.

NOTE 2—An ISS transmit request occurs at a service access point when a service user (in this case the user of a PrY's Private Port) wishes to transmit a frame and the service provider (in this case a PrY) can accept that request. The ISS transmit request and receive indication parameters describe the information conveyed between peer service users each time the service is used, and exclude information used only within one of the communicating systems

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(IEEE Std 802.1AC). While a transmit request is commonly described as if it were an action taken unilaterally by the service user, it describes a system's externally observable behavior. If a service provider within a system was unable or unwilling to accept a transmit request, the request did not occur.

20.5.2 Stream-based Privacy Selection

A PrY may, in addition to supporting user priority based privacy selection via the Privacy Selection Table as specified in this standard, allow use of the stream identification function specified in IEEE Std 802.1CB.

NOTE 1—Configuration options for use with stream identification are not specified by this standard. However, where a Bridge supports both per-stream classification and metering (see IEEE Std 802.1Q) and MAC Privacy protection, it is not necessary for each PrY to duplicate the identification process. The stream_handle can be passed as a sub-parameter of the transmit request (see IEEE Std 802.1CB).

NOTE 2—While simple priority selection of MPPDU encapsulation (or not) is a coarse control, it can distinguish traffic for which the octet overhead of a Privacy Frame or the delay and jitter that can result from use of a Privacy Channel protection might be undesirable, e.g., priority 1 is commonly used for TSN streams with network bandwidth allocation.

20.6 Unprotected frame transmission

When the PrY's Private Port accepts a transmit request for a frame that is not to be privacy protected by encapsulation in an MPPDU, a corresponding transmit request is used to pass that unmodified user data frame to the service supporting the Controlled Port before any other frame transmitted as the result of accepting a later Private Port transmit request. The priority used to access the underlying service (the access priority) can differ from the priority requested by the PrY's user and is determined by the Privacy Selection Table.

NOTE 1—Ordering preservation for individually transmitted frames does not preclude support of preemption (e.g., as specified by IEEE Std 802.3 for Interspersing Express Traffic) as the underlying service (supporting the Controlled Port) can accept a transmit request for an express frame while physical transmission of an earlier preemptable request is still in progress.

NOTE 2—Use of a Privacy Selection Table determined priority to access an underlying service does not result in any change to user data. It does not change the value of any priority parameter encoded in that data, e.g., in a VLAN tag. The priority used to access the underlying service can be used to determine values in any tag added by that service.

20.7 Privacy Frame transmission

When the PrY's Private Port accepts a transmit request for a user data frame that is to be protected as an individual Privacy Frame, it is encoded as an MPPDU Encapsulated Frame (19.5.1).

The Privacy Selection Table is used to determine the MPPDU's access priority, whether padding is to be added to the MPPDU, and whether the user data frame's transmit request's drop_eligible parameter is hidden, i.e. always set False when making the Controlled Port transmit request for the MPPDU.

NOTE—Different values of PrY access priority are only relevant if they select differentiated services from a supporting SecY (between two transmit SCs, for example) or from the media access method or other protocol entity that supports the SecY's Controlled Port (providing frame preemption, for example). Similarly the drop_eligible parameter is only relevant if it is used by the underlying service. In a simple end station or single component bridge, it is not encoded with the Privacy Frame (any VLAN tag is encoded by support for the EISS, higher in the interface stack than a PrY or SecY which both support the ISS). However the network component of an EDE-CC (for example) can receive the internally communicated drop_eligible parameter for use in its subsequent Provider Network Port VLAN tag encoding.

If the privacyPadding parameter is non-zero, a Trailing Pad (19.5.2) is added (if required) to increase the size of the MPPDU (excluding its source and destination MAC addresses) to four octets (to allow for the MAC Privacy protection EtherType and MPPCI) plus a multiple of 16, 32, or 64 octets. If the revealDE parameter is Visible (True) the user data frame's transmit request's drop_eligible parameter is passed unchanged to the Controlled Port.

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20.8 Privacy Channel transmission

The Privacy Channel transmission process supports two Privacy Channels, an Express Channel and a Preemptable Channel. A PrY can be configured to use both, one, or neither of these channels. Each channel is supported by its own instance of each of the following processes:

a) Privacy Channel MPPDU Generation (20.9)
b) Privacy Channel Encapsulation (20.10)

20.9 Privacy Channel MPPDU Generation

20.9.1 Objectives

Each instance of the Privacy Channel MPPDU generation process provides its channel's Privacy Channel MPPDU Encapsulation process with notionally empty MPPDUs that the latter can use to encapsulate user data frames. The objectives of the generation process(es) are to:

a) Initiate transmission of MPPDUs for the Privacy Channel at intervals that are independent of the user data frames and padding encoded in each of those MPPDUs.
b) Allow Privacy Channel MPPDUs to be padded to fixed sizes.
c) Support transmission scheduling decisions made by the processes that use the PrY's interface stack, e.g., transmission selection as specified for bridges by IEEE Std 802.1Q.
d) Minimize the impact of Privacy Channel encapsulation on total bandwidth and on the transit delay experienced by individual data flows.

Any particular Privacy Channel MPPDU generation algorithm necessarily reflects a balance between these objectives. This standard specifies a mandatory to implement default algorithm (20.9.4), suitable for use with the strict priority, credit-based shaper, enhanced transmission selection (ETS), and ATS transmission selection algorithms specified in IEEE Std 802.1Q. An optional algorithm, for use with the IEEE Std 802.1Q enhancements for scheduled traffic and the use of transmission gates is also specified (20.9.5).

20.9.2 Timeliness

Privacy Channel MPPDUs should be generated no earlier than required to provide the Controlled Port with transmit data at the maximum rate that port can support. Earlier Privacy Channel MPPDU generation can defeat the intended operation of the Controlled Port's user's transmission selection algorithm, by delaying a high priority frame that becomes available for transmission just prior to MPPDU transmission.

20.9.3 Competition for transmission bandwidth

Transmission of MPPDUs used to support a particular Privacy Channel can compete for transmission bandwidth with user data frames that are not privacy protected, are transmitted as individual Privacy Frames, or that are transmitted to support a second Privacy Channel. The SecY that supports a PrY's Controlled Port can also transmit frames sent by the users of its Uncontrolled Port, temporarily denying bandwidth to its Controlled Port and the PrY. All the above frames, including those conveying MPPDUs, can also be subject to later traffic shaping and consequent delay, e.g., by the Provider Network Port of an EDE-CC.

Provided that these delays and changes to Privacy Channel MPPDU transmission timing are independent of the content of Privacy Channel MPPDUs, they do not defeat the protection provided to user data frames conveyed by the Privacy Channel, except by signaling that not all traffic is carried by that channel. In particular traffic shaping and other frame timing changes can be made by the network component of an EDE-CC, EDE-SS, or EDE-CS.

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The Privacy Channel MPPDU generation algorithms described in this standard take account of the possible bandwidth competition and the desire to provide more or less constant bandwidth long term while avoiding bursts in channel transmission. Additional algorithms may be implemented, subject to documentation of the way that they meet the objectives described above (20.9.1) and of their additional objectives.

20.9.4 Default Privacy Channel MPPDU generation algorithm

By default, all the MPPDUs generated for a given Privacy Channel are of the same size. To simplify management, the configurable userDataFrameSize parameter is the size of the largest user data frame that MPPDU could convey, i.e. the number of octets in the user data frame's MAC Destination Address (DA), MAC Source Address (SA), MSDU, and 4-octet FCS (17.4.2). The userDataFrameSize, is used by the PrY to calculate a channelFrameSize in bits. The channelFrameSize calculation takes into account the 12 octets of the source and destination MAC addresses of the MPPDU, the 2 octets of the MAC Privacy protection EtherType, the 2 octets of the MPPCI used to encode an EncapsulatedFrame (19.5.1), the omission of the 4 octets of the user data frame FCS, and the frameTransmissionOverhead (in octets):

$$\text{channelFrameSize} = 8 \times (\text{userDataFrameSize} + 12 + \text{frameTransmissionOverhead})$$

The frameTransmissionOverhead includes the octets of SecTAG and ICV (9.2) added by MACsec (if the PrY's Controlled Port is supported by a SecY) and any other media independent or media dependent overhead added by interface stack components (e.g., frame preamble).

NOTE 1—The userDataFrameSize is 1522 octets for a maximum sized Ethernet frame with a single VLAN tag. Since the user data frame's FCS is not encoded in the MPPDU, transmission of such a frame will add only 1518 octets to the management count chOutUserOctets (20.14.1).

NOTE 2—See 3.2.7 and 1.4.245 of IEEE Std 802.3-2018 for applicable maximum frame sizes when Ethernet is used in conjunction with encapsulation protocols including MAC Security.

Privacy Channel MPPDUs are generated by a simple token bucket algorithm. The notional bucket contains a number of bits. When a Privacy Channel MPPDU is transmitted, channelFrameSize bits are subtracted from the bucket. Bits are added to the bucket at the manageable requestedKbitRate, expressed in bits per second, up to a maximum of channelBurstSize bits, calculated by the PrY from the manageable parameter userBurstOctets as:

$$\text{channelBurstSize} = \text{channelFrameSize} \times (1 + (\text{userBurstOctets})/(\text{userDataFrameSize}))$$

If a single Privacy Channel is configured for transmission, a channel MPPDU is generated for use by the Privacy Channel's MPPDU Encapsulation process when the following conditions are satisfied:

a) The Privacy Channel's token bucket contains at least channelFrameSize bits when the MPPDU is transmitted, and

b) The user of the Private Port has:

   1) no user data frame eligible for immediate transmission, or

   2) the highest access priority for a user data frame eligible for immediate transmission is not numerically higher than the access priority used by the Privacy Channel.

c) The timeliness condition (20.9.2, above) is satisfied, i.e. any later generation of the MPPDU could result in its later transmission on the medium supporting the PrY's interface stack.

If both an Express and a Preemptable Privacy Channel are configured, and conditions a) through c) in this subclause (20.9.4) are met for both Privacy Channels, a Channel MPPDU is only generated for the Express Privacy Channel.

NOTE 3—If delays in the Privacy Channel's MPPDU Encapsulation process, in the operation of a supporting SecY, and in the latter's use of the transmission medium are ignored, condition c) is equivalent to specifying that the transmission of the immediately prior MPPDU or any competing frame has been completed.

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20.9.5 Transmission gate MPPDU generation algorithm

A PrY that is part of an interface stack that supports the use of transmission gates and a gate control list (IEEE Std 802.1Q) may be configured to use the Privacy Channel MPPDU generation algorithm specified in this subclause (20.9.5).

20.9.5.1 Constraints

Successful operation of the algorithm constrains the assignment of user data frames to traffic class queues, the configuration of the gate control list (IEEE Std 802.1Q), and the configuration of the PrY's Privacy Selection Table as follows:

a) All frames queued for a given traffic class and thus subject to the operation of the same transmission gate are either all transmitted in a given Privacy Channel or are transmitted as individual frames.

b) The per traffic class transmission gates open at any given time are either all for user data frame transmission in a given Privacy Channel or all for transmission as individual frames.

c) Following execution of each entry in the gate control list that specifies an open state for a traffic class with Privacy Channel transmission, sufficient time elapses prior to the execution of the next entry in the list for transmission of an MPPDU containing:

    1) A Frame Fragment, if fragmentation is enabled for that channel, or

    2) An Encapsulated Frame of the maximum size that can be assigned to that traffic class.

20.9.5.2 Algorithm

A Privacy Channel MPPDU, for a given Privacy Channel, is transmitted when either of the following conditions apply:

a) An entry in the gate control list is executed and specifies that a transmission gate is open for a traffic class whose user data frames are encapsulated in that channel.

b) The transmission gate is open for such a traffic class, one or more prior MPPDUs have been generated as permitted by these conditions [a) and b)], generation and transmission of the MPPDU would not delay the subsequent transmission of an MPPDU or user data frame generated [see a)] or permitted by the execution of a gate control list entry, and later generation of the MPPDU could delay its transmission.

The size of each generated MPPDU is the maximum possible without delaying subsequent transmissions [see b) above] subject to an upper bound capable of conveying an unfragmented user data frame of userDataFrameSize octets, without any pad.

NOTE 1—If possible delays in the Privacy Channel's MPPDU Encapsulation process, in the operation of a supporting SecY, and in the latter's use of the transmission medium are ignored, condition b) is equivalent to the following: transmission of any prior MPPDU or competing frame has been completed, and the transmission of this MPPDU will be complete prior to execution of the next gate control list entry.

NOTE 2—The conditions [a) and b)] apply even if execution of a gate control list entry does not change the state of a transmission gate. The corresponding reduction in the MPPDU size (for specified control list entry execution intervals) can facilitate earlier reception and validation of the MACsec ICV protecting those MPPDUs and thus reduce the effective transit delay of the encapsulated user data frames.

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20.10 Privacy Channel Encapsulation

Once a Privacy Channel MPPDU has been generated and prior to its transmission through the PrY's Controlled Port, the PrY's Private Port and the encapsulation process accepts (and only accepts) transmit requests for user data frames that are to be conveyed by that Privacy Channel. The delay (if any) between the generation of an MPPDU and its subsequent transmission, following a transmit request to the Controlled Port, should be independent of the content (user data frames or padding) encapsulated in the MPPDU, or of any similar operation on any other port in the system of which the PrY is a part, in order to avoid disclosing the level of activity to an adversary.

A Privacy Channel encapsulation algorithm can fragment user data frames to make better use of Privacy Channel bandwidth. The Privacy Selection Table classifies each user data frame that can be conveyed by a Privacy Channel as an Express frame or as a Preemptable frame, and each encapsulated Frame Fragment is identified as an Express or a Preemptable frame fragment. After a Frame Fragment has been encoded in an MPPDU (19.5.4), the encapsulation process can hold the remainder of a single fragmented Express frame or of a single fragmented Preemptable frame (or the remainder of one user data frame of each classification, if a single Privacy Channel is configured) for later encoding in the same or a subsequent MPPDU.

Each Privacy Channel shall be capable of being configured to use the default encapsulation algorithm (20.10.1) and may be configured to support additional encapsulation algorithms as specified in (18.5). Fragmentation is expected to be used in most network scenarios but can be disabled on transmission, per Privacy Channel. A PrY is always capable of receiving and reassembling fragmented user data frames. Enabling or disabling transmit fragmentation can result in a temporary loss of connectivity with the PrY's Private Port's MAC_Operational parameter value temporarily false.

20.10.1 Default Privacy Channel encapsulation algorithm

The default encapsulation algorithm ensures that no fragment comprises less than 64 octets of a user data frame. A frame that can be fragmented (or the remainder of a frame than can be subject to further fragmentation) comprises at least 128 octets and its fragmentation removes the greatest multiple of 64 octets that can be encoded from its initial octets, leaving a remainder of at least 64 octets.

NOTE 1—An entire frame can comprise fewer than 64 octets, either as a result of its initial transmission using a medium access method other than that specified by IEEE Std 802.3 or after the removal of one or more tags (e.g., a VLAN tag).

NOTE 2—The minimum userDataFrameSize (20.9.4) that can be used with this algorithm is 132 octets, unless the maximum size of frames to be conveyed by the channel is restricted to be less than 130 octets.

When a new MPPDU is generated for a Privacy Channel that conveys both Express and Preemptable frames, user data frames (or fragments of those frames) are added in the following priority order:

a) The remaining octets (if any) of a previously fragmented Express Frame.

b) The highest priority user data frame (if any) that the PrY's user wishes to transmit that is identified (by the Privacy selection process, 20.5) as an Express frame.

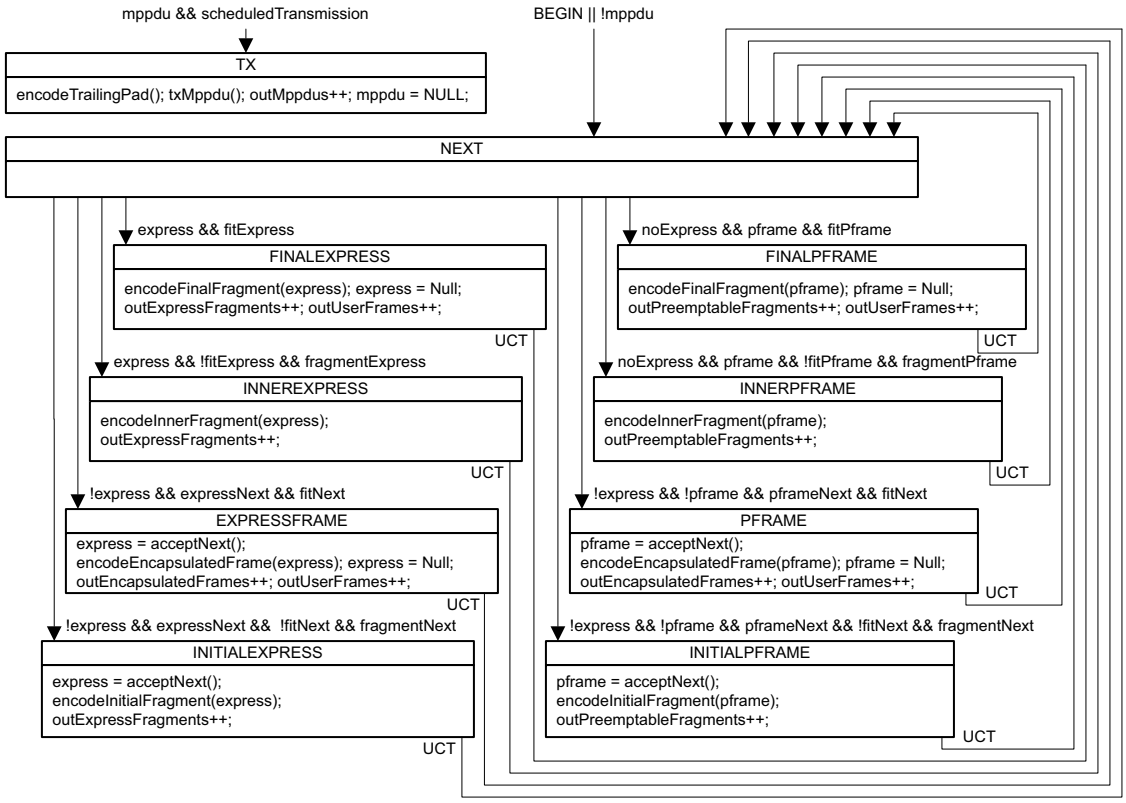
c) The remaining octets (if any) of a previously fragmented Preemptable Frame.

d) The highest priority user data frame (if any) that the PrY's user wishes to transmit that is identified (by the Privacy selection process, 20.5) as a Preemptable frame.

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Once encapsulation is complete, the value zero is encoded in each of the remaining octets of the MPPDU, and the MPPDU is transmitted.

Figure 20-3 specifies the default encapsulation algorithm using the state machine conventions used in this standard and takes precedence if there is any discrepancy between it and the text of this subclause.



State machine conditions, each True (not Null) iff (if and only if):

mppdu : An MPPDU has been generated and not yet transmitted.
express : The PrY is holding the remainder or all of an Express user data frame.
expressNext: The PrY's user has selected, and has available, an Express user data frame as the next frame for transmission.
noExpress: express and expressNext are both False.
pframe : The PrY is holding the remainder or all of a Preemptable user data frame.
pframeNext : The PrY's user has selected, and has available, a Preemptable user data frame as the next frame for transmission.
fitExpress : The Express frame (or all of the remainder of the fragmented Express frame) can be encoded in the remaining MPPDU octets.
fragmentExpress : The Express frame or its remainder can be fragmented, and the next fragment encoded in the remaining MPPDU octets.
fitPframe : The Preemptable frame (or all of the remainder of the fragmented Preemptable frame) can be encoded in the remaining MPPDU octets.
fragmentPframe : The Preemptable frame remainder can be fragmented, and the next fragment encoded in the remaining MPPDU octets.
fitNext : The next user data frame (Express or Preemptable) can be encoded in the remaining octets of the MPPDU without fragmentation.
fragmentNext : The next user data frame can be fragmented, and the first fragment encoded in the remaining MPPDU octets.
scheduledTransmission: The MPPDU will be eligible for transmission once the Trailing Pad, if any has been added.
Note: The 'open' transition to the TX state takes precedence over other transitions from NEXT, apart from BEGIN.

State machine procedures:

express = acceptNext() : Accept the next user data frame (an Express frame) for transmission, similarly frame = acceptNext() for a Preemptable frame.

encodeEncapsulatedFrame(express), encodeEncapsulatedFrame(pframe) : Encode the user data frame in the MPPDU, and add the number of user data octets encoded (not including the MPPCI) to outUserOctets.

encodeInitialFragment(express), encodeInitialFragment(pframe) : Encode an Initial Fragment, encapsulating the greatest multiple of 64 octets from the user data frame that will fit in the MPPDU leaving at least 64 octets of the user data frame as a remainder, and add the number of user data frame octets encoded (not including the MPPCI) to outUserOctets.

encodeInnerFragment(express), encodeInnerFragment(pframe) : Encode a Frame Fragment (with Initial and Final bits clear), encapsulating the greatest multiple of 64 octets that will fit in the MPPDU leaving at least 64 octets of the frame as a remainder, and add the number of user data frame octets encoded (not including the MPPCI) to outUserOctets.

encodeFinalFragment(express),encodeFinalFragment(pframe) : Encode the remainder of the user data frame in a Final Fragment.

encodeTrailing Pad() : Encode the value 0 in all the remaining octets (if any) of the MPPDU, add the number of pad octets to outPadOctets. Changes in the number of octets to be added shall not affect time at which the MPPDU is transmitted.

txMppdu() : Transmit (complete the transmission of) the MPPDU through the PrY's Controlled Port.

Figure 20-3—Privacy Channel Encapsulation state machine

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If a single Privacy Channel is configured, the encoding of an Express frame (or its fragments) can interrupt the encoding of a series of fragments of a default class frame (19.5.4). If two Privacy Channels are configured, conditions c) and d) in this subclause do not apply to the Express Privacy Channel and conditions a) and b) do not apply to the Preemptable Privacy Channel.

NOTE 3—When fixed sized Privacy Channel MPPDUs are to be sent at constant rate, the use of a single Privacy Channel can reduce the bandwidth required. Interrupting the encoding of a Preemptable frame to encode an Express frame can reduce the delay experienced by the latter if that allows its encoding to be completed in an earlier MPPDU. In contrast, the use of both Privacy Channels allows Express user data frames to use a higher access priority and take advantage of any preemption capabilities provided by the underlying service that supports the Controlled Port.

An implementation of the default encapsulation algorithm (Figure 20-3) can complete MPPDU encoding prior to transmission or can begin transmission while later octets of the MPPDU are still being encoded. An MPPDU component need only be available for encoding when that encoding takes place.

20.11 MPPDU reception and demultiplexing

The underlying service (typically that provided by a SecY) that supports a PrY's Controlled Port issues an ISS service indication to signal receipt of a frame. The frame is recognized as an MPPDU addressed to this PrY if, and only if both:

a) The destination MAC Address of the frame is:

   1) The address used by the PrY as the destination MAC address of transmitted MPPDUs (ifMppduDA, 22.6), and that address is a group MAC address; or

   2) The individual MAC Address used by the PrY as the source MAC address of transmitted MPPDUs (ifAddr, 22.6).

and

b) The first two data octets encode the MAC Privacy protection EtherType (19.3, Table 19-1).

MPPDUs addressed to this PrY are processed by the MPPDU component validation and extraction logic, unless MPPDU reception has been disabled by management, in which case they are discarded.

Otherwise receipt of the frame results in a corresponding ISS service indication at the PrY's Private Port to signal receipt of that (unmodified) frame to the PrY's user.

When the PrY's Controlled Port is directly supported by a SecY, the destination MAC address of transmitted MPPDUs is the group MAC Address used by the supporting PAE's KaY (IEEE Std 802.1X) to discover peer SecYs and distribute SAKs. SecY support of the PrY is reported by ifSecySupport (22.6).

NOTE 1—The PrY does not recognize or otherwise process received frames whose initial octets are those of a VLAN tag or any other type of tag. Similarly a PrY does not add VLAN tags to transmitted MPPDUs. VLAN tags can be added or removed by other protocol components in the system of which the PrY is part, e.g., an EDE.

Received frames that are not recognized as MPPDUs addressed to this PrY are delivered to the Private Port in the order that they are received from the Controlled Port. Their delivery is not suppressed if MPPDU reception and demultiplexing has been disabled for this PrY. The relative order of the delivery of those frames and of frames decapsulated from received MPPDUs is not specified.

NOTE 2—An interface stack containing a PrY for which MPPDU reception has been disabled can be on the path between another pair of communicating PrYs. Its presence does not disrupt that communication unless they are using a destination address associated with the disabled PrY, or filtered by the system of which the PrY is a part.

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20.12 MPPDU component validation and extraction

MPPDUs received from the Controlled Port and addressed to the PrY are processed in the order received. Each MPPDU is processed, and its components extracted and validated, as specified in (19.7). Encapsulated Frames (19.5.1) and Frame Fragments (19.5.4) are both delivered to the reassembly process in the order they were encoded in each received MPPDU and in the order those MPPDUs are received.

20.13 Protected frame reception and reassembly

Each user data frame is delivered to the user of the PrY's Private Port when its reception by the PrY (in an Encapsulated Frame or in a set of Frame Fragments) is complete, and is delivered before any other user data frame whose complete reception depends on an Encapsulated Frame or a Frame Fragment encoded later in the same MPPDU or in a subsequently received MPPDU.

Delivery of a complete user data frame is not postponed pending the (possible) arrival of other Frame Fragments or Encapsulated Frames.

A PrY shall be capable of the simultaneous reassembly of one Express frame and one Preemptable frame. A PrY may be capable of reassembling fragmented frames from more than one peer PrY, and shall then support simultaneous reassembly of one Express and one Preemptable frame for each of a specified maximum number of peers (ifMaxPeers, 22.6). If the PrY is supported by a SecY, ifSecYSupport (22.6) is true and MKA updates the actual number (ifNumPeers, 22.6) and their source MAC addresses (peers, 22.6), avoiding the need for management configuration of that information.

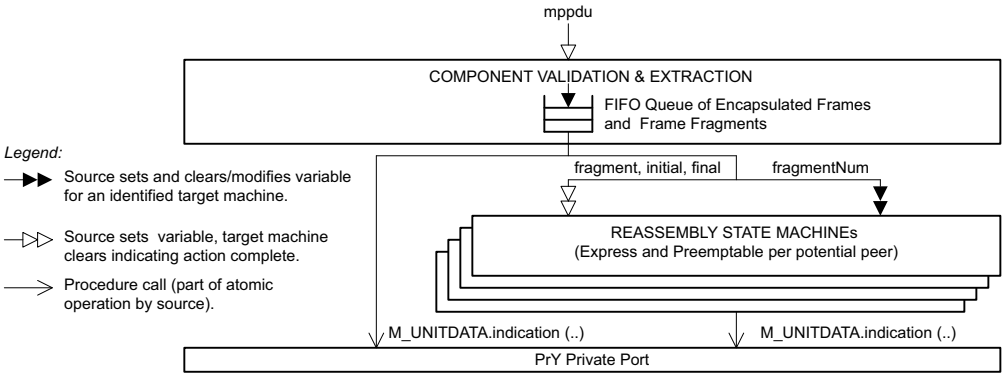
Any given frame shall be reassembled from fragments received in MPPDUs with same, verified, source MAC address, unless MKA has confirmed that there is a single peer. If fragments are received from more than the supported number of peers, all, some, or none of the frames encoded in a sequence of fragments are successfully reassembled. A failure to reassemble frame fragments shall not affect the successful reception of unfragmented frames. Any received fragment shall be retained for no longer than 0.1 second pending reassembly.

Figure 20-4 provides an overview of the relationship between the component validation and extraction process, delivery of unfragmented frames, and the specification of frame reassembly state machines.

This standard specifies a mandatory to implement default reassembly algorithm (20.13.1). This algorithm reassembles user data frames from Frame Fragments received in order from a given peer, as indicated by the relevant fragment sequence number (Express or Preemptable, 19.5.4), and discards out of order Frame Fragments. A PrY may support one or more additional algorithms subject to the constraints specified in this clause (20.13). Current use of the default algorithm is reported by ifDefaultReassembly (22.6).

NOTE 3—A privacy protection system that uses Link Aggregation and MACsec collocates PrYs with the SecYs (see 21.4) supporting individual links and transmits the fragments of any given user data frame on a single link with consecutive Frame Fragment (19.5.4) fragment sequence numbers. A supporting SecY that is configured to provide strict replay protection discards MPPDUs received out of order.

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MPPDU components are validated and extracted in MPPDU reception order, and in the order encoded in those MPPDUs as specified in 19.7. Padding and unrecognized components are extracted and discarded. The attributes of each received frame fragment are communicated to the appropriate reassembly state machine using the state machine variables **fragment**, **initial**, **final**, and **fragmentNum**. If there is no such machine (e.g., if the receiving PrY supports only point-to-point connectivity, but has more than one peer PrY), the frame fragment is discarded. The variable **fragment** references an implementation dependent construct for the user data frame fragment, and is PtrToNull (0, 'clear', and FALSE) when no fragment is referenced. User data frames received in Encapsulated Frames are delivered direct to the user of the PrY's Private Port.

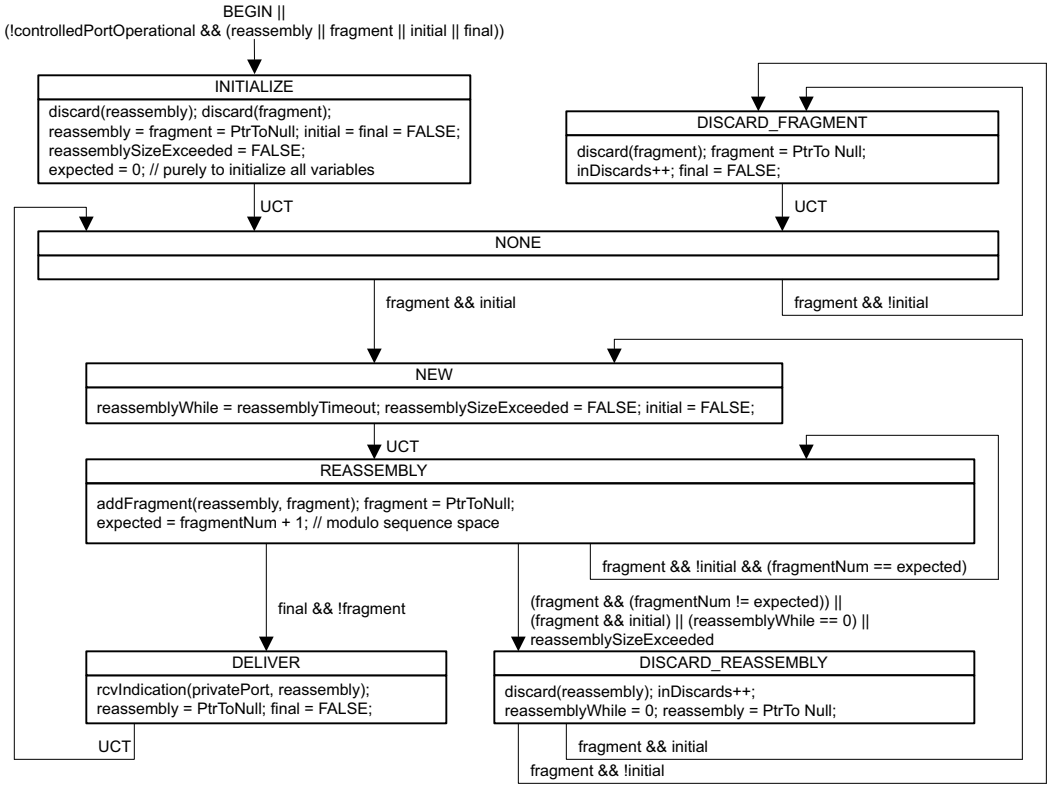
The default reassembly algorithm state machine (20.13.1, Figure 20-5) clears **fragment** and **initial** when the fragment is added to a user data frame reassembly or discarded, and clears **final** when the fragment is discarded or delivered to the PrY's Private Port as part of a successfully reassembled frame. The component validation and extraction process maintains the required user data frame delivery order by not presenting any following fragment to a reassembly state machine, or delivering a user data frame direct to the Private Port until **fragment**, **initial**, and **final** for each machine are clear.

Figure 20-4—Protected frame reception and reassembly

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20.13.1 Default reassembly algorithm

Figure 20-5 specifies the mandatory to implement default reassembly algorithm using the state machine conventions used in this standard, IEEE Std 802.1X and IEEE Std 802.1Q. This state machine takes precedence if there is any discrepancy between it and the text of this clause (20.13).



The entry conditions to any state are true when the actions in that state are executed and those actions are executed atomically (with respect to any other state machine or process) on entry.
The component validation process sets the state machine variables **fragment**, **initial**, and **final** atomically to pass a fragment (with its initial and final indications) to the appropriate reassembly state machine, and does not pass its next Frame Fragment or Encapsulated Frame to the same or any other state machine or the Private Port's user until **fragment**, **initial**, and **final** are all False (zero or PtrToNull).
The **fragment** and **reassembly** state machine variables reference implementation dependent constructs for the fragment or reassembled frame, and are PtrToNull (0 and FALSE) if there is no current fragment or reassembly.
The **addFragment(reassembly, fragment)** procedure adds the fragment to the current reassembly, and transfers responsibility for tracking any resources previously associated with **fragment** to **reassembly**. If the size limit on the reassembled frame would be exceeded the state machine variable **reassemblySizeExceeded** is set. The **reassemblyWhile** timer comprises an implementation dependent **reassemblyTimeout** number of clock ticks equal to 0.1 second, and decremented by the system.
The **discard(fragment)** procedure recovers any associated resources associated with **fragment**.
The **discard(reassembly)** procedure discards the current reassembly and recovers any associated resource, including any associated with a fragment that has caused reassemblySizeExceeded to be set.
If the reassembly state machine is reponsible for reassembling Express Frame Fragments, the action shown as **inDiscards++** increments the inExpressDiscards count. If Preemptable Frame Fragments are being ressembled, inPreemptableDiscards is incremented.

Figure 20-5—Reassembly state machine

If (for a given peer and given class of frames) a fragment is received that is not the next in sequence, the results of any reassembly already in progress is discarded. If there is no reassembly in progress, the received fragment shall also be discarded unless it is marked as an initial fragment. Otherwise the received fragment is added to the reassembly. If the received fragment was marked as a final fragment, the reassembled frame is delivered to the Private Port.

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20.14 PrY management

The PrY management process controls, monitors, and reports on the operation of the PrY, providing access to operational controls and statistics for network management through the LMI. It:

a) Maintains the MAC status and point-to-point MAC parameters (20.4) for the Private Port.

NOTE 1—Interface status reporting can be management protocol specific. See 22.3.2 for IF-MIB related detail.

b) Allows the encapsulation of user data frames in MPPDUs, and the decapsulation of received user data frames to be separately enabled or disabled for the PrY as a whole (20.5, 22.6, 20.11).

c) Allows the destination MAC Address of the PrY to be explicitly configured [5.10 n), p)].

NOTE 2—A PrY that is directly supported by a SecY uses the PAE Group Address used to support that SecY as the destination MPPDU MAC Address and does not need to be separately configured.

d) Allows the source MAC addresses used by peer PrYs to be explicitly configured, if the PrY supports (20.13) more than one peer [5.11 a)].

NOTE 3—When a PrY supports more than one peer [5.11 a)] and the connectivity provided by the Controlled Port is not point-to-point, the source MAC address of each peer is used in fragment reassembly (20.13). A PrY that is directly supported by a SecY uses the initial 6 octets of each of the peer SCIs discovered by MKA, and does not need to be separately configured.

e) Supports configuration of the Privacy Selection Table (20.5, 17.4, Figure 17-3), including the privacyType (None, PrivacyFrame, ExpressChannel, PreemptableChannel) and the parameters for each table entry that specifies Privacy Frame transmission (privacyPadding, revealDE, accessPriority), see 20.7.

f) Allows the use of the PrY's Express Channel and Preemptable Channel to be separately enabled or disabled (20.8, 17.4.2).

g) Supports configuration of each Privacy Channel's userDataFrameSize (17.4.2, 20.9.4, 20.9.5), and allows the use of user data frame fragmentation with the channel to be enabled or disabled (20.10).

h) Supports selection of the default Privacy Channel MPPDU generation algorithm (20.9.4), configuration of its requestedKbitRate and userDataBurstSize parameters, and reporting of the channelFrameSize and channelBurstSize calculated by the PrY for use with that algorithm.

i) Supports selection of the transmission gate Privacy Channel MPPDU generation (20.9.5) algorithm.

j) Maintains transmission statistics for the PrY and its Private Port interface (20.14.1).

k) Maintains reception statistics for the PrY and its Private Port interface (20.14.2).

Figure 20-6 illustrates the management information that represents a PrY's capabilities and provides control over and reporting on its operation. For convenience the figure uses UML 2.0 conventions together with C++ language constructs. For an explanation of these conventions, see *UML Distilled: A Brief Guide to the Standard Object Modeling Language, Third Edition* [B1].

Figure 20-6 reflects the model of SecY operation described in this clause (Clause 20). It includes data made available, through the LMI, to a PrY by other components (notably the PAE and the KaY) within the same system, in addition to operations and parameters directly accessible by remote management protocols. Security considerations can restrict management access, and the use of a PrY can depend on the configuration of those other system components. For particular managed object specifications based on this clause and figure, see Clause 13 (MAC Security Entity MIB) and Clause 23 (YANG Data Models).

NOTE 4—Figure 20-6 is aligned with the operational model, while remote management models group objects for ease of access. Object names are chosen to make correspondence clear where there are organizational differences.

In Figure 20-6 the management information for each PrY is shown as indexed by privatePortNumber within a PrY system. This containment relationship complements that specified in 10.4, where management information for each SecY is indexed by controlledPortNumber. In the PrY MIB (Clause 22), each PrY is identified by an interface index (ifIndex). In the YANG module (Clause 23), each PrY is indexed by an interfaceName.

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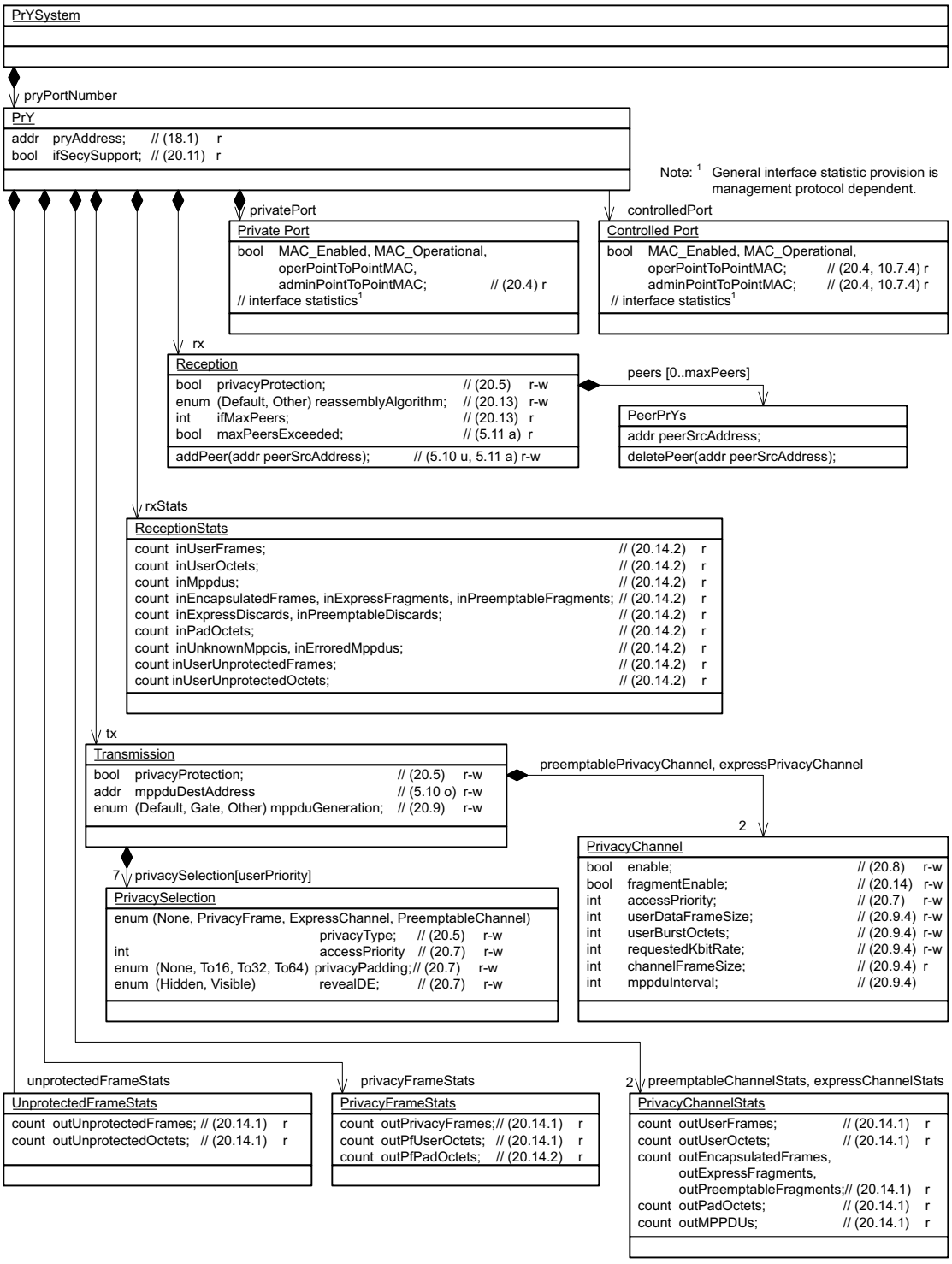


Figure 20-6—PrY Managed objects

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A managed system is responsible for setting the values of certain PrY management objects when possible. When the appropriate values are available from other system components, this reduces the requirement for explicit network management of PrYs and the chance of configuration error. The values of objects that are writable by network management, and not updated by the system, shall be persistent and remain unchanged across re-initialization of the system and its management entity.

20.14.1 PrY transmission statistics

PrY transmission statistics support assessment of the efficiency of PrY operation and of the adequacy of Privacy Channel provision.

The following transmission count totals are recorded for Privacy Frames:

a) outPrivacyFrames (22.6), the number of user data frames in Privacy Frames (equally, the number of Privacy Frame MPPDUs and the Encapsulated Frame components in those MPPDUs).

b) outPfUserOctets (22.6), the number of user data frame octets.

c) outPfPadOctets (22.6), the number of padding octets.

The following transmission count totals are recorded separately for each Privacy Channel:

d) chOutUserFrames (22.6), Encapsulated Frames and Frame Fragments with the Final (F) bit set.

NOTE—A Frame Fragment with the F bit set updates both frame and fragment counts.

e) chOutUserOctets (22.6), the number of user data frame octets.

f) chOutPadOctets (22.6), the number of padding octets added to Privacy Channel MPPDUs.

g) chOutMppdus (22.6), the number of transmitted MPPDUs for the Privacy Channel.

h) chOutEncapFrames (22.6), the number of Encapsulated Frames (19.5.1) encoded in MPPDUs.

i) chOutExpFragments (22.6), the number of Express Frame Fragments (19.5.4) encoded.

j) chOutPreFragments (22.6), the number of Preemptable Frame Fragments (19.5.4) encoded.

The following transmission count totals are recorded -for user data frames that are not privacy protected:

k) outUnprotectedFrames (22.6), the number of user data frames transmitted unprotected.

l) outUnprotectedOctets (22.6), the number of octets in those user data frames.

Counts of user data frame octets include the user MAC DA and SA, but not any encapsulating MPPDU's MAC DA and SA, MAC Privacy protection EtherType, MPPCIs, or FCS. The octets of each user data frame's FCS (which is not encoded in an MPPDU) are also excluded. Counts of padding octets include the MPPCI octets for Trailing Pad and Explicit Pad components. An Explicit Pad that is the last component in an MPPDU can have a *following length* that goes beyond the end of the MPPDU, only those octets that are actually sent are included in this count.

NOTE 1—The names of management counters are those used in Figure 10-5 and Figure 20-6. In the MIB modules these are preceded by a short names for the module and for the group or table of which they are part, and can be abbreviated to satisfy maximum name length conventions. In YANG hyphenation rather than case stropping is used to identify name components. The description clauses in both MIB and YANG modules provides a full identification of each object.

In the context of a particular system, some of this statistical information can be obtained from other system components for PrY reporting. If, for example, the user of the PrY's Private Port maintains separate frame and octet transmission counters for each user priority, then all of the counts a), b), c), d), e), k), and l) can be obtained by an appropriate summation of those user priority counters when the configuration of the Privacy Selection Table is taken into account. The sum of outMppdus and outUserFrames [g), a), and k)] is equal to the number of frames transmitted through the Controlled Port. If that count is available from its supporting protocol entity, outUserFrames [k)] need not be separately counted.

SMIv2 IF-MIB (see 22.3.2) and YANG interface statistics (see 23.4.2) can be obtained from the above counts [d) through l)].

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20.14.2 PrY reception statistics

PrY reception statistics can be used, in conjunction with transmission statistics from peer PrYs, to detect network loss, peer PrY configuration errors, and PrY implementation errors. See 19.4, 19.7, 20.11, and 20.12 for additional detail.

The following counts are recorded, in total, for received MPPDUs:

a) inUserFrames (22.6), the number of user data frames successfully received from correctly encoded Encapsulated Frames or reassembled from Frame Fragments [e), f) and g), below].

b) inUserOctets (22.6), the number of octets in those user data frames.

c) inPadOctets (22.6), pad octets received in MPPDUs.

d) inMppdus (22.6), the number of MPPDUs received.

e) inEncapsulatedFrames (22.6), valid Encapsulated Frames (19.5.1, 19.7).

f) inExpressFragments (22.6), valid Frame Fragments with the Express (E) bit set (19.5.4, 19.7).

g) inPreemptableFragments (22.6), valid Frame Fragments with the E bit clear (19.5.4, 19.7).

h) inExpressDiscards (22.6), the number of times reassembly (20.13) discards an Express Frame Fragment or an in progress user data frame reassembly of those Frame Fragments.

i) inPreemptableDiscards (22.6), the number of times reassembly (20.13) discards a Preemptable Frame Fragment or an in progress user data frame reassembly of those Frame Fragments

NOTE 1—A reassembly state machine is not required to maintain a count of fragments previously added to an in progress reassembly, so discard counters are not necessarily a count of received but discarded fragments. Discarding can result in the loss of fragments previously successfully added to an in progress reassembly.

j) inUnknownMppcis (22.6), MPPDU components that are not recognized (19.5.5)

k) inErroredMppdus (22.6), MPPDUs containing an incorrectly encoded component (19.5.5)

The following counts are recorded, in total, for user data frames that are not privacy protected:

l) inUserUnprotectedFrames (22.6), received user data frames that were not privacy protected.

m) inUserUnprotectedOctets (22.6), the number of octets in those user data frames.

The sums of the inUserFrames and inUserUnprotectedFrames [a) and l)] and the inUserOctets and the inUserUnprotectedOctets counts [b) and m)] reflect number of user data frames delivered to the user of the PrY's Private Port. In the absence of frame loss and network congestion due to increased use of bandwidth they should be unaffected by the decision to privacy protect, in a Privacy Channel or a Privacy Frame, any particular user data frames.

In the context of a particular system, some of these reception statistics can be obtained from other system components for PrY reporting.

The sum of inMppdus [d)] and inUserUnprotectedFrames is the number of frames received from the Controlled Port. If that total is available from the protocol entity supporting the Controlled Port, one of those counts need not be separately maintained by the PrY, although it is reported as part of the reception statistics.

If the inUnknownMppcis count is non-zero, the peer PrY has either been incorrectly implemented or is using a MPPDU component identifier reserved for future standardization.

20.15 PrY performance requirements

Time-sensitive networking (TSN) applications can benefit from or further constrain delays and delay variances experienced by relayed and transmitted frames (see IEEE Std 802.1AS, IEEE Std 802.1Q).

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21. MAC Privacy protection in Systems

This clause specifies how MAC Privacy protection is supported by the following:

a) Interface stacks in general (21.1), and end station (21.2) and bridge interfaces (21.3) in particular.
b) Link Aggregation (21.4)
c) Ethernet Data Encryption devices (EDEs, Clause 15) and related systems (21.5)
d) Shared media (21.6)
e) Multi-access LANs (21.7)
f) MACsec and MAC Privacy protection in separate systems (21.8)

Interoperability between systems requires not only interoperability between MAC Privacy protection Entity (PrY) implementations and use of the same LAN MAC technology, but also use of the same, or compatible, functions in the same relative position within interface stacks (21.1), as specified in this clause.

NOTE 1—An understanding of architectural concepts common to this and other IEEE 802.1 standards is essential to understanding this standard's specification of MAC Privacy protection. The reader is encouraged to review Clause 7 of IEEE Std 802.1AC-2016 and Clause 11 of this standard. Clause 17 provides an overview of MAC Privacy protection, and Clause 20 specifies the internal operation of PrYs. Some, but not all, of the provisions of those clauses are repeated to reduce the burden on the reader of frequent cross-referencing.

NOTE 2—The list of systems described in this clause is not intended to be exhaustive. Privacy protection can be provided wherever MACsec is used.

21.1 MAC Privacy protection interface stacks

Each PrY uses a MAC Internal Sublayer Service (ISS, IEEE Std 802.1AC) access point and provides the ISS at its secure Private Port. This allows use of MAC Privacy protection with other media-independent functions. Privacy protection depends not only on the operation of the MAC Privacy protection Entities (PrYs) that generate and validate the MPPDUs that convey user data frames, but also on the confidentiality, data integrity, and data origin authenticity of those MPPDUs. Each PrY should be directly supported by a MAC Security Entity (SecY) as shown in Figure 21-1. A PrY and its supporting SecY can also be located in adjacent systems (21.8) to support deployment of MAC Privacy protection in networks where MACsec is already deployed, or MACsec protection is separately managed.

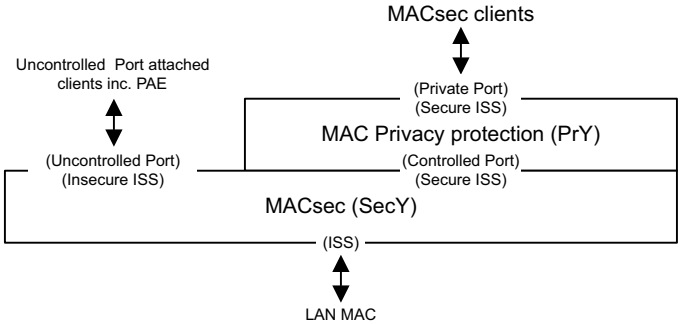


Figure 21-1—A MAC Privacy protection interface stack

Where the PrY is directly supported by a SecY (as shown in Figure 21-1), all MAC Clients that would otherwise be attached to the SecY's Controlled Port should be attached to the PrY's Private Port. The overhead of privacy protection is acceptable for most protocols (see 17.3 for discussion of the Quality of Service impact and mitigation) and a receiving peer PrY delivers the original user data frame to its client.

NOTE 1—The transmission order of frames is determined by the PrY's client. The PrY does not buffer user data frames, except as required to fill the next MPPDU to be transmitted, with the possibility of a fragment that does not fit being held over to a subsequent MPPDU. However if the PrY's client requests transmission of an express user frame before the PrY

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has completed transmission of the fragments of a default user frame, the express user frame can be encoded in an MPPDU that is transmitted prior to the MPPDU used to encode those fragments. This handling of express user frames supports use of the preemption as provided by the IEEE Std 802.3 MAC or equivalent capabilities.

A transmitting PrY can selectively protect protocols (17.2) without the need to coordinate the selection of those protocols with a peer PrY: a PrY that receives a user data frame that has been validated by its SecY but is not identified as an MPPDU to be processed by that PrY is simply passed to its client. Once a received MPPDU addressed to a PrY has been validated by its SecY, that PrY's processing of each user data frame component (an entire frame or a fragment) conveyed by the MPPDU is independent of the processing of other components in that MPPDU (with the exception of fragment reassembly) and independent of the addressing and priority parameters of the MPPDU. This independence allows a transmitting PrY to use a range of privacy protection and encapsulation strategies, again without the need to coordinate their selection with a peer PrY. This standard specifies management controls based on the user priority of each data frame, as the user priority is commonly used to distinguish between broad application classes (see IEEE Std 802.1Q) such as network critical frames or frames associated with flows requiring bandwidth allocation. A PrY may support additional management controls, including the use of flow classification (IEEE Std 802.1Q) to selectively or differentially protect user data frames of different protocols. Protection parameters can be associated with a stream_handle sub-parameter of the ISS connection_identifier parameter (see IEEE Std 802.1CB). The PICS should include Additional Information (A.3.2) for any additional management controls provided by a PrY implementation.

NOTE 2—The use of selective protection, or of varying MPPDU parameters (priority and size), can compromise privacy by revealing the characteristics of, and changes in, protected traffic. Maintaining a constant frequency, size, and bandwidth for observable traffic types can have a greater impact on quality of service than treating those types as an undifferentiated whole. This standard accordingly specifies the use of at most two Privacy Channels to control transmission timing and fragmentation of user data frames.

The use of MAC Privacy protection does not change the use or selection of MAC Clients attached to the SecY's Uncontrolled Port.

21.1.1 PrY Addressing

System interoperability also depends on the suitable selection of the destination address of the MAC Privacy protection Data Units (MPPDUs) that each PrY uses to encapsulate user data frames. The operation of networks depends on access by the intermediate systems that make up that network (bridges, routers, and switches) to the destination and source addresses of frames to be forwarded. Forwarding systems can require access to the user data conveyed to associate frames with data flows and bandwidth reservations, and to participate in control protocols. The intermediate systems' use of MACsec to validate frames, confining frames that have been corrupted or introduced by an attacker to individual LANs and protecting control protocol operation, also means that those systems have access to user data and addresses. Privacy protection depends not only on the operation of the MAC Privacy protection Entities (PrYs) that generate and validate the MPPDUs that convey user data frames, but also on the MACsec confidentiality-protection of those MPPDUs. The extent of each privacy protected region of a network is thus aligned with the MACsec protection of that region, which is typically an individual LAN but can be an equivalent LAN service supported by a provider network. To maintain this alignment without depending on additional configuration, the destination MAC address of each MPPDU is the PAE Group Address (Table 15-1) configured for MKA support of that SecY, except as specified for non-collocated SecY and PrY (21.8).

21.2 Privacy protection for end station interfaces

An end station can use the interface stack illustrated by Figure 21-1. If the end station transmits frames that are VLAN tagged, tagging (for transmission) and de-tagging (on reception) is carried out by the MACsec Client, just as if privacy protection was not being provided.

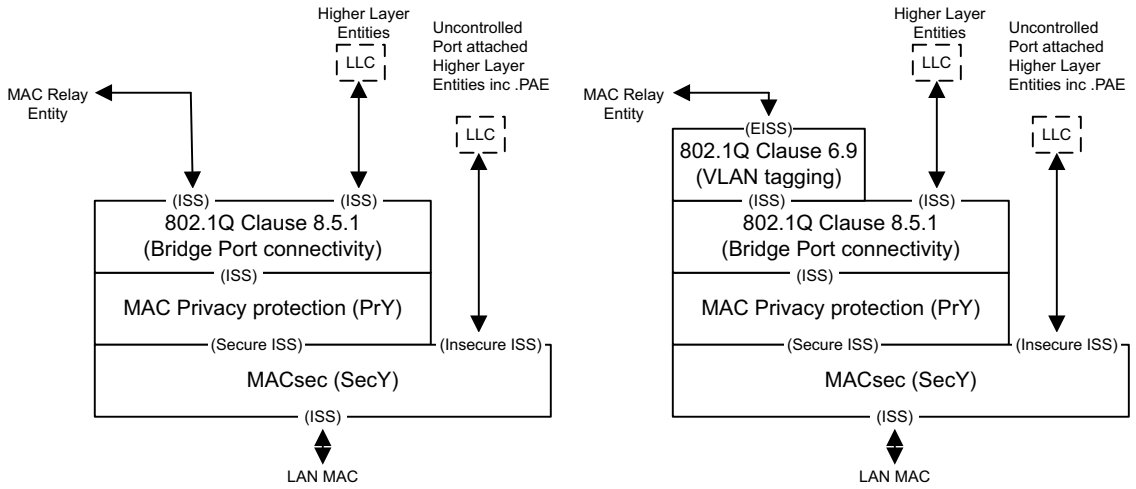
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In network scenarios where the end station identity is not strongly correlated with location, and all frames are privacy protected, the use of a locally assigned MPPDU source MAC address can help to maintain privacy. The encapsulated user data frame's source address can continue to be used (as is common practice) by a system that incorporates the receiving PrY to index end station related information.

NOTE—A media access control method can use control frames, or have other operational properties that are correlatable with station identity. See IEEE Std 802E-2020 [B5].

21.3 MAC Privacy protection for bridge interfaces

MAC Bridges are specified in IEEE Std 802.1Q. The MAC Relay Entity forwards frames between the interface stacks supported by each of the Bridge Ports. Figure 21-2 shows MAC privacy protection interface stacks for a VLAN-unaware Bridge (on the left) and a VLAN-aware Bridge (on the right), compare with Figure 11-5 and Figure 11-7.



Numeric references in this figure are to 8.5.1 and 6.9 of IEEE Std 802.1Q-2022.

Figure 21-2—MAC Privacy protected Bridge Ports

NOTE—If the MAC Bridge aggregates multiple LANs to support a single Bridge Port, each individual LAN supports its own PrY and SecY, which together provide a secure privacy protected MAC Service to the Link Aggregation sublayer, as specified in 21.4. Each aggregated port then provides that service to the Bridge Port transmit and receive functions.

Figure 17-2 shows the MPPDU frame format supported by these interface stacks, including the relative placement of the PrY MAC Addresses, MAC Privacy protection EtherType, SecTAG, and user data frame (including the VLAN tag, if present). The use of MACsec and privacy protection is not necessarily enabled for, or supported by, all ports of any given bridge. The scope of privacy protection is, as for MACsec, from a bridge port to its nearest peer neighbor, e.g., from Customer Bridge Port to Customer Bridge Port, from Customer Bridge Port to an end system, or from Provider Bridge Port to Provider Bridge Port.

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21.4 Privacy protection for Link Aggregation

Link Aggregation is specified in IEEE Std 802.1AX™. The service provided by two separate point-to-point LANs is combined to provide a single service interface. To provide MAC Security for such a system, two independent SecYs operate below the link aggregation sublayer (see 11.5). Privacy protection is provided for each SecY by a PrY that is a user of the secure MAC Service (ISS) provided by that SecY, and in turn supports the link aggregation sublayer (Figure 21-3, compare with Figure 11-8). Privacy Channel(s) for each PrY can be configured to schedule MPPDU transmission, eliminating or reducing the potential correlation between MPPDU transmission and changes in the traffic flows that link aggregation assigns to one individual link or another.

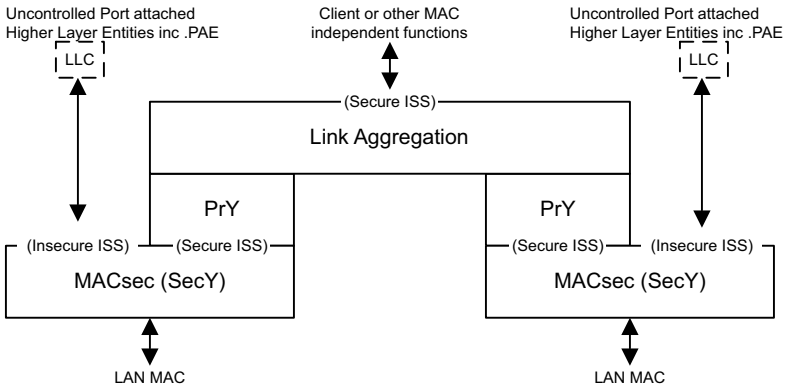


Figure 21-3—Privacy protection and Link Aggregation

If a given PrY fragments a user data frame, all the fragments of that frame are conveyed in MPPDUs protected by the SecY that supports that PrY. This facilitates the simple reassembly function (20.13.1) mandated for fragmentation support, assuming that MPPDUs transmitted with a given access priority (and thus all MPPDUs for a given Privacy Channel) and protected by a given SecY are received in order.

NOTE—MPPDU fragment encoding (19.5.4) can support reassembly of out of order fragments, but this is not required PrY reception capability (20.13).

21.5 EDEs with MAC Privacy protection

An EDE (Clause 15) can provide MAC Privacy protection for traffic transiting a Provider Bridged Network (PBN) by incorporating a PrY in the interface stack with a SecY, as illustrated in Figure 21-4 for an EDE-CC (compare with Figure 15-9).

Figure 21-4 provides a sectional view of the interface stacks, showing a single Provider Edge Port (PEP. An EDE-CC can include multiple PEPs so the PBN can use each frame's C-TAG to select the destination interface. In Figure 21-5, the member set for each VID (as specified for the Forwarding Process of a VLAN Bridge component by IEEE Std 802.1Q) includes at most one PEP so the edge C-VLAN Bridge component (shown on the left of EDE-CC1, connecting the CEP and PEPs) can forward a frame received from bridge B1 to just one PEP. That PEP can then protect the user data frame, encapsulating it within an MPPDU that is integrity and confidentiality protected by MACsec before it is forwarded to one of the Customer Network Ports (CNP) of the network C-VLAN Bridge component (shown on the right of EDE-CC1, connecting the CNPs and the Provider Network Port).

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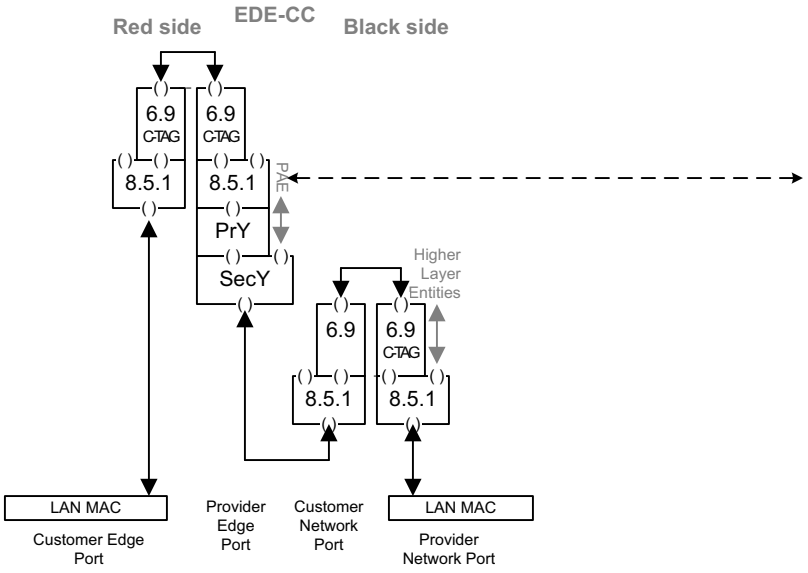


Figure 21-4—EDE-CC with MAC privacy protection

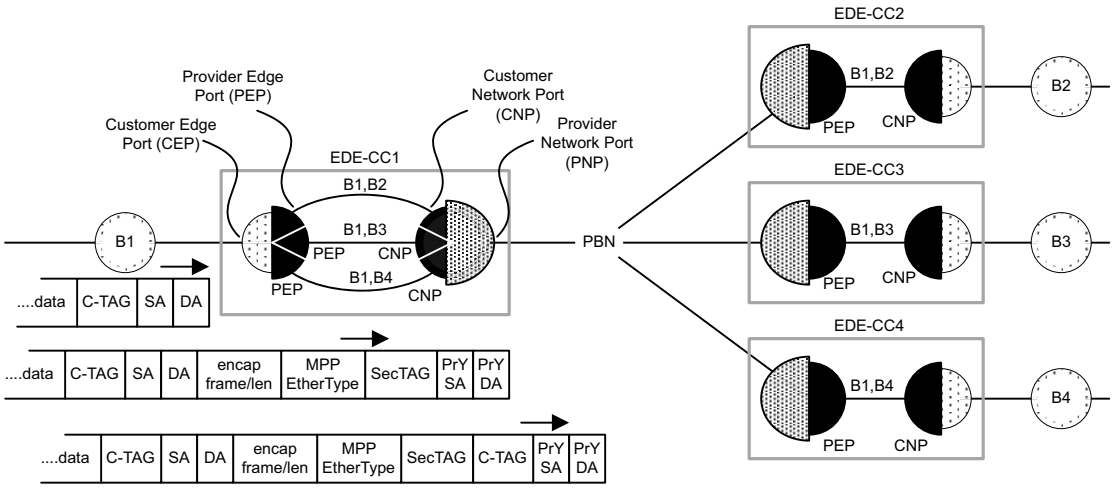


Figure 21-5—EDE-CCs communicating over a PBN

The initial octets of the protected MPPDU compose the SecTAG added by the PEP, so the CNP assigns it to the C-VLAN identified by the Port VLAN Identifier (PVID, as specified by IEEE Std 802.1Q for Support of the EISS) for that Bridge Port. The frame is then transmitted, C-tagged with that PVID, through the Provider Network Port (PNP). The PBN can then use that outer C-TAG to identify the destination PBN interface, modifying that tag if necessary. The original user data frame C-TAG is delivered, privacy protected within the MPPDU, to the distant customer equipment.

NOTE 1—IEEE Std 802.1AEcg-2017 and IEEE Std 802.1AE-2018 required an EDE-CC and EDE-SS to tag frames transmitted by the PNP with the same VID as received by the CEP [5.8 e) and 5.9 e) of IEEE Std 802.1AE-2018]. This configuration constraint simplified management, and avoided potential network forwarding changes as a result of enabling or disabling MACsec protection. EDEs implementing this constraint remain conformant to IEEE Std 802.1AE, but user data frames for different customer VLANs that are destined for the same remote PBN interface can also be forwarded through the same PEP by a conformant EDE. Those user data frames can then be encapsulated in a common set of MPDUs, and use the same SC and outer VID in the interests of privacy.

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When an EDE-CC's PNP receives a frame, its network C-VLAN component uses the member set for the VID in the outer C-TAG to select a PNP. The frame is transmitted untagged by that PNP, allowing the SecY for the attached PEP to recognize the frame as MACsec protected and to validate the frame. An associated PrY, above the SecY in the PEP interface stack can then decapsulate the original user data frames (including their original VLAN tags, if present) and pass them up the interface stack to the edge component's MAC Relay Entity.

Privacy protection can be enabled or disabled on a PEP by PEP basis: the customer equipment associated with a remote PBN interface accessible through a given EDE is not all necessarily capable of privacy protection, and can have MACsec disabled.

NOTE 2—The use of two bridge components to model the EDE behavior follows the approach used in IEEE Std 802.1Q to model Provider Edge Bridges. In the simplest cases this is elaborate, but has the advantage that a system whose behavior can always be explained in terms of the valid interconnection of existing network components does not introduce new interoperability challenges. In other words, bundling any part of a valid network of components into a distinct physical system always yields a valid (if not necessarily desirable) system. The model has the further advantage of explaining how further functionality, standardized separately for individual bridge components, can be added to an EDE or EDE-like system, and what the resulting externally observable behavior of the system should be. Connectivity Fault Management, as specified by IEEE Std 802.1Q, is one example of such functionality. The internal implementation of an EDE is not over-constrained by the two component model, conformance to this standard and to the provisions of IEEE Std 802.1Q is only to the externally observable behavior implied by the model.

NOTE 3—This standard's specification of EDEs is not intended to encompass all the possible functionality of a two bridge component device. Restricting the edge component's member set for each C-VLAN to include at most one PEP means that the edge component need not learn from customer MAC Addresses (see the enhanced filtering utility criteria specified by IEEE Std 802.1Q) and that the outer VID is determined by the inner VID. More general functionality could include learning the association of MAC Addresses with a particular PEP, with the possibility of dynamic determination of the PEP and thus of the outer VID in the VLAN tag added prior to transmission by the PNP.

21.6 Privacy protection with shared media

Privacy protection can be used with shared media and group CAs. The use of virtual shared media with a single VLAN and explicitly configured PrY addresses to optimize delivery of MPPDUs is not recommended, as enabling or disabling privacy protection would then result in a change in the underlying connectivity, possibly introducing or revealing configuration problems and making their diagnosis more difficult. If the simple use of VLANs to partition the connectivity provided by the shared media is not sufficient to meet scaling goals, the use of Provider Bridging or Provider Backbone Bridging as specified by IEEE Std 802.1Q is indicated, with privacy protection provided by the Provider Edge Port (PEP) interface stacks.

21.7 Privacy protection and multi-access LANs

Privacy protection can be used in conjunction with MACsec to support multi-access LANs (see 11.8). The MAC Address used by each PrY as the source MAC address of transmitted MPDUs is that used by protocols (EAPOL and MKA, see IEEE Std 802.1X) using the Uncontrolled Port to establish secure connectivity.

NOTE—As with end station interfaces in general (21.2) MAC Privacy protection does not prevent an observer from determining which MPPDUs are destined for which station, but does allow the end station user and a service provider to continue to use a permanent address associated with the station to identify information (e.g. access rights) for the station, without exposing that address to an unauthorized observer.

21.8 Separate privacy protection devices

MPPDUs are identified by a distinct EtherType (19.3) and can be transmitted and received without MACsec protection, so that protection and validation can be performed by separate devices or systems, possibly already deployed in a network or separately administered. Figure 21-6 illustrates the encapsulation of a user data frame, shown for simplicity as the initial component of an MPPDU, by a pair of such separate systems each on the red-side of a pair of interconnected EDEs. These systems can be modeled in the same way as a two bridge-component EDE, with the substitution (rather than the addition) of a PrY for the SecY in each

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PEP interface stack. The use of these separate privacy protecting systems is predicated on a desire to maintain separate MACsec protecting and validating systems for communication between all the connected PBN interfaces: if the associated EDE retains its default configuration it adds a copy of the outer C-TAG after the SecTAG prepended to each MPPDU transmitted across the PBN (compare Figure 21-6 with Figure 21-5), and does not inter-operate with an EDE that has both PrY and SecY in the PEP interface stack. An MPPDU destined to the group MAC Address used by the MACsec SecY is filtered by the receiving PEP, so a PrY in a privacy protecting system can be configured to use the individual MAC Address of its peer as the destination PrY MAC Address. Since an existing EDE or system on the path between the receiving EDE and the destination decapsulating device can be presumed not to recognize MPPDUs, the EDE and that device are best placed adjacent to one another, without intervening devices.

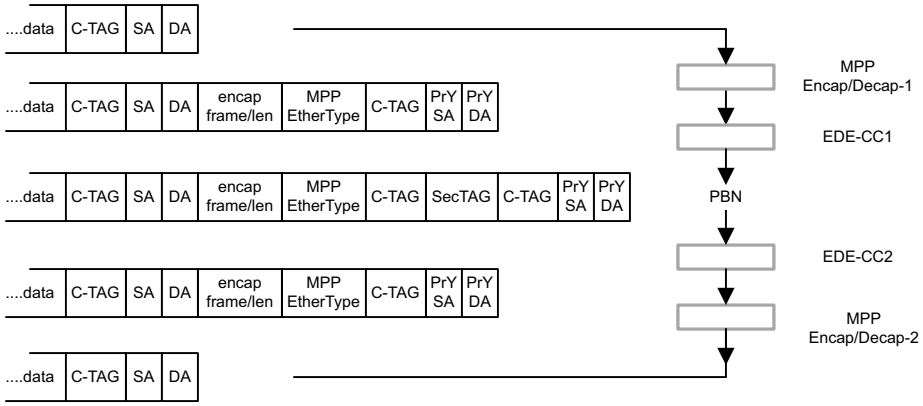


Figure 21-6—MAC Privacy protection using existing EDEs

The addressing and MPPDU format constraints described in this clause (21.7) for the deployment of MAC Privacy protection and MACsec confidentiality protection in separate systems do not apply when the functionality is provided in a single system comprising multiple components. The information that a receiving SecY needs to pass to its associated PrY is all present as parameters of an ISS indication, and each receive indication from the PrY to its client comprises the parameters expected with receipt of a user data frame. Individual MPPDU components (19.5) are self describing and can be extracted and processed separately by the PrY, with the exception of frame fragments received from a group CA. In that case the PrY needs to use the MPPDU's PrY source MAC Address to reassemble user data frame fragments received from different members of the CA.

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22. MAC Privacy protection Entity (Pry) MIB

22.1 Introduction

This clause contains an SMIv2 Management Information Base (MIB) module, based on the specification in Clause 20, 20.14, and Figure 20-6, for managing the operation of a MAC Privacy protection Entity (PrY).

22.2 The Internet-Standard Management Framework

For a detailed overview of the documents that describe the current Internet-Standard Management Framework, please refer to section 7 of IETF RFC 3410 [B8].

Managed objects are accessed via a virtual information store, termed the Management Information Base or MIB. MIB objects are generally accessed through the Simple Network Management Protocol (SNMP). Objects in the MIB are defined using the mechanisms defined in the Structure of Management Information (SMI). This MIB module is compliant to SMIv2, as described in IETF RFC 2578, IETF RFC 2579, and IETF RFC 2580.

22.3 Relationship to other MIBs

22.3.1 System MIB Group

It is assumed that a system implementing this MIB also implements the “system” group defined in IETF RFC 3418 (or at least that subset of the system group defined in IETF RFC 1213).

22.3.2 Relationship to the Interfaces MIB

It is assumed that a system implementing this MIB module implements the “interfaces” group defined in IETF RFC 2863, the Interfaces Group MIB. This MIB includes the clarifications mandated by IETF RFC 2863 for any MIB that is medium-specific or an adjunct of the Interfaces Group MIB.

A MAC Privacy protection Entity (PrY) is a shim layer in an interface stack, and is integrated with the IF-MIB as shown in Figure 22-1.

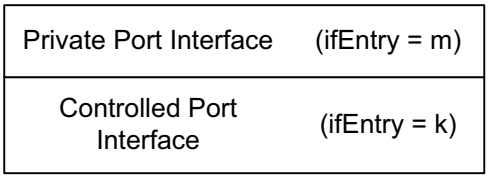


Figure 22-1—PrY Interfaces

The PrY’s Private Port is a service access point that provides one instance of the secure privacy protected MAC Service. Managed objects in this PrY MIB are indexed by the ifIndex for the PrY’s Private Port. The interface type for the Private Port is macSecControlledIF(231).

The PrY’s Controlled Port is a service access point that supports the PrY with a secure confidentiality protected instance of the MAC Service. Management parameters for the Controlled Port are as specified for the protocol entity supporting that Controlled Port (typically a SecY, see Clause 10, 10.7, and Clause 13), and are referenced by the ifIndex for that interface.

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The ifTable and ifStackTable makes the interface stack layering shown in Figure 22-1 explicit, with separate interfaces and interface indices for the Private and the Controlled Port, to allow the interface statistics for both Ports to accurately represent the numbers of packets they send and receive. These statistics can differ significantly: if a Privacy Channel (17.4.2, 20.9) is configured, ifOutOctets and ifOutUcastPkts for the Private Port remain unchanged for periods during which the PrY's user is not transmitting frames while the same statistics for the Controlled Port continue to reflect the regular transmission of Privacy Channel MPPDUs. The Private Port's interface statistics are of interest when considering the load presented by processes sending user data frames, while the Controlled Port's interface statistics reflect the load place on the supporting network elements.

The Private Port's ifAdminStatus is read-only and is `up(1)` if, and only if, MAC_Enabled (6.4, 20.4) is True. Otherwise ifAdminStatus is `down(2)`. The ifOperStatus is read-only and is `up(1)` if, and only if, MAC_Operational (6.4, 20.4) is True. Otherwise ifOperStatus is `lowerLayerDown(7)`.

NOTE 1—The Private Port's MAC_Enabled and MAC_Operational parameter values are the same as those for the Controlled Port.

The attributes in Table 22-1 are part of the required ifGeneralInformationGroup object group specified in IETF RFC 2863, and are not duplicated in the PrY MIB.

Table 22-1—Use of ifGeneralInformationGroup Objects

ifGeneralInformationGroup Objects	Use for PrY
ifDescr	See IETF RFC 2863.
ifType	macSecControlledIF(231).
ifSpeed	same as the Controlled Port interface ifSpeed.
ifPhysAddress	This object should have an octet string with zero length.
ifAdminStatus	Read-only, see above (22.3.2).
ifOperStatus	Read-only, see above (22.3.2).
ifLastChange	See IETF RFC 2863.
ifName	See IETF RFC 2863.
ifLinkUpDownTrapEnable	See IETF RFC 2863. Default: disabled(2).
ifHighSpeed	Same as the Controlled Port's ifHighSpeed.
ifConnectorPresent	Read-only: false(2).
ifAlias	See IETF RFC 2863.
ifTableLastChange	See IETF RFC 2863.

The attributes in Table 22-2 are part of the required ifCounterDiscontinuityGroup object group specified in IETF RFC 2863 and are not duplicated in the PrY MIB.

Table 22-2—Use of ifCounterDiscontinuityGroup Object

ifCounterDiscontinuityGroup Object	Use for MACsec
ifCounterDiscontinuityTime	See IETF RFC 2863. Always 0, no discontinuity.

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The ifStackTable is used to identify the layer relationships between PrY's Private Port (upper) interface, with an ifIndex value that identifies the PrY's managed objects, and the PrY's Controlled Port (lower) interface (with an ifIndex that identifies the protocol entity and managed objects supporting that port.

The use of the ifPacketGroup object group specified in IETF RFC 2863 is described in 20.14.

The ifRcvAddressTable is not applicable to PrY operation, as a PrY is always capable of receiving user data frames that are not MAC Privacy protected. The MAC Destination Address of those frames is the MAC address of the intended recipient, and not necessarily an address associated with the PrY or with the interface stack of which the PrY is a part.

NOTE 2—As specified, a PrY always operates in 'promiscuous mode'. As specified in IETF RFC 2863, entries in the ifRcvAddressTable are only required for those addresses for which the system would receive frames were it not operating in promiscuous mode. Other components in the system of which the PrY is a part can be configured to selectively forward, filter, or receive frames.

22.4 Security considerations

MAC Privacy protection is supported by the MAC Security protocol (MACsec) which, when properly deployed, can help to protect against inadvertent transmission to or from unauthenticated and unauthorized parties. Unauthorized access to management objects defined in this MIB module with MAX-ACCESS read-only, as well as read-write or read-create, can compromise privacy as follows:

a) The `ieee8021PryOutTable`, `ieee8021PryChannelOutTable`, and `ieee8021PryInTable` together provide direct statistical information (though not real-time timing information) about the true level of data traffic passing through the interface.

b) The configuration of the `ieee8021PrySelectionTable` , `ieee8021PryFrameTable`, and the `ieee8021PryChannelTable` can reveal information that can be correlated with the intended use of the interface, even if it is not carrying that traffic at present.

c) The `ieee8021PrySelectionTable`, in combination with the `ieee8021PryFrameTable` and the `ieee8021PryChannelTable` provides detailed information about the Privacy Frame, Privacy Channel, and access priority (potentially visible in a VLAN tag added by an EDE-CC or EDE-SS) support of user data frames of each priority. This information can be useful to an adversary seeking information that is correlated with the use and changing use of particular applications.

Some of this information, and further detailed information, can also be obtained by accessing read-only, as well as read-write or read-create, objects in other MIB modules supported by the system. It is important, if privacy its to be protected, that all MIB module access is restricted to authorized parties.

Access to read-write objects can compromise basic network operation as well as privacy, as follows:

d) Privacy Channels can be configured, through the `ieee8021PryChannelTable`, to use excessive network bandwidth or to use so little bandwidth that service is effectively denied to some or all traffic passing through the interface. If the PrY is not collocated with a SecY, the `ieee8021PryIfMppduDA` in the `ieee8021PryIfTable` can be configured to transmit excess traffic throughout the network.

e) If privacy protection is not desired, it is advisable to disable both `ieee8021PryIfTxProtection` and `ieee8021PryIfRxProtection` in `ieee8021PryIfTable`. If an adversary can enable receive privacy protection and can transmit a data frame that is relayed through the network (as if from an authorized source) and addressed to an PrY, that frame can be processed by that PrY and forwarded with a destination or source MAC address that would have resulted in prior frame filtering if not hidden by privacy protection.

f) The `ieee8021PryFrameTable` can be configured to change, or even reorder, the priority used to transmit user data frames, impacting network performance.

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This clause (22.4) does not provide a comprehensive description of privacy, security, and operational exposures that can result from unauthorized access to the PrY MIB and other system MIB objects. Additional vulnerabilities of greater or lesser significance can exist.

22.5 Structure of the MIB module

A single MIB module is defined in this clause (Clause 22). In the MIB module each PrY is identified by the InterfaceIndex used by the Interfaces MIB (22.3.2, Figure 22-1) for its Private Port. This facilitates identification of the PrY when investigating an interface stack, and discovery of the other interface stack entities (via the Interfaces MIB) related to a particular PrY, including a supporting SecY (if present) and its associated PAE. Figure 22-1 illustrates the structure of the MIB module.

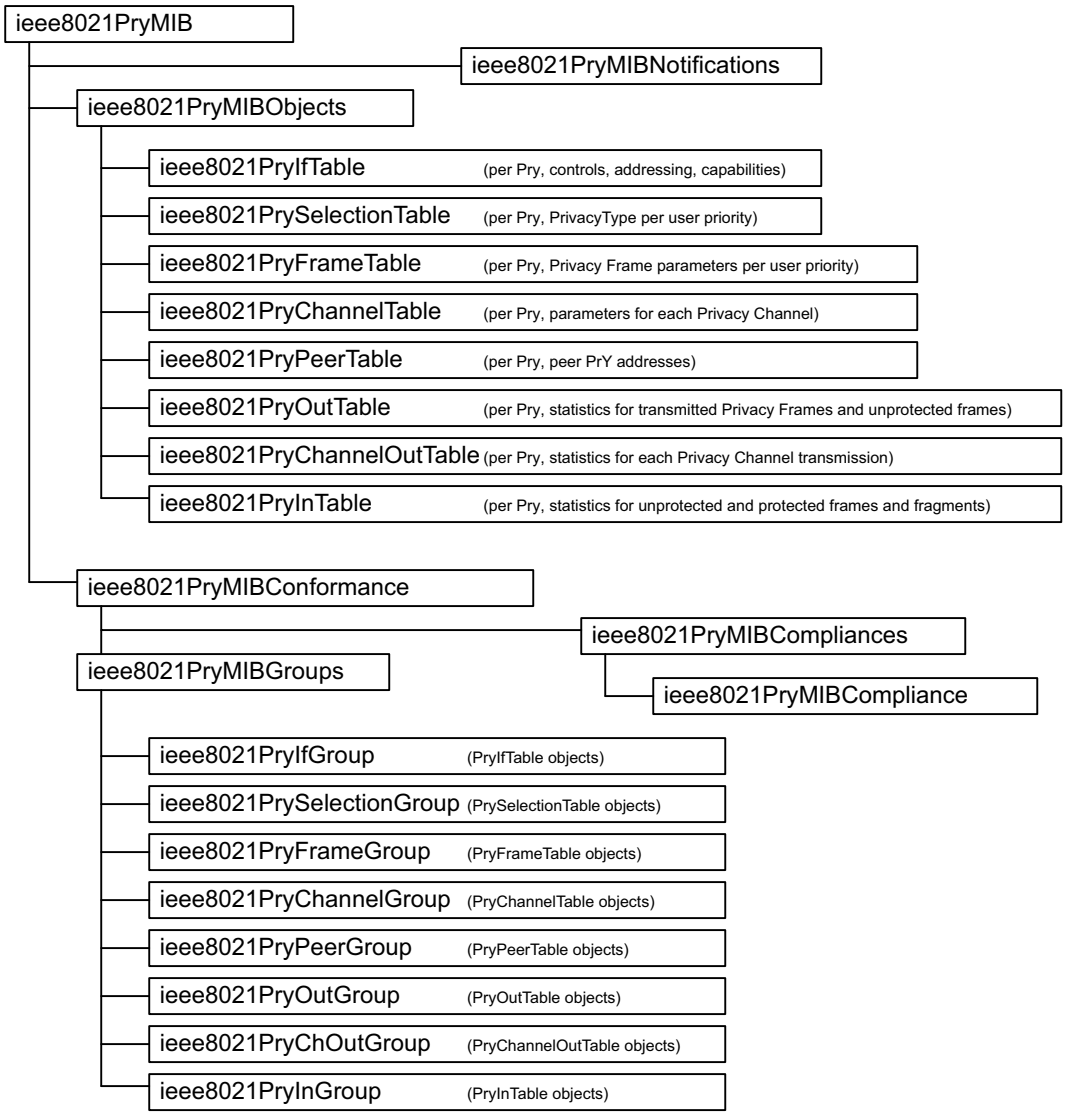


Figure 22-2—PrY MIB structure

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At the top level, the MIB module comprises management and statistics objects, both with an initial OID (object identifier) of ieee8021PryMIBObjects, and MIB conformance information with an initial OID of ieee8021PryMIBConformance. No notifications are defined.MIB objects are arranged in a number of tables (in MIB terms a SEQUENCE OF entries) with each entry in the table comprising an number of basic objects (in MIB terms a SEQUENCE OF objects such as truth values, integers, text strings).

The entries in each of the management and statistics objects tables are indexed in one of the following ways:

a) By the interface index, if the objects in each entry are for the PrY identified by that index).
b) By the interface index and user priority, for PrY objects that depend on user priority based Privacy Selection.
c) By the interface index and Privacy Channel (Express or Preemptable) for Privacy Channel management and statistics.

NOTE 1—Privacy Channel objects only apply to PrY transmission. A PrY's peers' transmit behavior is only constrained by this standard as needed for interoperability (17.4, 18.5).

d) By interface index and MAC Address to identify the peers of a given PrY.

The MIB conformance objects are organized into compliance statements and conformance groups.

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22.6 MAC Privacy protection Entity (PrY) MIB definition[1,2]

```
IEEE8021-PRY-MIB DEFINITIONS ::= BEGIN

-- ============================================================================
-- IEEE802.1AE MAC Privacy protection Entity (PrY) MIB
-- ============================================================================

IMPORTS
    MODULE-IDENTITY, OBJECT-TYPE,
    Integer32, Unsigned32, Counter64
        FROM SNMPv2-SMI
--  SnmpAdminString
--      FROM SNMP-FRAMEWORK-MIB
    MODULE-COMPLIANCE, OBJECT-GROUP
        FROM SNMPv2-CONF
    MacAddress, TruthValue, RowStatus
        FROM SNMPv2-TC
    InterfaceIndex
        FROM IF-MIB
    IEEE8021PriorityValue
        FROM IEEE8021-TC-MIB
;
ieee8021PryMIB MODULE-IDENTITY
    LAST-UPDATED "202210310000Z" -- October 31, 2022
    ORGANIZATION "IEEE 802.1 Working Group"
    CONTACT-INFO
       "WG-URL: http://www.ieee802.org/1
        WG-EMail: stds-802-1-L@ieee.org/1/
         Contact: IEEE 802.1 Working Group Chair
          Postal: C/O IEEE 802.1 Working Group
                  IEEE Standards Association
                  445 Hoes Lane
                  Piscataway, NJ 08854
                  USA
          E-mail: stds-802-1-chairs@ieee.org"
    DESCRIPTION
       "The MAC Privacy protection Entity (PrY) MIB module.

        Unless otherwise indicated, the references in this MIB module are to
        IEEE Std 802.1AE-2018 as amended by IEEE Std 802.1AEdk-2023.

        Copyright (C) IEEE (2022).
        This MIB module is part of IEEE Std 802.1AE; see that standard and its
        amendments for full legal notices.

        A MAC Privacy protection Entity (PrY) is a protocol shim in an
        interface stack that encapsulates user data frames in MAC Privacy
        protection Data Units (MPPDUs). Once those MPPDUs are confidentiality
        protected by MACsec, the ability of potential adversaries to draw
        conclusions from the source and destination MAC addresses, sizes, and
        transmission timing and frequency of user data frames is reduced or
        eliminated.

        Each PrY in a system and its managed objects is indexed by the
        InterfaceIndex(ifIndex) of its upper interface (Private Port), which
        provides a privacy protected service to its user, typically a Bridge
        Port(IEEE Std 802.1Q) or an end station protocol stack. Object names
        can be conveniently pronounced by rendering 'Pry' as 'Privacy ', and
        'If' as 'Interface '.
        "
```

[1] *Copyright release for MIBs:* Users of this standard may freely reproduce the MIB modules in this standard so that they can be used for their intended purpose.

[2] The MIB text in this clause includes clickable cross-references to the other clauses of this standard and to MIB objects (highlighted). A plain text (UTF-8) version of this MIB is attached to the PDF version of this standard, and can be obtained by Web browser from the IEEE 802.1 Website at https://1.ieee802.or/mib-modules/.

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```
    REVISION "202210310000Z" -- October 31, 2022
    DESCRIPTION "Initial Revision"
    ::= { iso (1) iso-identified-organization (3) ieee (111)
    standards-association-numbered-series-standards (2) lan-man-stds (802)
    ieee802dot1(1) ieee802dot1mibs(1) 36 }
-- ============================================================================
-- subtrees in the PrY MIB

ieee8021PryMIBNotifications OBJECT IDENTIFIER ::= { ieee8021PryMIB 1 }
ieee8021PryMIBObjects       OBJECT IDENTIFIER ::= { ieee8021PryMIB 2 }
ieee8021PryMIBConformance   OBJECT IDENTIFIER ::= { ieee8021PryMIB 3 }
-- ============================================================================
--ieee8021PryMIBObjects
--  ieee8021PryIfTable
--  ieee8021PrySelectionTable
--  ieee8021PryFrameTable
--  ieee8021PryChannelTable
--  ieee8021PryPeerTable
--  ieee8021PryOutTable
--  ieee8021PryChannelOutTable
--  ieee8021PryInTable
-- ============================================================================
--ieee8021PryIfTable
ieee8021PryIfTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021PryIfEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
    "A table with entries for each MAC Privacy protection capable interface in
    the system. Configured values of writable objects in table entries MUST be
    persistent and remain unchanged across re-initialization of the system's
    management entity."
    REFERENCE "20.14, "
    ::= { ieee8021PryMIBObjects 1 }

--ieee8021PryIfEntry
ieee8021PryIfEntry OBJECT-TYPE
    SYNTAX     Ieee8021PryIfEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A table entry with privacy controls for a particular PrY."
    INDEX { ieee8021PryIfIndex }
        ::= { ieee8021PryIfTable 1 }

--Ieee8021PryIfEntry
Ieee8021PryIfEntry ::=
   SEQUENCE {
        ieee8021PryIfIndex                  InterfaceIndex,
        ieee8021PryIfRxProtection           TruthValue,
        ieee8021PryIfTxProtection           TruthValue,
        ieee8021PryIfSecySupport            TruthValue,
        ieee8021PryIfAddr                   MacAddress,
        ieee8021PryIfMppduDA                MacAddress,
        ieee8021PryIfDefaultReassembly      TruthValue,
        ieee8021PryIfMaxPeers               Integer32,
        ieee8021PryIfNumPeers               Integer32
    }

--ieee8021PryIfIndex
ieee8021PryIfIndex OBJECT-TYPE
    SYNTAX      InterfaceIndex
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "The Interface Index for this PrY's Private Port."
    REFERENCE   "20.14"
    ::= { ieee8021PryIfEntry 1 }
```


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```
--ieee8021PryIfRxProtection
ieee8021PryIfRxProtection OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "When true(1), the PrY processes received MPPDUs addressed to
                 ieee8021PryAddr and ieee8021IfMppduDA (if that is a Group
                 address). When false(2) they are passed directly to the PrY's
                 Private Port. All other  MPPDUs are passed to the Private
                 Port, unprocessed, irrespective of this control's value."
    REFERENCE   "20.11"
    DEFVAL      { true }
    ::= { ieee8021PryIfEntry 2 }

--ieee8021PryIfTxProtection
ieee8021PryIfTxProtection OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "When true(1), the PrY protects transmitted user data frames
                 as configured in the Privacy Selection Table. When false(2),
                 all user data frames are passed directly to the PrY's
                 Controlled Port."
    REFERENCE   "20.5"
    DEFVAL      { true }
    ::= { ieee8021PryIfEntry 3 }

--ieee8021PryIfSecySupport
ieee8021PryIfSecySupport OBJECT-TYPE
  SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Set true(1) by the system if the PrY is directly supported by
                 a SecY and MKA, and false(2) otherwise. When true, the value
                 of ieee8021PryIfMppduDA and the entries in the PrY's peer
                 address table (ieee8021PryPeerTable) are determined by the
                 Key Agreement Entity (KaY) operating MKA, and are not
                 writable by network management."
    REFERENCE   "18.1, 20.11"
    DEFVAL      { true }
    ::= { ieee8021PryIfEntry 4 }

--ieee8021PryIfAddr
ieee8021PryIfAddr OBJECT-TYPE
    SYNTAX      MacAddress
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The individual MAC address associated with the PrY and other
                 components of the PrY's interface stack. Allocated by the
                 system. Used by PrY as the source address of MPPDUs and by a
                 supporting SecY (if present) for SCI assignment. The PrY
                 receives and processes MPPDUs with this destination address."
    REFERENCE   "18.1"
    ::= { ieee8021PryIfEntry 5 }

--ieee8021PryIfMppduDA
ieee8021PryIfMppduDA OBJECT-TYPE
    SYNTAX      MacAddress
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "The destination MAC address used to transmit MPPDUs.
                 Also used to receive MPPDUs (if a Group address) when
                 ieee8021PryIfRxProtection is true. Set by the KaY if
                 ieee8021PryIfSecySupport is true, otherwise writable. If
                 ieee8021PryIfSecySupport transitions from true to false,
                 defaults to the Nearest non-TPMR Bridge Group address."
    REFERENCE   "18.1, 20.11"
    ::= { ieee8021PryIfEntry 6 }

```


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```
--ieee8021PryIfDefaultReassembly
ieee8021PryIfDefaultReassembly OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Set true(1) by the system to indicate that the default
                 reassembly algorithm is used. Set false(2), otherwise. If
                 the system supports additional reassembly algorithms it shall
                 also support selection of the default algorithm. The maximum
                 size of the user data frame (DA, SA, MSDU) that can be
                 reassembled for delivery to the Private Port is the value of
                 ifMtu (as provided by the IF-MIB plus 22 octets."
    REFERENCE   "20.13, 20.13.1"
    ::= { ieee8021PryIfEntry 7 }

--ieee8021PryIfMaxPeers
ieee8021PryIfMaxPeers OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The maximum number of peer PrYs supported by the configured
                 reassembly algorithm."
    REFERENCE   "20.13"
    ::= { ieee8021PryIfEntry 8 }

--ieee8021PryIfNumPeers
ieee8021PryIfNumPeers OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of peer PrYs detected by the system."
    REFERENCE   "20.13"
    ::= { ieee8021PryIfEntry 9 }

-- ============================================================================
--ieee8021PrySelectionTable
ieee8021PrySelectionTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021PrySelectionEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A table for Privacy selection by transmit request priority"
    REFERENCE   "17.4, 17.4.3, 20.5"
    ::= { ieee8021PryMIBObjects 2 }

--ieee8021PrySelectionEntry
ieee8021PrySelectionEntry OBJECT-TYPE
    SYNTAX      Ieee8021PrySelectionEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "Privacy type selection for a given user priority"
    REFERENCE   "17.4.3, 20.5"
    INDEX { ieee8021PryIfIndex, ieee8021PrySelectionPriority }
      ::= { ieee8021PrySelectionTable 1 }

--Ieee8021PrySelectionEntry
Ieee8021PrySelectionEntry ::=
    SEQUENCE {
        ieee8021PrySelectionPriority    IEEE8021PriorityValue,
        ieee8021PrySelectionPrivacyType INTEGER
    }

--ieee8021PrySelectionPriority
ieee8021PrySelectionPriority OBJECT-TYPE
    SYNTAX      IEEE8021PriorityValue
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "Transmit request user priority."
    REFERENCE   "17.4.3, 20.5"
    ::= { ieee8021PrySelectionEntry 1 }

```


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```
--ieee8021PrySelectionPrivacyType
ieee8021PrySelectionPrivacyType OBJECT-TYPE
    SYNTAX      INTEGER {
                    none (1),
                    privacyFrame (2),
                    preemptableChannel (3),
                    expressChannel (4)
                }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "Privacy protection: none, privacyFrame, preemptableChannel,
                 or expressChannel."
    REFERENCE   "17.4.3, 20.5"
    ::= { ieee8021PrySelectionEntry 2 }

-- ============================================================================
--ieee8021PryFrameTable
ieee8021PryFrameTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021PryFrameEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A table of Privacy Frame parameters for each possible value
                 of Controlled Port transmission priority (PrY user priority).
                 Each table entry can be configured even if Privacy Frame
                 transmission is not currently selected for user data frames
                 of that user priority."
    REFERENCE   "17.4.3, 20.7"
    ::= { ieee8021PryMIBObjects 3 }

--ieee8021PryFrameEntry
ieee8021PryFrameEntry OBJECT-TYPE
    SYNTAX      Ieee8021PryFrameEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "Privacy Frame parameters for a given user priority."
    REFERENCE   "20.14.4"
    AUGMENTS { ieee8021PrySelectionEntry }
         ::= { ieee8021PryFrameTable 1 }

--Ieee8021PryFrameEntry
Ieee8021PryFrameEntry ::=
    SEQUENCE {
        ieee8021PryFrameAccessPriority   IEEE8021PriorityValue,
        ieee8021PryFrameRevealDE         TruthValue,
        ieee8021PryFramePadding          INTEGER
    }

--ieee8021PryFrameAccessPriority
ieee8021PryFrameAccessPriority OBJECT-TYPE
    SYNTAX      IEEE8021PriorityValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "The Controlled Port priority (access priority) used to
                 transmit Privacy Frames with the Private Port transmission
                 priority (user priority) that selects this table entry."
    REFERENCE   "17.4.3, 20.7"
    ::= { ieee8021PryFrameEntry 1 }

```


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```
--ieee8021PryFrameRevealDE
ieee8021PryFrameRevealDE OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "When true(1), the drop_eligible parameter value for
                 Controlled Port transmission of a Privacy Frame of the
                 specified user priority matches that used for the
                 encapsulated user data frame's Private Port transmit request.
                 Otherwise the Controlled Port transmit request has
                 drop_eligible false, and the DEI bit of a VLAN tag added as a
                 consequence of that transmit request (e.g., by the network
                 component of an EDE-CC) will be clear."
    REFERENCE   "20.7"
    ::= { ieee8021PryFrameEntry 2 }

--ieee8021PryFramePadding
ieee8021PryFramePadding OBJECT-TYPE
    SYNTAX      INTEGER {
                  one (1),
                  sixteen (16),
                  thirtyTwo(32),
                  sixtyFour(64)
                 }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "Specifies padding of the Privacy Frame MPPDU (excluding
                 its source and destination MAC addresses) to four octets
                 (to allow for the MAC Privacy protection EtherType and the
                 MPPCI for an Encapsulated Frame) plus the nearest
                 multiple of one(1) (for no padding), sixteen(16),
                 thirtyTwo(32), or sixtyFour (64) octets. The specified size
                 excludes any octets to be added by supporting components
                 lower in the interface stack (e.g. a MACsec SecTAG and ICV,
                 and the Ethernet FCS) or other bridge components (e.g. an
                 outer VLAN tag added by an EDE's network component)."
    REFERENCE   "17.4.3, 20.7"
    ::= { ieee8021PryFrameEntry 3 }

-- ============================================================================
--ieee8021PryChannelTable
ieee8021PryChannelTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021PryChannelEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A table with entries for the Express and the Preemptable
                 Privacy Channel's parameters."
    REFERENCE   "20.8, 20.9, 20.10"
    ::= { ieee8021PryMIBObjects 4 }

--ieee8021PryChannelEntry
ieee8021PryChannelEntry OBJECT-TYPE
    SYNTAX      Ieee8021PryChannelEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "Parameters for a given Privacy Channel."
    INDEX { ieee8021PryIfIndex, ieee8021PryChType }
        ::= { ieee8021PryChannelTable 1 }
```


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```
--Ieee8021PryChannelEntry
Ieee8021PryChannelEntry ::=
    SEQUENCE {
        ieee8021PryChType              INTEGER,
        ieee8021PryChEnable            TruthValue,
        ieee8021PryChFragmentEnable    TruthValue,
        ieee8021PryChAccessPriority    IEEE8021PriorityValue,
        ieee8021PryChUserDataFrameSize Integer32,
        ieee8021PryChMppduGeneration   INTEGER,
        ieee8021PryChRequestedKbitRate      Unsigned32,
        ieee8021PryChMppduBitsOnWire   Unsigned32,
        ieee8021PryChMppduInterval     Unsigned32,
        ieee8021PryChUserBurstOctets   Unsigned32
    }

--ieee8021PryChType
ieee8021PryChType OBJECT-TYPE
    SYNTAX      INTEGER {
                    express (1),
                    preemptable (2)
                }
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "Privacy Channel Type, express(1) or preemptable(2)."
    REFERENCE   "20.8"
    ::= { ieee8021PryChannelEntry 1 }

--ieee8021PryChEnable
ieee8021PryChEnable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "When true(1), user data frames assigned to this Privacy
                 Channel by a ieee8021PrySelectionEntry are transmitted using
                 this channel's parameters. When false(2), they are
                 transmitted using the other channel if ieee8021PryChEnable
                 is true for that channel and transmitted as Privacy Frames
                 using the relevant ieee8021PryFrameEntry otherwise."
    REFERENCE   "20.8"
    ::= { ieee8021PryChannelEntry 2 }

--ieee8021PryChFragmentEnable
ieee8021PryChFragmentEnable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "When true(1) permits user data frame fragmentation in this
                 Privacy Channel. Should be true, for bandwidth efficiency and
                 delay minimization. Provided to allow simple performance
                 testing and fragmentation benefit analysis."
    REFERENCE   "20.10"
    ::= { ieee8021PryChannelEntry 3 }

--ieee8021PryChAccessPriority
ieee8021PryChAccessPriority OBJECT-TYPE
    SYNTAX      IEEE8021PriorityValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "The Controlled Port priority (access priority) used to
                 transmit MPPDUs for this Privacy Channel."
    REFERENCE   "20.8, 20.9.3"
    ::= { ieee8021PryChannelEntry 4 }

```


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```
--ieee8021PryChUserDataFrameSize
ieee8021PryChUserDataFrameSize OBJECT-TYPE
    SYNTAX      Integer32 (128 .. 32768)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "User specification of MPPDU size. The largest user data
                 frame, at the Private Port interface (i.e. prior to MAC
                 Privacy protection)that can be transmitted as an MPPDU
                 Encapsulated Frame without fragmentation. Includes the user
                 data frame DA, SA, MSDU with EtherType, and a four octet FCS
                 allowance.  Excludes octets subsequently added by MACsec, or
                 other supporting interface stack components. Physical media,
                 and the configuration of other system components can impose
                 an upper bound lower than the configured value of this
                 parameter."
    REFERENCE   "20.9.3"
    ::= { ieee8021PryChannelEntry 5 }

--ieee8021PryChMppduGeneration
ieee8021PryChMppduGeneration OBJECT-TYPE
    SYNTAX      INTEGER { default (1), transmissionGate (2), other (3)}
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "The MPPDU generation algorithm for this Privacy Channel. When
                 default(1), the (maximum) bandwidth is requested, with
                 a catch up (burst)parameter to recovers lost bandwidth if
                 an MPPDU transmission has been delayed by another frame sent
                 with higher access priority or by another component of the
                 same interface stack. When transmissionGate(2), MPPDU
                 transmission timing is gated, see 20.9.5 and IEEE Std 802.1Q."
    REFERENCE   "20.9, 20.9.4, 20.9.5"
    ::= { ieee8021PryChannelEntry 6 }

--ieee8021PryChRequestedKbitRate
ieee8021PryChRequestedKbitRate OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "The physical medium bit rate (kilobits per second) to be
                 used by this Privacy Channel and the default MPPDU generation
                 algorithm in the absence of higher priority traffic or other
                 resource competition."
    REFERENCE   "20.9.4"
    ::= { ieee8021PryChannelEntry 7 }

--ieee8021PryChMppduBitsOnWire
ieee8021PryChMppduBitsOnWire OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The number of bit times required to transmit an MPPDU that
                 conveys a single, Private Port transmitted, user data frame
                 of ieee8021PryChUserDataFrameSize encoded as an Encapsulated
                 Frame(19.5.1). Calculated by the system, including all fields
                 added by the interface stack."
    REFERENCE   "20.9.4"
    ::= { ieee8021PryChannelEntry 8 }

--ieee8021PryChMppduInterval
ieee8021PryChMppduInterval OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "The approximate interval (as calculated by the system) in
                 nanoseconds between the transmission of MPPDUs for this
                 Privacy Channel, in the absence of competing higher priority
                 traffic or other resource competition.
"
    REFERENCE   "20.9.4"
    ::= { ieee8021PryChannelEntry 9 }

```


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```
--ieee8021PryChUserBurstOctets
ieee8021PryChUserBurstOctets OBJECT-TYPE
    SYNTAX      Unsigned32
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION "The number of additional user data frame burst for use by the
                 default MPPDU generation algorithm to recover channel
                 bandwidth lost to competing higher priority traffic."
    REFERENCE   "20.9.4"
    ::= { ieee8021PryChannelEntry 10 }

-- ============================================================================
--ieee8021PryPeerTable
ieee8021PryPeerTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021PryPeerEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "A table of peer PrYs. Frame Fragments received in MPPDUs with
                 source MAC addresses not in this table are discarded. When
                 ieee8021PryIfSecySupport is true, table entries are created
                 and deleted by the supporting Key Agreement Entity. When false
                 the system automatically creates an entry for
                 ieee8021PryIfMppduDA if that is not a Group address, and other
                 entries can be created by management."
    REFERENCE   "20.13"
    ::= { ieee8021PryMIBObjects 5 }

--ieee8021PryPeerEntry
ieee8021PryPeerEntry OBJECT-TYPE
    SYNTAX     Ieee8021PryPeerEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "An entry in the peer PrY table for one peer."
    INDEX { ieee8021PryIfIndex, ieee8021PryPeerAddr }
    ::= { ieee8021PryPeerTable 1 }

--Ieee8021PryPeerEntry
Ieee8021PryPeerEntry ::=
    SEQUENCE {
        ieee8021PryPeerAddr       MacAddress,
        ieee8021PryPeerRowStatus  RowStatus
    }

--ieee8021PryPeerAddr
ieee8021PryPeerAddr OBJECT-TYPE
    SYNTAX      MacAddress
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "MAC address of a peer PrY."
    REFERENCE   "20.13"
    ::= { ieee8021PryPeerEntry 1 }

--ieee8021PryPeerRowStatus
ieee8021PryPeerRowStatus OBJECT-TYPE
    SYNTAX      RowStatus
    MAX-ACCESS  read-create
    STATUS      current
    DESCRIPTION "Indicates the status of an entry in the ieee8021PryPeerTable,
                 can be used to create entries if ieee8021PryIfSecySupport
                 is false."
    REFERENCE   "20.13"
    ::= { ieee8021PryPeerEntry 2 }

```


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```
-- ============================================================================
--ieee8021PryOutTable
ieee8021PryOutTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021PryOutEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "Transmission statistics for Private Port user data frames
                 transmitted as Privacy Frames or unprotected."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryMIBObjects 6 }

--ieee8021PryOutEntry
ieee8021PryOutEntry OBJECT-TYPE
    SYNTAX      Ieee8021PryOutEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "Transmission statistics, for frames not in a Privacy Channel."
    AUGMENTS { ieee8021PryIfEntry }
        ::= { ieee8021PryOutTable 1 }

--Ieee8021PryOutEntry
Ieee8021PryOutEntry ::=
    SEQUENCE {
        ieee8021PryOutPrivacyFrames Counter64,
        ieee8021PryOutPfUserOctets  Counter64,
        ieee8021PryOutPfPadOctets   Counter64,
        ieee8021PryOutUnprtFrames   Counter64,
        ieee8021PryOutUnprtOctets   Counter64
    }

--ieee8021PryOutPrivacyFrames
ieee8021PryOutPrivacyFrames OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Total number of user data frames sent as Privacy Frames (each
                 in a separate MPPDU)."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryOutEntry 1 }

--ieee8021PryOutPfUserOctets
ieee8021PryOutPfUserOctets OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Total number of user data octets sent in Privacy Frames (each
                 user data frame in a separate MPPDU). Not counting pad octets."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryOutEntry 2 }

--ieee8021PryOutPfPadOctets
ieee8021PryOutPfPadOctets OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Total number of pad octets sent in Privacy Frames (each
                 conveying a single Private Port user data frame)."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryOutEntry 3 }

--ieee8021PryOutUnprtFrames
ieee8021PryOutUnprtFrames OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Total number of MAC Privacy PDUs sent."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryOutEntry 4 }

```


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```
--ieee8021PryOutUnprtOctets
ieee8021PryOutUnprtOctets OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Total number of MAC Privacy PDUs sent."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryOutEntry 5 }

-- ============================================================================
--ieee8021PryChannelOutTable
ieee8021PryChannelOutTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021PryChannelOutEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "Transmission statistics for the Express and Preemptable
                 Privacy Channels."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryMIBObjects 7 }

--ieee8021PryChannelOutEntry
ieee8021PryChannelOutEntry OBJECT-TYPE
    SYNTAX      Ieee8021PryChannelOutEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "Transmission statistics for a Privacy Channel."
    AUGMENTS { ieee8021PryChannelEntry }
        ::= { ieee8021PryChannelOutTable 1 }

--Ieee8021PryChannelOutEntry
Ieee8021PryChannelOutEntry ::=
    SEQUENCE {
        ieee8021PryChOutUserFrames      Counter64,
        ieee8021PryChOutUserOctets      Counter64,
        ieee8021PryChOutPadOctets       Counter64,
        ieee8021PryChOutMppdus          Counter64,
        ieee8021PryChOutEncapFrames     Counter64,
        ieee8021PryChOutExpFragments    Counter64,
        ieee8021PryChOutPreFragments    Counter64
    }

--ieee8021PryChOutUserFrames
ieee8021PryChOutUserFrames OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Number of user data frames sent in this Privacy Channel."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryChannelOutEntry 1 }

--ieee8021PryChOutUserOctets
ieee8021PryChOutUserOctets OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Number of user data octets sent in this Privacy Channel.
                 Not counting pad octets."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryChannelOutEntry 2 }

--ieee8021PryChOutPadOctets
ieee8021PryChOutPadOctets OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Number of pad octets sent in this Privacy Channel."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryChannelOutEntry 3 }

```


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```
--ieee8021PryChOutMppdus
ieee8021PryChOutMppdus OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Number of MPPDUs sent in this Privacy Channel."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryChannelOutEntry 4 }

--ieee8021PryChOutEncapFrames
ieee8021PryChOutEncapFrames OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Number of Encapsulated Frames encoded for this Privacy
                 Channel."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryChannelOutEntry 5 }

--ieee8021PryChOutExpFragments
ieee8021PryChOutExpFragments OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Number of Express Fragments encoded for this Privacy Channel."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryChannelOutEntry 6 }

--ieee8021PryChOutPreFragments
ieee8021PryChOutPreFragments OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Number of Preemptable Fragments encoded for this Privacy
                 Channel."
    REFERENCE   "20.14.1"
    ::= { ieee8021PryChannelOutEntry 7 }

-- ============================================================================
--ieee8021PryInTable
ieee8021PryInTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021PryInEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "Reception statistics for all user data frames, MPPDUs, their
                 components,and fragmented user data frame reassembly."
    REFERENCE   "20.14.2"
    ::= { ieee8021PryMIBObjects 8 }

--ieee8021PryInEntry
ieee8021PryInEntry OBJECT-TYPE
    SYNTAX      Ieee8021PryInEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION "Transmission statistics, for frames not in a Privacy Channel."
    AUGMENTS { ieee8021PryIfEntry }
        ::= { ieee8021PryInTable 1 }
```


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```
--Ieee8021PryInEntry
Ieee8021PryInEntry ::=
    SEQUENCE {
        ieee8021PryInUserFrames       Counter64,
        ieee8021PryInUserOctets       Counter64,
        ieee8021PryInPadOctets        Counter64,
        ieee8021PryInMppdus           Counter64,
        ieee8021PryInEncapFrames      Counter64,
        ieee8021PryInExpFragments     Counter64,
        ieee8021PryInPreFragments     Counter64,
        ieee8021PryInExpDiscards      Counter64,
        ieee8021PryInPreDiscards      Counter64,
        ieee8021PryInUnknownMppcis     Counter64,
        ieee8021PryInErroredMppdus      Counter64,
        ieee8021PryInUnprtFrames      Counter64,
        ieee8021PryInUnprtOctets      Counter64
    }

--ieee8021PryInUserFrames
ieee8021PryInUserFrames OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Total number of protected user data frames received in
                 MPPDUs, encoded as Encapsulated Frames or reassembled from
                 Frame Fragments."
    REFERENCE   "20.14.2"
    ::= { ieee8021PryInEntry 1 }

--ieee8021PryInUserOctets
ieee8021PryInUserOctets OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Total number of user data frame octets received. Excludes
                 padding."
    REFERENCE   "20.14.2"
    ::= { ieee8021PryInEntry 2 }

--ieee8021PryInPadOctets
ieee8021PryInPadOctets OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Number of pad octets received in MPPDUs."
    REFERENCE   "20.14.2"
    ::= { ieee8021PryInEntry 3 }

--ieee8021PryInMppdus
ieee8021PryInMppdus OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Total number of MAC Privacy protection PDUs received."
    REFERENCE   "20.14.2"
    ::= { ieee8021PryInEntry 4 }

--ieee8021PryInEncapFrames
ieee8021PryInEncapFrames OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Number of Encapsulated Frame components received in MPPDUs."
    REFERENCE   "20.14.2"
    ::= { ieee8021PryInEntry 5 }
```


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```
--ieee8021PryInExpFragments
ieee8021PryInExpFragments OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Total number of correctly encoded Express Frame Fragments
                 received in MPPDUs. Includes fragments discarded by
                 reassembly (unknown peer, too many peers, out of order,
                 reassembled frame too large)."
    REFERENCE   "20.14.2"
    ::= { ieee8021PryInEntry 6 }

--ieee8021PryInPreFragments
ieee8021PryInPreFragments OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Total number of correctly encoded Preemptable Frame Fragments
                 received in MPPDUs. Includes fragments discarded by
                 reassembly (unknown peer, too many peers, out of order,
                 reassembled frame too large)."
    REFERENCE   "20.14.2"
    ::= { ieee8021PryInEntry 7 }

--ieee8021PryInExpDiscards
ieee8021PryInExpDiscards OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Number of Express Frame Fragment discard events (discarding a
                 fragment and/or a partially reassembled user data frame)."
    REFERENCE   "20.14.2"
    ::= { ieee8021PryInEntry 8 }

--ieee8021PryInPreDiscards
ieee8021PryInPreDiscards OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Number of Preemptable Frame Fragment discard events
                 (discarding a fragment and/or a partially reassembled
                 user data frame)."
    REFERENCE   "20.14.2"
    ::= { ieee8021PryInEntry 9 }

--ieee8021PryInUnknownMppcis
ieee8021PryInUnknownMppcis OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Number of of unknown MPPDU components received."
    REFERENCE   "20.14.2"
    ::= { ieee8021PryInEntry 10 }

--ieee8021PryInErroredMppdus
ieee8021PryInErroredMppdus OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Number of received MPPDUs containing an incorrectly
                 encoded component."
    REFERENCE   "20.14.2"
    ::= { ieee8021PryInEntry 11 }

```


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```
--ieee8021PryInUnprtFrames
ieee8021PryInUnprtFrames OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Total number of MAC Privacy user frames received."
    REFERENCE   "20.14.2"
    ::= { ieee8021PryInEntry 12 }

--ieee8021PryInUnprtOctets
ieee8021PryInUnprtOctets OBJECT-TYPE
    SYNTAX      Counter64
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION "Total number of MAC Privacy user octets received. Does not
                 include padding"
    REFERENCE   "20.14.2"
    ::= { ieee8021PryInEntry 13 }

-- ============================================================================
--ieee8021PryMIBConformance

ieee8021PryMIBCompliances  OBJECT IDENTIFIER ::= { ieee8021PryMIBConformance 1 }
ieee8021PryMIBGroups       OBJECT IDENTIFIER ::= { ieee8021PryMIBConformance 2 }

-- ============================================================================
--ieee8021PryMIBCompliances

--ieee8021PryMIBCompliance
ieee8021PryMIBCompliance MODULE-COMPLIANCE
    STATUS current
    DESCRIPTION "IEEE8021-PRY-MIB compliance statement, IEEE Std 802.1AEdk."
    MODULE  -- this module
        MANDATORY-GROUPS {
            ieee8021PryIfGroup,
            ieee8021PrySelectionGroup,
            ieee8021PryFrameGroup,
            ieee8021PryChannelGroup,
            ieee8021PryPeerGroup,
            ieee8021PryOutGroup,
            ieee8021PryChOutGroup,
            ieee8021PryInGroup
        }
    ::= { ieee8021PryMIBCompliances 1 }

-- ============================================================================
--ieee8021PryMIBGroups
--ieee8021PryIfGroup
ieee8021PryIfGroup OBJECT-GROUP
    OBJECTS {
        ieee8021PryIfRxProtection,
        ieee8021PryIfTxProtection,
        ieee8021PryIfSecySupport,
        ieee8021PryIfAddr,
        ieee8021PryIfMppduDA,
        ieee8021PryIfDefaultReassembly,
        ieee8021PryIfMaxPeers,
        ieee8021PryIfNumPeers,
        ieee8021PryIfNumPeers
    }
    STATUS      current
    DESCRIPTION "PrY service management (mandatory ieee8021PryIfTable objects)."
    ::= { ieee8021PryMIBGroups 1 }

```


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```
--ieee8021PrySelectionGroup
ieee8021PrySelectionGroup    OBJECT-GROUP
    OBJECTS {
        ieee8021PrySelectionPrivacyType
    }
    STATUS      current
    DESCRIPTION "Privacy type selection, by Private Port transmitted frame
                 priority."
    ::= { ieee8021PryMIBGroups 2 }

--ieee8021PryFrameGroup
ieee8021PryFrameGroup OBJECT-GROUP
    OBJECTS {
        ieee8021PryFrameAccessPriority,
        ieee8021PryFrameRevealDE,
        ieee8021PryFramePadding
    }
    STATUS      current
    DESCRIPTION "Parameters for individual Privacy Frame transmission."
    ::= { ieee8021PryMIBGroups 3 }

--ieee8021PryChannelGroup
ieee8021PryChannelGroup OBJECT-GROUP
    OBJECTS {
        ieee8021PryChEnable,
        ieee8021PryChFragmentEnable,
        ieee8021PryChAccessPriority,
        ieee8021PryChUserDataFrameSize,
        ieee8021PryChMppduGeneration,
        ieee8021PryChRequestedKbitRate,
        ieee8021PryChMppduBitsOnWire,
        ieee8021PryChMppduInterval,
        ieee8021PryChUserBurstOctets
    }
    STATUS      current
    DESCRIPTION "PrY service management Group"
    ::= { ieee8021PryMIBGroups 4 }

--ieee8021PryPeerGroup
ieee8021PryPeerGroup OBJECT-GROUP
    OBJECTS {
        ieee8021PryPeerRowStatus
    }
    STATUS      current
    DESCRIPTION "PrY peer addresses."
    ::= { ieee8021PryMIBGroups 5 }

--ieee8021PryOutGroup
ieee8021PryOutGroup OBJECT-GROUP
    OBJECTS {
        ieee8021PryOutPrivacyFrames,
        ieee8021PryOutPfUserOctets,
        ieee8021PryOutPfPadOctets,
        ieee8021PryOutUnprtFrames,
        ieee8021PryOutUnprtOctets
    }
    STATUS      current
    DESCRIPTION "PrY service management Group"
    ::= { ieee8021PryMIBGroups 6 }

```


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```
--ieee8021PryChOutGroup
ieee8021PryChOutGroup OBJECT-GROUP
    OBJECTS {
        ieee8021PryChOutUserFrames,
        ieee8021PryChOutUserOctets,
        ieee8021PryChOutPadOctets,
        ieee8021PryChOutMppdus,
        ieee8021PryChOutEncapFrames,
        ieee8021PryChOutExpFragments,
        ieee8021PryChOutPreFragments
    }
    STATUS      current
    DESCRIPTION "PrY service management Group"
    ::= { ieee8021PryMIBGroups 7 }

--ieee8021PryInGroup
ieee8021PryInGroup OBJECT-GROUP
    OBJECTS {
        ieee8021PryInUserFrames,
        ieee8021PryInUserOctets,
        ieee8021PryInPadOctets,
        ieee8021PryInMppdus,
        ieee8021PryInEncapFrames,
        ieee8021PryInExpFragments,
        ieee8021PryInPreFragments,
        ieee8021PryInExpDiscards,
        ieee8021PryInPreDiscards,
        ieee8021PryInUnknownMppcis,
        ieee8021PryInErroredMppdus,
        ieee8021PryInUnprtFrames,
        ieee8021PryInUnprtOctets
    }
    STATUS      current
    DESCRIPTION "PrY service management Group"
    ::= { ieee8021PryMIBGroups 8 }

END
```


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23. YANG Data Models

This clause specifies YANG (IETF RFC 7950) data models that provide control and monitoring of systems and system components that implement functionality specified in this standard. These data models are based on the managed objects and their functionality specified in Clause 10 and Clause 20, and their use as specified in Clause 11, Clause 15, and Clause 21. The specifications in those clauses take precedence if there is any discrepancy with the text of this clause (Clause 23).

This clause

a) Introduces the YANG framework that governs the naming and hierarchy of configuration and operational data structures in the data models, and the modeling of network interfaces (23.1).
b) Describes each data model and its relationship to other YANG data models, and to the operational processes and managed objects specified in the other clauses of this standard (23.2, 23.4, 23.8).
c) Reviews security considerations particular to each of the data models, with specific reference to data nodes in the YANG modules that compose the model (23.3, 23.5, 23.9)
d) Includes a schema tree diagram for each YANG module (23.10)
e) Includes each YANG module (23.11)

YANG data models are specified for the addition of MAC Security (23.2) and MAC Privacy protection (23.4) capabilities to general interface stacks. Both explicit and augmented interface stack models are supported, and relevant modeling considerations described (23.6). These general interface models are used in models for EDEs and to support the addition of MAC Security and MAC Privacy protection to other systems as described in 23.8.1, Clause 11 and Clause 21.

The MIB modules specified in Clause 13 and Clause 22 were similarly derived from the other clauses of this standard. Consequently, the capabilities and structure of the YANG data models are closely aligned with those MIB modules. However, the YANG data modules were not derived from the MIB modules and do not include data or modeling constructs particular to MIB module specification and not in the information model. A system may support MACsec and MAC Privacy protection management using both MIB and YANG modules. However any given system is expected to be managed using either YANG or SNMP, rather than a combination of the two.

The YANG modules defined in this clause are designed to be accessed via a network configuration protocol, e.g., NETCONF protocol (IETF RFC 6241 [B13]). In the case of NETCONF, the lowest NETCONF layer is the secure transport layer and the mandatory to implement secure transport is SSH (IETF RFC 6242 [B14]). The NETCONF access control model (IETF RFC 6536 [B15]) provides the means to restrict access for particular NETCONF users to a preconfigured subset of all available NETCONF protocol operations and content. It is the responsibility of a system's implementor and administrator to ensure that the protocol entities in the system that support NETCONF, and any other remote configuration protocols that make use of these YANG modules, are properly configured to allow access only to those principals (users) that have legitimate rights to read or write data nodes. This standard does not specify how the credentials of those users are to be stored or validated.

23.1 YANG Framework

This clause has been developed in accordance with the YANG guidelines published in IETF RFC 6087 [B12] as applicable to IEEE standards.

The YANG framework applies hierarchy in the following areas:

1) The uniform resource name (URN), as specified in IEEE Std 802d–2017.
2) The YANG objects form a hierarchy of configuration and operational data structures that define the YANG model.

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Figure 23-1 outlines the organization, within the YANG hierarchy, of data models and objects that support and complement data models supported by this standard.

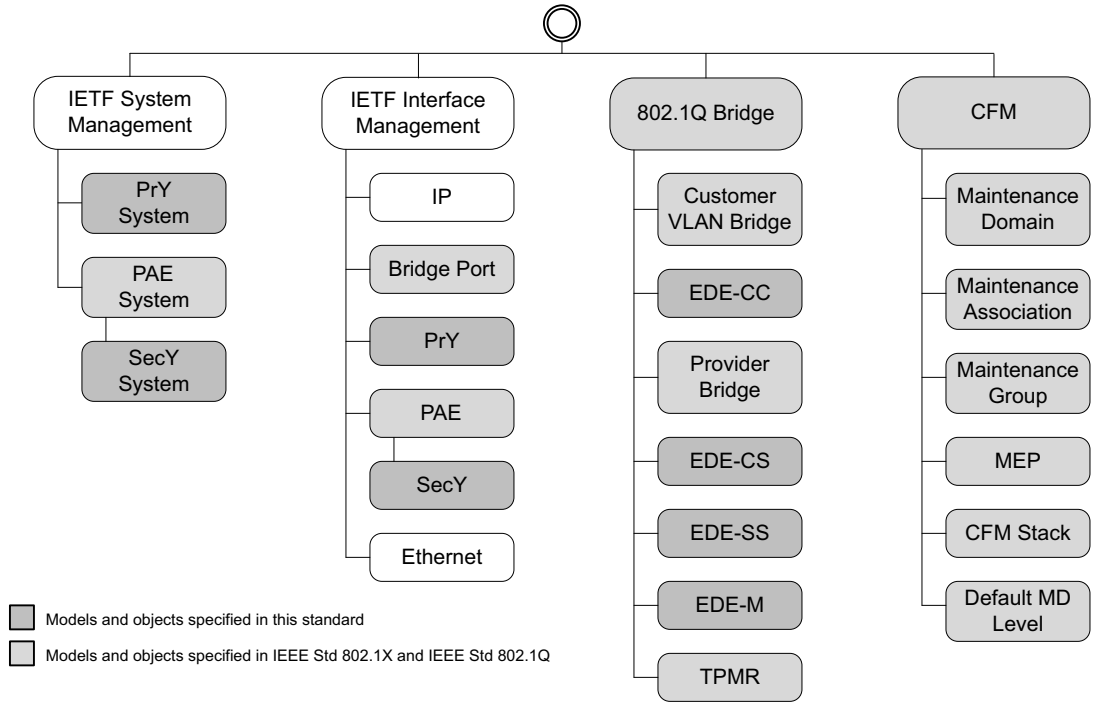


Figure 23-1—YANG hierarchy, models and objects

Figure 23-1 is not intended to be comprehensive. It shows data models and objects specified or referenced by this standard. The YANG modules specified augment both the IETF System Management YANG data model (IETF RFC 7317), with a hierarchy of nodes under **/ietf-system:system**, and the IETF Interface Management YANG data model (IETF RFC 8343), with a hierarchy of nodes under **/ietf-interfaces:interfaces/ietf-interfaces:interface**. EDE specifications make use of Bridge components specified in IEEE Std 802.1Q, with YANG data nodes under **/ieee:std:802.1Q:yang:ieee802-dot1q-bridge**.

NOTE—The prefixes **if** and **sys** are used to abbreviate **/ietf-interfaces:interfaces/ietf-interfaces:interface** to **/if:interfaces/if:interface** and **/ietf-system:system** to **/sys:system** in the YANG modules.

IEEE Std 802.1Q and IEEE Std 802.1X provide additional information on the IETF System Management and Interface Management models and their relationship to data models specified by IEEE 802.1 standards.

23.2 MAC Security Entity (SecY) model

The SecY YANG data model is based on Figure 10-5 [3], which summarizes the information model used by the SecY management process (10.7). Each SecY supports port-based network access control for a particular port and is associated with a Port Access Entity (PAE, IEEE Std 802.1X) which includes the SecY's Key Agreement Entity (KaY) as described in 7.1. The SecY model augments the PAE YANG data model,

[3] Figure 10-5 and figures in this clause (Clause 23) use UML [B1] conventions and C++ language constructs to present information in a common compact way, independent of its inclusion in MIB modules, YANG data models, or operational use within a system. These figures are not intended to support automated object encoding, and are simplified to allow significant information and relationships to be presented in a single figure, reducing the need to temporarily memorize detail when reading the standard.

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specified in IEEE Std 802.1X, with two primary YANG sub-trees. The **secy-system** sub-tree augments **pae-system**, which augments /**sys:system**. The **secy** sub-tree augments **pae**, which augments **/if:interfaces/if:interface**.

The SecY data model supports explicit and augmented interface stacks, including interfaces that include an augmentation for a protocol shim above a SecY. When the SecY is the only or the top-most protocol shim in an interface, nodes in the basic IETF interface management model (in **/if:interfaces/if:interface**) can duplicate or summarize information from the **secy** sub-tree (see 23.2.2, 23.6).

The PAE YANG data model includes nodes that provide information about or control the operation of MACsec capable interfaces (see 23.2.3). The **secy-system** and **secy** sub-trees do not duplicate this information.

Figure 23-2 shows the **secy-system** and **pae-system** nodes that support the SecY model, and references to those nodes from **secy** and **pae** interface nodes, as follows:

— **pae-system/system-access-control** allows a network manager to enable or disable port-based network access control for a system as a whole. When disabled this control overrides **secy/verification/validate-frames** and **secy/generation/protect-fram**es causing all SecYs to behave as if it were ‘null’.

— **secy-system/cipher-suites** lists Cipher Suites that might be used by the system, and their attributes. The Cipher Suites that a given SecY supports are referenced by **secy/cipher-suite-control**, and that currently used by **secy/current-ciphersuite**.

— **pae-system/nid-group** lists the NIDs supported by the system. On system initialization and prior to (or in the absence of) supporting authentication and authorization protocol, the use of MACsec on a particular interface can be controlled by a **pae/logon-nid.selected** reference to a particular NID (Network Identity). The scope of the NID’s identifier (**pae-system/nid-group.nid**) is that of the **/sys:system,** so a simple system can use a null NID for a configuration applicable to all MACsec capable interfaces, while a system that naturally has ports of particular types (such as an EDE-CC) can use a local convention to identify the configuration for ports of that type. Equally a network manager can use a network wide NID naming scheme.

— **pae-system** also contains EAPOL and MKA version information.

NOTE—Objects in the SecY YANG model are named using the normal YANG lower case convention with hyphens separating words or acronyms. These names correspond simply to those in the Clause 10 protocol independent description, which use ‘camelCase’ (capitalizing the start of each word other than the first and omitting hyphens).

Figure 23-3 shows the **secy**, **/if:interfaces/if:interface** and **pae** nodes that support the SecY model and their mutual dependencies, as follows:

— **pae/port-capabilities/macsec** and **pae/port-capabilities/mka** are true if an interface supports MACsec and MKA respectively. A PAE can, but does not necessarily, have other capabilities (such as support for authentication using EAP).

The relationship between the SecY data model and the IETF System Management, IETF Interface Management, and PAE models is further detailed in 23.2.2, 23.4, and 23.6 below.

The SecY data model does not include objects and operations from Figure 10-5 that are only made available by the SecY to other components (notably the Key Agreement Entity, KaY) within the same system. These inaccessible items include cryptographic keys (see 23.3) and their purely local key identifiers, controls over individual receive SCs and SAs [enableReceive, updtNextPN, updtLowestPN(10.7.15)], and controls over individual transmit SCs and SAs.

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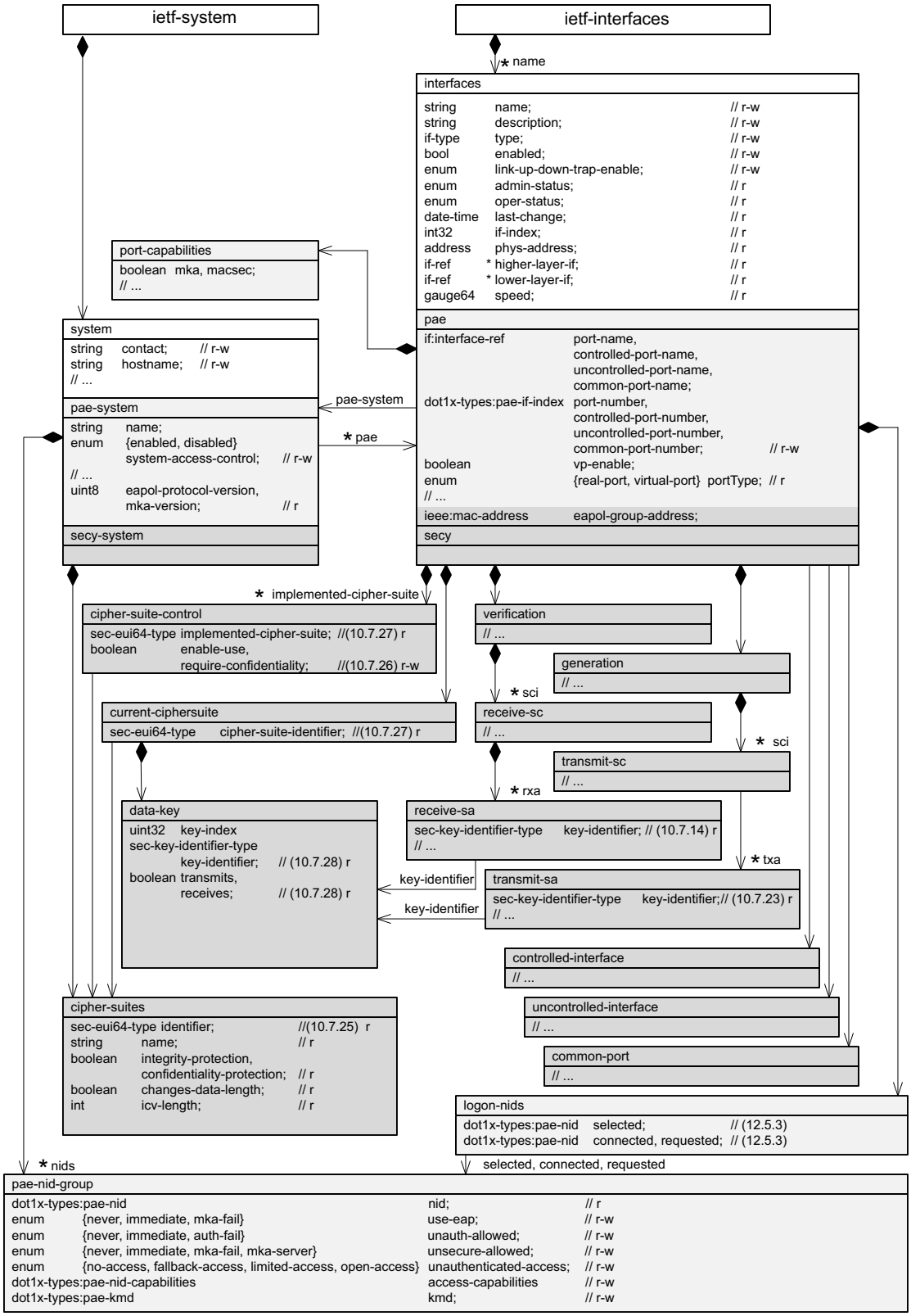
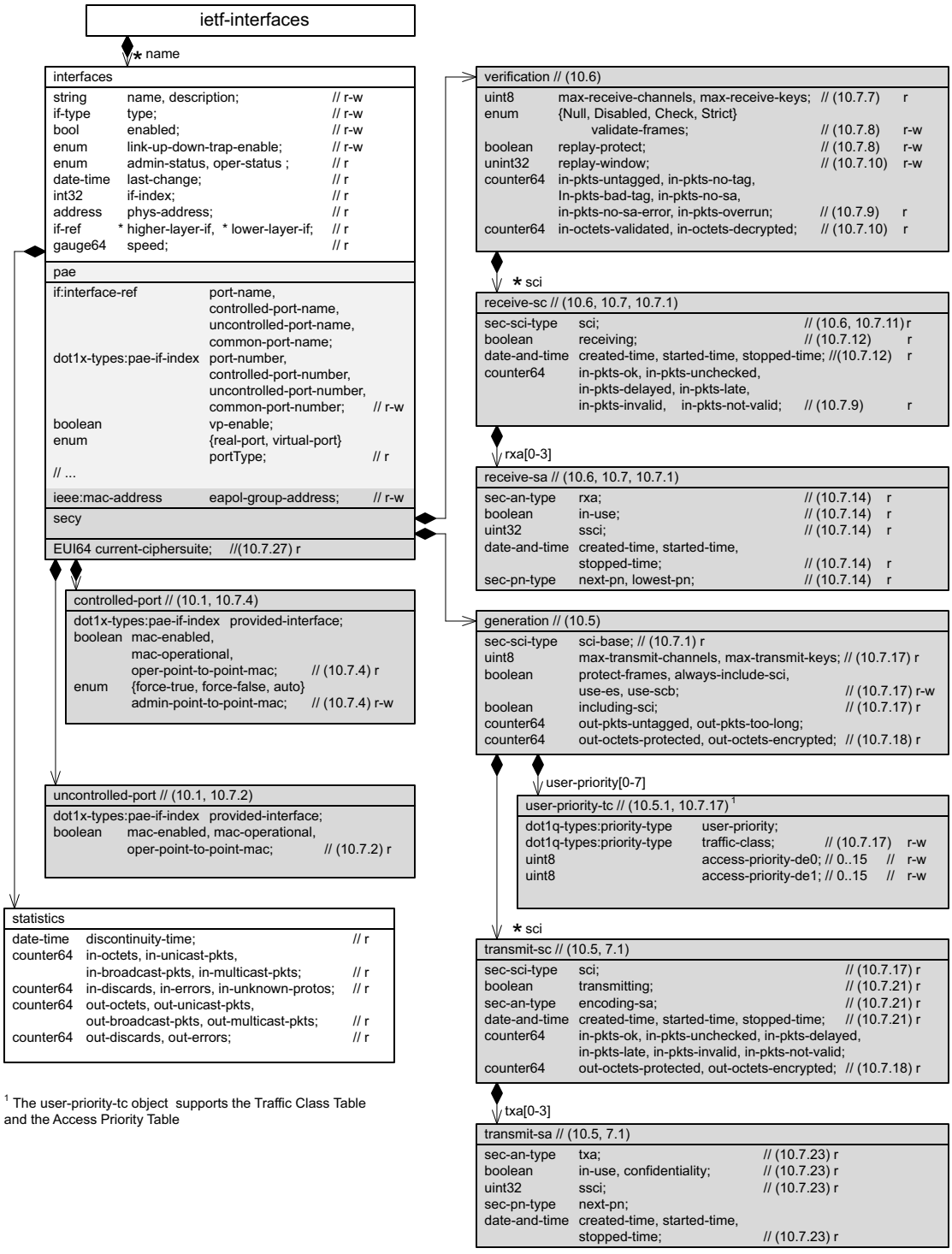


Figure 23-2—SecY model system nodes and references

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[1] The user-priority-tc object supports the Traffic Class Table and the Access Priority Table

Figure 23-3—SecY model interface nodes

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23.2.1 Relationship to the IETF System Management model

IEEE Std 802.1X describes the relationship between the IETF System Management model and the **pae-system**, which augments **/sys:system**. The same considerations that also apply to the **secy-system** which augments the **pae-system**. In particular a system that implements the SecY YANG data model shall also implement the IETF System Management data model defined in IETF RFC 7317).

23.2.2 Relationship to the IETF Interface Management model

IEEE Std 802.1X describes the relationship between the IETF Interface Management model and the **secy** sub-tree augments **pae**, which augments **/if:interfaces/if:interface**. The same considerations that also apply to the **secy-system** which augments the **pae-system**. In particular a system that implements the SecY YANG data model shall also implement the IETF Interface Management data model defined in IETF RFC 8343) for each interface that is MACsec capable.

IEEE Std 802.1X further describes both explicit and augmented interface stack models with both PAE and SecY functionality. An explicitly modeled interface represents a single shim or sublayer in an interface stack. An augmented interface combines adjacent shims or sublayers into a single interface to simplify configuration. In both cases the **/if:interfaces/if:interface/statistics** in-octets, in-unicast-pkt, in-broadcast-pkts, in-multicast-pkts, out-octets, out-unicast-pkt, out-broadcast-pkts, and out-multicast-pkts, reflect the contents of frames transmitted and received by the user(s) of the uppermost shim or sublayer, which might or might not be the SecY. The SecY's Controlled Port interface statistics are always available from the **secy/controlled-interface** container. Statistics for the SecY's Common Port are recorded by shim or interface directly supporting that port, and are not separately recorded.

In both explicit and augmented interface stacks, a SecY and its associated PAE augment the same interface, as described in IEEE Std 802.1X and as a consequence of the **secy** node augmenting its associated **pae** node. The PAE uses the SecY's Uncontrolled Port to transmit and receive EAPOL PDUs. Detailed statistics for those EAPOL frames are provided by the **pae/eapol-statistics** container, and are not otherwise broken down by unicast, broadcast, and multicast or recorded by the SecY model.

Unless otherwise specified, and unlike a PAE, a protocol entity that is represented by an augment or reference (e.g. in the case of a CFM MIP or MEP shim) to an interface that is augmented by a SecY uses the service provided by the SecY's Controlled Port (possibly indirectly if the interface has been augmented by a number of shims). An instance of LLDP (for example), when represented by and configured as an augment to the same interface as a SecY, transmits frames that are protected and validated by that SecY. Interface stack configurations are described in 23.6.

To assist in the isolation and diagnosis of connectivity issues, the current status of MAC_Enabled and MAC_Operational (6.4) are reported for the SecY's Controlled Port and Common Port in **secy/controlled-interface** and **secy/common-interface**. The value of MAC_Enabled for a port (interface) reflects the state of all the administrative controls for sublayers in the interface that support each port. The values of MAC_Operational and MAC_Enabled for the Uncontrolled Port are the same as those for the Common Port.

23.2.3 Relationship to the PAE model

The **secy-system** and **secy** nodes augment the PAE model as described above (23.2, 23.2.2). A system that implements the SecY YANG data model shall also implement the PAE model defined in IEEE Std 802.1X. The **pae-system** and **pae** node objects implemented as part of that model can be as simple as those required to identify particular interfaces as MACsec capable, and to support MKA operation with pre-shared CAKs. The PAE model can also support EAP operation (as a Supplicant or an Authenticator, or both) and the transmission and reception of announcements. IEEE Std 802.1X further describes the relationship of the PAE model to other models for authentication, authorization, and the use of credentials.

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23.2.4 Structure of the SecY model

The SecY model comprises the **ieee802-dot1ae-secy** (23.11.1) and **ieee802-dot1x-eapol** (23.11.3) YANG modules specified in this standard, the **ieee802-dot1x** YANG module specified in IEEE Std 802.1X, the **ietf-system** and **ietf-interfaces** YANG modules, and the types modules that they import.

NOTE—The ieee802-dot1x-eapol module is specified in this standard to provide access the EAPOL address used by a PAE. A future revision of the ieee802-dot1x module could add that missing functionality.

23.3 Security considerations for the SecY model

A SecY exists to secure communication, providing connectionless user data confidentiality, frame data integrity, and data origin authenticity (1.2, Clauses 6, 7, and 8). All the data nodes in the SecY model and supporting data nodes in the PAE model that are writable/creatable/deletable (i.e., config true) exist to control the way that the SecY does (or does not) secure communication and need to be considered to be sensitive in all network environments. In particular, config access can either compromise security directly or disrupt network operation as part of an effort to mislead a network administrator into compromising security, as follows:

- **pae-system/system-access-control** changes can disable the use of MACsec.
- **pae-system/pae-nid-group** and/or **pae/logon-nids** changes can disable or constrain the use of MACsec on particular ports.
- **pae.eapol-group-address** change can cause MKA operation to fail, preventing the use of a MACsec protected link.
- **secy/verification/validate-frames** and **secy/generation/protect-frames** changes can disable MACsec protection and validation of received frames.
- **secy/verification/replay-window** increases can render the network vulnerable to overload from repeated frames while decreases can reduce performance on links that reorder frames.

An implementation can remove or restrict access to these nodes (e.g., by requiring direct access to the system rather than through a network configuration protocol). Read access to these nodes can also allow an adversary to choose indirect attacks.

The SecY YANG model does not provide access to cryptographic keys, authentication credentials, and authorization parameters that would allow an adversary to introduce apparently valid frames on a secured network link.

Access to data nodes that are specified as readable (i.e., config false) in the SecY model does not elicit information that an adversary can use directly to compromise network operation. In principle that information could be gathered by an adversary that has direct access to the MACsec protected link. However access using a network configuration protocol can provide that information to a remote adversary.

23.4 MAC Privacy protection (PrY) model

The PrY YANG data model is based on Figure 20-6, which summarizes the information model used by the PrY management process (20.14). Each PrY provides MAC Privacy protection for a particular port. The PrY model augments the IETF Interface Management model: the **pry** sub-tree augments **pae**, which augments **/if:interfaces/if:interface**.

Each PrY is supported by a SecY [or by an equivalent protocol entity providing connectionless user data confidentiality, frame data integrity, and data integrity for MAC Clients (1.2)]. The supporting SecY can be part of the same interface stack or located in an adjacent system and connected by a physically secure

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connection. When the SecY is part of the same interface stack, the PrY is modeled as part of (i.e. an augment of) the same interface and supported directly by the SecY, i.e. the PrY's Controlled Port is the SecY's Controlled Port (20.1, Figure 20-1).

Figure 23-4 shows the **pry** nodes augmenting the **interface** nodes, including the following:

— **pry/reception/privacy-protection** and **pry/transmission/privacy-protection** allow a network manager to enable MAC privacy protection on reception and transmission independently.

— **pry/if-secy-support** is true if the PrY is directly supported by a SecY.

— **pry/pry-address** is the address the PrY uses as the source address of MPPDUs, **pry/pry-mppdu-dest-address** is the destination address used for transmitted MPPDU, and **pry/peer-entry*peer-src-address** lists the MAC addresses of the PrY's peers (MPPDUs received with unrecognized addresses are discarded). If **pry/if-secy-support** is true these addresses are supplied by MKA and are not configurable in the data model.

— **pry/transmission/privacy-selection** provides per-user priority control (privacy-type) of the way MAC Privacy protection is applied to any given user data frame on transmission (by encapsulation in a Privacy Channel, by transmission as an individual Privacy Frame, or not at all). For Privacy Frame transmission it also provides control over the access priority, whether the user data frame DE bit value is revealed, and the use of padding.

— **pry/transmission/channel*channel-id** allows each Privacy Channel (Express, Preemptable) and its use of user data fragmentation to be enabled or disabled, and controls Privacy Channel parameters (MPPDU size, bandwidth utilization).

— **pry/transmission/frame-tx-statistics**, **pry/transmission/channel*channel-id/tx-statistics**, and **pry/reception/rx-statistics** provide counts of transmitted and received user data frames, user data octets, padding octets, unprotected frames, MPPDUs, and user data frame fragments.

23.4.1 Relationship to the IETF System Management model

The IETF System Management model defines the scope of the if-index parameter values that identify each interface that is augmented by a **pry**. The PrY model is not otherwise dependent on **/sys:system**.

23.4.2 Relationship to the IETF Interface Management model

The **pry** sub-tree augments **/if:interfaces/if:interface**. Unless otherwise specified, and unlike a PAE, a protocol entity that is represented by an augment or reference to an interface that is augmented by a PrY, or both a PrY and a SecY, uses the service provided by the PrY's Private Port (possibly indirectly if the interface has been augmented by a number of shims). Interface stack configurations are described in 23.6.

23.4.3 Relationship to the PAE and SecY models

If the PrY is not in the same system as its supporting SecY, the PrY model is independent of PAE and SecY models, or rather it is the responsibility of the network administrator to ensure compatible configuration of the two systems. From a practical point of view this means that **pry/transmission/privacy-protection** and **pry/reception/privacy/protection** can only be enabled if the connectivity provided by the SecY is protected by MACsec. The network administrator also has to choose and configure appropriate PrY source MAC address, PrY peer source MAC addresses, and MPPDU destination MAC addresses.

If the PrY is in the same system (and thus in the same interface) as its supporting SecY, those MAC addresses are automatically configured by the SecY's supporting MKA instance (in the PAE model) and are config false in the PrY model. MAC Privacy protection (on transmission and reception) is only enabled if the SecY is protecting transmitted frames, i.e. **secy/generation.protect-frames** is true, and is validating all received frames, i.e. **secy/verification/validate-frames** is Strict. Changes to the use of MAC Privacy protection resulting from changes to MACsec protection of transmitted and received frames do not change the configured values of **pry/reception/privacy-protection** and **pry/transmission/privacy-protection**.

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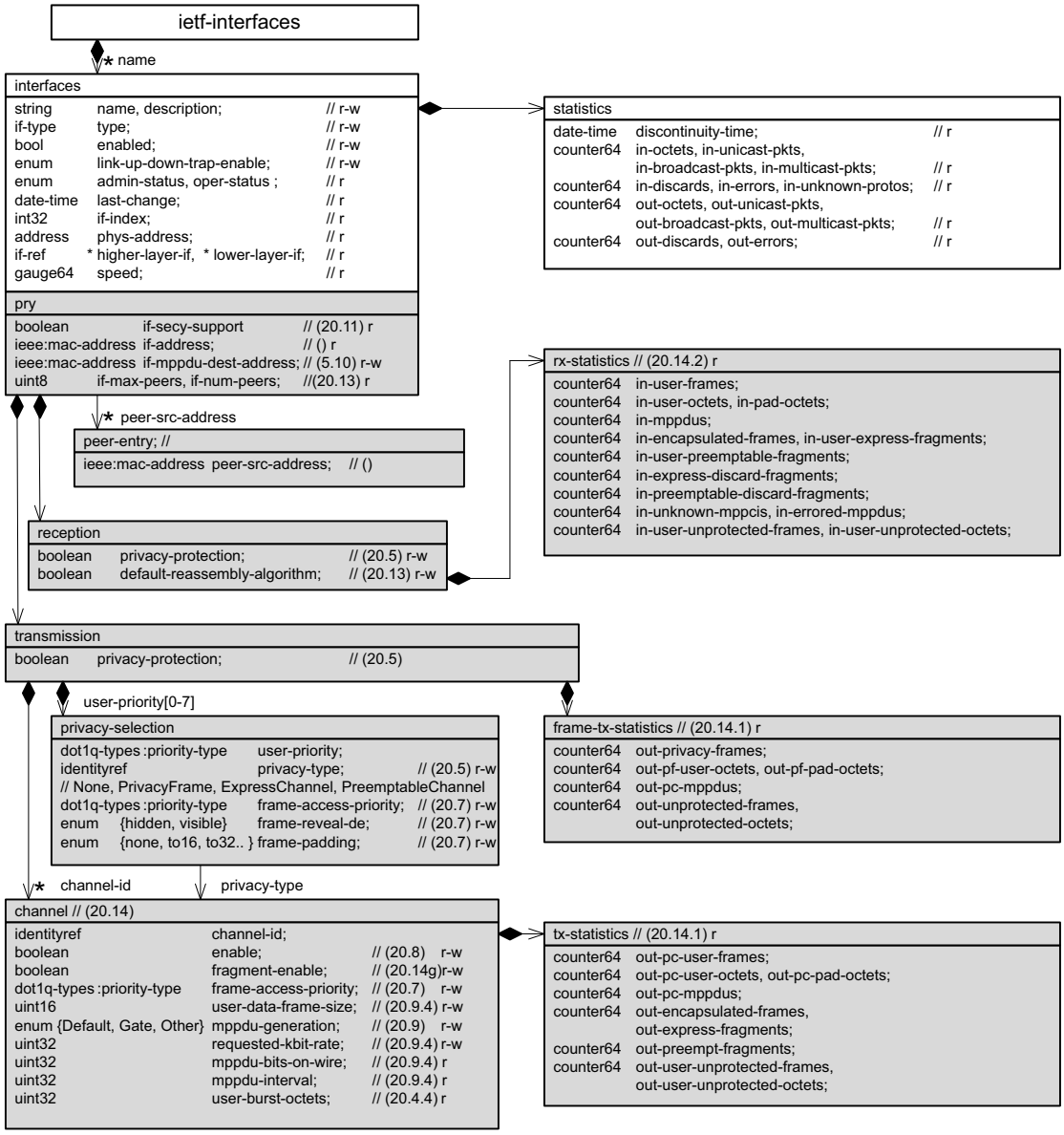


Figure 23-4—PrY model interface nodes

23.4.4 Structure of the PrY model

The PrY model comprises the **ieee802-dot1ae-pry** (23.11.2) YANG module specified in this standard, the **ieee802-dot1x** YANG module specified in IEEE Std 802.1X, the **ietf-system** and **ietf-interfaces** YANG modules, and the types modules that they import.

23.5 Security considerations for the PrY model

A PrY, when used in conjunction with MACsec, reduces the ability of adversaries to correlate the MAC addresses, sizes, and timing of user data frames with users and their use of the network (1.1). MACsec is responsible for frame data confidentiality and integrity protection, and data origin authenticity. Accordingly

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while access to the data and controls provided by the PrY data model does not allow an adversary to compromise security directly, by injecting frames, modifying frame data, or accessing that data, the model includes information that the PrY is intended to keep private. This private data includes the following read-only (config false) objects:

— **pry/transmission/frame-tx-statistics**, **pry/transmission/channel*channel-id/tx-statistics**, and **pry/reception/rx-statistics** counts.

— **/if:interfaces/if:interface/statistics**.

In addition any frame or octet statistics maintained by data models for entities higher in the PrY's interface stack, or in other interface stacks to which frames from that interface stack are forwarded (in the case of a bridge or router), or in higher layer entities in the system, can include information that the network administrator intends to be private.

Configuration (read-write) access to the following objects can also remove or reduce the level of privacy protection by allowing frames of different user priorities and sizes to be more clearly distinguished:

— **pry/transmission*user-priority/privacy-selection** (all objects)

— **pry/transmission/channel*channel-id** (enable, frame-access-priority, user-data-frame-size, and to a lesser extent other objects at this node).

The data model includes controls that an adversary can use to impact network performance and capability, as well as revealing correlatable information, including the following:

— **pry/transmission/privacy-selection/frame-padding**, if increased can reduce the available network bandwidth.

— **pry/transmission/channel*channel-id/requested-kbit-rate** controls the bandwidth available to user data frames allocated to the Privacy Channel when the default MPPDU generation algorithm is used. This bandwidth is used irrespective of the bandwidth taken by those user data frames. If set so high as to result in the MPPDU loss for the Privacy Channel, communicating user entities can reduce the number of user data frames they send but this 'back-off' does not reduce the bandwidth used by the Privacy Channel and the consequent probability of frame loss.

NOTE—A PrY using the default (20.9) MPPDU generation algorithm does not attempt to send MPPDUs faster than they can be transmitted by the interface of which it is part. Therefore, if connectivity to peer PrYs capable of line rate reception is provided by a simple LAN without intermediate buffering, Privacy Channel MPPDUs are not lost. However their transmission can preclude the transmission of other, lower priority frames.

— **pry/transmission*user-priority/privacy-selection/privacy-type**, if changed can change the access priority selected for MPPDUs for frames of a given user priority or change their allocation to a Privacy Channel, Privacy Frame, or unprotected transmission.

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23.6 Interface stack models

The SecY and PrY YANG models support both explicit and augmented interface models. An explicitly modeled interface represents a single shim or sublayer in an interface stack. An augmented interface combines adjacent shims or sublayers into a single interface to simplify configuration. IEEE Std 802.1X provides both general examples and examples that are particular to the use of port-based network access control supported by MACsec, and the same notation is used in the following description.

Figure 23-5 illustrates these modeling choices for an end station interface with both MACsec or MAC Privacy protection. The small alphanumeric circles (labelled A, B, C) in the figures represent the if-index used to identify each interface (represented by the larger circles).

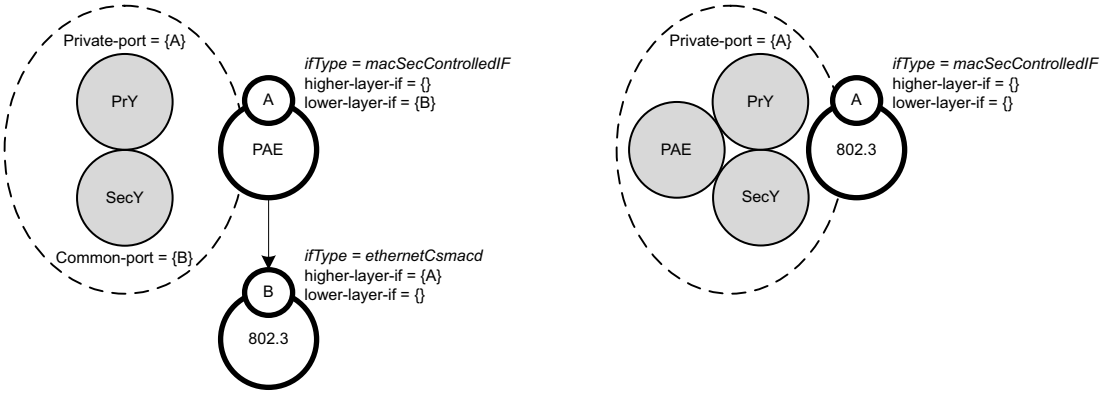


Figure 23-5—Explicit and augmented interface stack models for an end station

Figure 23-6 illustrates two further modeling choices for the interface.

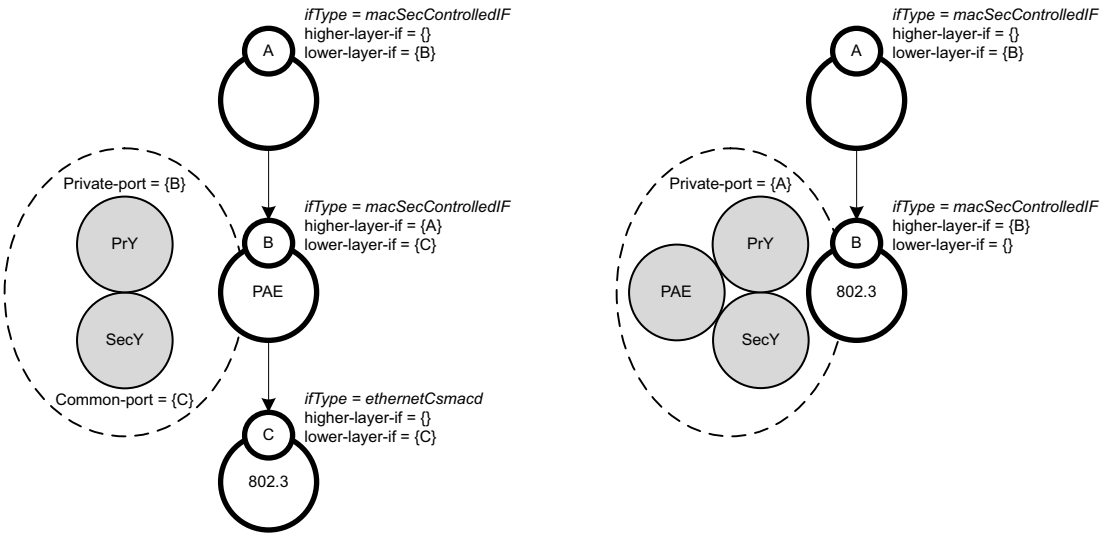


Figure 23-6—Two further interface stack modeling choices

The controller (e.g., NETCONF client) can dictate which interface stack model (explicit or augmented) is used. Consideration needs to be given to selecting the interface stack model that is right for the usage scenario. For example, starting with either of the two models shown in Figure 23-5 and transitioning to an interface stack that supports IEEE Std 802.1AX Link Aggregation (see Figure 23-7), requires the initial

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interfaces to be torn down (e.g., deleted) and re-instantiated (e.g., created). This is not the case when starting from either of the two models shown in Figure 23-6 (see, for example, Figure 23-7).

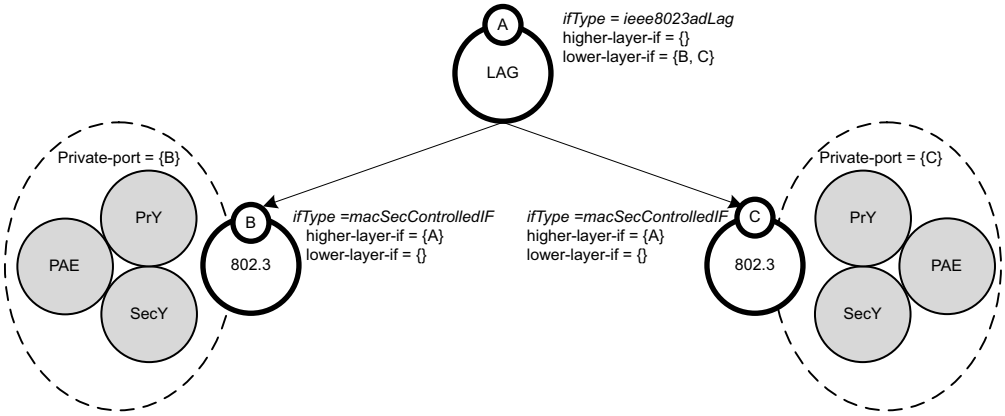


Figure 23-7—An interface stack model for link aggregation and MACsec

In Figure 23-5, 23-6, and 23-7, it is assumed that the network manager has overwritten the ifType to reflect the service provided by the top-most element of the interface stack. These models could be augmented by the Bridge Port model specified in IEEE Std 802.1Q, with an ifType of **bridge** for interface A.

Unless otherwise specified, and unlike a PAE, a protocol entity that is represented by an augment or reference to an interface that is augmented by a PrY uses the service provided by the PrY's Private Port (possibly indirectly if the interface has been augmented by additional shims). Similarly a protocol entity represented by an augment or reference to an interface augmented by a SecY, but not by a PrY, uses the secure MAC Service supported by that SecY. Some protocol entities, notably LLDP, can be usefully instantiated to make use of the insecure MAC service provided by a SecY's Common Port or the secure MAC Service provided by its Controlled Port. Figure 23-8 shows an interface stack supporting both.

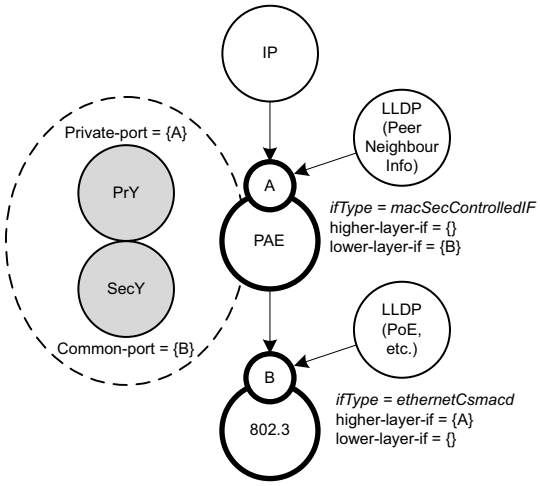


Figure 23-8—An interface stack with LLDP instances

In Figure 23-8, the lower instance of LLDP supports Power over Ethernet (PoE) operation prior to the establishment of MACsec secured connectivity which is used, by the upper LLDP instance, to share neighbor configuration information.

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23.7 Security considerations for interface stack models

The security considerations that apply to use of interface stacks incorporating SecYs or PrYs are generally those applicable to those interface stacks (23.6) and the basic SecY and PrY models (23.3, 23.5), with the additions and exceptions described in this clause (23.7).

23.7.1 Interface stacks with link aggregation

When an interface stack includes link aggregation as specified by IEEE Std 802.1AX, the SecY and PrY managed objects for each Aggregation Port (e.g. the subinterface stacks below interfaces B and C in Figure 23-7) need to have different values of the Aggregation Key (so they are not attached to the same Aggregator) or identical values for the following objects:

— **pae/logon-nids/selected**

— **pae/logon-nids/connected**

— **pae/logon-nids/requested**

Different values for Aggregation Ports attached to the same Aggregator can compromise the security by allowing a port with different (or no) authentication, confidentiality, and integrity requirements to carry part of the data traffic for an Aggregator that had supported stricter security requirements. The **pae** for simple systems might only support a single null NID (identified by the null string), in which case the above object values are necessarily the same for all Aggregation Ports.

23.8 System models

A given system can comprise end station functionality, one or more bridge components, and other protocol components (e.g., supporting IP routing).

In principle each of the interfaces within a system can be supported by MACsec, MAC Privacy protection, or both. However the utility of such support depends on the connectivity from the SecY and PrY instances to their potential peers, and the existence of those peers. Correct configuration of each SecY's associated **pae/eapol-group-address**, together with the frame filtering provided by VLAN and MAC Bridge component Reserved Addresses (IEEE Std 802.1Q) guards against the attempted creation of inappropriate secured connectivity and (where a PrY is directly supported by a SecY) inappropriately instantiating privacy protection.

Where a system comprises multiple groups of interfaces each with naturally or potentially different SecY and PrY characteristics, each of those groups can be identified by a NID (**dot1x-types:pae-nid**) with **pae-system/pae-nid-group*nids** objects (possibly augmented) listing the attributes for each group. The attributes associated with the Customer Edge Ports of a Provider Edge Bridge (PEB) could, e.g., be associated with a NID that is the value of the **pae/logon-nids/selected** object for each of those port's PAEs. If a particular PEB supports distinct groups of Customer Edge Ports, e.g. in locations with different physical access characteristics, the **pae/logon-nids/selected** object could then differ accordingly. A **pae-nid** is a UTF-8 string and can be chosen to match a **pae/controlled-port-name** or a prefix that is part of the system's port naming convention for ports of a particular type.

This clause (23.8) further describes aspects of the YANG model for particular systems, including those for the various types of EDE as specified in Clause 15 of this standard.

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23.8.1 EDE models

An EDE-M is modeled (15.2, 15.4) as specified by IEEE Std 802.1Q for a VLAN-unaware MAC Bridge, with a SecY, and optionally a PrY, in the black-side interface stack (5.5). An EDE-CS is modeled (15.5) as specified by IEEE Std 802.1Q for a Provider Edge Bridge, with a SecY, and optionally a PrY, in the Provider Edge Port (PEP) interface stack (5.5). An EDE-CC and EDE-SS uses the IEEE Std 802.1Q Provider Edge Port model with the exception that both VLAN Bridge components are C-VLAN components in the EDE-CC model and both are S-VLAN components in the EDE-SS model.

23.9 Security considerations for system models

The general security considerations for interface stack models (23.7) apply to each interface that is MACsec or MAC Privacy protection capable. Additional consideration for EDEs are specified below (23.9.1).

23.9.1 EDE model security considerations

23.9.1.1 Use of C-VLAN multiplexing for MAC Privacy protection

An EDE-CC that also supports MAC Privacy protection is not constrained to use the same value for the outer VLAN tag added to a frame transmitted through its Provider Network Port [5.8 g), 15.6, Figure 15-8]. This optional capability is provided by managing the PVID (assigning the VID for that outer VLAN tag) for the internal Customer Network Port attached to the Provider Edge Port (PEP) that provides edge component egress for frames with the mapped inner VID. The purpose of this capability is to enhance privacy when frames for two or more different C-VLANs (as received at the Customer Edge Port) are to be conveyed across the Provider Bridged Network (PBN) to the same destination. However if MACsec is not operational for that PEP, a SecTAG is not added by the PEP so an outer VLAN tag is not added. Not only is integrity, confidentiality, and privacy not provided in the absence of MACsec (as would be expected) but the customer's C-VLANs (as opposed to an outer C-VLAN added by the EDE) are used by the service provider to forward traffic. Configuring **pae-system/pae-nid-group*nids/unsecure-allowed** to never for each PEP's **pae/logon-nids/selected** can guard against both possibilities.

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23.10 YANG module schema

The YANG data modules specified by this standard are summarized by simple tree diagrams with the following notation:

- Brackets "[" and "]" enclose list keys.
- Abbreviations before data node names: "rw" means configuration (read-write), and "ro" means state data (read-only).
- Symbols after data node names: "?" means an optional node, "!" means a presence container, and "*" denotes a list and leaf-list.
- Parentheses enclose choice and case nodes, and case nodes are also marked with a colon (":").
- Ellipsis ("...") stands for contents of subtrees that are not shown.

23.10.1 ieee802-dot1ae-secy schema

```
module: ieee802-dot1ae-secy

  augment /if:interfaces/if:interface/dot1x:pae:
    +--rw secy
       +--rw controlled-port
       |  +--ro mac-enabled?               boolean
       |  +--ro mac-operational?           boolean
       |  +--ro oper-point-to-point-mac?   boolean
       |  +--rw admin-point-to-point-mac?  enumeration
       |  +--ro controlled-port-enabled?   boolean
       +--rw uncontrolled-port
       |  +--ro mac-enabled?               boolean
       |  +--ro mac-operational?           boolean
       |  +--ro oper-point-to-point-mac?   boolean
       |  +--rw admin-point-to-point-mac?  enumeration
       +--rw verification
       |  +--ro max-receive-channels?   uint8
       |  +--ro max-receive-keys?       uint8
       |  +--rw validate-frames?        enumeration
       |  +--rw replay-protect?         boolean
       |  +--rw replay-window?          uint32
       |  +--ro in-pkts-untagged?       yang:counter64
       |  +--ro in-pkts-no-tag?         yang:counter64
       |  +--ro in-pkts-bad-tag?        yang:counter64
       |  +--ro in-pkts-no-sa?          yang:counter64
       |  +--ro in-pkts-no-sa-error?    yang:counter64
       |  +--ro in-pkts-overrun?        yang:counter64
       |  +--ro in-octets-validated?    yang:counter64
       |  +--ro in-octets-decrypted?    yang:counter64
       |  +--ro receive-sc* [sci]
       |     +--ro sci                  sec-sci-type
       |     +--ro created-time?        yang:date-and-time
       |     +--ro started-time?        yang:date-and-time
       |     +--ro stopped-time?        yang:date-and-time
       |     +--ro receiving?           boolean
       |     +--ro in-pkts-ok?          yang:counter64
       |     +--ro in-pkts-unchecked?   yang:counter64
       |     +--ro in-pkts-delayed?     yang:counter64
       |     +--ro in-pkts-late?        yang:counter64
       |     +--ro in-pkts-invalid?     yang:counter64
       |     +--ro in-pkts-not-valid?   yang:counter64
       |     +--ro receive-sa* [rxa]
       |        +--ro in-use?           boolean
       |        +--ro ssci?             uint32
       |        +--ro next-pn?          sec-pn-type
       |        +--ro created-time?     yang:date-and-time
       |        +--ro started-time?     yang:date-and-time
       |        +--ro stopped-time?     yang:date-and-time
       |        +--ro rxa               sec-an-type
       |        +--ro lowest-pn?        sec-pn-type
       |        +--ro key-identifier?   sec-key-identifier-type
       +--rw generation
```


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```
      |  +--ro sci-base?                sec-sci-type
      |  +--rw max-transmit-channels?   uint8
      |  +--rw max-transmit-keys?       uint8
      |  +--rw protect-frames?          boolean
      |  +--rw always-include-sci?      boolean
      |  +--rw use-es?                  boolean
      |  +--rw use-scb?                 boolean
      |  +--ro including-sci?           boolean
      |  +--ro out-pkts-untagged?       yang:counter64
      |  +--ro out-pkts-too-long?       yang:counter64
      |  +--ro out-octets-protected?    yang:counter64
      |  +--ro out-octets-encrypted?    yang:counter64
      |  +--rw user-priority-tc* [user-priority]
      |  |  +--rw user-priority       dot1q-types:priority-type
      |  |  +--rw traffic-class?      dot1q-types:priority-type
      |  |  +--rw access-class-de0?   uint8
      |  |  +--rw access-class-de1?   uint8
      |  +--ro transmit-sc* [sci]
      |     +--ro sci                   sec-sci-type
      |     +--ro created-time?         yang:date-and-time
      |     +--ro started-time?         yang:date-and-time
      |     +--ro stopped-time?         yang:date-and-time
      |     +--ro transmitting?         boolean
      |     +--ro encoding-sa?          sec-an-type
      |     +--ro out-pkts-protected?   yang:counter64
      |     +--ro out-pkts-encrypted?   yang:counter64
      |     +--ro transmit-sa* [txa]
      |        +--ro in-use?            boolean
      |        +--ro ssci?              uint32
      |        +--ro next-pn?           sec-pn-type
      |        +--ro created-time?      yang:date-and-time
      |        +--ro started-time?      yang:date-and-time
      |        +--ro stopped-time?      yang:date-and-time
      |        +--ro txa                sec-an-type
      |        +--ro confidentiality?   boolean
      |        +--ro key-identifier?    sec-key-identifier-type
      +--rw current-cipher-suite
      |  +--rw cipher-suite-identifier?   sec-eui64-type
      |  +--rw data-key* [key-index]
      |     +--rw key-index         uint32
      |     +--ro key-identifier?   sec-key-identifier-type
      |     +--ro transmits?        boolean
      |     +--ro receives?         boolean
      +--rw cipher-suite-control* [implemented-cipher-suite]
         +--rw implemented-cipher-suite    sec-eui64-type
         +--rw enable-use?                 boolean
         +--rw require-confidentiality?    boolean
augment /sys:system/dot1x:pae-system:
  +--rw secy-system
     +--rw cipher-suites* [cipher-suite]
        +--rw cipher-suite                  sec-eui64-type
        +--ro name?                         string
        +--ro integrity-protection?         boolean
        +--ro confidentiality-protection?   boolean
        +--ro changes-data-length?          boolean
        +--ro icv-length?                   uint16
```

23.10.2 ieee802-dot1ae-pry schema

```
module: ieee802-dot1ae-pry

augment /if:interfaces/if:interface:
  +--rw pry {macsec-priv}?
     +--ro secy-support?             boolean
     +--ro pry-address?              ieee:mac-address
     +--rw pry-mppdu-dest-address?   ieee:mac-address
     +--rw max-peers?                uint8
     +--ro num-peers?                uint8
     +--ro peer-entry*               ieee:mac-address
     +--rw reception
```


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```
      |  +--rw privacy-protection?             boolean
      |  +--ro default-reassembly-algorithm?   boolean
      |  +--rw rx-statistics
      |     +--ro in-user-frames?                      yang:counter64
      |     +--ro in-user-octets?                      yang:counter64
      |     +--ro in-pad-octets?                       yang:counter64
      |     +--ro in-mppdus?                           yang:counter64
      |     +--ro in-encapsulated-frames?              yang:counter64
      |     +--ro in-user-express-fragments?           yang:counter64
      |     +--ro in-user-preemptable-fragments?       yang:counter64
      |     +--ro in-express-discard-fragments?        yang:counter64
      |     +--ro in-preemptable-discard-fragments?    yang:counter64
      |     +--ro in-unknown-mppcis?                   yang:counter64
      |     +--ro in-errored-mppdus?                   yang:counter64
      |     +--ro in-user-unprotected-frames?          yang:counter64
      |     +--ro in-user-unprotected-octets?          yang:counter64
      +--rw transmission
         +--rw privacy-protection?    boolean
         +--rw privacy-selection* [user-priority]
         |  +--rw user-priority            dot1q-types:priority-type
         |  +--rw privacy-type             identityref
         |  +--rw frame-access-priority?   dot1q-types:priority-type
         |  +--rw frame-reveal-de?         enumeration
         |  +--rw frame-padding?           enumeration
         +--rw channel* [channel-id]
         |  +--rw channel-id               identityref
         |  +--rw enable?                  boolean
         |  +--rw fragment-enable?         boolean
         |  +--rw access-priority?         dot1q-types:priority-type
         |  +--rw user-data-frame-size?    uint16
         |  +--rw mppdu-generation?        enumeration
         |  +--rw requested-kbit-rate      uint32
         |  +--ro mppdu-bits-on-wire?      uint32
         |  +--ro mppdu-interval?          uint32
         |  +--rw user-burst-octets?       uint32
         |  +--rw tx-statistics
         |     +--ro out-ch-user-frames?         yang:counter64
         |     +--ro out-ch-user-octets?         yang:counter64
         |     +--ro out-ch-pad-octets?          yang:counter64
         |     +--ro out-mppdus?                 yang:counter64
         |     +--ro out-encapsulated-frames?    yang:counter64
         |     +--ro out-express-fragments?      yang:counter64
         |     +--ro out-preempt-fragments?      yang:counter64
         +--rw frame-tx-statistics
            +--ro out-pf-user-frames?       yang:counter64
            +--ro out-pf-user-octets?       yang:counter64
            +--ro out-pf-pad-octets?        yang:counter64
            +--ro out-unprotected-frames?   yang:counter64
            +--ro out-unprotected-octets?   yang:counter64

notifications:
  +---n pry-max-peers-exceeded
     +--ro pry-interface?   -> /if:interfaces/interface/name
```

23.10.3 ieee802-dot1x-eapol schema

```
module: ieee802-dot1x-eapol

  augment /if:interfaces/if:interface/dot1x:pae:
    +--rw eapol-group-address?   ieee:mac-address
```


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23.11 YANG modules[4] [5] [6]

23.11.1 ieee802-dot1ae-secy YANG module

```
module ieee802-dot1ae-secy {
  yang-version 1.1;
  namespace "urn:ieee:std:802.1AE:yang:ieee802-dot1ae-secy";
  prefix secy;

  import ietf-interfaces {
    prefix if;
  }
  import ietf-yang-types {
    prefix yang;
  }
  import ietf-system {
    prefix sys;
  }
  import ieee802-dot1q-types {
    prefix dot1q-types;
  }
  import ieee802-dot1x {
    prefix dot1x;
  }

  organization
    "Institute of Electrical and Electronics Engineers";
  contact
    "WG-URL: http://ieee802.org/1/
     WG-EMail: stds-802-1-l@ieee.org

     Contact: IEEE 802.1 Working Group Chair
     Postal: C/O IEEE 802.1 Working Group
           IEEE Standards Association
           445 Hoes Lane
           Piscataway, NJ 08855
           USA

     E-mail: stds-802-1-chairs@ieee.org";
  description
    "The MAC security entity (SecY) YANG module. A SecY is a protocol
     shim providing MAC Security (MACsec) in an interface stack.

     Each SecY transmits MACsec protected frames on one or more Secure
     Channels (SCs) to each of the other SecYs attached to the same LAN
     and participating in the same Secure Connectivity Association
     (CA). The CA is a security relationship, that is established and
     maintained by key agreement protocols and supported by MACsec to
     provide full connectivity between its participants. Each SC
     provides unidirectional point to multipoint connectivity from one
     participant to all the others and is supported by a succession of
     similarly point to multipoint Secure Associations (SAs). The
     Secure Association Key (SAK) used to protect frames is changed as
     an SA is replaced by its (overlapping) successor so fresh keys can
     be used without disrupting a long lived SC and CA.

     Two different upper interfaces, a Controlled Port (for frames
     protected by MACsec, providing an instance of the secure MAC
     service) and an Uncontrolled Port (for frames not requiring
     protection, like the key agreement frames used to establish the CA
     and distribute keys) are associated with a SecY shim.";

```

[4] Copyright release for YANG: Users of this standard may freely reproduce the YANG modules contained in this standard so that they can be used for their intended purpose.

[5] An ASCII version of each YANG module is attached to the PDF of this standard and can also be obtained from the IEEE 802.1 Website at https://1.ieee802.org/yang-modules/.

[6] References in this standard's YANG module definitions are not clickable, as each module has been incorporated unchanged after development and verification using YANG tools.

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```
revision 2022-06-14 {
  description
    "The following reference statement identifies each referenced IEEE
     Standard as updated by applicable amendments.";
  reference
    "IEEE Std 802.1AE Media Access Control (MAC) Security:
     IEEE Stds 802.1AE-2018, 802.1AE-2018-Cor1-2020, 802.1AEdk-2022.
     IEEE Std 802.1X Port-Based Network Access Control:
     IEEE Std 802.1X-2020.
     IEEE Std 802.1AC Media Access Control (MAC) Service Definition:
     IEEE Stds 802.1AC-2016, 802.1AC-2016-Cor1-2018.";
}

/* ----------------------------------------
 *  Typedefs
 * ----------------------------------------
 */

typedef sec-an-type {
  type uint8 {
    range "0..3";
  }
  description
    "A 2-bit number that is concatenated with a MACsec Secure
     Channel Identifier to identify a Secure Association. Indicates
     an Association Number (AN) assigned by the Key Server for use
     with the key number for transmission.

     Each SC is comprised of a succession of SAs, each with a
     different SAK, identified by a Secure Association Identifier
     (SAI) comprising an SCI concatenated with a two-bit AN. The SAI
     is unique for SAs used by SecYs participating in a given CA at
     any instant.";
  reference
    "9.6 of IEEE Std 802.1AE";
}

typedef sec-pn-type {
  type uint64;
  description
    "The Packet Number (PN). A 32-bit or 64-bit unsigned value.
     A monotonically increasing value that is guaranteed unique
     for each MACsec frame transmitted using a given Secure
     Association Key (SAK).";
  reference
    "9.8 of IEEE Std 802.1AE";
}

typedef sec-sci-type {
  type string {
    pattern '[0-9a-fA-F]{2}(-[0-9a-fA-F]{2}){5}-[0-9a-fA-F]{4}';
  }
  description
    "The Secure Channel Identifier (SCI). An 8 octet binary
     number, where the first (most significant) 6 octets
     represent the MAC Address (in canonical format), and the
     next 2 octets represents the Port Identifier. Integers can
     be entered as hexadecimal.";
  reference
    "9.9 of IEEE Std 802.1AE, 10.7.14, 10.7.23 and
     9.8 of IEEE Std 802.1X";
}

typedef sec-eui64-type {
  type uint64;
  description
    "A 64 bit identifier.";
  reference
    "10.7.25 of IEEE Std 802.1AE";
}

typedef sec-key-identifier-type {
```


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```
    type string {
      length "0..32";
    }
    description
      "The sec-key-identifier-type is an octet string, whose
       format and interpretation depends on the key agreement
       protocol in use. It does not contain any information about
       the SAK other than that explicitly chosen by the key
       agreement protocol to publicly identify the key. If MKA is
       being used, it is the 128-bit Key Identifier (KI)
       specified by IEEE Std 802.1X encoded in an octet string as
       specified by that standard.";
    reference
      "10.7.14, 10.7.23 and
       9.8 of IEEE Std 802.1X";
  }

  /* ---------------------------------------------------
   * Group objects used by IEEE Std 802.1AE YANG module
   * ---------------------------------------------------
   */

  grouping mac-status {
    description
      "This holds statistics for the Provided interface ports both the
       controlled port and the uncontrolled port.";
    leaf mac-enabled {
      type boolean;
      config false;
      description
        "The mac-enabled parameter is True if use of the service is
         permitted and is otherwise False. The value of this parameter
         is determined by administrative controls specific to the
         entity providing the service.";
      reference
        "6.4 of IEEE Std 802.1AE,
        11.2 and 11.3 of IEEE  Std 802.1AC";
    }
    leaf mac-operational {
      type boolean;
      config false;
      description
        "The mac-operational parameter is True if, and only if,
         service requests can be made and service indications can
         occur.";
      reference
        "6.4 of IEEE Std 802.1AE";
    }
    leaf oper-point-to-point-mac {
      type boolean;
      config false;
      description
        "If the oper-point-to-point-mac parameter is True, the
         service is used as if it provides connectivity to at
         most one other system; if False, the service is used as
         if it can provide connectivity to a number of systems.";
      reference
        "6.5 of IEEE Std 802.1AE";
    }
    leaf admin-point-to-point-mac {
      type enumeration {
        enum force-true {
          value 1;
          description
            "If admin-point-to-point-mac is set to force-true
             oper-point-to-point-mac shall be True, regardless of any
             indications to the contrary generated by the entity
            providing the service.";
          reference
            "6.5, 10.7.4 of IEEE Std 802.1AE";
        }
        enum force-false {
```


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```
          value 2;
          description
            "If admin-point-to-point-mac is set to force-false
             oper-point-to-point-mac shall be False.";
          reference
            "6.5, 10.7.4 of IEEE Std 802.1AE";
        }
        enum auto {
          value 3;
          description
            "If admin-point-to-point-mac is set to auto
             oper-point-to-point-mac is as currently determined by the
             the entity providing the service.";
          reference
            "6.5, 10.7.4 of IEEE Std 802.1AE";
        }
      }
      default "auto";
      description
        "Each service access point can make available status
         parameters that reflect the point-to-point status for
         the service instance provided, and that allow
         administrative control over the use of that
         information. The admin-point-to-point-mac parameter can
         take one of three values.";
      reference
        "6.5, 10.7.4 of IEEE Std 802.1AE";
    }
  }

  //end provided-interface-grouping
  /* common SC items */

  grouping secy-secure-channel-grouping {
    description
      "The secy-secure-channel grouping contains configuration and
       state common to both transmit and receive SCs.";
    leaf created-time {
      type yang:date-and-time;
      config false;
      description
        "The system time when the SC was created.";
      reference
        "10.7.12 of IEEE Std 802.1AE";
    }
    leaf started-time {
      type yang:date-and-time;
      config false;
      description
        "The system time when receiving last became True for
         the SC.";
      reference
        "10.7.12 of IEEE Std 802.1AE";
    }
    leaf stopped-time {
      type yang:date-and-time;
      config false;
      description
        "The system time when receiving last became False for
         the SC.";
      reference
        "10.7.12 of IEEE Std 802.1AE";
    }
  }

  // end secy-secure-channel-grouping
  /* common SA items */

  grouping secy-secure-association-grouping {
    description
      "The secy-secure-association grouping contains configuration and
       state common to both transmit and receive Security
```


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```
       Associations(SAs).";
    leaf in-use {
      type boolean;
      config false;
      description
        "If in-use is True, and MAC_Operational is True for the Common
         Port, the SA can receive and transmit frames.";
      reference
        "10.7.14, 10.7.23 of IEEE Std 802.1AE";
    }
    leaf ssci {
      type uint32;
      config false;
      description
        "Short Secure Channel Identifier for the Send and Transmit SA";
      reference
        "10.7.14, 10.7.23 of IEEE Std 802.1AE";
    }
    leaf next-pn {
      type sec-pn-type;
      config false;
      description
        "The Next Packet Number, one more than the highest PN conveyed
         in the SecTAG of successfully validates frames received on
         this SA.";
      reference
        "10.7.14, 10.7.23 of IEEE Std 802.1AE";
    }
    leaf created-time {
      type yang:date-and-time;
      config false;
      description
        "The system time when the SA was created.";
      reference
        "10.7.14, 10.7.23 of IEEE Std 802.1AE";
    }
    leaf started-time {
      type yang:date-and-time;
      config false;
      description
        "The system time when in-use last became True for the
         SA.";
      reference
        "10.7.14 of IEEE Std 802.1AE";
    }
    leaf stopped-time {
      type yang:date-and-time;
      config false;
      description
        "The system time when in-use last became False for the
         SA.";
      reference
        "10.7.14 of IEEE Std 802.1AE";
    }
  }

  // end secy-secure-association-grouping
  /* ------------------------------------------------
   * Configuration objects used by IEEE Std 802.1AE YANG module
   * ------------------------------------------------
   */

  augment "/if:interfaces/if:interface/dot1x:pae" {
    description
      "SecY augments a PAE under an ietf interface.";
    container secy {
      description
        "Augment interface with 802.1 SecY configuration nodes. The
         management information for each SecY is indexed by
         controlled-port-number within a SecY System. This containment
         relationship complements that specified in IEEE Std 802.1X,
         where the management information for each PAE is indexed by
```


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```
    portNumber within a PAE System.";
  reference
    "10.7 of IEEE Std 802.1AE";
  container controlled-port {
    description
      "Controlled port control and status.";
    uses mac-status;
    leaf controlled-port-enabled {
      type boolean;
      config false;
      description
        "By setting controlled-port-enabled False, the KaY can
         prohibit use of the Controlled Port until the secure
         connectivity required has been configured.";
      reference
        "10.7.6 of IEEE Std 802.1AE";
    }
  }
  container uncontrolled-port {
    description
      "Uncontrolled port control and status.";
    uses mac-status;
  }
  container verification {
    description
      "The Verification controls for validation and replay
       protect for a given secy.";
    reference
      "10.6 of IEEE Std 802.1AE";
    leaf max-receive-channels {
      type uint8;
      config false;
      description
        "Specifies maximum number of receive channels for a SecY.";
      reference
        "10.7.7 of IEEE Std 802.1AE";
    }
    leaf max-receive-keys {
      type uint8;
      config false;
      description
        "Specifies maximum number of receive keys for a SecY.";
      reference
        "10.7.7 of IEEE Std 802.1AE";
    }
    leaf validate-frames {
      type enumeration {
        enum disabled {
          value 1;
          description
            "Frame Verification is disabled. Remove SecTAGs and
             ICVs (if present) from received frames.";
        }
        enum check {
          value 2;
          description
            "Frame Verification is enabled. Do not discard invalid
             frames.";
        }
        enum strict {
          value 3;
          description
            "Frame Verification is enabled and strictly enforced.
             Discard any invalid frames.";
        }
        enum null {
          value 4;
          description
            "No Frame Verification is performed, do not
             remove-secTags or ICVs.";
        }
      }
```


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```
      default "strict";
      description
        "Controls the frame verification settings. If the
         management control validate-frames is not Strict, frames
         without a SecTAG are received, counted, and delivered to
         the Controlled Port; otherwise, they are counted and
         discarded. If validate-frames is Disabled, cryptographic
         validation is not applied to tagged frames, but frames
         whose original service user data can be recovered are
         delivered. Frames with a SecTAG that has the TCI E bit set
         but the C bit clear are discarded, as this reserved
         encoding is used to identify frames with a SecTAG that are
         not to be delivered to the Controlled Port. If
         validate-frames is Null, all received frames are delivered
         to the Controlled Port without modification, irrespective
         of the absence, presence, or validity of a SecTAG.";
      reference
        "10.7.8, Figure 10-4 of IEEE Std 802.1AE";
    }
    leaf replay-protect {
      type boolean;
      default "true";
      description
        "If the Packet Number (PN) of the received frame is less
         than the lowest acceptable packet number for the SA, and
         replay-protect is enabled, the frame is discarded and the
         in-pkts-late counter incremented. The replay-protect and
         replay-window controls allows replay protection to be
         disabled, to operate on a packet number window, or to
         enforce strict frame order. If replay-protect is set but
         the replay-window is not zero, frames within the window can
         be received out of order; however, they are not replay
         protected.";
      reference
        "10.6.2, 10.4 of IEEE Std 802.1AE";
    }
    leaf replay-window {
      type uint32;
      default "0";
      description
        "Controls the replay-window size in packets that supports
         media access control methods and provider networks that
         can misorder frames with different priorities and/or
         addresses.";
      reference
        "10.7.8 of IEEE Std 802.1AE";
    }
    leaf in-pkts-untagged {
      type yang:counter64;
      config false;
      description
        "The number of packets received without the MACsec tag
         (SecTAG) received while validate-frames was not strict.";
      reference
        "10.7.9 of IEEE Std 802.1AE";
    }
    leaf in-pkts-no-tag {
      type yang:counter64;
      config false;
      description
        "The number of packets received without the MACsec tag
         (SecTAG) discarded because validate-frames was set to
         strict.";
      reference
        "10.7.9 of IEEE Std 802.1AE";
    }
    leaf in-pkts-bad-tag {
      type yang:counter64;
      config false;
      description
        "The number of received packets discarded with an invalid
         MACsec tag (SecTAG), zero value PN, or invalid ICV.";
```


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```
      reference
        "10.7.9 of IEEE Std 802.1AE";
    }
    leaf in-pkts-no-sa {
      type yang:counter64;
      config false;
      description
        "The number of received packets discarded with an unknown
         SCI or for an unused SA.";
      reference
        "10.7.9 of IEEE Std 802.1AE";
    }
    leaf in-pkts-no-sa-error {
      type yang:counter64;
      config false;
      description
        "The number of packets discarded because the received SCI
         is unknown or the SA is not in use.";
      reference
        "10.7.9 of IEEE Std 802.1AE";
    }
    leaf in-pkts-overrun {
      type yang:counter64;
      config false;
      description
        "The number of packets discarded because they exceeded
         cryptographic performance capabilities.";
      reference
        "10.7.9 of IEEE Std 802.1AE";
    }
    leaf in-octets-validated {
      type yang:counter64;
      config false;
      description
        "The number of plaintext octets recovered from packets
         that were integrity protected but not encrypted.";
      reference
        "10.6, 10.6.3 of IEEE Std 802.1AE";
    }
    leaf in-octets-decrypted {
      type yang:counter64;
      config false;
      description
        "The number of plaintext octets recovered from packets
         that were integrity protected and encrypted.";
      reference
        "10.6, 10.6.3 of IEEE Std 802.1AE";
    }
    list receive-sc {
      key "sci";
      config false;
      description
        "The Receive Security Channel Status for a given
         secure channel identifier.";
      reference
        "10.7.9 of IEEE Std 802.1AE";
      leaf sci {
        type sec-sci-type;
        description
          "Each SecY transmits frames conveying secure MAC Service
           requests of any given priority on a single SC. Each SC
           provides unidirectional point-to-multipoint
           communication, and it can be long lived, persisting
           through SAK changes. Each SC is identified by a Secure
           Channel Identifier (SCI) comprising a 48-bit MAC address
           concatenated with a 16-bit Port Identifier.";
        reference
          "7.1.2 and figure 7.7 of IEEE Std 802.1AE";
      }
      uses secy-secure-channel-grouping;
      leaf receiving {
        type boolean;
```


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```
    config false;
    description
      "Receiving is True if in-use is True for any of the SAs
       for the SC, and False otherwise.";
    reference
      "10.7.12 of IEEE Std 802.1AE";
  }
  leaf in-pkts-ok {
    type yang:counter64;
    config false;
    description
      "For this SC, the number of validated packets.";
    reference
      "10.6.5, 10.7.9 of IEEE Std 802.1AE";
  }
  leaf in-pkts-unchecked {
    type yang:counter64;
    config false;
    description
      "For this SC, the number of packets while
       validate-frames was disabled.";
    reference
      "10.6.5, 10.7.9 of IEEE Std 802.1AE";
  }
  leaf in-pkts-delayed {
    type yang:counter64;
    config false;
    description
      "For this SC, the number of received packets, with
       Packet Number (PN) lower than the lowest acceptable PN
       lowest-pn and replay-protect is False.";
    reference
      "10.6.5, 10.7.9 of IEEE Std 802.1AE";
  }
  leaf in-pkts-late {
    type yang:counter64;
    config false;
    description
      "For this SC, the number of discarded packets, because
       the Packet Number (PN) was lower than the lowest
       acceptable PN lowest-pn and replay-protect is True.";
    reference
      "10.7.9 of IEEE Std 802.1AE";
  }
  leaf in-pkts-invalid {
    type yang:counter64;
    config false;
    description
      "For this SC, the number packets that failed validation
       but could be received because validate-frames was
       'check' and the data was not encrypted (so the original
       frame could be recovered).";
    reference
      "10.7.9 of IEEE Std 802.1AE";
  }
  leaf in-pkts-not-valid {
    type yang:counter64;
    config false;
    description
      "For this SC, the number of packets discarded because
       validation failed and validate-frames was 'strict' or
       the data was encrypted (so the original frame could not
       be recovered).";
    reference
      "10.7.9 of IEEE Std 802.1AE";
  }
  list receive-sa {
    key "rxa";
    description
      "The Receive Security Association (SA) Status for
       this association.";
    uses secy-secure-association-grouping;
```


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```
        leaf rxa {
          type sec-an-type;
          description
            "The Association Number for this Receiving SA.";
          reference
            "10.7.13 of IEEE Std 802.1AE";
        }
        leaf lowest-pn {
          type sec-pn-type;
          config false;
          description
            "The lowest acceptable packet number. A received frame
             with a lower PN is discarded if replay-protect is
             enabled.";
          reference
            "10.7.14 of IEEE Std 802.1AE";
        }
        leaf key-identifier {
          type sec-key-identifier-type;
          config false;
          description
            "The key-identifier is an octet string, whose format
             and interpretation depends on the key agreement
             protocol in use. It does not contain any information
             about the SAK other than that explicitly chosen by the
             key agreement protocol to publicly identify the key.
             If MKA is being used, it is the 128-bit Key Identifier
             (KI) specified by IEEE Std 802.1X encoded in an octet
             string as specified by that standard.";
          reference
            "10.7.14, 10.7.24, of IEEE Std 802.1AE and
            9.8 of IEEE Std 802.1X";
        }
      }
    }
  }
  container generation {
    description
      "The Generation controls for given secy.";
    reference
      "10.5 of IEEE Std 802.1AE";
    leaf sci-base {
      type sec-sci-type;
      config false;
      description
        "The base for a set of secure channels Security
         Channel Identifier.";
      reference
        "7.1.2, 10.7.17 of IEEE Std 802.1AE";
    }
    leaf max-transmit-channels {
      type uint8;
      description
        "Number of transmit channels.";
      reference
        "10.7.16 of IEEE Std 802.1AE";
    }
    leaf max-transmit-keys {
      type uint8;
      description
        "Number of transmit keys.";
      reference
        "10.7.16 of IEEE Std 802.1AE";
    }
    leaf protect-frames {
      type boolean;
      default "true";
      description
        "The protect-frames control is provided to facilitate
         deployment.";
      reference
        "10.7.17 of IEEE Std 802.1AE";
```


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```
      }
      leaf always-include-sci {
        type boolean;
        default "false";
        description
          "Mandates inclusion of an explicit SCI in the SecTAG when
           transmitting protected frames.";
        reference
          "10.5.3, 10.7.17 of IEEE Std 802.1AE";
      }
      leaf use-es {
        type boolean;
        default "false";
        description
          "Enables use of the ES bit in the SecTAG when transmitting
           protected frames.";
        reference
          "10.5.3, 10.7.17 of IEEE Std 802.1AE";
      }
      leaf use-scb {
        type boolean;
        default "false";
        description
          "Enables use of the SCB bit in the SecTAG when
           transmitting protected frames.";
        reference
          "10.5.3, 10.7.17 of IEEE Std 802.1AE";
      }
      leaf including-sci {
        type boolean;
        config false;
        description
          "True if an explicit SCI is included in the SecTAG when
           transmitting protected frames.";
        reference
          "10.5.3, 10.7.17 of IEEE Std 802.1AE";
      }
      leaf out-pkts-untagged {
        type yang:counter64;
        config false;
        description
          "The number of packets transmitted without a SecTAG
           because protect-frames is configured False.";
        reference
          "10.7.18 of IEEE Std 802.1AE";
      }
      leaf out-pkts-too-long {
        type yang:counter64;
        config false;
        description
          "The number of transmit packets discarded because their
           length is greater than the ifMtu of the Common Port.";
        reference
          "10.7.18 of IEEE Std 802.1AE";
      }
      leaf out-octets-protected {
        type yang:counter64;
        config false;
        description
          "The number of plain text octets integrity protected but
           not encrypted in transmitted frames.";
        reference
          "10.7.9 of IEEE Std 802.1AE";
      }
      leaf out-octets-encrypted {
        type yang:counter64;
        config false;
        description
          "The number of plain text octets integrity protected and
           encrypted in transmitted frames.";
        reference
          "10.7.9 of IEEE Std 802.1AE";
```


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```
      }
      list user-priority-tc {
        key "user-priority";
        description
          "Each entry in the Traffic Class Table is a traffic class,
           represented by an integer from 0 (default) through 7 that also
           comprises the numeric value of the four most significant bits
           of the Port Identifier component of the SCI for the selected
           SC. The default for this table is every row has a non-mapping
           priority with the first row having all zeros, the second row
           having all ones etc. up to the last row having all sevens.";
        reference
          "10.7.17 of IEEE Std 802.1AE";
        leaf user-priority {
          type dot1q-types:priority-type;
          description
            "The User Priority.";
          reference
            "10.7.17 of IEEE Std 802.1AE";
        }
        leaf traffic-class {
          type dot1q-types:priority-type;
          description
            "The traffic class that maps to the four most significant
             bits of the Port Identifier component of the SCI for the
             selected SC.";
          reference
            "10.7.17 of IEEE Std 802.1AE";
        }
        leaf access-class-de0 {
          type uint8 {
            range "0..15";
          }
          description
            "The access priority when not drop eligible.";
          reference
            "10.7.17 of IEEE Std 802.1AE";
        }
        leaf access-class-de1 {
          type uint8 {
            range "0..15";
          }
          description
            "The access priority when drop eligible.";
          reference
            "10.7.17 of IEEE Std 802.1AE";
        }
      }
      list transmit-sc {
        key "sci";
        config false;
        description
          "The transmit Security Channel, status for a given
           Security Channel Identifier.";
        reference
          "10.7.1 of IEEE Std 802.1AE";
        leaf sci {
          type sec-sci-type;
          description
            "Each SecY transmits frames conveying secure MAC Service
             requests of any given priority on a single SC. Each SC
             provides unidirectional point-to-multipoint
             communication, and it can be long lived, persisting
             through SAK changes. Each SC is identified by a Secure
             Channel Identifier (SCI) comprising a 48-bit MAC address
             concatenated with a 16-bit Port Identifier.";
          reference
            "7.1.2 and figure 7.7 of IEEE Std 802.1AE";
        }
        uses secy-secure-channel-grouping;
        leaf transmitting {
          type boolean;
```


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```
      config false;
      description
        "True if in-use is True for any of the SAs for the SC,
         and False otherwise.";
      reference
        "10.7.21 of IEEE Std 802.1AE";
    }
    leaf encoding-sa {
      type sec-an-type;
      config false;
      description
        "The current value of the encoding-sa variable for the
         selected transmit SC.";
      reference
        "10.7.24 of IEEE Std 802.1AE";
    }
    leaf out-pkts-protected {
      type yang:counter64;
      config false;
      description
        "The number of integrity protected but not encrypted
         packets for this transmit SC.";
      reference
        "10.7.18, Figure 10-3 of IEEE Std 802.1AE";
    }
    leaf out-pkts-encrypted {
      type yang:counter64;
      config false;
      description
        "The number of integrity protected and encrypted packets
         for this transmit SC.";
      reference
        "10.7.18, Figure 10-3 of IEEE Std 802.1AE";
    }
    list transmit-sa {
      key "txa";
      config false;
      description
        "The transmit security association status for a
         given association number.";
      uses secy-secure-association-grouping;
      leaf txa {
        type sec-an-type;
        config false;
        description
          "The association number for the SA.";
        reference
          "10.7.23 of IEEE Std 802.1AE";
      }
      leaf confidentiality {
        type boolean;
        config false;
        description
          "True if the SA provides confidentiality as well as
           integrity for transmitted frames.";
        reference
          "10.7.23 of IEEE Std 802.1AE";
      }
      leaf key-identifier {
        type sec-key-identifier-type;
        config false;
        description
          "The key-identifier is an octet string, whose format
           and interpretation depends on the key agreement
           protocol in use. It does not contain any information
           about the SAK other than that explicitly chosen by the
           key agreement protocol to publicly identify the key.
           If MKA is being used, it is the 128-bit Key Identifier
           (KI) specified by IEEE Std 802.1X encoded in an octet
           string as specified by that standard.";
        reference
          "10.7.14, 14.7, 14.8 of IEEE Std 802.1AE,
```


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```
            9.8 of IEEE Std 802.1X";
        }
      }
    }
  }
  // end generation
  container current-cipher-suite {
    description
      "The current-cipher-suite is selected by the KaY.
       The Current Cipher Suite may also be selected and keys
       created by management, but a conformant implementation
       shall provide a mechanism to allow such selection
       and creation by network management to be disabled.";
    leaf cipher-suite-identifier {
      type sec-eui64-type;
      description
        "The Cipher Suite currently used by this SecY.";
      reference
        "10.7.27 of IEEE Std 802.1AE";
    }
    list data-key {
      key "key-index";
      description
        "An index of Keys Used.";
      leaf key-index {
        type uint32;
        description
          "Numeric key number used as index.";
        reference
          "10.7.27 of IEEE Std 802.1AE";
      }
      leaf key-identifier {
        type sec-key-identifier-type;
        config false;
        description
          "Key Identifier (KI), comprising the Key Server's MI
           (providing the more significant bits) and a 32-bit Key
           Number (KN) assigned by that Key Server (sequentially,
           beginning with 1). Each KI is used to identify the
           corresponding SAK for the purposes of SAI assignment,
           and appears in the clear in MKPDUs, so network
           management equipment and personnel can observe and
           diagnose MKA operation (if necessary) without having
           access to any secret key.";
        reference
          "10.7.28 of IEEE Std 802.1AE";
      }
      leaf transmits {
        type boolean;
        config false;
        description
          "Transmits True means key is used for transmitting
           direction.";
        reference
          "10.5 of IEEE Std 802.1AE";
      }
      leaf receives {
        type boolean;
        config false;
        description
          "Receives True means key is used for receiving
           direction.";
        reference
          "10.5 of IEEE Std 802.1AE";
      }
    }
  }
  // end current-cipher-suite
  list cipher-suite-control {
    key "implemented-cipher-suite";
    description
      "The MKA Key Server selects the Cipher Suite to be used to
```


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```
          protect communication within a CA. If enable-use is False
          for the selected Cipher Suite, the SecY does not participate
          in the CA and MAC_Operational for the Controlled Port
          remains False. If the MKA Key Server has selected integrity
          protection and enable-use and require-confidentiality are
          both True for the selected Cipher Suite, confidentiality
          protection is used.";
        leaf implemented-cipher-suite {
          type sec-eui64-type;
          description
            "cipher suite identifier (EUI-64)";
          reference
            "10.7.26 of IEEE Std 802.1AE";
        }
        leaf enable-use {
          type boolean;
          default "true";
          description
            "Enables use of the Cipher Suite by this SecY.";
          reference
            "10.7.26 of IEEE Std 802.1AE";
        }
        leaf require-confidentiality {
          type boolean;
          default "true";
          description
            "True if confidentiality protection is required if
             this Cipher Suite is used.";
          reference
            "10.7.26 of IEEE Std 802.1AE";
        }
      }
    }
  }

  // end secy augment interfaces
  /*
   Secy System
   */

  augment "/sys:system/dot1x:pae-system" {
    description
      "Augment system with 802.1AE MACSec System Cipher Suites nodes.";
    container secy-system {
      description
        "Augment system with 802.1AE SecY configuration nodes.";
      list cipher-suites {
        key "cipher-suite";
        description
          "A list of configuration parameters and operational state
           associated with a cipher suite.";
        leaf cipher-suite {
          type sec-eui64-type;
          description
            "A globally unique 64-bit (EUI-64) identifier for this
             cipher suite.";
          reference
            "10.7.25 of IEEE Std 802.1AE";
        }
        leaf name {
          type string {
            length "1..254";
          }
          config false;
          description
            "Cipher Suite Name, a human readable and displayable UTF-8
             (IETF RFC 2279) string.";
          reference
            "10.7.25 of IEEE Std 802.1AE";
        }
        leaf integrity-protection {
          type boolean;
```


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```
          config false;
          description
            "True if integrity protection without confidentiality can
             be provided.";
          reference
            "10.7.25 of IEEE Std 802.1AE";
        }
        leaf confidentiality-protection {
          type boolean;
          config false;
          description
            "True if confidentiality with integrity protection can be
             provided.";
          reference
            "10.7.25 of IEEE Std 802.1AE";
        }
        leaf changes-data-length {
          type boolean;
          config false;
          description
            "Indicates that the cipher suite changes the data length.";
          reference
            "10.7.25 of IEEE Std 802.1AE";
        }
        leaf icv-length {
          type uint16;
          config false;
          description
            "The number of octets in the ICV.";
          reference
            "10.7.25 of IEEE Std 802.1AE";
        }
      }
    }
  } // end /sys:system

}

```


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23.11.2 ieee802-dot1ae-pry YANG module

```
module ieee802-dot1ae-pry {
  yang-version 1.1;
  namespace "urn:ieee:std:802.1AE:yang:ieee802-dot1ae-pry";
  prefix pry;

  import ietf-interfaces {
    prefix if;
  }
  import ietf-yang-types {
    prefix yang;
  }
  import ieee802-dot1q-types {
    prefix dot1q-types;
  }
  import ieee802-types {
    prefix ieee;
  }
  import iana-if-type {
    prefix ianaift;
  }

  organization
    "IEEE 802.1 Working Group";
  contact
    "WG-URL: http:ieee802.org/1/
     WG-EMail: stds-802-1-l@ieee.org

     Contact: IEEE 802.1 Working Group Chair
     Postal: C/O IEEE 802.1 Working Group
           IEEE Standards Association
           445 Hoes Lane
           Piscataway, NJ 08855
           USA

     E-mail: stds-802-1-chairs@ieee.org";
  description
    "This YANG module augments the configuration and operational state
     data for interfaces for the MAC Privacy project: Std 802.1AE; see
     that standard and its amendments for full legal notices.

     A MAC Privacy protection Entity (PrY) is a protocol shim in an
     interface stack that encapsulates user data frames in MAC Privacy
     protection Data Units (MPPDUs). Once those MPPDUs are
     confidentiality protected by MACsec, the ability of potential
     adversaries to draw conclusions from the source and destination
     MAC addresses, sizes, and transmission timing and frequency of
     user data frames is reduced or eliminated.

     Each PrY in a system and its managed objects augments its upper
     interface (Private Port), which provides a privacy protected
     service to its user, typically a Bridge Port (IEEE Std 802.1Q) or
     an end station protocol stack. Object names can be conveniently
     pronounced by rendering Pry as Privacy.";

  revision 2022-06-17 {
    description
      "The following reference statement identifies each referenced IEEE
       Standard as updated by applicable amendments.";
    reference
      "IEEE Std 802.1AE Media Access Control (MAC) Security:
       IEEE Stds 802.1AE-2018, 802.1AE-2018-Cor1-2020, 802.1AEdk-2022.
       IEEE Std 802.1X Port-Based Network Access Control:
       IEEE Std 802.1X-2020.
       IEEE Std 802.1Q Bridges and Bridged Networks:
       IEEE Std 802.Q-2022";
  }

  /*--------------------*/
  /* Feature            */
```


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```
/*--------------------*/

feature macsec-priv {
  description
    "Feature MAC Privacy.";
}

/*--------------------*/
/*   identities       */
/*--------------------*/

identity priority-map-identity {
  description
    "Base identity for assigning a priority to a Privacy type.";
}

identity channel-identity {
  description
    "Base identity for privacy channel.";
}

identity express-channel {
  base channel-identity;
  base priority-map-identity;
  description
    "This is the express privacy channel frame designation.";
  reference
    "20.13.4 of IEEE Std 802.1AE";
}

identity preemptable-channel {
  base channel-identity;
  base priority-map-identity;
  description
    "This is the preemptable privacy channel designation.";
  reference
    "20.13.4 of IEEE Std 802.1AE";
}

identity frame-identity {
  description
    "Base identity for privacy frame.";
}

identity privacy-frame {
  base frame-identity;
  base priority-map-identity;
  description
    "This is a privacy frame designation.";
  reference
    "3 of IEEE Std 802.1AE";
}

identity none-identity {
  description
    "Base identity for privacy frame.";
}

identity none {
  base none-identity;
  base priority-map-identity;
  description
    "This is no privacy encapsulation. Frames mapped to this
     identity are forwarded directly without MAC privacy
     encapsulation.";
  reference
    "17 of IEEE Std 802.1AE";
}

/*-------------------------*/
/* Notification statements */
/*-------------------------*/
```


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```
notification pry-max-peers-exceeded {
  description
    "A max-peers-exceeded notification is sent when the value
     of if-num-peers exceeds if-max-peers. This is triggered
     only on the transition to the exceeded state and reset
     when the if-num peers is less than or equal to
     if-num-peers.";
  leaf pry-interface {
    type leafref {
      path "/if:interfaces/if:interface/if:name";
    }
    description
      "Contains the interface name containing the PrY that has
       exceeded the number of peers.";
  }
}

/*--------------------*/
/* Configuration Data */
/*--------------------*/

augment "/if:interfaces/if:interface" {
  when "if:type = 'ianaift:ethernetCsmacd' or if:type = "
     + "'ianaift:ilan' or if:type = 'ianaift:macSecControlledIF' or "
     + "if:type = 'ianaift:ptm' or if:type = 'ianaift:bridge'" {
    description
      "Augment interfaces with 802.1ae MACSec System specific
       configuration nodes.";
  }
  if-feature "macsec-priv";
  description
    "MACsec Privacy Mode.";
  container pry {
    description
      "Configure the MAC Privacy Options.";
    leaf secy-support {
      type boolean;
      config false;
      description
        "Set True by the system if the PrY is directly supported
         by a SecY and MKA, and False otherwise.  When True, the
         value of if-mppdu-dest-address and the entries in the
         PrYs peer address table (perr-entry list) are determined
         by the Key Agreement Entity (KaY) operating MKA, and are
         not writable by network management.";
      reference
        "23.4, of IEEE Std 802.1AE
         11.1.1 IEEE Std 802.1X";
    }
    leaf pry-address {
      type ieee:mac-address;
      config false;
      description
        "The individual MAC address associated with the PrY and
         other components of the PrYs interface stack.  Allocated
         by the system. Used by PrY as the source address of
         MPPDUs and by a supporting SecY (if present) for SCI
         assignment. The PrY will receive and process MPPDUs with
         this destination address.";
      reference
        "18.1, 23.4 of IEEE Std 802.1AE";
    }
    leaf pry-mppdu-dest-address {
      type ieee:mac-address;
      description
        "The destination MAC address used by the PrY to transmit
         MPPDUs. Also used to receive MPPDUs (if a Group address)
         when  reception privacy-protection  is True. Set by the
         KaY if if-secy-support is True, otherwise writable. If
         if-secy-support transitions from True to False, defaults
         to the Nearest non-TPMR Bridge Group address.";
```


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```
      reference
        "18.1, 20.11 of IEEE Std 802.1AE
         11.1.1 IEEE Std 802.1X";
    }
    leaf max-peers {
      type uint8;
      description
        "The maximum number of peer PrYs supported by the
         configured reassembly algorithm.";
      reference
        "20.11, 20.13, 23.8 of IEEE Std 802.1AE";
    }
    leaf num-peers {
      type uint8;
      config false;
      description
        "The number of peer PrYs detected by the system. This
         value may be greater than if-max-peers a notification is
         raised when this value exceeds if-max-peers.";
      reference
        "20.13, of IEEE Std 802.1AE";
    }
    leaf-list peer-entry {
      type ieee:mac-address;
      config false;
      description
        "A list of peer PrYs. Frame Fragments received in MPPDUs
         with source MAC addresses not in this table are
         discarded.  When if-secy-support is True, table entries
         are created and deleted by the supporting Key Agreement
         Entity. When False the system automatically creates an
         entry for if-mppdu-dest-address if that is not a Group
         address, and other entries can be created by management.";
      reference
        "20.13, of IEEE Std 802.1AE";

    }
    container reception {
      description
        "Configure the MAC Privacy Reception.";
      leaf privacy-protection {
        type boolean;
        default "true";
        description
          "MACSec Privacy Reception Enable - True or False. When
           True the PrY processes received MPPDUs addressed to
           pry-address and if-mppdu-dest-address (if that is a Group
           address).  When False they are passed directly to the
           PrY's Private Port. All other MPPDUs are passed to the
           Private Port, unprocessed, irrespective of this control’s
           value.";
        reference
          "20.11 of IEEE Std 802.1AE";
      }
      leaf default-reassembly-algorithm {
        type boolean;
        config false;
        description
          "Set True by the system to indicate that the default
           reassembly algorithm is used. Set False, otherwise.  If
           the system supports additional reassembly algorithms it
           shall also support selection of the default algorithm.
           The maximum size of the user data frame (DA, SA, MSDU)
           that can be reassembled for delivery to the Private Port
           is the value of if-mtu (as provided by the IF-MIB plus 22
           octets).";
        reference
          "20.13, 20.13.1 of IEEE Std 802.1AE";
      }
      container rx-statistics {
        description
          "Configure the MAC Privacy Reception Statistics.";
```


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```
leaf in-user-frames {
  type yang:counter64;
  config false;
  description
    "Total number of protected user data frames received in
     MPPDUs, encoded as Encapsulated Frames or reassembled
     from Frame Fragments.";
  reference
    "20.14.2 of IEEE Std 802.1AE";
}
leaf in-user-octets {
  type yang:counter64;
  config false;
  description
    "Total number of user data frame octets received.
     Excludes padding.";
  reference
    "20.14.2 of IEEE Std 802.1AE";
}
leaf in-pad-octets {
  type yang:counter64;
  config false;
  description
    "Number of pad octets received in MPPDUs. This
     includes MPPDU overhead bytes.";
  reference
    "20.14.2 of IEEE Std 802.1AE";
}
leaf in-mppdus {
  type yang:counter64;
  config false;
  description
    "Total number of MAC Privacy PDUs received.";
  reference
    "20.14.2 of IEEE Std 802.1AE";
}
leaf in-encapsulated-frames {
  type yang:counter64;
  config false;
  description
    "Total number of MAC Privacy user frames received
     that were not fragmented.";
  reference
    "20.14.2 of IEEE Std 802.1AE";
}
leaf in-user-express-fragments {
  type yang:counter64;
  config false;
  description
    "Total number of correctly encoded Express Frame
     Fragments received in MPPDUs. Includes fragments
     discarded by reassembly (unknown peer, too many peers,
     out of order, reassembled frame too large).";
  reference
    "20.14.2 of IEEE Std 802.1AE";
}
leaf in-user-preemptable-fragments {
  type yang:counter64;
  config false;
  description
    "Total number of correctly encoded Preemptable Frame
     Fragments received in MPPDUs. Includes fragments
     discarded by reassembly (unknown peer, too many peers,
     out of order, reassembled frame too large).";
  reference
    "20.14.2 of IEEE Std 802.1AE";
}
leaf in-express-discard-fragments {
  type yang:counter64;
  config false;
  description
    "Number of Express Frame Fragment discard events
```


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```
            (discarding a fragment and/or a partially reassembled
            user data frame).";
          reference
            "20.14.2 of IEEE Std 802.1AE";
        }
        leaf in-preemptable-discard-fragments {
          type yang:counter64;
          config false;
          description
            "Number of Preemptable Frame Fragment discard events
             (discarding a fragment and/or a partially reassembled
             user data frame).";
          reference
            "20.14.2 of IEEE Std 802.1AE";
        }
        leaf in-unknown-mppcis {
          type yang:counter64;
          config false;
          description
            "Number of of unknown MPPDU components received.";
          reference
            "20.14.2 of IEEE Std 802.1AE";
        }
        leaf in-errored-mppdus {
          type yang:counter64;
          config false;
          description
            "Number of received MPPDUs containing an incorrectly
             encoded component.";
          reference
            "20.14.2 of IEEE Std 802.1AE";
        }
        leaf in-user-unprotected-frames {
          type yang:counter64;
          config false;
          description
            "Total number of frames with no privacy protection
             received.";
          reference
            "20.14.2 of IEEE Std 802.1AE";
        }
        leaf in-user-unprotected-octets {
          type yang:counter64;
          config false;
          description
            "Total number of octets with no privacy protection
             received.";
          reference
            "20.14.2 of IEEE Std 802.1AE";
        }
      }
    }
    container transmission {
      description
        "Configure the MAC Privacy Transmission.";
      leaf privacy-protection {
        type boolean;
        default "true";
        description
          "MACSec Privacy Enable - True or False. When True, the PrY
           protects transmitted user data frames as configured in
           the Privacy Selection list. When False, all user data
           frames are passed directly to the PrY's Controlled
           Port.";
        reference
          "20.5 of IEEE Std 802.1AE";
      }
      list privacy-selection {
        key "user-priority";
        description
          "User priority is mapped to privacy channels express or
           preemptable or to privacy frames.";
```


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```
        reference
            "17.4, 17.4.3, 20.5 of IEEE Std 802.1AE";
        leaf user-priority {
          type dot1q-types:priority-type {
            range "0..7";
          }
          description
            "Transmit request user priority. There are eight values
             of User Priority that map to ether a priority channel,
             a priority frame or to none.";
          reference
            "17.4.3, 20.5 of IEEE Std 802.1AE";
        }
        leaf privacy-type {
          type identityref {
            base priority-map-identity;
          }
          mandatory true;
          description
            "An identity associated with the privacy channel or
             frame. Privacy protection type: none, privacy-frame,
             preemptable-channel, or express-channel.";
          reference
            "17.4.3, 20.5 of IEEE Std 802.1AE";
        }
        leaf frame-access-priority {
          type dot1q-types:priority-type;
          description
            "The Controlled Port priority (access priority) used to
             transmit Privacy Frames with the Private Port
             transmission priority (user priority) that selects this
             table entry.";
          reference
            "17.4.3, 20.7 of IEEE Std 802.1AE";
        }
        leaf frame-reveal-de {
          type enumeration {
            enum hidden {
              value 0;
              description
                "Set to zero to hide (clear) drop_eligible for Privacy Frames
                 transmission.";
                }
            enum visible {
              value 1;
              description
                "Set to one to use (make visible) the drop_eligible value
                 provided by the PrY's user for Privacy Frame transmission.";
                }
          }
          default "hidden";
          description
            "frame-reveal-de allows the drop_eligible parameter accompanying
             Privacy Frame transmission to be as supplied by the PrY's user or
             hidden.";
          reference
            "17.4.1, 20.7 of IEEE Std 802.1AE";
        }
        leaf frame-padding {
          type enumeration {
            enum none {
              value 1;
              description
                "Set to none when no extra pad octets are added.";
            }
            enum to-16 {
              value 16;
              description
                "Set to 16 when padding out to the nearest 16 octet
                 boundary.";
            }
            enum to-32 {
```


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```
            value 32;
            description
              "Set to 32 when padding out to the nearest 32 octet
               boundary.";
          }
          enum to-64 {
            value 64;
            description
              "Set to 64 when padding out to the nearest 64 octet
               boundary.";
          }
        }
        default "to-64";
        description
          "Specifies padding of the Privacy Frame MPPDU (excluding
           its source and destination MAC addresses) to four
           octets (to allow for the MAC Privacy protection
           EtherType and the MPPCI for an Encapsulated Frame) plus
           the nearest multiple of one(1) (for no padding),
           sixteen(16), thirty two(32), or sixty four (64) octets.
           The specified size excludes any octets to be added by
           supporting components lower in the interface stack
           (e.g. a MACsec SecTAG and ICV, and the Ethernet FCS) or
           other bridge components (e.g. an outer VLAN tag added
           by an EDE's network component).";
        reference
          "17.4.2, 20.7 of IEEE Std 802.1AE";
      }
    }
    list channel {
      key "channel-id";
      description
        "List of Channels supported with their corresponding per
         channel configuration Note both channels are forced to be
         configured.";
      reference
        "20.13.6 of IEEE Std 802.1AE";
      leaf channel-id {
        type identityref {
          base channel-identity;
        }
        description
          "The Channel may be express or preemptable. If only one is active
           then all traffic maps to the active channel and the express
           indication bit is set.";
      }
      leaf enable {
        type boolean;
        default "false";
        description
          "When True, user data frames assigned to this Privacy
           Channel a privacy-selection are transmitted using this
           channel's parameters. When False, they are transmitted
           using the other channel if enable is True
           for that channel and transmitted as Privacy Frames
           using the relevant frame privacy-type otherwise.";
        reference
          "20.8 of IEEE Std 802.1AE";
      }
      leaf fragment-enable {
        type boolean;
        default "true";
        description
          "When True permits user data frame fragmentation in this
           Privacy Channel. Should be True, for bandwidth
           efficiency and delay minimization. Provided to allow
           simple performance testing and fragmentation benefit
           analysis.";
        reference
          "20.10 of IEEE Std 802.1AE";
      }
      leaf access-priority {
```


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```
    type dot1q-types:priority-type;
    description
      "The Controlled Port priority (access priority) used to
       transmit MPPDUs for this Privacy Channel.";
    reference
      "20.8 20.5.9.1 of IEEE Std 802.1AE";
  }
  leaf user-data-frame-size {
    type uint16 {
      range "128 .. 32768";
    }
    units "octets";
    default "1522";
    description
      "The largest user data frame, at the Private Port interface (i.e.
       prior to MAC Privacy protection) that can be transmitted as an
       MPPDU Encapsulated Frame without fragmentation.  Default allows
       for a standard Ethernet frame with a single VLAN tag. (The
       number of octets in an encapsulated frame component after the
       'following length' will be 1518 as the FCS is not encoded).  The
       user data frame size excludes octets subsequently added by
       MACsec, or other supporting interface stack components.
       Physical media, and the configuration of other system components
       can impose an upper bound lower than the configured value of
       this parameter.";
    reference
      "20.9.4 of IEEE Std 802.1AE";
  }
  leaf mppdu-generation {
    type enumeration {
      enum default {
        value 1;
        description
          "Default represents a regular timed delivery of
           a Privacy Channel based on requested-kbit-rate
           and mppdu-bits-on-wire.";
      }
      enum transmission-gate {
        value 2;
        description
         "Transmission-gate specifies transmission gate
          control of Privacy Channel MPPDU transmission.";
      }
      enum other {
        value 3;
        description
          "Optional other timing.";
      }
    }
    default "default";
    description
      "The MPPDU generation algorithm for this Privacy
       Channel. When default (fixed-rate), the (maximum)
       bandwidth is requested, with a catch up
       (burst)parameter to recover lost bandwidth if an MPPDU
       transmission has been delayed by another frame sent
       with higher access priority or by another component of
       the same interface stack. When transmission-gate, MPPDU
       transmission timing is gated.";
    reference
      "20.9, 20.9.4, 20.9.5 of IEEE Std 802.AE.
       IEEE Std 802.1Q";
  }
  leaf requested-kbit-rate {
    type uint32;
    units "kbit/s";
    mandatory true;
    description
      "The physical medium bit rate (kilobits per second) to
       be used by this Privacy Channel and the default MPPDU
       generation algorithm in the absence of higher priority
       traffic or other resource competition.";
```


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```
      reference
        "23.5 of IEEE Std 802.1AE";
    }
    leaf mppdu-bits-on-wire {
      type uint32;
      units "octets";
      config false;
      description
        "The number of bit times required to transmit an MPPDU
         that conveys a single, Private Port transmitted, user
         data frame of user-data-frame-size encoded as an
         Encapsulated Frame(19.5.1). Calculated by the system,
         including all fields added by the interface stack.";
      reference
        "20.9.4 of IEEE Std 802.1AE";
    }
    leaf mppdu-interval {
      type uint32;
      units "nanoseconds";
      config false;
      description
        "The approximate interval (as calculated by the system)
         in nanoseconds between the transmission of MPPDUs for
         this Privacy Channel, in the absence of competing
         higher priority traffic or other resource competition.";
      reference
        "20.9.4 of IEEE Std 802.1AE";
    }
    leaf user-burst-octets {
      type uint32;
      description
        "The number of additional user data frame burst for use
         by the default MPPDU generation algorithm to recover
         channel bandwidth lost to competing higher priority
         traffic.";
      reference
        "20.9.4 of IEEE Std 802.1AE";
    }
    container tx-statistics {
      description
        "Transmission statistics for a Privacy Channel.";
      leaf out-ch-user-frames {
        type yang:counter64;
        config false;
        description
          "Number of user data frames sent in this Privacy
           Channel.";
        reference
          "20.14.1 of IEEE Std 802.1AE";
      }
      leaf out-ch-user-octets {
        type yang:counter64;
        config false;
        description
          "Number of user data octets sent in this Privacy
           Channel.  Not counting pad octets.";
        reference
          "20.14.1 of IEEE Std 802.1AE";
      }
      leaf out-ch-pad-octets {
        type yang:counter64;
        config false;
        description
          "Number of pad octets sent in this Privacy Channel.
           This includes MPPDU overhead.";
        reference
          "20.14.1 of IEEE Std 802.1AE";
      }
      leaf out-mppdus {
        type yang:counter64;
        config false;
        description
```


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```
            "Number of MPPDUs sent in this Privacy Channel.";
          reference
            "20.14.1 of IEEE Std 802.1AE";
        }
        leaf out-encapsulated-frames {
          type yang:counter64;
          config false;
          description
            "Number of Encapsulated Frames encoded for this
             Privacy Channel.";
          reference
            "20.14.1 of IEEE Std 802.1AE";
        }
        leaf out-express-fragments {
          type yang:counter64;
          config false;
          description
            "Number of Express Fragments encoded for this Privacy
             Channel.";
          reference
            "20.14.1 of IEEE Std 802.1AE";
        }
        leaf out-preempt-fragments {
          type yang:counter64;
          config false;
          description
            "Number of Preemptable Fragments encoded for this
             Privacy Channel.";
          reference
            "20.14.1 of IEEE Std 802.1AE";
        }
      }
    }
    container frame-tx-statistics {
      description
        "Frame Transmission stats.";
      leaf out-pf-user-frames {
        type yang:counter64;
        config false;
        description
          "Total number of user data frames sent as Privacy Frames
           (each in a separate MPPDU).";
        reference
          "20.14.1 of IEEE Std 802.1AE";
      }
      leaf out-pf-user-octets {
        type yang:counter64;
        config false;
        description
          "Total number of user data octets sent in Privacy Frames
           (each user data frame in a separate MPPDU). Not
           counting pad octets.";
        reference
          "20.14.1 of IEEE Std 802.1AE";
      }
      leaf out-pf-pad-octets {
        type yang:counter64;
        config false;
        description
          "Total number of pad octets sent in Privacy Frames (each
           conveying a single Private Port user data frame).  This
           includes MPPDU overhead.";
        reference
          "20.14.1 of IEEE Std 802.1AE";
      }
      leaf out-unprotected-frames {
        type yang:counter64;
        config false;
        description
          "Total number of user frames sent that are not privacy
           protected. These frames are mapped to none and these
           frames are not MPPDU encapsulated.";
```


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```
            reference
              "20.14.1 of IEEE Std 802.1AE";
          }
          leaf out-unprotected-octets {
            type yang:counter64;
            config false;
            description
              "Total number of user octets sent that are not privacy
               protected. These octets are from the frames that are
               mapped to none and these frames are not MPPDU
               encapsulated.";
            reference
              "20.14.1 of IEEE Std 802.1AE";
          }
        }
      }
    }
  }
}

```


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23.11.3 ieee802-dot1x-eapol YANG module

```
module ieee802-dot1x-eapol {
  yang-version 1.1;
  namespace "urn:ieee:std:802.1X:yang:ieee802-dot1x-eapol";
  prefix eapol;

  import ietf-interfaces {
    prefix if;
  }
  import ieee802-dot1x {
    prefix dot1x;
  }
  import ieee802-types {
    prefix ieee;
  }

  organization
    "Institute of Electrical and Electronics Engineers";
  contact
    "WG-URL: http://ieee802.org/1/
     WG-EMail: stds-802-1-l@ieee.org

     Contact: IEEE 802.1 Working Group Chair
     Postal: C/O IEEE 802.1 Working Group
           IEEE Standards Association
           445 Hoes Lane
           Piscataway, NJ 08855
           USA

     E-mail: stds-802-1-chairs@ieee.org";
  description
    "Augment to be added to 802.1X on next revision:
     Missing Eapol Address for 802.1X-2020";

  revision 2022-05-25 {
    description
      "The following reference statement identifies each referenced IEEE
       Standard as updated by applicable amendments.";
    reference
      "IEEE Std 802.1X Port-Based Network Access Control:
       IEEE Std 802.1X-2020.";
  }

  augment "/if:interfaces/if:interface/dot1x:pae" {
    leaf eapol-group-address {
      type ieee:mac-address;
      description
        "The destination Group MAC Address used by this PAE
         when transmitting EAPOL frames.";
      reference
        "12.9, and Figure 12-3 of IEEE Std 802.1X";
    }
    description
      "The destination Group MAC Address augmentation for
       transmitting EAPOL frames.";
  }
}
```


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Annex A

(normative)

PICS proforma[7]

A.1 Introduction

The supplier of a protocol implementation which is claimed to conform to this standard shall complete the following Protocol Implementation Conformance Statement (PICS) proforma.

A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of which capabilities and options of the protocol have been implemented. The PICS can have a number of uses, including use

a) By the protocol implementor, as a checklist to reduce the risk of failure to conform to the standard through oversight;
b) By the supplier and acquirer—or potential acquirer—of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PICS proforma;
c) By the user—or potential user—of the implementation, as a basis for initially checking the possibility of interworking with another implementation (note that, while interworking can never be guaranteed, failure to interwork can often be predicted from incompatible PICSs);
d) By a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

A.2 Abbreviations and special symbols

A.2.1 Status symbols

M mandatory
O optional
O.n optional, but support of at least one of the group of options labelled by the same numeral is required
X prohibited
pred:conditional-item symbol, including predicate identification: see A.3.4
¬ logical negation, applied to a conditional item’s predicate

A.2.2 General abbreviations

N/A not applicable
PICSProtocol Implementation Conformance Statement

[7]*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

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A.3 Instructions for completing the PICS proforma

A.3.1 General structure of the PICS proforma

The first part of the PICS proforma, implementation identification and protocol summary, is to be completed as indicated with the information necessary to identify fully both the supplier and the implementation.

The main part of the PICS proforma is a fixed-format questionnaire, divided into several subclauses, each containing a number of individual items. Answers to the questionnaire items are to be provided in the rightmost column, either by simply marking an answer to indicate a restricted choice (usually Yes or No), or by entering a value or a set or range of values. (Note that there are some items where two or more choices from a set of possible answers can apply; all relevant choices are to be marked.)

Each item is identified by an item reference in the first column. The second column contains the question to be answered; the third column records the status of the item—whether support is mandatory, optional, or conditional; see also A.3.4 below. The fourth column contains the reference or references to the material that specifies the item in the main body of this standard, and the fifth column provides the space for the answers.

A supplier may also provide (or be required to provide) further information, categorized as either Additional Information or Exception Information. When present, each kind of further information is to be provided in a further subclause of items labelled *Ai* or *Xi,* respectively, for cross-referencing purposes, where *i* is any unambiguous identification for the item (e.g., simply a numeral). There are no other restrictions on its format and presentation.

A completed PICS proforma, including any Additional Information and Exception Information, is the Protocol Implementation Conformation Statement for the implementation in question.

NOTE—Where an implementation is capable of being configured in more than one way, a single PICS may be able to describe all such configurations. However, the supplier has the choice of providing more than one PICS, each covering some subset of the implementation's configuration capabilities, in case that makes for easier and clearer presentation of the information.

A.3.2 Additional information

Items of Additional Information allow a supplier to provide further information intended to assist the interpretation of the PICS. It is not intended or expected that a large quantity will be supplied, and a PICS can be considered complete without any such information. Examples might be an outline of the ways in which a (single) implementation can be set up to operate in a variety of environments and configurations, or information about aspects of the implementation that are outside the scope of this standard but that have a bearing upon the answers to some items.

References to items of Additional Information may be entered next to any answer in the questionnaire, and may be included in items of Exception Information.

A.3.3 Exception information

It may occasionally happen that a supplier will wish to answer an item with mandatory status (after any conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer will be found in the Support column for this: instead, the supplier shall write the missing answer into the Support column, together with an *Xi* reference to an item of Exception Information, and shall provide the appropriate rationale in the Exception item itself.

An implementation for which an Exception item is required in this way does not conform to this standard.

NOTE—A possible reason for the situation described above is that a defect in this standard has been reported, a correction for which is expected to change the requirement not met by the implementation.

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A.3.4 Conditional status

A.3.4.1 Conditional items

The PICS proforma contains a number of conditional items. These are items for which both the applicability of the item itself, and its status if it does apply—mandatory or optional—are dependent upon whether or not certain other items are supported.

Where a group of items is subject to the same condition for applicability, a separate preliminary question about the condition appears at the head of the group, with an instruction to skip to a later point in the questionnaire if the "Not Applicable" answer is selected. Otherwise, individual conditional items are indicated by a conditional symbol in the Status column.

A conditional symbol is of the form "**pred:** S" where **pred** is a predicate as described in A.3.4.2 below, and S is a status symbol, M or 0.

If the value of the predicate is True (see A.3.4.2), the conditional item is applicable, and its status is indicated by the status symbol following the predicate: the answer column is to be marked in the usual way. If the value of the predicate is False, the "Not Applicable" (N/A) answer is to be marked.

A.3.4.2 Predicates

A predicate is one of the following:

a) An item-reference for an item in the PICS proforma: the value of the predicate is True if the item is marked as supported, and is False otherwise;

b) A predicate-name, for a predicate defined as a Boolean expression constructed by combining item-references using the Boolean operator OR: the value of the predicate is True if one or more of the items is marked as supported;

c) A predicate-name, for a predicate defined as a Boolean expression constructed by combining item-references using the boolean operator AND: the value of the predicate is True if all of the items are marked as supported;

d) The logical negation symbol "¬" prefixed to an item-reference or predicate-name: the value of the predicate is True if the value of the predicate formed by omitting the "¬" symbol is False, and vice versa.

Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for grouped conditional items, is indicated by an asterisk in the Item column.

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A.4 PICS proforma for IEEE Std 802.1AE

A.4.1 Implementation identification

Supplier	
Contact point for queries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification—e.g., name(s) and version(s) of machines and/or operating system names	
NOTE 1—Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirement for full identification. NOTE 2—The terms *Name* and *Version* should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

A.4.2 Protocol summary, IEEE Std 802.1AE

Identification of protocol specification	IEEE Std 802.1AE-2018, IEEE Standard for Local and Metropolitan Area Networks: Media Access Control (MAC) Security
Identification of amendments and corrigenda to the PICS proforma which have been completed as part of the PICS	Amd. : Corr. : Amd. : Corr. :
Have any Exception items been required? (See A.3.3: the answer Yes means that the implementation does not conform to IEEE Std 802.1AE-2018.)	No [ ] Yes [ ]

Date of Statement	

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A.5 Major capabilities

Item	Feature	Status	References	Support
SAP	Does the implementation of each MAC Security Entity support the Controlled and Uncontrolled Ports, and use a Common Port as specified in Clause 10?	M	5.3(a), Clause 10, A.6	Yes [ ]
STAT	Does the implementation support the MAC status and point-to-point parameters for the Controlled and Uncontrolled Ports as specified in 6.4, 6.5, and 10.7?	M	5.3(b), 6.4, 6.5, 10.7, A.7	Yes [ ]
GEN	Does the implementation process transmit requests from the Controlled Port as required by the specification of Secure Frame Generation (10.5)?	M	5.3(c), 10.5, A.8	Yes [ ]
VER	Does the implementation process receive indications from the Common Port as required by the specification of Secure Frame Verification (10.6), prior to causing receive indications at the Controlled Port?	M	5.3(d), 10.6, A.9	Yes[ ]
FMT	Does the implementation encode and decode MACsec PDUs as specified in Clause 9?	M	5.3(e), Clause 9, A.10	Yes [ ]
SCI	Does the implementation use a 48-bit MAC Address and a 16-bit Port Identifier unique within the scope of that address assignment to identify each transmit SCI, as specified in 8.2.1?	M	5.3(f), 8.2.1	Yes [ ]
PERF	Does the implementation satisfy the performance requirements specified in Table 10-4 and 8.2.2?	M	5.3(g), 10.1, Table 10-4, 8.2.2	Yes [ ]
FCS	Does the implementation introduce an undetected frame error rate greater than that achievable by preserving the original FCS, as required by 10.4?	X	5.3(n), 10.4, 6.10	No [ ]
KAY	Does the implementation support the LMI operations required by the Key Agreement Entity as specified in Clause 10?	M	5.3(h), Clause 10, A.11	Yes [ ]
MGT	Does the implementation provide the management functionality specified in 10.7?	M	5.3(i), 10.7, A.12	Yes [ ]
MIB	Does the implementation support access to MACsec parameters by a network management using SNMPv3 and the MIB module specified in Clause 13?	O	5.3(a), Clause 13	Yes [ ] No[ ]
SNMX	Does the implementation support access to MACsec parameters using any version of SNMP prior to SNMPv3?	X	5.3(p)	No[ ]
MSC	Does the implementation support more than one receive SC?	O	5.4(b)	Yes [ ] No[ ]
MSAK	Does the implementation support more than two receive SAKs?	O	5.4(c)	Yes [ ] No[ ]
CSI	Does the implementation support Integrity Protection using the Default Cipher Suite?*	O.1	5.3(k), 5.4(e), Clause 14, 14.5	Yes [ ]
CSC	Does the implementation support Confidentiality Protection using the Default Cipher Suite without a Confidentiality Offset?	**CSI**:M	5.4(e), Clause 14, 14.5	Yes [ ]

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A.5 Major capabilities *(continued)*

Item	Feature	Status	References	Support
CSO	Does the implementation support Confidentiality Protection using the Default Cipher Suite with a Confidentiality Offset?	**CSI**:O	5.4(f), Clause 14, 14.5	Yes [ ]
ACSI	Does the implementation support Integrity Protection using the Ascon-XPN-128 Cipher Suite?[a]	O.1		
ACSC	Does the implementation support Confidentiality Protection using the Ascon-XPN-128 Cipher Suite?	**ACSC**:O		
CSA	Does the implementation include additional Cipher Suites specified in Clause 14?[a]	O	5.4(g), A.16	Yes [ ] No[ ]
CSX	Does the implementation include any Cipher Suite that is additional to those specified in Clause 14 and does not meet all the criteria specified in 14.2, 14.3, 14.4.1?	X	5.3(o), 14.2, 14.3, 14.4.1	No[ ]
CSV	Does the implementation include Cipher Suites other than those specified in Clause 14, but meeting the criteria specified in 14.2, 14.3, 14.4.1?†	O	5.4(i), A.17	Yes [ ] No[ ]
CSR	Does the implementation support a minimum of one receive SC, two receive SAKs, one transmit SC, and one of the two receive SAKs at a time for transmission as specified in 5.3(l), for each Cipher Suite implemented?	M	5.3(l), Clause 14	Yes [ ]
CSS	Does this completed PICS specify the maximum number of receive SCs, receive SAKS, and transmit SCs for each Cipher Suite implemented?	M	5.3(m), A.16, A.17	Yes [ ]
TC	Does the implementation support more than one transmit SC for any Cipher Suite?	O	5.4(d)	Yes [ ] No [ ]
TCT	Is a Traffic Class Table implemented?	**TC**:M	5.4(i), 10.7.17	Yes [ ] N/A[ ]
TCAPT	Is an Access Priority Table implemented?	**TC**:M	5.4(i), 10.7.17,	Yes [ ] N/A[ ]
FULL	Is a claim for full conformance being made for the implementation?	**CSV**:X ¬**CSV**:O	5.3	Yes [ ] No[ ]
VAR	Is a claim for conformance with cipher suite variance being made for the implementation?		5.3	Yes [ ] No[ ]

[*] This PICS requires the completion of a copy of Table A.16 for each Clause 14 Cipher Suite implemented.
[†] This PICS requires the completion of a copy of Table A.17 for each variant Cipher Suite implemented.

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A.6 Support and use of Service Access Points

Item	Feature	Status	References	Support
SAP-1	Does each transmit request from the Uncontrolled Port result in a single request to the Common Port with the same parameters?	M	10.4	Yes [ ]
SAP-2	Does each receive indication from the Common Port result in a single indication to the Uncontrolled Port with the same parameters if any of the users of the Common Port wishes to receive the indication?	M	10.4	Yes [ ]
SAP-3	Does each transmit request from the Controlled Port result in at most one request to the Common Port?	M	10.4	Yes [ ]
SAP-4	Does each receive indication from the Common Port result in at most one indication to the Controlled Port?	M	10.4	Yes [ ]
SAP-5	Are any transmit requests made to the Common Port that do not correspond to requests made at the Uncontrolled or Controlled Port?	X	10.4	No [ ]
SAP-6	Are any receive indications caused at the Uncontrolled or Controlled Port that do not correspond to indications from the Common Port?	X	10.4	No [ ]
SAP-7	Is the order of requests made at the Common Port unchanged from the order of corresponding requests from the Uncontrolled Port?	M	10.4	Yes [ ]
SAP-8	Is the order of requests made at the Common Port unchanged from the order of corresponding requests from the Controlled Port?	M	10.4	Yes [ ]
SAP-9	Is the order of receive indications caused at the Uncontrolled Port the same as the order of reception from the Common Port?	M	10.4	Yes [ ]
SAP-10	Is each transmit request from the Controlled Port processed in accordance with the specification of the Secure Frame Generation process, prior to discarding the request or making a corresponding request to the Common Port?	M	10.4, 10.5	Yes [ ]
SAP-11	Is each receive indication from the Common Port processed in accordance with the specification of the Secure Frame Verification process prior to causing a possible corresponding indication at the Controlled Port?	M	10.4, 10.6	Yes [ ]

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A.7 MAC status and point-to-point parameters

Item	Feature	Status	References	Support
STAT-1	Are the values for MAC_Operational and operPointToPointMAC for the Uncontrolled Port identical to those for the Common Port?	M	6.4, 10.7.2	Yes [ ]
STAT-2	Is MAC_Operational False for the Controlled Port, and frames neither accepted or delivered on the port, if the SA identified by the encodingSA is not available for use and protectFrames is set?	M	6.4, 10.5.1, 7.1	Yes [ ]
STAT-3	Is MAC_Operational False for the Controlled Port and frames neither accepted nor delivered, if the nextPN for the encodingSA is zero or 2^{32}?	M	6.4, 10.5.2	Yes [ ]
STAT-4	Is MAC_Operational True only if MAC_Enabled is True and MAC_Operational for the Common Port is True?	M	6.4, 10.7.4	Yes [ ]
STAT-5	Is the value of operPointToPointMAC for the Controlled Port always as specified in 10.7.4.	M	6.5, 10.7.4	Yes [ ]

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A.8 Secure Frame Generation

Item	Feature	Status	References	Support
GEN-1	Does each transmit request from the Controlled Port result in an identical transmit request at the Common Port if the management control protectFrames is False?	M	10.5	Yes [ ]
GEN-2	Does each transmit request at the Common Port resulting from a request at the Common Port convey request parameters, i.e., a frame, protected in accordance with Clause 10.5 if the management control protectFrames is True?	M	10.5	Yes[ ]
GEN-3	Is each protected frame assigned to the SA with AN corresponding to the current value of encodingSA as specified by the KaY?	M	10.5.1	Yes[ ]
GEN-4	Are frames to be protected discarded if the assigned SA cannot be used?	M	10.5.1	Yes[ ]
GEN-5	Is the PN value of zero used?	X	10.5.2	No [ ]
GEN-6	Following assignment of a PN to a protected frame, is the next frame to be protected for the same SA assigned the next higher value of PN?	M	10.5.2	Yes[ ]
GEN-7	Is the SecTAG encoded as specified in Clause 9?	M	10.5.3, Clause 9	
GEN-8	Is the ES bit set or clear as required by the management controls useES and alwaysIncludeSCI?	M	10.5.3	
GEN-9	Is the SC bit set or clear and the SCI explicitly encoded or not as required by the management controls useES, use SCB, alwaysIncludeSCI, and by the number of receive SCs?	M	10.5.3	
GEN-10	Is the SCB bit set or clear as required by the management controls useSCB and alwaysIncludeSCI?	M	10.5.3	
GEN-11	Is the E bit set if the frame is confidentiality protected, and clear otherwise?	M	9.5	
GEN-12	Is the C bit set if the octets of the Secure Data differ from those of the User Data or the ICV is not 16 octets, and clear otherwise?	M	9.5	
GEN-13	Is each frame transmitted from the Controlled Port protected using a Cipher Suite as specified in Clause 14 if protectFrames is set?	M	10.5	
GEN-14	Is OutOctetsEncrypted incremented by the number of octets in the User Data if confidentiality protections is provided, and OutOctetsProtected incremented otherwise?	M	10.5.4	
GEN-15	Is the protected frame transmitted if the MACsec PDU (SecTAG, Secure Data, and ICV) does not exceed the maximum data unit size supported by the Common Port and discarded otherwise?	M	10.5.5	

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A.9 Secure Frame Verification

Item	Feature	Status	References	Support
VER-1	For each receive indication, does the Secure Frame Verification process examine the user data for a SecTAG and validate frames with a SecTAG as specified in 9.12, extracting and decoding the SecTAG as specified in 9.3 through 9.9, and extracting the User Data and ICV as specified in 9.10 and 9.11?	M	10.6, 9.3 through 9.9, 9.10, 9.11, 9.12	Yes[ ]
VER-2	Is a received frame without a SecTAG delivered to the Controlled Port if validateFrames is not Strict, and discarded otherwise?	M	10.6	Yes[ ]
VER-3	Is a received frame with the SecTAG E bit set and C bit clear discarded and not delivered to the Controlled Port?	M	10.6	Yes[ ]
VER-4	Is the received frame discarded if the SC is unknown and validateFrames is Strict or the C bit is set, and delivered to the Controlled Port otherwise?	M	10.6.1	Yes[ ]
VER-5	Is the received frame discarded if the SA is unused and validateFrames is Strict or the C bit is set, and delivered to the Controlled Port otherwise?	M	10.6.1	Yes[ ]
VER-6	Is the received frame discarded if the PN is less than the lowest acceptable packet number for the SA and replayProtect is enabled?	M	10.6.2, 10.6.4	Yes[ ]
VER-7	Is the InPktsOverrun counter incremented if a received frame is discarded for reasons not attributed to the data conveyed?	M	10.6.3	Yes[ ]
VER-8	If validateFrames is Disabled, is Cipher Suite validation omitted and a received frame delivered to the Controlled Port if the C bit is not set?	M	10.6.3, 10.6.5	Yes[ ]
VER-9	If validateFrames is not Disabled is the Cipher Suite used to validated the received frame?	M	10.6.3	Yes[ ]
VER-10	Are frames that are not successfully validated discarded if validateFrames is Strict or the C bit is set?	M	10.6.5	Yes[ ]
VER-11	Are the values for the next expected and lowest acceptable PN updated as specified in 10.6.5 following receipt of a MACsec PDU successfully validate by the Cipher Suite, and not modified by received frames otherwise?	M	10.6.5	Yes[ ]
VER-12	Are received frames not discarded by Secure Frame Verification delivered to the Controlled Port after removal of a SecTAG and ICV?	M	10.6	Yes[ ]
VER-13	Are all received frames delivered to Controlled Port unmodified if validateFrames is Null?	M	10.6	Yes[ ]
VER-14	Is protectFrames set False if validateFrames is set to Null?	M	10.6	Yes[ ]

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A.10 MACsec PDU encoding and decoding

Item	Feature	Status	References	Support
FMT-1	Does each MACsec PDU transmitted contain an integral number of octets?	M	9.1	Yes[ ]
FMT-2	Does each MACsec PDU transmitted comprise a SecTAG, formatted as specified in Clause 9, one or more octets of Secure Data, and an ICV of the length specified by the Cipher Suite in use?	M	9.1, 9.2, 9.3, Figure 9-1, 10.5.3	Yes[ ]
FMT-3	Is the EtherType encoded in the SecTAG the value specified in Table 9-1?	M	9.3, 9.4	Yes[ ]
FMT-4	Is the version number in the SecTAG encoded as zero?	M	9.5	Yes[ ]
FMT-5	Is the SC bit clear and the SCI not explicitly encoded if the ES bit is set?	M	9.5	Yes[ ]
FMT-6	Is the SC bit set if an SCI is explicitly encoded and clear otherwise?	M	9.5	Yes[ ]
FMT-7	Is the SC bit clear if the SCB bit is set?	M	9.5	Yes[ ]
FMT-8	Are bits 7 and 8 of octet 4 of the SecTAG zero?	M	9.7	Yes[ ]
FMT-9	Is each received MACsec PDU validated as specified in 9.12.	M	9.5	Yes[ ]

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A.11 Key Agreement Entity LMI

Item	Feature	Status	References	Support
KAY-1	Does the implementation allow the KaY to read the values of the MAC_Enabled, MAC_Operational, and operPointToPointMAC parameters?	M	10.7.2	Yes[ ]
KAY-2	Does the implementation allow the KaY to set and clear the ControlledPortEnabled parameter, acting on the parameter as specified?	M	10.7.4, 10.7.5	Yes[ ]
KAY-3	Does the implementation allow the KaY to discover which Cipher Suites are implemented and how many receive SCs each can support?	M	10.2, 10.7.7, 10.7.16, 10.7.25	Yes[ ]
KAY-4	Does the implementation allow the KaY to create a receive SC?	M	10.6.1, 10.7.11	Yes[ ]
KAY-5	Does the implementation allow the KaY to create receive SAs as specified in 10.7.13?	M	10.7.13	Yes[ ]
KAY-6	Does the implementation allow the KaY to control the use of each receive SA and to update the values of the next expected PN and lowest acceptable PN as specified in 10.7.15?	M	10.7.15	Yes[ ]
KAY-7	Does the implementation allow the KaY to create transmit SAs as specified in 10.7.22?	M	10.7.22, 10.5.2	Yes[ ]
KAY-8	Does the implementation allow the KaY to control the use of each transmit SA as specified in 10.7.24?	M	10.7.24, 10.5.1, 10.5.2	Yes[ ]
KAY-9	Does the implementation allow the KaY to monitor the nextPN associated with each transmit SA in order to create a new SA with a fresh SAK prior to PN exhaustion?	M	10.7.2	Yes[ ]
KAY-10	Does the implementation allow the KaY to select the Current Cipher Suite as specified in 10.7.27?	M	10.7.27	Yes[ ]
KAY-11	Does the implementation allow the KaY to create and control an SAK as specified in 10.7.26 and 10.7.28?	M	10.7.26, 10.7.28	Yes[ ]

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A.12 Management—control and status information

Item	Feature	Status	References	Support
Can each of the following parameter values be read by management?				
MGT1-1	The SCI for the SecY	M	10.7.1	Yes[ ]
MGT1-2	MAC_Enabled, MAC_Operational, and operPointToPointMAC for the Uncontrolled Port	M	10.7.2	Yes[ ]
MGT1-3	MAC_Enabled, MAC_Operational, and operPointToPointMAC for the Controlled Port	M	10.7.4	Yes[ ]
MGT1-4	The maximum number of receive SCs and SAKs that can be in simultaneous use	M	10.7.7	Yes[ ]
MGT1-5	validateFrames, replayProtect, and replayWindow	M	10.7.8	Yes[ ]
MGT1-6	The SCI, receiving, createdTime, startedTime, and stoppedTime for each receive SC	M	10.7.12	Yes[ ]
MGT1-7	inUse, nextPN, lowestPN, createdTime, startedTime, stoppedTime, and Key Identifier for each receive SA	M	10.7.14	Yes[ ]
MGT1-8	The maximum number of SAKs that can be in simultaneous use for transmission	M	10.7.16	Yes[ ]
MGT1-9	protectFrames, useES, useSCB, and alwaysIncludeSCI	M	10.7.17	Yes[ ]
MGT1-10	transmitting, createdTime, startedTime, and stoppedTime for the transmit SC	M	10.7.21	Yes[ ]
MGT1-11	inUse, nextPN, lowestPN, createdTime, startedTime, stoppedTime, and Key Identifier for each transmit SA	M	10.7.23	Yes[ ]
MGT1-12	The currentCipherSuite identifier and the confidentialityOffset for frames with confidentiality protection	M	10.7.27	Yes[ ]
MGT1-13	transmits, receives, and createdTime for each SAK	M	10.7.29	Yes[ ]
MGT1-14	Can the management information for each implemented Cipher Suite be read?	M	10.7.25	Yes[ ]

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A.13 Management—basic controls

Item	Feature	Status	References	Support
Can the following parameters be written by management, independently for each Controlled Port?				
MGT2-1	validateFrames	O	10.7.8, 10.6	Yes[ ] No [ ]
MGT2-2	replayProtect	O	10.7.8, 10.6.2, 10.6.4	Yes[ ] No [ ]
MGT2-3	replayWindow	O	10.7.8, 10.6.5	Yes[ ] No [ ]
MGT2-4	protectFrames	O	10.7.17, 10.5	Yes[ ] No [ ]
MGT2-5	useES	O	10.7.17, 10.5.3	Yes[ ] No [ ]
MGT2-6	useSCB	O	10.7.17, 10.5.3	Yes[ ] No [ ]
MGT2-7	alwaysIncludeSCI	O	10.7.17, 10.5.3	Yes[ ] No [ ]
Can the following parameters be written by management, independently for each Controlled Port for each CipherSuite implemented ?				
MGT2-15	enableUse	O	10.7.26	Yes[ ] No [ ]
MGT2-16	requireConfidentiality	O	10.7.26	Yes[ ] No [ ]
Can write access by management to each of the following parameters be disabled individually?				
MGT2-8	validateFrames	MGT2-1:M	10.7.8	Yes[ ]
MGT2-9	replayProtect	MGT2-2:M	10.7.8	Yes[ ]
MGT2-10	replayWindow	MGT2-3:M	10.7.8	Yes[ ]
MGT2-11	protectFrames	MGT2-4:M	10.7.17	Yes[ ]
MGT2-12	useES	MGT2-5:M	10.7.17	Yes[ ]
MGT2-13	useSCB	MGT2-6:M	10.7.17	Yes[ ]
MGT2-14	alwaysIncludeSCI	MGT2-7:M	10.7.17	Yes[ ]
Can write access by management to each of the following CipherSuite use parameters be disabled individually for each Controlled Port?				
MGT2-17	enableUse	MGT2-15:M	10.7.26	Yes[ ]
MGT2-18	requireConfidentiality	MGT2-16:M	10.7.26	Yes[ ]

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A.14 Management—control over secure communication

Item	Feature	Status	References	Support
Can the following be created, controlled, or selected by management?				
MGT3-1	Receive SCs and SAs	O	10.7.11, 10.7.13, 10.7.15	Yes[ ] No [ ]
MGT3-2	Transmit SAs	O	10.7.22, 10.7.24	Yes[ ] No [ ]
MGT3-3	The current CipherSuite	O	10.7.27	Yes[ ] No [ ]
MGT3-4	confidentialityOffset	O	10.7.27	Yes[ ] No [ ]
MGT3-5	SAKs	O	10.7.28, 10.7.29	Yes[ ] No [ ]
Can creation, control, or selection by management of the following be disabled individually?				
MGT3-1	Receive SCs and SAs	MGT3-1:M	10.7.11	Yes[ ]
MGT3-2	Transmit SAs	MGT3-2:M	10.7.22, 10.7.24	Yes[ ]
MGT3-3	The current CipherSuite	MGT3-3:M	10.7.27	Yes[ ]
MGT3-4	confidentialityOffset	MGT3-4:M	10.7.27	Yes[ ]
MGT3-5	SAKs	MGT3-5:M	10.7.27	Yes[ ]

A.15 Management—statistics

Item	Feature	Status	References	Support
Are each of the following interface statistics provided for the Controlled Port as specified in 10.7.6?				
MGT4-1	ifInOctets	M	10.7.6	Yes[ ]
MGT4-2	ifInUcastPkts, ifInMulticastPkts, ifInBroadcastPkts	M	10.7.6	Yes[ ]
MGT4-3	ifInDiscards	M	10.7.6	Yes[ ]
MGT4-4	ifInErrors	M	10.7.6	Yes[ ]
MGT4-5	ifOutOctets	M	10.7.6	Yes[ ]
MGT4-6	ifOutUcastPkts, ifOutMulticastPkts, ifOutBroadcastPkts	M	10.7.6	Yes[ ]
MGT4-7	ifOutErrors	M	10.7.6	Yes[ ]
Are each of the following frame verification statistics recorded as specified in 10.6 and maintained for the frame verification process as a whole?				
MGT4-8	InPktsUntagged	M	10.7.9, 10.6 Figure 10-4	Yes[ ]
MGT4-9	InPktsNoTag	M	10.7.9, 10.6 Figure 10-4	Yes[ ]
MGT4-10	InPktsBadTag	M	10.7.9, 10.6 Figure 10-4	Yes[ ]
MGT4-11	InPktsNoSARcv	M	10.7.9, 10.6.1	Yes[ ]
MGT4-12	InPktsNoSADiscard	M	10.7.9, 10.6.1	Yes[ ]

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A.15 Management—statistics *(continued)*

Item	Feature	Status	References	Support
MGT4-13	InPktsOverrun	M	10.7.9, 10.6.3	Yes[ ]
Are each of the following frame verification statistics recorded as specified in 10.6 and maintained for each receive SC?				
MGT4-14	InPktsUnchecked	M	10.7.9, 10.6.5	Yes[ ]
MGT4-15	InPktsDelayed	M	10.7.9, 10.6.5	Yes[ ]
MGT4-16	InPktsLate	M	10.7.9, 10.6.2, 10.6.4	Yes[ ]
MGT4-17	InPktsOK	M	10.7.9, 10.6.5	Yes[ ]
MGT4-18	InPktsInvalid	M	10.7.9, 10.6.5	Yes[ ]
MGT4-19	InPktsNotValid	M	10.7.9, 10.6.5	Yes[ ]
Are each of the following frame validation statistics recorded as specified in 10.7?				
MGT4-22	InOctetsValidated	M	10.7.10	Yes[ ]
MGT4-23	InOctetsDecrypted	M	10.7.10	Yes[ ]
Are each of the following frame generation statistics recorded as specified in 10.5 and maintained for the frame verification process as a whole?				
MGT4-24	OutPktsUntagged	M	10.7.18, 10.5	Yes[ ]
MGT4-25	OutPktsTooLong	M	10.7.18, 10.5.5, Figure 10-3	Yes[ ]
Are each of the following frame generation statistics recorded as specified in 10.5 and maintained for each transmit SC?				
MGT4-26	OutPktsProtected	M	10.7.18, 10.5.4	Yes[ ]
MGT4-27	OutPktsEncrypted	M	10.7.18, 10.5.4	Yes[ ]
Are each of the following frame protection statistics recorded as specified in 10.7?				
MGT4-28	OutOctetsProtected	M	10.7.19	Yes[ ]
MGT4-29	OutOctetsEncrypted	M	10.7.19	Yes[ ]

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A.16 Fully conformant Cipher Suite capabilities

Item	Feature	Status	References	Support
CSA-1	Name of Cipher Suite as specified in Clause 14. _			
CSA-2	Does the Cipher Suite implementation provide integrity without confidentiality?	M	14.2(a)	Yes[ ] No [ ]
CSA-3	Does the Cipher Suite implementation provide confidentiality for all the octets of the User Data?	O	14.2(d), 14.3(c)	Yes[ ] No [ ]
CSA-5	What is the maximum number of receive SCs supported by the Cipher Suite implementation? _ _ _ _ _		5.3(m)	
CSA-6	What is the maximum number of receive SAKs supported by the Cipher Suite implementation? _ _ _ _ _		5.3(m)	
CSA-7	What is the maximum number of transmit SCs supported by the Cipher Suite implementation? _ _ _ _ _		5.3(m)	

A.17 Additional variant Cipher Suite capabilities

Item	Feature	Status	References	Support
CSV-1	Name of Cipher Suite or other commonly used identification (to be supplied) _			
CSV-2	Identify the specification(s) of the Cipher Suite, including any additional information necessary to acquire the specification(s) (supply items of Additional Information if necessary) _ _ _ _	M	14.3	
CSV-3	Does the specification include interoperable protection and verification procedures specified in terms of the parameters of 14.1?	M	14.3, 14.1	Yes [ ]
CSV-4	Does the specification state: Whether confidentiality of the User Data is provided? The maximum difference in the lengths of the User Data and Secure Data? The length of the ICV? The length and properties of the keys required, including assumptions of the scope and uniqueness?	M	 14.3(a) 14.3(b) 14.3(c) 14.3(d)	 Yes [ ] Yes [ ] Yes [ ] Yes [ ]
CSV-5	Do the Cipher Suite algorithms have an effective key length of at least 128 bits, and does any block cipher used have a block width of at least 128 bits?	M	14.4.1(a)	Yes[ ]

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A.17 Additional variant Cipher Suite capabilities *(continued)*

Item	Feature	Status	References	Support
CSV-6	If serviced by separate algorithms, the properties of the authentication and confidentiality mechanisms are combinable in accordance with well-established security results?	M	14.4.1(b)	Yes[ ]
CSV-7a	Is the underlying cryptographic cipher approved by either a national or international standards body or a government agency?	O.1	14.4.1(c)(1)	Yes[ ] No[ ]
CSV-7b	Does the additional Cipher Suite meet the conditions expressed in 14.4.1(c)(2)?	O.1	14.4.1(c)(2)	Yes[ ] No[ ]
CSV-8	Does the Cipher Suite satisfy the message authentication requirements of 14.4.1? Identify the proof of security, including any additional information necessary to acquire the proof _ _	CSV-7b:M	14.4.1(c)(2)(i)	Yes [ ]
CSV-9	Does the Cipher Suite satisfy the confidentiality requirements of 14.4.1? Identify the proof of security, including any additional information necessary to acquire the proof _ _	CSV-7b:M	14.4.1(c)(2)(ii)	Yes [ ]
CSV-10	Does the Cipher Suite use mechanisms for confidentiality and authentication in a way that is consistent with the proofs of security?	CSV-7b:M	14.4.1(c)(2)(iii), 14.4.1(c)(2)(iv),	Yes[ ]
CSV-11	Does the Cipher Suite provide integrity protection for the SCI, PN, Source Address, Destination Address, SecTAG, and User Data?	M	14.2(a)	Yes[ ]
CSV-12	Does the Cipher Suite provide protection for at least 2^{32}–1 invocations without requiring a fresh SAK?	M	14.2(b)	Yes[ ]
CSV-13	Does the Cipher Suite generate a predictable number of octets of Secure Data and ICV given any specific number of octets of User Data?	M	14.2(c)	Yes[ ]
CSV-14	Does the maximum difference in length of the User Data and the Secure Data plus ICV exceed 896 octets?	X	14.2(f)	Yes[ ]
CSV-15	What is the maximum difference in length of the User Data and the Secure Data? _ _ _ _ _ octets		14.3(b)	
CSV-16	What is the length of the ICV _ _ _ _ _ octets		14.3(e)	
CSV-17	Does the specification specify the length and properties of the keys required, including assumptions of the scope of uniqueness?	M	14.3(f)	Yes[ ]
CSV-18	Does the Cipher Suite implementation provide confidentiality for all the octets of the User Data?	¬CSV-19: O CSV-19:M	14.2(d), 14.3(c)	Yes[ ] No [ ]

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A.17 Additional variant Cipher Suite capabilities *(continued)*

Item	Feature	Status	References	Support
CSV-19	Does the Cipher Suite implementation provide offset confidentiality for the User Data?	O	14.2(e), 14.3(c)	Yes[ ] No [ ]
CSV-20	Does the Cipher Suite modify or constrain the values of the SCI, PN, Source Address, Destination Address, or SecTAG fields other than as specified in Clause 14?	X	14.2(g)	No [ ]
CSV-21	Does the Cipher Suite require an SAK exceeding 1024 bits long?	X	14.2(h)	No [ ]
CSV-22	Does the Cipher Suite require different keys for the protect and validate operations?	X	14.2(i)	No [ ]
CSV-23	What is the maximum number of receive SCs supported by the Cipher Suite implementation? _ _ _ _ _		5.3(m)	
CSV-24	What is the maximum number of receive SAKs supported by the Cipher Suite implementation? _ _ _ _ _		5.3(m)	
CSV-25	What is the maximum number of transmit SCs supported by the Cipher Suite implementation? _ _ _ _ _		5.3(m)	

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Annex B

(informative)

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Annex C

(informative)

GCM Cipher Suite test vectors

This annex provides test case examples of the use of MACsec. Each example shows an unprotected frame that could be transmitted as a result of a MAC Service request (with a given set of parameters) and the corresponding MACsec protected frame (with a given set of MACsec SecY parameters). Test cases include the use of integrity protection without confidentiality (authenticated, but unencrypted) and the use of both integrity protection and confidentiality (authenticated and encrypted).

The test cases use a number of different unprotected frame sizes. Two correspond to common sizes of internet packets, 54 octets and 60 octets—two common representations of a TCP/IP SYN packet. A TCP SYN comprises 40 octets plus 14 octets of MAC DA+SA+EtherType. The frame could be padded to 60 octets to meet minimum Ethernet frame length requirements prior to MACsec processing. The remaining frame sizes represent “corner cases” of the GCM padding algorithm. A 61-octet frame, when encrypted, has a 49-octet payload, which results in the maximum 15 octets of padding for ICV calculation. When integrity protection is provided but confidentiality is not (i.e., when the user data is not encrypted) a 65-octet frame also requires that maximum padding. A 75-octet frame has a 63 octet payload, requiring 1 octet of padding for ICV calculation, as does a 79-octet frame that is integrity protected without confidentiality. The zero-octet padding case is covered by the 60-octet frame, above. MACsec processing is performed above the media-dependent functions of media access control, so all frame sizes given are prior to the addition of the 32-bit CRC or other media dependent fields.

Test cases are provided for the Default Cipher Suite (GCM-AES-128, 14.5), GCM-AES-256 (14.6), GCM-AES-XPN-256 (14.7), GCM-AES-XPN-256 (14.8). The notation used in this annex is that specified in Clause 14 (Cipher Suites) and NIST SP 800-38D. Fields in the MACsec header are specified in Clause 9. Summaries of the computation and intermediate outputs are provided.

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C.1 Integrity protection (54-octet frame)

The MAC Destination Address, MAC Source Address, and MAC Service Data Unit (MSDU, User Data) of a MAC Service data request and a corresponding data indication are shown in Table C-1. These comprise the octets of an unprotected frame when concatenated in the order given (with the addition of any media dependent additional fields such as padding). The User Data shown includes the IP EtherType.

Table C-1—Unprotected frame (example)

Field	Value
MAC DA	D6 09 B1 F0 56 63
MAC SA	7A 0D 46 DF 99 8D
User Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 00 01

The MAC Security TAG (SecTAG) comprises the MACsec EtherType, the TCI, the AN, the SL, the PN (32 least significant bits for Cipher Suites using extended packet numbering), and the (optional) SCI. The PN differs for each protected frame transmitted with any given SAK (*K*) and has been arbitrarily chosen (for this and in other examples) as have the other parameter values. The fields of the protected frame are shown (in the order transmitted) in Table C-2.

Table C-2—Integrity protected frame (example)

Field	Value
MAC DA	D6 09 B1 F0 56 63
MAC SA	7A 0D 46 DF 99 8D
MACsec EtherType	88 E5
TCI and AN	22
SL	2A
PN	B2 C2 84 65
SCI	12 15 35 24 C0 89 5E 81
Secure Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 00 01
ICV	Cipher Suite and Key (SAK) dependent (see Table C-3, Table C-4, Table C-5, and Table C-6)

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C.1.1 GCM-AES-128 (54-octet frame integrity protection)

Table C-3 specifies an arbitrary 128-bit key (SAK) and the ICV generated by the GCM-AES-128 Cipher Suite when that key is used in conjunction with the frame field data of Table C-2. The GCM parameter A, the additional data to be authenticated, is formed by concatenating the MAC DA, the MAC SA, the SecTAG, and the User Data. This input is then processed through the authentication only operation of the GCM module. The SCI and the PN are concatenated (in that order) to form the 96-bit IV used by GCM. The computed GCM parameter T is the ICV. Details of the computation follow the table.

Table C-3—GCM-AES-128 Key and calculated ICV (example)

Field	Value
Key (SAK)	AD7A2BD03EAC835A6F620FDCB506B345
ICV	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

```
key size = 128 bits
P:     0 bits
A:     560 bits
IV:    96 bits
ICV:   128 bits
K:     AD7A2BD03EAC835A6F620FDCB506B345
P:
A:     D609B1F056637A0D46DF998D88E5222A
       B2C2846512153524C0895E8108000F10
       1112131415161718191A1B1C1D1E1F20
       2122232425262728292A2B2C2D2E2F30
       313233340001
IV:    12153524C0895E81B2C28465
GCM-AES Authentication
H:     73A23D80121DE2D5A850253FCF43120E
Y[0]:  12153524C0895E81B2C2846500000001
E(K,Y[0]): EB4E051CB548A6B5490F6F11A27CB7D0
X[1]:  6B0BE68D67C6EE03EF7998E399C01CA4
X[2]:  5AABADF6D7806EC0CCCB028441197B22
X[3]:  FE072BFE2811A68AD7FDB0687192D293
X[4]:  A47252D1A7E09B49FB356E435DBB4CD0
X[5]:  18EBF4C65CE89BF69EFB4981CEE13DB9
GHASH(H,A,C): 1BDA7DB505D8A165264986A703A6920D
C:
T:     F09478A9B09007D06F46E9B6A1DA25DD
```


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C.1.2 GCM-AES-256 (54-octet frame integrity protection)

Table C-4 specifies an arbitrary 256-bit key (SAK) and the ICV generated by the GCM-AES-256 Cipher Suite when that key is used in conjunction with the frame field data of Table C-2. The GCM parameter A, the additional data to be authenticated, is formed by concatenating the MAC DA, the MAC SA, the SecTAG, and the User Data. This input is then processed through the authentication only operation of the GCM module. The SCI and the PN are concatenated (in that order) to form the 96-bit IV used by GCM. The computed GCM parameter T is the ICV. Details of the computation follow the table.

Table C-4—GCM-AES-256 Key and calculated ICV (example)

Field	Value
Key (SAK)	E3C08A8F06C6E3AD95A70557B23F7548 3CE33021A9C72B7025666204C69C0B72
ICV	2F 0B C5 AF 40 9E 06 D6 09 EA 8B 7D 0F A5 EA 50

```
key size = 256 bits
P:     0 bits
A:     560 bits
IV:    96 bits
ICV:   128 bits
K:     E3C08A8F06C6E3AD95A70557B23F7548
       3CE33021A9C72B7025666204C69C0B72
P:
A:     D609B1F056637A0D46DF998D88E5222A
       B2C2846512153524C0895E8108000F10
       111213141516171819 1A1B1C1D1E1F20
       212223242526272829 2A2B2C2D2E2F30
       313233340001
IV:    12153524C0895E81B2C28465
GCM-AES Authentication
H:     286D73994EA0BA3CFD1F52BF06A8ACF2
Y[0]:  12153524C0895E81B2C2846500000001
E(K,Y[0]): 714D54FDCFCEE37D5729CDDAB383A016
X[1]:  BA7C26F578254853CF321281A48317CA
X[2]:  2D0DF59AE78E84ED64C3F85068CD9863
X[3]:  702DE0382ABF4D42DD62B8F115124219
X[4]: DAED65979342F0D155BFDFE362132078
X[5]: 9AB4AFD6344654B2CD23977E41AA18B3
GHASH(H,A,C): 5E4691528F50E5AB5EC346A7BC264A46
C:
T: 2F0BC5AF409E06D609EA8B7D0FA5EA50
```


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C.1.3 GCM-AES-XPN-128 (54-octet frame integrity protection)

Table C-5 specifies an arbitrary value for the SSCI, the 32 most significant bits of the 64-bit PN (the 32 least significant bits are those of the PN field in the SecTAG), a 96-bit Salt, and 128-bit key (SAK), with the ICV generated by the GCM-AES-XPN-128 Cipher Suite when that key is used in conjunction with the foregoing and the frame field data of Table C-2. The GCM parameter *A*, the additional data to be authenticated, is formed by concatenating the MAC DA, the MAC SA, the SecTAG, and the User Data. This input is then processed through the authentication only operation of the GCM module. The 32 most significant bits of the 96-bit *IV* are the octets of the SSCI, encoded as a binary number (9.1) and exclusive-or'd with the 32 most significant bits of the Salt. The 64 least significant bits of the 96-bit *IV* are the octets of the PN, encoded as a binary number (9.1) and exclusive-or'd with the 64 least significant bits of the Salt. The computed GCM parameter *T* is the ICV. Details of the computation follow the table.

Table C-5—GCM-AES-XPN-128 Key and calculated ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	AD7A2BD03EAC835A6F620FDCB506B345
ICV	17 FE 19 81 EB DD 4A FC 50 62 69 7E 8B AA 0C 23

```
key size = 128 bits
P:     0 bits
A:     560 bits
IV:    96 bits
ICV:   128 bits
K:     AD7A2BD03EAC835A6F620FDCB506B345
P:
A:     D609B1F056637A0D46DF998D88E5222A
       B2C2846512153524C0895E8108000F10
       1112131415161718191A1B1C1D1E1F20
       2122232425262728292A2B2C2D2E2F30
       313233340001
IV:    9C002902F801C33EAEA47E08
GCM-AES Authentication
H:     73A23D80121DE2D5A850253FCF43120E
Y[0]:  9C002902F801C33EAEA47E0800000001
E(K,Y[0]): 0C246434EE05EB99762BEFD9880C9E2E
X[1]:  6B0BE68D67C6EE03EF7998E399C01CA4
X[2]:  5AABADF6D7806EC0CCCB028441197B22
X[3]:  FE072BFE2811A68AD7FDB0687192D293
X[4]:  A47252D1A7E09B49FB356E435DBB4CD0
X[5]:  18EBF4C65CE89BF69EFB4981CEE13DB9
GHASH(H,A,C): 1BDA7DB505D8A165264986A703A6920D
C:
T:     17FE1981EBDD4AFC5062697E8BAA0C23
```


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C.1.4 GCM-AES-XPN-256 (54-octet frame integrity protection)

Table C-6 specifies an arbitrary value for the SSCI, the 32 most significant bits of the 64-bit PN (the 32 least significant bits are those of the PN field in the SecTAG), a 96-bit Salt, and 256-bit key (SAK), with the ICV generated by the GCM-AES-XPN-256 Cipher Suite when that key is used in conjunction with the foregoing and the frame field data of Table C-2. The GCM parameter *A*, the additional data to be authenticated, is formed by concatenating the MAC DA, the MAC SA, the SecTAG, and the User Data. This input is then processed through the authentication only operation of the GCM module. The 32 most significant bits of the 96-bit *IV* are the octets of the SSCI, encoded as a binary number (9.1) and exclusive-or'd with the 32 most significant bits of the Salt. The 64 least significant bits of the 96-bit *IV* are the octets of the PN, encoded as a binary number (9.1) and exclusive-or'd with the 64 least significant bits of the Salt. The computed GCM parameter *T* is the ICV. Details of the computation follow the table.

Table C-6—GCM-AES-XPN-256 Key and calculated ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	E3C08A8F06C6E3AD95A70557B23F7548 3CE33021A9C72B7025666204C69C0B72
ICV	4D BD 2F 6A 75 4A 6C F7 28 CC 12 9B A6 93 15 77

```
key size = 256 bits
P:     0 bits
A:     560 bits
IV:    96 bits
ICV:   128 bits
K:     E3C08A8F06C6E3AD95A70557B23F7548
       3CE33021A9C72B7025666204C69C0B72
P:
A:     D609B1F056637A0D46DF998D88E5222A
       B2C2846512153524C0895E8108000F10
       1112131415161718191A1B1C1D1E1F20
       2122232425262728292A2B2C2D2E2F30
       313233340001
IV:    9C002902F801C33EAEA47E08
GCM-AES Authentication
H:     286D73994EA0BA3CFD1F52BF06A8ACF2
Y[0]:  9C002902F801C33EAEA47E0800000001
E(K,Y[0]): 13FBBE38FA1A895C760F543C1AB55F31
X[1]:  BA7C26F578254853CF321281A48317CA
X[2]:  2D0DF59AE78E84ED64C3F85068CD9863
X[3]:  702DE0382ABF4D42DD62B8F115124219
X[4]:  DAED65979342F0D155BFDFE362132078
X[5]:  9AB4AFD6344654B2CD23977E41AA18B3
GHASH(H,A,C): 5E4691528F50E5AB5EC346A7BC264A46
C:
T:     4DBD2F6A754A6CF728CC129BA6931577
```


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C.2 Integrity protection (60-octet frame)

The MAC Destination Address, MAC Source Address, and MAC Service Data Unit (MSDU, User Data) of a MAC Service data request and a corresponding data indication are shown in Table C-7. These comprise the octets of an unprotected frame when concatenated in the order given (with the addition of any media dependent additional fields such as padding). The User Data shown includes the IP EtherType.

Table C-7—Unprotected frame (example)

Field	Value
MAC DA	E2 01 06 D7 CD 0D
MAC SA	F0 76 1E 8D CD 3D
User Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 00 03

The MAC Security TAG (SecTAG) comprises the MACsec EtherType, the TCI, the AN, the SL, and the PN. In this example the optional SCI has been omitted. The fields of the protected frame are shown (in the order transmitted) in Table C-8.

Table C-8—Integrity protected frame (example)

Field	Value
MAC DA	E2 01 06 D7 CD 0D
MAC SA	F0 76 1E 8D CD 3D
MACsec EtherType	88 E5
TCI and AN	40
SL	00
PN	76 D4 57 ED
Secure Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 00 03
ICV	Cipher Suite and Key (SAK) dependent (see Table C-9, Table C-10, Table C-11, and Table C-12)

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C.2.1 GCM-AES-128 (60-octet frame integrity protection)

Table C-9 specifies an arbitrary 128-bit key (SAK) and the ICV generated by the GCM-AES-128 Cipher Suite when that key is used in conjunction with the frame field data of Table C-8. Details of the computation follow the table.

Table C-9—GCM-AES-128 Key and calculated ICV (example)

Field	Value
Key (SAK)	071B113B0CA743FECCCF3D051F737382
ICV	0C 01 7B C7 3B 22 7D FC C9 BA FA 1C 41 AC C3 53

```
key size = 128 bits
P:     0 bits
A:     544 bits
IV:    96 bits
ICV:   128 bits
K:     071B113B0CA743FECCCF3D051F737382
P:
A:     E20106D7CD0DF0761E8DCD3D88E54000
       76D457ED08000F101112131415161718
       191A1B1C1D1E1F202122232425262728
       292A2B2C2D2E2F303132333435363738
       393A0003
IV:    F0761E8DCD3D000176D457ED
GCM-AES Authentication
H:     E4E01725D724C1215C7309AD34539257
Y[0]:  F0761E8DCD3D000176D457ED00000001
E(K,Y[0]): FC25539100959B80FE3ABED435E54CAB
X[1]:  8DAD4981E33493018BB8482F69E4478C
X[2]:  5B0BFA3E67A3E080CB60EA3D523C734A
X[3]:  051F8D267A68CF88748E56C5F64EF503
X[4]:  4187F1240DB1887F2A92DDAB8903A0F6
X[5]:  C7D64941A90F02FA9FCDECC083B4B276
GHASH(H,A,C): F02428563BB7E67C378044C874498FF8
C:
T:     0C017BC73B227DFCC9BAFA1C41ACC353
```


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C.2.2 GCM-AES-256 (60-octet frame integrity protection)

Table C-10 specifies an arbitrary 256-bit key (SAK) and the ICV generated by the GCM-AES-256 Cipher Suite when that key is used in conjunction with the frame field data of Table C-8. Details of the computation follow the table.

Table C-10—GCM-AES-256 Key and calculated ICV (example)

Field	Value
Key (SAK)	691D3EE909D7F54167FD1CA0B5D76908 1F2BDE1AEE655FDBAB80BD5295AE6BE7
ICV	35 21 7C 77 4B BC 31 B6 31 66 BC F9 D4 AB ED 07

```
key size = 256 bits
P:     0 bits
A:     544 bits
IV:    96 bits
ICV:   128 bits
K:     691D3EE909D7F54167FD1CA0B5D76908
       1F2BDE1AEE655FDBAB80BD5295AE6BE7
P:
A:     E20106D7CD0DF0761E8DCD3D88E54000
       76D457ED08000F101112131415161718
       191A1B1C1D1E1F202122232425262728
       292A2B2C2D2E2F303132333435363738
       393A0003
IV:    F0761E8DCD3D000176D457ED
GCM-AES Authentication
H:     1E693C484AB894B26669BC12E6D5D776
Y[0]:  F0761E8DCD3D000176D457ED00000001
E(K,Y[0]): 87E183649AE3E7DBF725659152C39A22
X[1]:  20107B262134C35B60499E905C532004
X[2]:  D7A468F455F09F947884E35A2C80CD7F
X[3]:  A82D607070F2E4470FD94C0EECA9FCC1
X[4]: 03C3C8725883EB355963BD53B515C82D
X[5]: 8FF6F0311DDE274FFA936965C0C905B4
GHASH(H,A,C): B2C0FF13D15FD66DC643D96886687725
C:
T: 35217C774BBC31B63166BCF9D4ABED07
```


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C.2.3 GCM-AES-XPN-128 (60-octet frame integrity protection)

Table C-11 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN (the 32 least significant bits are those of the PN field in the SecTAG), 96-bit Salt, and 128-bit key (SAK), with the ICV generated by the GCM-AES-XPN-128 Cipher Suite when that key is used in conjunction with the foregoing and the frame field data of Table C-8.

Table C-11—GCM-AES-XPN-128 Key and calculated ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	071B113B0CA743FECCCF3D051F737382
ICV	AB C4 06 85 A3 CF 91 1D 37 87 E4 9D B6 A7 26 5E

```
key size = 128 bits
P:     0 bits
A:     544 bits
IV:    96 bits
ICV:   128 bits
K:     071B113B0CA743FECCCF3D051F737382
P:
A:     E20106D7CD0DF0761E8DCD3D88E54000
       76D457ED08000F101112131415161718
       191A1B1C1D1E1F202122232425262728
       292A2B2C2D2E2F303132333435363738
       393A0003
IV:    9C002902F801C33E6AB2AD80
GCM-AES Authentication
H:     E4E01725D724C1215C7309AD34539257
Y[0]:  9C002902F801C33E6AB2AD8000000001
E(K,Y[0]): 5BE02ED3987877610007A055C2EEA9A6
X[1]:  8DAD4981E33493018BB8482F69E4478C
X[2]:  5B0BFA3E67A3E080CB60EA3D523C734A
X[3]:  051F8D267A68CF88748E56C5F64EF503
X[4]:  4187F1240DB1887F2A92DDAB8903A0F6
X[5]:  C7D64941A90F02FA9FCDECC083B4B276
GHASH(H,A,C): F02428563BB7E67C378044C874498FF8
C:
T:     ABC40685A3CF911D3787E49DB6A7265E
```


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C.2.4 GCM-AES-XPN-256 (60-octet frame integrity protection)

Table C-12 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN (the 32 least significant bits are those of the PN field in the SecTAG), a 96-bit Salt, and 256-bit key (SAK), with the ICV generated by the GCM-AES-XPN-256 Cipher Suite when that key is used in conjunction with the foregoing and the frame field data of Table C-8.

Table C-12—GCM-AES-XPN-256 Key and calculated ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	691D3EE909D7F54167FD1CA0B5D76908 1F2BDE1AEE655FDBAB80BD5295AE6BE7
ICV	AC 21 95 7B 83 12 AB 3C 99 AB 46 84 98 79 C3 F3

```
key size = 256 bits
P:     0 bits
A:     544 bits
IV:    96 bits
ICV:   128 bits
K:     691D3EE909D7F54167FD1CA0B5D76908
       1F2BDE1AEE655FDBAB80BD5295AE6BE7
P:
A:     E20106D7CD0DF0761E8DCD3D88E54000
       76D457ED08000F101112131415161718
       191A1B1C1D1E1F202122232425262728
       292A2B2C2D2E2F303132333435363738
       393A0003
IV:    9C002902F801C33E6AB2AD80
GCM-AES Authentication
H:     1E693C484AB894B26669BC12E6D5D776
Y[0]:  9C002902F801C33E6AB2AD8000000001
E(K,Y[0]): 1EE16A68524D7D515FE89FEC1E11B4D6
X[1]:  20107B262134C35B60499E905C532004
X[2]:  D7A468F455F09F947884E35A2C80CD7F
X[3]:  A82D607070F2E4470FD94C0EECA9FCC1
X[4]:  03C3C8725883EB355963BD53B515C82D
X[5]:  8FF6F0311DDE274FFA936965C0C905B4
GHASH(H,A,C): B2C0FF13D15FD66DC643D96886687725
C:
T: AC21957B8312AB3C99AB46849879C3F3
```


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C.3 Integrity protection (65-octet frame)

The MAC Destination Address, MAC Source Address, and MAC Service Data Unit (MSDU, User Data) of a MAC Service data request and a corresponding data indication are shown in Table C-13. These comprise the octets of an unprotected frame when concatenated in the order given (with the addition of any media dependent additional fields such as padding). The User Data shown includes the IP EtherType.

Table C-13—Unprotected frame (example)

Field	Value
MAC DA	84 C5 D5 13 D2 AA
MAC SA	F6 E5 BB D2 72 77
User Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 3B 3C 3D 3E 3F 00 05

The MAC Security TAG (SecTAG) comprises the MACsec EtherType, the TCI, the AN, the SL, the PN, and the (optional) SCI. The fields of the protected frame are shown (in the order transmitted) in Table C-14.

Table C-14—Integrity protected frame (example)

Field	Value
MAC DA	84 C5 D5 13 D2 AA
MAC SA	F6 E5 BB D2 72 77
MACsec EtherType	88 E5
TCI and AN	23
SL	00
PN	89 32 D6 12
SCI	7C FD E9 F9 E3 37 24 C6
Secure Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 3B 3C 3D 3E 3F 00 05
ICV	(see Table C-15, Table C-16, Table C-17, and Table C-18)

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C.3.1 GCM-AES-128 (65-octet frame integrity protection)

Table C-15 specifies an arbitrary 128-bit key (SAK) and the ICV generated by the GCM-AES-128 Cipher Suite when that key is used in conjunction with the frame field data of Table C-14. Details of the computation follow the table.

Table C-15—GCM-AES-128 Key and calculated ICV (example)

Field	Value
Key (SAK)	013FE00B5F11BE7F866D0CBBC55A7A90
ICV	21 78 67 E5 0C 2D AD 74 C2 8C 3B 50 AB DF 69 5A

```
key size = 128 bits
P:     0 bits
A:     648 bits
IV:    96 bits
ICV:   128 bits
K:     013FE00B5F11BE7F866D0CBBC55A7A90
P:
A:     84C5D513D2AAF6E5BBD2727788E52300
       8932D6127CFDE9F9E33724C608000F10
       1112131415161718191A1B1C1D1E1F20
       2122232425262728292A2B2C2D2E2F30
       3132333435363738393A3B3C3D3E3F00
       05
IV:    7CFDE9F9E33724C68932D612
GCM-AES Authentication
H:     EB28DCB361EE1110F98CA0C9A07C88F7
Y[0]:  7CFDE9F9E33724C68932D61200000001
E(K,Y[0]): 4EAAF8E4DF948ACAC7F3349C1006A91F
X[1]:  279344E391DB8834EFA68FD3F1BA5CD8
X[2]:  DC35B123F4D387BBB076D0822BD60816
X[3]:  8AB3B52963CC15C9C2DB3E4C801CB65A
X[4]:  CAB6A261225F42578E6B86ABA9F0DD18
X[5]:  6ABDBB3ECAC0458F116A82AA0DAC563F
X[6]:  8F39EF45985C691E35814202B6BB6EF6
GHASH(H,A,C): 6FD29F01D3B927BE057F0FCCBBD9C045
C:
T:     217867E50C2DAD74C28C3B50ABDF695A
```


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C.3.2 GCM-AES-256 (65-octet frame integrity protection)

Table C-16 specifies an arbitrary 256-bit key (SAK) and the ICV generated by the GCM-AES-256 Cipher Suite when that key is used in conjunction with the frame field data of Table C-14. Details of the computation follow the table.

Table C-16—GCM-AES-256 Key and calculated ICV (example)

Field	Value
Key (SAK)	83C093B58DE7FFE1C0DA926AC43FB360 9AC1C80FEE1B624497EF942E2F79A823
ICV	6E E1 60 E8 FA EC A4 B3 6C 86 B2 34 92 0C A9 75

```
key size = 256 bits
P:     0 bits
A:     648 bits
IV:    96 bits
ICV:   128 bits
K:     83C093B58DE7FFE1C0DA926AC43FB360
       9AC1C80FEE1B624497EF942E2F79A823
P:
A:     84C5D513D2AAF6E5BBD2727788E52300
       8932D6127CFDE9F9E33724C608000F10
       1112131415161718191A1B1C1D1E1F20
       2122232425262728292A2B2C2D2E2F30
       3132333435363738393A3B3C3D3E3F00
       05
IV:    7CFDE9F9E33724C68932D612
GCM-AES Authentication
H:     D03D3B51FDF2AACB3A165D7DC362D929
Y[0]:  7CFDE9F9E33724C68932D61200000001
E(K,Y[0]): E97EA8EE4455AE79EC4225CAC340E326
X[1]:  22C28F4DF8D09267EA3E11F019F5932C
X[2]:  3D02CFE5FC6A8A9E65B8FFD63E525083
X[3]:  78466AE4A3490819A08645DDC95B143B
X[4]: 6FE4921A6F0A1D5DD90A100A40206142
X[5]: C880DEC2FF2C44F8AD611692AF6D1069
X[6]:  CF4D709A4D020BA876F4371BAA788444
GHASH(H,A,C): 879FC806BEB90ACA80C497FE514C4A53
C:
T: 6EE160E8FAECA4B36C86B234920CA975
```


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C.3.3 GCM-AES-XPN-128 (65-octet frame integrity protection)

Table C-17 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN (the 32 least significant bits are those of the PN field in the SecTAG), 96-bit Salt, and 128-bit key (SAK), with the ICV generated by the GCM-AES-XPN-128 Cipher Suite when that key is used in conjunction with the foregoing and the frame field data of Table C-14.

Table C-17—GCM-AES-XPN-128 Key and calculated ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	013FE00B5F11BE7F866D0CBBC55A7A90
ICV	67 85 59 B7 E5 2D B0 06 82 E3 B8 30 34 CE BE 59

```
key size = 128 bits
P:     0 bits
A:     648 bits
IV:    96 bits
ICV:   128 bits
K:     013FE00B5F11BE7F866D0CBBC55A7A90
P:
A:     84C5D513D2AAF6E5BBD2727788E52300
       8932D6127CFDE9F9E33724C608000F10
       1112131415161718191A1B1C1D1E1F20
       2122232425262728292A2B2C2D2E2F30
       3132333435363738393A3B3C3D3E3F00
       05
IV:    9C002902F801C33E95542C7F
GCM-AES Authentication
H:     EB28DCB361EE1110F98CA0C9A07C88F7
Y[0]:  9C002902F801C33E95542C7F00000001
E(K,Y[0]): 0857C6B6369497B8879CB7FC8F177E1C
X[1]:  279344E391DB8834EFA68FD3F1BA5CD8
X[2]:  DC35B123F4D387BBB076D0822BD60816
X[3]:  8AB3B52963CC15C9C2DB3E4C801CB65A
X[4]:  CAB6A261225F42578E6B86ABA9F0DD18
X[5]:  6ABDBB3ECAC0458F116A82AA0DAC563F
X[6]:  8F39EF45985C691E35814202B6BB6EF6
GHASH(H,A,C): 6FD29F01D3B927BE057F0FCCBBD9C045
C:
T:     678559B7E52DB00682E3B83034CEBE59
```


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C.3.4 GCM-AES-XPN-256 (65-octet frame integrity protection)

Table C-18 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN (the 32 least significant bits are those of the PN field in the SecTAG), a 96-bit Salt, and 256-bit key (SAK), with the ICV generated by the GCM-AES-XPN-256 Cipher Suite when that key is used in conjunction with the foregoing and the frame field data of Table C-14.

Table C-18—GCM-AES-XPN-256 Key and calculated ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	83C093B58DE7FFE1C0DA926AC43FB360 9AC1C80FEE1B624497EF942E2F79A823
ICV	84 BA C8 E5 3D 1E A3 55 A5 C7 D3 34 84 0A E9 62

```
key size = 256 bits
P:     0 bits
A:     648 bits
IV:    96 bits
ICV:   128 bits
K:     83C093B58DE7FFE1C0DA926AC43FB360
       9AC1C80FEE1B624497EF942E2F79A823
P:
A:     84C5D513D2AAF6E5BBD2727788E52300
       8932D6127CFDE9F9E33724C608000F10
       1112131415161718191A1B1C1D1E1F20
       2122232425262728292A2B2C2D2E2F30
       3132333435363738393A3B3C3D3E3F00
       05
IV:    9C002902F801C33E95542C7F
GCM-AES Authentication
H:     D03D3B51FDF2AACB3A165D7DC362D929
Y[0]:  9C002902F801C33E95542C7F00000001
E(K,Y[0]): 032500E383A7A99F250344CAD546A331
X[1]:  22C28F4DF8D09267EA3E11F019F5932C
X[2]:  3D02CFE5FC6A8A9E65B8FFD63E525083
X[3]:  78466AE4A3490819A08645DDC95B143B
X[4]:  6FE4921A6F0A1D5DD90A100A40206142
X[5]:  C880DEC2FF2C44F8AD611692AF6D1069
X[6]:  CF4D709A4D020BA876F4371BAA788444
GHASH(H,A,C): 879FC806BEB90ACA80C497FE514C4A53
C:
T: 84BAC8E53D1EA355A5C7D334840AE962
```


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C.4 Integrity protection (79-octet frame)

The MAC Destination Address, MAC Source Address, and MAC Service Data Unit (MSDU, User Data) of a MAC Service data request and a corresponding data indication are shown in Table C-19. These comprise the octets of an unprotected frame when concatenated in the order given (with the addition of any media dependent additional fields such as padding). The User Data shown includes the IP EtherType.

Table C-19—Unprotected frame (example)

Field	Value
MAC DA	68 F2 E7 76 96 CE
MAC SA	7A E8 E2 CA 4E C5
User Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 3B 3C 3D 3E 3F 40 41 42 43 44 45 46 47 48 49 4A 4B 4C 4D 00 07

The MAC Security TAG (SecTAG) comprises the MACsec EtherType, the TCI, the AN, the SL, and the PN. In this example the optional SCI has been omitted. The fields of the protected frame are shown (in the order transmitted) in Table C-20.

Table C-20—Integrity protected frame (example)

Field	Value
MAC DA	68 F2 E7 76 96 CE
MAC SA	7A E8 E2 CA 4E C5
MACsec EtherType	88 E5
TCI and AN	41
SL	00
PN	2E 58 49 5C
Secure Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 3B 3C 3D 3E 3F 40 41 42 43 44 45 46 47 48 49 4A 4B 4C 4D 00 07
ICV	(see Table C-21, Table C-22, Table C-23, and Table C-24)

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C.4.1 GCM-AES-128 (79-octet frame integrity protection)

Table C-21 specifies an arbitrary 128-bit key (SAK) and the ICV generated by the GCM-AES-128 Cipher Suite when that key is used in conjunction with the frame field data of Table C-20. Details of the computation follow the table.

Table C-21—GCM-AES-128 Key and calculated ICV (example)

Field	Value
Key (SAK)	88EE087FD95DA9FBF6725AA9D757B0CD
ICV	07 92 2B 8E BC F1 0B B2 29 75 88 CA 4C 61 45 23

```
key size = 128 bits
P:     0 bits
A:     696 bits
IV:    96 bits
ICV:   128 bits
K:     88EE087FD95DA9FBF6725AA9D757B0CD
P:
A:     68F2E77696CE7AE8E2CA4EC588E54100
       2E58495C08000F101112131415161718
       191A1B1C1D1E1F202122232425262728
       292A2B2C2D2E2F303132333435363738
       393A3B3C3D3E3F404142434445464748
       494A4B4C4D0007
IV:    7AE8E2CA4EC500012E58495C
GCM-AES Authentication
H:     AE19118C3B704FCE42AE0D15D2C15C7A
Y[0]:  7AE8E2CA4EC500012E58495C00000001
E(K,Y[0]): D2521AABC48C06033E112424D4A6DF74
X[1]:  CA0CAE2BEE8F19845DCB7FE3C5E713AB
X[2]:  5D3F9C7A3BC869457EA5FDFD404A415F
X[3]:  760E6A2873ACC0515D4901B5AC1C85E4
X[4]:  5A40A8425165E3D1978484F07AFC70D8
X[5]:  D9687630FC4436EE582A90A8E4AFC504
X[6]:  311CE361065F86403CDA5DB00798B961
GHASH(H,A,C): D5C03125787D0DB11764ACEE98C79A57
C:
T:     07922B8EBCF10BB2297588CA4C614523
```


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C.4.2 GCM-AES-256 (79-octet frame integrity protection)

Table C-22 specifies an arbitrary 256-bit key (SAK) and the ICV generated by the GCM-AES-256 Cipher Suite when that key is used in conjunction with the frame field data of Table C-20. Details of the computation follow the table.

Table C-22—GCM-AES-256 Key and calculated ICV (example)

Field	Value
Key (SAK)	4C973DBC7364621674F8B5B89E5C1551 1FCED9216490FB1C1A2CAA0FFE0407E5
ICV	00 BD A1 B7 E8 76 08 BC BF 47 0F 12 15 7F 4C 07

```
key size = 256 bits
P:     0 bits
A:     696 bits
IV:    96 bits
ICV:   128 bits
K:     4C973DBC7364621674F8B5B89E5C1551
       1FCED9216490FB1C1A2CAA0FFE0407E5
P:
A:     68F2E77696CE7AE8E2CA4EC588E54100
       2E58495C08000F101112131415161718
       191A1B1C1D1E1F202122232425262728
       292A2B2C2D2E2F303132333435363738
       393A3B3C3D3E3F404142434445464748
       494A4B4C4D0007
IV:    7AE8E2CA4EC500012E58495C
GCM-AES Authentication
H:     9A5E559A96459C21E43C0DFF0FA426F3
Y[0]:  7AE8E2CA4EC500012E58495C00000001
E(K,Y[0]): 316F5EDB0829AC9271A6AFF79F3600BF
X[1]:  06A9019B44B76FFEC18978E8B21513E2
X[2]:  89A6401E39EAB6EE5B8159570139F54D
X[3]:  0A5E22BA54F282CE464C334D1AF598EF
X[4]: 4514D8A5C15E15CABC3D2A0E24FC758E
X[5]: 6F98DE3369B88F25AACBF3A993003E78
X[6]:  8183B21C0A932A2D5F598E1B2967564B
GHASH(H,A,C): 31D2FF6CE05FA42ECEE1A0E58A494CB8
C:
T: 00BDA1B7E87608BCBF470F12157F4C07
```


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C.4.3 GCM-AES-XPN-128 (79-octet frame integrity protection)

Table C-23 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN (the 32 least significant bits are those of the PN field in the SecTAG), 96-bit Salt, and 128-bit key (SAK), with the ICV generated by the GCM-AES-XPN-128 Cipher Suite when that key is used in conjunction with the foregoing and the frame field data of Table C-20.

Table C-23—GCM-AES-XPN-128 Key and calculated ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	88EE087FD95DA9FBF6725AA9D757B0CD
ICV	D0 DC 89 6D C8 37 98 A7 9F 3C 5A 95 BA 3C DF 9A

```
key size = 128 bits
P:     0 bits
A:     696 bits
IV:    96 bits
ICV:   128 bits
K:     88EE087FD95DA9FBF6725AA9D757B0CD
P:
A:     68F2E77696CE7AE8E2CA4EC588E54100
       2E58495C08000F101112131415161718
       191A1B1C1D1E1F202122232425262728
       292A2B2C2D2E2F303132333435363738
       393A3B3C3D3E3F404142434445464748
       494A4B4C4D0007
IV:    9C002902F801C33E323EB331
GCM-AES Authentication
H:     AE19118C3B704FCE42AE0D15D2C15C7A
Y[0]:  9C002902F801C33E323EB33100000001
E(K,Y[0]): 051CB848B04A95168858F67B22FB45CD
X[1]:  CA0CAE2BEE8F19845DCB7FE3C5E713AB
X[2]:  5D3F9C7A3BC869457EA5FDFD404A415F
X[3]:  760E6A2873ACC0515D4901B5AC1C85E4
X[4]:  5A40A8425165E3D1978484F07AFC70D8
X[5]:  D9687630FC4436EE582A90A8E4AFC504
X[6]:  311CE361065F86403CDA5DB00798B961
GHASH(H,A,C): D5C03125787D0DB11764ACEE98C79A57
C:
T:     D0DC896DC83798A79F3C5A95BA3CDF9A
```


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C.4.4 GCM-AES-XPN-256 (79-octet frame integrity protection)

Table C-24 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN (the 32 least significant bits are those of the PN field in the SecTAG), a 96-bit Salt, and 256-bit key (SAK), with the ICV generated by the GCM-AES-XPN-256 Cipher Suite when that key is used in conjunction with the foregoing and the frame field data of Table C-20.

Table C-24—GCM-AES-XPN-256 Key and calculated ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	4C973DBC7364621674F8B5B89E5C1551 1FCED9216490FB1C1A2CAA0FFE0407E5
ICV	04 24 9A 20 8A 65 B9 6B 3F 32 63 00 4C FD 86 7D

```
key size = 256 bits
P:     0 bits
A:     696 bits
IV:    96 bits
ICV:   128 bits
K:     4C973DBC7364621674F8B5B89E5C1551
       1FCED9216490FB1C1A2CAA0FFE0407E5
P:
A:     68F2E77696CE7AE8E2CA4EC588E54100
       2E58495C08000F101112131415161718
       191A1B1C1D1E1F202122232425262728
       292A2B2C2D2E2F303132333435363738
       393A3B3C3D3E3F404142434445464748
       494A4B4C4D0007
IV:    9C002902F801C33E323EB331
GCM-AES Authentication
H:     9A5E559A96459C21E43C0DFF0FA426F3
Y[0]:  9C002902F801C33E323EB33100000001
E(K,Y[0]): 35F6654C6A3A1D45F1D3C3E5C6B4CAC5
X[1]:  06A9019B44B76FFEC18978E8B21513E2
X[2]:  89A6401E39EAB6EE5B8159570139F54D
X[3]:  0A5E22BA54F282CE464C334D1AF598EF
X[4]:  4514D8A5C15E15CABC3D2A0E24FC758E
X[5]:  6F98DE3369B88F25AACBF3A993003E78
X[6]:  8183B21C0A932A2D5F598E1B2967564B
GHASH(H,A,C): 31D2FF6CE05FA42ECEE1A0E58A494CB8
C:
T: 04249A208A65B96B3F3263004CFD867D
```


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C.5 Confidentiality protection (54-octet frame)

The MAC Destination Address, MAC Source Address, and MAC Service Data Unit (MSDU, User Data) of a MAC Service data request and a corresponding data indication are shown in Table C-25. These comprise the octets of an unprotected frame when concatenated in the order given (with the addition of any media dependent additional fields such as padding). The User Data shown includes the IP EtherType.

Table C-25—Unprotected frame (example)

Field	Value
MAC DA	E2 01 06 D7 CD 0D
MAC SA	F0 76 1E 8D CD 3D
User Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 00 04

The MAC Security TAG (SecTAG) comprises the MACsec EtherType, the TCI, the AN, the SL, and the PN. In this example the optional SCI has been omitted. The fields of the protected frame are shown (in the order transmitted) in Table C-26.

Table C-26—Confidentiality protected frame (example)

Field	Value
MAC DA	E2 01 06 D7 CD 0D
MAC SA	F0 76 1E 8D CD 3D
MACsec EtherType	88 E5
TCI and AN	4C
SL	2A
PN	76 D4 57 ED
Secure Data	Cipher Suite and Key (SAK) dependent (see Table C-27, Table C-28, Table C-29, and Table C-30)
ICV	Cipher Suite and Key (SAK) dependent (see Table C-27, Table C-28, Table C-29, and Table C-30)

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C.5.1 GCM-AES-128 (54-octet frame confidentiality protection)

Table C-27 specifies an arbitrary 128-bit key (SAK), the Secure Data, and the ICV generated by the GCM-AES-128 Cipher Suite when that key is used in conjunction with the frame field data of Table C-25 and Table C-26. The GCM parameter P, the data to be encrypted, is the User Data. The additional data A to be authenticated is formed by concatenating the MAC DA, the MAC SA, and the SecTAG. The SCI and the PN are concatenated (in that order) to form the 96-bit IV used by GCM. The computed GCM parameter T is the ICV. Details of the computation follow the table.

Table C-27—GCM-AES-128 Key, Secure Data, and ICV (example)

Field	Value
Key (SAK)	071B113B0CA743FECCCF3D051F737382
Secure Data	13 B4 C7 2B 38 9D C5 01 8E 72 A1 71 DD 85 A5 D3 75 22 74 D3 A0 19 FB CA ED 09 A4 25 CD 9B 2E 1C 9B 72 EE E7 C9 DE 7D 52 B3 F3
ICV	D6 A5 28 4F 4A 6D 3F E2 2A 5D 6C 2B 96 04 94 C3

```
key size = 128 bits
P:     336 bits
A:     160 bits
IV:    96 bits
ICV:   128 bits
K:     071B113B0CA743FECCCF3D051F737382
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F30313233340004
A:     E20106D7CD0DF0761E8DCD3D88E54C2A
       76D457ED
IV:    F0761E8DCD3D000176D457ED
GCM-AES Encryption
H:             E4E01725D724C1215C7309AD34539257
Y[0]:          F0761E8DCD3D000176D457ED00000001
E(K,Y[0]):     FC25539100959B80FE3ABED435E54CAB
Y[1]:          F0761E8DCD3D000176D457ED00000002
E(K,Y[1]):     1BB4C83B298FD6159B64B669C49FBECF
C[1]:          13B4C72B389DC5018E72A171DD85A5D3
Y[2]:          F0761E8DCD3D000176D457ED00000003
E(K,Y[2]):     683C6BF3813BD8EEC82F830DE4B10530
C[2]:          752274D3A019FBCAED09A425CD9B2E1C
Y[3]:          F0761E8DCD3D000176D457ED00000004
E(K,Y[3]):     B65CC1D7F8EC4E66B3F7182C2E358591
C[3]:          9B72EEE7C9DE7D52B3F3
X[1]:          A0AE6DFAE25C0AE80E9A1AAC0D5123D3
X[2]:          EAEA2A767986B7D5B9E6ED37A3CBC63B
X[3]:          8809F1263C02DC9BD09FDF0F34575BA6
X[4]:          A173C5A2C03DE08C025C93945B2E74B7
X[5]:          65D113682551614E556BFAA80AA2FA7A
GHASH(H,A,C): 2A807BDE4AF8A462D467D2FFA3E1D868
C:     13B4C72B389DC5018E72A171DD85A5D3
       752274D3A019FBCAED09A425CD9B2E1C
       9B72EEE7C9DE7D52B3F3
T:     D6A5284F4A6D3FE22A5D6C2B960494C3
```


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C.5.2 GCM-AES-256 (54-octet frame confidentiality protection)

Table C-28 specifies an arbitrary 256-bit key (SAK), the Secure Data, and the ICV generated by the GCM-AES-256 Cipher Suite when that key is used in conjunction with the frame field data of Table C-25 and Table C-26. The GCM parameter P, the data to be encrypted, is the User Data. The additional data A to be authenticated is formed by concatenating the MAC DA, the MAC SA, and the SecTAG. The SCI and the PN are concatenated (in that order) to form the 96-bit IV used by GCM. Details of the computation follow the table.

Table C-28—GCM-AES-256 Key, Secure Data, and ICV (example)

Field	Value
Key (SAK)	691D3EE909D7F54167FD1CA0B5D76908 1F2BDE1AEE655FDBAB80BD5295AE6BE7
Secure Data	C1 62 3F 55 73 0C 93 53 30 97 AD DA D2 56 64 96 61 25 35 2B 43 AD AC BD 61 C5 EF 3A C9 0B 5B EE 92 9C E4 63 0E A7 9F 6C E5 19
ICV	12 AF 39 C2 D1 FD C2 05 1F 8B 7B 3C 9D 39 7E F2

```
key size = 128 bits
P:     336 bits
A:     160 bits
IV:    96 bits
ICV:   128 bits
K:     691D3EE909D7F54167FD1CA0B5D76908
       1F2BDE1AEE655FDBAB80BD5295AE6BE7
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F30313233340004
A:     E20106D7CD0DF0761E8DCD3D88E54C2A
       76D457ED
IV:    F0761E8DCD3D000176D457ED
GCM-AES Encryption
H:             1E693C484AB894B26669BC12E6D5D776
Y[0]:          F0761E8DCD3D000176D457ED00000001
E(K,Y[0]):     87E183649AE3E7DBF725659152C39A22
Y[1]:          F0761E8DCD3D000176D457ED00000002
E(K,Y[1]):     C9623045621E80472581BAC2CB4C7F8A
C[1]:          C1623F55730C93533097ADDAD2566496
Y[2]:          F0761E8DCD3D000176D457ED00000003
E(K,Y[2]):     7C3B2A0B628F8F9944E3C812E02170C2
C[2]:          6125352B43ADACBD61C5EF3AC90B5BEE
Y[3]:          F0761E8DCD3D000176D457ED00000004
E(K,Y[3]):     BFB2CB533F95AC58E51D6608DBEBDBC2
C[3]:          929CE4630EA79F6CE519
X[1]:          F268EF5B38A96261A139D06CD7F43A33
X[2]:          9AE3BF42A20F4FB773EEFD5B5C5DBDD3
X[3]:          22A7FA0F7E5FC49715374D6B72EC7FBB
X[4]:          2FE103C6651C845A71217C1C7E80D559
X[5]:          FA94D93A0A7D235AEED7891F5E381A17
GHASH(H,A,C): 954EBAA64B1E25DEE8AE1EADCFFAE4D0
C:     C1623F55730C93533097ADDAD2566496
       6125352B43ADACBD61C5EF3AC90B5BEE
       929CE4630EA79F6CE519
T:     12AF39C2D1FDC2051F8B7B3C9D397EF2
```


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C.5.3 GCM-AES-XPN-128 (54-octet frame confidentiality protection)

Table C-29 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN (the 32 least significant bits are those of the PN field in the SecTAG), 96-bit Salt, and 128-bit key (SAK), with the Secure Data and ICV generated by the GCM-AES-XPN-128 Cipher Suite when used in conjunction with the foregoing and the frame field data of Table C-25 and Table C-26. The GCM parameter P, the data to be encrypted, is the User Data. The additional data A to be authenticated is formed by concatenating the MAC DA, the MAC SA, and the SecTAG. The computed GCM parameter T is the ICV.

Table C-29—GCM-AES-XPN-128 Key, Secure Data, and ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	071B113B0CA743FECCCF3D051F737382
Secure Data	9C A4 69 84 43 02 03 ED 41 6E BD C2 FE 26 22 BA 3E 5E AB 69 61 C3 63 83 00 9E 18 7E 9B 0C 88 56 46 53 B9 AB D2 16 44 1C 6A B6
ICV	F0 A2 32 E9 E4 4C 97 8C F7 CD 84 D4 34 84 D1 01

```
key size = 128 bits
P:     336 bits
A:     160 bits
IV:    96 bits
ICV:   128 bits
K:     071B113B0CA743FECCCF3D051F737382
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F30313233340004
A:     E20106D7CD0DF0761E8DCD3D88E54C2A
       76D457ED
IV:    9C002902F801C33E6AB2AD80
GCM-AES Encryption
H:            E4E01725D724C1215C7309AD34539257
Y[0]:         9C002902F801C33E6AB2AD8000000001
E(K,Y[0]):    5BE02ED3987877610007A055C2EEA9A6
Y[1]:         9C002902F801C33E6AB2AD8000000002
E(K,Y[1]):    94A46694521010F95478AADAE73C39A6
C[1]:         9CA46984430203ED416EBDC2FE2622BA
Y[2]:         9C002902F801C33E6AB2AD8000000003
E(K,Y[2]):    2340B44940E140A725B83F56B226A37A
C[2]:         3E5EAB6961C36383009E187E9B0C8856
Y[3]:         9C002902F801C33E6AB2AD8000000004
E(K,Y[3]):    6B7D969BE32477286AB2194F5E91341E
C[3]:         4653B9ABD216441C6AB6
X[1]:         A0AE6DFAE25C0AE80E9A1AAC0D5123D3
X[2]:         EAEA2A767986B7D5B9E6ED37A3CBC63B
X[3]:         2B263EA98B4A3CDAC1039172AD286472
X[4]:         7A8F8EDACACADACB31FC58F3C1750828
X[5]:         3B9CA06E32AF0F5A1F6A49B5F38EC6C0
GHASH(H,A,C): AB421C3A7C34E0EDF7CA2481F66A78A7
C:     9CA46984430203ED416EBDC2FE2622BA
       3E5EAB6961C36383009E187E9B0C8856
       4653B9ABD216441C6AB6
T:     F0A232E9E44C978CF7CD84D43484D101
```


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C.5.4 GCM-AES-XPN-256 (54-octet frame confidentiality protection)

Table C-30 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN (the 32 least significant bits are those of the PN field in the SecTAG), a 96-bit Salt, and 256-bit key (SAK), with the Secure Data and ICV generated by the GCM-AES-XPN-256 Cipher Suite when used in conjunction with the foregoing and the frame field data of Table C-25 and Table C-26. The GCM parameter P, the data to be encrypted, is the User Data. The additional data A to be authenticated is formed by concatenating the MAC DA, the MAC SA, and the SecTAG. The computed GCM parameter T is the ICV.

Table C-30—GCM-AES-XPN-256 Key, Secure Data, and ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	691D3EE909D7F54167FD1CA0B5D76908 1F2BDE1AEE655FDBAB80BD5295AE6BE7
Secure Data	88 D9 F7 D1 F1 57 8E E3 4B A7 B1 AB C8 98 93 EF 1D 33 98 C9 F1 DD 3E 47 FB D8 55 3E 0F F7 86 EF 56 99 EB 01 EA 10 42 0D 0E BD
ICV	39 A0 E2 73 C4 C7 F9 5E D8 43 20 7D 7A 49 7D FA

```
key size = 256 bits        P:     336 bits      A:     160 bits
IV:    96 bits             ICV:   128 bits
K:     691D3EE909D7F54167FD1CA0B5D76908
       1F2BDE1AEE655FDBAB80BD5295AE6BE7
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F30313233340004
A:     E20106D7CD0DF0761E8DCD3D88E54C2A
       76D457ED
IV:    9C002902F801C33E6AB2AD80
GCM-AES Encryption
H:             1E693C484AB894B26669BC12E6D5D776
Y[0]:          9C002902F801C33E6AB2AD8000000001
E(K,Y[0]):     1EE16A68524D7D515FE89FEC1E11B4D6
Y[1]:          9C002902F801C33E6AB2AD8000000002
E(K,Y[1]):     80D9F8C1E0459DF75EB1A6B3D18288F3
C[1]:          88D9F7D1F1578EE34BA7B1ABC89893EF
Y[2]:          9C002902F801C33E6AB2AD8000000003
E(K,Y[2]):     002D87E9D0FF1D63DEFE721626DDADC3
C[2]:          1D3398C9F1DD3E47FBD8553E0FF786EF
Y[3]:          9C002902F801C33E6AB2AD8000000004
E(K,Y[3]):     7BB7C431DB2271390EB9457F85679B03
C[3]:          5699EB01EA10420D0EBD
X[1]:          F268EF5B38A96261A139D06CD7F43A33
X[2]:          9AE3BF42A20F4FB773EEFD5B5C5DBDD3
X[3]:          A4C4D446D50875C031E49F1039DA6E86
X[4]:          CA6D1BC70BFCAF8B7D1D2279E57D6B45
X[5]:          A76805EBCFDBB50AB9D9C198701753C0
GHASH(H,A,C): 2741881B968A840F87ABBF916458C92C
C:     88D9F7D1F1578EE34BA7B1ABC89893EF
       1D3398C9F1DD3E47FBD8553E0FF786EF
       5699EB01EA10420D0EBD
T:     39A0E273C4C7F95ED843207D7A497DFA
```


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C.6 Confidentiality protection (60-octet frame)

The MAC Destination Address, MAC Source Address, and MAC Service Data Unit (MSDU, User Data) of a MAC Service data request and a corresponding data indication are shown in Table C-31. These comprise the octets of an unprotected frame when concatenated in the order given (with the addition of any media dependent additional fields such as padding). The User Data shown includes the IP EtherType.

Table C-31—Unprotected frame (example)

Field	Value
MAC DA	D6 09 B1 F0 56 63
MAC SA	7A 0D 46 DF 99 8D
User Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 00 02

The MAC Security TAG (SecTAG) comprises the MACsec EtherType, the TCI, the AN, the SL, the PN, and the (optional) SCI. The fields of the protected frame are shown (in the order transmitted) in Table C-32.

Table C-32—Confidentiality protected frame (example)

Field	Value
MAC DA	D6 09 B1 F0 56 63
MAC SA	7A 0D 46 DF 99 8D
MACsec EtherType	88 E5
TCI and AN	2E
SL	00
PN	B2 C2 84 65
SCI	12 15 35 24 C0 89 5E 81
Secure Data	Cipher Suite and Key (SAK) dependent (see Table C-33, Table C-34, Table C-35, and Table C-36)
ICV	Cipher Suite and Key (SAK) dependent (see Table C-33, Table C-34, Table C-35, and Table C-36)

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C.6.1 GCM-AES-128 (60-octet frame confidentiality protection)

Table C-33 specifies an arbitrary 128-bit key (SAK), the Secure Data, and the ICV generated by the GCM-AES-128 Cipher Suite when that key is used in conjunction with the frame field data of Table C-31 and Table C-32. Details of the computation follow the table.

Table C-33—GCM-AES-128 Key, Secure Data, and ICV (example)

Field	Value
Key (SAK)	AD7A2BD03EAC835A6F620FDCB506B345
Secure Data	70 1A FA 1C C0 39 C0 D7 65 12 8A 66 5D AB 69 24 38 99 BF 73 18 CC DC 81 C9 93 1D A1 7F BE 8E DD 7D 17 CB 8B 4C 26 FC 81 E3 28 4F 2B 7F BA 71 3D
ICV	4F 8D 55 E7 D3 F0 6F D5 A1 3C 0C 29 B9 D5 B8 80

```
key size = 128 bits
P:     384 bits
A:     224 bits
IV:    96 bits
ICV:   128 bits
K:     AD7A2BD03EAC835A6F620FDCB506B345
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F303132333435363738393A0002
A:     D609B1F056637A0D46DF998D88E52E00
       B2C2846512153524C0895E81
IV:    12153524C0895E81B2C28465
GCM-AES Encryption
H:            73A23D80121DE2D5A850253FCF43120E
Y[0]:         12153524C0895E81B2C2846500000001
E(K,Y[0]):    EB4E051CB548A6B5490F6F11A27CB7D0
Y[1]:         12153524C0895E81B2C2846500000002
E(K,Y[1]):    781AF50CD12BD3C370049D7E44B17238
C[1]:         701AFA1CC039C0D765128A665DAB6924
Y[2]:         12153524C0895E81B2C2846500000003
E(K,Y[2]):    2587A05339EEFFA5ECB53A895694A5F1
C[2]:         3899BF7318CCDC81C9931DA17FBE8EDD
Y[3]:         12153524C0895E81B2C2846500000004
E(K,Y[3]):    5039E4BB7D14CFB5D61E78134680713F
C[3]:         7D17CB8B4C26FC81E3284F2B7FBA713D
X[1]:         9CABBD91899C1413AA7AD629C1DF12CD
X[2]:         B99ABF6BDBD18B8E148F8030F0686F28
X[3]:         8B5BD74B9A65A459150392C3872BCE7F
X[4]:         934E9D58C59230EE652675D0FF4FB255
X[5]:         4738D208B10FAFF24D6DFBDDC916DC44
GHASH(H,A,C): A4C350FB66B8C960E83363381BA90F50
C:     701AFA1CC039C0D765128A665DAB6924
       3899BF7318CCDC81C9931DA17FBE8EDD
       7D17CB8B4C26FC81E3284F2B7FBA713D
T:     4F8D55E7D3F06FD5A13C0C29B9D5B880
```


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C.6.2 GCM-AES-256 (60-octet frame confidentiality protection)

Table C-34 specifies an arbitrary 256-bit key (SAK), the Secure Data, and the ICV generated by the GCM-AES-256 Cipher Suite when that key is used in conjunction with the frame field data of Table C-31 and Table C-32. Details of the computation follow the table.

Table C-34—GCM-AES-256 Key, Secure Data, and ICV (example)

Field	Value
Key (SAK)	E3C08A8F06C6E3AD95A70557B23F7548 3CE33021A9C72B7025666204C69C0B72
Secure Data	E2 00 6E B4 2F 52 77 02 2D 9B 19 92 5B C4 19 D7 A5 92 66 6C 92 5F E2 EF 71 8E B4 E3 08 EF EA A7 C5 27 3B 39 41 18 86 0A 5B E2 A9 7F 56 AB 78 36
ICV	5C A5 97 CD BB 3E DB 8D 1A 11 51 EA 0A F7 B4 36

```
key size = 256 bits
P:     384 bits
A:     224 bits
IV:    96 bits
ICV:   128 bits
K:     E3C08A8F06C6E3AD95A70557B23F7548
       3CE33021A9C72B7025666204C69C0B72
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F303132333435363738393A0002
A:     D609B1F056637A0D46DF998D88E52E00
       B2C2846512153524C0895E81
IV:    12153524C0895E81B2C28465
GCM-AES Encryption
H:             286D73994EA0BA3CFD1F52BF06A8ACF2
Y[0]:          12153524C0895E81B2C2846500000001
E(K,Y[0]):     714D54FDCFCEE37D5729CDDAB383A016
Y[1]:          12153524C0895E81B2C2846500000002
E(K,Y[1]):     EA0061A43E406416388D0E8A42DE02CB
C[1]:          E2006EB42F5277022D9B19925BC419D7
Y[2]:          12153524C0895E81B2C2846500000003
E(K,Y[2]):     B88C794CB37DC1CB54A893CB21C5C18B
C[2]:          A592666C925FE2EF718EB4E308EFEAA7
Y[3]:          12153524C0895E81B2C2846500000004
E(K,Y[3]):     E8091409702AB53E6ED49E476F917834
C[3]:          C5273B394118860A5BE2A97F56AB7836
X[1]:          D62D2B0792C282A27B82C3731ABCB7A1
X[2]:          841068CDEDA878030E644F03743927D0
X[3]:          224CE5247BE62FB2AC5932EFAC5D1991
X[4]:          EB66718E589AB6472880D1A2C908CB72
X[5]:          6D109A3C7F34085754FDDFF0EB5D4595
GHASH(H,A,C): 2DE8C33074F038F04D389C30B9741420
C:     E2006EB42F5277022D9B19925BC419D7
       A592666C925FE2EF718EB4E308EFEAA7
       C5273B394118860A5BE2A97F56AB7836
T:     5CA597CDBB3EDB8D1A1151EA0AF7B436
```


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C.6.3 GCM-AES-XPN-128 (60-octet frame confidentiality protection)

Table C-35 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN, 96-bit Salt, and 128-bit key (SAK), with the Secure Data and ICV generated by the GCM-AES-XPN-128 Cipher Suite when used with the frame field data of Table C-31 and Table C-32.

Table C-35—GCM-AES-XPN-128 Key, Secure Data, and ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	AD7A2BD03EAC835A6F620FDCB506B345
Secure Data	07 12 D9 80 CA 50 BB ED 35 A0 FA 56 63 38 72 9F FA 16 D1 9F FC F0 7B 3A 1E 79 19 B3 77 6A AC EC 8A 59 37 20 8B 48 3A 76 91 98 4D 38 07 92 E0 7F
ICV	C2 C3 C7 9F 26 3F A6 BF F8 E7 58 1E 2C E4 5A F8

```
key size = 128 bits
P:     384 bits
A:     224 bits
IV:    96 bits
ICV:   128 bits
K:     AD7A2BD03EAC835A6F620FDCB506B345
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F303132333435363738393A0002
A:     D609B1F056637A0D46DF998D88E52E00
       B2C2846512153524C0895E81
IV:    9C002902F801C33EAEA47E08
GCM-AES Encryption
H:            73A23D80121DE2D5A850253FCF43120E
Y[0]:         9C002902F801C33EAEA47E0800000001
E(K,Y[0]):    0C246434EE05EB99762BEFD9880C9E2E
Y[1]:         9C002902F801C33EAEA47E0800000002
E(K,Y[1]):    0F12D690DB42A8F920B6ED4E7A226983
C[1]:         0712D980CA50BBED35A0FA566338729F
Y[2]:         9C002902F801C33EAEA47E0800000003
E(K,Y[2]):    E708CEBFDDD2581E3B5F3E9B5E4087C0
C[2]:         FA16D19FFCF07B3A1E7919B3776AACEC
Y[3]:         9C002902F801C33EAEA47E0800000004
E(K,Y[3]):    A7771810BA7A0942A4AE7A003EA8E07D
C[3]:         8A5937208B483A7691984D380792E07F
X[1]:         9CABBD91899C1413AA7AD629C1DF12CD
X[2]:         B99ABF6BDBD18B8E148F8030F0686F28
X[3]:         D51D27D562BEC296CAA7989F501D7438
X[4]:         ED6EBBA57C47B9A98F94037EAA603CF7
X[5]:         09D09142DBDE8105AB991E09076D6399
GHASH(H,A,C): CEE7A3ABC83A4D268ECCB7C7A4E8C4D6
C:     0712D980CA50BBED35A0FA566338729F
       FA16D19FFCF07B3A1E7919B3776AACEC
       8A5937208B483A7691984D380792E07F
T:     C2C3C79F263FA6BFF8E7581E2CE45AF8
```


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C.6.4 GCM-AES-XPN-256 (60-octet frame confidentiality protection)

Table C-36 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN, 96-bit Salt, and 256-bit key (SAK), with the Secure Data and ICV generated by the GCM-AES-XPN-256 Cipher Suite when used with the frame field data of Table C-31 and Table C-32.

Table C-36—GCM-AES-XPN-256 Key, Secure Data, and ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	E3C08A8F06C6E3AD95A70557B23F7548 3CE33021A9C72B7025666204C69C0B72
Secure Data	3E B0 4A 4B BF 54 C6 EB 12 22 A9 AE A0 0C 38 68 7F 6C 35 20 D9 76 A3 B6 94 80 06 50 CE 65 85 E6 20 A4 19 19 17 D2 A6 05 D8 70 C7 8D 27 52 CE 49
ICV	3B 44 2A C0 C8 16 D7 AB D7 0A D6 5C 25 D4 64 13

```
key size = 256 bits
P:     384 bits
A:     224 bits
IV:    96 bits
ICV:   128 bits
K:     E3C08A8F06C6E3AD95A70557B23F7548
       3CE33021A9C72B7025666204C69C0B72
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F303132333435363738393A0002
A:     D609B1F056637A0D46DF998D88E52E00
       B2C2846512153524C0895E81
IV:    9C002902F801C33EAEA47E08
GCM-AES Encryption
H:            286D73994EA0BA3CFD1F52BF06A8ACF2
Y[0]:         9C002902F801C33EAEA47E0800000001
E(K,Y[0]):    13FBBE38FA1A895C760F543C1AB55F31
Y[1]:         9C002902F801C33EAEA47E0800000002
E(K,Y[1]):    36B0455BAE46D5FF0734BEB6B9162374
C[1]:         3EB04A4BBF54C6EB1222A9AEA00C3868
Y[2]:         9C002902F801C33EAEA47E0800000003
E(K,Y[2]):    62722A00F8548092B1A62178E74FAECA
C[2]:         7F6C3520D976A3B694800650CE6585E6
Y[3]:         9C002902F801C33EAEA47E0800000004
E(K,Y[3]):    0D8A362926E09531ED46F0B51E68CE4B
C[3]:         20A4191917D2A605D870C78D2752CE49
X[1]:         D62D2B0792C282A27B82C3731ABCB7A1
X[2]:         841068CDEDA878030E644F03743927D0
X[3]:         EF4AD8D95E4309D95F8E5F73533BAED7
X[4]:         152F572D6A56D8F0F70E77BE99BAC80D
X[5]:         A9BE13F28CD8F4CCFF1870E5EBB5A9D6
GHASH(H,A,C): 28BF94F8320C5EF7A10582603F613B22
C:     3EB04A4BBF54C6EB1222A9AEA00C3868
       7F6C3520D976A3B694800650CE6585E6
       20A4191917D2A605D870C78D2752CE49
T:     3B442AC0C816D7ABD70AD65C25D46413
```


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C.7 Confidentiality protection (61-octet frame)

The MAC Destination Address, MAC Source Address, and MAC Service Data Unit (MSDU, User Data) of a MAC Service data request and a corresponding data indication are shown in Table C-37. These comprise the octets of an unprotected frame when concatenated in the order given (with the addition of any media dependent additional fields such as padding). The User Data shown includes the IP EtherType.

Table C-37—Unprotected frame (example)

Field	Value
MAC DA	84 C5 D5 13 D2 AA
MAC SA	F6 E5 BB D2 72 77
User Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 3B 00 06

The MAC Security TAG (SecTAG) comprises the MACsec EtherType, the TCI, the AN, the SL, the PN, and the (optional) SCI. The fields of the protected frame are shown (in the order transmitted) in Table C-38.

Table C-38—Confidentiality protected frame (example)

Field	Value
MAC DA	84 C5 D5 13 D2 AA
MAC SA	F6 E5 BB D2 72 77
MACsec EtherType	88 E5
TCI and AN	2F
SL	00
PN	89 32 D6 12
SCI	7C FD E9 F9 E3 37 24 C6
Secure Data	Cipher Suite and Key (SAK) dependent (see Table C-39, Table C-40, Table C-41, and Table C-42)
ICV	Cipher Suite and Key (SAK) dependent (see Table C-39, Table C-40, Table C-41, and Table C-42)

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C.7.1 GCM-AES-128 (61-octet frame confidentiality protection)

Table C-39 specifies an arbitrary 128-bit key (SAK), the Secure Data, and the ICV generated by the GCM-AES-128 Cipher Suite when that key is used in conjunction with the frame field data of Table C-37 and Table C-38. Details of the computation follow the table.

Table C-39—GCM-AES-128 Key, Secure Data, and ICV (example)

Field	Value
Key (SAK)	013FE00B5F11BE7F866D0CBBC55A7A90
Secure Data	3A 4D E6 FA 32 19 10 14 DB B3 03 D9 2E E3 A9 E8 A1 B5 99 C1 4D 22 FB 08 00 96 E1 38 11 81 6A 3C 9C 9B CF 7C 1B 9B 96 DA 80 92 04 E2 9D 0E 2A 76 42
ICV	BF D3 10 A4 83 7C 81 6C CF A5 AC 23 AB 00 39 88

```
key size = 128 bits
P:     392 bits
A:     224 bits
IV:    96 bits
ICV:   128 bits
K:     013FE00B5F11BE7F866D0CBBC55A7A90
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F303132333435363738393A3B00
       06
A:     84C5D513D2AAF6E5BBD2727788E52F00
       8932D6127CFDE9F9E33724C6
IV:    7CFDE9F9E33724C68932D612
GCM-AES Encryption
H:             EB28DCB361EE1110F98CA0C9A07C88F7
Y[0]:          7CFDE9F9E33724C68932D61200000001
E(K,Y[0]):     4EAAF8E4DF948ACAC7F3349C1006A91F
Y[1]:          7CFDE9F9E33724C68932D61200000002
E(K,Y[1]):     324DE9EA230B0300CEA514C137F9B2F4
C[1]:          3A4DE6FA32191014DBB303D92EE3A9E8
Y[2]:          7CFDE9F9E33724C68932D61200000003
E(K,Y[2]):     BCAB86E16C00D82C25B0C61038AB4110
C[2]:          A1B599C14D22FB080096E13811816A3C
Y[3]:          7CFDE9F9E33724C68932D61200000004
E(K,Y[3]):     B1B5E04C2AA9A5EEB5A433DAA4341176
C[3]:          9C9BCF7C1B9B96DA809204E29D0E2A76
Y[4]:          7CFDE9F9E33724C68932D61200000005
E(K,Y[4]):     44491285F0FCF957EB73F79AC5D4E273
C[4]:          42
X[1]:          BA7749648FCB954F95B5933AC87D5AA3
X[2]:          A78C78463850956BF8939E6D8314DED1
X[3]:          18EB5A2C2541C14DD668468C26D2CD8A
X[4]:          32C49AA9AD2B7025767B14F37740A2E8
X[5]:          59CEE3A487F7ACAA9531883B31B11561
X[6]:          3FC125EEEC404708A0D8B9998FE0DE9B
GHASH(H,A,C): F179E8405CE80BA6085698BFBB069097
C:     3A4DE6FA32191014DBB303D92EE3A9E8
       A1B599C14D22FB080096E13811816A3C
       9C9BCF7C1B9B96DA809204E29D0E2A76
       42
T:     BFD310A4837C816CCFA5AC23AB003988
```


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C.7.2 GCM-AES-256 (61-octet frame confidentiality protection)

Table C-40 specifies an arbitrary 256-bit key (SAK), the Secure Data, and the ICV generated by the GCM-AES-256 Cipher Suite when that key is used in conjunction with the frame field data of Table C-37 and Table C-38. Details of the computation follow the table.

Table C-40—GCM-AES-256 Key, Secure Data, and ICV (example)

Field	Value
Key (SAK)	83C093B58DE7FFE1C0DA926AC43FB360 9AC1C80FEE1B624497EF942E2F79A823
Secure Data	11 02 22 FF 80 50 CB EC E6 6A 81 3A D0 9A 73 ED 7A 9A 08 9C 10 6B 95 93 89 16 8E D6 E8 69 8E A9 02 EB 12 77 DB EC 2E 68 E4 73 15 5A 15 A7 DA EE D4
ICV	A1 0F 4E 05 13 9C 23 DF 00 B3 AA DC 71 F0 59 6A

```
key size = 256 bits
P:     392 bits
A:     224 bits
IV:    96 bits
ICV:   128 bits
K:     83C093B58DE7FFE1C0DA926AC43FB360
       9AC1C80FEE1B624497EF942E2F79A823
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F303132333435363738393A3B00
       06
A:     84C5D513D2AAF6E5BBD2727788E52F00
       8932D6127CFDE9F9E33724C6
IV:    7CFDE9F9E33724C68932D612
GCM-AES Encryption
H:             D03D3B51FDF2AACB3A165D7DC362D929
Y[0]:          7CFDE9F9E33724C68932D61200000001
E(K,Y[0]):     E97EA8EE4455AE79EC4225CAC340E326
Y[1]:          7CFDE9F9E33724C68932D61200000002
E(K,Y[1]):     19022DEF9142D8F8F37C9622C98068F1
C[1]:          110222FF8050CBECE66A813AD09A73ED
Y[2]:          7CFDE9F9E33724C68932D61200000003
E(K,Y[2]):     678417BC3149B6B7AC30A9FEC143A585
C[2]:          7A9A089C106B959389168ED6E8698EA9
Y[3]:          7CFDE9F9E33724C68932D61200000004
E(K,Y[3]):     2FC53D47EADE1D5CD14522622C9DE1EE
C[3]:          02EB1277DBEC2E68E473155A15A7DAEE
Y[4]:          7CFDE9F9E33724C68932D61200000005
E(K,Y[4]):     D2541F9E6E5ABAB19C0341912287646B
C[4]:          D4
X[1]:          0B75EC495656426640FD4E24ABA3ED1E
X[2]:          4BC3618F5864A86E9F4EE84504DE347C
X[3]:          F67E393EC69D2D6FFD54C4EFA6F5FF88
X[4]:          C7FE302C946CC29D1EFAAA22B7F587DD
X[5]:          87FCCA374A2EAFC6FD08FE08F919FB8E
X[6]:          0A648461F8E051A0B03165459D5E6F59
GHASH(H,A,C): 4871E6EB57C98DA6ECF18F16B2B0BA4C
C:     110222FF8050CBECE66A813AD09A73ED
       7A9A089C106B959389168ED6E8698EA9
       02EB1277DBEC2E68E473155A15A7DAEE
       D4
T:     A10F4E05139C23DF00B3AADC71F0596A
```


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C.7.3 GCM-AES-XPN-128 (61-octet frame confidentiality protection)

Table C-41 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN, 96-bit Salt, and 128-bit key (SAK), with the Secure Data and ICV generated by the GCM-AES-XPN-128 Cipher Suite when used with the frame field data of Table C-37 and Table C-38.

Table C-41—GCM-AES-XPN-128 Key, Secure Data, and ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	013FE00B5F11BE7F866D0CBBC55A7A90
Secure Data	14 C1 76 93 BC 82 97 EE 6C 47 C5 65 CB E0 67 9E 80 F0 0F CA F5 92 C9 AA 04 73 92 8E 7F 2F 21 6F F5 A0 33 DE C7 51 3F 45 D3 4C BB 98 1C 5B D6 4E 8B
ICV	D8 4B 8E 2A 78 E7 4D AF EA A0 38 46 FE 93 0C 0E

```
key size = 128 bits        P:     392 bits     A:     224 bits
IV:    96 bits             ICV:   128 bits
K:     013FE00B5F11BE7F866D0CBBC55A7A90
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F303132333435363738393A3B00
       06
A:     84C5D513D2AAF6E5BBD2727788E52F00
       8932D6127CFDE9F9E33724C6
IV:    9C002902F801C33E95542C7F
GCM-AES Encryption
H:             EB28DCB361EE1110F98CA0C9A07C88F7
Y[0]:          9C002902F801C33E95542C7F00000001
E(K,Y[0]):     0857C6B6369497B8879CB7FC8F177E1C
Y[1]:          9C002902F801C33E95542C7F00000002
E(K,Y[1]):     1CC17983AD9084FA7951D27DD2FA7C82
C[1]:          14C17693BC8297EE6C47C565CBE0679E
Y[2]:          9C002902F801C33E95542C7F00000003
E(K,Y[2]):     9DEE10EAD4B0EA8E2155B5A656050A43
C[2]:          80F00FCAF592C9AA0473928E7F2F216F
Y[3]:          9C002902F801C33E95542C7F00000004
E(K,Y[3]):     D88E1CEEF6630C71E67A8CA02561ED4E
C[3]:          F5A033DEC7513F45D34CBB981C5BD64E
Y[4]:          9C002902F801C33E95542C7F00000005
E(K,Y[4]):     8D2F0332C7B929F6A40244B1750EDD0A
C[4]:          8B
X[1]:          BA7749648FCB954F95B5933AC87D5AA3
X[2]:          A78C78463850956BF8939E6D8314DED1
X[3]:          A751317726E0F1E84315A9C743DF0C4F
X[4]:          767D6E085166E75CAEAB804D781C2415
X[5]:          5047BCF3D97EADA35994813E3373B800
X[6]:          FA31D67AA9192FB24E5491D4FE366987
GHASH(H,A,C): D01C489C4E73DA176D3C8FBA71847212
C:     14C17693BC8297EE6C47C565CBE0679E
       80F00FCAF592C9AA0473928E7F2F216F
       F5A033DEC7513F45D34CBB981C5BD64E
       8B
T:     D84B8E2A78E74DAFEAA03846FE930C0E
```


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C.7.4 GCM-AES-XPN-256 (61-octet frame confidentiality protection)

Table C-42 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN, 96-bit Salt, and 256-bit key (SAK), with the Secure Data and ICV generated by the GCM-AES-XPN-256 Cipher Suite when used with the frame field data of Table C-37 and Table C-38.

Table C-42—GCM-AES-XPN-256 Key, Secure Data, and ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	83C093B58DE7FFE1C0DA926AC43FB360 9AC1C80FEE1B624497EF942E2F79A823
Secure Data	09 96 E0 C9 A5 57 74 E0 A7 92 30 4E 7D C1 50 BD 67 FD 74 7D D1 B9 41 95 94 BF 37 3D 4A CE 8F 87 F5 C1 34 9A FA C4 91 AA 0A 40 D3 19 90 87 B2 9F DF
ICV	80 2F 05 0E 69 1F 11 A2 D9 B3 58 F6 99 41 84 F5

```
key size = 256 bits        P:     392 bits     A:     224 bits
IV:    96 bits             ICV:   128 bits
K:     83C093B58DE7FFE1C0DA926AC43FB360
       9AC1C80FEE1B624497EF942E2F79A823
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F303132333435363738393A3B00
       06
A:     84C5D513D2AAF6E5BBD2727788E52F00
       8932D6127CFDE9F9E33724C6
IV:    9C002902F801C33E95542C7F
GCM-AES Encryption
H:             D03D3B51FDF2AACB3A165D7DC362D929
Y[0]:          9C002902F801C33E95542C7F00000001
E(K,Y[0]):     032500E383A7A99F250344CAD546A331
Y[1]:          9C002902F801C33E95542C7F00000002
E(K,Y[1]):     0196EFD9B44567F4B284275664DB4BA1
C[1]:          0996E0C9A55774E0A792304E7DC150BD
Y[2]:          9C002902F801C33E95542C7F00000003
E(K,Y[2]):     7AE36B5DF09B62B1B199101563E4A4AB
C[2]:          67FD747DD1B9419594BF373D4ACE8F87
Y[3]:          9C002902F801C33E95542C7F00000004
E(K,Y[3]):     D8EF1BAACBF6A29E3F76E421A9BD899F
C[3]:          F5C1349AFAC491AA0A40D3199087B29F
Y[4]:          9C002902F801C33E95542C7F00000005
E(K,Y[4]):     D965B16B455F2E1F14A50977DF3CAB5E
C[4]:          DF
X[1]:          0B75EC495656426640FD4E24ABA3ED1E
X[2]:          4BC3618F5864A86E9F4EE84504DE347C
X[3]:          CBBE203CFB8356BE9E2454FFE5A1C9AA
X[4]:          78CD005E32EC7ECB0BDFF959C34FA917
X[5]:          C2B1847755117D0352CD4E6C77FF618C
X[6]:          8CA987756B41595D96458142BBBF57C3
GHASH(H,A,C): 830A05EDEAB8B83DFCB01C3C4C0727C4
C:     0996E0C9A55774E0A792304E7DC150BD
       67FD747DD1B9419594BF373D4ACE8F87
       F5C1349AFAC491AA0A40D3199087B29F
       DF
T:     802F050E691F11A2D9B358F6994184F5
```


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C.8 Confidentiality protection (75-octet frame)

The MAC Destination Address, MAC Source Address, and MAC Service Data Unit (MSDU, User Data) of a MAC Service data request and a corresponding data indication are shown in Table C-43. These comprise the octets of an unprotected frame when concatenated in the order given (with the addition of any media dependent additional fields such as padding). The User Data shown includes the IP EtherType.

Table C-43—Unprotected frame (example)

Field	Value
MAC DA	68 F2 E7 76 96 CE
MAC SA	7A E8 E2 CA 4E C5
User Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 3B 3C 3D 3E 3F 40 41 42 43 44 45 46 47 48 49 00 08

The MAC Security TAG (SecTAG) comprises the MACsec EtherType, the TCI, the AN, the SL, and the PN. The optional SCI has been omitted. The fields of the protected frame are shown (in the order transmitted) in Table C-44.

Table C-44—Confidentiality protected frame (example)

Field	Value
MAC DA	68 F2 E7 76 96 CE
MAC SA	7A E8 E2 CA 4E C5
MACsec EtherType	88 E5
TCI and AN	4D
SL	00
PN	2E 58 49 5C
Secure Data	Cipher Suite and Key (SAK) dependent (see Table C-45, Table C-46, Table C-47, and Table C-48)
ICV	Cipher Suite and Key (SAK) dependent (see Table C-45, Table C-46, Table C-47, and Table C-48)

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C.8.1 GCM-AES-128 (75-octet frame confidentiality protection)

Table C-45 specifies an arbitrary 128-bit key (SAK), the Secure Data, and the ICV generated by the GCM-AES-128 Cipher Suite when that key is used in conjunction with the frame field data of Table C-43 and Table C-44. Details of the computation follow the table.

Table C-45—GCM-AES-128 Key, Secure Data, and ICV (example)

Field	Value
Key (SAK)	88EE087FD95DA9FBF6725AA9D757B0CD
Secure Data	C3 1F 53 D9 9E 56 87 F7 36 51 19 B8 32 D2 AA E7 07 41 D5 93 F1 F9 E2 AB 34 55 77 9B 07 8E B8 FE AC DF EC 1F 8E 3E 52 77 F8 18 0B 43 36 1F 65 12 AD B1 6D 2E 38 54 8A 2C 71 9D BA 72 28 D8 40
ICV	88 F8 75 7A DB 8A A7 88 D8 F6 5A D6 68 BE 70 E7

```
key size = 128 bits
P:     504 bits
A:     160 bits
IV:    96 bits
ICV:   128 bits
K:     88EE087FD95DA9FBF6725AA9D757B0CD
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F303132333435363738393A3B3C
       3D3E3F404142434445464748490008
A:     68F2E77696CE7AE8E2CA4EC588E54D00
       2E58495C
IV:    7AE8E2CA4EC500012E58495C
GCM-AES Encryption
H:             AE19118C3B704FCE42AE0D15D2C15C7A
Y[0]:          7AE8E2CA4EC500012E58495C00000001
E(K,Y[0]):     D2521AABC48C06033E112424D4A6DF74
Y[1]:          7AE8E2CA4EC500012E58495C00000002
E(K,Y[1]):     CB1F5CC98F4494E323470EA02BC8B1FB
C[1]:          C31F53D99E5687F7365119B832D2AAE7
Y[2]:          7AE8E2CA4EC500012E58495C00000003
E(K,Y[2]):     1A5FCAB3D0DBC18F117350B32EA493D2
C[2]:          0741D593F1F9E2AB3455779B078EB8FE
Y[3]:          7AE8E2CA4EC500012E58495C00000004
E(K,Y[3]):     81F1C32FBF0C6143CD2E3C7B0F255E2E
C[3]:          ACDFEC1F8E3E5277F8180B43361F6512
Y[4]:          7AE8E2CA4EC500012E58495C00000005
E(K,Y[4]):     908F526E7916C96834DBFD3A61D848B2
C[4]:          ADB16D2E38548A2C719DBA7228D840
X[1]:          A9845CAED3E164079E217A8D26A600DA
X[2]:          09410740B1204002F754119A976F31C8
X[3]:          CB897D3B71442B121E77CEA5416D3931
X[4]:          5F3A6A2D049FF2337096523ECAA1BD30
X[5]:          0C95908AEEBDAF1B1C279837AE498000
X[6]:          1ACA99E1E46D2395BC610D21BB4216A0
GHASH(H,A,C): 5AAA6FD11F06A18BE6E77EF2BC18AF93
C:     C31F53D99E5687F7365119B832D2AAE7
       0741D593F1F9E2AB3455779B078EB8FE
       ACDFEC1F8E3E5277F8180B43361F6512
       ADB16D2E38548A2C719DBA7228D840
T:     88F8757ADB8AA788D8F65AD668BE70E7
```


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C.8.2 GCM-AES-256 (75-octet frame confidentiality protection)

Table C-46 specifies an arbitrary 256-bit key (SAK), the Secure Data, and the ICV generated by the GCM-AES-256 Cipher Suite when that key is used in conjunction with the frame field data of Table C-43 and Table C-44. Details of the computation follow the table.

Table C-46—GCM-AES-256 Key, Secure Data, and ICV (example)

Field	Value
Key (SAK)	4C973DBC7364621674F8B5B89E5C1551 1FCED9216490FB1C1A2CAA0FFE0407E5
Secure Data	BA 8A E3 1B C5 06 48 6D 68 73 E4 FC E4 60 E7 DC 57 59 1F F0 06 11 F3 1C 38 34 FE 1C 04 AD 80 B6 68 03 AF CF 5B 27 E6 33 3F A6 7C 99 DA 47 C2 F0 CE D6 8D 53 1B D7 41 A9 43 CF F7 A6 71 3B D0
ICV	26 11 CD 7D AA 01 D6 1C 5C 88 6D C1 A8 17 01 07

```
key size = 256 bits
P:     504 bits
A:     160 bits
IV:    96 bits
ICV:   128 bits
K:     4C973DBC7364621674F8B5B89E5C1551
       1FCED9216490FB1C1A2CAA0FFE0407E5
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F303132333435363738393A3B3C
       3D3E3F404142434445464748490008
A:     68F2E77696CE7AE8E2CA4EC588E54D00
       2E58495C
IV:    7AE8E2CA4EC500012E58495C
GCM-AES Encryption
H:             9A5E559A96459C21E43C0DFF0FA426F3
Y[0]:          7AE8E2CA4EC500012E58495C00000001
E(K,Y[0]):     316F5EDB0829AC9271A6AFF79F3600BF
Y[1]:          7AE8E2CA4EC500012E58495C00000002
E(K,Y[1]):     B28AEC0BD4145B797D65F3E4FD7AFCC0
C[1]:          BA8AE31BC506486D6873E4FCE460E7DC
Y[2]:          7AE8E2CA4EC500012E58495C00000003
E(K,Y[2]):     4A4700D02733D0381D12D9342D87AB9A
C[2]:          57591FF00611F31C3834FE1C04AD80B6
Y[3]:          7AE8E2CA4EC500012E58495C00000004
E(K,Y[3]):     452D80FF6A15D5070A904BA1E37DF9CC
C[3]:          6803AFCF5B27E6333FA67C99DA47C2F0
Y[4]:          7AE8E2CA4EC500012E58495C00000005
E(K,Y[4]):     F3E8B2135A9502ED0689B0EE383BD81D
C[4]:          CED68D531BD741A943CFF7A6713BD0
X[1]:          1F7477283AA77457BD0C161CB6F179C5
X[2]:          617F112B72DF67BC42218163B73AF025
X[3]:          20A91ADD33433324DBE7822A5BC98013
X[4]:          84D320FCB3B7AF10A66A48BADD00CFA1
X[5]:          52F52D34BC031431185DB9A617FCE98C
X[6]:          57E7CFDDBA0BA07415FD58BCEE906CAC
GHASH(H,A,C): 177E93A6A2287A8E2D2EC236372101B8
C:     BA8AE31BC506486D6873E4FCE460E7DC
       57591FF00611F31C3834FE1C04AD80B6
       6803AFCF5B27E6333FA67C99DA47C2F0
       CED68D531BD741A943CFF7A6713BD0
T:     2611CD7DAA01D61C5C886DC1A8170107
```


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C.8.3 GCM-AES-XPN-128 (75-octet frame confidentiality protection)

Table C-47 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN, 96-bit Salt, and 128-bit key (SAK), with the Secure Data and ICV generated by the GCM-AES-XPN-128 Cipher Suite when used with the frame field data of Table C-43 and Table C-44.

Table C-47—GCM-AES-XPN-128 Key, Secure Data, and ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	88EE087FD95DA9FBF6725AA9D757B0CD
Secure Data	EA EC C6 AF 65 12 FC 8B 6C 8C 43 BC 55 B1 90 B2 62 6D 07 D3 D2 18 FA F5 DA A7 D8 F8 00 A5 73 31 EB 43 B5 A1 7A 37 E5 B1 D6 0D 27 5C CA F7 AC D7 04 CC 9A CE 2B F8 BC 8B 9B 23 B9 AD F0 2F 87
ICV	34 6B 96 D1 13 6A 75 4D F0 A6 CD E1 26 C1 07 F8

```
key size = 128 bits
P:     504 bits
A:     160 bits
IV:    96 bits
ICV:   128 bits
K:     88EE087FD95DA9FBF6725AA9D757B0CD
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F303132333435363738393A3B3C
       3D3E3F404142434445464748490008
A:     68F2E77696CE7AE8E2CA4EC588E54D00
       2E58495C
IV:    9C002902F801C33E323EB331
GCM-AES Encryption
H:             AE19118C3B704FCE42AE0D15D2C15C7A
Y[0]:          9C002902F801C33E323EB33100000001
E(K,Y[0]):     051CB848B04A95168858F67B22FB45CD
Y[1]:          9C002902F801C33E323EB33100000002
E(K,Y[1]):     E2ECC9BF7400EF9F799A54A44CAB8BAE
C[1]:          EAECC6AF6512FC8B6C8C43BC55B190B2
Y[2]:          9C002902F801C33E323EB33100000003
E(K,Y[2]):     7F7318F3F33AD9D1FF81FFD0298F581D
C[2]:          626D07D3D218FAF5DAA7D8F800A57331
Y[3]:          9C002902F801C33E323EB33100000004
E(K,Y[3]):     C66D9A914B05D685E33B1064F3CD97EB
C[3]:          EB43B5A17A37E5B1D60D275CCAF7ACD7
Y[4]:          9C002902F801C33E323EB33100000005
E(K,Y[4]):     39F2A58E6ABAFFCFDE65FEE5B92F8F8E
C[4]:          04CC9ACE2BF8BC8B9B23B9ADF02F87
X[1]:          A9845CAED3E164079E217A8D26A600DA
X[2]:          09410740B1204002F754119A976F31C8
X[3]:          AEEAC0AE90A8F750E3328F7EC27BC7C9
X[4]:          0AC259846D1B384C53E945A6EFFFD3B3
X[5]:          05E38C36366E7137A9CAB89B45CDCE1A
X[6]:          868E036CF44752D418D368EA772C9CE4
GHASH(H,A,C): 31772E99A320E05B78FE3B9A043A4235
C:     EAECC6AF6512FC8B6C8C43BC55B190B2
       626D07D3D218FAF5DAA7D8F800A57331
       EB43B5A17A37E5B1D60D275CCAF7ACD7
       04CC9ACE2BF8BC8B9B23B9ADF02F87
T:     346B96D1136A754DF0A6CDE126C107F8
```


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C.8.4 GCM-AES-XPN-256 (75-octet frame confidentiality protection)

Table C-48 specifies arbitrary values for the SSCI, the 32 most significant bits of the 64-bit PN, 96-bit Salt, and 256-bit key (SAK), with the Secure Data and ICV generated by the GCM-AES-XPN-256 Cipher Suite when used with the frame field data of Table C-43 and Table C-44.

Table C-48—GCM-AES-XPN-256 Key, Secure Data, and ICV (example)

Field	Value
SSCI	7A30C118
PN (ms 32 bits)	B0DF459C
Salt	E630E81A48DE86A21C66FA6D
Key (SAK)	4C973DBC7364621674F8B5B89E5C1551 1FCED9216490FB1C1A2CAA0FFE0407E5
Secure Data	B0 FE A3 63 18 B9 B3 64 66 C4 6E 9E 1B DA 1A 26 68 58 19 6E 7E 70 D8 82 AE 70 47 56 68 CD E4 EC 88 3F 6A C2 36 9F 28 4B ED 1F E3 2F 42 09 2F DF F5 86 8A 3C 64 E5 61 51 92 A7 A3 76 0B 34 BC
ICV	85 69 2C D8 15 B6 64 71 1A EF 91 1D F7 8D 7F 46

```
key size = 256 bits        P:     504 bits      A:     160 bits
IV:    96 bits             ICV:   128 bits
K:     4C973DBC7364621674F8B5B89E5C1551
       1FCED9216490FB1C1A2CAA0FFE0407E5
P:     08000F101112131415161718191A1B1C
       1D1E1F202122232425262728292A2B2C
       2D2E2F303132333435363738393A3B3C
       3D3E3F404142434445464748490008
A:     68F2E77696CE7AE8E2CA4EC588E54D00
       2E58495C
IV:    9C002902F801C33E323EB331
GCM-AES Encryption
H:             9A5E559A96459C21E43C0DFF0FA426F3
Y[0]:          9C002902F801C33E323EB33100000001
E(K,Y[0]):     35F6654C6A3A1D45F1D3C3E5C6B4CAC5
Y[1]:          9C002902F801C33E323EB33100000002
E(K,Y[1]):     B8FEAC7309ABA07073D2798602C0013A
C[1]:          B0FEA36318B9B36466C46E9E1BDA1A26
Y[2]:          9C002902F801C33E323EB33100000003
E(K,Y[2]):     7546064E5F52FBA68B56607E41E7CFC0
C[2]:          6858196E7E70D882AE70475668CDE4EC
Y[3]:          9C002902F801C33E323EB33100000004
E(K,Y[3]):     A51145F207AD1B7FD829D4177B3314E3
C[3]:          883F6AC2369F284BED1FE32F42092FDF
Y[4]:          9C002902F801C33E323EB33100000005
E(K,Y[4]):     C8B8B57C25A72215D7E1E43E4234B450
C[4]:          F5868A3C64E5615192A7A3760B34BC
X[1]:          1F7477283AA77457BD0C161CB6F179C5
X[2]:          617F112B72DF67BC42218163B73AF025
X[3]:          0ECB1CA029F4B30D352C800C284B6BAD
X[4]:          3E312F6336A81FEF782782F906EEBC0E
X[5]:          17322A4719E50FA1E2082E54CB12CB49
X[6]:          44D9DB0896C5D819DA6C9B40330B8BAC
GHASH(H,A,C): B09F49947F8C7934EB3C52F83139B583
C:     B0FEA36318B9B36466C46E9E1BDA1A26
       6858196E7E70D882AE70475668CDE4EC
       883F6AC2369F284BED1FE32F42092FDF
       F5868A3C64E5615192A7A3760B34BC
T:     85692CD815B664711AEF911DF78D7F46
```


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Annex D

(normative)

PICS proforma for an Ethernet Data Encryption device[17]

D.1 Introduction

The supplier of a protocol implementation which is claimed to conform to this standard's provisions for an Ethernet Data Encryption device (EDE) shall complete the following Protocol Implementation Conformance Statement (PICS) proforma.

A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of which capabilities and options of the protocol have been implemented. The PICS can have a number of uses, including use

a) By the protocol implementor, as a checklist to reduce the risk of failure to conform to the standard through oversight;
b) By the supplier and acquirer—or potential acquirer—of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PICS proforma;
c) By the user—or potential user—of the implementation, as a basis for initially checking the possibility of interworking with another implementation (note that, while interworking can never be guaranteed, failure to interwork can often be predicted from incompatible PICSs);
d) By a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

D.2 Abbreviations and special symbols

D.2.1 Status symbols

M mandatory
O optional
O.n optional, but support of at least one of the group of options labelled by the same numeral is required
X prohibited
pred: conditional-item symbol, including predicate identification: see D.3.4
¬ logical negation, applied to a conditional item's predicate

D.2.2 General abbreviations

N/A not applicable
PICS Protocol Implementation Conformance Statement

[17] *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

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D.3 Instructions for completing the PICS proforma

D.3.1 General structure of the PICS proforma

The first part of the PICS proforma, implementation identification and protocol summary, is to be completed as indicated with the information necessary to identify fully both the supplier and the implementation.

The main part of the PICS proforma is a fixed-format questionnaire, divided into several subclauses, each containing a number of individual items. Answers to the questionnaire items are to be provided in the rightmost column, either by simply marking an answer to indicate a restricted choice (usually Yes or No), or by entering a value or a set or range of values. (Note that there are some items where two or more choices from a set of possible answers can apply; all relevant choices are to be marked.)

Each item is identified by an item reference in the first column. The second column contains the question to be answered; the third column records the status of the item—whether support is mandatory, optional, or conditional; see also D.3.4. The fourth column contains the reference or references to the material that specifies the item in the main body of this standard, and the fifth column provides the space for the answers.

A supplier may also provide (or be required to provide) further information, categorized as either Additional Information or Exception Information. When present, each kind of further information is to be provided in a further subclause of items labelled *Ai* or *Xi,* respectively, for cross-referencing purposes, where *i* is any unambiguous identification for the item (e.g., simply a numeral). There are no other restrictions on its format and presentation.

A completed PICS proforma, including any Additional Information and Exception Information, is the Protocol Implementation Conformation Statement for the implementation in question.

NOTE—Where an implementation is capable of being configured in more than one way, a single PICS may be able to describe all such configurations. However, the supplier has the choice of providing more than one PICS, each covering some subset of the implementation's configuration capabilities, in case that makes for easier and clearer presentation of the information.

D.3.2 Additional information

Items of Additional Information allow a supplier to provide further information intended to assist the interpretation of the PICS. It is not intended or expected that a large quantity will be supplied, and a PICS can be considered complete without any such information. Examples might be an outline of the ways in which a (single) implementation can be set up to operate in a variety of environments and configurations, or information about aspects of the implementation that are outside the scope of this standard but that have a bearing upon the answers to some items.

References to items of Additional Information may be entered next to any answer in the questionnaire, and may be included in items of Exception Information.

D.3.3 Exception information

It may occasionally happen that a supplier will wish to answer an item with mandatory status (after any conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer will be found in the Support column for this: instead, the supplier shall write the missing answer into the Support column, together with an *Xi* reference to an item of Exception Information, and shall provide the appropriate rationale in the Exception item itself.

An implementation for which an Exception item is required in this way does not conform to this standard.

NOTE—A possible reason for the situation described above is that a defect in this standard has been reported, a correction for which is expected to change the requirement not met by the implementation.

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D.3.4 Conditional status

D.3.4.1 Conditional items

The PICS proforma contains a number of conditional items. These are items for which both the applicability of the item itself, and its status if it does apply—mandatory or optional—are dependent upon whether or not certain other items are supported.

Where a group of items is subject to the same condition for applicability, a separate preliminary question about the condition appears at the head of the group, with an instruction to skip to a later point in the questionnaire if the "Not Applicable" answer is selected. Otherwise, individual conditional items are indicated by a conditional symbol in the Status column.

A conditional symbol is of the form "**pred:** S" where **pred** is a predicate as described in D.3.4.2, and S is a status symbol, M or 0.

If the value of the predicate is True (see D.3.4.2), the conditional item is applicable, and its status is indicated by the status symbol following the predicate: the answer column is to be marked in the usual way. If the value of the predicate is False, the "Not Applicable" (N/A) answer is to be marked.

D.3.4.2 Predicates

A predicate is one of the following:

a) An item-reference for an item in the PICS proforma: the value of the predicate is True if the item is marked as supported, and is False otherwise;
b) A predicate-name, for a predicate defined as a Boolean expression constructed by combining item-references using the Boolean operator OR: the value of the predicate is True if one or more of the items is marked as supported;
c) A predicate-name, for a predicate defined as a Boolean expression constructed by combining item-references using the Boolean operator AND: the value of the predicate is True if all of the items are marked as supported;
d) The logical negation symbol "¬" prefixed to an item-reference or predicate-name: the value of the predicate is True if the value of the predicate formed by omitting the "¬" symbol is False, and vice versa.

Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for grouped conditional items, is indicated by an asterisk in the Item column.

D.4 PICS proforma for IEEE Std 802.1AE EDE

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D.4.1 Implementation identification

Supplier	
Contact point for queries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification—e.g., name(s) and version(s) of machines and/or operating system names	
NOTE 1—Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirement for full identification. NOTE 2—The terms *Name* and *Version* should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

D.4.2 Protocol summary, IEEE Std 802.1AE EDE

Identification of protocol specification	IEEE Std 802.1AE-2018, IEEE Standard for Local and Metropolitan Area Networks: Media Access Control (MAC) Security			
Identification of amendments and corrigenda to the PICS proforma which have been completed as part of the PICS	Amd.	:	Corr.	: :
	Amd.	:	Corr.	:
Have any Exception items been required? (See D.3.3: the answer Yes means that the implementation does not conform to IEEE Std 802.1AE-2018.)	No [ ]		Yes [ ]	

Date of Statement	

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D.5 EDE type and common requirements

Item	Feature	Status	References	Support	
EDEM	Does the implementation conform to the specification for an EDE-M?	O.1	5.2(a)(b)(c), 5.5, 5.6, 15.4, D.7	Yes [ ]	
EDECS	Does the implementation conform to the specification for an EDE-CS?	O.1	5.2(a)(b)(d), 5.5, 5.7, 15.5, D.7	Yes [ ]	
EDECC	Does the implementation conform to the specification for an EDE-CC?	O.1	5.2(a)(b)(e), 5.5, 5.8, 15.6, D.7	Yes[ ]	
EDESS	Does the implementation conform to the specification for an EDE-SS?	O.1	5.2(a)(b)(f), 5.5, 5.9, 15.7, 5.5, 5.9, 15.7, D.7	Yes [ ]	
TWOP	Does the EDE have two and only two externally accessible ports identified as red-side and black-side?	M	5.5(a)	Yes [ ]	
SECY	Is an MKA-capable PAE associated with each SecY?	M	5.5(b)	Yes [ ]	
PRY	Does the EDE provide MAC Privacy protection	O	5.5(c)	Yes [ ]	No [ ]
SECB	Does the EDE incorporate a SecY in the black-side port interface stack?	EDEM:M	5.6(b)	Yes [ ]	
SECP	Does the EDE incorporate a SecY in each Provider Edge Port interface stack?	EDECS:M EDECC:M EDESS:M	5.7(b), 5.8(b), 5.9(b)	Yes [ ]	

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D.6 EDE-M Configuration

Item	Feature	Status	References	Support
MC	Can the EDE-M be configured to secure connectivity within a customer or provider network? When so configured:	EDEM:M	5.6(c), 15.2	Yes [ ]
MCadd	Does the PAE supporting the black-side port SecY use the Nearest non-TPMR Bridge group address to transmit and receive EAPOL frames?		15.2	Yes [ ]
	Does the EDE's relay entity:			
MCftr	Filter frames whose destination MAC address is a TPMR component Reserved Address or the Nearest non-TPMR Bridge group address?		15.2	Yes [ ]
MCfwd	Forward frames whose destination MAC address is a Reserved Address that is not one of the above?		15.2	Yes [ ]
MB	Can the EDE-M be configured to secure connectivity across a provider network? When so configured:	EDEM:M	5.6(d), 15.4, 15.5	Yes [ ]
MBadd	Does the PAE supporting the black-side port SecY use the Nearest Customer Bridge group address to transmit and receive EAPOL frames?		15.4	Yes [ ]
	Does the EDE's relay entity:			
MBftr	Filter frames whose destination MAC address is a C-VLAN component Reserved Address with the exception of the Nearest Customer Bridge group address?		15.4	Yes [ ]
MBfwd	Forward frames whose destination MAC address is the Nearest Customer Bridge group address?		15.4	Yes [ ]
MM	Can the EDE-M be configured to secure connectivity across a provider network to an EDE-CC? When so configured:	EDEM:O	5.6(e), 15.6	Yes [ ]
MMadd	Does the PAE supporting the black-side port SecY use the EDE-CC PAE group address to transmit and receive EAPOL frames?		15.6	Yes [ ]
	Does the EDE's relay entity:			
MMftr	Filter frames whose destination MAC address is a C-VLAN component Reserved Address or the EDE-CC PAE group address?		15.6	Yes [ ]
Mpri	Can the EDE-M be configured to recover priority from C-VLAN tagged frames transmitted by the black-side port? When so configured:	EDEM:O	5.6(f), 15.4	Yes [ ]
Mprit	Can the black-side port be configured to priority tag or not priority tag transmitted frames?	MPri:M	5.6(f), 15.4	Yes [ ]

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D.7 EDE-CS Configuration

Item	Feature	Status	References	Support
CS	Does the EDE-CS comprise a Provider Edge Bridge with a single C-VLAN component?	EDECS:M	5.7(a), 15.5	Yes [ ]
CSadd	Can the PAE supporting each PEP's SecY use the Nearest Customer Bridge group address to transmit and receive EAPOL frames?		5.7(c), 15.5	Yes [ ]

D.8 EDE-CC Configuration

Item	Feature	Status	References	Support	
CC	Does the EDE-CC comprise two C-VLAN components internally connected as specified in 15.6?	EDECC:M	5.8(a), 15.6	Yes [ ]	
CCadd	Can the PAE supporting each PEP's SecY use the EDE-CC PAE group address to transmit and receive EAPOL frames?		5.8(c), 15.6	Yes [ ]	
CCftr	Does the EDE's edge component relay entity filter frames whose destination MAC address is a C-VLAN component Reserved Address or the EDE-CC PAE group address?		5.8(d), 15.6	Yes [ ]	
CCrlyu	Are frames received untagged on the red-side port and relayed through the black-side port transmitted untagged?	EDECC AND ¬PRY:M	5.8(e), 15.6	Yes [ ]	
CCrlypu	Can the EDE be configured to relay frames received untagged on the red-side port through the black-side port untagged?	EDECC AND PRY:M	5.8(f), 15.6	Yes [ ]	
CCrlyt	Are frames received C-tagged on the red-side port and relayed through the black-side port transmitted C-tagged with the received C-VID?	EDECC AND ¬PRY:M	5.8(e), 15.6	Yes [ ]	
CCrlypt	Can the EDE be configured to relay frames received C-tagged on the red-side port through the black-side port C-tagged with the received C-VID?	EDECC AND PRY:M	5.8(f), 15.6	Yes [ ]	
CCrlypm	Can the EDE be configured to use the C-VID of a frame received on the red-side port to determine the C-VID when relaying that frame through the black-side port?	EDECC AND PRY:O	5.8(g), 15.6	Yes [ ]	No [ ]

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D.9 EDE-SS Configuration

Item	Feature	Status	References	Support
SS	Does the EDE-SS comprise two S-VLAN components internally connected as specified in 15.7?	EDESS:M	5.9(a), 15.7	Yes [ ]
SSadd	Can the PAE supporting each PEP's SecY use the EDE-SS PAE group address to transmit and receive EAPOL frames?		5.9(c), 15.7	Yes [ ]
SSftr	Does the EDE's edge component relay entity filter frames whose destination MAC address is an S-VLAN component Reserved Address or the EDE-SS PAE group address?		5.9(d), 15.7	Yes [ ]
SSrlyu	Are frames received untagged on the red-side port and relayed through the black-side port transmitted untagged?		5.8(e), 15.7	Yes [ ]
SSrlyt	Are frames received S-tagged on the red-side port and relayed through the black-side port transmitted S-tagged with the received S-VID?		5.8(e), 15.7	Yes [ ]

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Annex E

(informative)

MKA operation for multiple transmit SCs

As specified in this standard (7.1.2) when a SecY uses multiple transmit SCs, each SCI is represented by a separate MKA participant that sends and receives MKPDUs to and from all the other participants just as if it were representing a separate SecY in the same group CA. This annex provides tutorial information that might prove useful when implementing a KaY that supports multiple participants. Conformance to this standard remains strictly in terms of externally observable behavior and does not depend on any implementation suggestion in this Annex. Refer to IEEE Std 802.1X for the specification of the normative externally observable behavior of an MKA participant and for the definition of additional terms and acronyms used in this Annex.

All potential participants in the same CA necessarily possess the same secure Connectivity Association Key (CAK) identified by the same secure Connectivity Association Key Name (CKN). Whether this information is derived from an authentication protocol exchange conducted by the PAE or is pre-shared keying material it applies equally to a KaY's multiple traffic class SCI MKA participants just as it does for a participant representing a single SCI, and there are no additional authentication protocol requirements. Similarly there are no additional requirements for group CAK distribution. Although the use of multiple transmit SCIs by a SecY participating in a CA is in many respects similar to that of a group with multiple SecYs, each with a single transmit SC, there is no need to use or distribute an explicit group CAK if only two KaYs participate in the CA; those KaYs simply make the same keying material available to all their MKA participants.

If an implementation of a KaY operates by instantiating separate MKA participants, the implementer needs to be aware that each needs to receive MKPDUs transmitted by other KaYs and by each other. It is also possible to mimic two or more externally visible participants by extending the internal operation of a single participant's implementation, as in the following description. This has advantages including the following:

— Not complicating transmission and reception and the associated scheduling.
— Preserving the existing operation of the PAE's Logon Process[18] and its interaction with the PAE's CP state machine and the KaY.
— Using a single instance of the CP state machine, thus avoiding any need to make decisions about distributing the responsibility for managing receive SAs, avoiding introducing complications into SAK and CAK rollover resulting from dependencies between the participants.[19]

Each participant uses a separate MI and its own MN. The conditions under which either chooses a new MI remain unchanged, though each new MI should also be checked (and reselected if necessary) against those of its co-resident participants (representing the SCIs for other traffic classes).

The KaY maintains just one Live Peers List, and one Potential Peers List, using received MKPDUs to add to each in the usual way. When a participant transmits an MKPDU, its co-resident participants (representing the SCIs for other traffic classes) are added to the transmitted Live Peer List.

Only one participant, that associated with the default traffic class SC, should advertise itself as a potential Key Server, with the others advertising a Key Server Priority of 0xFF.[20] Similarly only the default traffic class participant should distribute keys in MKPDUs (if selected).

[18] See 12.5 and Clause 12 of IEEE Std 802.1X-2010.

[19] If the actors are implemented and scheduled separately and one of them is a potential key server, the best results will generally be obtained by scheduling that actor first on MKPDU reception.

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In the CP state machine, RECEIVE state createSAs(lki) refers to the SAs used to receive from other members of the CA and all the transmit SAs used by the co-resident participants. Naturally there is no need to instantiate receive SAs for the latter. The 'electedSelf ...' transition from CP:RECEIVING is taken if satisfied by default traffic class participant.[21] In CP:TRANSMIT all the traffic class group transmit SAs are enabled.

The retireWhen timer in CP:TRANSMITTING takes care of the fact that a preempted frame may complete its arrival after a preempting frame has been received on a new SA from its transmitting port. This is legitimate when strict replay protection is not enforced.

[20] See 9.5 of IEEE Std 802.1X-2010. The default traffic class SC should be retained even if SCIs for other traffic class groups are added to or deleted from a running system, so this decision minimizes the potential disruption as well as avoiding the need for a management variable for the other SCs. IEEE Std 802.1Xbx-2014 clarifies a number of other issues related to Key Server selection that are equally applicable to the single and multiple participant cases.

[21] None of the other co-resident participants can be elected Key Server in preference.

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Annex F

(informative)

EDE Interoperability and PAE addresses

This annex discusses interoperability between EDEs of the same and different types, and between EDEs and MACsec-capable bridges, providing some background and analysis that goes beyond the normative requirements of this standard.

Each of the types of EDE is naturally designed to interoperate with one or more EDEs of the same type, given suitable connectivity. For example, an EDE-M can interoperate with another EDE-M, or an EDE-CC with one or more EDE-CCs attached to the same PBN. Other interoperability scenarios are also possible, an EDE-M can interoperate with a MACsec-capable Customer Bridge, for example. In all scenarios, the secure Connectivity Associations (CAs) that they create need to have an appropriate scope. If, for example, two EDEs on either side of a Customer Bridge were to create a CA that passed through that bridge, it would then be unable to peer with its neighboring Customer Bridge. Configuration frames, such as RSTP BPDUs, originated by the bridge would be discarded by the EDEs, while frames destined for it would have an additional SecTAG and might be encrypted, rendering them indecipherable.

IEEE Std 802.1Q specifies Reserved Addresses, group MAC destination addresses used by MAC sublayer configuration and discovery protocols, that are selectively filtered by bridge components of various types. This filtering ensures that each protocol is confined to an appropriate scope. For example, when the Link Layer Discovery Protocol (LLDP, IEEE Std 802.1AB) is used to manage Power over Ethernet, it uses the Individual LAN (01-80-C2-00-00-0E) group address which is filtered by all bridges, including TPMRs. The LLDP frames thus reflect the power available from, and power drawn by, the devices attached to a single Ethernet LAN and do not erroneously include devices attached to other, bridged, LANs that do not affect the power budget. LLDP can also be used with different addresses. For example, the Provider Bridge Group Address (01-80-C2-00-00-08) is not filtered by TPMRs but is filtered by Provider Bridges, so it can be used by a Provider Bridge to discover its neighboring Provider Bridges as part of constructing a map of the provider network topology, without reaching the erroneous conclusion that distant bridges are locally connected. Similarly, a Provider Bridge Port that uses MACsec to secure connectivity can use the Nearest non-TPMR Bridge group address as the destination address of the EAPOL frames that it sends and receives to support peer authentication and key agreement—thus ensuring that the scope of any resulting CA does not extend beyond bridges that need to interpret the protected frames.

Note that it is the destination MAC address that determines the extent of the propagation (through various types of bridge) of protocol frames, and not the protocol type[22] or (when the frame is being transmitted by a bridge port) the type of bridge that originally transmits the frame. Separate instances of the same protocol can operate independently of each other, each within its own scope, without having to allocate additional protocol types for each possible application. Each bridge can enforce the desired scoping by correctly filtering a small number of group addresses without having to know every one of a larger (and increasing number) of protocol types, and without having to access the protocol type in (potentially encrypted) frames. A Customer Bridge Port can, for example, use LLDP frames with the Individual LAN group address to manage power (as described in the previous paragraph) and LLDP frames with the Nearest Customer Bridge group address to discover its peers. Similarly, the scope of one MACsec protected CA (between two EDE-Ms that form part of the connectivity between two neighboring Customer Bridges, for example) might be nested within the scope of another (between those bridges), with the organizations responsible for securing the two CAs being quite separate, not depending on or even necessarily being aware of each other.[23]

[22] Most commonly the EtherType of the frame.

[23] Clause 11 provides further examples of nested scopes (11.7).

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The EAPOL destination address used by an EDE's Provider Edge Port PAE is naturally chosen to match the scope of the desired CA. The edge component of EDE-CCs, EDE-CSs, and EDE-SSs, and the MAC Bridge component of EDE-Ms, also enforce that scope by filtering their PAE addresses—precluding the choice of an address that supports an enclosing, more extensive, scope.

— The PAE of an EDE-M that is configured to secure simple LAN connectivity uses the Nearest non-TPMR group address (15.2).

— The PAE of an EDE-M that is configured to secure connectivity across a PBN uses the Nearest Customer Bridge group address (15.4). Frames with this destination address are forwarded by PBN port-based interfaces, S-tagged interfaces, and C-tagged interfaces that provide access to a single service instances, and the scope of the CA created can encompass the provider network service instance as the PBN has no need to interpret the protected frame. The PAE will be unable to establish secure connectivity if the EDE-M is attached to a PBN C-tagged interface that supports multiple service instances, as that interface will discard the PAE's EAPOL frames. The EDE will also not secure connectivity if the EAPOL frames received from the interface are tagged—it transmits EAPOL frames untagged or priority tagged.

  Frames destined to the Nearest Customer Bridge group address are forwarded by the EDE-M provided that the requirements for connectivity through the associated SecY's Controlled Port are met—its operation is transparent to that of Customer Bridges using that address for other protocols. In principle, two nested MACsec secured CAs could be created, both supported by EAPOL frames using the Nearest Customer Bridge address: an inner CA between a pair of EDE-Ms, and an outer CA between MACsec capable Customer Bridges attached to those EDEs. EAPOL frames associated with the outer CA would not be forwarded by the inner CA until the latter had established secure connectivity, when their initial octets would be a SecTAG so they would not be recognized by the EDE PAEs as EAPOL frames. If one of the EDE-Ms (M1, say) has been configured to allow insecure connectivity, the other EDE-M (M2) might establish a CA with M1's attached Customer Bridge (C1) if common PSKs have been (inadvisably) configured or authorization controls omitted from EAP authentication. This might or might not be viewed as a problem, though deliberate creation of this scenario is to be avoided. If the inner CA is subsequently established, the situation will eventually resolve into the two nested CAs.

— An EDE-CS's PAEs use an address that is forwarded by the S-VLAN components in Provider Bridges and is filtered (preferably) by C-VLAN components in Customer Bridges and Provider Edge Bridges, and by VLAN-unaware MAC Bridges. The Nearest Customer Bridge address can be used, allowing interoperability in a hub-and-spoke configuration with EDE-Ms or MACsec-capable Customer Bridges attached to PBN port-based interfaces.

— An EDE-CC's PAE's address has to be forwarded by PEBs that support multiple provider service instances, ruling out use of any of the Reserved Addresses specified by IEEE Std 802.1Q. This standard specifies the use of the EDE-CC PEP Address (01-80-C2-00-00-1F) to facilitate interoperability. It also recommends filtering of that address by bridges in the customer network, to reduce the chance of creating an inappropriately scoped CA that extends from an EDE-CC, through a PBN, across part of a customer network, through the same or another PBN, and back to another EDE-CC.

— An EDE-SS's PAEs have the same component filtering requirements as an EDE-CS with two differences.

  First, the address used should not be the same as the EDE-CS PAE Address. An EDE-SS PAE and EDE-CS PAE can communicate, the EAPOL frames that each transmits and receive are simply S-tagged. However, the purpose of the EDE-CS and that of the EDE-SS differ: the former accepts and delivers C-tagged frames to and from its Customer Edge Port; the latter accepts and delivers S-tagged frames. Just as there is little point in using either EDE to support a single untagged VLAN, so there is little point to both participating in the same CA, while preventing that participation will guard against the unintended S-tagged/C-tagged swap.

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Second, a Reserved address that is not the Nearest Customer Bridge should be used, as the PAE address should be filtered by EDE-SSs and by all C-VLAN components, making it impossible to reach the EDE-SS PAE with an untagged EAPOL frame from a bridge or end station attached to a C-tagged PBN interface, even if frames from the external bridge or end station are not C-tagged when mapped to a provider service instance.

Table F-1 summarizes the scenarios described in this annex and others detailed in this standard, with recommended PAE addresses.

Table F-1—Interoperability scenarios and PAE Addresses

System(s)	Peer System(s)	Connectivity	PAE Address
MAC or Customer Bridge Port — EDE-M	MAC or Customer Bridge Port — EDE-M	Port i/f–PBN–Port i/f	Nearest Customer Bridge[a]
		LAN	Nearest non-TPMR Bridge[b]
	PEB Customer Edge Port — Provider Bridge Port		
	EDE-CS	Port i/f–PBN-S-tagged i/f[c]	Nearest Customer Bridge
	EDE-CC	Port i/f–PBN—C-tagged[c]	EDE-CC PEP Address[d]
EDE-CC	EDE-CC	C-tagged i/f–PBN–C-tagged i/f	EDE-CC PEP Address
EDE-CC	EDE-CS	C-tagged i/f-PBN-S-tagged i/f	EDE-CC PEP Address[e]
EDE-CS	EDE-CS	C-tagged i/f–PBN–S-tagged i/f	Nearest Customer Bridge
EDE-SS	EDE-SS	PBN or PBBN S-tagged i/f–PBN–PBN or PBBN S-tagged i/f — PBBN transparent i/f–PBBN transparent i/f	EDE-SS PEP Address[f]
Provider Bridge Port	Provider Bridge Port	LAN	Nearest non-TPMR

[a]01-80-C2-00-00-00

[b]01-80-C2-00-00-03

[c]Hub-and-spoke scenarios with an EDE hub, with the PBN configured to deliver a single service instance with no outer VLAN tag at the port based interface.

[d]01-80-C2-00-00-1F

[e]This configuration requires that the PBN service instances not deliver the service selecting C-tag used by the EDE-CC at the S-tagged interfaces used by the EDE-CS.

[f]01-80-C2-00-00-0B

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Annex G

(informative)

SecY Management and MIB revisions

Prior to IEEE Std 802.1AEcg-2017,[24] the SecY MIB module in this standard had not been revised since the initial version published as part of IEEE Std 802.1AE-2006 and identified as MIB-2006 in the following discussion. The revised MIB-2016 includes an updated compliance statement. Implementations that do not require MIB-2016's additional capabilities can continue to conform to the standard's (optional) requirement for remote management by continuing to support MIB-2006—subject to future revision as new implementations are expected to find MIB-2016 easier to support, whatever their functionality. Table 13-5, Table 13-6, Table 13-7, and Table 13-8 list both current and deprecated MIB objects. Where possible, MIB-2016 uses tables and objects already defined by MIB-2006. However, when the need to index for multiple transmit SCs required the creation of a new table, new objects were added only to that table and not to its single transmit SC predecessor.

IEEE Std 802.1AEcg-2017 added support for multiple, per traffic class, transmit SCs so that frames misordered within a traffic class can be discarded and strict replay protection provided even when the MAC service reorders frames of different classes or priorities. Such reordering services include Provider Bridged Networks and the use of IEEE 802.3 frame preemption on individual LANs. Management changes were necessary (for both control and reporting) to take full advantage of this new capability. Distinct traffic classes use distinct SCIs (secyIfSCI added to secyIfTable, as the SCI for the default traffic class) and distinct transmit SCs and SAs (secyTSCTable and secyTSATable derived from secyTxSCTable and secyTxSATable respectively, with an additional index of secyTSCI for the traffic class for each table entry). Control over the mapping of frames (with a given user priority) to traffic class SCs, and over the mapping of user Priority Code Point (PCP) (comprising priority bits and the discard eligible bit) to access PCP values, is provided a new Traffic Class Table (secyIfTCTable) and an Access Priority Table (secyIfAPTable).

The use of multiple transmit SCs multiplies the number of both receive and transmit per SC and SA counters. MIB-2006 was developed prior to the standardization of the MACsec Key Agreement protocol (MKA) in IEEE Std 802.1X-2010, and the desired counter capability extended to diagnosing manual or ad-hoc key distribution failures. MKA provides better management visibility without requiring counting at packet data rates, and MIB-2016 includes a text string Key Identifier (matching that originally called for in Clause 10 of IEEE Std 802.1AE-2006) that links MKA operation to the creation and use of SecY SAs. Any alternate key agreement or key distribution protocol should provide equivalent functionality, so a number of counters have been deprecated and reorganized. This annex describes the rationale for the changes (G.1). It also describes how the counts collected by an implementation supporting the 2006 MIB can be used to support the 2016 MIB.

Finally, since these changes involve a significant MIB revision and a new compliance statement, maintenance changes that by themselves would not have justified a MIB update have been included, rather than delay these to a separate later revision with a possible further change to compliance. The general industry move to YANG means that these might be the last changes to this MIB. The maintenance changes included making the full extended PN and the SSCI visible for XPN-capable Cipher Suites, and rectifying discrepancies between the MIB and normative text of the rest of the standard.

[24] This revision, IEEE Std 802.1AE-202X, incorporates the text of IEEE Std 802.1AE-2006 and amendments IEEE Std 802.1AEbn-2011, IEEE Std 802.1AEbw-2013, IEEE Std 802.1AEcg-2017, and IEEE Std 802.1AEdk-2023.

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G.1 Counter changes

IEEE Std 802.1AEcg-2017 made changes to the specification of management counters with a primary goal of enabling cost reduction in real implementations, both by pointing out where implementation dependent cost reduction opportunities exist, and by reducing the number of formal counters.

Whether a verification counter can or cannot be incremented by a received frame depends on the setting of the management control validateFrames. While distinctly named counters are provided for clarity and can all accumulate counts over the long term (as validateFrames can be changed while an SC persists), the real time counting requirement can be reduced. NOTE 1 to Figure 10-4 is intended to make this clear.

Similarly real-time generation counts can be derived from the value of nextPN. See NOTE 1 to Figure 10-3. MIB-2016 does not explicitly include per SA OutPktsEncrypted and OutPktsProtected counts, removing the need for MIB-2006's secyTxSAStatsTable. If necessary this information can be retrieved from the secyTSATable.

MIB-2016's addition of the secyTSAKeyIdentifier and secyRxSAKeyIdentifier provide an explicit indication of whether or not two system's SAKs match, removing the need to use per SA receive counters to analyze synchronization issues. SA status (including SAK use) is also conveyed explicitly by MKA, so that MIB-2016 no longer requires per-SA receive counters (other than nextPN and lowestPN, which are required for basic operation). The use of XPN-capable Cipher Suites also makes it easier to identify SA and SAK rollover related issues, since SAK changes are now required only to meet key lifetime policy requirements (typically one week or longer) even on the fastest links, rather than being forced by PN exhaustion. MIB-2006's unknown SCI and per SC and SA unused SA counts have been combined into a single per SecY MIB-2016 count, as have the no SCI and per SC and SA not using SA counts. This combination also addresses the issue of attributing a precise error to a packet whose integrity cannot be validated.

MIB-2006 included per transmit SA and per receive SC octet counters (distinguishing protected/validated/encrypted/decrypted octets) not called for in Clause 10. These were deprecated in MIB-2016. Subclauses 10.7.10, 10.7.19, and Figure 10-5 describe using counts to investigate or validate cryptographic performance under the assumption that protection resources are dedicated to the SecY as a whole, not to individual SCs or SAs. MIB-2016 includes these per SecY counts, which are a simple sum of the MIB-2006 counts.

Packet counters required for MIB-2016 current compliance and not in MIB-2006 can be derived from those for the latter as follows:

— secyStatsRxNoSAPkts (per SecY) is the sum of
  secyStatsRxUnknownSCIPkts (per SecY),
  secyRxSCStatsUnusedSAPkts (per SC), and
  secyRxSAStatsUnusedSAPkts (per SA)
— secyStatsErrNoSAPkts is the sum of
  secyStatsRxNoSCIPkts (per SecY),
  secyRxSCStatsNotUsingSAPkts (per SC), and
  secyRxSAStatsNotUsingSAPkts (per SA)

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Two of the IETF RFC 2863 Interfaces Group MIB counts do not have to be counted separately, but can be derived from MIB-2016 counts as follows:

— ifInErrors ("the number of inbound packets that contained errors preventing them from being deliverable") is the sum of
  secyStatsRxNoTagPkts (per SecY),
  secyStatsRxBadTagPkts (per SecY),
  secyStatsRxNoSAErrorPkts (per SecY),
  secyRxSCStatsLatePkts (per SC), and
  secyRxSCStatsNotValidPkts (per SC)

— ifInDiscards ("the number of inbound packets which were chosen to be discarded even though no errors had been detected") is the same (provided that no other SecY resource limitation prevents frame delivery) as
  secyStatsRxOverrunPkts (per Secy)

G.2 Available Cipher Suites

MIB-2006 provided a list of Cipher Suites for a system, with limited control over the use of each Cipher Suite, and a way for network management to add to the list. It did not provide a way of controlling or limiting Cipher Suite use per Controlled Port, nor did it support systems with different Cipher Suite capabilities on different ports (which might occur when cards for a chassis based system are procured over time). MIB-2016 added a per Controlled Port list that provides control over the use of each implemented Cipher Suite (secyIfCipherEnableUse, secyIfCipherRqConfidentiality) together with an index for the Cipher Suite entry in the secyCipherSuiteTable. The latter contains information that does not change per Controlled Port and was carried over from MIB-2006, though the use of the RowStatus object was deprecated as it is the system or subsystem that determines whether a Cipher Suite is implemented and potentially available for use, and not a network manager.

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Annex H

(normative)

PICS proforma for MAC Privacy protection[1]

H.1 Introduction

The supplier of a protocol implementation which is claimed to conform to this standard's provisions for MAC Privacy protection shall complete the following Protocol Implementation Conformance Statement (PICS) proforma.

A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of which capabilities and options of the protocol have been implemented. The PICS can have a number of uses, including use

a) By the protocol implementor, as a checklist to reduce the risk of failure to conform to the standard through oversight;
b) By the supplier and acquirer—or potential acquirer—of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PICS proforma;
c) By the user—or potential user—of the implementation, as a basis for initially checking the possibility of interworking with another implementation (note that, while interworking can never be guaranteed, failure to interwork can often be predicted from incompatible PICSs);
d) By a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

H.2 Abbreviations and special symbols

H.2.1 Status symbols

M mandatory
O optional
O.n optional, but support of at least one of the group of options labelled by the same numeral is required
X prohibited
pred: conditional-item symbol, including predicate identification: see H.3.4
¬ logical negation, applied to a conditional item's predicate

H.2.2 General abbreviations

N/A not applicable
PICS Protocol Implementation Conformance Statement

[1] *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

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H.3 Instructions for completing the PICS proforma

H.3.1 General structure of the PICS proforma

The first part of the PICS proforma, implementation identification and protocol summary, is to be completed as indicated with the information necessary to identify fully both the supplier and the implementation.

The main part of the PICS proforma is a fixed-format questionnaire, divided into several subclauses, each containing a number of individual items. Answers to the questionnaire items are to be provided in the rightmost column, either by simply marking an answer to indicate a restricted choice (usually Yes or No), or by entering a value or a set or range of values. (Note that there are some items where two or more choices from a set of possible answers can apply; all relevant choices are to be marked.)

Each item is identified by an item reference in the first column. The second column contains the question to be answered; the third column records the status of the item—whether support is mandatory, optional, or conditional; see also H.3.4 below. The fourth column contains the reference or references to the material that specifies the item in the main body of this standard, and the fifth column provides the space for the answers.

A supplier may also provide (or be required to provide) further information, categorized as either Additional Information or Exception Information. When present, each kind of further information is to be provided in a further subclause of items labelled *Ai* or *Xi,* respectively, for cross-referencing purposes, where *i* is any unambiguous identification for the item (e.g., simply a numeral). There are no other restrictions on its format and presentation.

A completed PICS proforma, including any Additional Information and Exception Information, is the Protocol Implementation Conformation Statement for the implementation in question.

NOTE—Where an implementation is capable of being configured in more than one way, a single PICS may be able to describe all such configurations. However, the supplier has the choice of providing more than one PICS, each covering some subset of the implementation's configuration capabilities, in case that makes for easier and clearer presentation of the information.

H.3.2 Additional information

Items of Additional Information allow a supplier to provide further information intended to assist the interpretation of the PICS. It is not intended or expected that a large quantity will be supplied, and a PICS can be considered complete without any such information. Examples might be an outline of the ways in which a (single) implementation can be set up to operate in a variety of environments and configurations, or information about aspects of the implementation that are outside the scope of this standard but that have a bearing upon the answers to some items.

References to items of Additional Information may be entered next to any answer in the questionnaire, and may be included in items of Exception Information.

H.3.3 Exception information

It may occasionally happen that a supplier will wish to answer an item with mandatory status (after any conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer will be found in the Support column for this: instead, the supplier shall write the missing answer into the Support column, together with an *Xi* reference to an item of Exception Information, and shall provide the appropriate rationale in the Exception item itself.

An implementation for which an Exception item is required in this way does not conform to this standard.

NOTE—A possible reason for the situation described above is that a defect in this standard has been reported, a correction for which is expected to change the requirement not met by the implementation.

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H.3.4 Conditional status

H.3.4.1 Conditional items

The PICS proforma contains a number of conditional items. These are items for which both the applicability of the item itself, and its status if it does apply—mandatory or optional—are dependent upon whether or not certain other items are supported.

Where a group of items is subject to the same condition for applicability, a separate preliminary question about the condition appears at the head of the group, with an instruction to skip to a later point in the questionnaire if the "Not Applicable" answer is selected. Otherwise, individual conditional items are indicated by a conditional symbol in the Status column.

A conditional symbol is of the form "**pred:** S" where **pred** is a predicate as described in H.3.4.2 below, and S is a status symbol, M or 0.

If the value of the predicate is True (see H.3.4.2), the conditional item is applicable, and its status is indicated by the status symbol following the predicate: the answer column is to be marked in the usual way. If the value of the predicate is False, the "Not Applicable" (N/A) answer is to be marked.

H.3.4.2 Predicates

A predicate is one of the following:

a) An item-reference for an item in the PICS proforma: the value of the predicate is True if the item is marked as supported, and is False otherwise;

b) A predicate-name, for a predicate defined as a Boolean expression constructed by combining item-references using the Boolean operator OR: the value of the predicate is True if one or more of the items is marked as supported;

c) A predicate-name, for a predicate defined as a Boolean expression constructed by combining item-references using the boolean operator AND: the value of the predicate is True if all of the items are marked as supported;

d) The logical negation symbol "¬" prefixed to an item-reference or predicate-name: the value of the predicate is True if the value of the predicate formed by omitting the "¬" symbol is False, and vice versa.

Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for grouped conditional items, is indicated by an asterisk in the Item column.

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H.4 PICS proforma for IEEE Std 802.1AE MAC Privacy protection

H.4.1 Implementation identification

Supplier	
Contact point for queries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification—e.g., name(s) and version(s) of machines and/or operating system names	
NOTE 1—Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirement for full identification. NOTE 2—The terms *Name* and *Version* should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

H.4.2 Protocol summary, IEEE Std 802.1AE MAC Privacy protection

Identification of protocol specification	IEEE Std 802.1AEdk-202x, IEEE Standard for Local and Metropolitan Area Networks: Media Access Control (MAC) Security : MAC Privacy protection
Identification of amendments and corrigenda to the PICS proforma which have been completed as part of the PICS	Amd. : Corr. : Amd. : Corr. :
Have any Exception items been required? (See H.3.3: the answer Yes means that the implementation does not conform to IEEE Std 802.1AE.)	No [ ] Yes [ ]

Date of Statement	

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H.5 Mandatory capabilities

Item	Feature	Status	References	Support
Does the implementation of each MAC Privacy protection entity support:				
PrySAPs	The PrivatePort and Controlled Ports as specified in Clause 20?	M	5.10 a), Clause 20	Yes [ ]
PrySTAT	The MAC status and point-to-point parameters for the Private Port as specified in 6.4, 6.5, and 20.4?	M	5.10 b), 20.4	Yes [ ]
PryENC	MPPDU encoding constraints as specified in 19.6?	M	19.6	Yes [ ]
PryTX	Transmission from the Private Port as specified in 20.5 through 20.10?	M	5.10 d), 20.5, 20.6, 20.7, 20.8, 20.9, 20.10	Yes [ ]
PryPP	Per priority transmitted user data frame privacy selection (unprotected, as an individual Privacy Frame, or in a Privacy Channel)?	M	5.10 e), 20.5	Yes[ ]
PryUNP	Unprotected user data frame transmission, with the ability to manage transmission access priority?	M	5.10 f), 20.6	Yes [ ]
PryPF	Individual Privacy Frame transmission, with the ability to manage transmission access priority?	M	5.10 g), 20.7	Yes [ ]
PryCH	Configuration of up to two Privacy Channels?	M	5.10 h), 20.8	Yes [ ]
PryGEN	Generation of Privacy Channel MPPDUs as specified?	M	5.10 i), 20.9	Yes [ ]
PryGEND	Use of the default Privacy Channel MPPDU generation algorithm?	M	5.10 j), 20.9.4	Yes [ ]
PryCHA	Accepting user data frames for Privacy Channel encapsulation as specified in 20.10?	M	5.10 k), 20.10	Yes [ ]
PryCHE	Use of the default Privacy Channel MPPDU encapsulation algorithm?	M	5.10 l), 20.10.1	Yes [ ]
PryDA	Transmit all MPPDUs using the same destination MAC Address?	M	5.10 m), 18.1	Yes [ ]
PryPAE	Use of the PAE Group Address used by MKA as the destination MAC Address of MPPDUs?	M	5.10 n), 17.4, 18.1, 21.1.1	Yes [ ]
PrySRC	Use of the MAC Address used to generate MACsec SCI(s) as the source address of transmitted MPPDU when directly supported by a SecY?	M	5.10 o), 18.1	Yes [ ]
PryDAC	Administrative configuration of the destination MAC Address used to transmit MPPDUs?	M	5.10 p), 20.14	Yes [ ]
PryRX	Recognize received MPPDUs as specified?	M	5.10 r), 20.11	Yes [ ]
PryRXV	Validate and extract received MPPDU components as specified?	M	5.10 s), 20.12, 19.7	Yes [ ]
PryRE	Support simultaneous reassembly of Express and Preemptable frames?	M	5.10 t), 20.13	Yes [ ]
PryRDV	Deliver user data frames received unfragmented and fragmented in the order specified?	M	5.10 u), 20.13	Yes [ ]
PryRED	Use of the default algorithm to reassemble frames from a single peer using one or two Privacy Channels?	M	5.10 v), 20.10.1	Yes [ ]

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H.6 Optional capabilities

Item	Feature	Status	References	Support
Does the implementation of each MAC Privacy protection entity support:				
PryREP	Simultaneous frame reassembly from both Express and Preemptable Frame Fragments from more than one peer PrY?	O	5.11 a), Clause 20	Max peers: _____
PrySPS	Stream-based Privacy Selection?	O	5.11 k), 20.5.2	Yes [ ] No [ ]

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Annex I

(informative)

Privacy considerations in bridged networks

This informative annex describes privacy considerations related to the use, design, and deployment of bridged networks based on IEEE Std 802.1Q and related standards (IEEE Std 802.1X, IEEE Std 802.1AB, IEEE Std 802.1AE, IEEE Std 802.1AR, IEEE Std 802.1AS, IEEE Std 802.1AX, IEEE Std 802.1BA, IEEE Std 802.1BR, IEEE Std 802.1CB, and IEEE Std 802.1CM).

The unintentional or unauthorized disclosure of personal information arises from a combination of the following factors:

a) The use of personal devices that are attached to, or form part of, the network (I.1)
b) The type of information that adversaries might wish to acquire (I.2)
c) The efficient operation and management of the network (I.3)
d) The use of security protocols for authentication, authorization, integrity, and confidentiality (I.5)
e) The frame fields that contain information useful to an adversary, the sophistication of, and the network access afforded to, that adversary (I.5)

Privacy considerations particular to a given referenced standard are discussed in I.6.

This annex is informative. It does not modify the mandatory or optional provisions or the recommendations contained in any referenced standard.

I.1 Personal devices

Privacy, in the context of bridged networks, relates to the use of personal devices i.e. devices used by one person or a small group of people. Information that identifies a personal device or is associated with that device identification can thus yield information about the location and activities of a person.

Shared service devices, in contrast, support applications for a large enough group of people such that correlation between any given person and the observable behavior of the device is weak. Other devices, e.g. sensors in industrial networks, have no direct correlation with a person.

In general IEEE 802.1 standards are applicable to both personal devices and shared service and other devices. However some protocol roles, e.g. Grandmaster in IEEE Std 802.1AS Timing and Synchronization for Time-Sensitive Applications, are unlikely to be associated with personal devices in other than the smallest bridged networks, and are even more unlikely to be associated with mobile personal devices.

I.2 Goals of adversaries

An adversary can be interested in the following personal information:

a) Who is using a personal device (identification)
b) Where are they (location, and location tracking)
c) What are they doing (activity, application use)
d) With whom are they associated (communicating, shared interest).

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The information on all, or indeed on any of these, need not be complete to be useful to an adversary. The adversary can, for example, be interested in facts such as:

- an identified person appears to be engaging in the same, unknown, activity as a group of unknown persons at another identified location
- there appears to be no one at an identified location.

The information obtained need not be particularly accurate to be useful to an adversary. It is sufficient that the cost of acquiring the information is less than the benefit expected from its use, allowing for the probability that it is incorrect and any costs associated with the use of incorrect information. Use of incorrect information can negatively affect a targeted person.

I.3 Network operation

Bridged networks support frame based transmission, with variable length frames and without requiring attached stations (except for certain time-sensitive network applications) to adhere rigidly to a clocked transmission schedule. Stations are not obliged to transmit when there is nothing to transmit and frames are not all padded to the same length, so the use of network resources benefits from statistical multiplexing. At the same time some network applications have requirements for timely delivery that cannot be met simply by relying on that multiplexing and increasing transmission speeds but require signaling, to bridges in the network, of the differential service requirements of individual frames. Time-sensitive network applications with more stringent delivery requirements require bandwidth allocation, supported by end station protocols or management configuration, and sufficient information in individual frames for bridges to associate each frame with an allocation (and thus with an individual end station and a particular type of end station application). Bridged networks provide more bandwidth than is available from each of their constituent individual LANs by restricting data frames to paths to their intended destinations. One of a number of alternate paths to a given destination end station or set of end stations can be used to further increase the available bandwidth, but common network application frame ordering requirements constrain the distribution of frames amongst such paths to those that bridges can distinguish as belonging to separate application flows.

Bridges in the network can distinguish between application flows using each frame’s destination MAC address (DA), source MAC address (SA), the VLAN identifier (VID) and priority code point (PCP encoded in the VLAN tag (if present), the EtherType (or LSAP) identifying the higher layer protocol conveyed by the frame, and the initial fields of that protocol. Protocols that operate over the bridged network and are used by personal devices to support network applications and to communicate with application servers and other devices (as opposed to reserving network resources for that communication) typically use the Internet Protocol (IP). It is rare for two personal devices to communicate without transmitting frames via one or more intervening routers. Any given IP subnet is often supported by a single VLAN, so bridges that support parallel paths for routed application flows from individual end stations typically use the source and destination IP addresses, the conveyed protocol type (IP, UDP, or SCTP), and source and destination ports for that protocol (see 9.1.5 of IEEE Std 802.1CB-2017).

Some protocols, e.g. IEEE Std 802.1Q Stream Reservation Protocol (SRP), transmit frames with a group destination MAC address that identifies the type of the intended recipient protocol entity and allows bridges to use address filtering to restrict those frames to an appropriate scope, reaching only the nearest bridge, for example. Some group addresses support a particular type of application, and thus associate the source MAC address (and the station that is using it) with that application.

The deployment and operational costs of bridged networks have been considerably reduced by the use of protocols that volunteer device information (e.g. IEEE Std 802.1AB) even when those protocols are not used to support full ‘plug-and-play’ operation. Management and troubleshooting of faulty devices or apparently incorrect network behavior depends on the recording of device location and gathering statistics on network

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use. Stations implementing protocols whose operation depends on the presence of a reachable collaborating peer (e.g. IEEE Std 802.1AS gPTP time synchronization) typically advertise their capabilities, either by using IEEE Std 802.1AB or by sending their own messages.

I.4 Network security and privacy

As described above (I.3) efficient use of network resources, particularly for data frames that require other than best effort delivery, depends on the bridges in the network being able to identify end stations and (for some applications) service characteristics (priority, bandwidth and delay) required by their network applications. Where physical access and attachment to the whole or part of a bridged network is restricted to authorized personnel, confidentiality protection can be limited to that provided by higher layer protocols, notably TLS or IPsec. This leaves all the identifying information specified in IEEE 802.1 standards exposed to an adversary that does gain access to the network media.

The end stations and bridges in bridged networks are typically connected by IEEE Std 802.3 Ethernet links. MACsec (IEEE Std 802.1AE) can be used to provide both confidentiality and integrity protection hop by hop, leaving (in the most common configuration) just the MAC source and destination addresses, frame length, and frame transmission timing visible to an adversary with access to the network media. MACsec adds fields to each frame, but an adversary can recover the original frame length. MACsec operation can affect frame timing, but implementations suitable for use in time-sensitive networks impose a small fixed delay so as not to degrade the operation of IEEE Std 802.1AS time synchronization or IEEE Std 802.1Q timing gates supporting traffic shaping and bandwidth allocation. The frame to frame timing relationships that an adversary might observe remain unaltered. Where MACsec is used with Ethernet frame preemption and in-order delivery of preemptable and (separately) of preempting frames is enforced, an observer can distinguish these two classes of frames. Privacy considerations particular to IEEE Std 802.1X (Port-Based Network Access Control) support of MACsec are described below (I.6.2).

Unlike IEEE Std 802.11 operation in which a mobile end station participates in observable protocol to discover and select a suitable service it is rare for an end station to be connected to an Ethernet link that does not provide the expected service. Authentication, authorization, and confidentiality protection of subsequent data frames, if required, typically occur before additional end station information is disclosed. For exceptions see I.6.2, I.6.3.

I.5 Privacy exposures

A personal device can be identified explicitly by a single frame field, notably by using a universal MAC address as the source address of transmitted frames.

A station can use a locally assigned MAC address, as described in IEEE Std 802 and IEEE Std 802c. These addresses can be chosen randomly, or explicitly assigned by a higher layer protocol. A local assignment can be drawn from the entire local address space, or from one of the subsets described in IEEE Std 802c. However once a local MAC address has been assigned to a station and is being used to support higher layer protocols (such as IP), to restrict data frames to the path to that station, and to reserve resources in bridges along the path, any further MAC address change can be expected to interrupt or degrade the MAC Service. Moreover the disappearance of one address coupled with rapid appearance of another facilitates correlation of the two addresses and cannot be expected to reduce an adversary’s ability to infer information from the frame fields and other characteristics of persistent flows.

Where an individual frame field does not directly identify a personal device, either persistently as in the case of a universal MAC address or temporarily while the device is continuously active, an adversary can correlate those frame fields and other frame characteristics to identify (to an acceptable probability) the frames and frame flows associated with a single device and even to ascribe a permanent identity to that device or the particular network applications and activities supported by the device. Such a correlation is

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called a ‘fingerprint’, and the process of obtaining it ‘fingerprinting’. Fingerprinting does not necessarily require a detailed understanding of the protocols used by a device, but can use general correlation and machine learning techniques to find any persistent pattern in the behavior of a device. Indeed a fingerprint can use device characteristics, such as the persistent scheduling of a transmission by one activity immediately after transmission for another activity, that do not appear in protocol specifications. In the absence of information that all the personal devices of a given type in widespread use consistently use the same network applications in the same way (and consequently exhibit indistinguishable network behavior) it has to be assumed that devices and activities can be distinguished by a sufficiently interested adversary.

The pattern of frame sizes transmitted and received by a personal device can fingerprint application activity and reveal details of that activity. The Ethernet MAC does, however, impose a minimum frame size, and higher layer protocols include fields that allow them to determine the applicable data length. To support Ethernet bridging of frames to and from media without the minimum size requirement, MACsec can encode the short length of those frames, but short frames that have been padded prior to being protected with MACsec appear to be of uniform length, thus depriving an adversary of the opportunity of fingerprinting application types using the small frame sizes that can be used in initial capability advertisement.

NOTE 1—Frame size patterns have been used to identify banking applications for specific financial institutions, approximate account balances, and whether money is being added to or removed from the account.

Static personal devices, e.g. desktop computers and home routers, typically connect to a bridged network using an individual wired IEEE Std 802.3 Ethernet connection. An adversary that can gain access to that wired connection has usually already identified (knowledge of home occupancy, etc.) the person or people associated with such a device and there is no question of tracking device movement. However the pattern of device activity (e.g. turning on security cameras when there is nobody at home) can reveal important personal location information.

NOTE 2—This annex does not detail privacy exposures resulting from media access control method operation, but notes that they can exist. For example, PoE (Power over Ethernet) use can reveal the identity and software version of some consumer electronics devices even when the adversary is restricted to observing the neighboring electromagnetic field.

Bridged networks are typically intraconnected with Ethernet links. Where these are wholly on private premises, access by an adversary can be prevented or at least made so difficult and expensive as to limit the targets to previously identified persons. Where personal device traffic to and from those private premises passes through an IP router, the privacy considerations are those applicable to the use of IP.

NOTE 3—At the time of preparation of this annex, discussion of the extension of TSN capabilities beyond the scope of bridged networks to the use of IP under the heading of ‘DetNet’ (deterministic networks) was still at an early stage. The privacy impacts of explicit flow identification and resource allocation described in this annex can be expected to apply.

IEEE Std 802.11, non-standard wireless connectivity, and in-home electrical power wiring can also be used to connect devices to personal bridged networks and to connect bridges within those networks. Where IEEE Std 802.11 is used to connect to an access point (AP) operating as an IP router, the security considerations applicable to IEEE Std 802.11 and IP apply. Where non-standard wireless connectivity and electrical power wiring are used, an adversary located sufficiently close as to be able to intercept the wireless signal or access power wiring outside possibly secured premises can be assumed to have access to MAC address, frame size, and frame timing information at a minimum with the further possibility of access to all the resource allocation and flow identification information conveyed. Frames with specific group and individual MAC addresses can be filtered by bridges in the network and do not necessarily traverse those links.

An adversary with management access to bridges in the network has access to resource allocation and flow identification information, but not (at least with standardized objects) the sizes of specific frames and their transmission timing. Such adversaries can include organizations that have a business relationship with the targeted person and are considered trustworthy by that person.

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I.6 Standard specific considerations

This clause (I.6) summarizes particular ways in which each of the bridged network related standards can, when supporting personal devices, expose information that can be used to fingerprint the device's identity or use of network applications. Unless otherwise stated the general considerations described above (I.3, I.4, I.5) also apply to the use of each standard. The brief summary of each standard's capabilities is intended to provide the context for privacy considerations, and is not a substitute for the text of each referenced standard.

I.6.1 IEEE Std 802.1Q Bridges and bridged networks

The general considerations described above (I.3, I.4, I.5) all apply to the use of IEEE Std 802.1Q.

I.6.2 IEEE Std 802.1X Port-Based Network Access Control

IEEE Std 802.1X specifies a general method controlling access to a network, both by systems that are the source and destination of frames carried by the network and by relay systems that are to be connected to multiple other systems in the network and that forward frames between those connections. In both cases each of the system's ports either participates in a mutual authentication exchange with the neighboring system or proves the success of past authentication and authorization to access the network. This clause discusses potential privacy exposures arising from the use of the media-independent capabilities of IEEE Std 802.1X with Ethernet, for privacy considerations related to the use of IEEE 802.11 connections to or within bridged networks see IEEE Std 802.11.

Extensible Authentication Protocol (EAP, IETF RFC 3748 [B9]) messages are encapsulated in EAP over LANs (EAPOL) PDUs so they can be sent between a Supplicant port (also referred to as a Peer in IETF EAP RFCs), that wishes to gain access to the network, and an Authenticator port, on a system that provides network access. EAP is an authentication framework, not a specific authentication mechanism, and more than 40 specific authentication methods have been defined. An Authenticator is typically supported by an Authentication Server (AS) that executes the particular method or sequence of methods selected. The authentication credentials supported by different methods can differ, as can the degree to which they expose the identity claimed by a Supplicant. EAP messages between the Authenticator and the Authentication Server are typically encapsulated in the RADIUS (IETF RFC 3579) or Diameter (IETF RFC 4072) protocols. IEEE Std 802.1X mandates the use of mutual authentication methods, and requires support for EAP-TLS (IETF RFC 5216) if integration with the use of IEEE Std 802.1AR is claimed.

Following EAP authentication, RADIUS or Diameter server can provide the Authenticator with attributes that include access controls appropriate to the authorization accorded to the Supplicant and information that supports subsequent reattachment of a device to the network without repetition of the full authentication exchange and authorization process. These attributes can include persistent identifiers, e.g. the EAP-Key-Name (the IEEE 802.1X secure Connectivity Association Key Name, CKN) and Network-Id-Name (2.2 and 2.7 of IETF RFC 7268). Privacy considerations relating to communication between the Authenticator, the Authentication Server, a RADIUS or Diameter Server, and any Online Certificate Status Protocol (OCSP) Server are described in the relevant IETF RFCs.

If data transmission, following successful authentication and authorization, between the Supplicant and Authenticator ports is protected by MACsec, the MACsec Key Agreement protocol (MKA) is used to distribute the succession of Secure Association Keys (SAKs) used to provide confidentiality and integrity protection. MKA uses keys derived from a secure Connectivity Association Key (CAK) and the CKN to integrity protect MKPDUs and to confidentiality and integrity protect (using AES Key Wrap) distributed SAKs. The contents of MKPDUs (other than distributed keys) are not confidentiality protected to support network monitoring and debugging without needing to share the CAK or derived keys. The CAK and CKN can be derived from an EAP authentication or can be pre-shared by other means, including local device

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management. The initial octets of each MKPDU contain the CKN, so a peer MACsec capable system knows which (if any) of its keys to use to verify that the MKPDU has been transmitted by a previously authenticated system. A device can be configured to attempt, or require, EAP authentication each time it is connected to the network, thus obtaining a fresh CAK and CKN. Shared service infrastructure devices typically need to be capable of restoring connectivity to their neighbors without re-authentication, since neither they nor their neighbors are guaranteed to have connectivity to an Authentication Server or other supporting services.

EAPOL frames, and integrity protected MKPDUs which are carried in EAPOL frames, can convey network announcements (IEEE Std 802.1X). These can be used by personal devices, but are expected to be transmitted by shared service devices.

I.6.3 IEEE Std 802.1AB Station and Media Access Control Connectivity Discovery

The Link Layer Discovery Protocol (LLDP) allows a station to advertise, to others attached to the same LAN, the station's management address and major capabilities. The receiving stations allow management access to received LLDP information to support network topology discovery and configuration checking. The point of LLDP would be lost if the advertised attributes were to be temporary or unavailable to intended recipients. Standard attributes include a system name and description. The range of attributes has been extended by other standards and organizations such as equipment suppliers.

LLDP is a one way protocol: it does not contain mechanisms for soliciting or confirming receipt of information. The destination address of each LLDPDU is usually one of the reserved group addresses specified in IEEE Std 802.1Q and filtered by bridges to limit the scope of its propagation through the network. This filtering allows a management application to use the information received by end stations and bridges in the network to build a map of the network topology. The filtering also restricts exposure of any station's advertised attributes to adversaries that have access to the individual LANs traversed by the LLDPDUs that station transmits, or that have management access to their recipients or to the management application. IEEE Std 802.1AB-2016 mandates support for the Nearest Bridge group address (01-80-C2-00-00-0E, also referred to as the Individual LAN Scope group address). This address is filtered by all bridges.

Where port access is controlled by IEEE Std 802.1X, IEEE Std 802.1AB mandates Controlled Port support for LLDP exchanges, thus providing confidentiality (on the LAN) if MACsec is used. Unprotected transmission using the Uncontrolled Port is permitted.

I.6.4 IEEE Std 802.1AE MAC Security

The exposure of personal information, including information that can contribute to fingerprinting a device or activity, conveyed in frames that are confidentiality protected by MAC Security (MACsec) can be reduced as described above (I.4). The potential exposure of personal device information by the supporting IEEE Std 802.1X MACsec Key Agreement protocol (MKA) is discussed in I.6.2.

MACsec protects communication between neighboring systems, but the scope of that protection depends on what each system considers to be a potential neighbor. By default frames conveyed by the IEEE Std 802.1X Port Access Control Protocol (PACP) that encapsulates the Extensible Authentication Protocol (EAP, IETF RFC 3748) are transmitted to the Nearest non-TPMR Bridge group address (also referred to as the IEEE Std 802.1X PAE address), so any intervening TPMR cannot access confidentiality protected frame fields (see I.4). However MACsec can also be used to secure a point-to-point connection across a Provider Bridge Network exposing any priority information required by PBN systems to provide the desired class of service, and to secure connectivity where the PBN uses VLAN tag information to select a provider service instance (15.4 and 15.5 of IEEE Std 802.1AE). Where a Provider Backbone Bridge (PBB) is used, the source MAC address of the originator of the frame is encapsulated and confidentiality protected. A PBB is not, itself, likely to be a personal device.

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MAC Privacy protection (Clause 17) can be used in conjunction with MACsec to further address Privacy aspects of Ethernet frames discussed in I.2. Specifically MAC Privacy protection addresses these issues as outlined in (17.2) by hiding user data frame attributes such as:

a) Source and destination MAC addresses,

b) Frame sizes.

c) Frame transmission timing.

I.6.5 IEEE Std 802.1AR Secure Device Identity

IEEE Std 802.1AR specifies Secure Device Identifiers (DevIDs) for use with IEEE Std 802.1X and other industry authentication, provisioning, and authorization protocols. Privacy consideration for use of DevIDs are discussed in 6.5 of IEEE Std 802.1AR-2018.

I.6.6 IEEE Std 802.1AS Timing and Synchronization for Time-Sensitive Applications in Bridged Local Area Networks

The generalized precision time protocol (gPTP), state machines, and algorithms specified in IEEE Std 802.1AS support time-sensitive applications such as audio, video, and time-sensitive control, by maintaining synchronized time across packet networks, including bridged networks, comprising interconnected time-aware systems. Each time-aware system exchanges messages with its immediate neighbor to measure the link propagation delay experienced by packets forwarded by that neighbor. Time-aware end stations receive time information, either directly or indirectly via one or more time-aware relay systems, from a grandmaster that is the source of time information in a network domain. Each system adjusts the time information received to account for the link propagation delay, and in the case of time-aware relays for the residence time of the information in the relay prior to forwarding. The current grandmaster, and the port used to receive information from that grandmaster, is selected by a best master clock algorithm (BMCA) that constructs a time-synchronization spanning tree throughout the network domain with a spanning tree priority vector that allows each time-aware system to select its best port for receiving (and in the case of a time-aware relay, as the basis for forwarding) timing information.

Each time-aware system port that supports gPTP is identified by a sourcePortIdentity, comprising a clockIdentity and a portNumber. The clockIdentity identifies the clock being used by a specific time-aware bridge or end station for a particular instance of distributed time and is constructed using an NUI-48 or NUI-64 (see IEEE Std 802c): i.e., while it is an identifier and not a protocol address it is constructed in the same way as a MAC address, is intended to be unique within a network, and it is possible to tell by examining one of the bits derived from the NUI in the construction (the bit corresponding to the U/L bit when the an NUI is used as a MAC Address) whether the clockIdentity is intended to be locally or globally unique.

The media-independent specification of gPTP is supported by media-dependent procedures. Neighboring time-aware systems connected by full-duplex point-to-point links, such as those specified by IEEE Std 802.3, use gPTP messages to measure the propagation delay and convey timing information. Each message includes the transmitter’s sourcePortIdentity. If the connection is confidentiality protected by MACsec, this message field is only visible to the communicating systems.

Neighboring IEEE Std 802.11 stations, whether AP capable or not, do not use gPTP messages to measure propagation delay and convey timing. They use the IEEE 802.11 MAC Layer Management Entity (MLME) which generates, timestamps, and consumes measurement frames to provide timing information.

NOTE 1—For privacy considerations related to the IEEE 802.11 MLME see IEEE Std 802.11.

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The BMCA spanning tree conveys a trace of each port's sourcePortIdentity on the best path (for timing distribution) from each potential grandmaster. A personal device attached to the network is thus aware of, and receives a permanent identifier for each system that is part of, that path. The BMCA protocol does not propagate path information in the reverse direction (i.e., towards the grandmaster root of a timing tree): the sourcePortIdentity of a personal device that has a single port attached to the network is only conveyed to its immediate neighbor.

NOTE 2—While use of the redundant grandmasters and the BMCA allows the precision timing service provided by IEEE Std 802.1AS to be resilient in the face of system and link failures, it is highly desirable that the network remain stable and the standard provides priority values for grandmaster selection and timing path selection that discriminate against devices that are not permanently part of the network and powered on. A personal device, and particularly a mobile personal device, is therefore unlikely to find itself in the position of propagating BMCA path trace information including the sourcePortIdentity of one of its ports, even if it has more than one port.

Time-aware stations connected by media for which gPTP is supported by media-independent procedures send Signaling messages (10.4 of IEEE Std 802.1AS-2011) that signal each station's ability to participate in the protocol together with station dependent parameters that control aspects of protocol operation (e.g., message interval request). For stations connected by IEEE Std 802.11 media this capability is provided by the IEEE 802.11 MLME.

I.6.7 IEEE Std 802.1AX Link Aggregation

Link Aggregation allows parallel point-to-point links to be aggregated to form a Link Aggregation Group (LAG) that is treated as a single link. A bridge or end station port generally distributes frames amongst the links so as to preserve frame ordering within flows (see I.3 above). The distribution algorithm and parameters for its use can be specified by using Conversation-sensitive Collection and Distribution (CSCD). A further capability Distributed Resilient Network Interface (DRNI), that provides system level redundancy by allowing two cooperating systems to mimic the behavior of a single system terminating a LAG, is unlikely to be used by personal devices.

The addition and removal of links to and from a LAG is facilitated by the operation of the Link Aggregation Control Protocol (LACP) in each of the systems they connect. To ensure that the candidate links for a given LAG do connect the same pair of systems, LACP exchanges a System Identifier that is a combination of System Priority and System MAC Address. This System MAC Address needs to be unique amongst any set system capable of aggregating links with each other, but does not have to be globally unique and is not necessarily (except for any conditions imposed to avoid the profligate assignment of unique identifiers) the address used as a source MAC address by transmitted frames originating from the system. Other LACP parameters, because of potential system to system differences, can contribute to system fingerprinting though not in such a clear way. LACPDUs are transmitted to a group address selected to limit their propagation within the network, typically the Nearest non-TPMR Bridge group address (01-80-C2-00-00-03).

When MACsec is used to protect communication between neighboring systems, the MAC Security Entity is instantiated in the interface stack associated with each of the individual aggregatable links (see 11.5) and thus can confidentiality protect both the conversations carried over those links and operation of LACP.

I.6.8 IEEE Std 802.1BA Audio Video Bridging (AVB) Systems

IEEE Std 802.1BA specifies the selection of specific features and options from IEEE Std 802.1Q, IEEE Std 802.1AS, and LAN MAC/PHY standards that facilitate manufacture of AVB-capable components. A person not skilled in networking can use those components to build networks that provide working audio and video services. This standard does not introduce additional privacy considerations beyond those inherent in the referenced standards.

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I.6.9 IEEE Std 802.1BR Virtual Bridged Local Area Networks—Bridge Port Extension

IEEE Std 802.1BR specifies a method for increasing the effective geographical extent of the control parameters of a single bridge by supporting multiple instances of the Enhanced Internal Sublayer Service, each associated with a single bridge port, over a single LAN connected to an External Bridge Port Extender that can support one or more ports attached to LANs, each serving a single end station, and zero or more ports connected to further External Bridge Port Extenders. Bridge Port Extenders also support frame replication for multicast. While Bridge Port Extenders extend the effective extent of a single bridge they do require port extender specific configuration to support time-sensitive network flows.

Bridge Port Extenders were standardized to meet data center bridging requirements and are not expected to be personal devices or to provide services directly to personal devices.

I.6.10 IEEE Std 802.1CB Frame Replication and Elimination for Reliability

Frame Replication and Elimination for Reliability (FRER) increases the probability that any given packet will be delivered by replicating each of an identifiable sequence of packets, transmitting the replicates on disjoint network paths, and eliminating duplicates where those paths meet. The sequence of replication, replicate transmission, and duplicate elimination, can be repeated between transmission by the original source of the packets and reception by the eventual destination(s). Resources can be reserved on each of the paths that support duplicate transmission so that TSN delivery objectives (timeliness and extremely low loss) can be met even if LANs or relay systems fail (on all but one of the potential paths between the original source and a destination).

FRER requires, at a minimum, the addition of a sequence number to each packet. IEEE Std 802.1CB specifies a redundancy tag (R-TAG) that adds just that sequence number, and also allows use of the High-availability Seamless Redundancy (HSR) sequence tag or the Parallel Redundancy Protocol (PRP) sequence trailer both specified by IEC 62439-3:2016 (7.8, 7.9, and 7.10 of IEEE Std 802.1CB-2016). None of these contribute significantly to an adversary’s ability to assign each packet to a stream or flow as is necessary, using the contents of other frame fields, by bridges and end stations supporting resource allocation and FRER. IEEE Std 802.1CB does not specify positioning of its processing relative to that carried out by the IEEE Std 802.1AE MAC Security Entity in interface stacks, but the usual considerations place the latter closer to the PHY. MACsec confidentiality protection, where used, applies to the FRER tags and trailer just as it would to the stream and flow identifying frame fields.

While IEEE Std 802.1CB addresses the requirements addressing from industrial networks it can be used to support personal devices.

I.6.11 IEEE Std 802.1CM Time-Sensitive Networking for Fronthaul

IEEE Std 802.1CM specifies the selection of specific features and options from IEEE Std 802.1Q, IEEE Std 802.1AC, IEEE Std 802.3, IEEE Std 1588, ITU-T G.8275.1, ITU-T G.8261, ITU-T G.8262, ITU-T G.8262.1, and ITU-T G.8264, to enable the transport of time-sensitive fronthaul streams in Ethernet bridged networks. This standard does not introduce additional privacy considerations beyond those inherent in the referenced standards.

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Annex J

(informative)

Ascon-XPN-128 Cipher Suite test vectors

This annex provides test vectors for the Ascon-XPN-128 Cipher Suite (14.9). Each shows an unprotected frame that could be transmitted as a result of a MAC Service request and corresponding MACsec protected frames (with a given set of MACsec SecY parameters). The notation specified in Clause 14 (Cipher Suites) and NIST SP 800-232 is used. MACsec header fields are specified in Clause 9.

Test vectors are provided for a number of unprotected frame sizes, including (for each size) the following:

— Integrity protection without confidentiality, with a 4-octet SecTAG (optional SCI field omitted).
— Integrity and confidentiality protection, with a 4-octet SecTAG (optional SCI field omitted).
— Integrity protection without confidentiality, with a 12-octet SecTAG (SCI field included).
— Integrity and confidentiality protection, with a 12-octet SecTAG (SCI field included).

NOTE—While the optional SCI field in the SecTAG is not included in the Cipher Suite associated data (A, 14.9), its presence or absence is indicated by the SC bit in the third octet of the SecTAG which is included in *A*. The SCI itself is included in the nonce (*N*, 14.9, 14.9) so the calculated ICVs for the four cases described reflect both the SCI value and its inclusion or not in the SecTAG.

The first block of associated data, *A*, comprises the 16 octets of MAC DA, SA, and the first 4 octets of the SecTAG (MACsec EtherType, TCI, AN, SL) for all cases: a single Ascon block (without padding).

Very short frames can be of interest for sensor applications. At least one octet of user data is present for all cases, and unprotected frames can include up to 16 octets of user data in a single Ascon block (padded if less than 16) : for a total of one 12-round permutation [initialization incorporating the Key (*K*) and nonce (N)], plus two 8-round permutations for the following blocks. No additional Ascon permutations are required if the SCI is explicitly included in the SecTAG. Test vectors are provided for 27-octet unprotected frames (15 octets of user data, necessitating a single octet pad in the last block) and 28-octet unprotected frame (16 octets of user data, no padding required).

Two sets of test vectors correspond to common sizes of internet packets, 54 octets and 60 octets—two common representations of a TCP/IP SYN packet. A TCP SYN comprises 40 octets plus 14 octets of MAC DA+SA+EtherType. The frame could be padded to 60 octets to meet minimum Ethernet frame length requirements prior to MACsec processing.

One set of test vectors, for a 61-octet unprotected frame address the case where the SL field is 0 and 15 octets of padding are required for the 49 octets of user data. The case of SL 0 and no padding is covered by the 60 octet frame.

The addition of unprotected frame fields after MACsec protection of the frame, whether these are added prior to the MAC Destination Address by media access method specific methods or a provider service supporting the MAC Service interfaces [such as a Provider Backbone Bridged Network, PBNN (IEEE Std 802.1Q)] or between the MAC SA and the MACsec EtherType [by, for example, an EDE--CC or EDE-CS (Clause 15) or a Provider Bridge Network, PBN, (IEEE Std 802.1Q)] does not change the protected user data on the protected frames ICV. The transmitting and receiving SecYs are protocol peers, and the added octets will be removed prior to validation by the receiving SecY.

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All the test cases in this Annex use the extended Packet Number and the MKA parameter values shown in Table J-1. These parameters contribute to the Cipher Suite nonce (14.9.3). The test case values match those given in the Table 14-2 Nonce construction example, and have been selected to show the range of each parameter's potential contribution: there is, for example, no suggestion that such a high Key Number value is likely to be used.

Table J-1—Test case parameters

SecY parameters	Value (applicable to protected frame) *
Key	AD7A 2BD0 3EAC 835A 6F62 0FDC B506 B345
PN	0000 0025 76D4 57ED
SCI	68F2 E776 96CE 0001 †
Nonce prior to Salt ‡	0100 CE96 76E7 F268 0000 0025 76D4 57ED
Key Number (KN)	0001 2853
Key Server MI	E630 E81A 48DE 85B4 6A21 C66F
Salt §	6B21 C66F E630 E81A 608D 85B4 6A21 C66F
Nonce **	6A21 08F9 90D7 1A72 608D 8591 1CF5 9182

* Integers, hexadecimal, most significant digits to the left, leading zero padding to notional size, spaces added to break into 16-bit blocks for readability.

† Comprising a MAC Address with a Port Identifier of 0001.

‡ Calculated from the PN and SCI.

§ Calculated from the KN and Key Server MI.

** Calculated from the Nonce prior to Salt and the Salt.

All the test cases also use an Association Number (AN) of 1, identifying the SA used to protect the frames, and hence (indirectly) the SAK. This value is used in these test vectors to guard against accidental bit-reversal of the TCI and AN octet in the SecTAG [bit 8, the version number bit, V, is 0 (9.5)]. The values used in the TCI and AN octet in these test vectors are:

- — 0x01 Integrity protection, SCI omitted, with AN 1.
- — 0x0D Integrity and confidentiality, SCI omitted, with AN 1.
- — 0x21 Integrity protection, SCI included, with AN 1.
- — 0x2D Integrity and confidentiality, SCI included, with AN 1.

Note—The test vector cases do not include setting of the ES (End Station) bit in the TCI field, which would be reflected in the TCI and AN octet as 0x41 (integrity, SCI necessarily omitted, AN 1) or 0x4D (integrity and confidentiality, SCI necessarily omitted, AN 1). This omission is not intended to deprecate the use of this capability, the test vector coverage of field omission or reversal is believed to be adequate without its inclusion.

Test vector results are 'to be supplied'. All 'results' coloured as per this reminder are temporary placeholders.

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J.1 27-octet frame examples

Table J-2—Unprotected frame (27-octet example)

Field	Value *
MAC DA	01 80 C2 00 00 0E
MAC SA	7A 0D 46 DF 99 8D
MSDU (User Data)	88 CC 01 07 04 7A 0D 46 DF 99 8D 02 02 07 31

* Octets in transmission order, left to right, top to bottom, with each octet in hexadecimal, most significant digit to the left.

Table J-3—Integrity protected, SCI omitted (27-octet example)

	Field	Value
	MAC DA	01 80 C2 00 00 0E
	MAC SA	7A 0D 46 DF 99 8D
SecTAG	MACsec EtherType	88 E5
	TCI and AN	01
	SL	0F
	PN (ls 32 bits)	76 D4 57 ED
	MSDU (Secure Data)	88 CC 01 07 04 7A 0D 46 DF 99 8D 02 02 07 31
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

Table J-4—Integrity and confidentiality protected, SCI omitted (27-octet example)

	Field	Value
	MAC DA	01 80 C2 00 00 0E
	MAC SA	7A 0D 46 DF 99 8D
SecTAG	MACsec EtherType	88 E5
	TCI and AN	0D
	SL	0F
	PN	76 D4 57 ED
	MSDU (Secure Data)	88 CC 01 07 04 7A 0D 46 DF 99 8D 02 02 07 31
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

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Table J-5—Integrity protected frame, SCI included (27-octet example)

	Field	Value
	MAC DA	01 80 C2 00 00 0E
	MAC SA	7A 0D 46 DF 99 8D
SecTAG	MACsec EtherType	88 E5
	TCI and AN	21
	SL	0F
	PN	76 D4 57 ED
	SCI	68 F2 E7 76 96 CE 00 01
	MSDU (Secure Data)	88 CC 01 07 04 7A 0D 46 DF 99 8D 02 02 07 31
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

Table J-6—Integrity and confidentiality, SCI included (27-octet example)

	Field	Value
	MAC DA	01 80 C2 00 00 0E
	MAC SA	7A 0D 46 DF 99 8D
SecTAG	MACsec EtherType	88 E5
	TCI and AN	2D
	SL	0F
	PN	76 D4 57 ED
	SCI	68 F2 E7 76 96 CE 00 01
	MSDU (Secure Data)	88 CC 01 07 04 7A 0D 46 DF 99 8D 02 02 07 31
	ICV	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

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J.2 28-octet frame examples

Table J-7—Unprotected frame (28-octet example)

Field	Value *
MAC DA	01 80 C2 00 00 0E
MAC SA	F0 76 1E 8D CD 3D
MSDU (User Data)	88 CC 01 07 04 7A 0D 46 DF 99 8D 02 03 07 30 31

* Octets in transmission order, left to right, top to bottom, with each octet in hexadecimal, most significant digit to the left.

Table J-8—Integrity protected, SCI omitted (28-octet example)

	Field	Value *
	MAC DA	01 80 C2 00 00 0E
	MAC SA	F0 76 1E 8D CD 3D
SecTAG	MACsec EtherType	88 E5
	TCI and AN	01
	SL	10
	PN (ls 32 bits)	76 D4 57 ED
	MSDU (Secure Data)	88 CC 01 07 04 7A 0D 46 DF 99 8D 02 03 07 30 31
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

* Octets in transmission order, left to right, top to bottom, with each octet in hexadecimal, most significant digit to the left.

Table J-9—Integrity and confidentiality, SCI omitted (28-octet example)

	Field	Value
	MAC DA	01 80 C2 00 00 0E
	MAC SA	F0 76 1E 8D CD 3D
SecTAG	MACsec EtherType	88 E5
	TCI and AN	0D
	SL	10
	PN	76 D4 57 ED
	MSDU (Secure Data)	88 CC 01 07 04 7A 0D 46 DF 99 8D 02 03 07 30 31
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

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Table J-10—Integrity protected frame, SCI included (28-octet example)

	Field	Value
	MAC DA	01 80 C2 00 00 0E
	MAC SA	F0 76 1E 8D CD 3D
SecTAG	MACsec EtherType	88 E5
	TCI and AN	21
	SL	10
	PN	76 D4 57 ED
	SCI	68 F2 E7 76 96 CE 00 01
	MSDU (Secure Data)	88 CC 01 07 04 7A 0D 46 DF 99 8D 02 03 07 30 31
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

Table J-11—Integrity and confidentiality, SCI included (28-octet example)

	Field	Value
	MAC DA	01 80 C2 00 00 0E
	MAC SA	F0 76 1E 8D CD 3D
SecTAG	MACsec EtherType	88 E5
	TCI and AN	2D
	SL	10
	PN	76 D4 57 ED
	SCI	68 F2 E7 76 96 CE 00 01
	MSDU (Secure Data)	88 CC 01 07 04 7A 0D 46 DF 99 8D 02 03 07 30 31
	ICV	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

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J.3 54-octet frame examples

Table J-12—Unprotected frame (54-octet example)

Field	Value *
MAC DA	84 C5 D5 13 D2 AA
MAC SA	F6 E5 BB D2 72 77
MSDU (User Data)	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 00 01

* Octets in transmission order, left to right, top to bottom, with each octet in hexadecimal, most significant digit to the left.

Table J-13—Integrity protected, SCI omitted (54-octet example)

	Field	Value *
	MAC DA	84 C5 D5 13 D2 AA
	MAC SA	F6 E5 BB D2 72 77
SecTAG	MACsec EtherType	88 E5
	TCI and AN	01
	SL	2A
	PN (ls 32 bits)	76 D4 57 ED
	MSDU (Secure Data)	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 00 01
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

* Octets in transmission order, left to right, top to bottom, with each octet in hexadecimal, most significant digit to the left.

Table J-14—Integrity and confidentiality, SCI omitted (54-octet example)

	Field	Value
	MAC DA	84 C5 D5 13 D2 AA
	MAC SA	F6 E5 BB D2 72 77
SecTAG	MACsec EtherType	88 E5
	TCI and AN	0D
	SL	2A
	PN	76 D4 57 ED
	MSDU (Secure Data)	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 00 01
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

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Table J-15—Integrity protected frame, SCI included (54-octet example)

	Field	Value
	MAC DA	84 C5 D5 13 D2 AA
	MAC SA	F6 E5 BB D2 72 77
SecTAG	MACsec EtherType	88 E5
	TCI and AN	21
	SL	2A
	PN	76 D4 57 ED
	SCI	68 F2 E7 76 96 CE 00 01
	MSDU (Secure Data)	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 00 01
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

Table J-16—Integrity and confidentiality, SCI included (54-octet example)

	Field	Value
	MAC DA	84 C5 D5 13 D2 AA
	MAC SA	F6 E5 BB D2 72 77
SecTAG	MACsec EtherType	88 E5
	TCI and AN	2D
	SL	2A
	PN	76 D4 57 ED
	SCI	68 F2 E7 76 96 CE 00 01
	MSDU (Secure Data)	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 00 01
	ICV	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

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J.4 60-octet frame examples

Table J-17—Unprotected frame (60-octet example)

Field	Value *
MAC DA	D6 09 B1 F0 56 63
MAC SA	7A E8 E2 CA 4E C5
MSDU (User Data)	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 00 03

* Octets in transmission order, left to right, top to bottom, with each octet in hexadecimal, most significant digit to the left.

Table J-18—Integrity protected, SCI omitted (60-octet example)

	Field	Value *
	MAC DA	D6 09 B1 F0 56 63
	MAC SA	7A E8 E2 CA 4E C5
SecTAG	MACsec EtherType	88 E5
	TCI and AN	01
	SL	00
	PN (ls 32 bits)	76 D4 57 ED
	MSDU (Secure Data)	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 00 03
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

* Octets in transmission order, left to right, top to bottom, with each octet in hexadecimal, most significant digit to the left.

Table J-19—Integrity and confidentiality, SCI omitted (60-octet example)

	Field	Value
	MAC DA	D6 09 B1 F0 56 63
	MAC SA	7A E8 E2 CA 4E C5
SecTAG	MACsec EtherType	88 E5
	TCI and AN	0D
	SL	00
	PN	76 D4 57 ED
	MSDU (Secure Data)	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 00 03
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

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Table J-20—Integrity protected frame, SCI included (60-octet example)

	Field	Value
	MAC DA	D6 09 B1 F0 56 63
	MAC SA	7A E8 E2 CA 4E C5
SecTAG	MACsec EtherType	88 E5
	TCI and AN	21
	SL	00
	PN	76 D4 57 ED
	SCI	68 F2 E7 76 96 CE 00 01
	MSDU (Secure Data)	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 00 03
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

Table J-21—Integrity and confidentiality, SCI included (60-octet example)

	Field	Value
	MAC DA	D6 09 B1 F0 56 63
	MAC SA	7A E8 E2 CA 4E C5
SecTAG	MACsec EtherType	88 E5
	TCI and AN	2D
	SL	00
	PN	76 D4 57 ED
	SCI	68 F2 E7 76 96 CE 00 01
	MSDU (Secure Data)	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 00 03
	ICV	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

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J.5 61-octet frame examples

Table J-22—Unprotected frame (61-octet example)

Field	Value
MAC DA	E2 01 06 D7 CD 0D
MAC SA	68 F2 E7 76 96 CE
User Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 3B 00 06

Table J-23—Integrity protected frame (61-octet example)

Field	Value
MAC DA	E2 01 06 D7 CD 0D
MAC SA	68 F2 E7 76 96 CE
MACsec EtherType	88 E5
TCI and AN	01
SL	00
PN	76 D4 57 ED
Secure Data	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 3B 00 06
ICV	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

Table J-24—Integrity and confidentiality, SCI omitted (61-octet example)

	Field	Value
	MAC DA	D6 09 B1 F0 56 63
	MAC SA	68 F2 E7 76 96 CE
SecTAG	MACsec EtherType	88 E5
	TCI and AN	0D
	SL	00
	PN	76 D4 57 ED
	MSDU (Secure Data)	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 3B 00 06
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

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Table J-25—Integrity protected frame, SCI included (61-octet example)

	Field	Value
	MAC DA	D6 09 B1 F0 56 63
	MAC SA	68 F2 E7 76 96 CE
SecTAG	MACsec EtherType	88 E5
	TCI and AN	21
	SL	00
	PN	76 D4 57 ED
	SCI	68 F2 E7 76 96 CE 00 01
	MSDU (Secure Data)	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 3B 00 06
	ICV (*T*)	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

Table J-26—Integrity and confidentiality, SCI included (61-octet example)

	Field	Value
	MAC DA	D6 09 B1 F0 56 63
	MAC SA	68 F2 E7 76 96 CE
SecTAG	MACsec EtherType	88 E5
	TCI and AN	2D
	SL	00
	PN	76 D4 57 ED
	SCI	68 F2 E7 76 96 CE 00 01
	MSDU (Secure Data)	08 00 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35 36 37 38 39 3A 3B 00 06
	ICV	F0 94 78 A9 B0 90 07 D0 6F 46 E9 B6 A1 DA 25 DD

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